



US012384741B2

(12) **United States Patent**
Horhota et al.

(10) **Patent No.:** **US 12,384,741 B2**

(45) **Date of Patent:** **Aug. 12, 2025**

(54) **LIPID NANOPARTICLE COMPOSITIONS
FOR DELIVERING CIRCULAR
POLYNUCLEOTIDES**

47/18 (2013.01); *A61K 47/183* (2013.01);
A61K 47/24 (2013.01); *A61K 47/28* (2013.01);
(Continued)

(71) Applicant: **Orna Therapeutics, Inc.**, Watertown,
MA (US)

(58) **Field of Classification Search**
CPC .. *C07C 229/24*; *A61K 9/1272*; *A61K 31/7105*
See application file for complete search history.

(72) Inventors: **Allen T. Horhota**, Cambridge, MA
(US); **Junghoon Yang**, Cambridge, MA
(US); **Kevin Kauffman**, Newton, MA
(US); **Thomas Barnes**, Cambridge, MA
(US); **Robert Alexander Wesselhoeft**,
Cambridge, MA (US); **Amy M. Becker**,
Cambridge, MA (US); **Gregory Motz**,
Cambridge, MA (US)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2024/0116852 A1 4/2024 Horhota et al.
2024/0158336 A1 5/2024 Horhota et al.
2024/0189340 A1 6/2024 Horhota et al.

(73) Assignee: **Orna Therapeutics, Inc.**, Watertown,
MA (US)

FOREIGN PATENT DOCUMENTS

WO WO 2017/049245 A1 3/2017
WO WO 2018/170306 A1 9/2018
(Continued)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

OTHER PUBLICATIONS

(21) Appl. No.: **18/501,945**

PCT International Search Report and Written Opinion, PCT Appli-
cation No. PCT/US2022/049313, Mar. 17, 2023, 11 pages.

(22) Filed: **Nov. 3, 2023**

(Continued)

(65) **Prior Publication Data**

US 2024/0190808 A1 Jun. 13, 2024

Related U.S. Application Data

(63) Continuation of application No.
PCT/US2022/049313, filed on Nov. 8, 2022.

(60) Provisional application No. 63/277,055, filed on Nov.
8, 2021.

Primary Examiner — Carlos A Azpuru

(74) *Attorney, Agent, or Firm* — Christopher K. Haley;
Carl A. Morales; Goodwin Procter LLP

(51) **Int. Cl.**
C07C 229/12 (2006.01)
A61K 9/1272 (2025.01)

(Continued)

(57) **ABSTRACT**

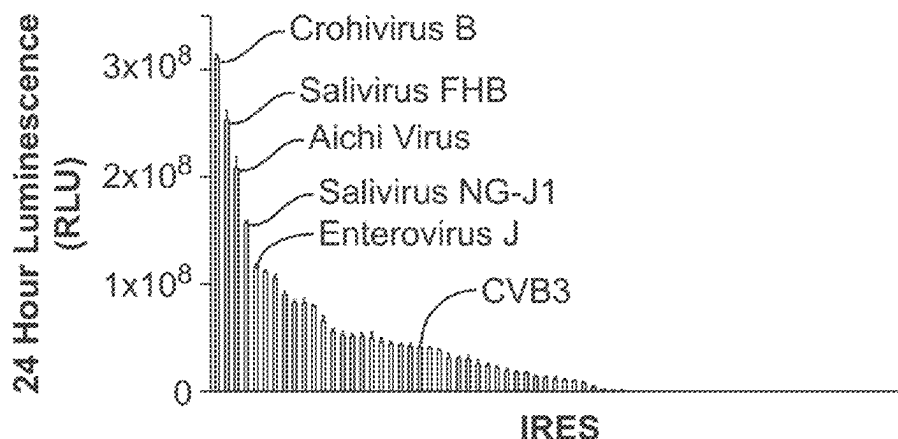
Disclosed herein are novel lipids that can be used in com-
bination with other lipid components, such as helper lipids,
structural lipids, and cholesterol, to form lipid nanoparticles
for delivery of therapeutic agents, such as nucleic acids (e.g.,
circular polynucleotides), both in vitro and in vivo.

(52) **U.S. Cl.**
CPC *C07C 229/12* (2013.01); *A61K 9/1272*
(2013.01); *A61K 31/7105* (2013.01); *A61K*
45/06 (2013.01); *A61K 47/10* (2013.01); *A61K*

28 Claims, 108 Drawing Sheets

Specification includes a Sequence Listing.

293



(51) **Int. Cl.**

A61K 31/7105 (2006.01)
A61K 45/06 (2006.01)
A61K 47/10 (2017.01)
A61K 47/18 (2017.01)
A61K 47/24 (2006.01)
A61K 47/28 (2006.01)
A61K 47/54 (2017.01)
A61K 48/00 (2006.01)
C07C 219/06 (2006.01)
C07C 229/22 (2006.01)
C07C 229/24 (2006.01)

(52) **U.S. Cl.**

CPC *A61K 47/543* (2017.08); *A61K 48/0033*
 (2013.01); *C07C 219/06* (2013.01); *C07C*
229/22 (2013.01); *C07C 229/24* (2013.01)

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

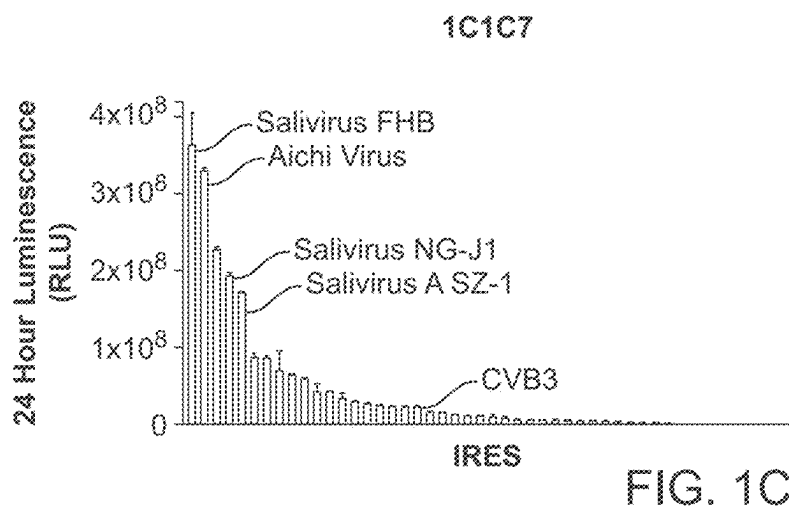
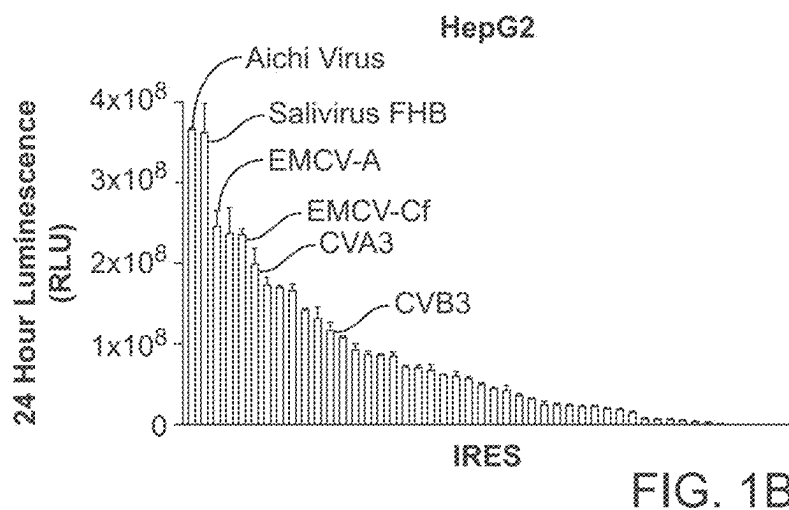
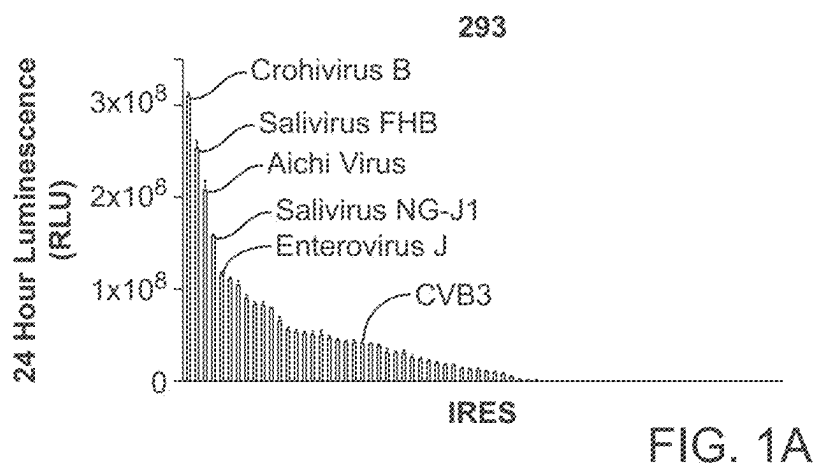
WO WO 2019/152557 A1 8/2019
 WO WO 2020/061367 A1 3/2020

WO WO 2020/257611 A1 12/2020
 WO 2021/142280 * 7/2021
 WO WO 2021/142280 A1 7/2021
 WO WO 2021/178396 A1 9/2021
 WO 2022/140252 * 6/2022
 WO WO 2022/140252 A1 6/2022
 WO 2022/247755 * 12/2022
 WO WO 2022/247755 A1 12/2022

OTHER PUBLICATIONS

PCT International Preliminary Report on Patentability, PCT Application No. PCT/US2022/049313, Feb. 23, 2024, 14 pages.
 PCT International Search Report and Written Opinion, PCT Application No. PCT/US2022/045408, Mar. 22, 2023, 20 pages.
 PCT Written Opinion of the International Preliminary Examining Authority, PCT Application No. PCT/US2022/049313, Nov. 29, 2023, eight pages.
 Zhao, X. et al. "Imidazole-Based Synthetic Lipidoids for In Vivo mRNA Delivery into Primary T Lymphocytes." *Angewandte Chemie International Edition*, vol. 59, No. 45, Nov. 2, 2020, pp. 20083-20089.

* cited by examiner



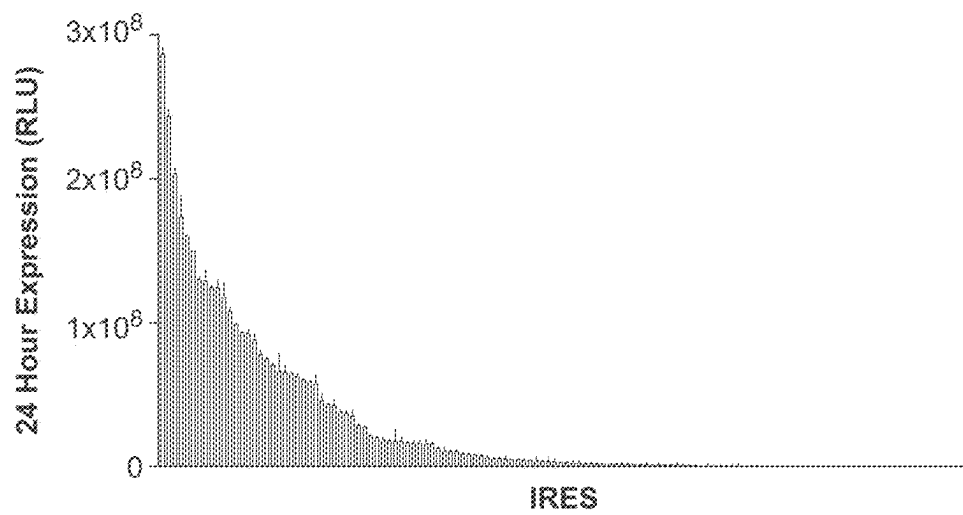


FIG. 1D

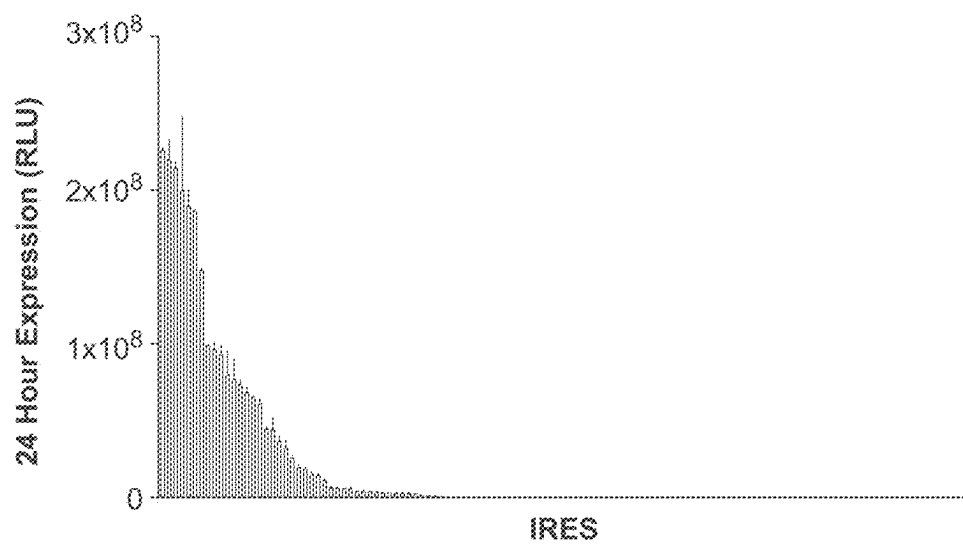


FIG. 1E

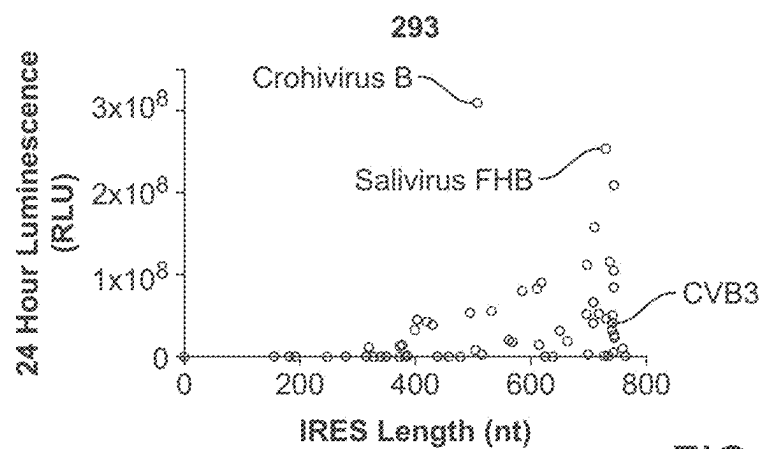


FIG. 2A

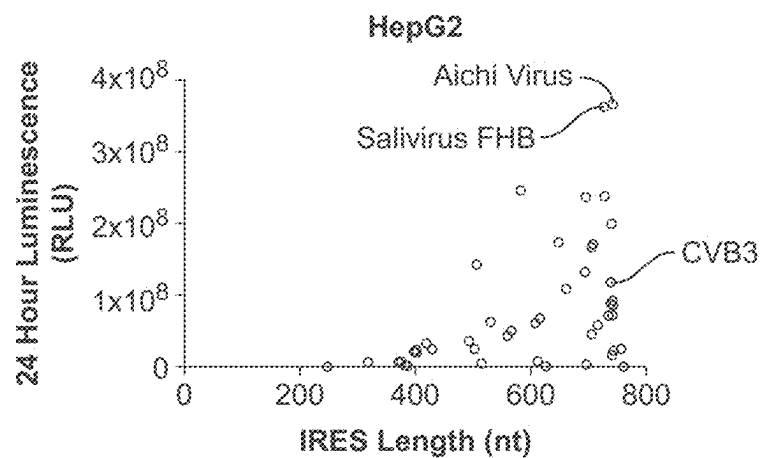


FIG. 2B

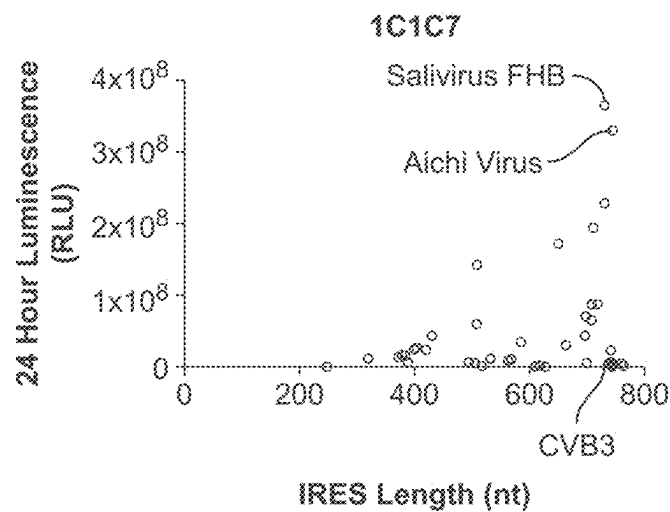


FIG. 2C

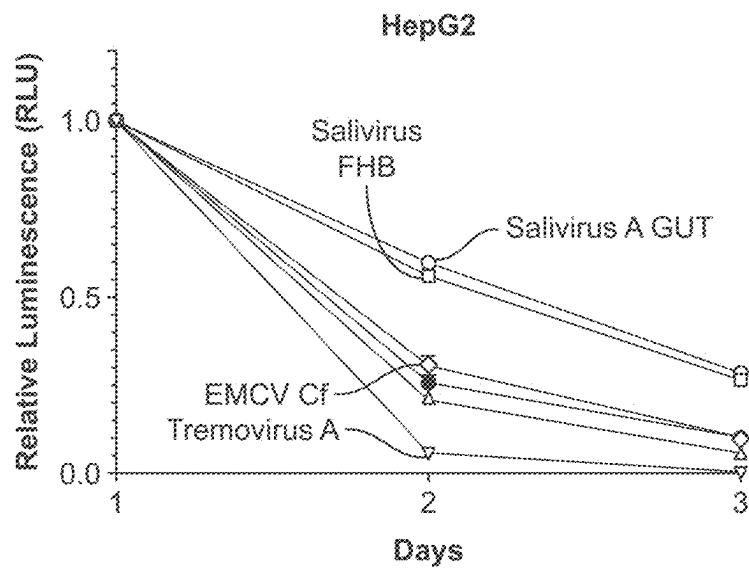


FIG. 3A

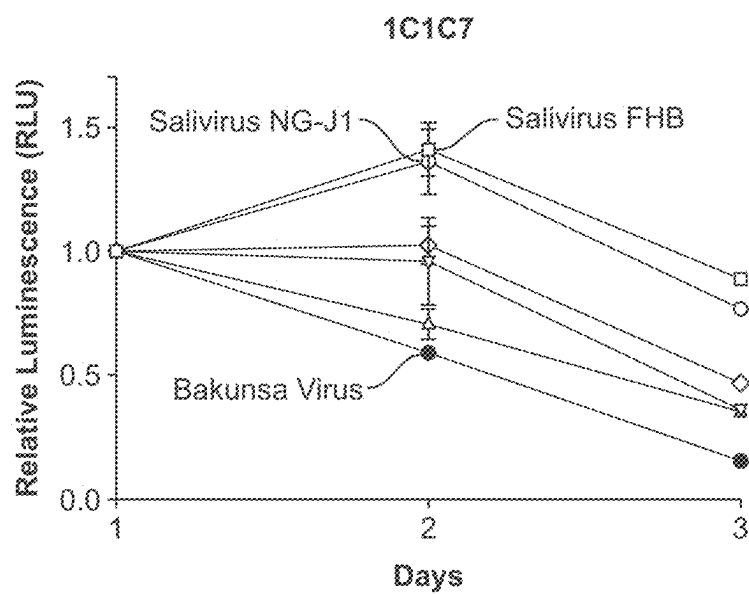


FIG. 3B

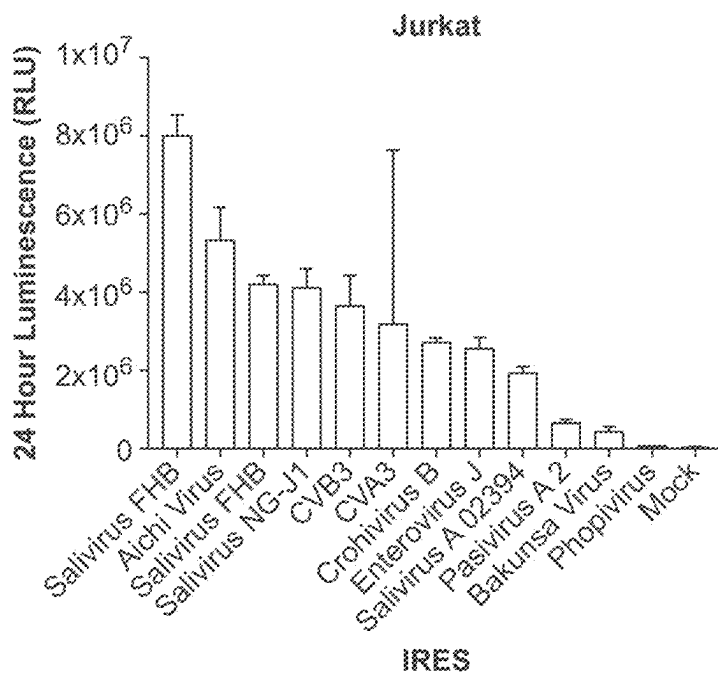


FIG. 4A

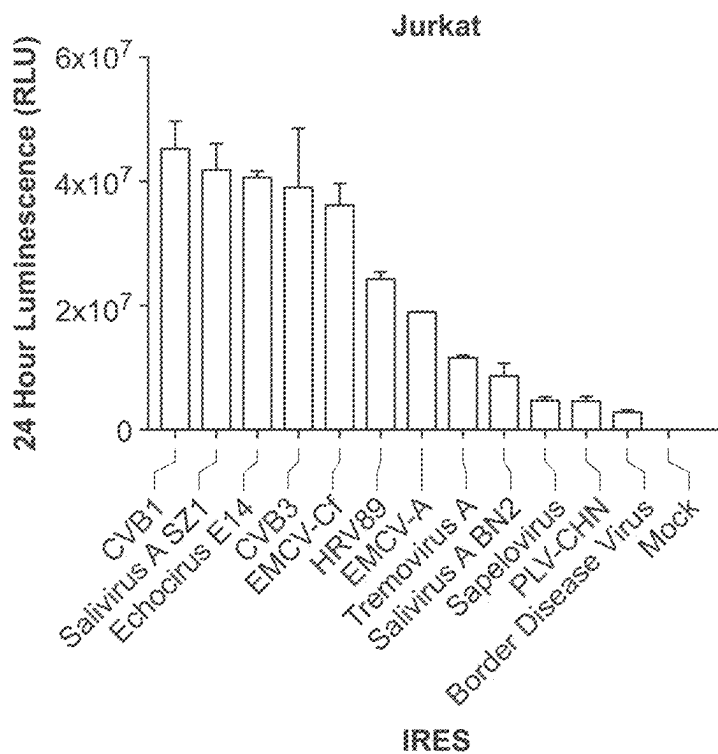


FIG. 4B

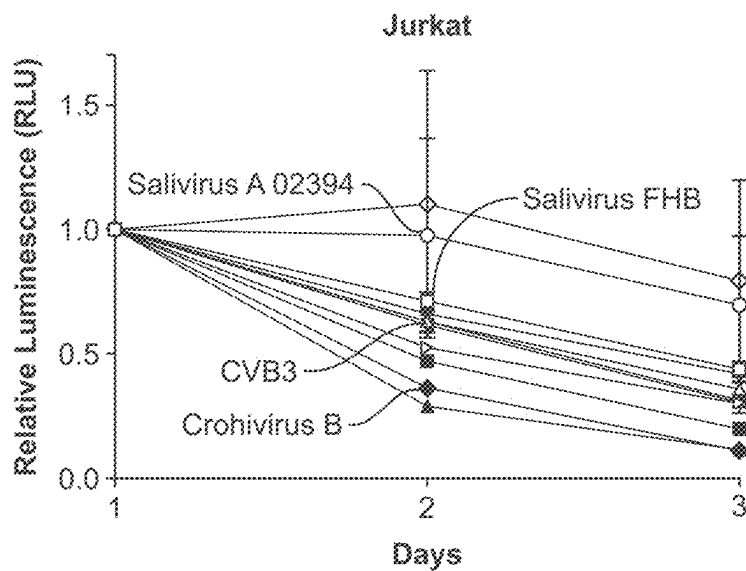


FIG. 5A

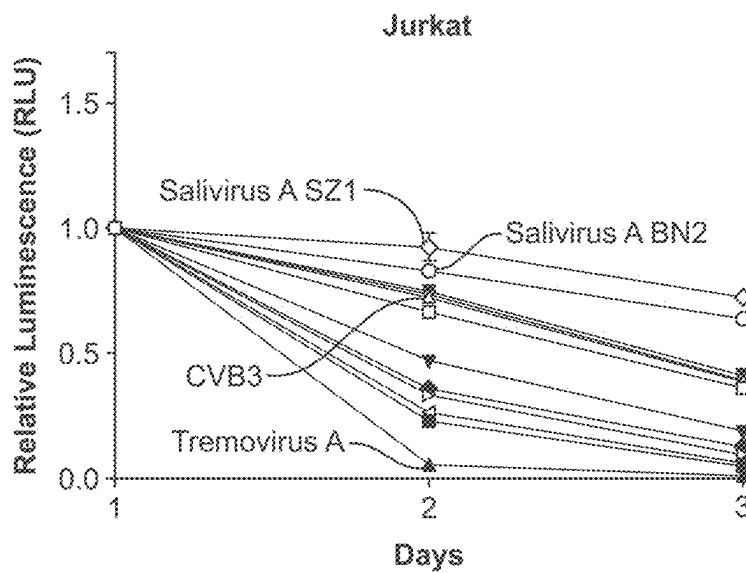


FIG. 5B

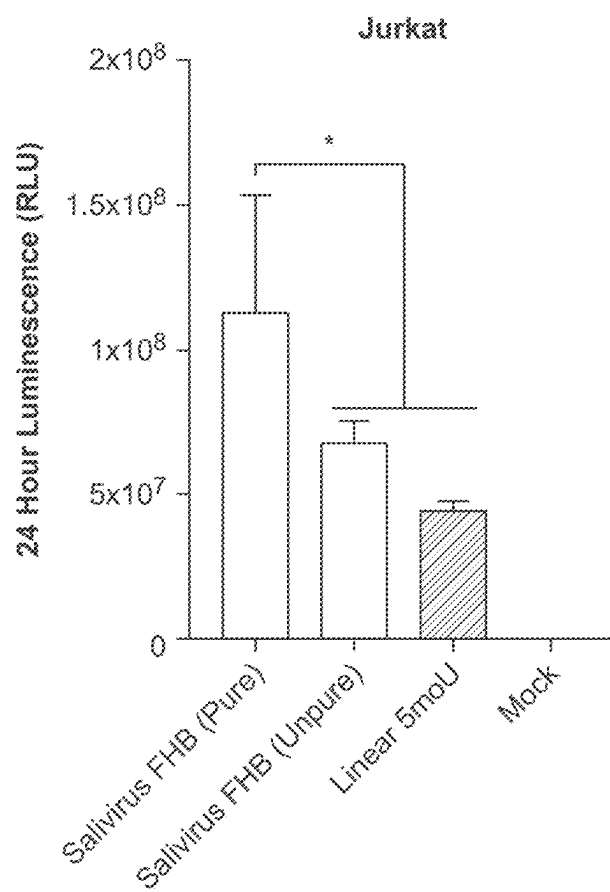


FIG. 6A

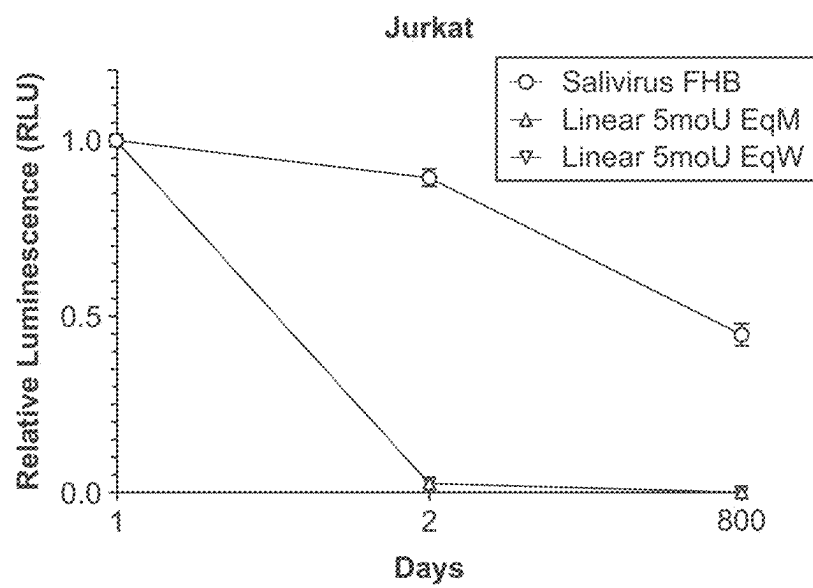


FIG. 6B

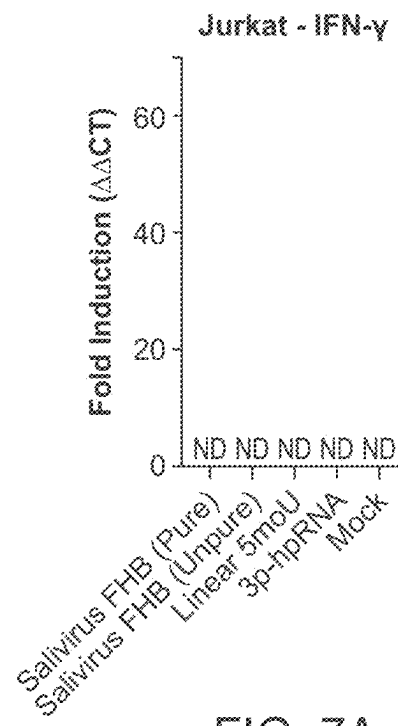


FIG. 7A

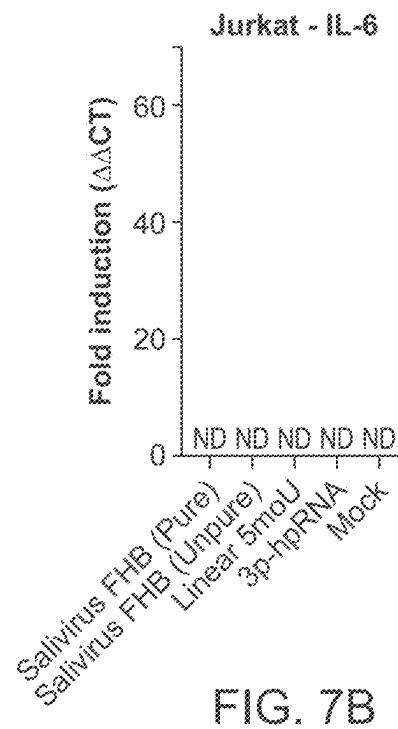


FIG. 7B

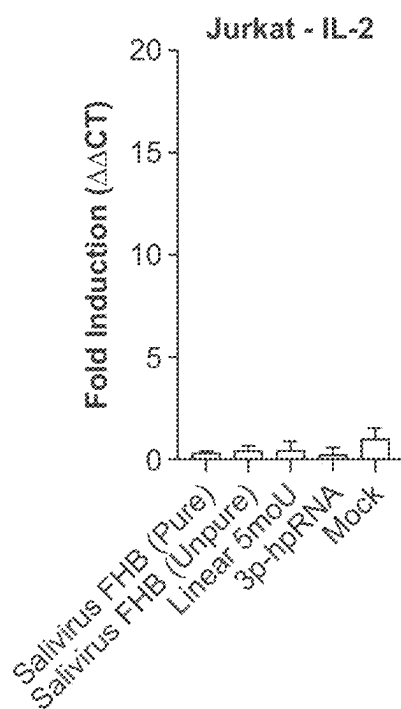


FIG. 7C

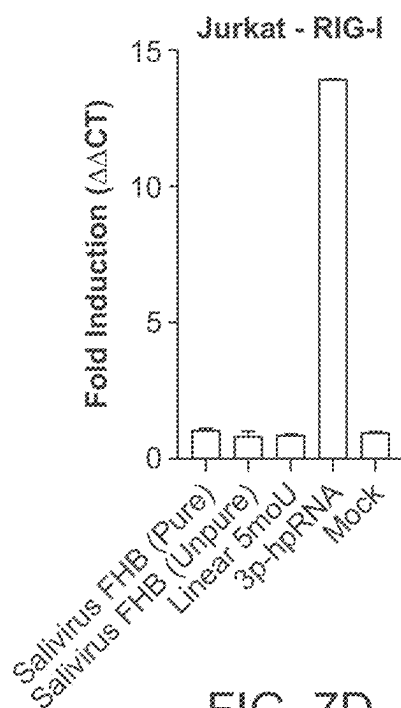


FIG. 7D

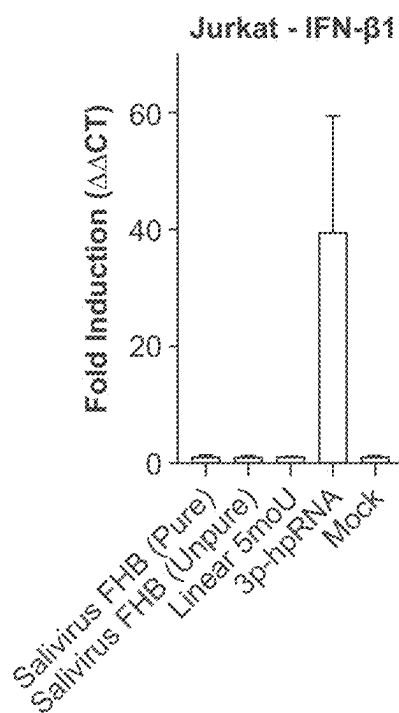


FIG. 7E

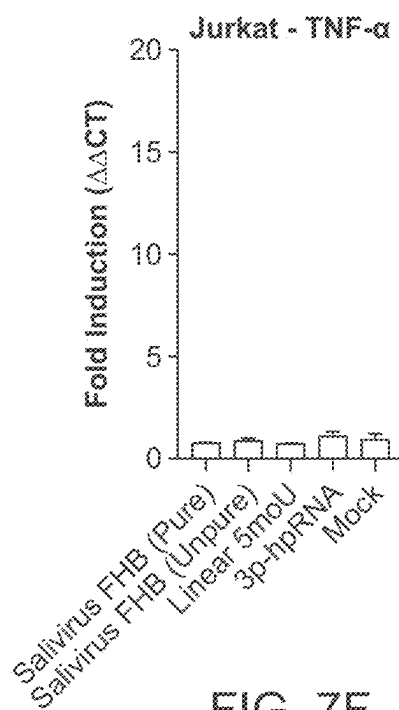


FIG. 7F

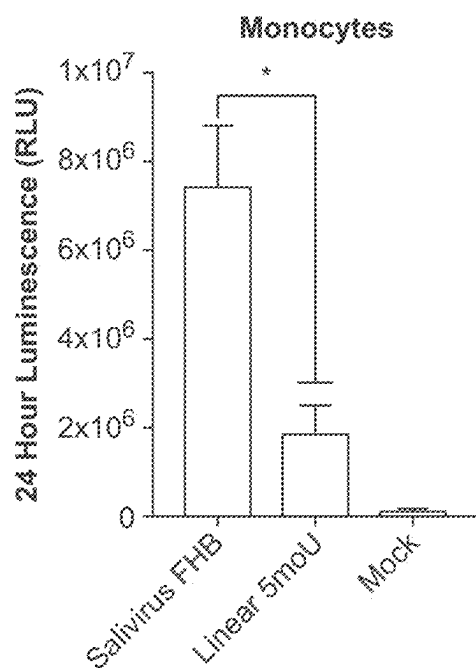


FIG. 8A

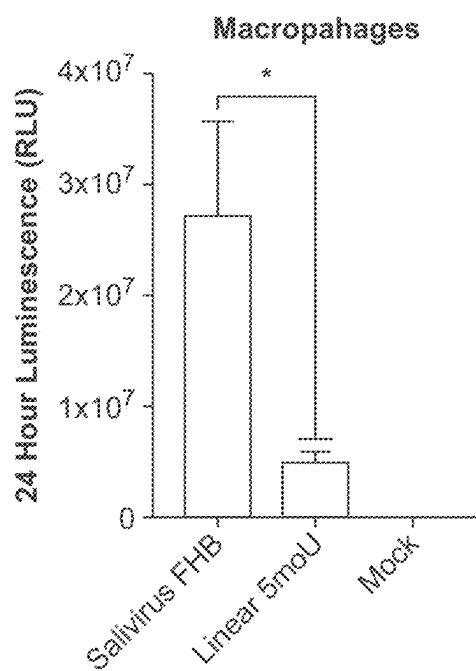


FIG. 8B

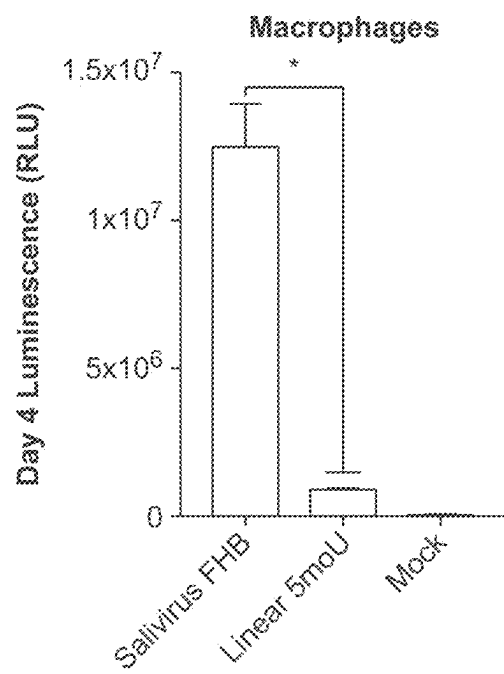


FIG. 8C

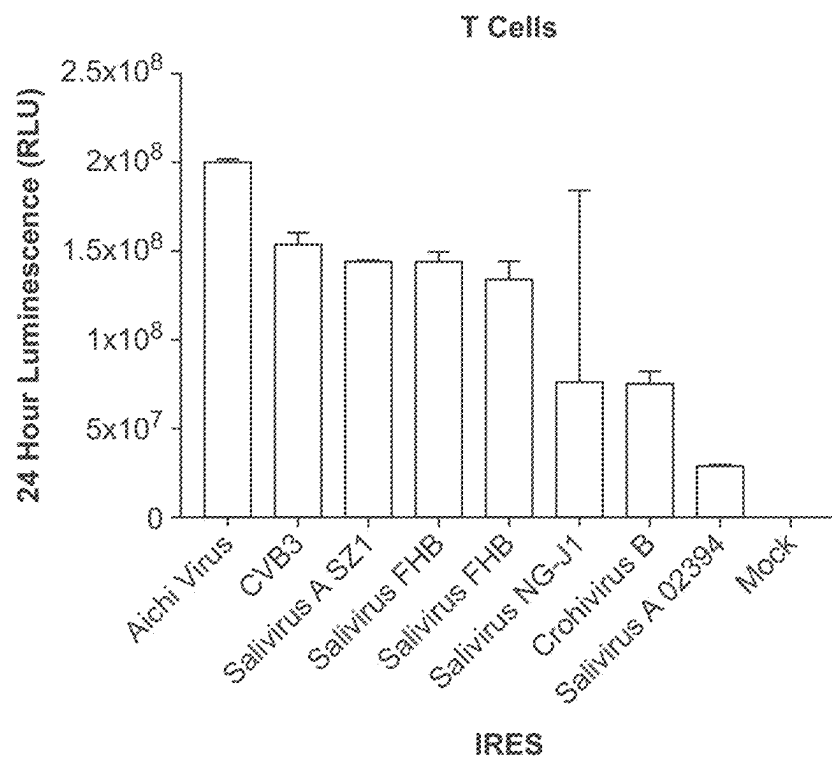


FIG. 9A

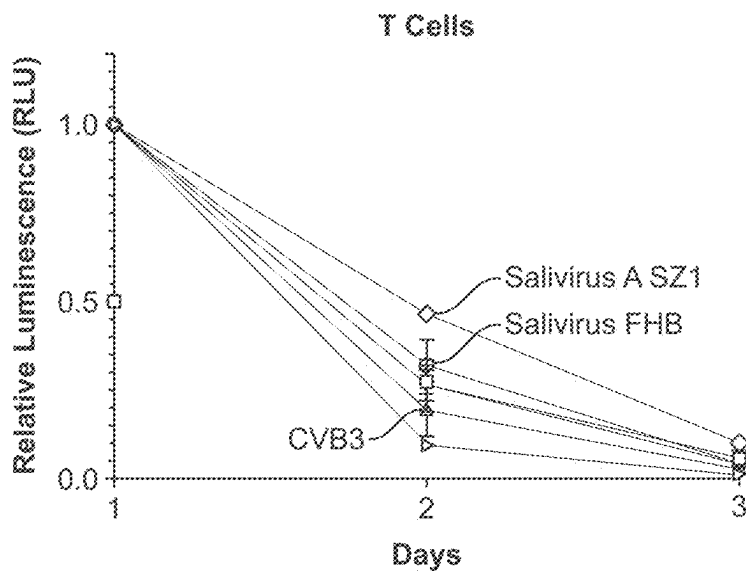


FIG. 9B

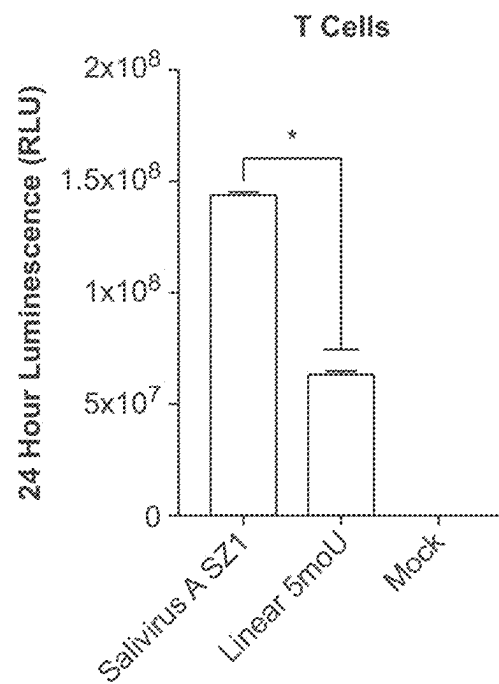


FIG. 10A

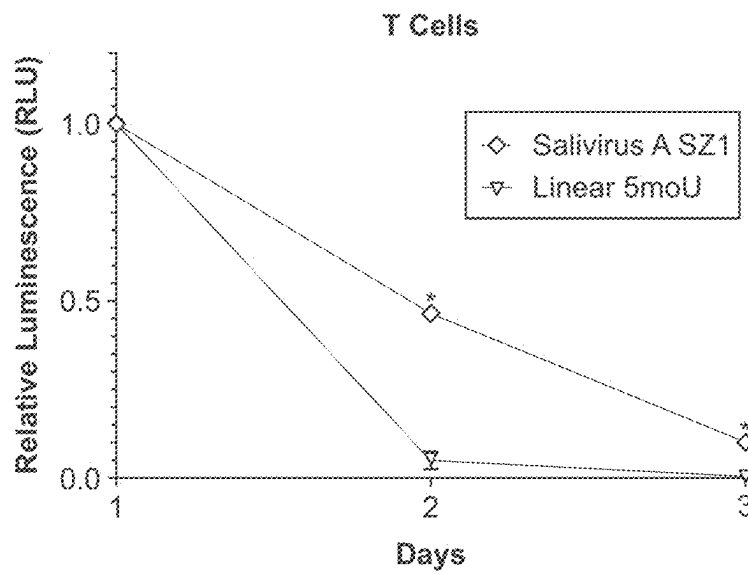
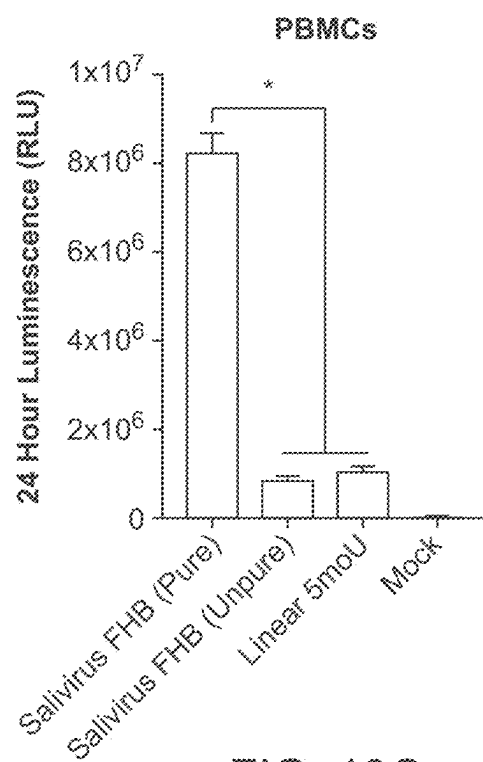


FIG. 10B



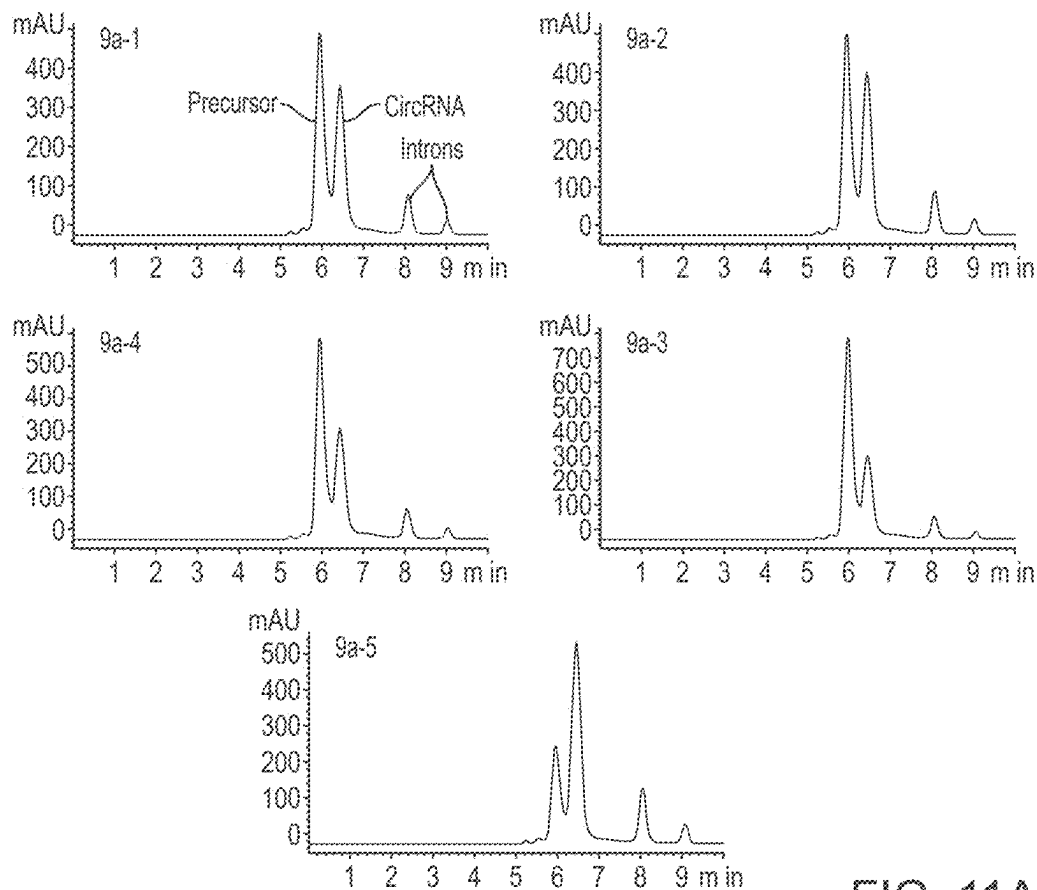


FIG. 11A

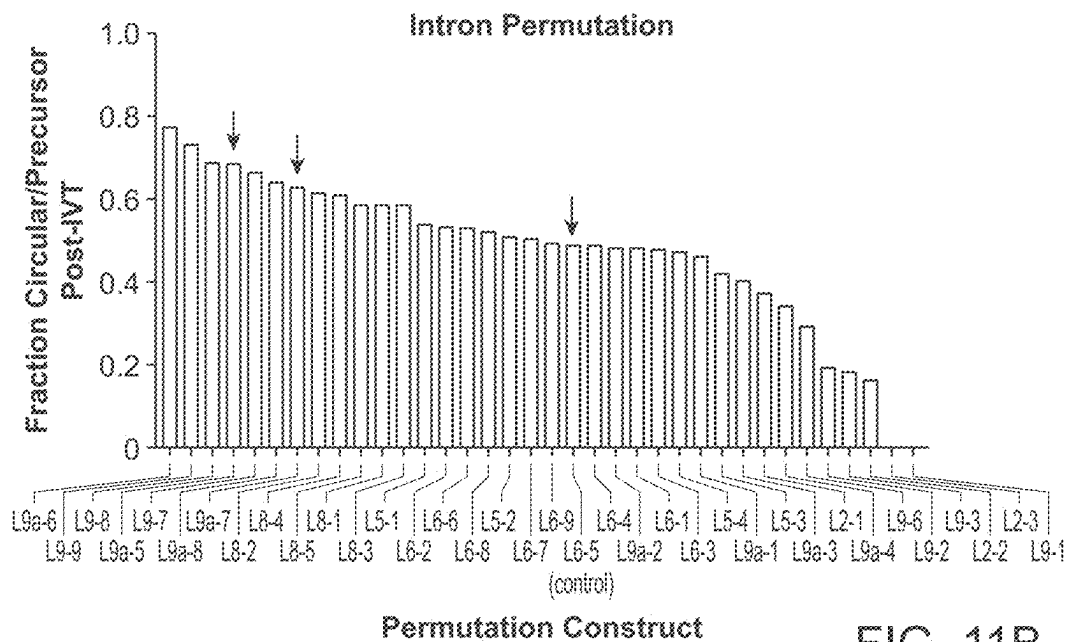


FIG. 11B

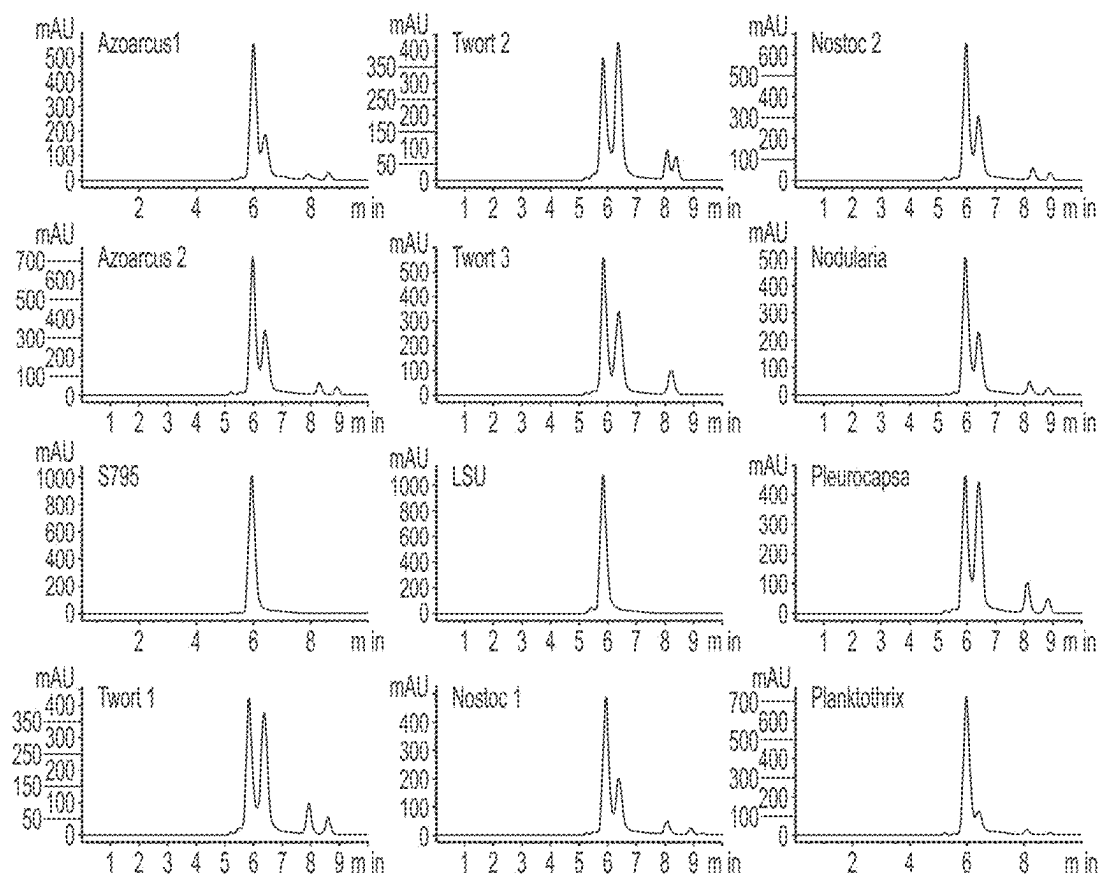


FIG. 12A

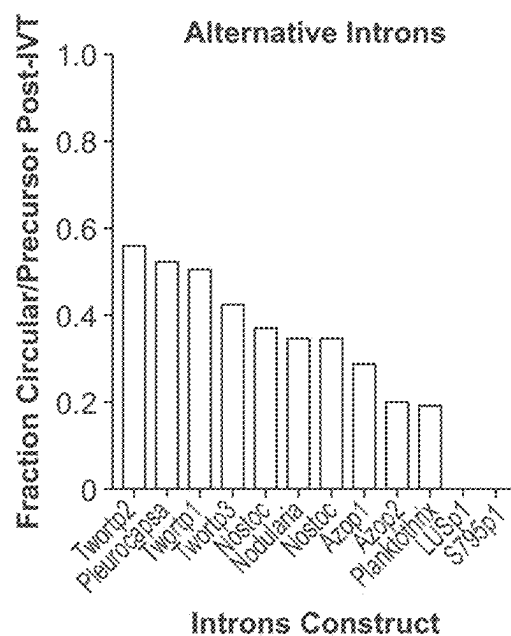


FIG. 12B

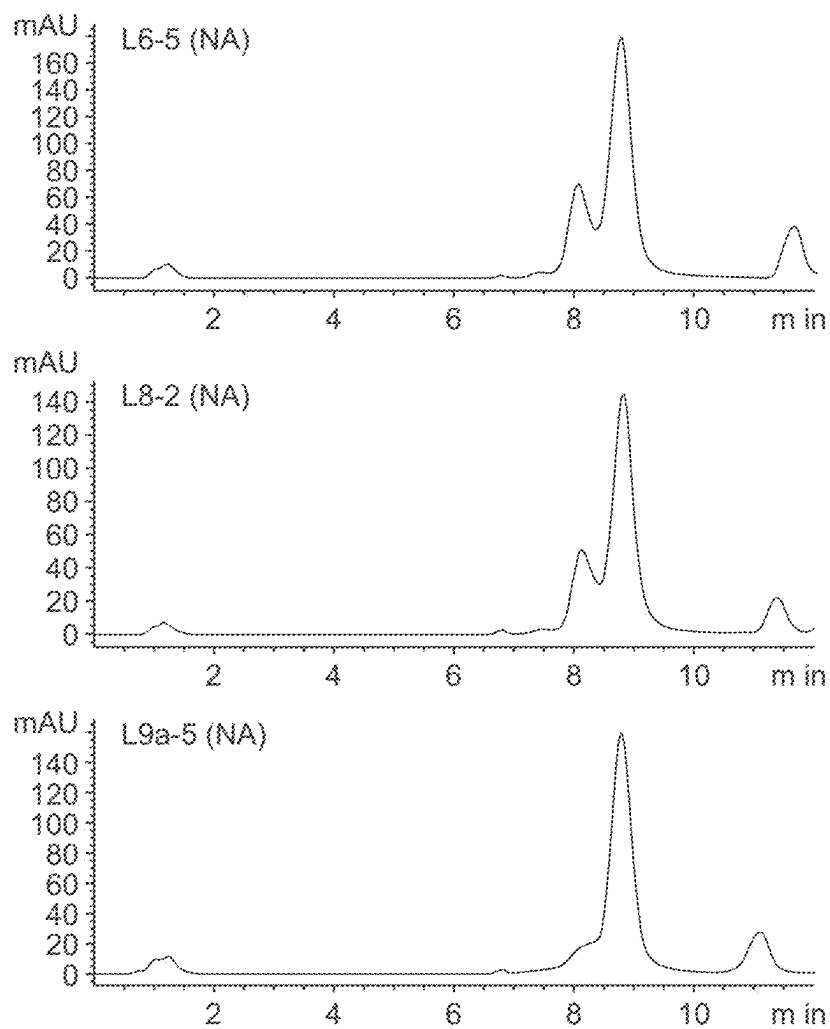


FIG. 13A

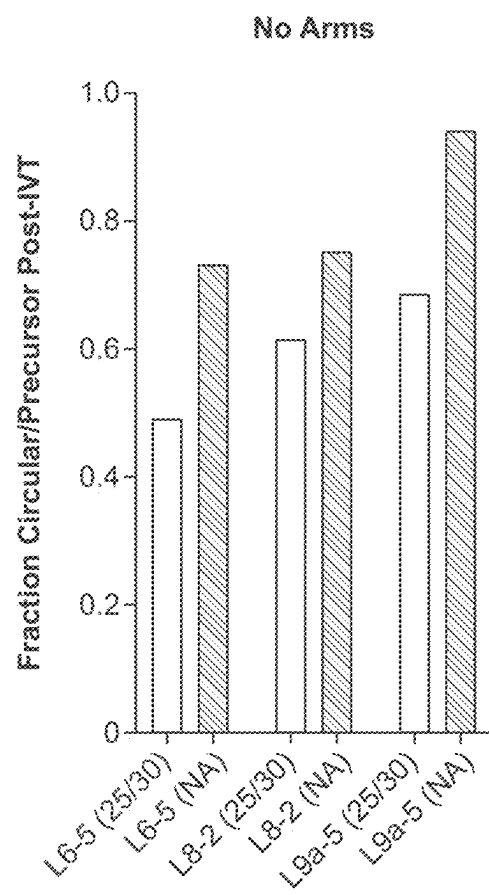


FIG. 13B

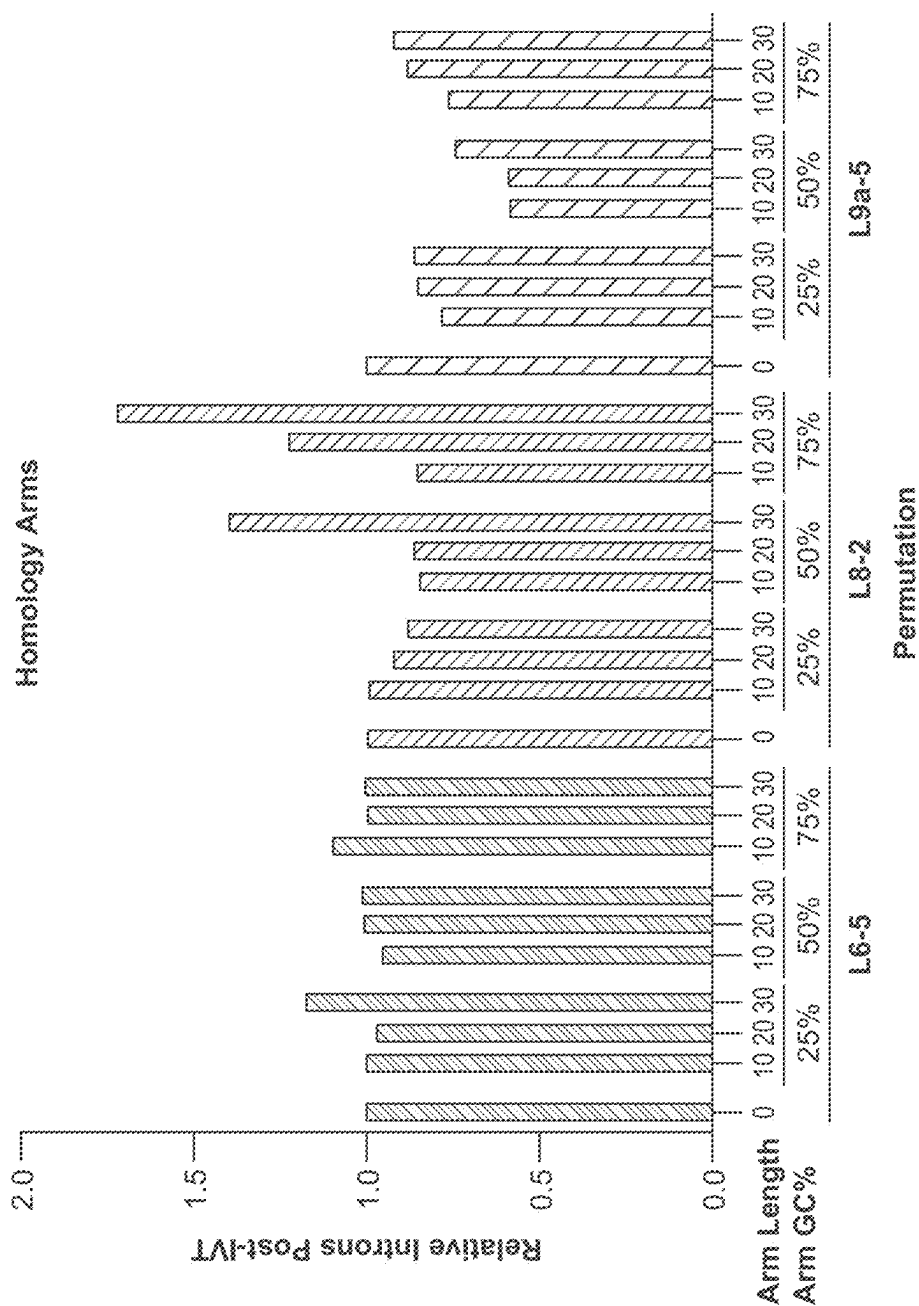


FIG. 14

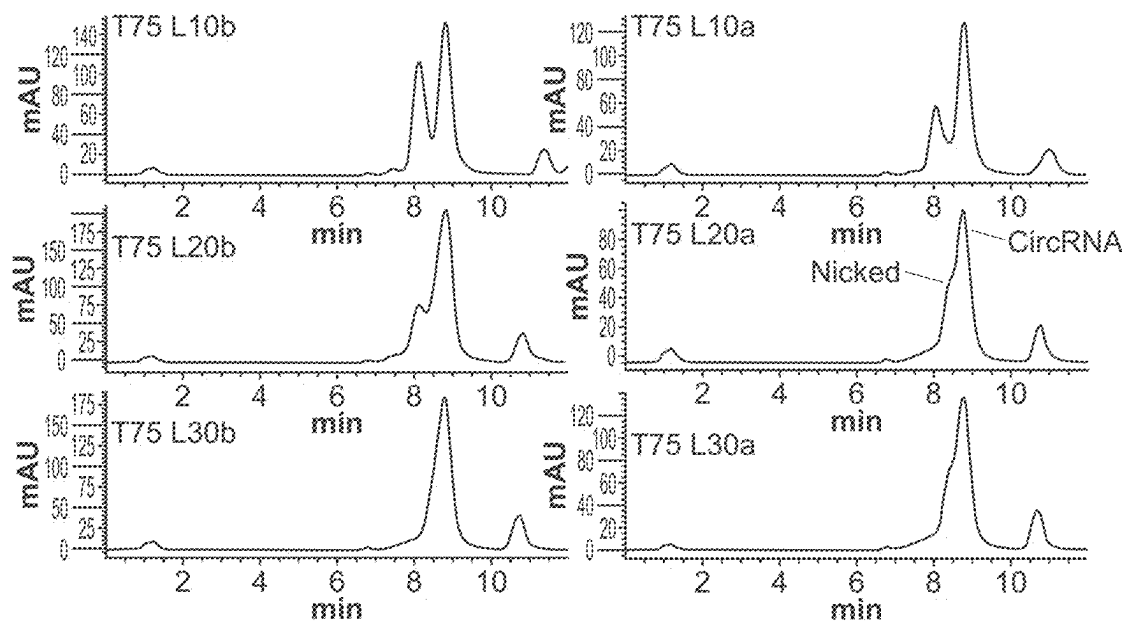


FIG. 15A

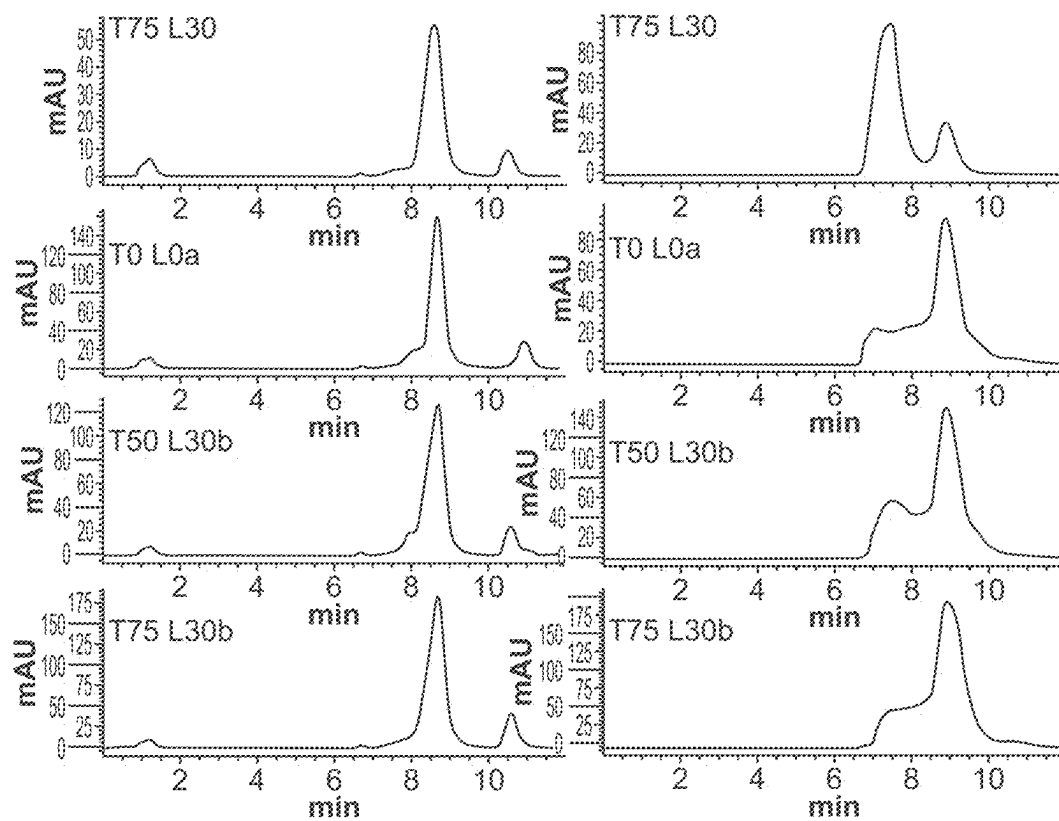


FIG. 15B

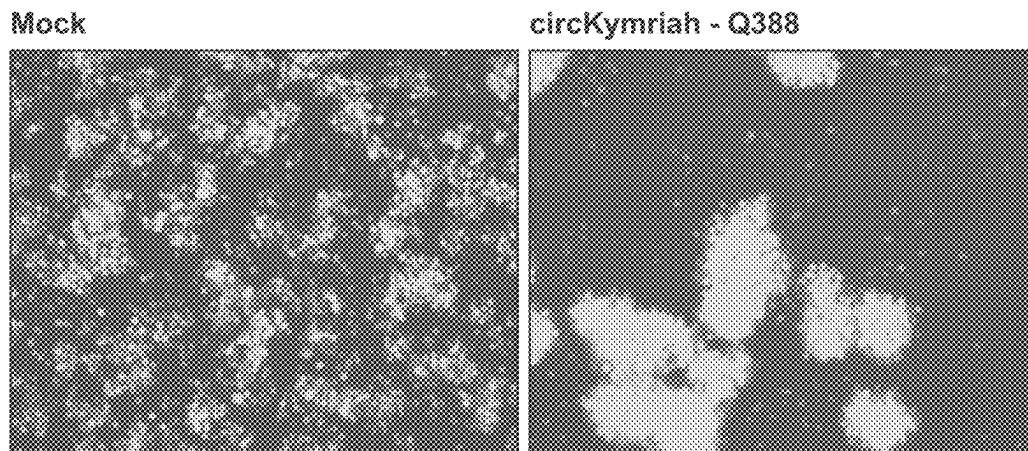


FIG. 16

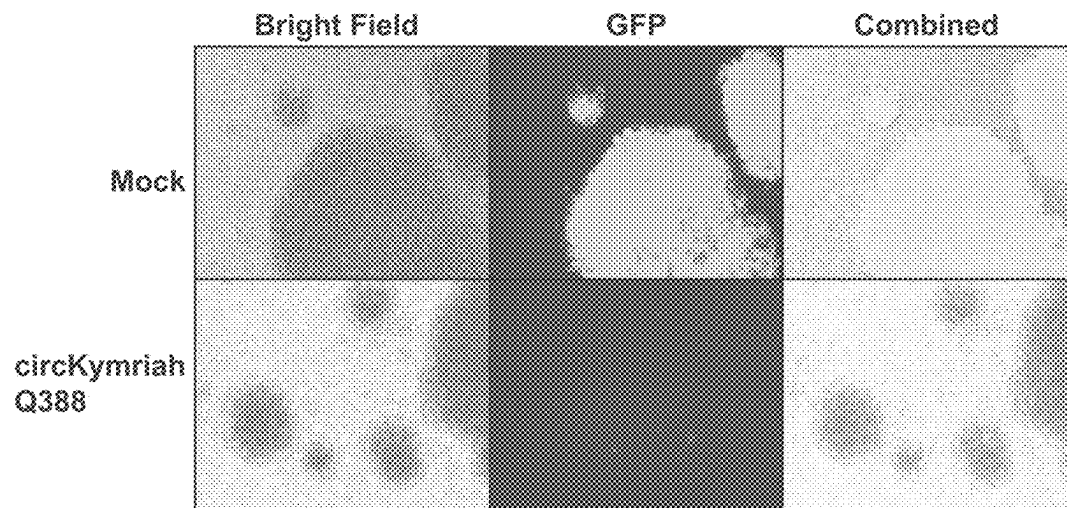


FIG. 17

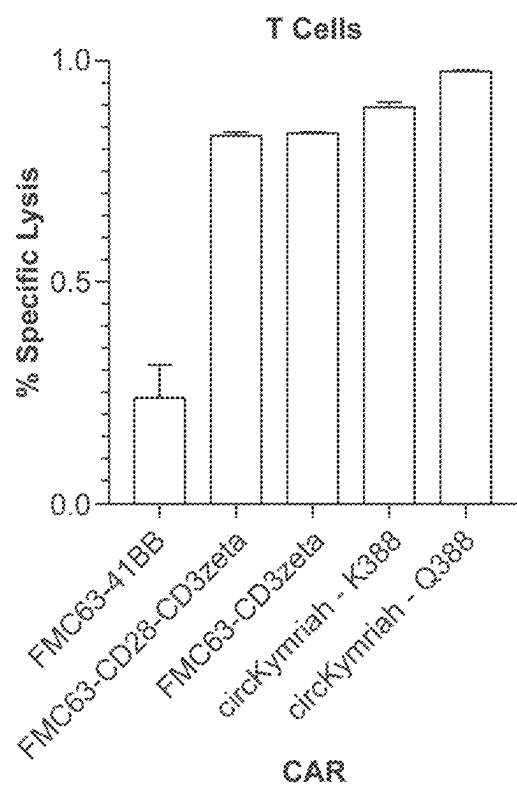


FIG. 18

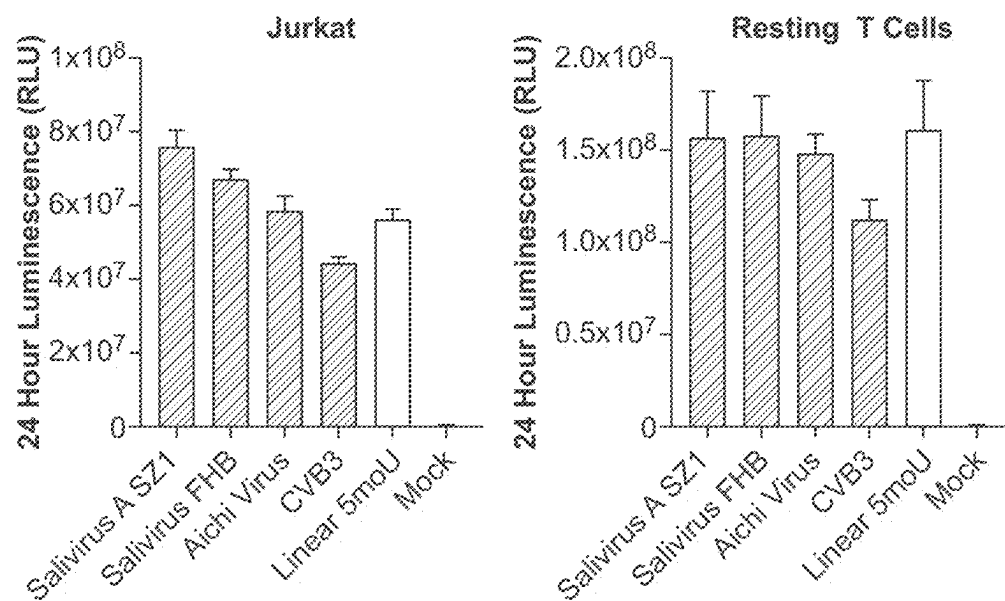


FIG. 19A

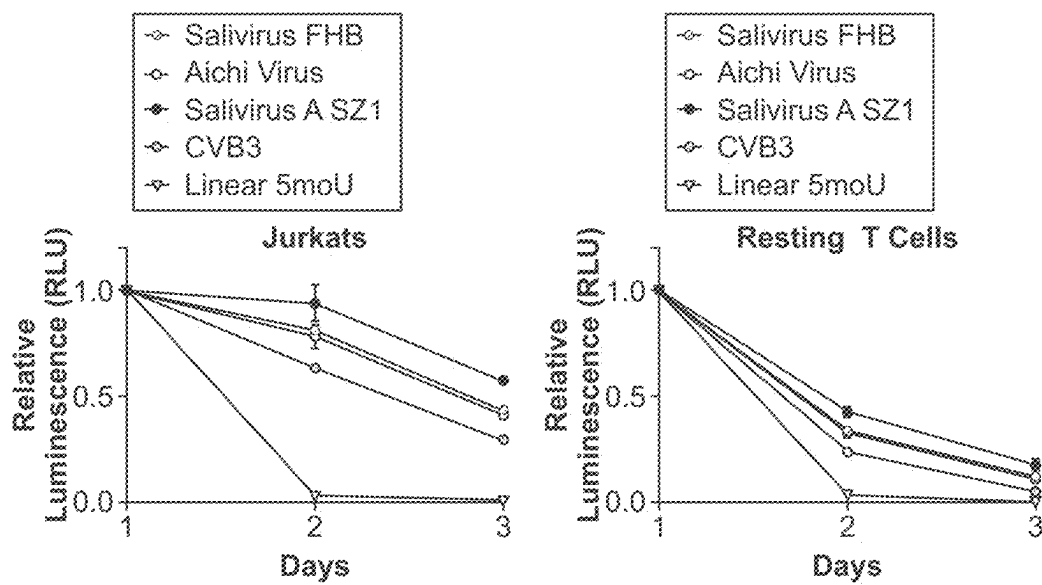


FIG. 19B

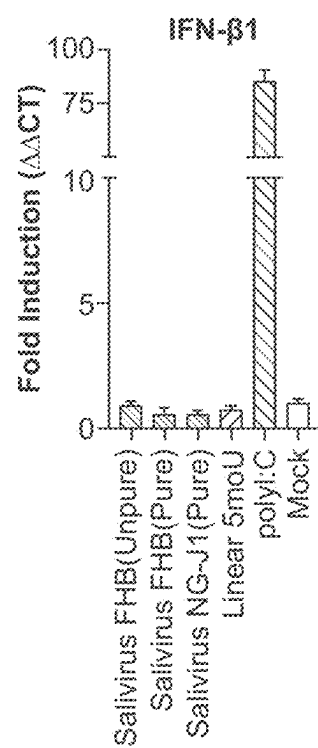


FIG. 20A

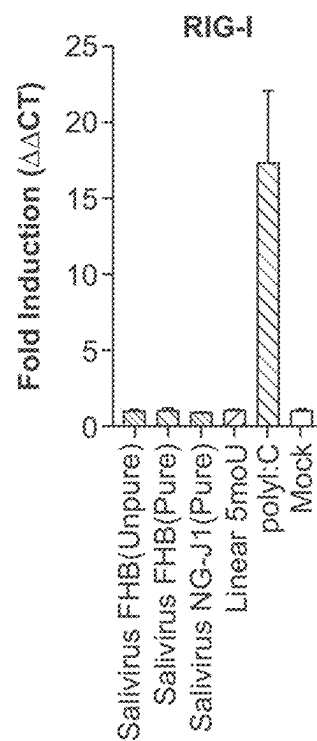


FIG. 20B

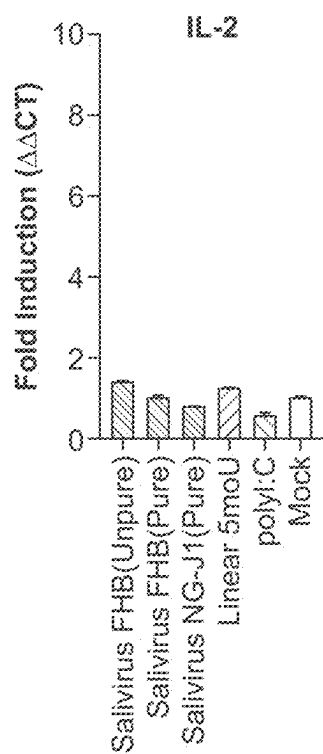


FIG. 20C

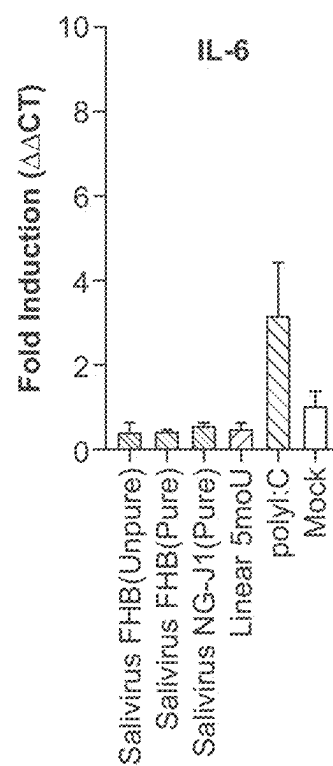


FIG. 20D

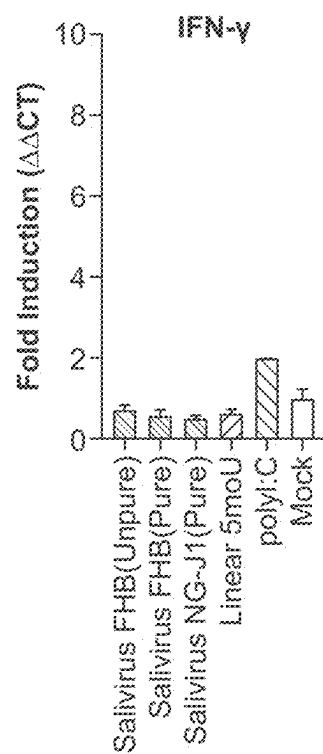


FIG. 20E

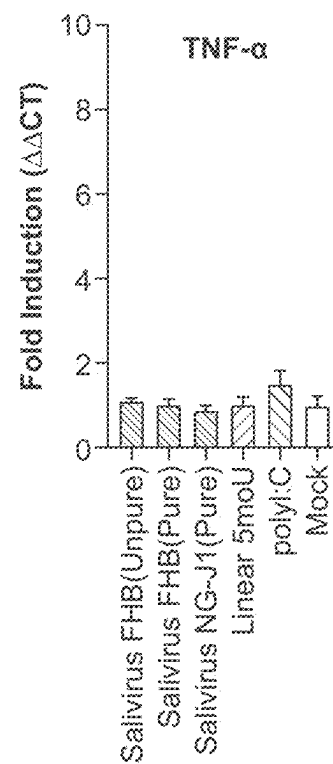


FIG. 20F

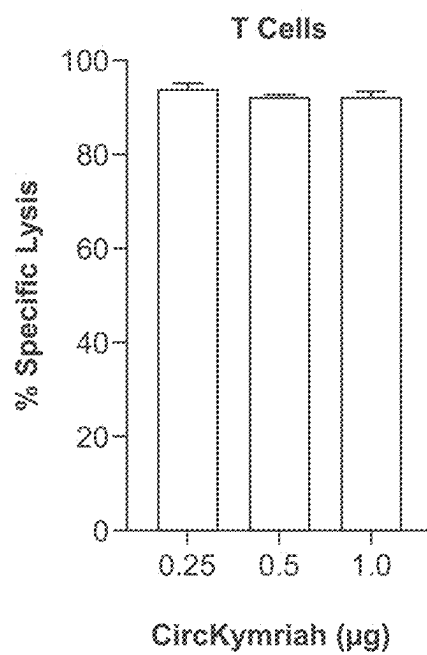


FIG. 21A

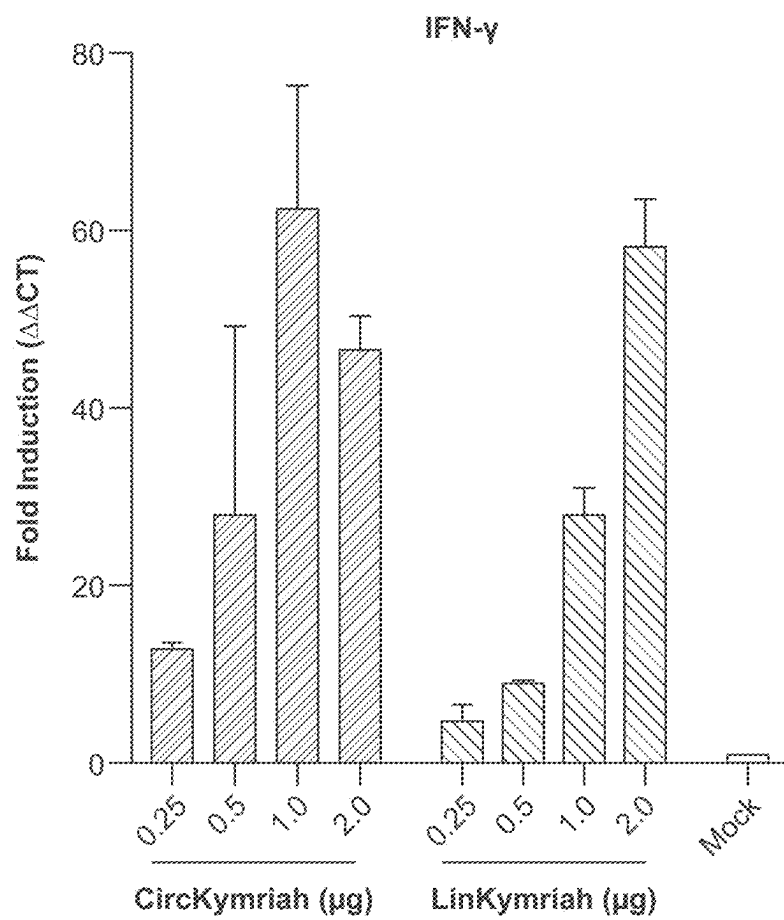


FIG. 21B

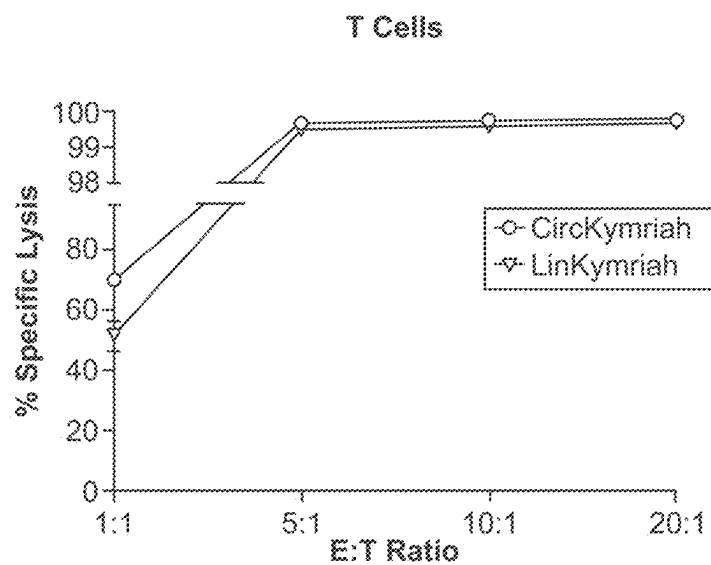


FIG. 22A

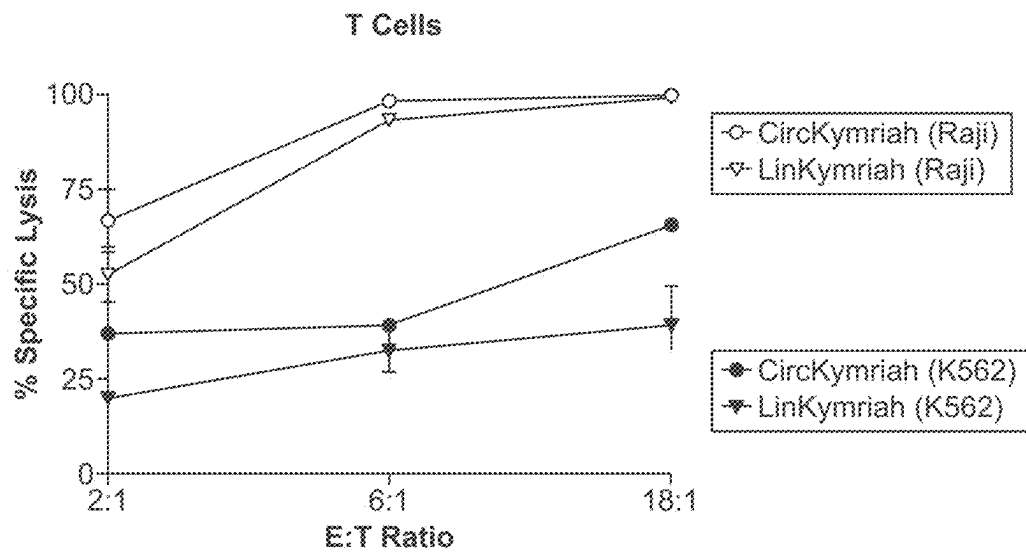


FIG. 22B

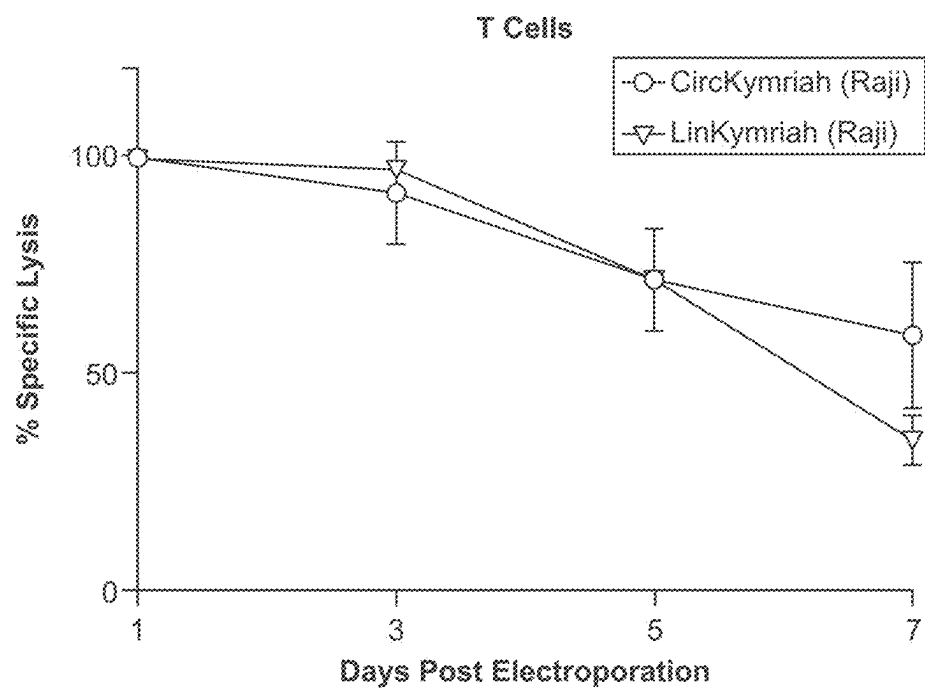


FIG. 23

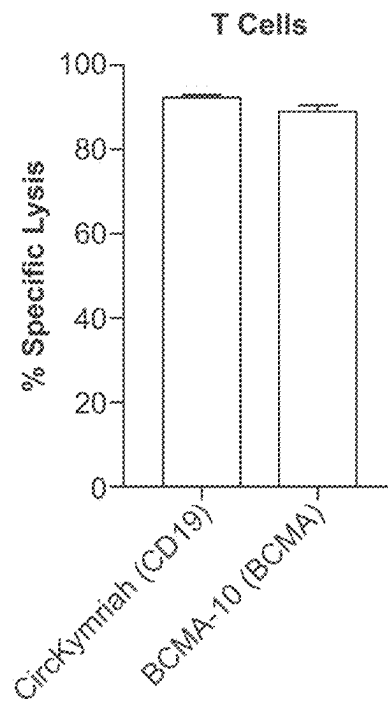


FIG. 24

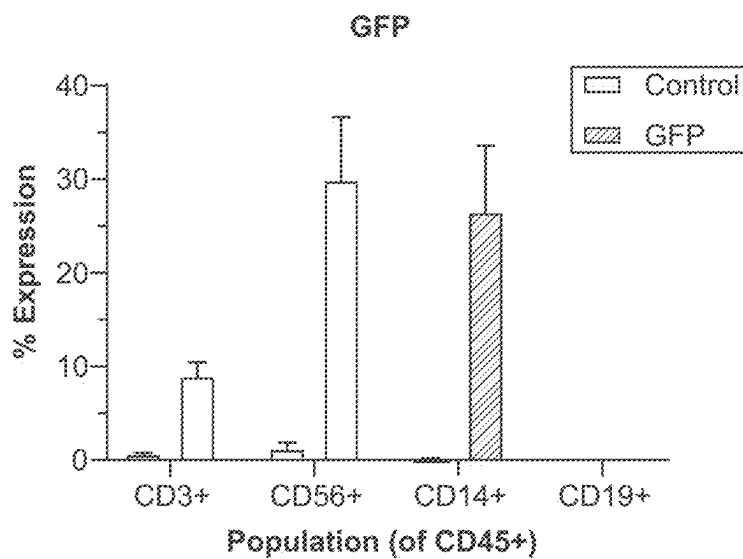


FIG. 25A

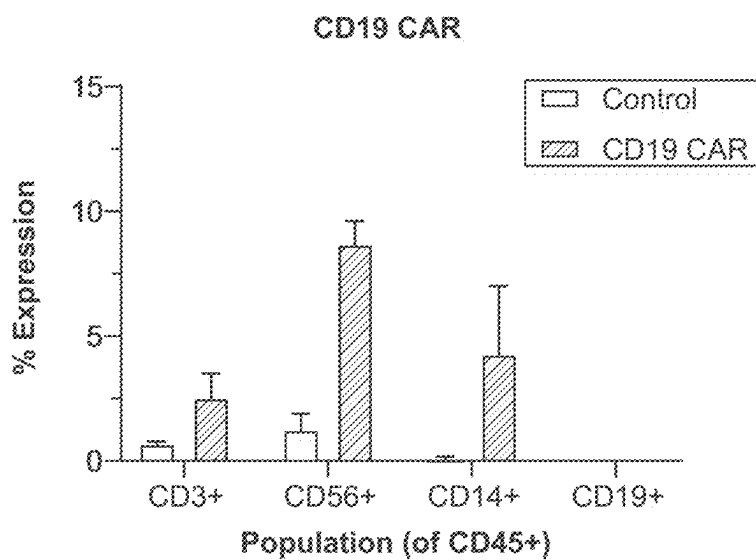


FIG. 25B

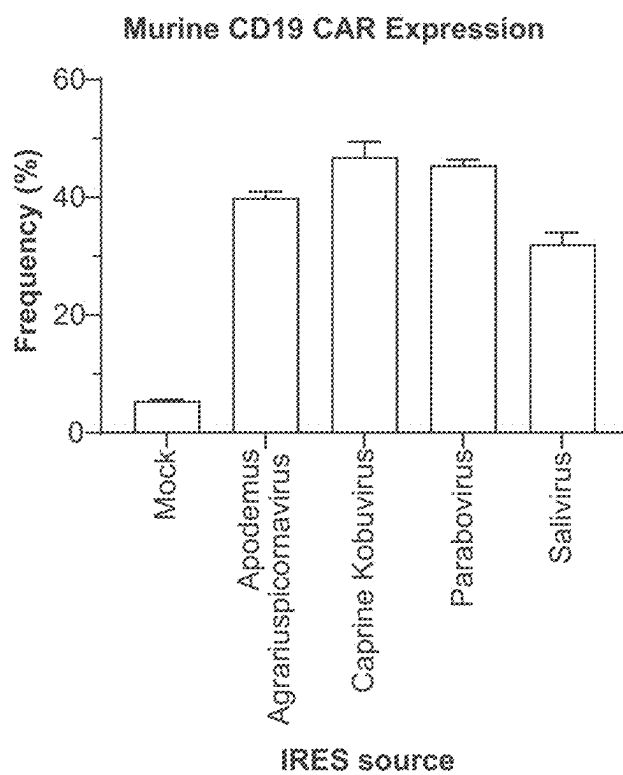


FIG. 26

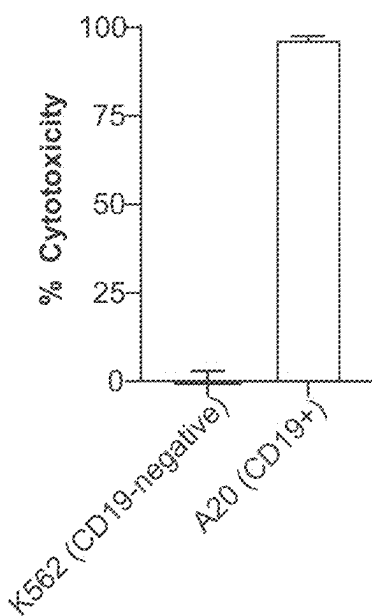


FIG. 27

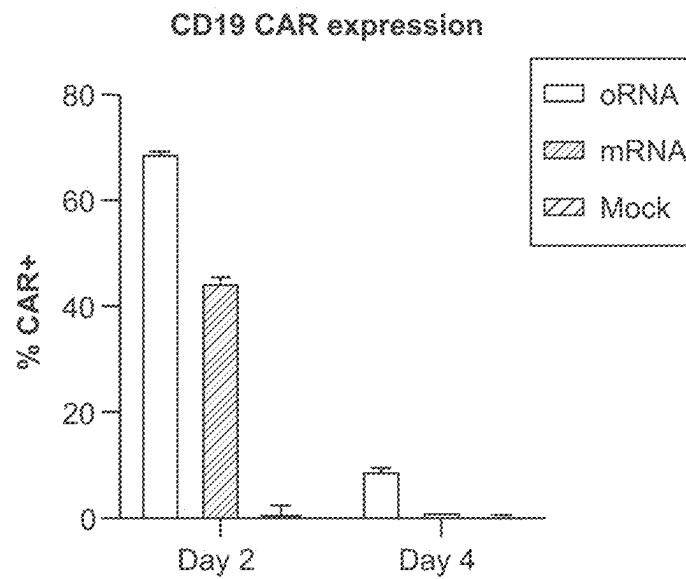


FIG. 28A

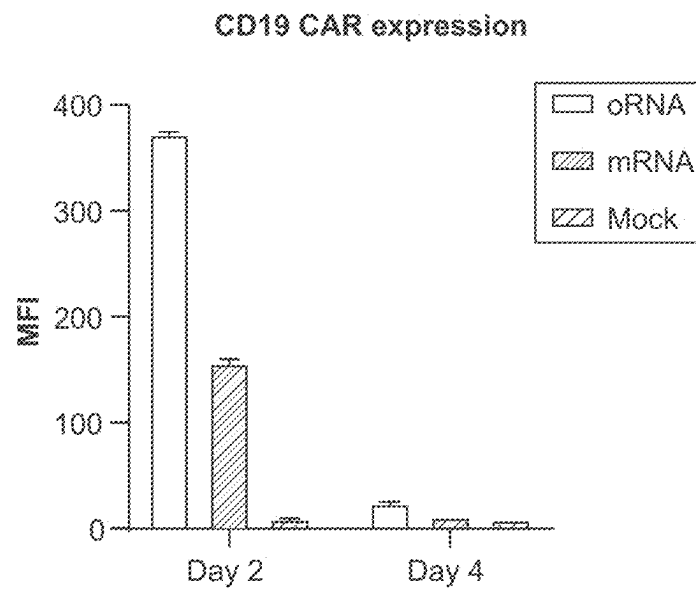


FIG. 28B

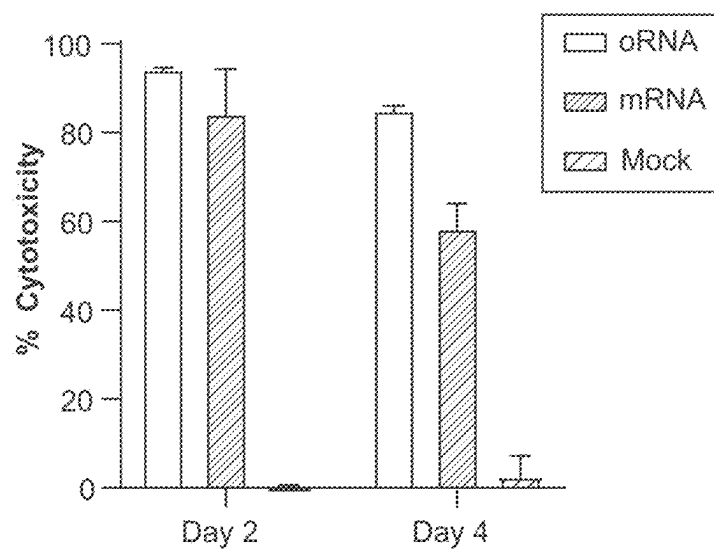


FIG. 29A

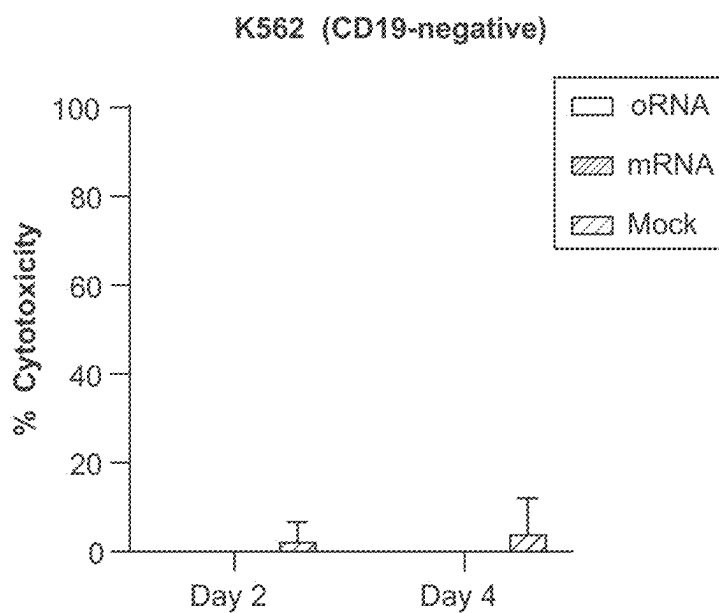


FIG. 29B

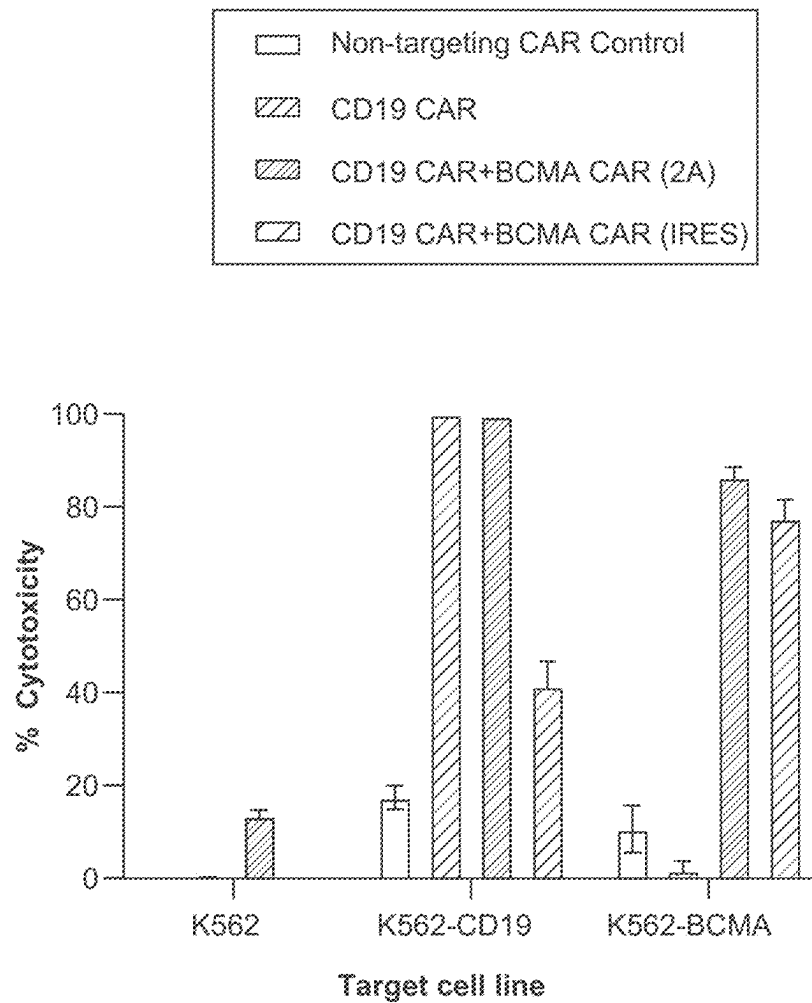


FIG. 30



FIG. 31A

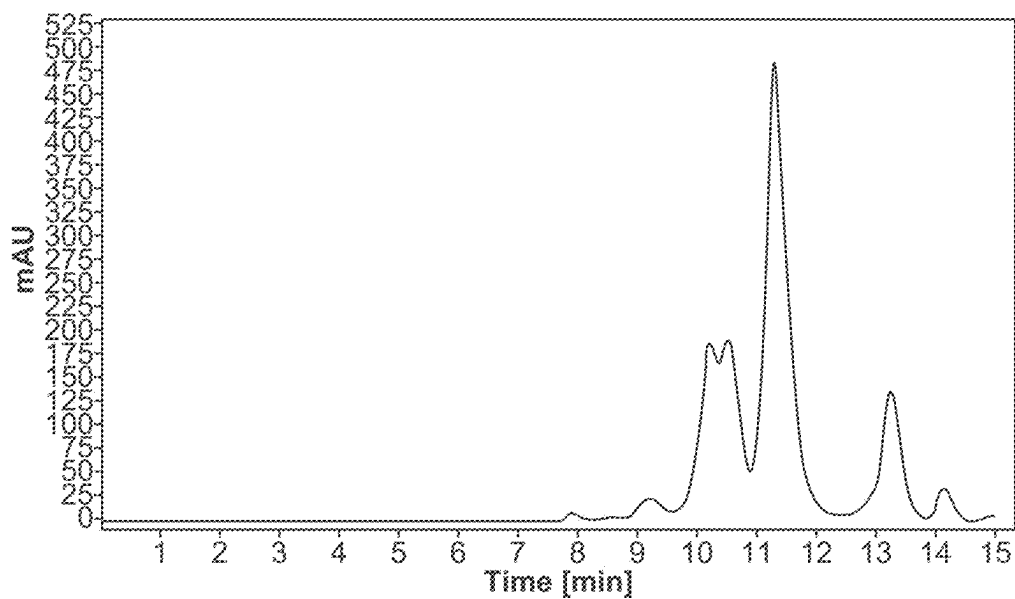


FIG. 31B

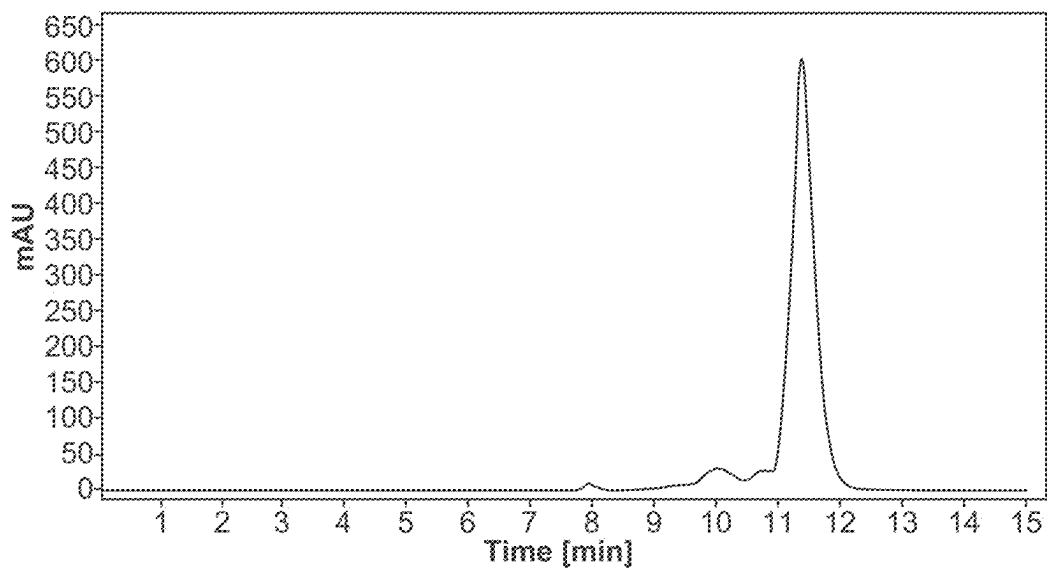


FIG. 31C

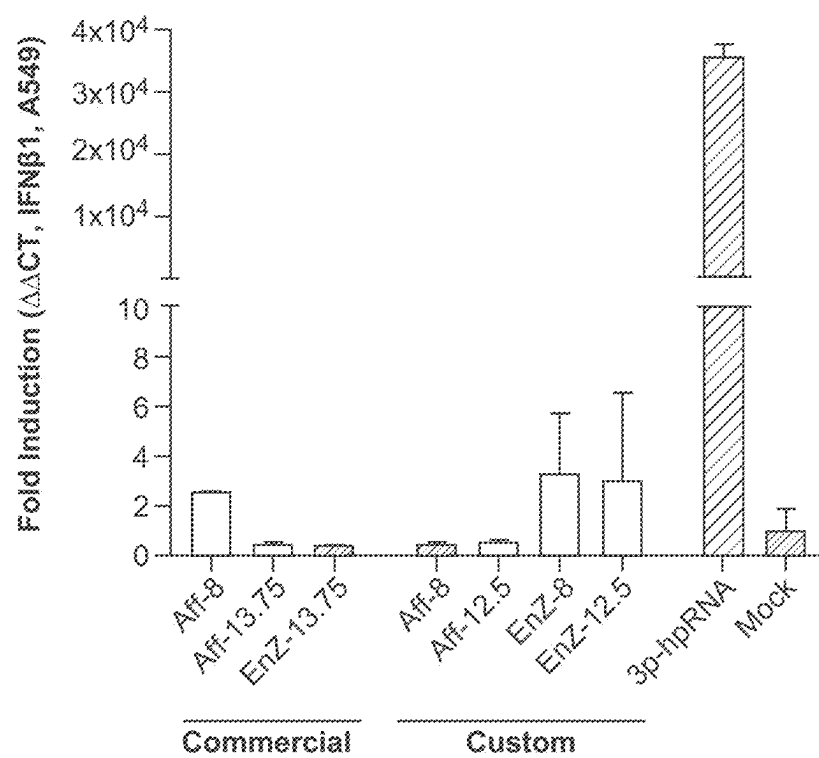


FIG. 31D

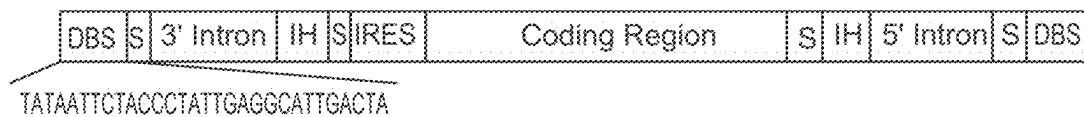


FIG. 32A

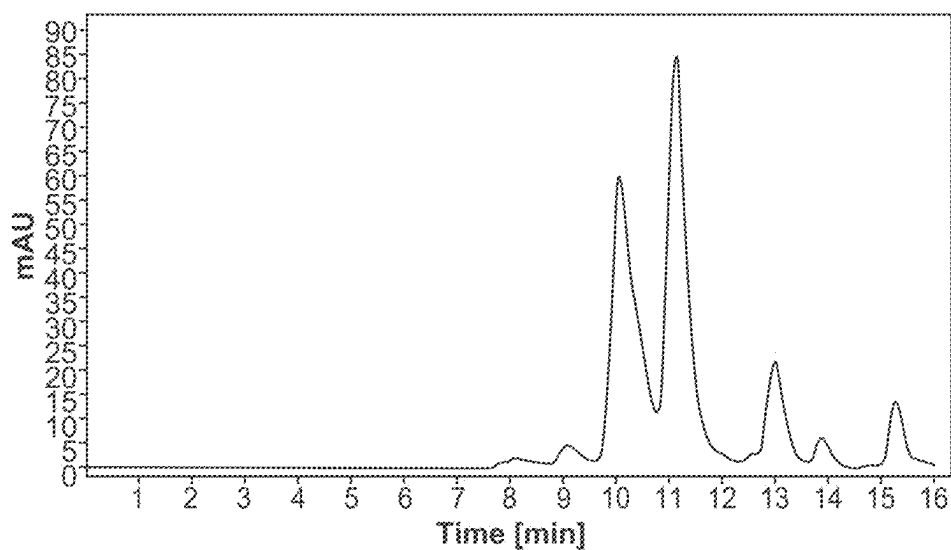


FIG. 32B

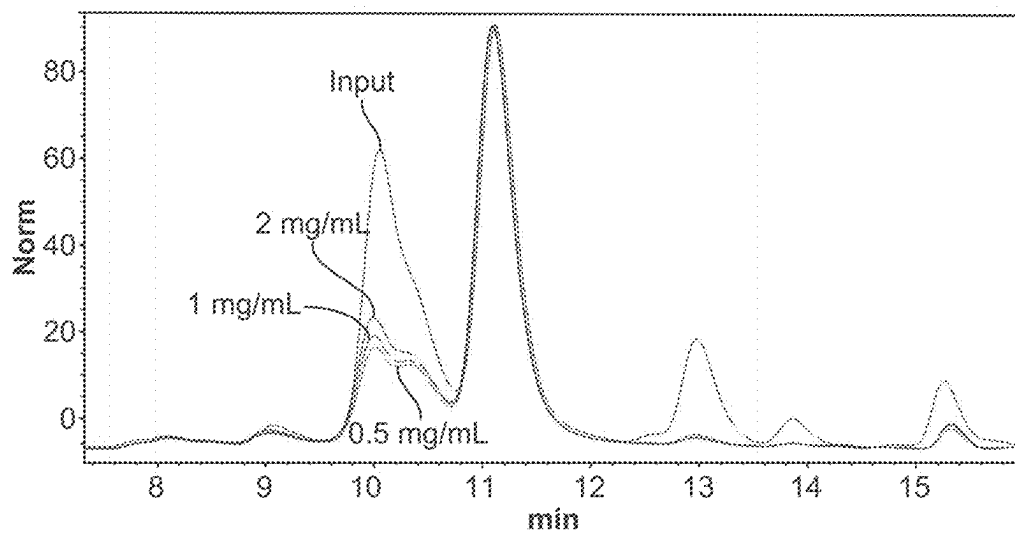


FIG. 32C

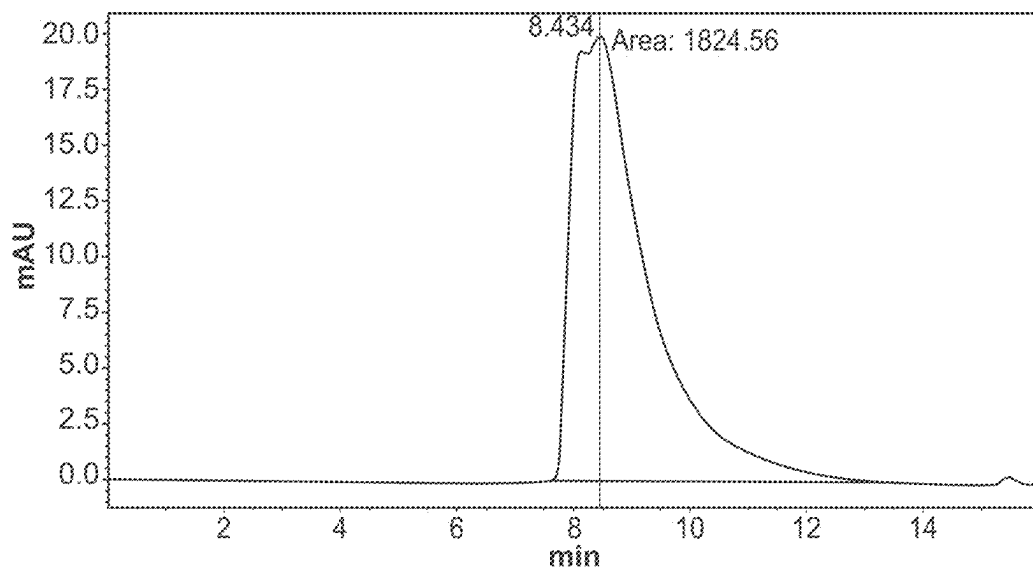


FIG. 33A

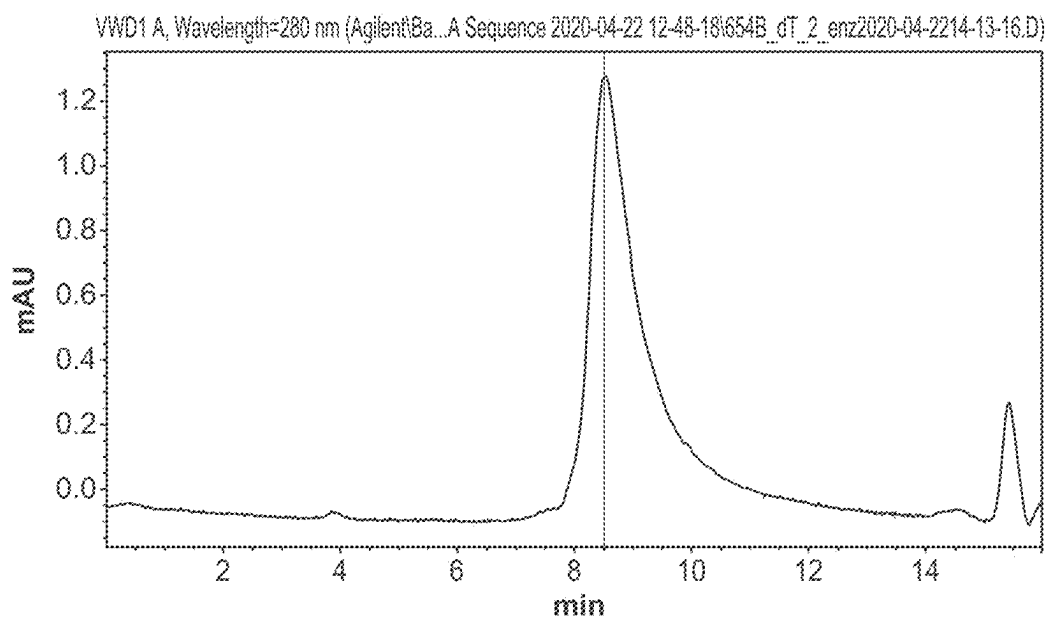


FIG. 33B

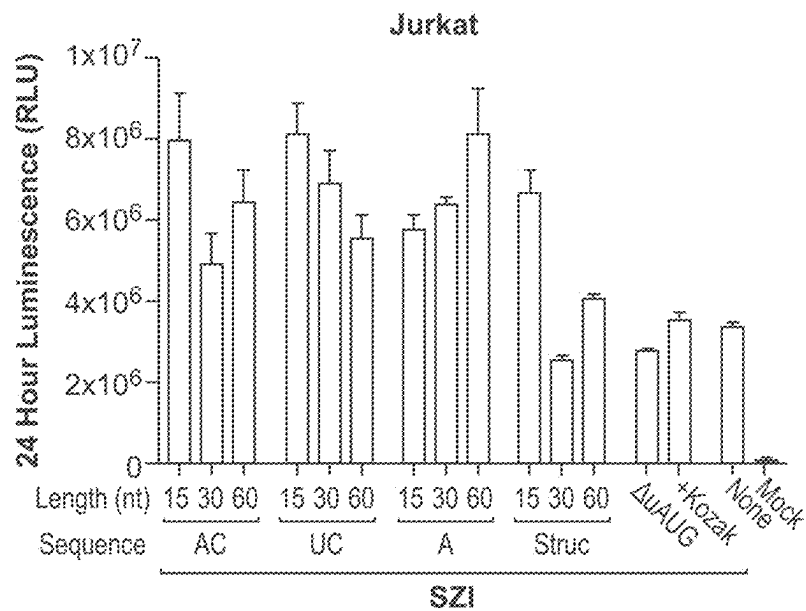


FIG. 34A

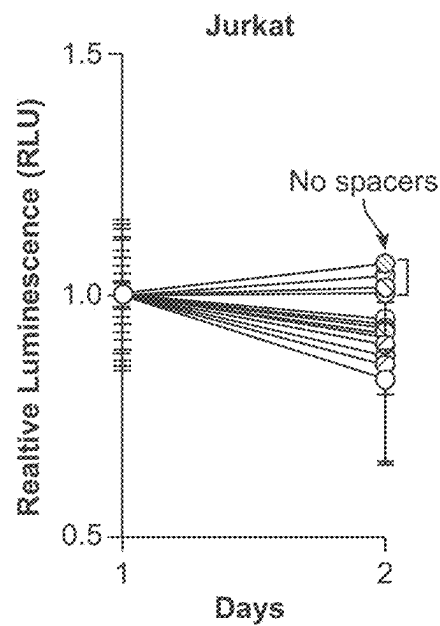


FIG. 34B

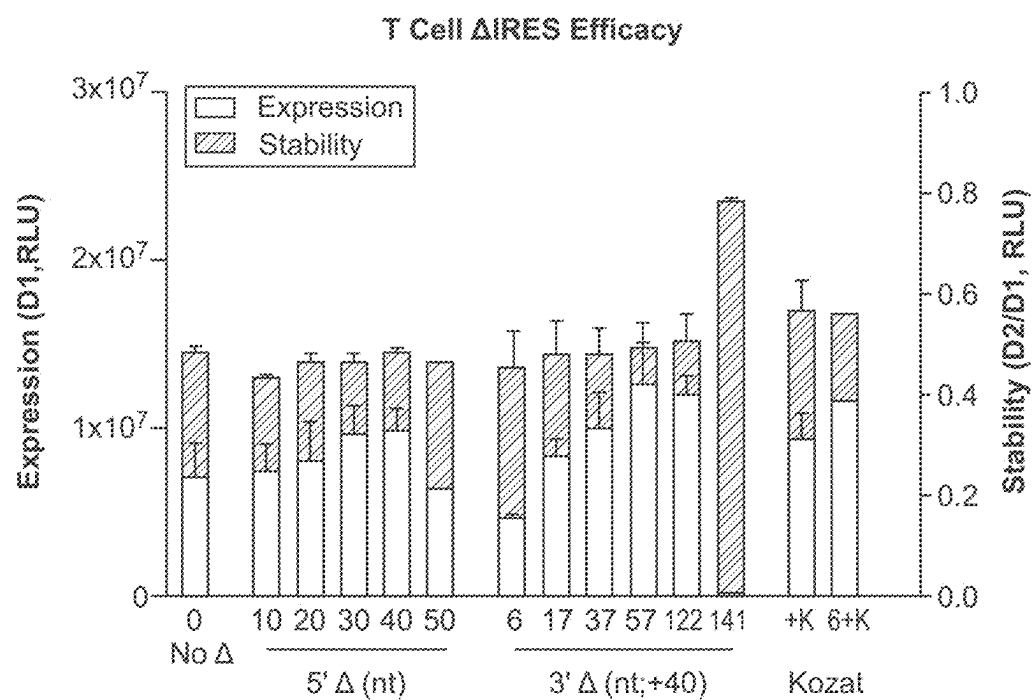


FIG. 35

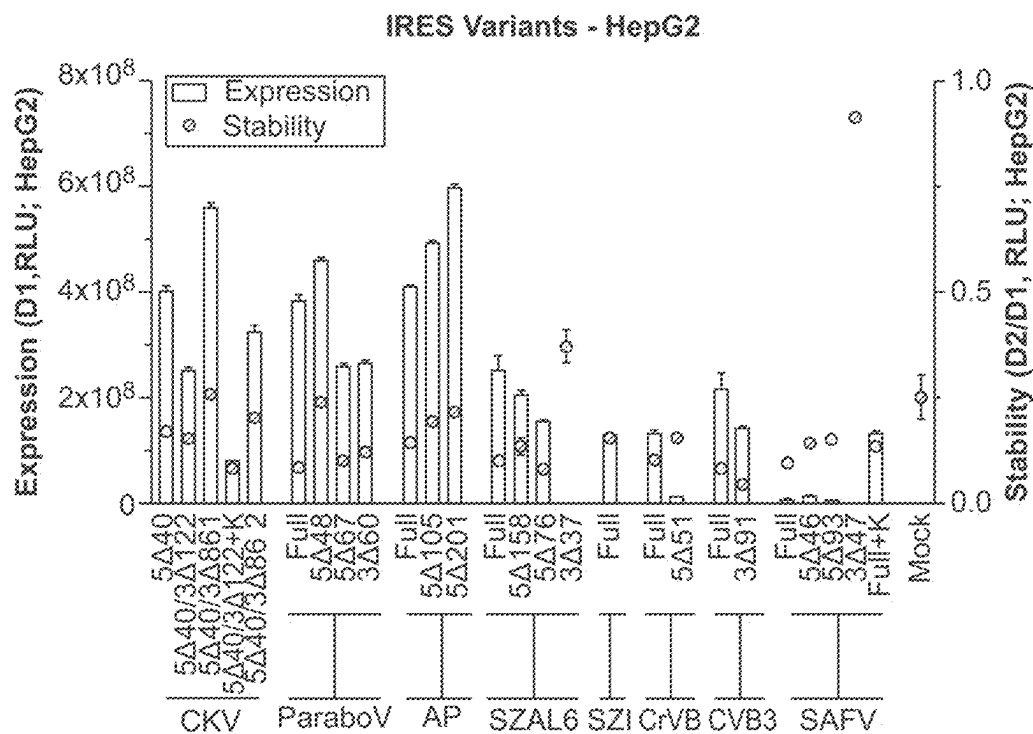


FIG. 36

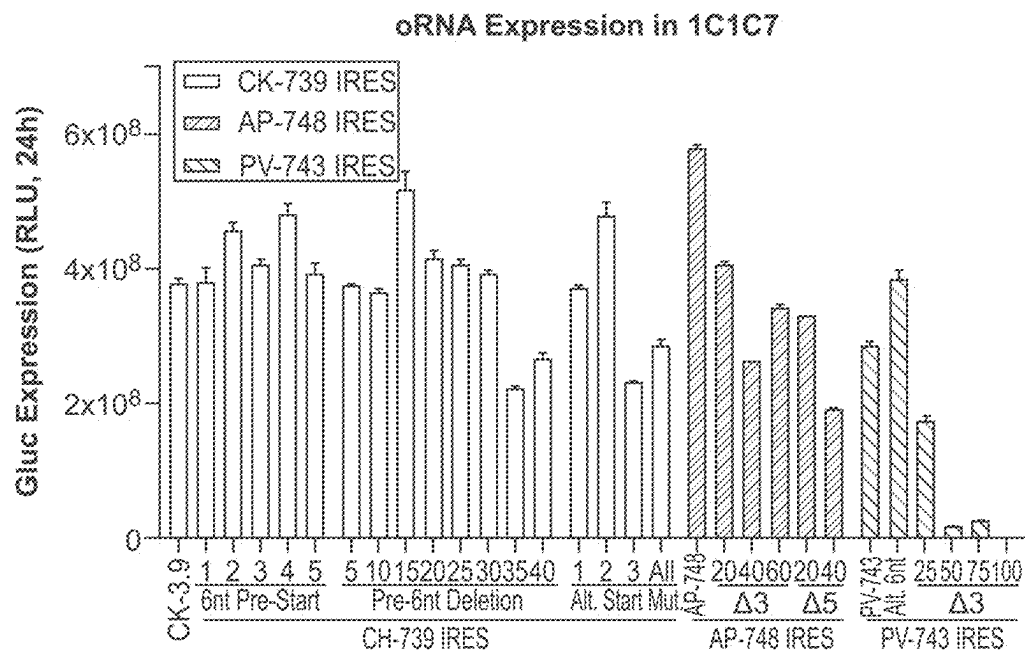


FIG. 37

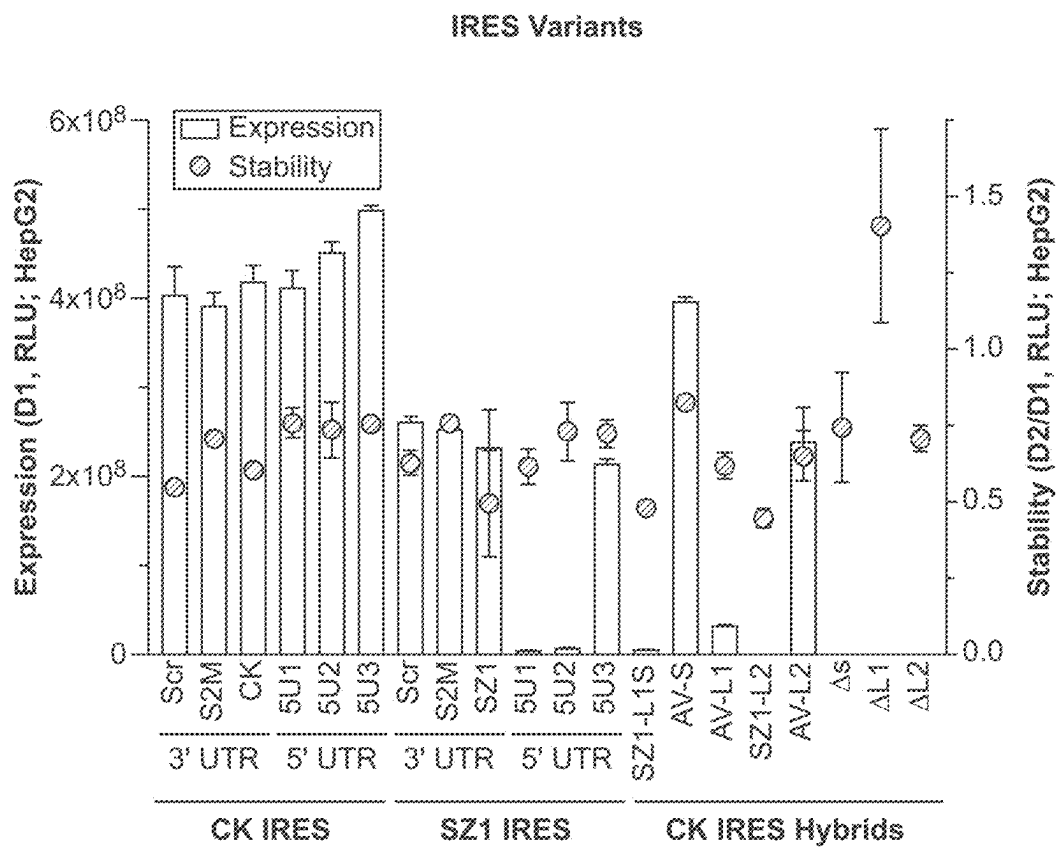


FIG. 38

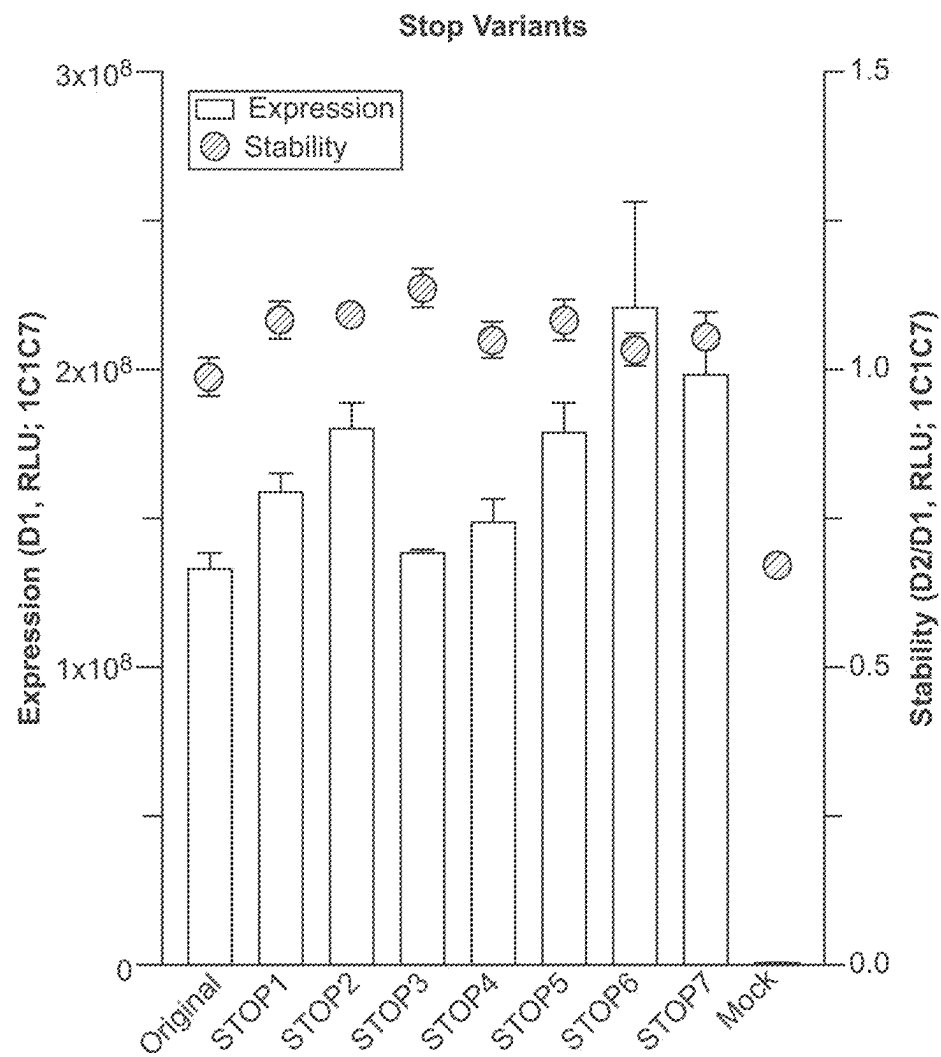


FIG. 39

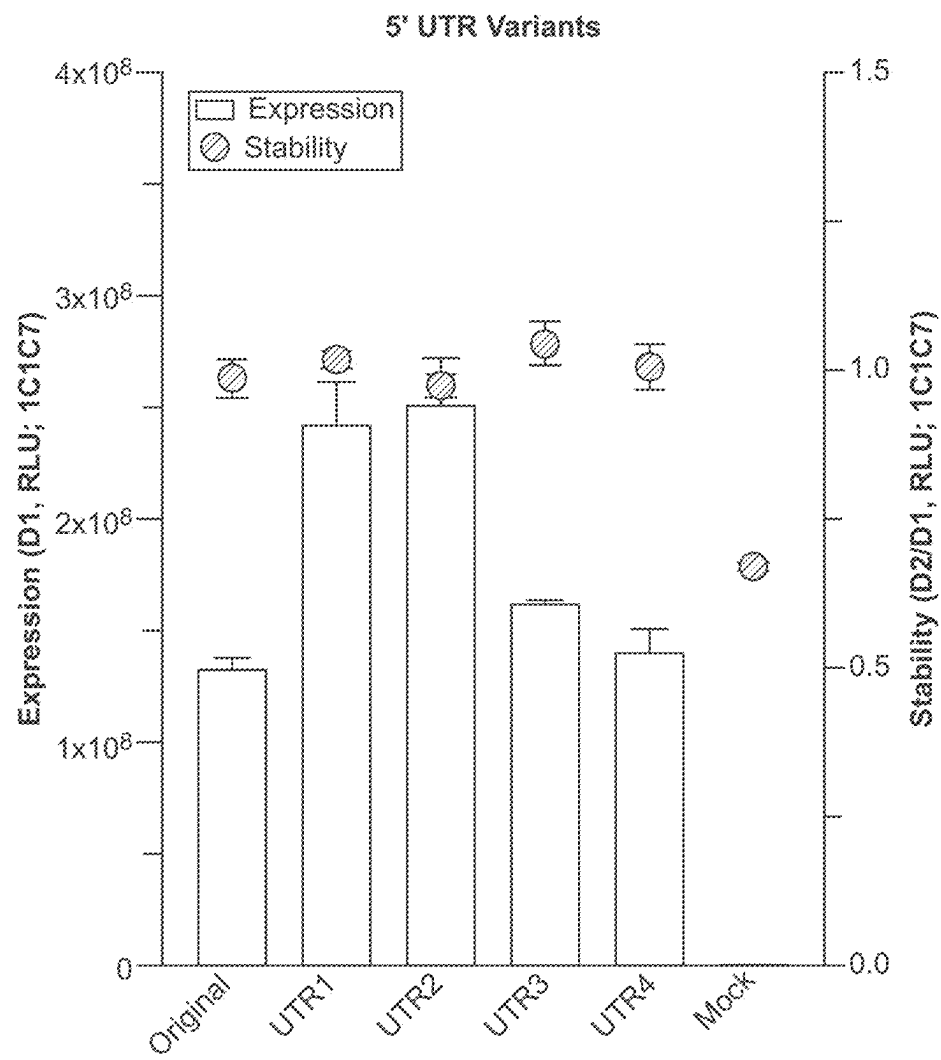


FIG. 40

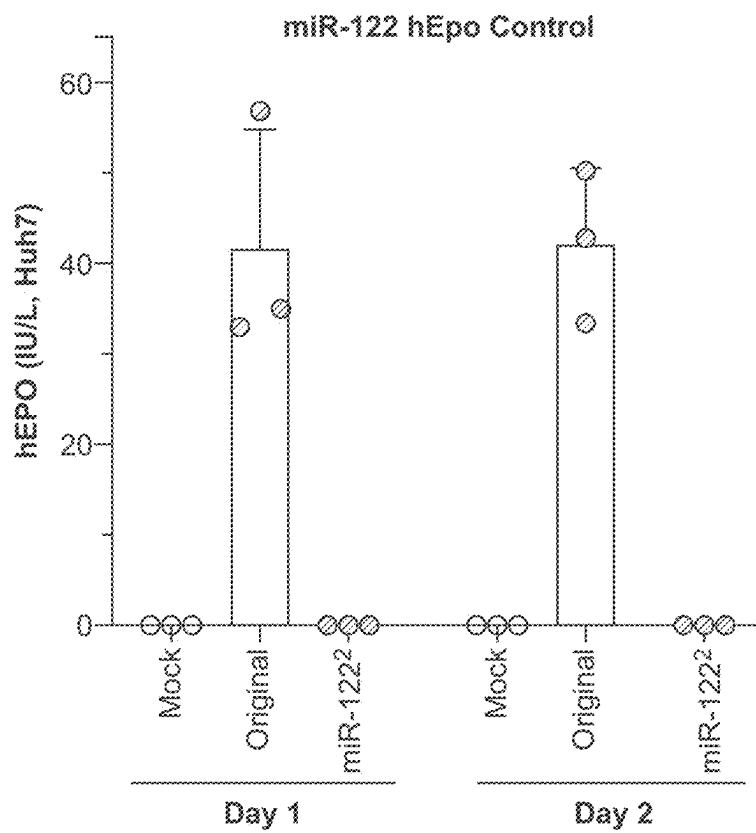


FIG. 41

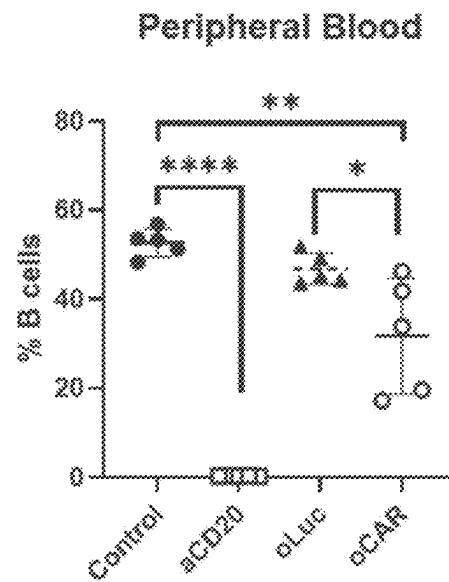


FIG. 42A

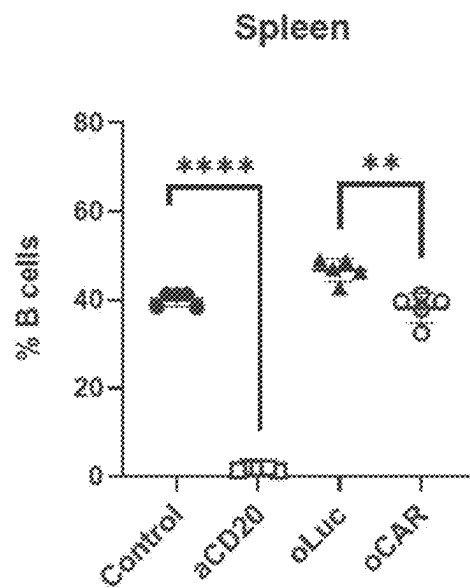


FIG. 42B

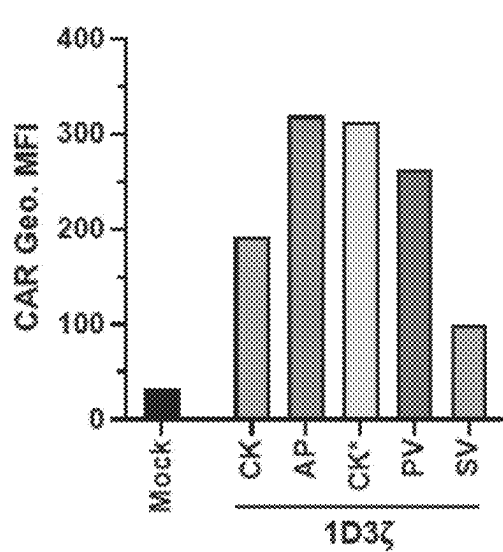


FIG. 43A

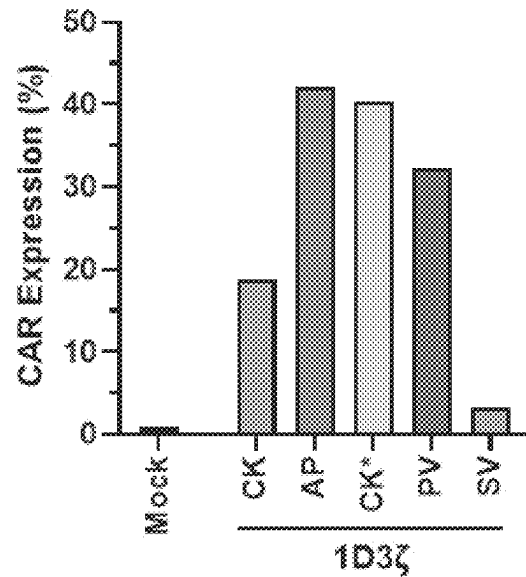


FIG. 43B

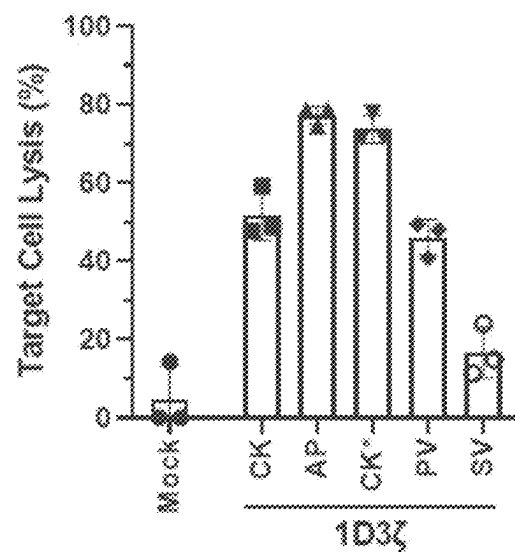


FIG. 43C

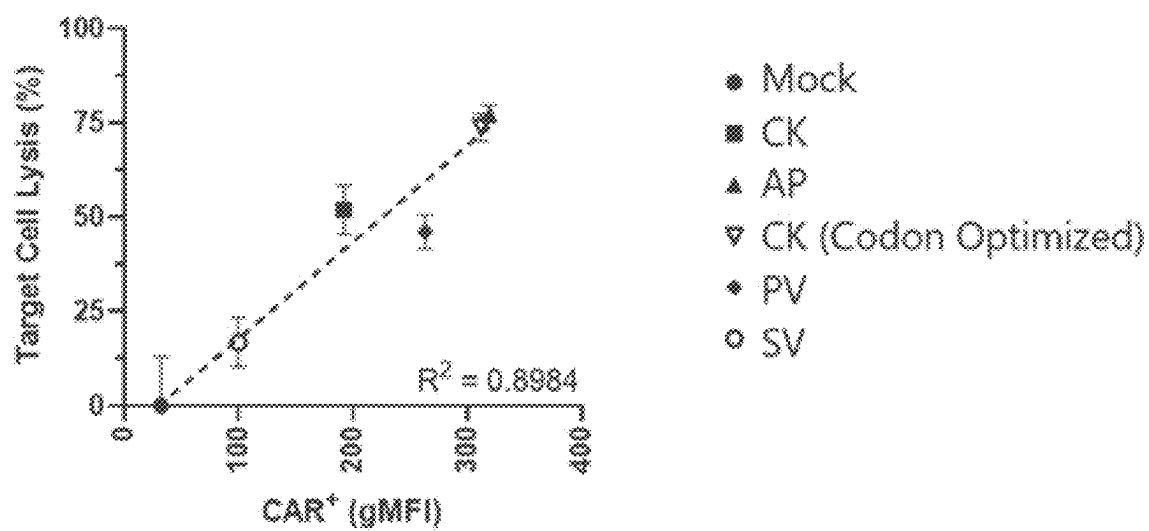


FIG. 44

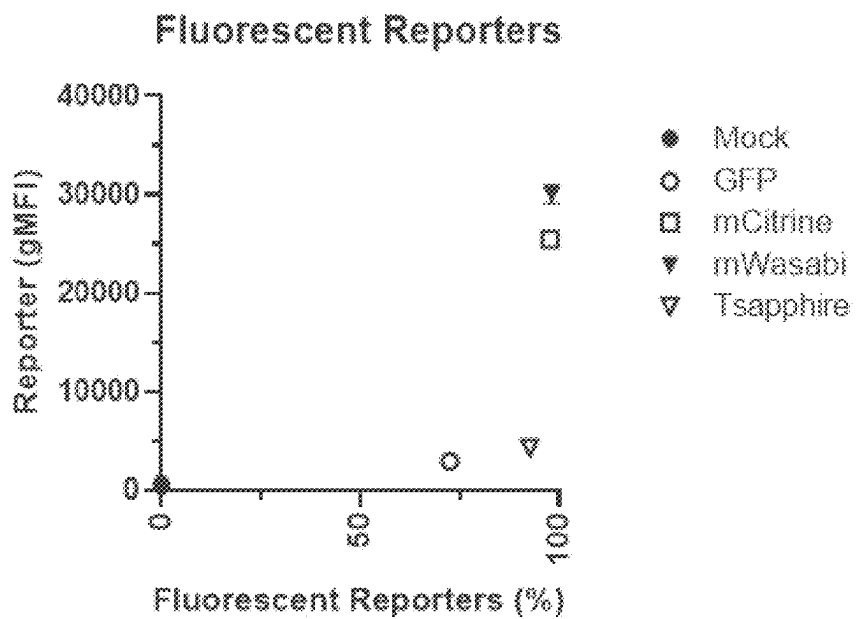


FIG. 45A

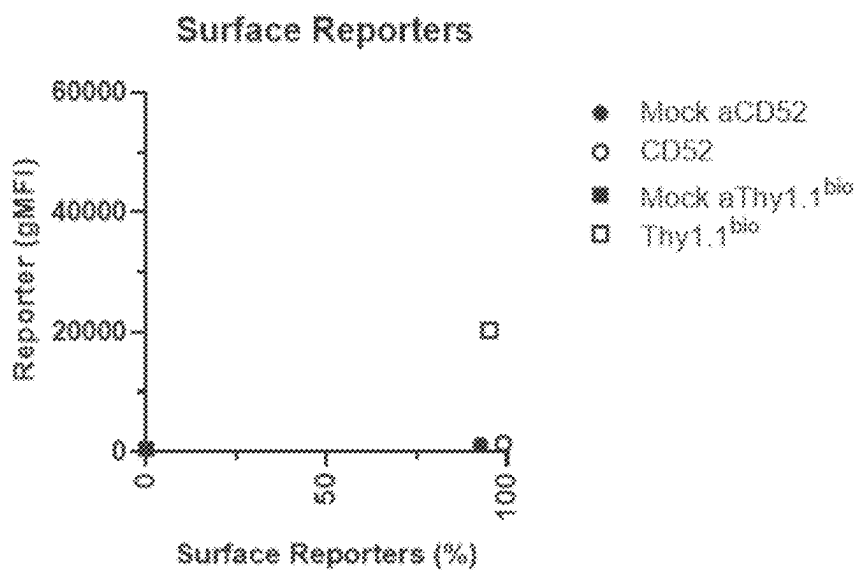


FIG. 45B

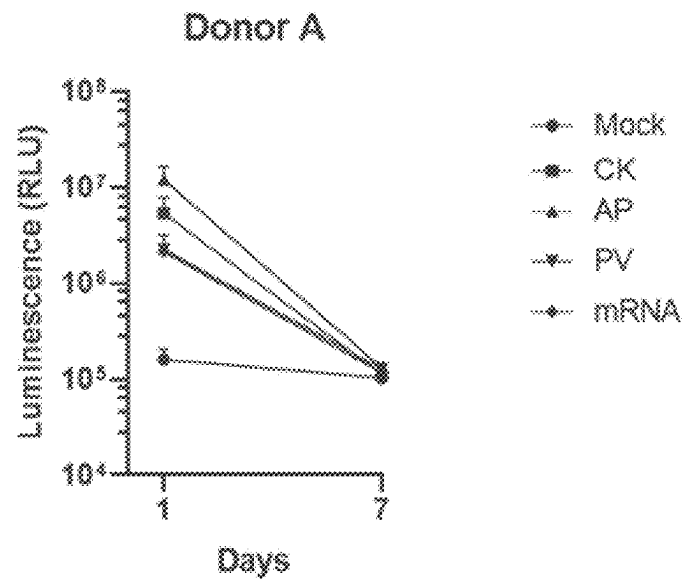


FIG. 46A

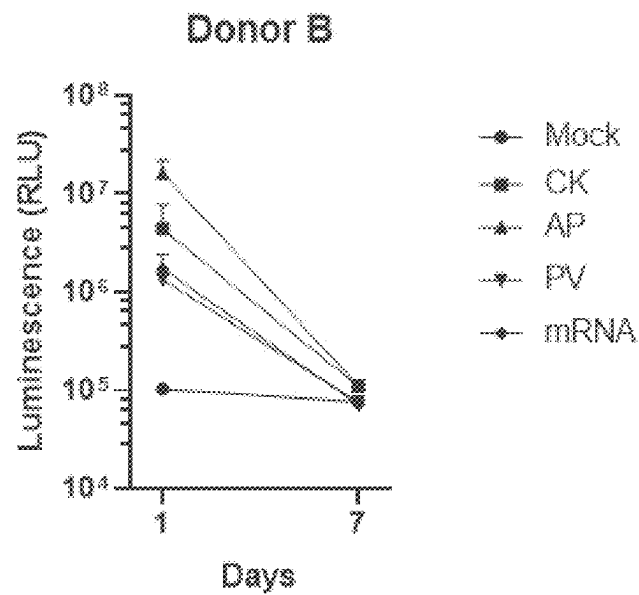


FIG. 46B

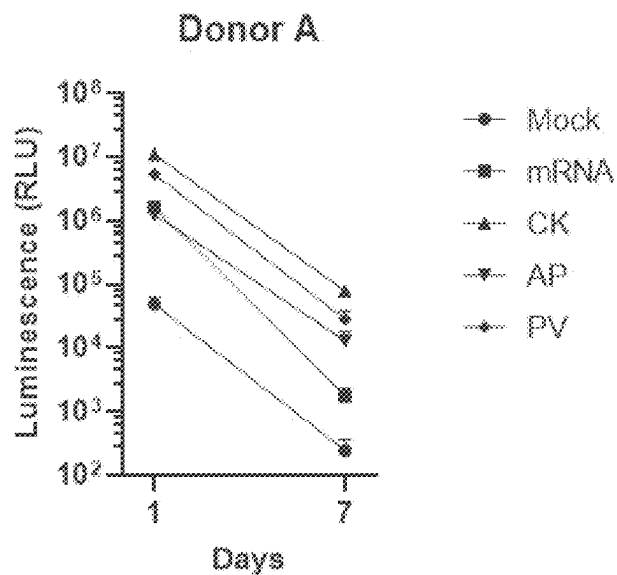


FIG. 46C

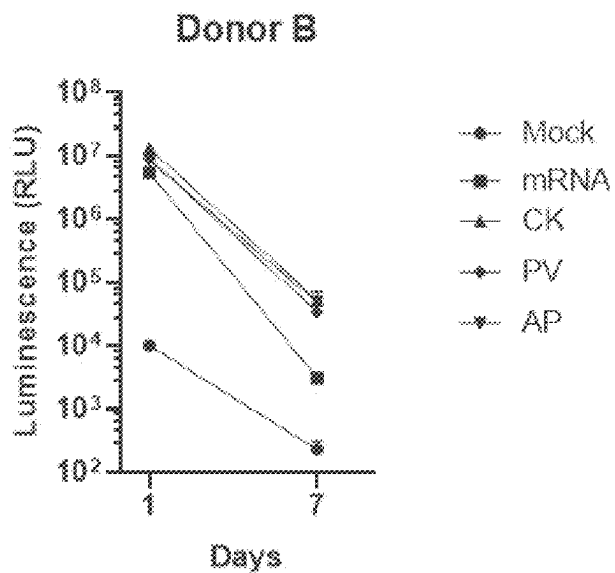


FIG. 46D

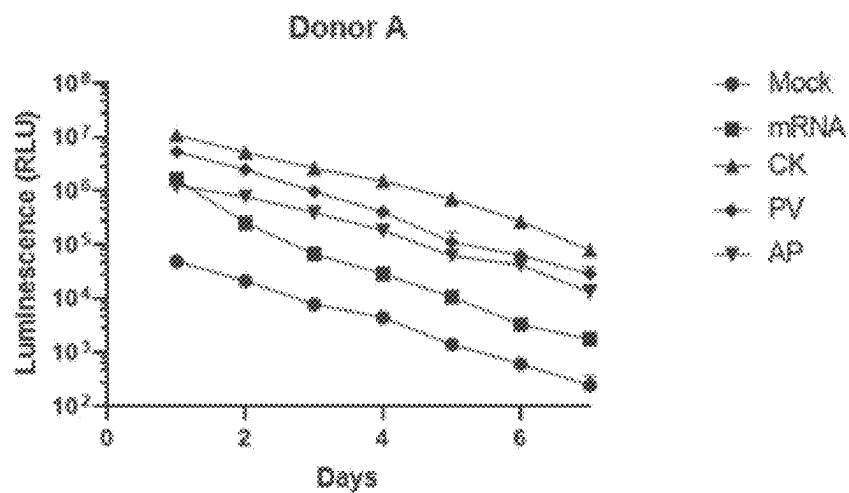


FIG. 46E

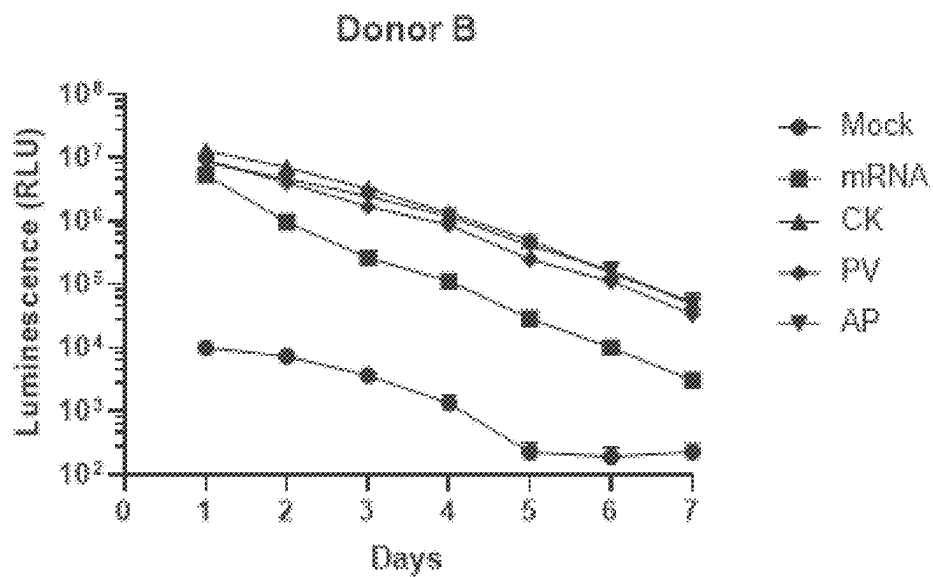


FIG. 46F

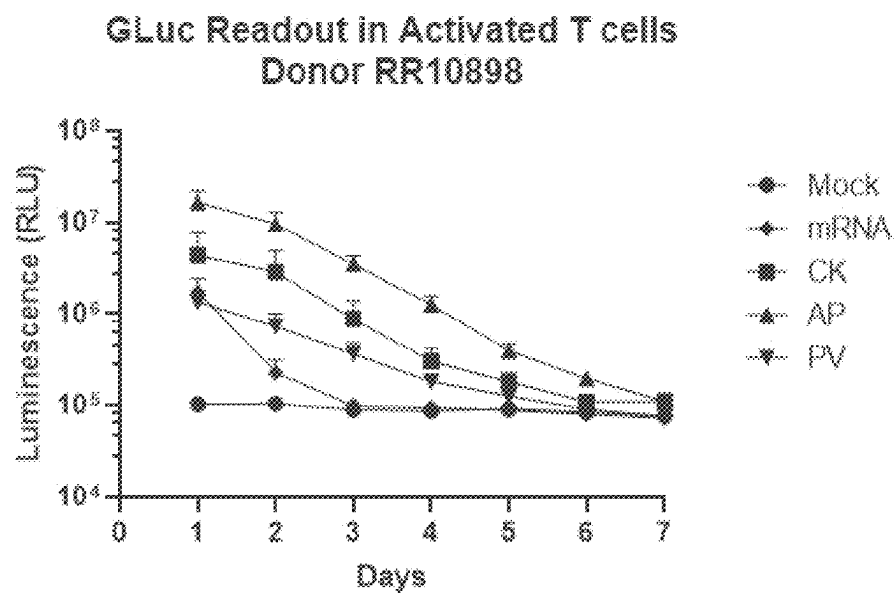


FIG. 46G

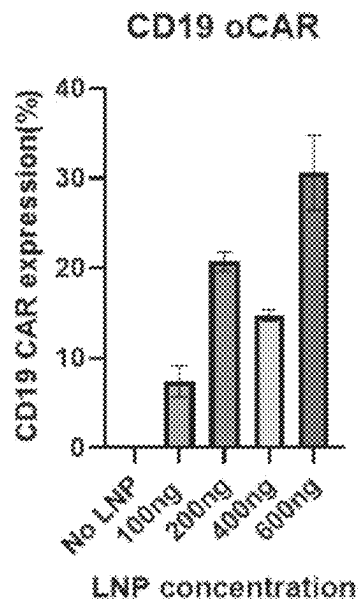


FIG. 47A

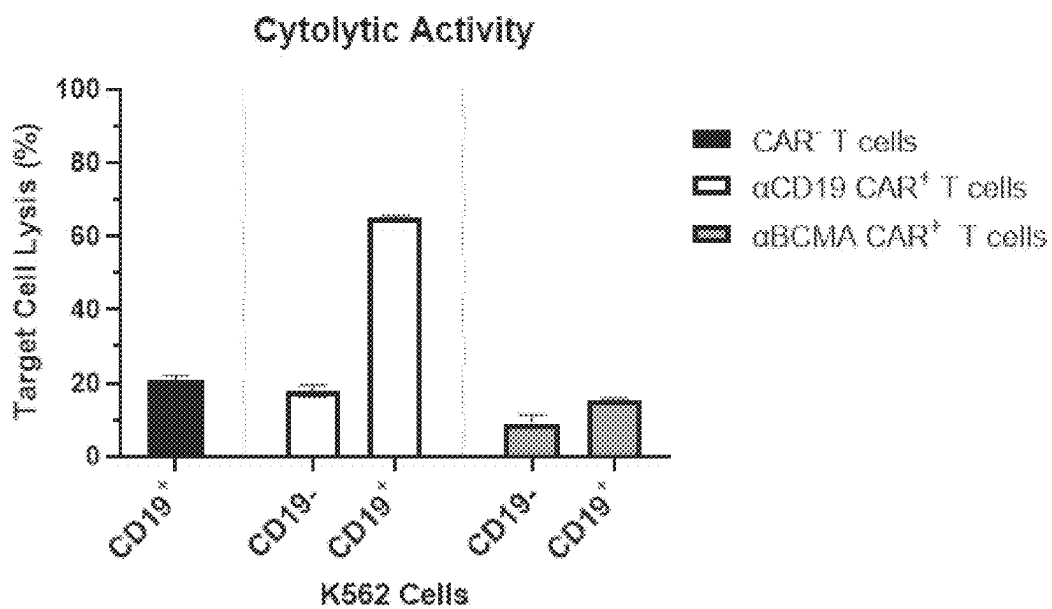


FIG. 47B

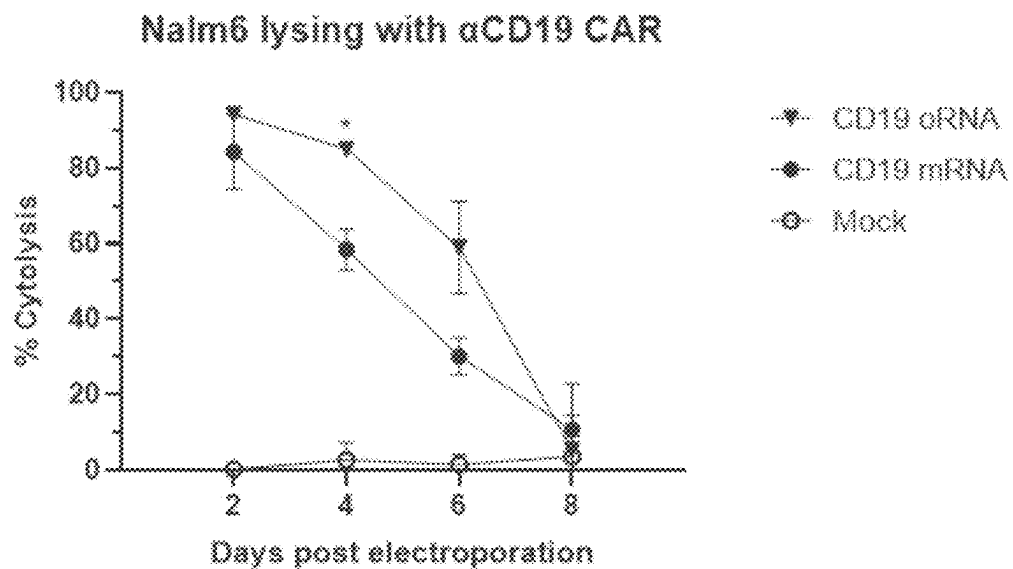


FIG. 48A

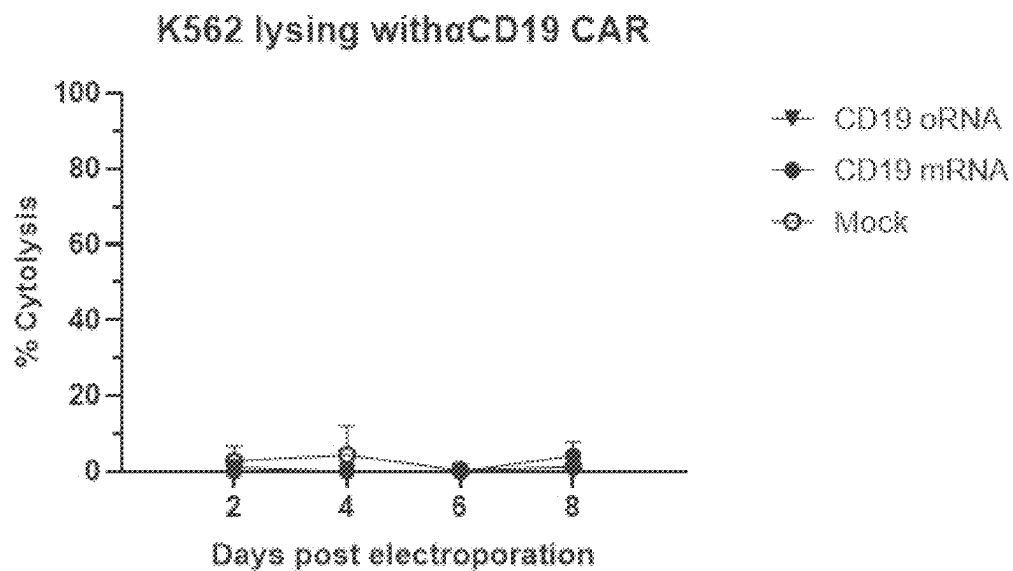
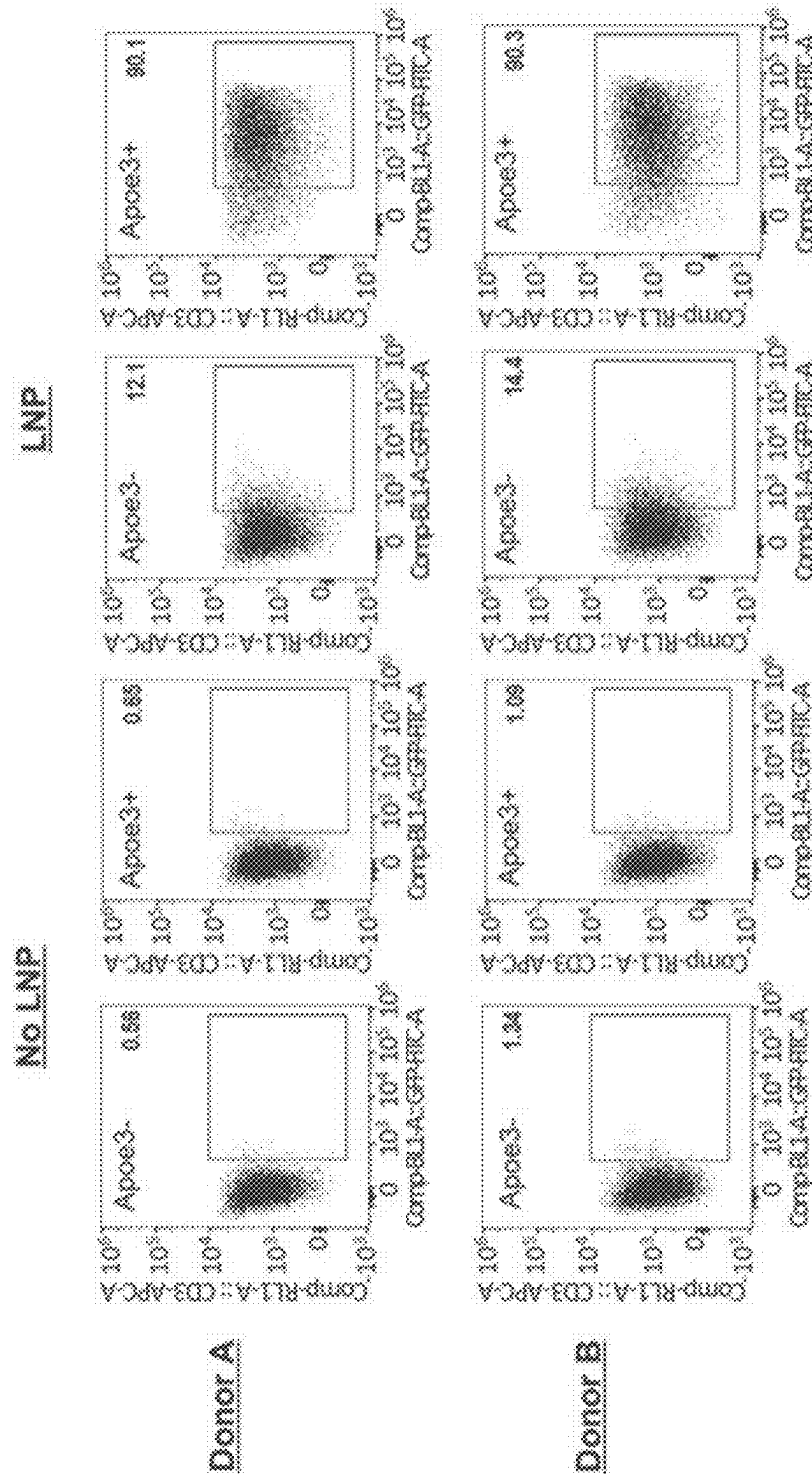


FIG. 48B



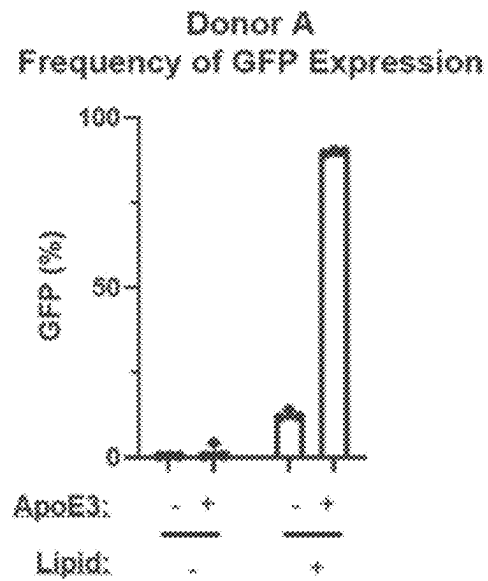


FIG. 49B

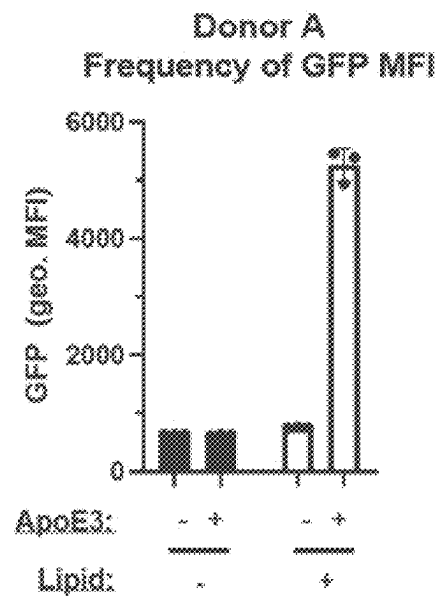


FIG. 49C

Donor B
Frequency of GFP Expression

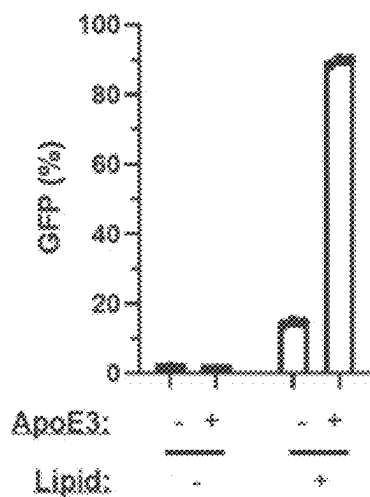


FIG. 49D

Donor B
Frequency of GFP MFI

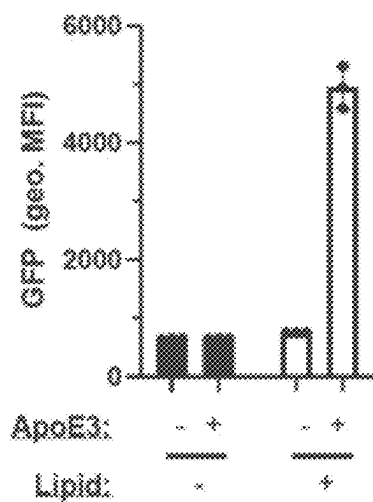


FIG. 49E

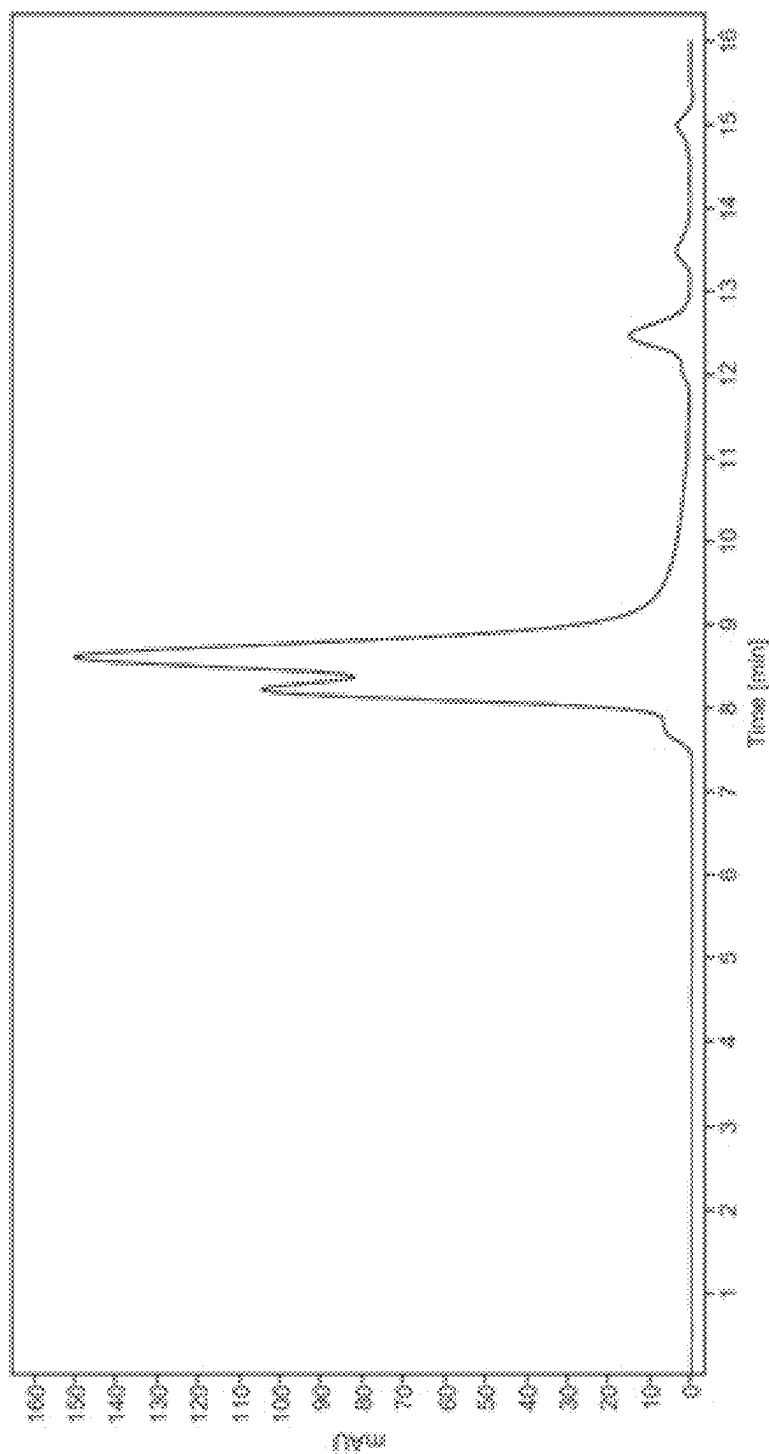


FIG. 50A

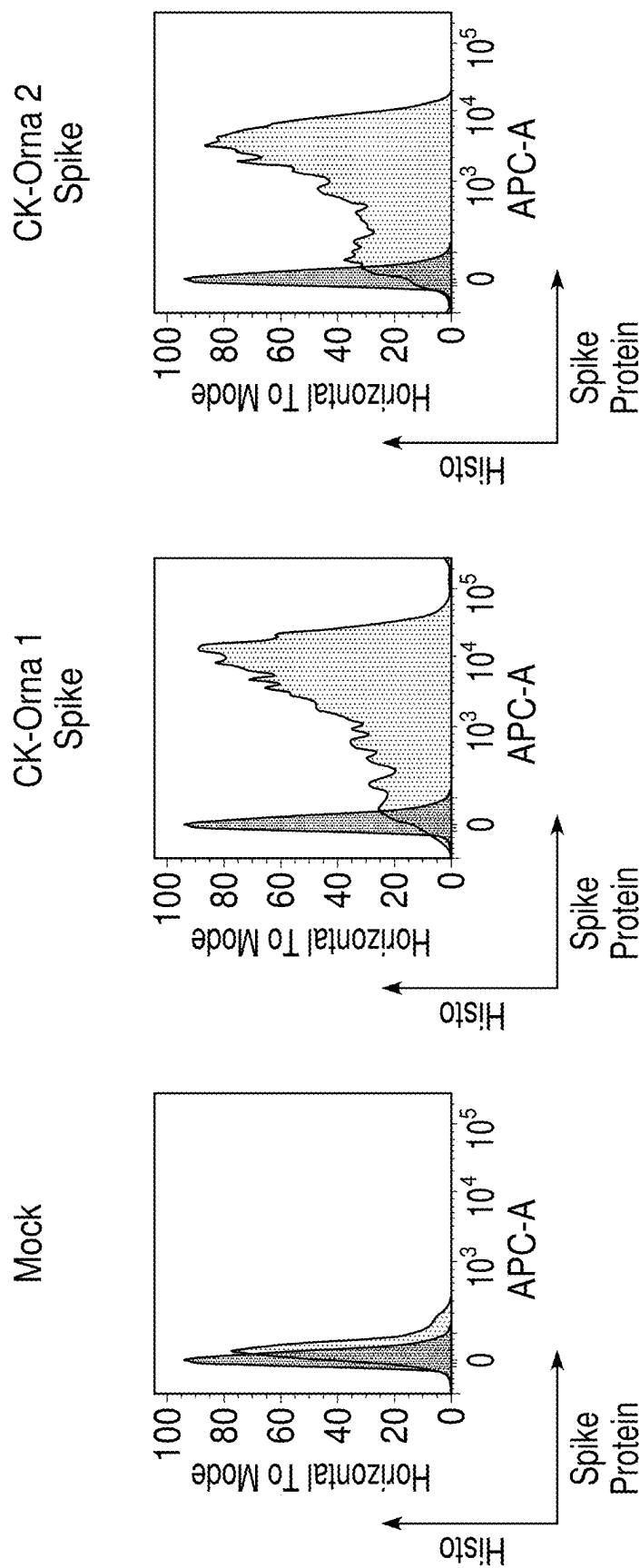


FIG. 50B

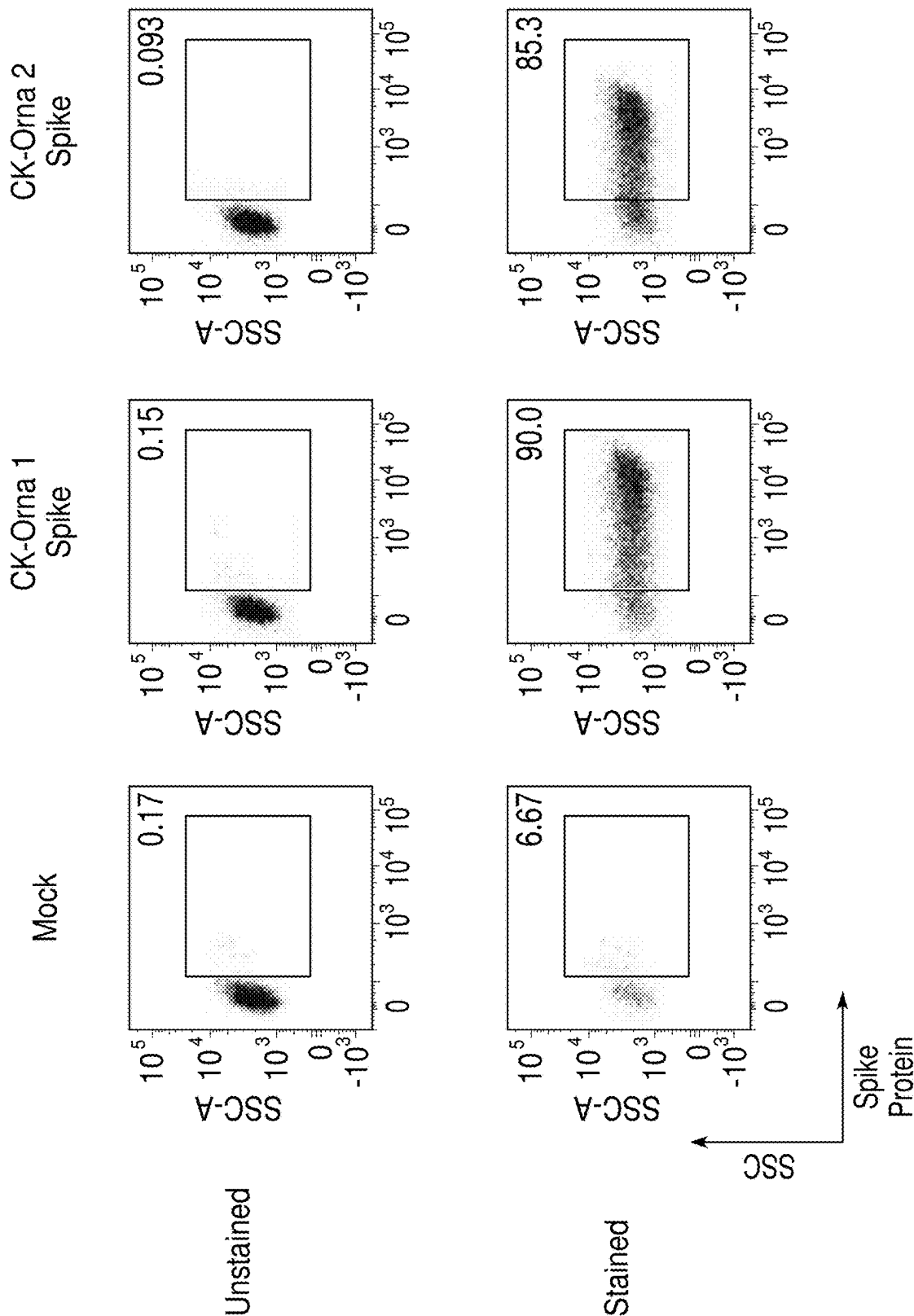


FIG. 50C

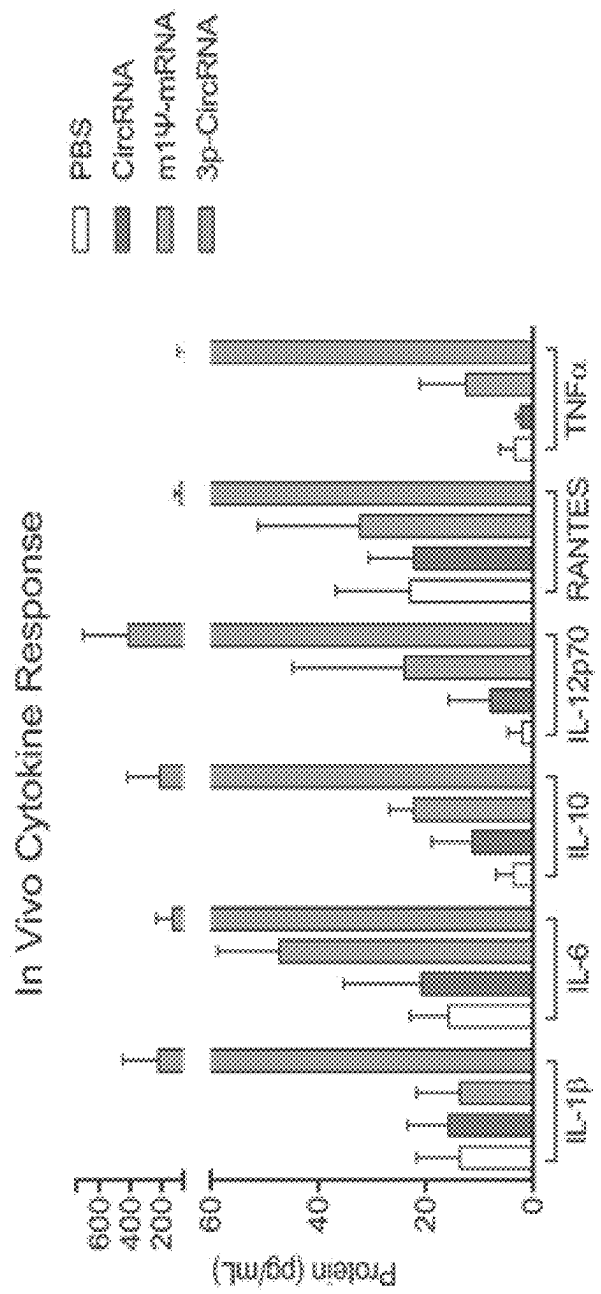


FIG. 51

Live Whole Body Flux 6 hr Post IM Injection

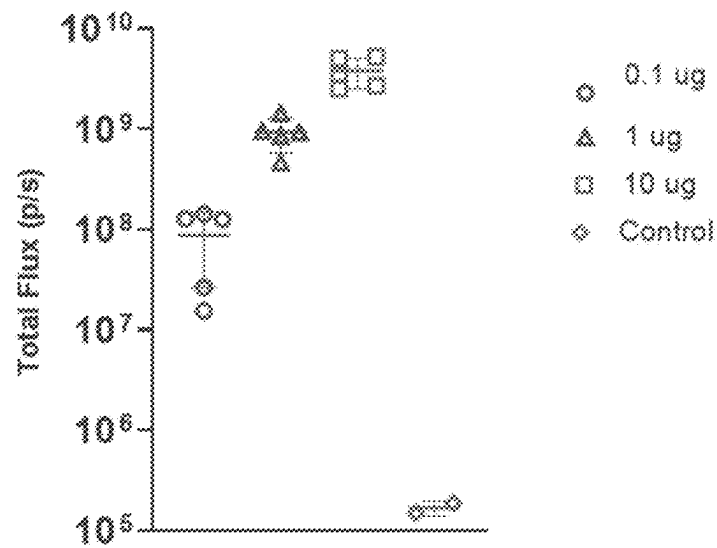


FIG. 52A

Whole Body IVIS:
1 ug dose IM at 6hr

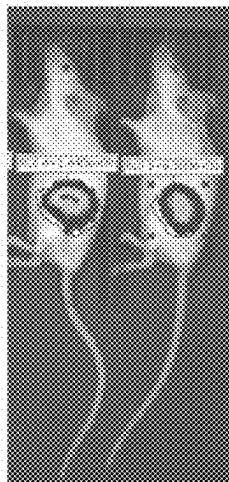


FIG. 52B

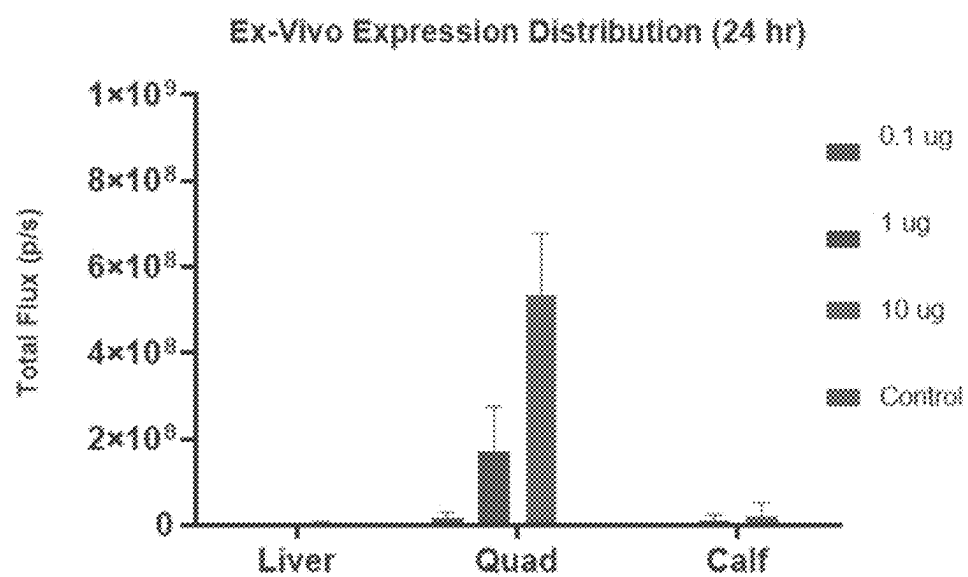


FIG. 52C

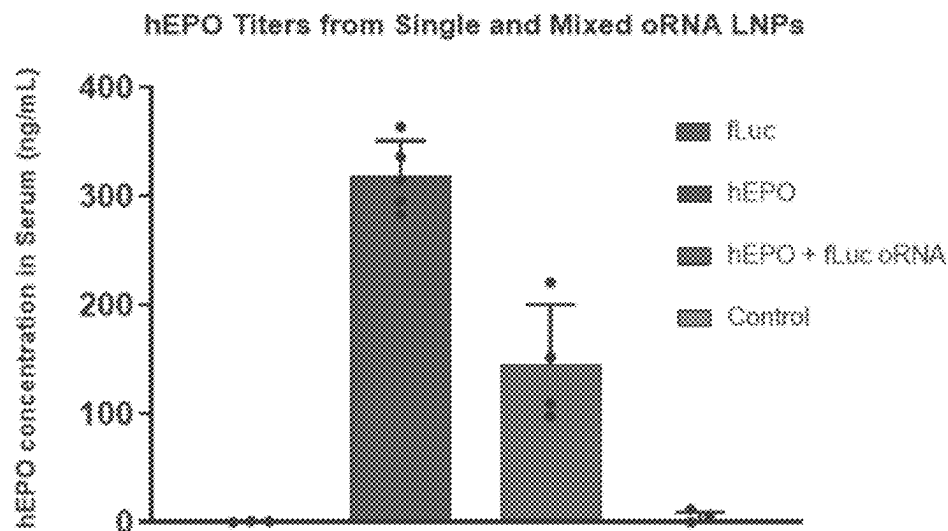


FIG. 53A

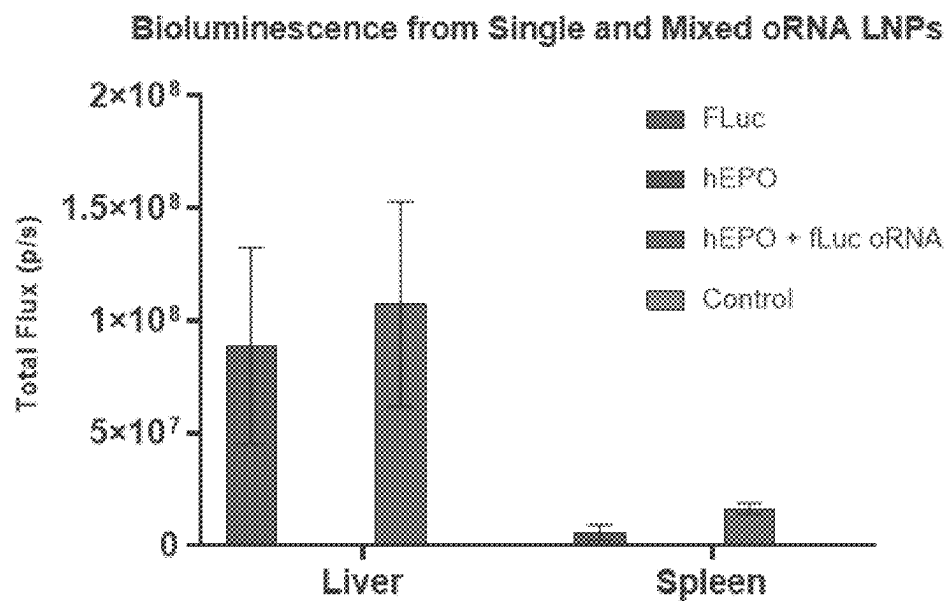


FIG. 53B

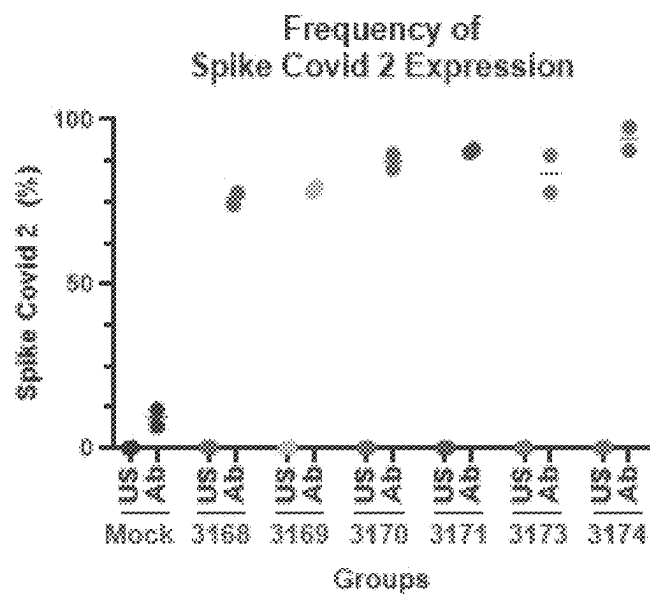


FIG. 54A

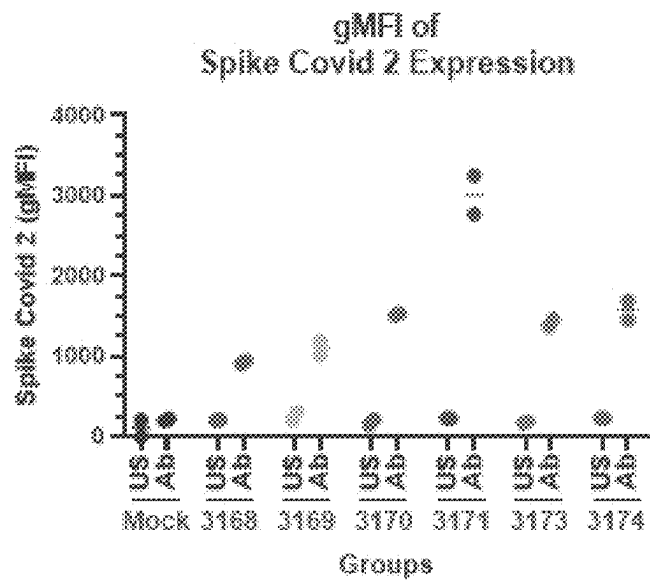


FIG. 54B

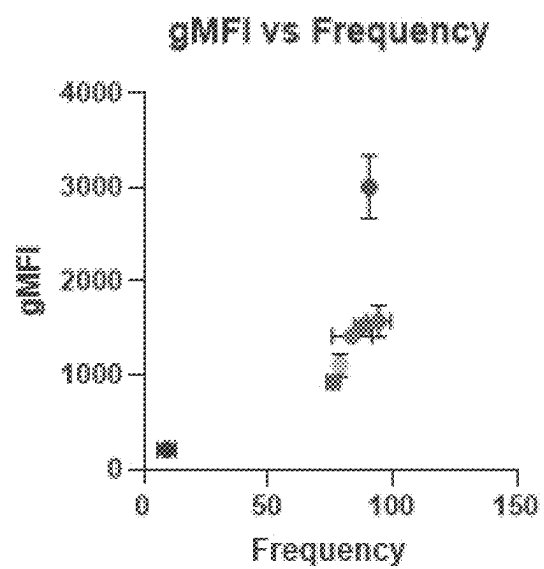


FIG. 54C

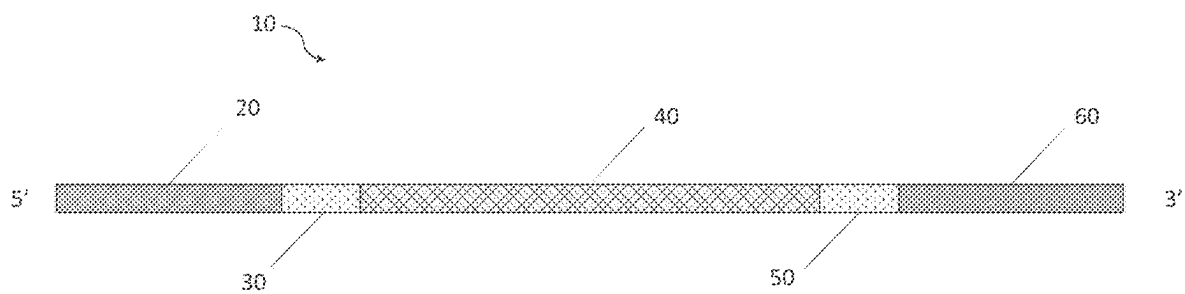


FIG. 55

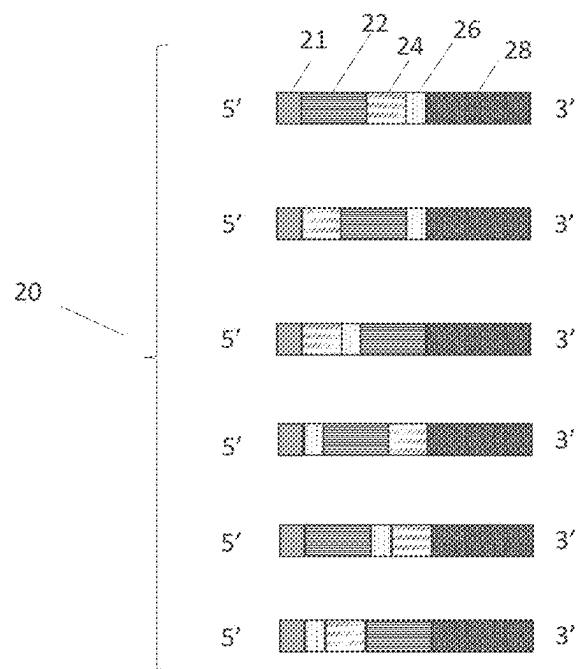


FIG. 56

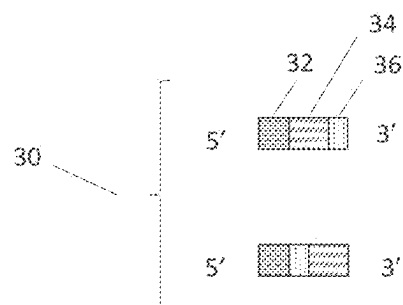


FIG. 57

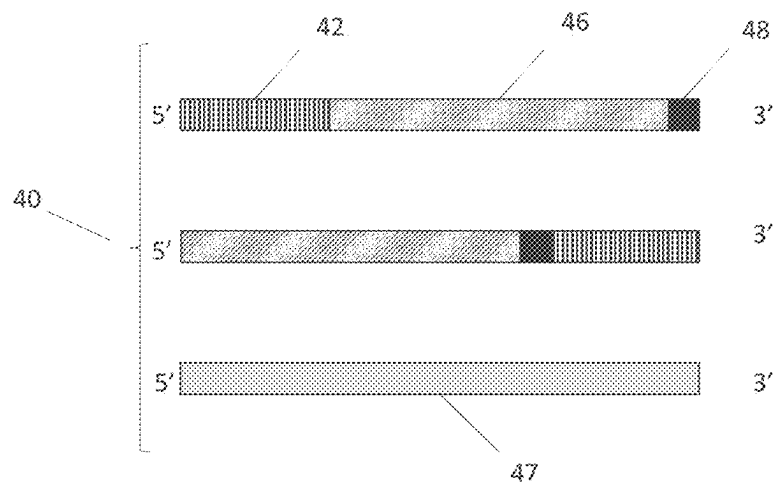


FIG. 58

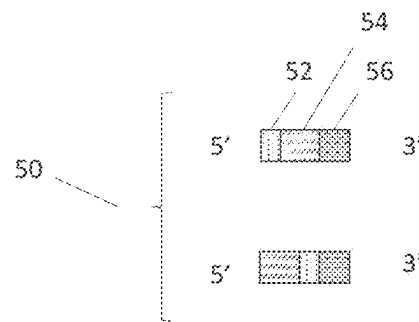


FIG. 59

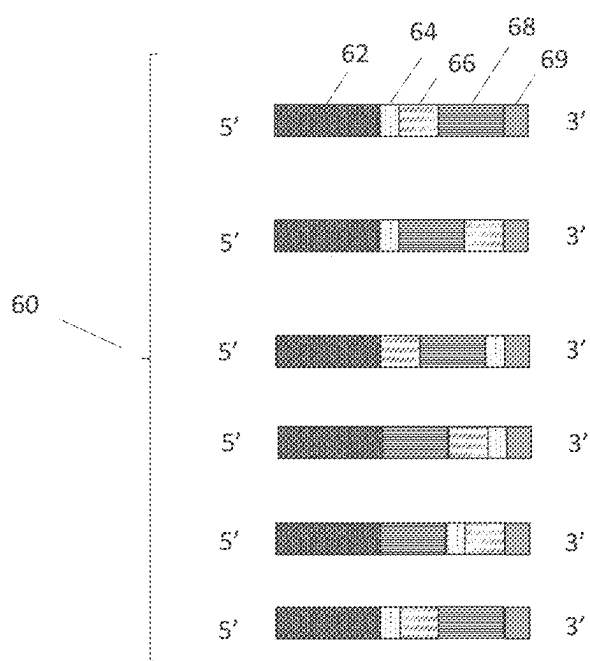


FIG. 60

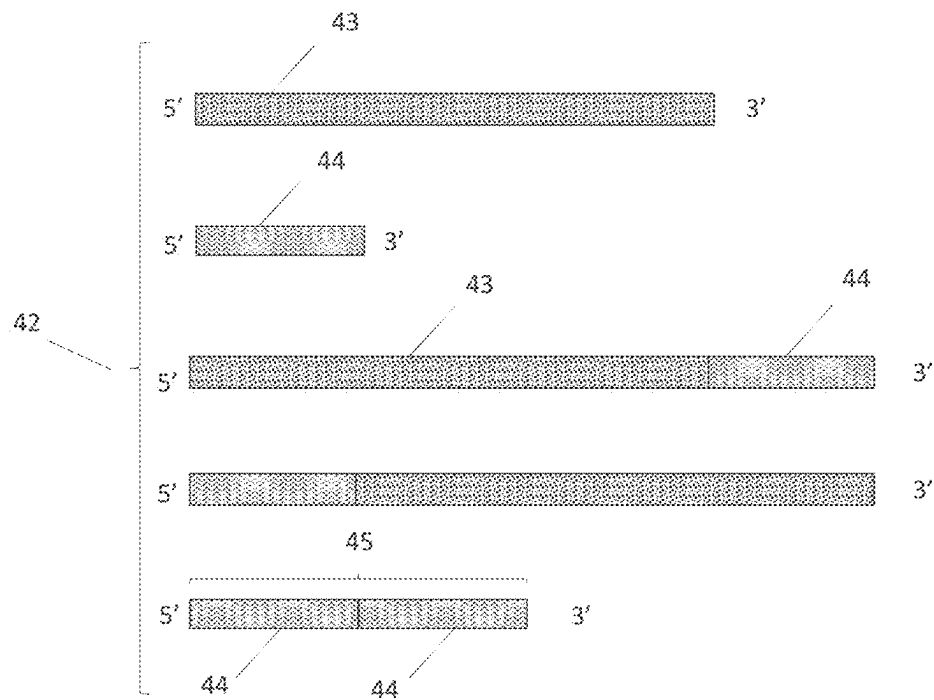


FIG. 61

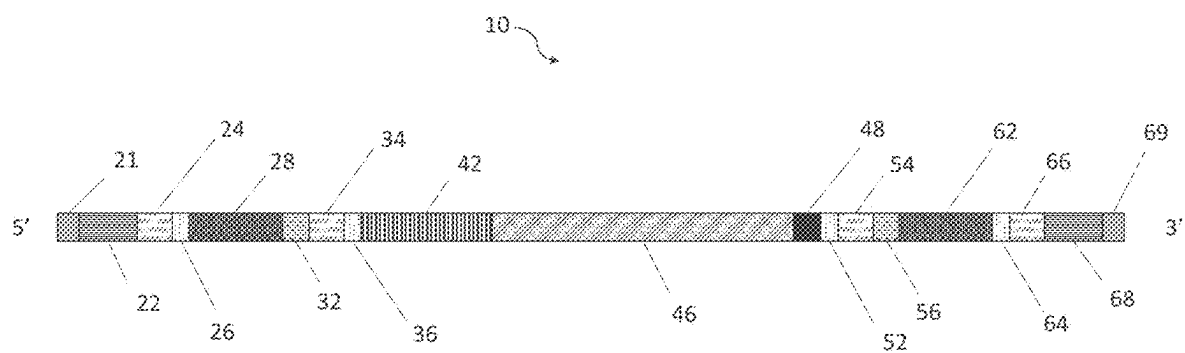


FIG. 62

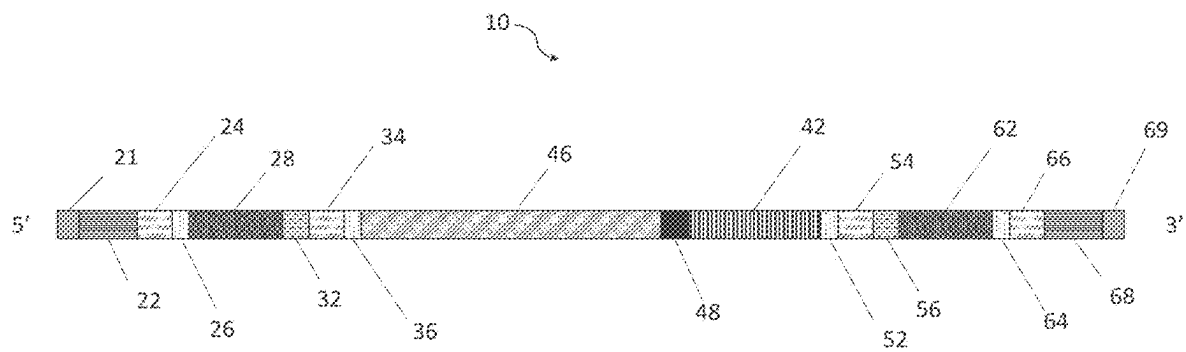


FIG. 63

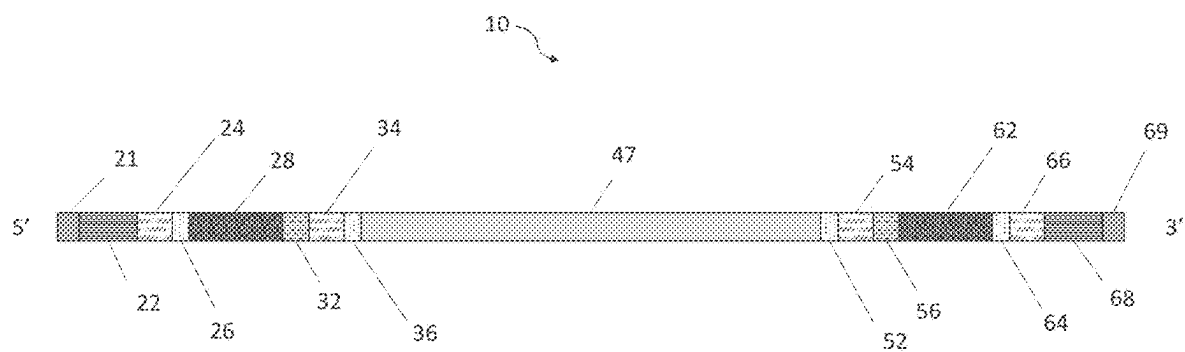


FIG. 64

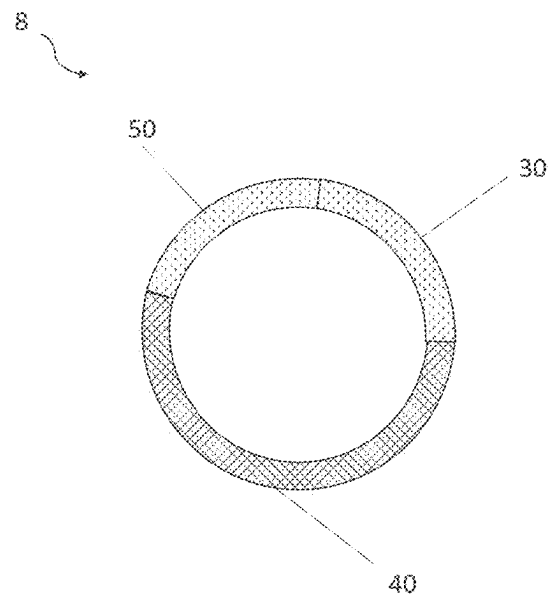


FIG. 65

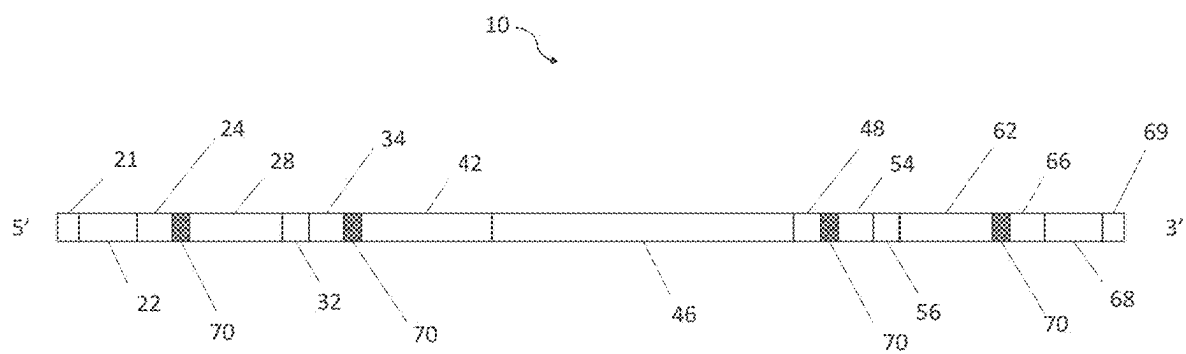


FIG. 66A

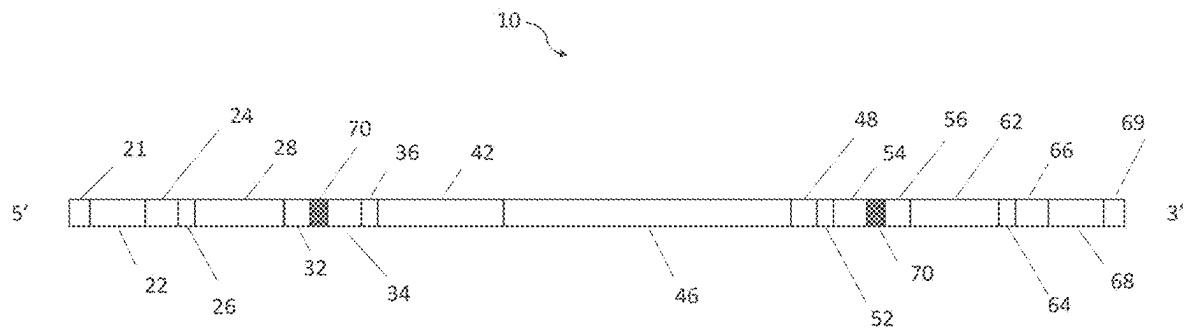


FIG. 66B

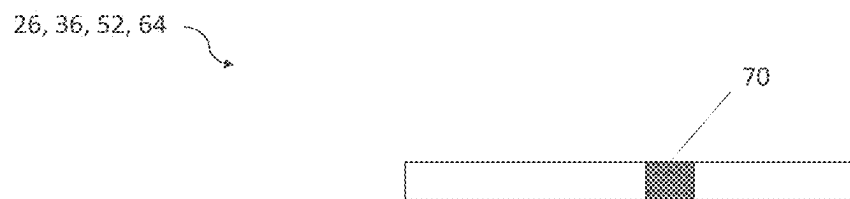


FIG. 66C

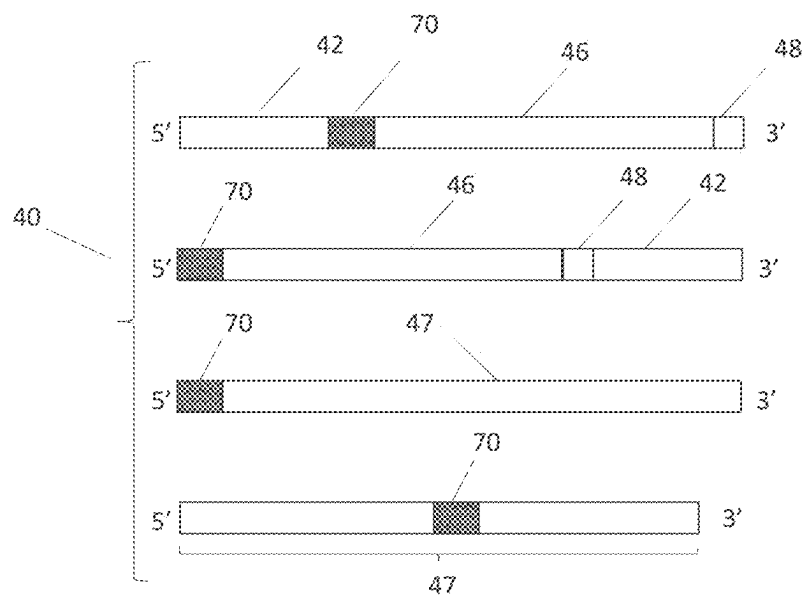


FIG. 66D

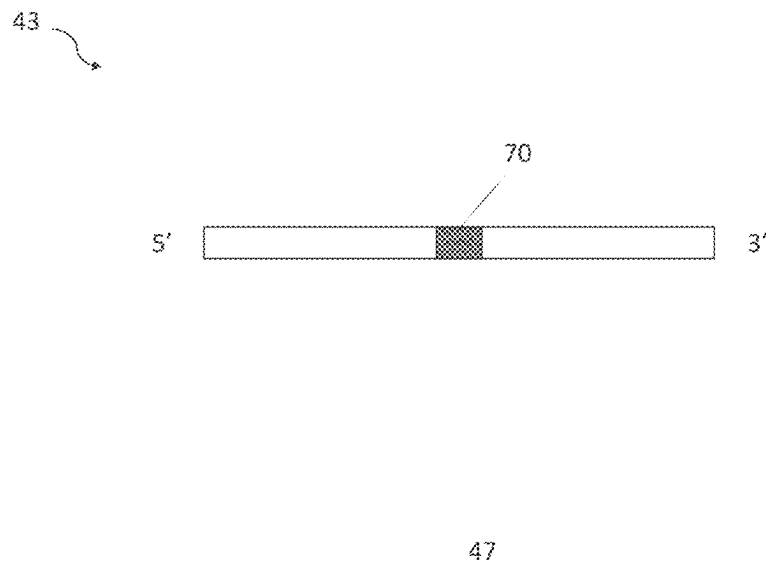


FIG. 66E

Human

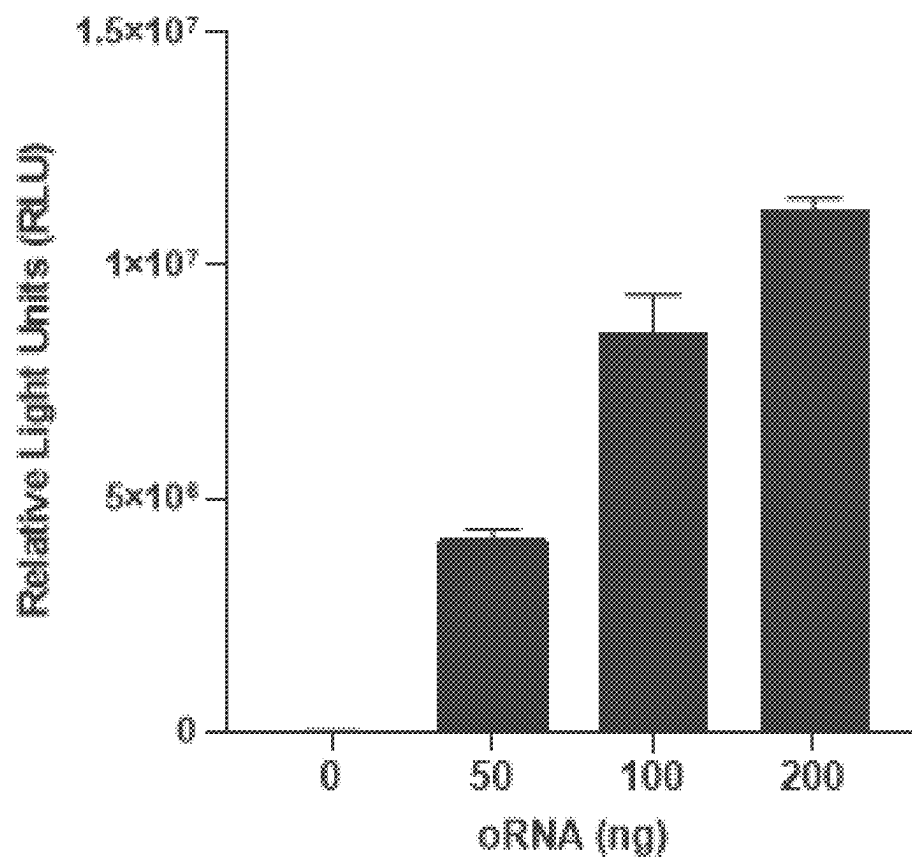


FIG. 67A

Mouse

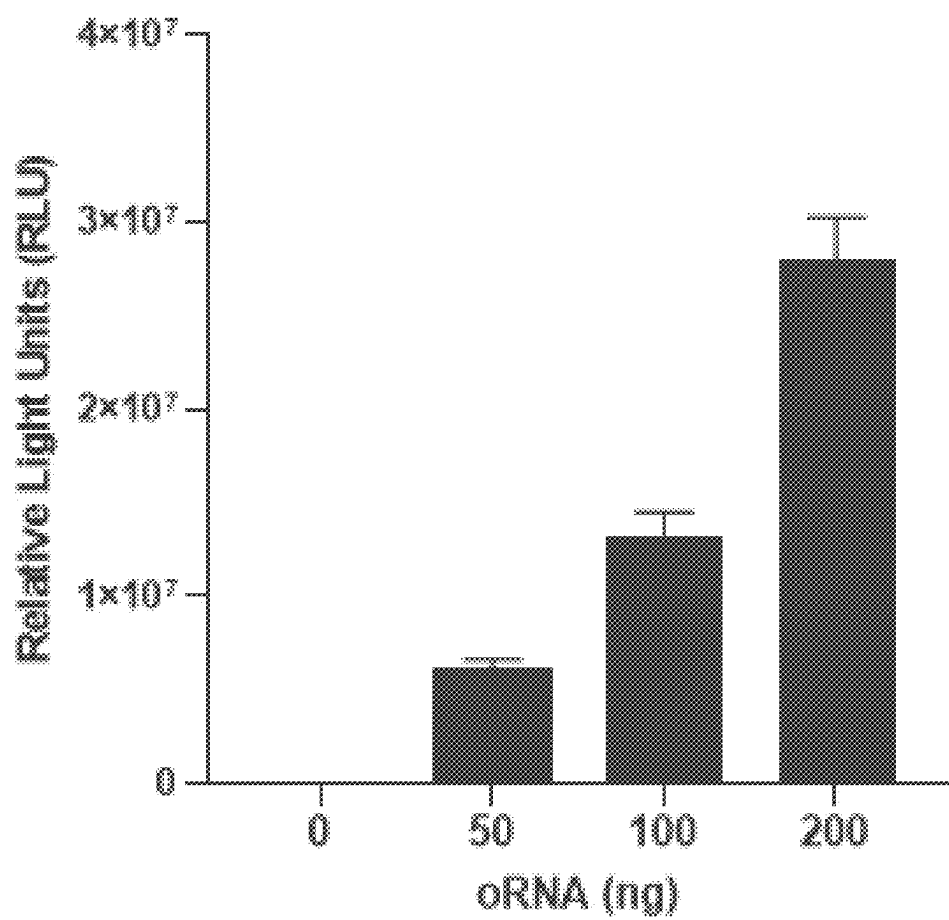


FIG. 67B

Cynomolgus Monkey

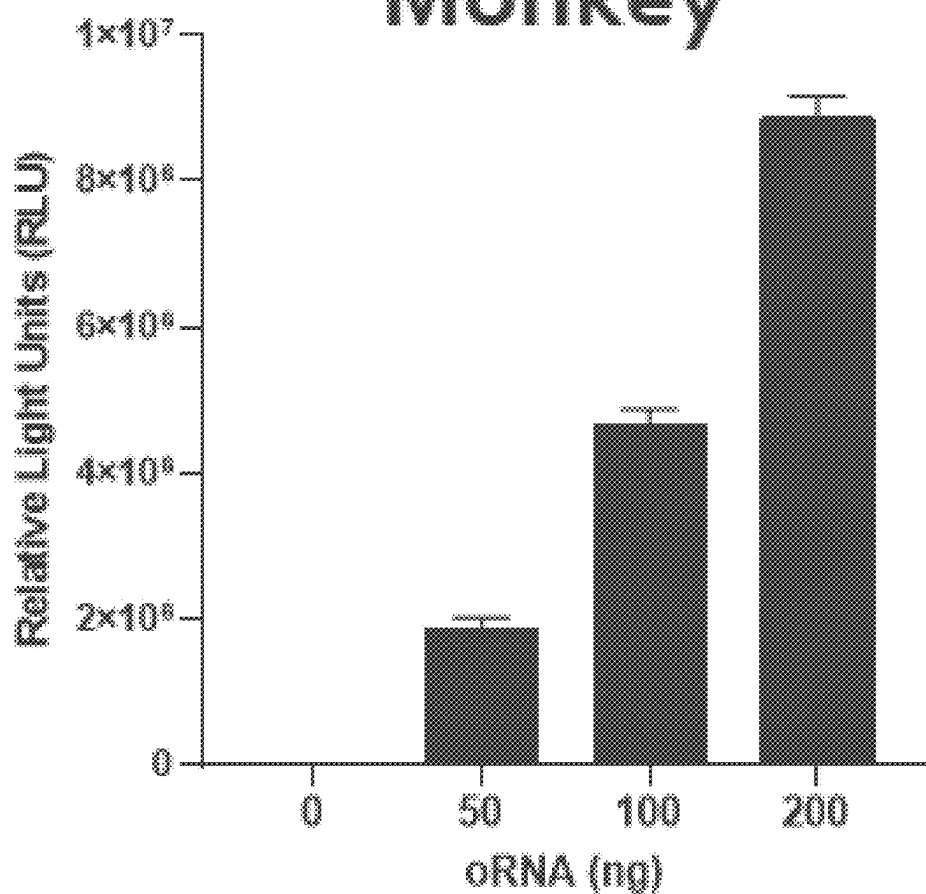


FIG. 67C

Donor 1

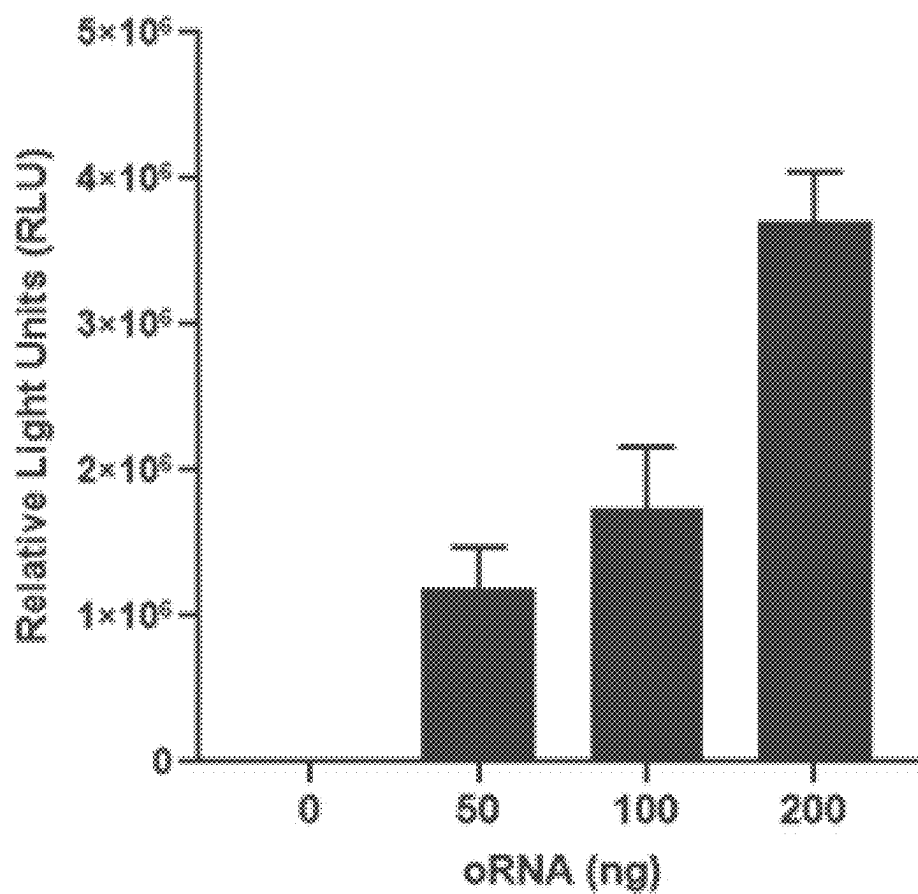


FIG. 68A

Donor 2

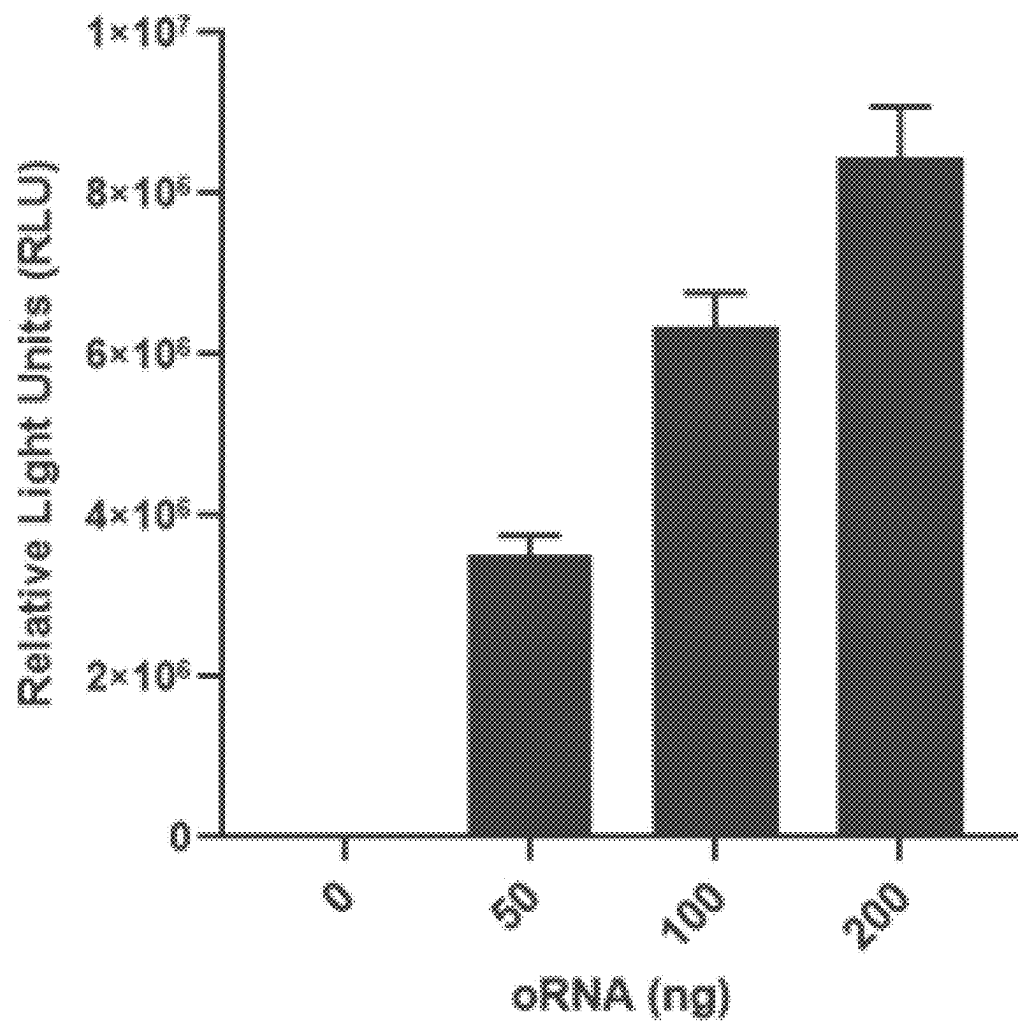


FIG. 68B

Donor 3

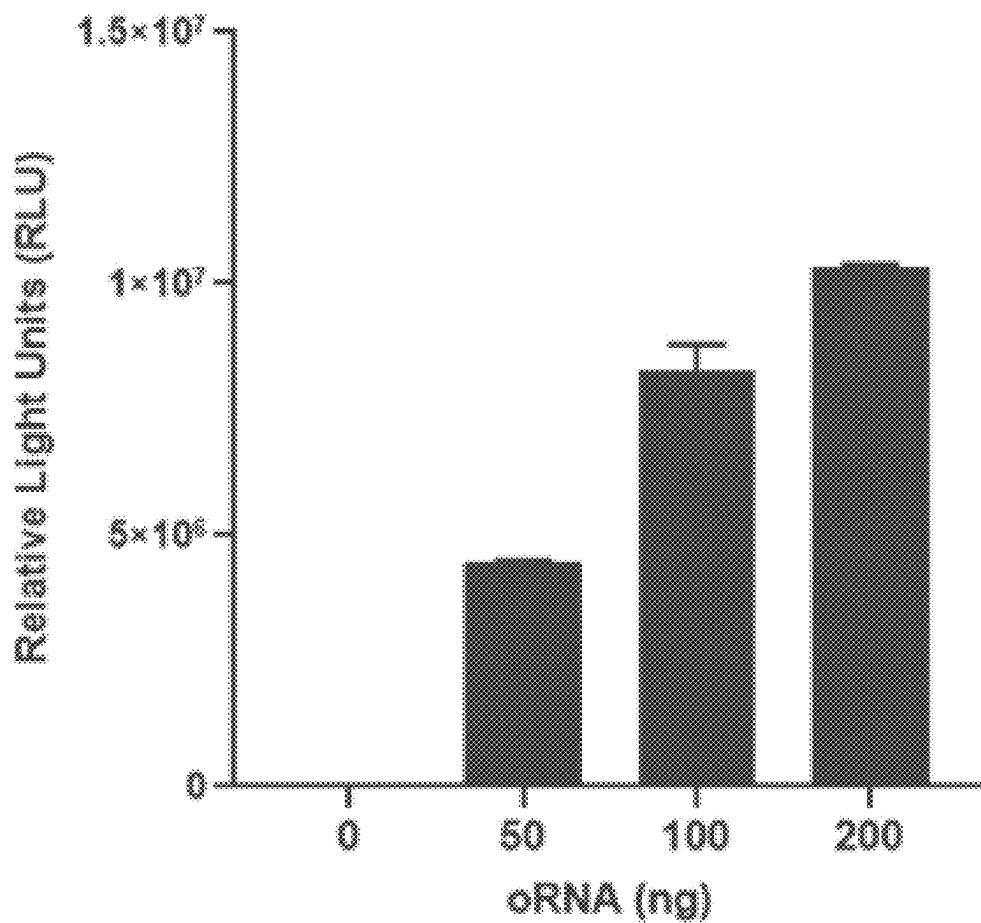


FIG. 68C

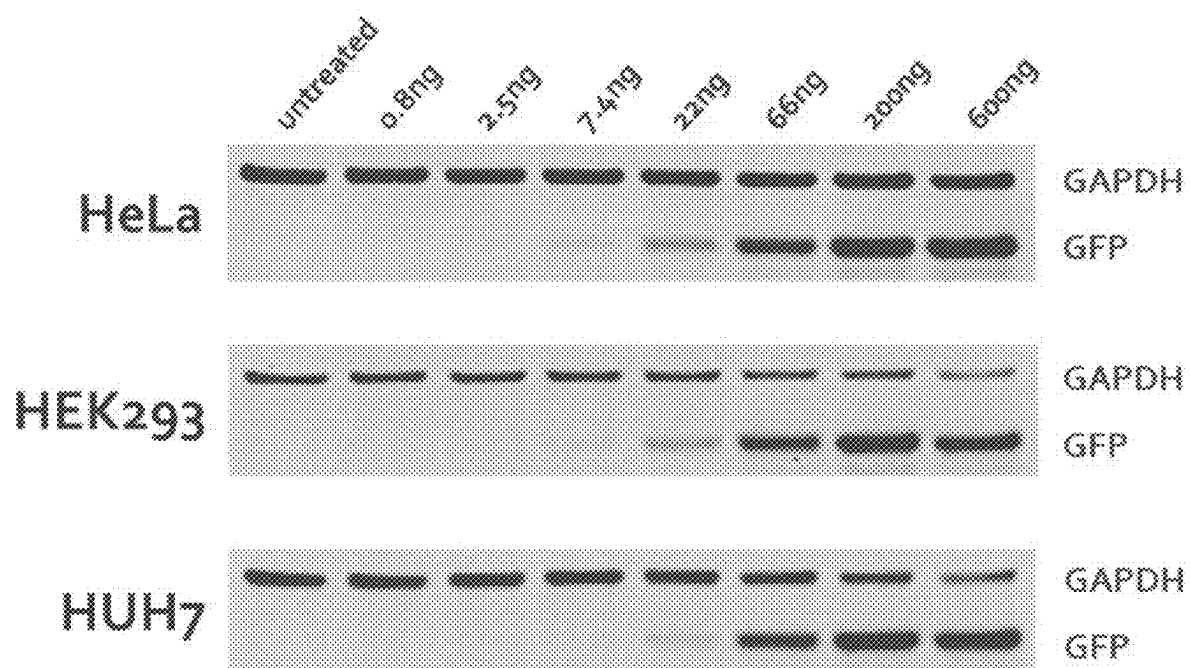


FIG. 69

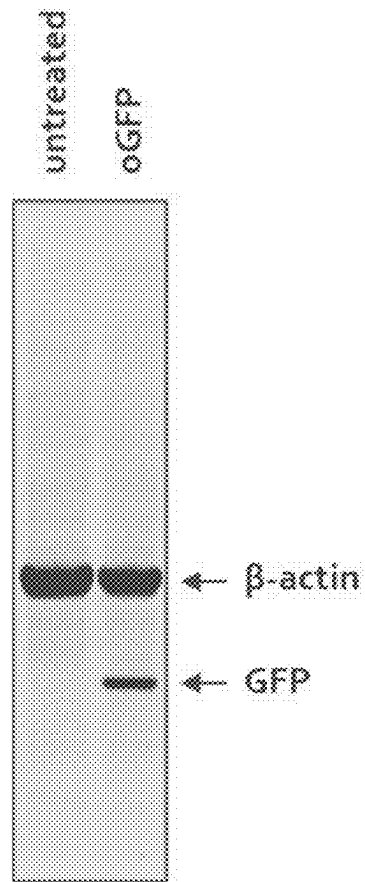


FIG. 70

Mouse C2C12 cells

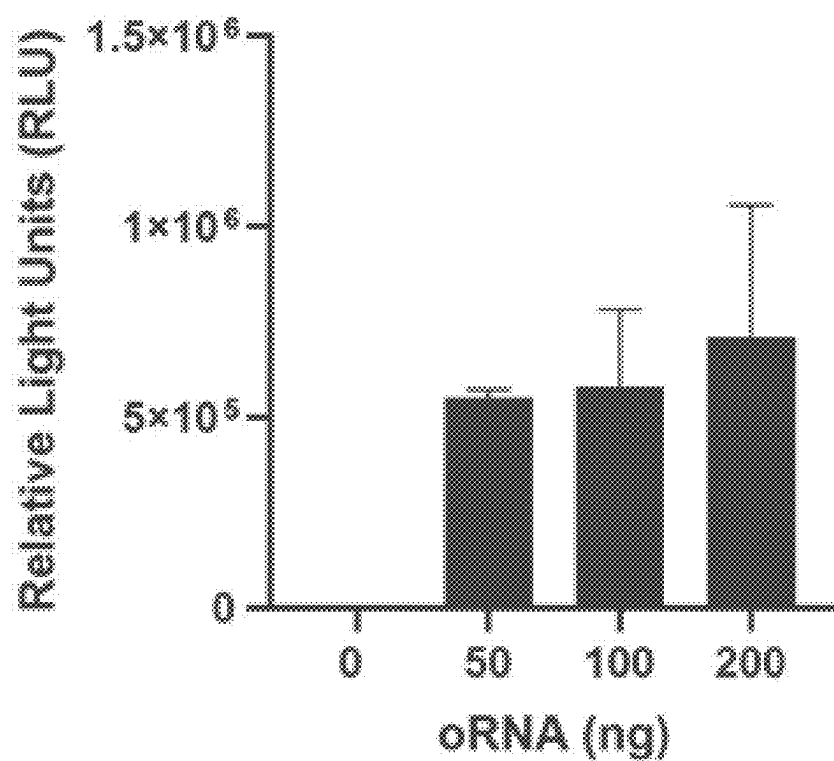


FIG. 71A

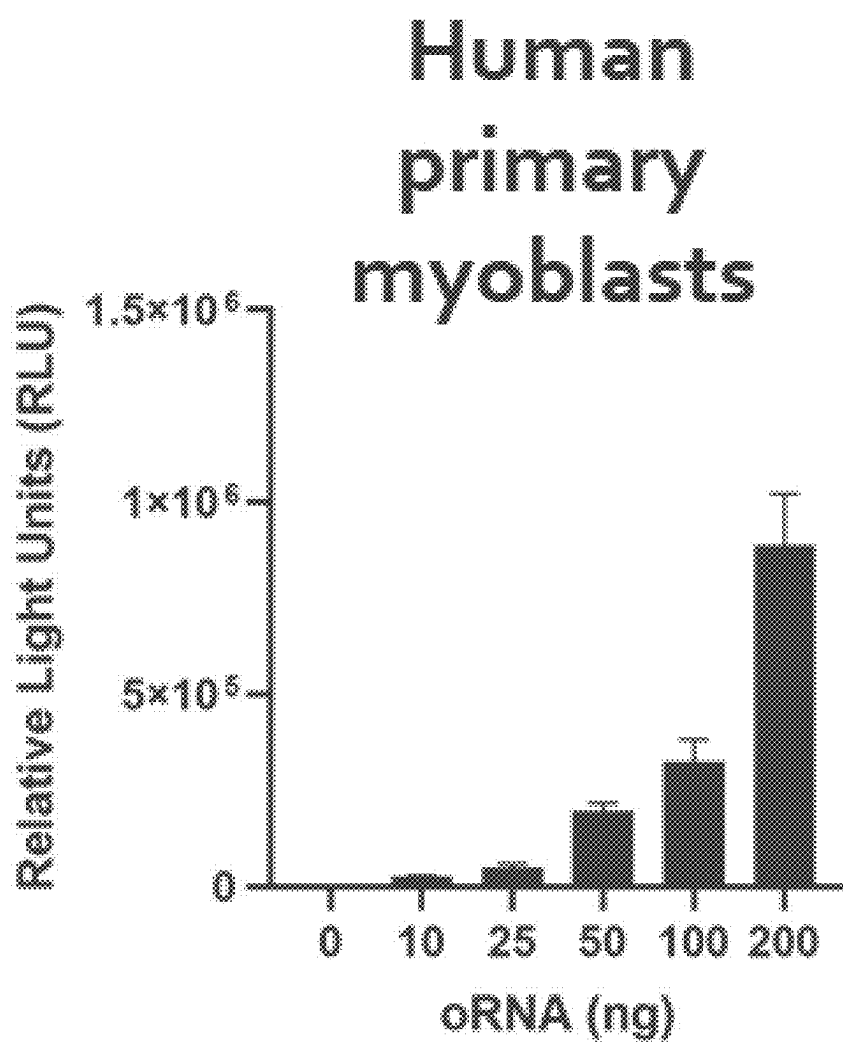


FIG. 71B

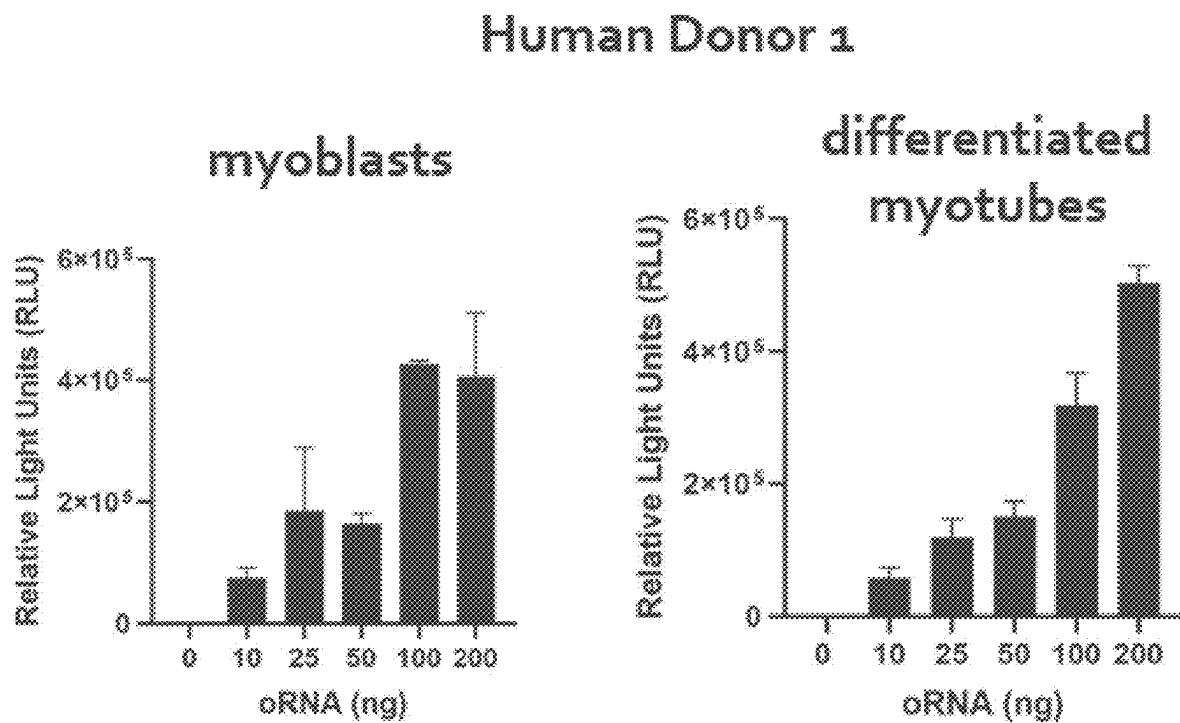


FIG. 72A

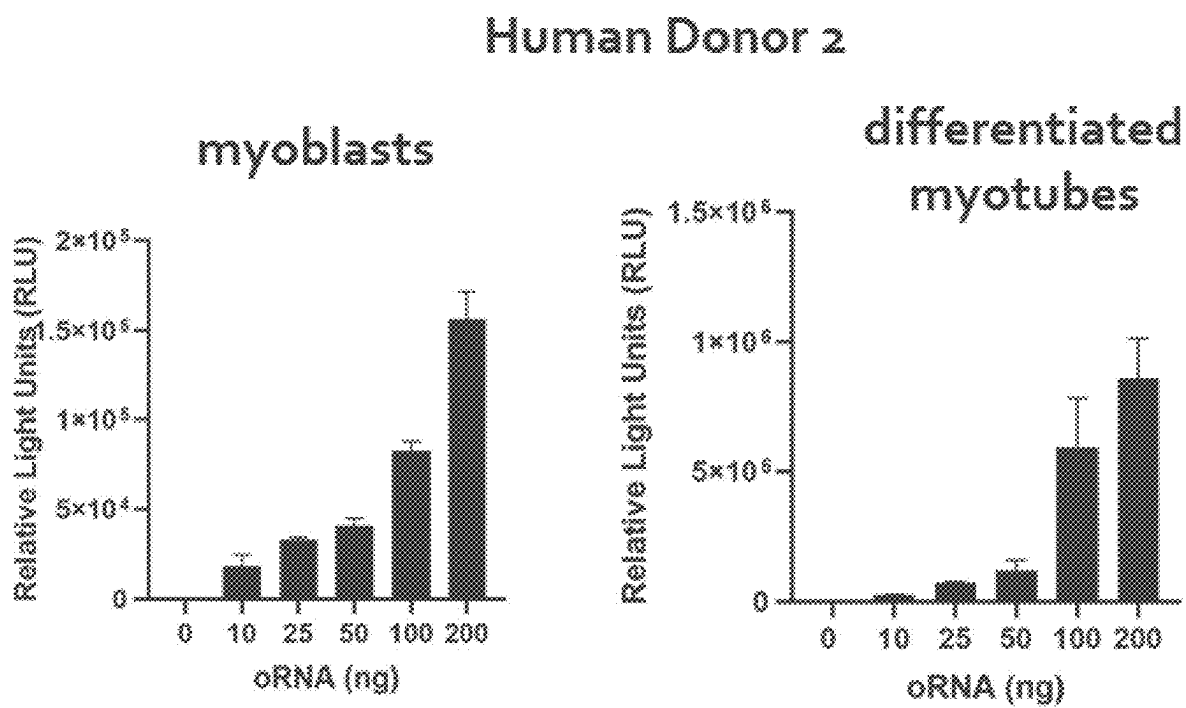


FIG. 72B

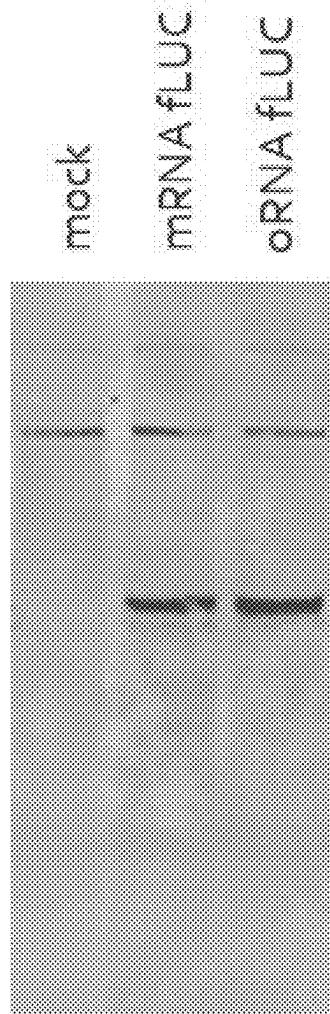


FIG. 73A

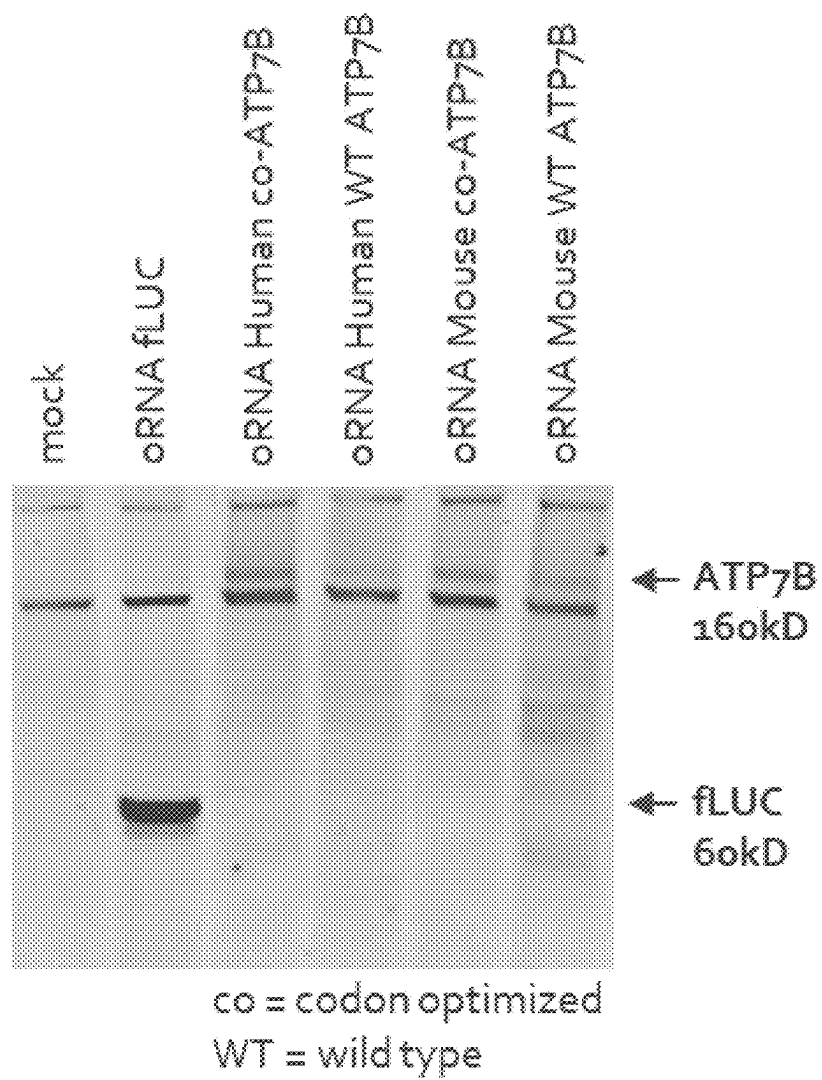


FIG. 73B

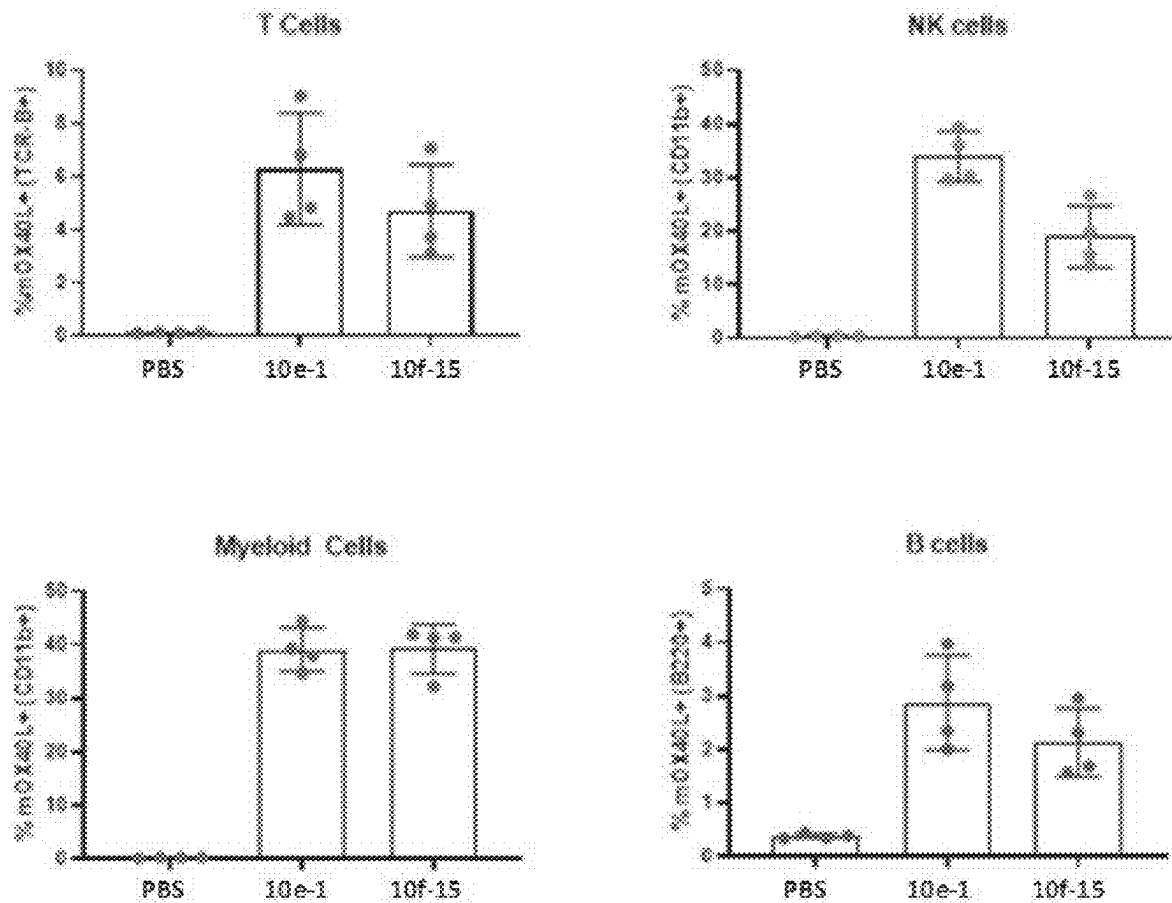


FIG. 74

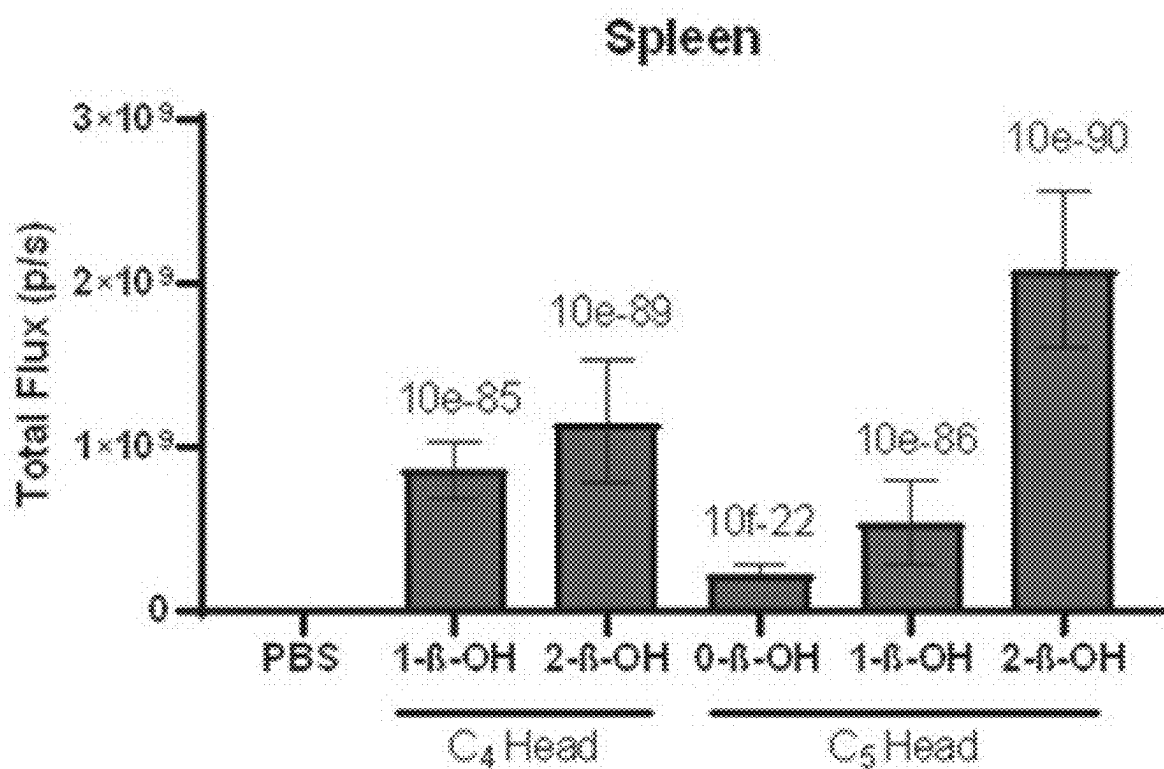


FIG. 75

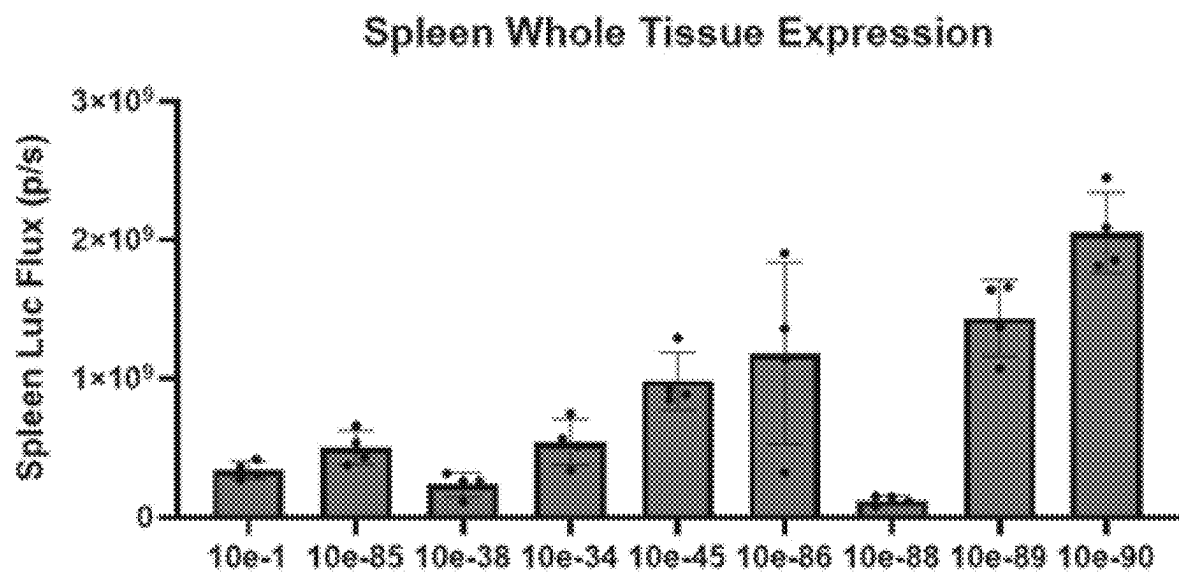


FIG. 76

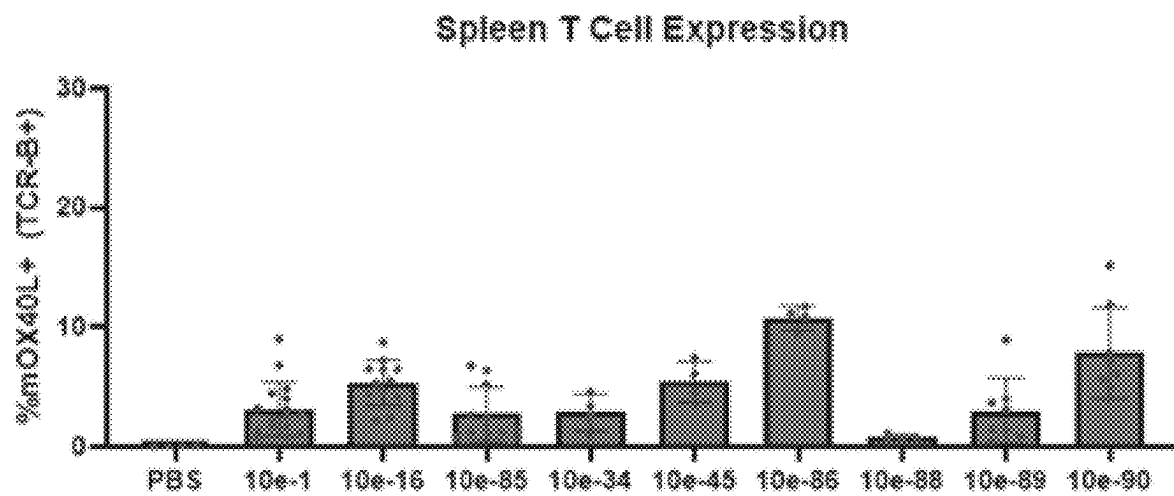


FIG. 77

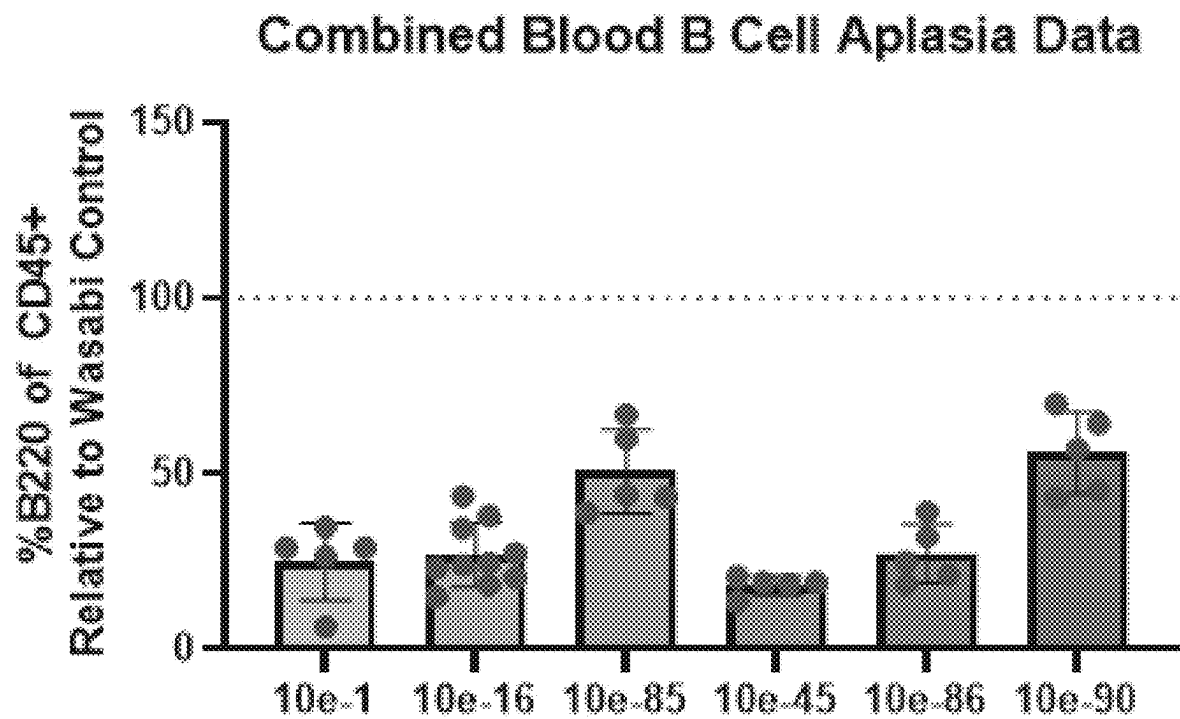


FIG. 78

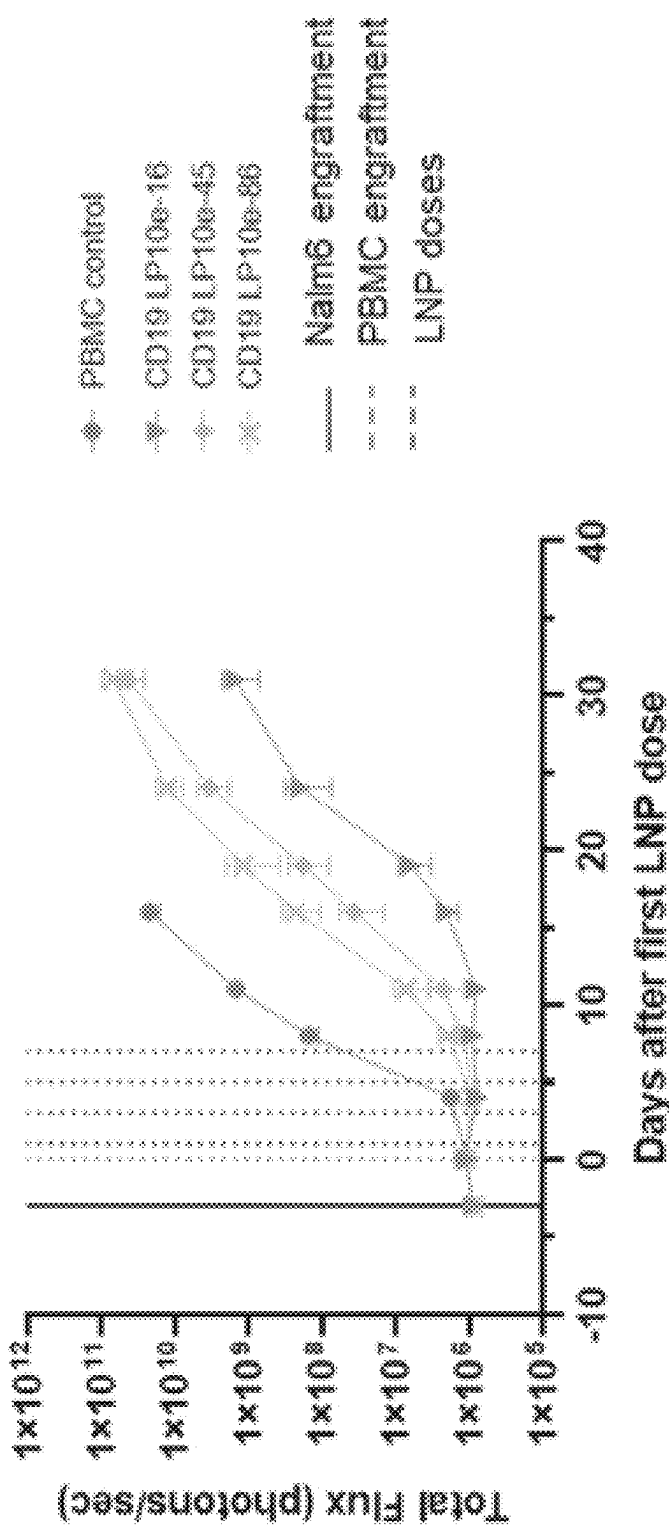


FIG. 79

1

LIPID NANOPARTICLE COMPOSITIONS FOR DELIVERING CIRCULAR POLYNUCLEOTIDES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of PCT/US2022/049313, filed Nov. 8, 2022, which claims the benefit of, and priority to, U.S. Provisional Application No. 63/277,055, filed on Nov. 8, 2021, the contents of which are hereby incorporated by reference in their entirety for all purposes.

FIELD OF THE INVENTION

The present invention generally relates to novel lipids that can be used in combination with other lipid components, such as helper lipids, structural lipids, and cholesterol, to form lipid nanoparticles for delivery of therapeutic agents, such as nucleic acids (e.g. circular polynucleotides), both in vitro and in vivo.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Feb. 24, 2023, is named OBS-028WO_SL.xml and is 159,393 bytes in size.

BACKGROUND OF THE INVENTION

In the past few decades, nucleic acid therapeutics has rapidly expanded and has become the basis for treating a wide variety of diseases. Nucleic acid therapies available include, but are not limited to, the use of DNA or viral vectors for insertion of desired genetic information into the host cell, and/or RNA constructed to encode for a therapeutic protein. DNA and viral vector deliveries carry their own setbacks and challenges that make them less favorable to RNA therapeutics. For example, the introduced DNA in some cases may be unintentionally inserted into an intact gene and result in a mutation that impede or even wholly eliminate the function of the endogenous gene leading to an elimination or deleteriously reduced production of an essential enzyme or interruption of a gene critical for the regulating cell growth. Viral vector-based therapies can result in an adverse immune response. Compared to DNA or viral vectors, RNA is substantially safer and more effective gene therapy agent due to its ability to encode for the protein outside of the nucleus to perform its function. With this, the RNA does not involve the risk of being stably integrated into the genome of the transfected cell.

RNA therapeutics conventionally has consisted of engineering linear messenger RNAs (mRNA). Although more effective than DNA or viral vectors, linear mRNAs have their own set of challenges regarding the stability, immunogenicity, translation efficiency, and delivery. Some of these challenges may lead to size restraints and/or destruction of the linear mRNA due to the challenges present with linear mRNAs' caps. To overcome these limitations, circular polynucleotides or circular RNAs may be used. Due to being covalently closed continuous loops, circular RNAs are useful in the design and production of stable forms of RNA. The circularization of an RNA molecule provides an advantage to the study of RNA structure and function, especially in the case of molecules that are prone to folding in an inactive

2

conformation (Wang and Ruffner, 1998). Circular RNA can also be particularly interesting and useful for in vivo applications, especially in the research area of RNA-based control of gene expression and therapeutics, including protein replacement therapy and vaccination.

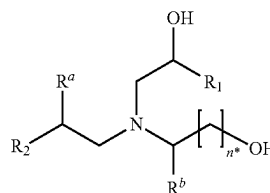
Further to promote an effective delivery of the RNA polynucleotides, nanoparticles delivery systems can be used. This invention disclosed herein provides a robust therapeutic using engineered polynucleotides and lipid nanoparticle compositions, comprising novel lipids.

SUMMARY

The present application provides ionizable lipids and related transfer vehicles, compositions, and methods. The transfer vehicles can comprise ionizable lipid (e.g., ionizable lipids disclosed herein), PEG-modified lipid, and/or structural lipid, thereby forming lipid nanoparticles encapsulating therapeutic agents (e.g., RNA polynucleotides such as circular RNAs). Pharmaceutical compositions comprising such circular RNAs and transfer vehicles are particularly suitable for efficient protein expression in immune cells in vivo. The present application also provides precursor RNAs and materials useful in producing the precursor or circular RNAs, which have improved circularization efficiency and/or are compatible with effective circular RNA purification methods.

In one aspect, provided herein is an ionizable lipid represented by Formula (13*):

Formula (13*)



or a pharmaceutically acceptable salt thereof, wherein:

n is an integer between 1 to 7;

R^a is hydrogen or hydroxyl;

R^b is hydrogen or C₁-C₆ alkyl;

R₁ and R₂ are each independently a linear or branched C₁-C₃₀ alkyl, C₂-C₃₀ alkenyl, or C₁-C₃₀ heteroalkyl, optionally substituted by one or more substituents selected from oxo, halo, hydroxy, cyano, alkyl, alkenyl, aldehyde, heterocyclalkyl, hydroxyalkyl, dihydroxyalkyl, hydroxyalkylaminoalkyl, aminoalkyl, alkylaminoalkyl, dialkylaminoalkyl, (heterocycl)(alkyl)aminoalkyl, heterocycl, heteroaryl, alkylheteroaryl, alkynyl, alkoxy, amino, dialkylamino, aminoalkylcarbonylamino, aminocarbonylalkylamino, (aminocarbonylalkyl)(alkyl)amino, alkenylcarbonylamino, hydroxycarbonyl, alkyloxy-carbonyl, alkylcarbonyloxy, alkylcarbonate, alkenyloxy-carbonyl, alkenylcarbonyloxy, alkenylcarbonate, alkynyloxy-carbonyl, alkynylcarbonyloxy,

3

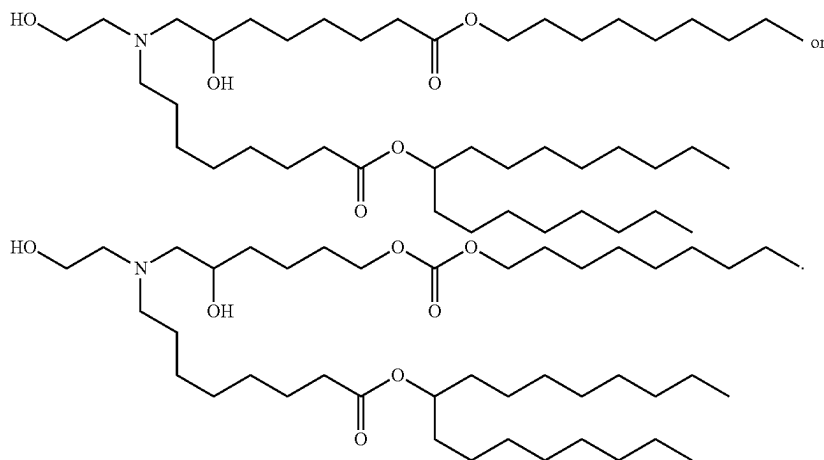
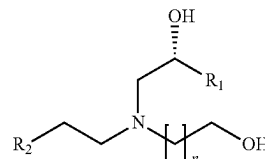
alkylcarbonate, aminocarbonyl, aminoalkylaminocarbonyl, alkylaminoalkylaminocarbonyl, dialkylaminoalkylaminocarbonyl, heterocyclalkylaminocarbonyl, (alkylaminoalkyl)(alkyl)aminocarbonyl, alkylaminoalkylcarbonyl, dialkylaminoalkylcarbonyl, heterocyclalkylcarbonyl, alkenylcarbonyl, alkynylcarbonyl, alkylsulfoxide, alkylsulfoxidealkyl alkylsulfonyl, and alkylsulfonealkyl;

with the proviso that the ionizable lipid is not

4

-continued

Formula (13a-3)

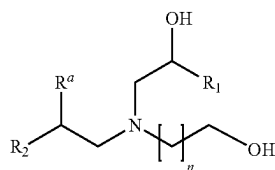


In some embodiments, R^b is C_1 - C_6 alkyl.
In some embodiments, R^b is H and the ionizable lipid is represented by Formula (13):

In some embodiments, R_a is hydroxyl. In some embodiments, the ionizable lipid is represented by Formula (13b-1), Formula (13b-2), or Formula (13b-3):

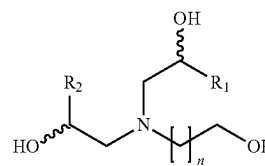
35

Formula (13)



Formula (13b-1)

40



45

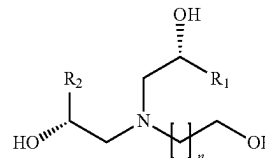
wherein n is an integer between 1 to 7.

In some embodiments, n is 1, 2, 3, or 4.

In some embodiments, R_z is hydrogen. In some embodiments, the ionizable lipid is represented by Formula (13a-1), Formula (13a-2), or Formula (13a-3):

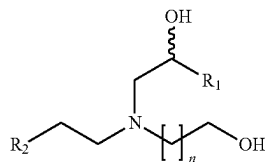
Formula (13b-2)

50



Formula (13a-1)

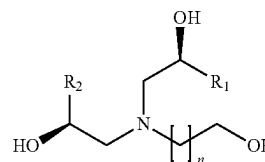
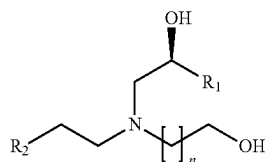
55



Formula (13b-3)

Formula (13a-2)

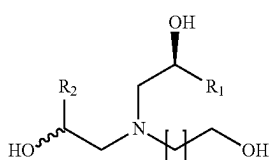
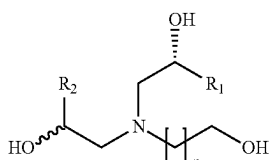
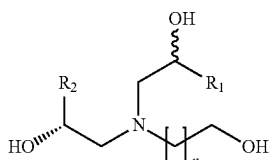
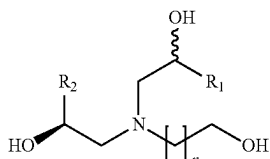
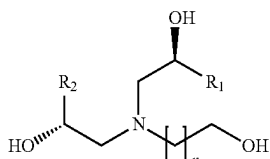
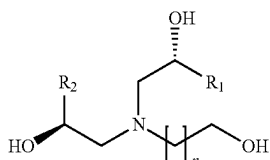
60



In some embodiments, the ionizable lipid is represented by Formula (13b-4), Formula (13b-5), Formula (13b-6), Formula (13b-7), Formula (13b-8), or Formula (13 b-9):

65

5



In some embodiments, R_1 and R_2 are independently a linear or branched C_1 - C_{20} alkyl, C_2 - C_{20} alkenyl, or C_1 - C_{20} heteroalkyl, optionally substituted by one or more substituents selected from C_1 - C_{20} alkoxy, C_1 - C_{20} alkylloxycarbonyl, C_1 - C_{20} alkylcarbonyloxy, C_1 - C_{20} alkylcarbonate, C_2 - C_{20} alkenyloxycarbonyl, C_2 - C_{20} alkenylcarbonyloxy, C_2 - C_{20} alkenylcarbonate, C_2 - C_{20} alkynyloxycarbonyl, C_2 - C_{20} alkynylcarbonyloxy, and C_2 - C_{20} alkynylcarbonate.

In some embodiments, at least one of R_1 and R_2 is an unsubstituted, linear or branched C_6 - C_{30} alkyl, C_6 - C_{30} alkenyl, or C_6 - C_{30} heteroalkyl. In some embodiments, at least one of R_1 and R_2 is a linear C_1 - C_{12} alkyl substituted by $-\text{OC}(\text{O})\text{R}^6$, $-\text{C}(\text{O})\text{OR}^6$, or $-\text{OC}(\text{O})\text{OR}^6$, wherein each R^6 is independently linear or branched C_1 - C_{20} alkyl or C_2 - C_{20} alkenyl. In some embodiments, R_1 and R_2 are each independently a linear C_1 - C_{12} alkyl substituted by $-\text{OC}(\text{O})\text{R}^6$, $-\text{C}(\text{O})\text{OR}^6$, or $-\text{OC}(\text{O})\text{OR}^6$, wherein each R^6 is independently linear or branched C_1 - C_{20} alkyl or C_2 - C_{20} alkenyl.

In some embodiments, the at least one of R_1 and R_2 is selected from: $-(\text{CH}_2)_q\text{C}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$, $-(\text{CH}_2)_q\text{OC}(\text{O})(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$, and $-(\text{CH}_2)_q\text{OC}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$, wherein:

6

q is an integer between 0 to 12,

r is an integer between 0 to 6,

R^8 is H or R^{10} , and

R^9 and R^{10} are independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_2 - C_{12} -alkenyl.

In some embodiments, R_1 and R_2 are each independently selected from: $-(\text{CH}_2)_q\text{C}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$, $-(\text{CH}_2)_q\text{OC}(\text{O})(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$, and $-(\text{CH}_2)_q\text{OC}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$,

wherein:

q is an integer between 0 to 12,

r is an integer between 0 to 6,

R^8 is H or R^{10} , and

R^9 and R^{10} are independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_2 - C_{12} -alkenyl.

In some embodiments, R_1 is unsubstituted, linear or branched C_6 - C_{30} alkyl. In some embodiments, R_1 is $-(\text{CH}_2)_q\text{C}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$. In some embodiments, R_1 is $-(\text{CH}_2)_q\text{OC}(\text{O})(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$. In some embodiments, R_1 is $-(\text{CH}_2)_q\text{OC}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$. In some embodiments, wherein R_2 is unsubstituted, linear or branched C_6 - C_{30} alkyl. In some embodiments, R_2 is $-(\text{CH}_2)_q\text{C}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$. In some embodiments, R_2 is $-(\text{CH}_2)_q\text{OC}(\text{O})(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$. In some embodiments, R_2 is $-(\text{CH}_2)_q\text{OC}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$.

In some embodiments, q is an integer between 1 to 6. In some embodiments, q is 3, 4, 5, or 6. In some embodiments, r is 0. In some embodiments, r is an integer between 1 to 6. In some embodiments, r is 1. In some embodiments, r is 2.

In some embodiments, R^8 is H. In some embodiments, R^8 is R^{10} .

In some embodiments, R^9 and R^{10} are each independently unsubstituted linear C_1 - C_{12} alkyl. In some embodiments, R^9 and R^{10} are each independently unsubstituted linear C_4 - C_8 alkyl. In some embodiments, R^9 and R^{10} are each independently unsubstituted linear C_6 - C_8 alkyl.

In some embodiments, R_1 and R_2 are each $-(\text{CH}_2)_m\text{-L-R}'$, wherein:

m is an integer from 0 to 10;

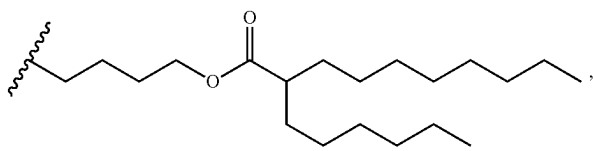
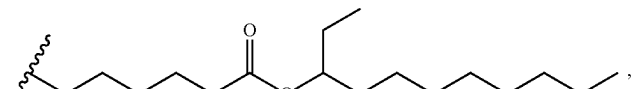
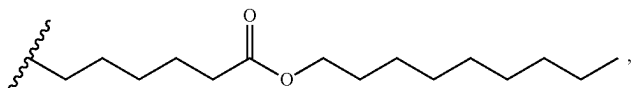
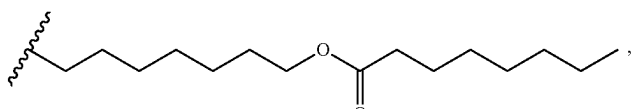
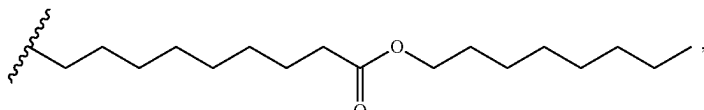
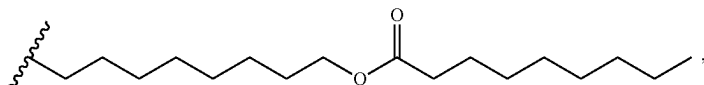
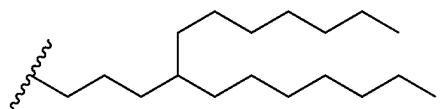
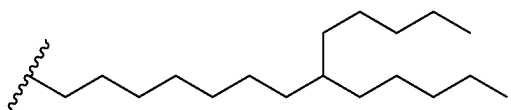
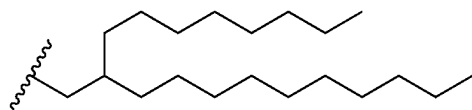
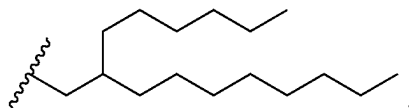
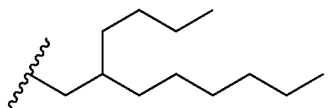
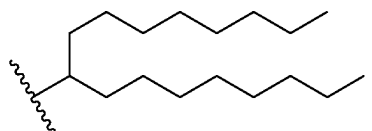
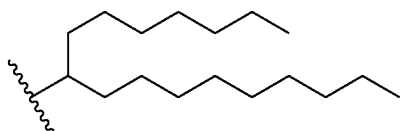
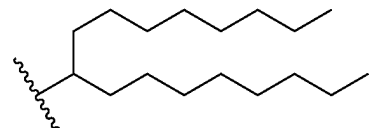
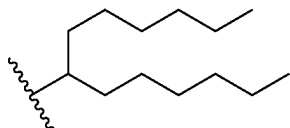
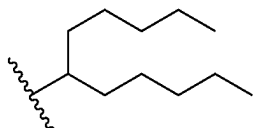
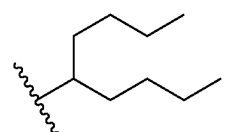
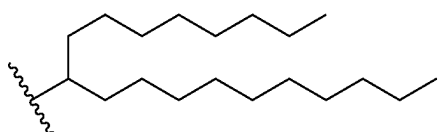
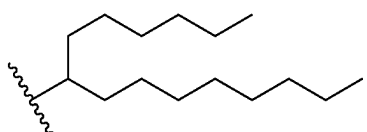
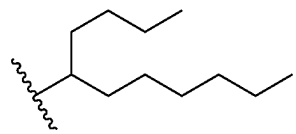
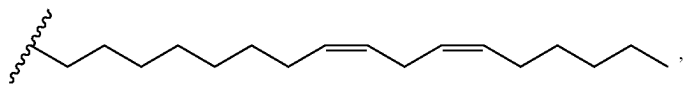
L is absent, $-\text{C}(\text{H})(\text{R}^L)-*$, $-\text{OC}(\text{O})-*$, or $-\text{C}(\text{O})\text{O}-*$, wherein “-*” indicates the attachment point to R' ;

R' is selected from the group consisting of: C_1 - C_{30} alkyl, C_2 - C_{30} alkenyl, C_1 - C_{30} alkoxy, 2-30-membered heteroalkylene, and 3-12-membered heterocyclyl, wherein 2-30-membered heteroalkylene is optionally substituted with one or more R'' , and 3-12-membered heterocyclyl is optionally substituted with one or more C_1 - C_{30} alkyl;

R'' is selected from the group consisting of: C_1 - C_{30} alkyl, C_2 - C_{30} alkenyl, C_1 - C_{30} alkoxy, 2-30-membered heteroalkylene, wherein 3-12-membered heteroalkylene is optionally substituted one or more with R ,

R'' is each independently selected from the group consisting of: oxo, C_1 - C_{30} alkoxy, $-\text{C}(\text{O})-\text{C}_1$ - C_{30} alkyl, $-\text{C}(\text{O})-\text{C}_1$ - C_{30} alkoxy, and $-\text{C}(\text{O})-\text{C}_1$ - C_{30} alkylene- $\text{C}(\text{O})-\text{C}_1$ - C_{30} alkoxy.

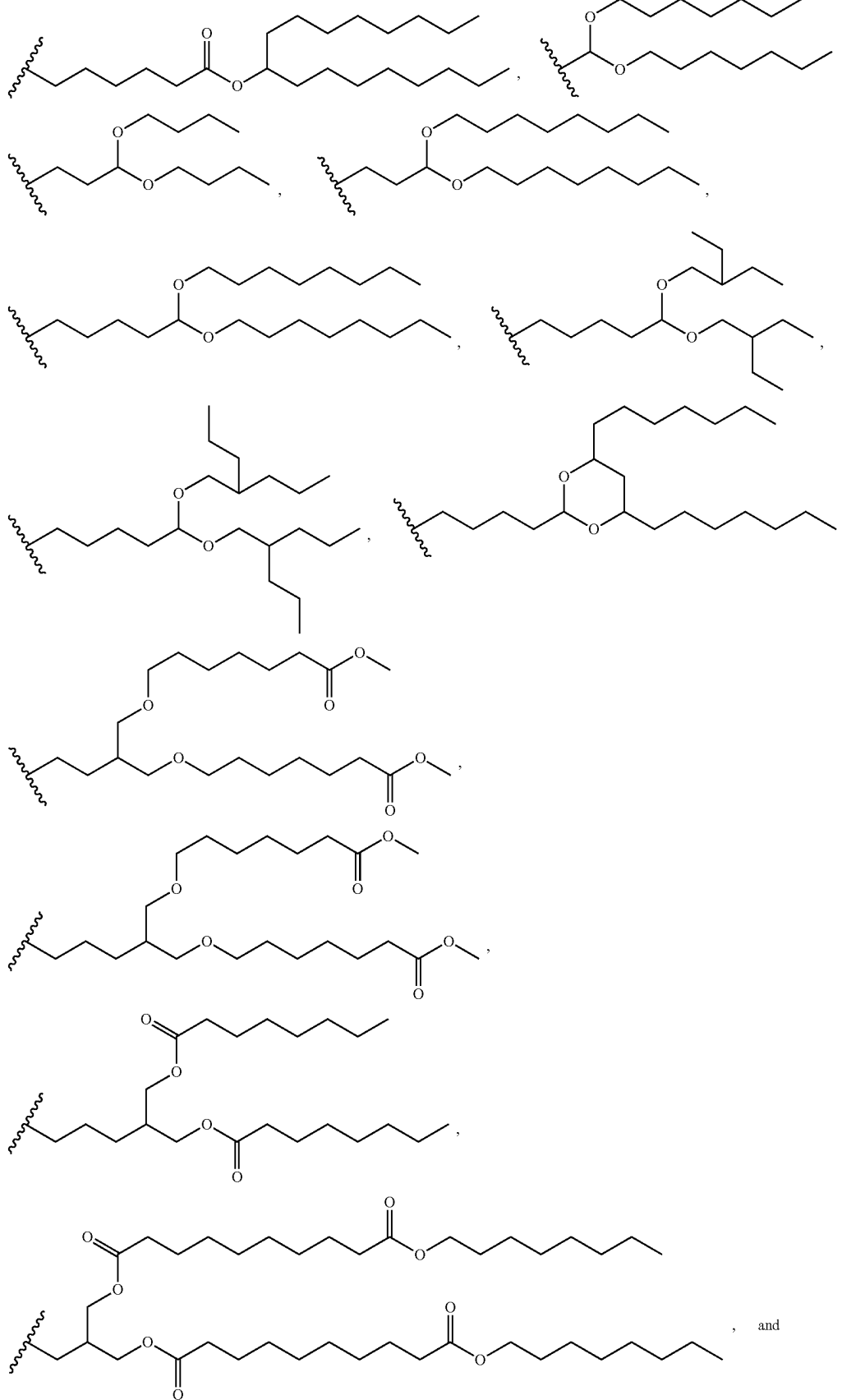
In some embodiments, R_1 and R_2 are each independently selected from the group consisting of:



9

10

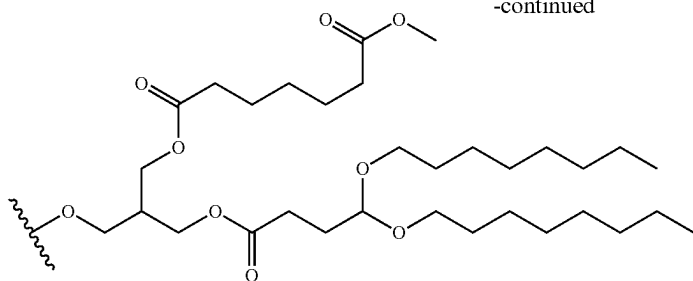
-continued



11

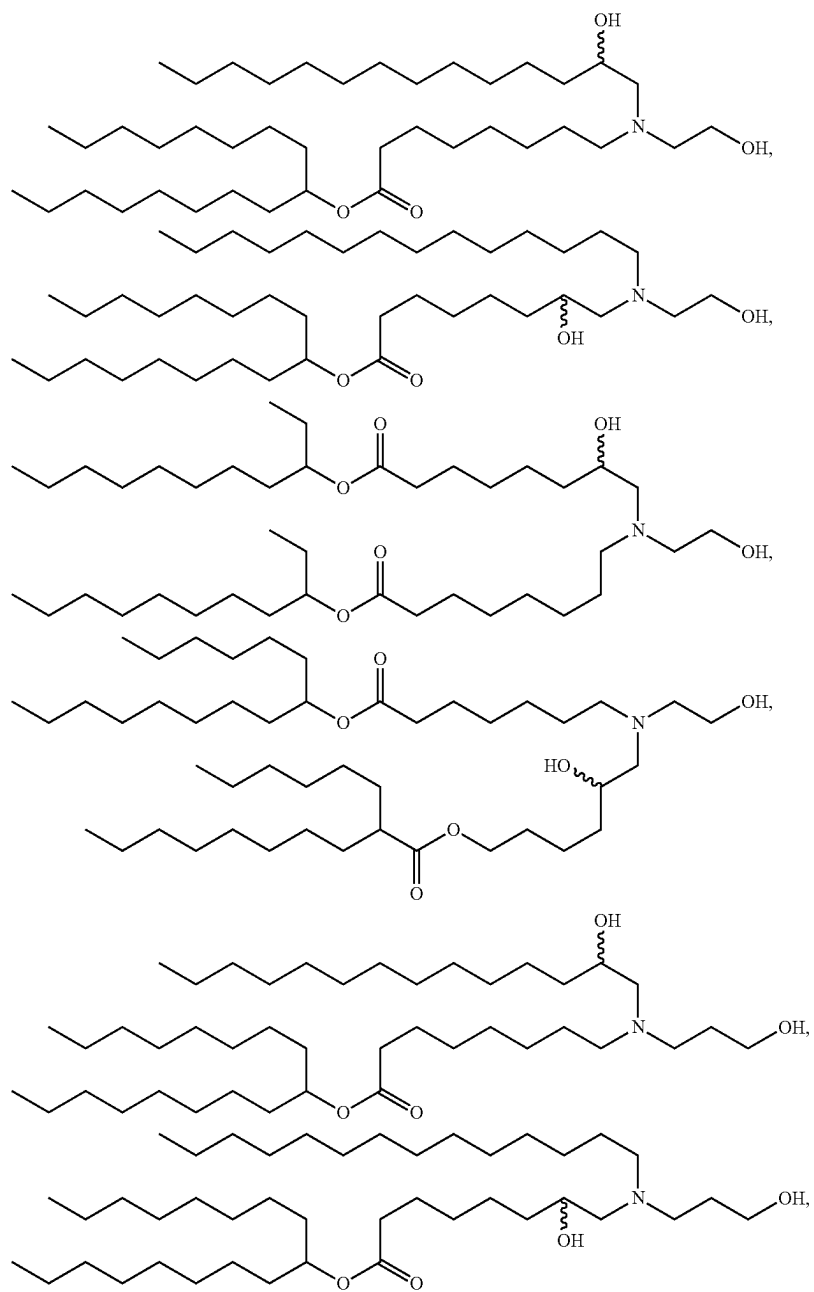
12

-continued



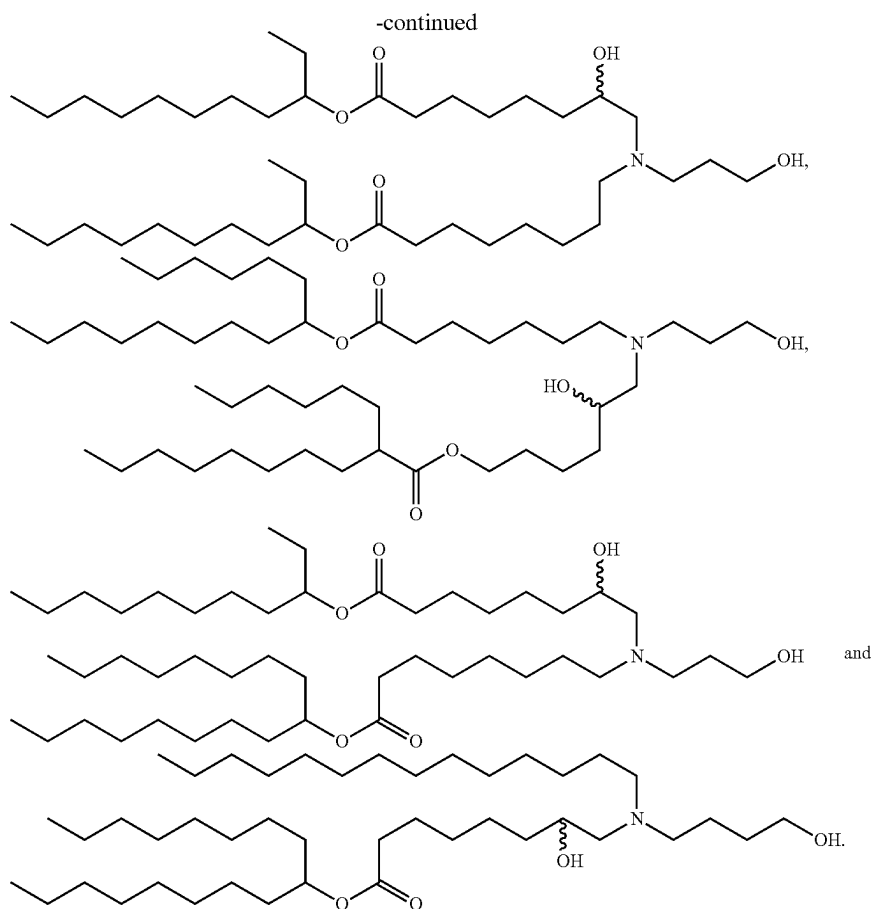
In some embodiments, R_1 and R_2 are the same. In some embodiments, R_1 and R_2 are different.

In some embodiments, the ionizable lipid is selected from the group consisting of

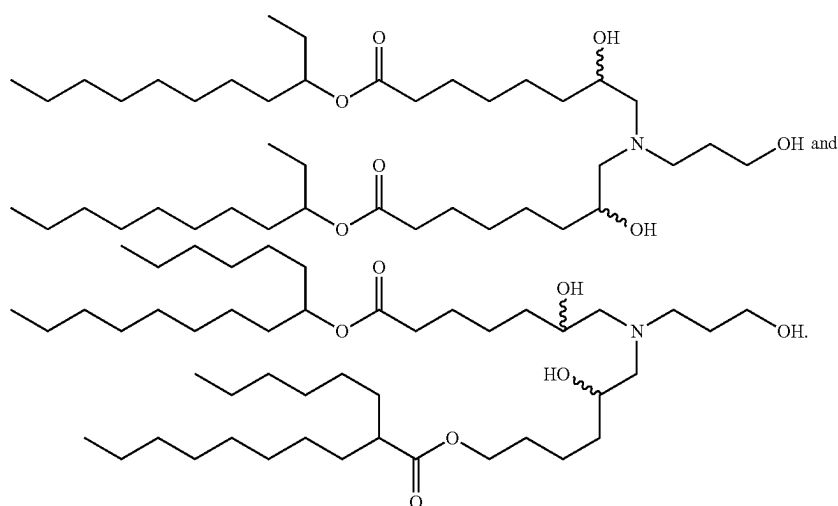


13

14



In some embodiments the ionizable lipid is selected from the group consisting of



In some embodiments, the ionizable lipid is selected from Table 10e.

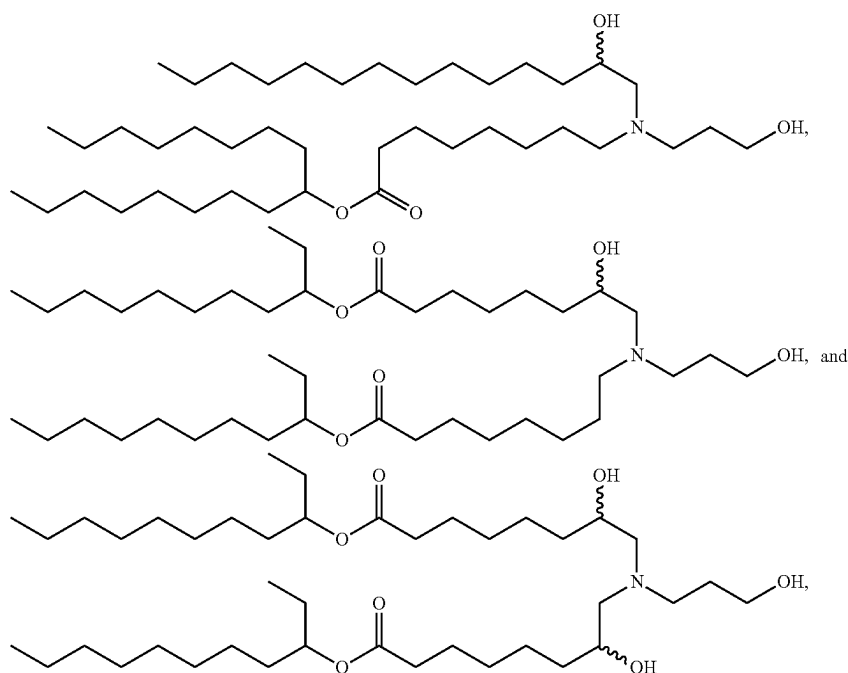
In another aspect, the present disclosure provides a pharmaceutical composition comprising a transfer vehicle, wherein the transfer vehicle comprises an ionizable lipid described above.

In some embodiments, the pharmaceutical composition further comprises an RNA polynucleotide. In some embodiments, the RNA polynucleotide is a linear or circular RNA polynucleotide. In some embodiments, the RNA polynucleotide is a circular RNA polynucleotide.

In some embodiments, the transfer vehicle comprises about 0.5% to about 4% PEG-modified lipids by molar ratio. In some embodiments, the transfer vehicle comprises about 1% to about 2% PEG-modified lipids by molar ratio.

17

In some embodiments, the transfer vehicle comprises:
a. an ionizable lipid selected from:



18

izable lipid: DSPC:cholesterol:DMG-PEG(2000) is about 45:9:44:2.

- or a mixture thereof,
- b. a helper lipid selected from DOPE or DSPC,
- c. cholesterol, and
- d. a PEG-lipid selected from DSPE-PEG(2000) or DMG-PEG(2000).

In some embodiments, the transfer vehicle comprises ionizable lipid, helper lipid, cholesterol, and PEG-lipid at the molar ratio of ionizable lipid:helper lipid:cholesterol:PEG-lipid is about 45:9:44:2, about 50:10:38.5:1.5, about 41:12:45:2, about 62:4:33:1, or about 53:5:41:1. In some embodiments, the molar ratio of each of the ionizable lipid, helper lipid, cholesterol, and PEG-lipid is within 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.9%, 0.8%, 0.7%, 0.6%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.09%, 0.08%, 0.07%, 0.06%, 0.05%, 0.04%, 0.03%, 0.02%, or 0.01% of the stated value.

In some embodiments, the transfer vehicle comprises the helper lipid of DOPE and the PEG-lipid of DMG-PEG(2000), and wherein the molar ratio of ionizable lipid:DOPE:cholesterol:DMG-PEG(2000) is about 45:9:44:2, about 50:10:38.5:1.5, about 41:12:45:2, about 62:4:33:1, or about 53:5:41:1. In some embodiments, the molar ratio of ionizable lipid:DOPE:cholesterol:DSPE-PEG(2000) is about 62:4:33:1. In some embodiments, the molar ratio of ionizable lipid:DOPE:cholesterol:DSPE-PEG(2000) is about 53:5:41:1.

In some embodiments, the transfer vehicle comprises the helper lipid of DSPC and the PEG-lipid of DMG-PEG(2000), and wherein the molar ratio of ionizable lipid: DSPC:cholesterol:DMG-PEG(2000) is about 45:9:44:2, about 50:10:38.5:1.5, about 41:12:45:2, about 62:4:33:1, or about 53:5:41:1. In some embodiments, the molar ratio of ionizable lipid: DSPC:cholesterol:DMG-PEG(2000) is about 50:10:38.5:1.5. In some embodiments, the molar ratio of ionizable lipid: DSPC:cholesterol:DMG-PEG(2000) is about 41:12:45:2. In some embodiments, the molar ratio of ion-

In some embodiments, the transfer vehicle comprises the helper lipid of DSPC and the PEG-lipid of DSPE-PEG(2000), and wherein the molar ratio of ionizable lipid: DSPC:cholesterol:DSPE-PEG(2000)) is about 45:9:44:2, about 50:10:38.5:1.5, about 41:12:45:2, about 62:4:33:1, or about 53:5:41:1.

In some embodiments, the transfer vehicle comprises the helper lipid of DOPE and the PEG-lipid is C14-PEG(2000), and wherein the molar ratio of ionizable lipid:DOPE:cholesterol:C14-PEG(2000) is about 45:9:44:2, about 50:10:38.5:1.5, about 41:12:45:2, about 62:4:33:1, or about 53:5:41:1.

In some embodiments, the transfer vehicle comprises the helper lipid of DOPE and the PEG-lipid of DMG-PEG(2000), wherein the molar ratio of ionizable lipid:DOPE:cholesterol:DMG-PEG(2000) is about 45:9:44:2, about 50:10:38.5:1.5, about 41:12:45:2, about 62:4:33:1, or about 53:5:41:1.

In some embodiments, a pharmaceutical composition of the present disclosure has a lipid to phosphate (IL:P) molar ratio of about 3 to about 9, such as about 3, about 4, about 4.5, about 5, about 5.5, about 5.7, about 6, about 6.2, about 6.5, or about 7.

In some embodiments, the transfer vehicle is formulated for endosomal release of the RNA polynucleotide. In some embodiments, the transfer vehicle is capable of binding to apolipoprotein E (APOE) or is substantially free of APOE binding sites. In some embodiments, the transfer vehicle is capable of low density lipoprotein receptor (LDLR) dependent uptake or LDLR independent uptake into a cell.

In some embodiments, the transfer vehicle has a diameter of less than about 120 nm and/or does not form aggregates with a diameter of more than 300 nm.

In some embodiments, a pharmaceutical composition of the present disclosure is substantially free of linear RNA.

In some embodiments, further comprising a targeting moiety operably connected to the transfer vehicle. In some embodiments, the targeting moiety specifically or indirectly

binds an immune cell antigen, wherein the immune cell antigen is a T cell antigen selected from the group consisting of CD2, CD3, CD5, CD7, CD8, CD4, beta7 integrin, beta2 integrin, and C1qR

In some embodiments, the targeting moiety is a small molecule. In some embodiments, the small molecule is mannose, a lectin, acivicin, biotin, or digoxigenin. In some embodiments, the small molecule binds to an ectoenzyme on an immune cell, wherein the ectoenzyme is selected from the group consisting of CD38, CD73, adenosine 2a receptor, and adenosine 2b receptor. In some embodiments, the targeting moiety is a single chain Fv (scFv) fragment, nanobody, peptide, peptide-based macrocycle, minibody, small molecule ligand such as folate, arginylglycylaspartic acid (RGD), or phenol-soluble modulin alpha 1 peptide (PSMA1), heavy chain variable region, light chain variable region or fragment thereof.

In some embodiments, a pharmaceutical composition of the present disclosure has less than 1%, by weight, of the polynucleotides in the composition are double stranded RNA, DNA splints, or triphosphorylated RNA. In some embodiments, the pharmaceutical composition has less than 1%, by weight, of the polynucleotides and proteins in the pharmaceutical composition are double stranded RNA, DNA splints, triphosphorylated RNA, phosphatase proteins, protein ligases, or capping enzymes.

In another aspect, provided herein is a method of treating or preventing a disease, disorder, or condition, comprising administering an effective amount of a pharmaceutical composition described above and herein.

In another aspect, provided herein is a method of treating a subject in need thereof comprising administering a therapeutically effective amount of the pharmaceutical composition described above and herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1E depict luminescence in supernatants of HEK293 (FIGS. 1A, 1D, and 1E), HepG2 (FIG. 1B), or 1C1C7 (FIG. 1C) cells 24 hours after transfection with circular RNA comprising a *Gaussia luciferase* expression sequence and various IRES sequences.

FIGS. 2A-2C depict luminescence in supernatants of HEK293 (FIG. 2A), HepG2 (FIG. 2B), or 1C1C7 (FIG. 2C) cells 24 hours after transfection with circular RNA comprising a *Gaussia luciferase* expression sequence and various IRES sequences having different lengths.

FIG. 3A and FIG. 3B depict stability of select IRES constructs in HepG2 (FIG. 3A) or 1C1C7 (FIG. 3B) cells over 3 days as measured by luminescence.

FIG. 4A and FIG. 4B depict protein expression from select IRES constructs in Jurkat cells, as measured by luminescence from secreted *Gaussia luciferase* in cell supernatants.

FIG. 5A and FIG. 5B stability of select IRES constructs in Jurkat cells over 3 days as measured by luminescence.

FIG. 6A and FIG. 6B depict comparisons of 24 hour luminescence (FIG. 6A) or relative luminescence over 3 days (FIG. 6B) of modified linear, unpurified circular, or purified circular RNA encoding *Gaussia luciferase*.

FIGS. 7A-7F depict transcript induction of IFN γ (FIG. 7A), IL-6 (FIG. 7B), IL-2 (FIG. 7C), RIG-I (FIG. 7D), IFN- β 1 (FIG. 7E), and TNF α (FIG. 7F) after electroporation of Jurkat cells with modified linear, unpurified circular, or purified circular RNA.

FIGS. 8A-8C depict a comparison of luminescence of circular RNA and modified linear RNA encoding *Gaussia*

luciferase in human primary monocytes (FIG. 8A) and macrophages (FIG. 8B and FIG. 8C).

FIG. 9A and FIG. 9B depict relative luminescence over 3 days (FIG. 9A) in supernatant of primary T cells after transduction with circular RNA comprising a *Gaussia luciferase* expression sequence and varying IRES sequences or 24 hour luminescence (FIG. 9B).

FIGS. 10A-10C depict 24 hour luminescence in supernatant of primary T cells (FIG. 10A) after transduction with circular RNA or modified linear RNA comprising a *Gaussia luciferase* expression sequence, or relative luminescence over 3 days (FIG. 10B), and 24 hour luminescence in PBMCs (FIG. 10C).

FIG. 11A and FIG. 11B depict HPLC chromatograms (FIG. 11A) and circularization efficiencies (FIG. 11B) of RNA constructs having different permutation sites.

FIG. 12A and FIG. 12B depict HPLC chromatograms (FIG. 12A) and circularization efficiencies (FIG. 12B) of RNA constructs having different introns and/or permutation sites.

FIG. 13A and FIG. 13B depict HPLC chromatograms (FIG. 13A) and circularization efficiencies (FIG. 13B) of 3 RNA constructs with or without homology arms.

FIG. 14 depicts circularization efficiencies of 3 RNA constructs without homology arms or with homology arms having various lengths and GC content.

FIGS. 15A and 15B depict HPLC HPLC chromatograms showing the contribution of strong homology arms to improved splicing efficiency, the relationship between circularization efficiency and nicking in select constructs, and combinations of permutations sites and homology arms hypothesized to demonstrate improved circularization efficiency.

FIG. 16 shows fluorescent images of T cells mock electroporated (left) or electroporated with circular RNA encoding a CAR (right) and co-cultured with Raji cells expressing GFP and firefly luciferase.

FIG. 17 shows bright field (left), fluorescent (center), and overlay (right) images of T cells mock electroporated (top) or electroporated with circular RNA encoding a CAR (bottom) and co-cultured with Raji cells expressing GFP and firefly luciferase.

FIG. 18 depicts specific lysis of Raji target cells by T cells mock electroporated or electroporated with circular RNA encoding different CAR sequences.

FIG. 19A and FIG. 19B depict luminescence in supernatants of Jurkat cells (left) or resting primary human CD3+ T cells (right) 24 hours after transduction with linear or circular RNA comprising a *Gaussia luciferase* expression sequence and varying IRES sequences (FIG. 19A), and relative luminescence over 3 days (FIG. 19B).

FIGS. 20A-20F depict transcript induction of IFN- β 1 (FIG. 20A), RIG-I (FIG. 20B), IL-2 (FIG. 20C), IL-6 (FIG. 20D), IFN γ (FIG. 20E), and TNF α (FIG. 20F) after electroporation of human CD3+ T cells with modified linear, unpurified circular, or purified circular RNA.

FIG. 21A and FIG. 21B depict specific lysis of Raji target cells by human primary CD3+ T cells electroporated with circRNA encoding a CAR as determined by detection of firefly luminescence (FIG. 21A), and IFN γ transcript induction 24 hours after electroporation with different quantities of circular or linear RNA encoding a CAR sequence (FIG. 21B).

FIG. 22A and FIG. 22B depict specific lysis of target or non-target cells by human primary CD3+ T cells electropo-

rated with circular or linear RNA encoding a CAR at different E:T ratios as determined by detection of firefly luminescence.

FIG. 23 depicts specific lysis of target cells by human CD3+ T cells electroporated with RNA encoding a CAR at 1, 3, 5, and 7 days post electroporation.

FIG. 24 depicts specific lysis of target cells by human CD3+ T cells electroporated with circular RNA encoding a CD19 or BCMA targeted CAR.

FIG. 25A and FIG. 25B show the expression of GFP (FIG. 25A) and CD19 CAR (FIG. 25B) in human PBMCs after incubating with testing lipid nanoparticle containing circular RNA encoding either GFP or CD19 CAR.

FIG. 26 depicts the expression of an anti-murine CD19 CAR in 1C1C7 cells lipotransfected with circular RNA comprising an anti-murine CD19 CAR expression sequence and varying IRES sequences.

FIG. 27 shows the cytotoxicity of an anti-murine CD19 CAR to murine T cells. The CD19 CAR is encoded by and expressed from a circular RNA, which is electroporated into the murine T cells.

FIG. 28A and FIG. 28B compare the expression level of an anti-human CD19 CAR expressed from a circular RNA with that expressed from a linear mRNA.

FIG. 29A and FIG. 29B compare the cytotoxic effect of an anti-human CD19 CAR expressed from a circular RNA with that expressed from a linear mRNA.

FIG. 30 depicts the cytotoxicity of two CARs (anti-human CD19 CAR and anti-human BCMA CAR) expressed from a single circular RNA in T cells.

FIG. 31A depicts an exemplary RNA construct design with built-in polyA sequences in the introns. FIG. 31B shows the chromatography trace of unpurified circular RNA. FIG. 31C shows the chromatography trace of affinity-purified circular RNA. FIG. 31D shows the immunogenicity of the circular RNAs prepared with varying IVT conditions and purification methods. (Commercial=commercial IVT mix; Custom=customerized IVT mix; Aff=affinity purification; Enz=enzyme purification: GMP:GTP ratio=8, 12.5, or 13.75).

FIG. 32A depicts an exemplary RNA construct design with a dedicated binding sequence as an alternative to polyA for hybridization purification. FIG. 32A discloses SEQ ID NO: 123. FIG. 32B shows the chromatography trace of unpurified circular RNA. FIG. 32C shows the chromatography trace of affinity-purified circular RNA.

FIG. 33A shows the chromatography trace of unpurified circular RNA encoding dystrophin. FIG. 33B shows the chromatography trace of enzyme-purified circular RNA encoding dystrophin.

FIG. 34A and FIG. 34B compare the expression (FIG. 34A) and stability (FIG. 34B) of purified circRNAs with different 5' spacers between the 3' intron fragment/5' internal duplex region and the IRES in Jurkat cells. (AC=only A and C were used in the spacer sequence: UC=only U and C were used in the spacer sequence.)

FIG. 35 shows luminescence expression levels and stability of expression in primary T cells from circular RNAs containing the original or modified IRES elements indicated.

FIG. 36 shows luminescence expression levels and stability of expression in HepG2 cells from circular RNAs containing the original or modified IRES elements indicated.

FIG. 37 shows luminescence expression levels and stability of expression in 1C1C7 cells from circular RNAs containing the original or modified IRES elements indicated.

FIG. 38 shows luminescence expression levels and stability of expression in HepG2 cells from circular RNAs

containing IRES elements with untranslated regions (UTRs) inserted or hybrid IRES elements. "Scr" means Scrambled, which was used as a control.

FIG. 39 shows luminescence expression levels and stability of expression in 1C1C7 cells from circular RNAs containing an IRES and variable stop codon cassettes operably linked to a gaussian luciferase coding sequence.

FIG. 40 shows luminescence expression levels and stability of expression in 1C1C7 cells from circular RNAs containing an IRES and variable untranslated regions (UTRs) inserted before the start codon of a gaussian luciferase coding sequence.

FIG. 41 shows expression levels of human erythropoietin (hEPO) in Huh7 cells from circular RNAs containing two miR-122 target sites downstream from the hEPO coding sequence.

FIG. 42A and FIG. 42B show CAR expression levels in the peripheral blood (FIG. 42A) and spleen (FIG. 42B) when treated with LNP encapsulating circular RNA that expresses anti-CD19 CAR. Anti-CD20 (aCD20) and circular RNA encoding luciferase (oLuc) were used for comparison.

FIGS. 43A-43C show the overall frequency of anti-CD19 CAR expression, the frequency of anti-CD19 CAR expression on the surface of cells and effect on anti-tumor response of IRES specific circular RNA encoding anti-CD19 CARs on T-cells. FIG. 43A shows anti-CD19 CAR geometric mean fluorescence intensity, FIG. 43B shows percentage of anti-CD19 CAR expression, and FIG. 43C shows the percentage target cell lysis performed by the anti-CD19 CAR. (CK=Caprine Kobuvirus; AP=Apodemus Picornavirus; CK*=Caprine Kobuvirus with codon optimization; PV=Parabovirus; SV=Salivirus.)

FIG. 44 shows CAR expression levels of A20 FLuc target cells when treated with IRES specific circular RNA constructs.

FIG. 45A and FIG. 45B show luminescence expression levels for cytosolic (FIG. 45A) and surface (FIG. 45B) proteins from circular RNA in primary human T-cells.

FIGS. 46A-46G show luminescence expression in human T-cells when treated with IRES specific circular constructs. Expression in circular RNA constructs were compared to linear mRNA. FIGS. 46A, FIG. 46B, and FIG. 46G provide *Gaussian luciferase* expression in multiple donor cells. FIGS. 46C, FIG. 46D, FIG. 46E, and FIG. 46F provides firefly luciferase expression in multiple donor cells.

FIG. 47A and FIG. 47B show anti-CD19 CAR and anti-BCMA CAR (FIG. 47B) expression in human T-cells following treatment of a lipid nanoparticle encompassing a circular RNA that encodes either an anti-CD19 or anti-BCMA CAR to a firefly luciferase expressing K562 cell.

FIG. 48A and FIG. 48B show anti-CD19 CAR expression levels resulting from delivery via electroporation in vitro of a circular RNA encoding an anti-CD19 CAR in a specific antigen-dependent manner. FIG. 48A shows Nalm6 cell lysing with an anti-CD19 CAR. FIG. 48B shows K562 cell lysing with an anti-CD19 CAR.

FIGS. 49A-49E show transfection of LNP mediated by use of ApoE3 in solutions containing LNP and circular RNA expressing green fluorescence protein (GFP). FIG. 49A showed the live-dead results. FIGS. 49B, FIG. 49C, FIG. 49D, and FIG. 49E provide the frequency of expression for multiple donors.

FIGS. 50A-50C show circularization efficiency of an RNA molecule encoding a stabilized (double proline mutant) SARS-CoV2 spike protein. FIG. 50A shows the in vitro transcription product of the -4.5kb SARS-CoV2 spike-encoding circRNA. FIG. 50B shows a histogram of spike

23

protein surface expression via flow cytometry after transfection of spike-encoding circRNA into 293 cells. Transfected 293 cells were stained 24 hours after transfection with CR3022 primary antibody and APC-labeled secondary antibody. FIG. 50C shows a flow cytometry plot of spike protein surface expression on 293 cells after transfection of spike-encoding circRNA. Transfected 293 cells were stained 24 hours after transfection with CR3022 primary antibody and APC-labeled secondary antibody.

FIG. 51 provides multiple controlled adjuvant strategies. CircRNA as indicated on the figure entails an unpurified sense circular RNA splicing reaction using GTP as an indicator molecule in vitro. 3p-circRNA entails a purified sense circular RNA as well as a purified antisense circular RNA mixed containing triphosphorylated 5' termini. FIG. 51 shows in vivo cytokine response to formulated circRNA generated using the indicated strategy.

FIGS. 52A-52C illustrate an intramuscular delivery of LNP containing circular RNA constructs. FIG. 52A provides a live whole body flux post a 6 hour period and 52B provides whole body IVIS 6 hours following a 1 pg dose of the LNP-circular RNA construct. FIG. 52C provides an ex vivo expression distribution over a 24-hour period.

FIG. 53A and FIG. 53B illustrate expression of multiple circular RNAs from a single lipid formulation. FIG. 53A provides hEPO titers from a single and mixed set of LNP containing circular RNA constructs, while FIG. 53B provides total flux of bioluminescence expression from single or mixed set of LNP containing circular RNA constructs.

FIGS. 54A-54C illustrate SARS-CoV2 spike protein expression of circular RNA encoding spike SARS-CoV2 proteins. FIG. 54A shows frequency of spike CoV2 expression;

FIG. 54B shows geometric mean fluorescence intensity (gMFI) of the spike CoV2 expression; and FIG. 54C compares gMFI expression of the construct to the frequency of expression.

FIG. 55 depicts a general sequence construct of a linear RNA polynucleotide precursor (10). The sequence as provided is illustrated in a 5' to 3' order of a 5' enhanced intron element (20), a 5' enhanced exon element (30), a core functional element (40), a 3' enhanced exon element (50) and a 3' enhanced intron element (60).

FIG. 56 depicts various exemplary iterations of the 5' enhanced exon element (20). As illustrated, one iteration of the 5' enhanced exon element (20) comprises in a 5' to 3' order in the following order: a leading untranslated sequence (21), a 5' affinity tag (22), a 5' external duplex region (24), a 5' external spacer (26), and a 3' intron fragment (28).

FIG. 57 depicts various exemplary iterations of the 5' enhanced exon element (30). As illustrated, one iteration of the 5' enhanced exon element (30) comprises in a 5' to 3' order: a 3' exon fragment (32), a 5' internal duplex region (34), and a 5' internal spacer (36).

FIG. 58 depicts various exemplary iterations of the core functional element (40). As illustrated, one iteration of the core functional element (40) comprises a TIE (42), a coding region (46) and a stop region (e.g., a stop codon or stop cassette) (48). Another iteration is illustrated to show the core functional element (47) comprising a noncoding region (47).

FIG. 59 depicts various exemplary iterations of the 3' enhanced exon element (50). As illustrated, one of the iterations of the 3' enhanced exon element (50) comprises, in the following 5' to 3' order: a 3' internal spacer (52), a 3' internal duplex region (54), and a 5' exon fragment (56).

24

FIG. 60 depicts various exemplary iterations of the 3' enhanced intron element (60). As illustrated, one of the iterations of the 3' enhanced intron element (60) comprises, in the following order, a 5' intron fragment (62), a 3' external spacer (64), a 3' external duplex region (66), a 3' affinity tag (68) and a terminal untranslated sequence (69).

FIG. 61 depicts various exemplary iterations a translation initiation element (TIE)(42). TIE (42) sequence as illustrated in one iteration is solely an IRES (43). In another iteration, the TIE (42) is an aptamer (44). In two different iterations, the TIE (42) is an aptamer (44) and IRES (43) combination. In another iteration, the TIE (42) is an aptamer complex (45).

FIG. 62 illustrates an exemplary linear RNA polynucleotide precursor (10) comprising in the following 5' to 3' order, a leading untranslated sequence (21), a 5' affinity tag (22), a 5' external duplex region (24), a 5' external spacer (26), a 3' intron fragment (28), a 3' exon fragment (32), a 5' internal duplex region (34), a 5' internal spacer (36), a TIE (42), a coding element (46), a stop region (48), a 3' internal spacer (52), a 3' internal duplex region (54), a 5' exon fragment (56), a 5' intron fragment (62), a 3' external spacer (64), a 3' external duplex region (66), a 3' affinity tag (68) and a terminal untranslated sequence (69).

FIG. 63 illustrates an exemplary linear RNA polynucleotide precursor (10) comprising in the following 5' to 3' order, a leading untranslated sequence (21), a 5' affinity tag (22), a 5' external duplex region (24), a 5' external spacer (26), a 3' intron fragment (28), a 3' exon fragment (32), a 5' internal duplex region (34), a 5' internal spacer (36), a coding element (46), a stop region (48), a TIE (42), a 3' internal spacer (52), a 3' internal duplex region (54), a 5' exon fragment (56), a 5' intron fragment (62), a 3' external spacer (64), a 3' external duplex region (66), a 3' affinity tag (68) and a terminal untranslated sequence (69).

FIG. 64 illustrates an exemplary linear RNA polynucleotide precursor (10) comprising in the following 5' to 3' order, a leading untranslated sequence (21), a 5' affinity tag (22), a 5' external duplex region (24), a 5' external spacer (26), a 3' intron fragment (28), a 3' exon fragment (32), a 5' internal duplex region (34), a 5' internal spacer (36), a noncoding element (47), a 3' internal spacer (52), a 3' internal duplex region (54), a 5' exon fragment (56), a 5' intron fragment (62), a 3' external spacer (64), a 3' external duplex region (66), a 3' affinity tag (68) and a terminal untranslated sequence (69).

FIG. 65 illustrates the general circular RNA (8) structure formed post splicing. The circular RNA as depicted includes a 5' exon element (30), a core functional element (40) and a 3' exon element (50).

FIGS. 66A-66E illustrate the various ways an accessory element (70) (e.g., a miRNA binding site) may be included in a linear RNA polynucleotide. FIG. 66A shows a linear RNA polynucleotide comprising an accessory element (70) at the spacer regions. FIG. 66B shows a linear RNA polynucleotide comprising an accessory element (70) located between each of the external duplex regions and the exon fragments. FIG. 66C depicts an accessory element (70) within a spacer. FIG. 66D illustrates various iterations of an accessory element (70) located within the core functional element. FIG. 66E illustrates an accessory element (70) located within an internal ribosome entry site (IRES).

FIGS. 67A-67C illustrate a screening of a LNP formulated with circular RNA encoding firefly luciferase and having a TIE in primary human (FIG. 67A), mouse (FIG. 67B), and cynomolgus monkey (FIG. 67C) hepatocyte with varying dosages in vitro.

25

FIGS. 68A-68C illustrates a screening of a LNP formulated with circular RNA encoding firefly luciferase and having a TIE, in primary human hepatocyte from three different donors with varying dosages in vitro.

FIG. 69 illustrates in vitro expression of LNP formulated with circular RNA encoding for GFP and having a TIE, in HeLa, HEK293, and HUH7 human cell models.

FIG. 70 illustrates in vitro expression of LNP formulated with circular RNAs encoding a GFO protein and having a TIE, in primary human hepatocytes.

FIG. 71A and FIG. 71B illustrate in vitro expression of circular RNA encoding firefly luciferase and having a TIE, in mouse myoblast (FIG. 71A) and primary human muscle myoblast (FIG. 71B) cells.

FIG. 72A and FIG. 72B illustrate in vitro expression of circular RNA encoding for firefly luciferase and having a TIE, in myoblasts and differentiated primary human skeletal muscle myotubes. FIG. 72A provides the data related to cells received from human donor 1; FIG. 72B provides the data related to cell received from human donor 2.

FIG. 73A and FIG. 73B illustrate cell-free in vitro translation of circular RNA of variable sizes. In FIG. 73A circular RNA encoding for firefly luciferase and linear mRNA encoding for firefly luciferase was tested for expression. In FIG. 73B, human and mouse cells were given circular RNAs encoding for ATP7B proteins. Some of the circular RNAs tested were codon optimized. Circular RNA expressing firefly luciferase was used for comparison.

FIG. 74 illustrates mOX40L expression in the spleen of mice from LNPs comprising either Lipid 1 of Table 10e (10e-1) or Lipid 15 of Table 10f (10f-15), at a lipid to phosphate ratio (IL:P) of 5.7 (5.7A parameters formulation) encapsulating circular RNA encoding for mOX40L.

FIG. 75 illustrates circular RNA expression of firefly luciferase delivered using LNPs formulated with ionizable lipids comprising varying numbers of β -hydroxyl groups or a negative control (PBS). Ionizable lipids used comprise left to right: Table 10e, Lipid 85 (10e-85); Table 10e, Lipid 89 (10e-89); Table 10f, Lipid 22 (10f-22); Table 10e, Lipid 86 (10e-86); and Table 10e, Lipid 90 (10e-90).

FIG. 76 illustrates oRNA expression of firefly luciferase in the spleen delivered using LNPs formulated with ionizable lipids from Table 10e (from left to right: lipids 1, 85, 38, 34, 45, 86, 88, 89, 90) post intravenous administration. Total luciferase flux was measured in the spleen.

FIG. 77 illustrates splenic T cell expression post intravenous administration of circular RNA encoding for mOX40L delivered using LNPs comprising an ionizable lipid from Table 10e (from left to right: Lipid 1, Lipid 85, Lipid 38, Lipid 34, Lipid 45, Lipid 86, Lipid 88, Lipid 89, Lipid 90).

FIG. 78 illustrates B cell depletion within mice when treated with a circular RNA encoding a CD-19 chimeric antigen receptor (CAR) protein encapsulated in mice. The circular RNAs were delivered via an LNP comprising an ionizable lipid from Table 10e (1, 16, 85, 45, 86, or 90). In FIG. 78, B cell aplasia was observed in blood cells. The dotted line on the figure indicates Wasabi control B cell aplasia. % B cells were normalized to the Wasabi control.

FIG. 79 illustrates tumor growth kinetics in a Nalm6 model post administration of LNP-oRNA constructs in Table 10e, lipids 16, 45, or 86. Total flux of the tested mice was measured.

DETAILED DESCRIPTION

The present invention provides, among other things, ionizable lipids and related transfer vehicles, compositions, and

26

methods. In some embodiments, the transfer vehicles comprise ionizable lipid (e.g., ionizable lipids disclosed herein), PEG-modified lipid, and/or structural lipid, thereby forming lipid nanoparticles suitable for delivering nucleic acids. In certain embodiments, the nucleic acid may be RNA, such as siRNA, mRNA or circular RNA. The nucleic acids may encode therapeutic agents. In some embodiments, the nucleic acids are encapsulated in the transfer vehicles.

Also disclosed herein is RNA therapy, along with associated compositions and methods. In some embodiments, the RNA therapy allows for increased RNA stability, expression, and prolonged half-life, among other things.

Also disclosed herein is a DNA template (e.g., a vector) for making circular RNA. In some embodiments, the DNA template comprises a 3' enhanced intron fragment, a 3' enhanced exon fragment, a core functional element, a 5' enhanced exon fragment, and a 5' enhanced intron fragment. In some embodiments, these elements are positioned in the DNA template in the above order.

Additional embodiments include circular RNA polynucleotides, including circular RNA polynucleotides (e.g., a circular RNA comprising 3' enhanced exon element, a core functional element, and a 5' enhanced exon element) made using the DNA template provided herein, compositions comprising such circular RNA, cells comprising such circular RNA, methods of using and making such DNA template, circular RNA, compositions and cells.

In some embodiments, provided herein are methods comprising administration of circular RNA polynucleotides provided herein into cells for therapy or production of useful proteins. In some embodiments, the method is advantageous in providing the production of a desired polypeptide inside eukaryotic cells with a longer half-life than linear RNA, due to the resistance of the circular RNA to ribonucleases.

Circular RNA polynucleotides lack the free ends necessary for exonuclease-mediated degradation, causing them to be resistant to several mechanisms of RNA degradation and granting extended half-lives when compared to an equivalent linear RNA. Circularization may allow for the stabilization of RNA polynucleotides that generally suffer from short half-lives and may improve the overall efficacy of exogenous mRNA in a variety of applications. In an embodiment, the functional half-life of the circular RNA polynucleotides provided herein in eukaryotic cells (e.g., mammalian cells, such as human cells) as assessed by protein synthesis is at least 20 hours (e.g., at least 80 hours).

Various aspects of the invention are described in detail in the following sections. The use of sections is not meant to limit the invention. Each section can apply to any aspect of the invention. In this application, the use of "or" means "and/or" unless stated otherwise.

1. Definitions

As used herein, the terms "circRNA" or "circular polyribonucleotide" or "circular RNA" or "oRNA" are used interchangeably and refers to a polyribonucleotide that forms a circular structure through covalent bonds.

As used herein, the term "DNA template" refers to a DNA sequence capable of transcribing a linear RNA polynucleotide. For example, but not intending to be limiting, a DNA template may include a DNA vector, PCR product or plasmid.

As used herein, the term "3' group I intron fragment" refers to a sequence with 75% or higher similarity to the 3'-proximal end of a natural group I intron including the splice site dinucleotide.

As used herein, the term “5' group I intron fragment” refers to a sequence with 75% or higher similarity to the 5'-proximal end of a natural group I intron including the splice site dinucleotide.

As used herein, the term “permutation site” refers to the site in a group I intron where a cut is made prior to permutation of the intron. This cut generates 3' and 5' group I intron fragments that are permuted to be on either side of a stretch of precursor RNA to be circularized.

As used herein, the term “splice site” refers to a dinucleotide that is partially or fully included in a group I intron and between which a phosphodiester bond is cleaved during RNA circularization. (As used herein, “splice site” refers to the dinucleotide or dinucleotides between which cleavage of the phosphodiester bond occurs during a splicing reaction. A “5' splice site” refers to the natural 5' dinucleotide of the intron e.g., group I intron, while a “3' splice site” refers to the natural 3' dinucleotide of the intron).

As used herein, the term “expression sequence” refers to a nucleic acid sequence that encodes a product, e.g., a peptide or polypeptide, regulatory nucleic acid, or non-coding nucleic acid. An exemplary expression sequence that codes for a peptide or polypeptide can comprise a plurality of nucleotide triads, each of which can code for an amino acid and is termed as a “codon.”

As used herein, “coding element” or “coding region” is region located within the expression sequence and encodings for one or more proteins or polypeptides (e.g., therapeutic protein).

As used herein, a “noncoding element” or “non-coding nucleic acid” is a region located within the expression sequence. This sequence, but itself does not encode for a protein or polypeptide, but may have other regulatory functions, including but not limited, allow the overall polynucleotide to act as a biomarker or adjuvant to a specific cell.

As used herein, the term “therapeutic protein” refers to any protein that, when administered to a subject directly or indirectly in the form of a translated nucleic acid, has a therapeutic, diagnostic, and/or prophylactic effect and/or elicits a desired biological and/or pharmacological effect.

As used herein, the term “immunogenic” refers to a potential to induce an immune response to a substance. An immune response may be induced when an immune system of an organism or a certain type of immune cells is exposed to an immunogenic substance. The term “non-immunogenic” refers to a lack of or absence of an immune response above a detectable threshold to a substance. No immune response is detected when an immune system of an organism or a certain type of immune cells is exposed to a non-immunogenic substance. In some embodiments, a non-immunogenic circular polyribonucleotide as provided herein, does not induce an immune response above a predetermined threshold when measured by an immunogenicity assay. In some embodiments, no innate immune response is detected when an immune system of an organism or a certain type of immune cells is exposed to a non-immunogenic circular polyribonucleotide as provided herein. In some embodiments, no adaptive immune response is detected when an immune system of an organism or a certain type of immune cell is exposed to a non-immunogenic circular polyribonucleotide as provided herein.

As used herein, the term “circularization efficiency” refers to a measurement of resultant circular polyribonucleotide as compared to its linear starting material.

As used herein, the term “translation efficiency” refers to a rate or amount of protein or peptide production from a ribonucleotide transcript. In some embodiments, translation

efficiency can be expressed as amount of protein or peptide produced per given amount of transcript that codes for the protein or peptide.

The term “nucleotide” refers to a ribonucleotide, a deoxyribonucleotide, a modified form thereof, or an analog thereof. Nucleotides include species that comprise purines, e.g., adenine, hypoxanthine, guanine, and their derivatives and analogs, as well as pyrimidines, e.g., cytosine, uracil, thymine, and their derivatives and analogs. Nucleotide analogs include nucleotides having modifications in the chemical structure of the base, sugar and/or phosphate, including, but not limited to, 5'-position pyrimidine modifications, 8'-position purine modifications, modifications at cytosine exocyclic amines, and substitution of 5-bromo-uracil; and 2'-position sugar modifications, including but not limited to, sugar-modified ribonucleotides in which the 2'-OH is replaced by a group such as an H, OR, R, halo, SH, SR, NH₂, NHR, NR₂, or CN, wherein R is an alkyl moiety as defined herein. Nucleotide analogs are also meant to include nucleotides with bases such as inosine, queuosine, xanthine; sugars such as 2'-methyl ribose; non-natural phosphodiester linkages such as methylphosphonate, phosphorothioate and peptide linkages. Nucleotide analogs include 5-methoxyuridine, 1-methylpseudouridine, and 6-methyladenosine.

The term “nucleic acid” and “polynucleotide” are used interchangeably herein to describe a polymer of any length, e.g., greater than about 2 bases, greater than about 10 bases, greater than about 100 bases, greater than about 500 bases, greater than 1000 bases, or up to about 10,000 or more bases, composed of nucleotides, e.g., deoxyribonucleotides or ribonucleotides, and may be produced enzymatically or synthetically (e.g., as described in U.S. Pat. No. 5,948,902 and the references cited therein), which can hybridize with naturally occurring nucleic acids in a sequence specific manner analogous to that of two naturally occurring nucleic acids, e.g., can participate in Watson-Crick base pairing interactions. Naturally occurring nucleic acids are comprised of nucleotides including guanine, cytosine, adenine, thymine, and uracil (G, C, A, T, and U respectively).

The terms “ribonucleic acid” and “RNA” as used herein mean a polymer composed of ribonucleotides.

The terms “deoxyribonucleic acid” and “DNA” as used herein mean a polymer composed of deoxyribonucleotides.

“Isolated” or “purified” generally refers to isolation of a substance (for example, in some embodiments, a compound, a polynucleotide, a protein, a polypeptide, a polynucleotide composition, or a polypeptide composition) such that the substance comprises a significant percent (e.g., greater than 1%, greater than 2%, greater than 5%, greater than 10%, greater than 20%, greater than 50%, or more, usually up to about 90%-100%) of the sample in which it resides. In certain embodiments, a substantially purified component comprises at least 50%, 80%-85%, or 90%-95% of the sample. Techniques for purifying polynucleotides and polypeptides of interest are well-known in the art and include, for example, ion-exchange chromatography, affinity chromatography and sedimentation according to density. Generally, a substance is purified when it exists in a sample in an amount, relative to other components of the sample, that is more than as it is found naturally.

The terms “duplexed,” “double-stranded,” or “hybridized” as used herein refer to nucleic acids formed by hybridization of two single strands of nucleic acids containing complementary sequences. In most cases, genomic DNA is double-stranded. Sequences can be fully complementary or partially complementary.

As used herein, “unstructured” with regard to RNA refers to an RNA sequence that is not predicted by the RNAFold software or similar predictive tools to form a structure (e.g., a hairpin loop) with itself or other sequences in the same RNA molecule. In some embodiments, unstructured RNA can be functionally characterized using nuclease protection assays.

As used herein, “structured” with regard to RNA refers to an RNA sequence that is predicted by the RNAFold software or similar predictive tools to form a structure (e.g., a hairpin loop) with itself or other sequences in the same RNA molecule.

As used herein, two “duplex sequences,” “duplex region,” “duplex regions,” “homology arms,” or “homology regions” may be any two regions that are thermodynamically favored to cross-pair in a sequence specific interaction. In some embodiments, two duplex sequences, duplex regions, homology arms, or homology regions, share a sufficient level of sequence identity to one another’s reverse complement to act as substrates for a hybridization reaction. As used herein polynucleotide sequences have “homology” when they are either identical or share sequence identity to a reverse complement or “complementary” sequence. The percent sequence identity between a homology region and a counterpart homology region’s reverse complement can be any percent of sequence identity that allows for hybridization to occur. In some embodiments, an internal duplex region of an inventive polynucleotide is capable of forming a duplex with another internal duplex region and does not form a duplex with an external duplex region.

As used herein, an “affinity sequence” or “affinity tag” is a region of polynucleotide sequences polynucleotide sequence ranging from 1 nucleotide to hundreds or thousands of nucleotides containing a repeated set of nucleotides for the purposes of aiding purification of a polynucleotide sequence. For example, an affinity sequence may comprise, but is not limited to, a polyA or poly AC sequence.

As used herein, a “spacer” refers to a region of a polynucleotide sequence ranging from 1 nucleotide to hundreds or thousands of nucleotides separating two other elements along a polynucleotide sequence. The sequences can be defined or can be random. A spacer is typically non-coding. In some embodiments, spacers include duplex regions.

Linear nucleic acid molecules are said to have a “5'-terminus” (5' end) and a “3'-terminus” (3' end) because nucleic acid phosphodiester linkages occur at the 5' carbon and 3' carbon of the sugar moieties of the substituent mononucleotides. The end nucleotide of a polynucleotide at which a new linkage would be to a 5' carbon is its 5' terminal nucleotide. The end nucleotide of a polynucleotide at which a new linkage would be to a 3' carbon is its 3' terminal nucleotide. A terminal nucleotide, as used herein, is the nucleotide at the end position of the 3'- or 5'-terminus.

As used herein, a “leading untranslated sequence” is a region of polynucleotide sequences ranging from 1 nucleotide to hundreds of nucleotides located at the upmost 5' end of a polynucleotide sequence. The sequences can be defined or can be random. An leading untranslated sequence is non-coding.

As used herein, a “leading untranslated sequence” is a region of polynucleotide sequences ranging from 1 nucleotide to hundreds of nucleotides located at the downmost 3' end of a polynucleotide sequence. The sequences can be defined or can be random. An leading untranslated sequence is non-coding.

“Transcription” means the formation or synthesis of an RNA molecule by an RNA polymerase using a DNA mol-

ecule as a template. The invention is not limited with respect to the RNA polymerase that is used for transcription. For example, in some embodiments, a T7-type RNA polymerase can be used.

“Translation” means the formation of a polypeptide molecule by a ribosome based upon an RNA template.

It is to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to “a cell” includes combinations of two or more cells, or entire cultures of cells; reference to “a polynucleotide” includes, as a practical matter, many copies of that polynucleotide. Unless specifically stated or obvious from context, as used herein, the term “or” is understood to be inclusive. Unless defined herein and below in the remainder of the specification, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the invention pertains.

Unless specifically stated or obvious from context, as used herein, the term “about” is understood as within a range of normal tolerance in the art, for example within 2 standard deviations of the mean. “About” can be understood as within 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.9%, 0.8%, 0.7%, 0.6%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.09%, 0.08%, 0.07%, 0.06%, 0.05%, 0.04%, 0.03%, 0.02%, or 0.01% of the stated value. Unless otherwise clear from the context, all numerical values provided herein are modified by the term “about.”

As used herein, the term “encode” refers broadly to any process whereby the information in a polymeric macromolecule is used to direct the production of a second molecule that is different from the first. The second molecule may have a chemical structure that is different from the chemical nature of the first molecule.

By “co-administering” is meant administering a therapeutic agent provided herein in conjunction with one or more additional therapeutic agents sufficiently close in time such that the therapeutic agent provided herein can enhance the effect of the one or more additional therapeutic agents, or vice versa.

The terms “treat,” and “prevent” as well as words stemming therefrom, as used herein, do not necessarily imply 100% or complete treatment or prevention. Rather, there are varying degrees of treatment or prevention of which one of ordinary skill in the art recognizes as having a potential benefit or therapeutic effect. The treatment or prevention provided by the method disclosed herein can include treatment or prevention of one or more conditions or symptoms of the disease. Also, for purposes herein, “prevention” can encompass delaying the onset of the disease, or a symptom or condition thereof.

As used herein, an “internal ribosome entry site” or “IRES” refers to an RNA sequence or structural element ranging in size from 10 nt to 1000 nt or more, capable of initiating translation of a polypeptide in the absence of a typical RNA cap structure. An IRES is typically about 500 nt to about 700 nt in length.

As used herein, “aptamer” refers in general to either an oligonucleotide of a single defined sequence or a mixture of said nucleotides, wherein the mixture retains the properties of binding specifically to the target molecule (e.g., eukaryotic initiation factor, 40S ribosome, polyC binding protein, polyA binding protein, polypyrimidine tract-binding protein, argonaute protein family. Heterogeneous nuclear ribo-

nucleoprotein K and La and related RNA-binding protein). Thus, as used herein “aptamer” denotes both singular and plural sequences of nucleotides, as defined hereinabove. The term “aptamer” is meant to refer to a single- or double-stranded nucleic acid which is capable of binding to a protein or other molecule. In general, aptamers preferably comprise about 10 to about 100 nucleotides, preferably about 15 to about 40 nucleotides, more preferably about 20 to about 40 nucleotides, in that oligonucleotides of a length that falls within these ranges are readily prepared by conventional techniques. Optionally, aptamers can further comprise a minimum of approximately 6 nucleotides, preferably 10, and more preferably 14 or 15 nucleotides, that are necessary to effect specific binding.

An “eukaryotic initiation factor” or “eIF” refers to a protein or protein complex used in assembling an initiator tRNA, 40S and 60S ribosomal subunits required for initiating eukaryotic translation.

As used herein, an “internal ribosome entry site” or “IRES” refers to an RNA sequence or structural element ranging in size from 10 nt to 1000 nt or more, capable of initiating translation of a polypeptide in the absence of a typical RNA cap structure. An IRES is typically about 500 nt to about 700 nt in length.

As used herein, a “miRNA site” refers to a stretch of nucleotides within a polynucleotide that is capable of forming a duplex with at least 8 nucleotides of a natural miRNA sequence.

As used herein, an “endonuclease site” refers to a stretch of nucleotides within a polynucleotide that is capable of being recognized and cleaved by an endonuclease protein.

As used herein, “bicistronic RNA” refers to a polynucleotide that includes two expression sequences coding for two distinct proteins. These expression sequences can be separated by a nucleotide sequence encoding a cleavable peptide such as a protease cleavage site. They can also be separated by a ribosomal skipping element.

As used herein, the term “ribosomal skipping element” refers to a nucleotide sequence encoding a short peptide sequence capable of causing generation of two peptide chains from translation of one RNA molecule. While not wishing to be bound by theory, it is hypothesized that ribosomal skipping elements function by (1) terminating translation of the first peptide chain and re-initiating translation of the second peptide chain: or (2) cleavage of a peptide bond in the peptide sequence encoded by the ribosomal skipping element by an intrinsic protease activity of the encoded peptide, or by another protease in the environment (e.g., cytosol).

As used herein, the term “co-formulate” refers to a nanoparticle formulation comprising two or more nucleic acids or a nucleic acid and other active drug substance. Typically, the ratios are equimolar or defined in the ratio-metric amount of the two or more nucleic acids or the nucleic acid and other active drug substance.

As used herein, “transfer vehicle” includes any of the standard pharmaceutical carriers, diluents, excipients, and the like, which are generally intended for use in connection with the administration of biologically active agents, including nucleic acids.

As used herein, the phrase “lipid nanoparticle” refers to a transfer vehicle comprising one or more lipids (e.g., in some embodiments, cationic lipids, non-cationic lipids, and PEG-modified lipids).

As used herein, the phrase “ionizable lipid” refers to any of a number of lipid species that carry a net positive charge

at a selected pH, such as physiological pH 4 and a neutral charge at other pHs such as physiological pH 7.

In some embodiments, a lipid, e.g., an ionizable lipid, disclosed herein comprises one or more cleavable groups. The terms “cleave” and “cleavable” are used herein to mean that one or more chemical bonds (e.g., one or more of covalent bonds, hydrogen-bonds, van der Waals’ forces and/or ionic interactions) between atoms in or adjacent to the subject functional group are broken (e.g., hydrolyzed) or are capable of being broken upon exposure to selected conditions (e.g., upon exposure to enzymatic conditions). In certain embodiments, the cleavable group is a disulfide functional group, and in particular embodiments is a disulfide group that is capable of being cleaved upon exposure to selected biological conditions (e.g., intracellular conditions). In certain embodiments, the cleavable group is an ester functional group that is capable of being cleaved upon exposure to selected biological conditions. For example, the disulfide groups may be cleaved enzymatically or by a hydrolysis, oxidation or reduction reaction. Upon cleavage of such disulfide functional group, the one or more functional moieties or groups (e.g., one or more of a head-group and/or a tail-group) that are bound thereto may be liberated. Exemplary cleavable groups may include, but are not limited to, disulfide groups, ester groups, ether groups, and any derivatives thereof (e.g., alkyl and aryl esters). In certain embodiments, the cleavable group is not an ester group or an ether group. In some embodiments, a cleavable group is bound (e.g., bound by one or more of hydrogen-bonds, van der Waals’ forces, ionic interactions and covalent bonds) to one or more functional moieties or groups (e.g., at least one head-group and at least one tail-group). In certain embodiments, at least one of the functional moieties or groups is hydrophilic (e.g., a hydrophilic head-group comprising one or more of imidazole, guanidinium, amino, imine, enamine, optionally-substituted alkyl amino and pyridyl).

As used herein, the term “hydrophilic” is used to indicate in qualitative terms that a functional group is water-prefering, and typically such groups are water-soluble. For example, disclosed herein are compounds that comprise a cleavable disulfide (S—S) functional group bound to one or more hydrophilic groups (e.g., a hydrophilic head-group), wherein such hydrophilic groups comprise or are selected from the group consisting of imidazole, guanidinium, amino, imine, enamine, an optionally-substituted alkyl amino (e.g., an alkyl amino such as dimethylamino) and pyridyl.

In certain embodiments, at least one of the functional groups of moieties that comprise the compounds disclosed herein is hydrophobic in nature (e.g., a hydrophobic tail-group comprising a naturally occurring lipid such as cholesterol). As used herein, the term “hydrophobic” is used to indicate in qualitative terms that a functional group is water-avoiding, and typically such groups are not water soluble. For example, disclosed herein are compounds that comprise a cleavable functional group (e.g., a disulfide (S—S) group) bound to one or more hydrophobic groups, wherein such hydrophobic groups comprise one or more naturally occurring lipids such as cholesterol, and/or an optionally substituted, variably saturated or unsaturated C₆-C₂₀ alkyl and/or an optionally substituted, variably saturated or unsaturated C₆-C₂₀ acyl.

Compound described herein may also comprise one or more isotopic substitutions. For example, H may be in any isotopic form, including ¹H, ²H (D or deuterium), and ³H (T or tritium); C may be in any isotopic form, including ¹²C,

13C, and 14C; O may be in any isotopic form, including 16O and 18O; F may be in any isotopic form, including 18F and 19F; and the like.

When describing the invention, which may include compounds and pharmaceutically acceptable salts thereof, pharmaceutical compositions containing such compounds and methods of using such compounds and compositions, the following terms, if present, have the following meanings unless otherwise indicated. It should also be understood that when described herein any of the moieties defined forth below may be substituted with a variety of substituents, and that the respective definitions are intended to include such substituted moieties within their scope as set out below. Unless otherwise stated, the term "substituted" is to be defined as set out below. It should be further understood that the terms "groups" and "radicals" can be considered interchangeable when used herein.

When a range of values is listed, it is intended to encompass each value and sub-range within the range. For example, "C1-6 alkyl" is intended to encompass, C1, C2, C3, C4, C5, C6, C1-6, C1-5, C1-4, C1-3, C1-2, C2-6, C2-5, C2-4, C2-3, C3-6, C3-5, C3-4, C4-6, C4-5, and C5-6 alkyl.

In certain embodiments, the compounds disclosed herein comprise, for example, at least one hydrophilic head-group and at least one hydrophobic tail-group, each bound to at least one cleavable group, thereby rendering such compounds amphiphilic. As used herein to describe a compound or composition, the term "amphiphilic" means the ability to dissolve in both polar (e.g., water) and non-polar (e.g., lipid) environments. For example, in certain embodiments, the compounds disclosed herein comprise at least one lipophilic tail-group (e.g., cholesterol or a C6-C20 alkyl) and at least one hydrophilic head-group (e.g., imidazole), each bound to a cleavable group (e.g., disulfide).

It should be noted that the terms "head-group" and "tail-group" as used describe the compounds of the present invention, and in particular functional groups that comprise such compounds, are used for ease of reference to describe the orientation of one or more functional groups relative to other functional groups. For example, in certain embodiments a hydrophilic head-group (e.g., guanidinium) is bound (e.g., by one or more of hydrogen-bonds, van der Waals' forces, ionic interactions and covalent bonds) to a cleavable functional group (e.g., a disulfide group), which in turn is bound to a hydrophobic tail-group (e.g., cholesterol).

As used herein, the term "alkyl" refers to both straight and branched chain C1-C40 hydrocarbons (e.g., C6-C20 hydrocarbons), and include both saturated and unsaturated hydrocarbons. In certain embodiments, the alkyl may comprise one or more cyclic alkyls and/or one or more heteroatoms such as oxygen, nitrogen, or sulfur and may optionally be substituted with substituents (e.g., one or more of alkyl, halo, alkoxy, hydroxy, amino, aryl, ether, ester or amide). In certain embodiments, a contemplated alkyl includes (9Z, 12Z)-octadeca-9,12-dien. The use of designations such as, for example, "C6-C20" is intended to refer to an alkyl (e.g., straight or branched chain and inclusive of alkenes and alkyls) having the recited range carbon atoms. In some embodiments, an alkyl group has 1 to 10 carbon atoms ("C1-10 alkyl"). In some embodiments, an alkyl group has 1 to 9 carbon atoms ("C1-9 alkyl"). In some embodiments, an alkyl group has 1 to 8 carbon atoms ("C1-8 alkyl"). In some embodiments, an alkyl group has 1 to 7 carbon atoms ("C1-7 alkyl"). In some embodiments, an alkyl group has 1 to 6 carbon atoms ("C1-6 alkyl"). In some embodiments, an alkyl group has 1 to 5 carbon atoms ("C1-5 alkyl"). In some embodiments, an alkyl group has 1 to 4 carbon atoms ("C1-4

alkyl"). In some embodiments, an alkyl group has 1 to 3 carbon atoms ("C1-3 alkyl"). In some embodiments, an alkyl group has 1 to 2 carbon atoms ("C1-2 alkyl"). In some embodiments, an alkyl group has 1 carbon atom ("C1 alkyl"). Examples of C1-6 alkyl groups include methyl, ethyl, propyl, isopropyl, butyl, isobutyl, pentyl, hexyl, and the like.

As used herein, "alkenyl" refers to a radical of a straight-chain or branched hydrocarbon group having from 2 to 20 carbon atoms, one or more carbon-carbon double bonds (e.g., 1, 2, 3, or 4 carbon-carbon double bonds), and optionally one or more carbon-carbon triple bonds (e.g., 1, 2, 3, or 4 carbon-carbon triple bonds)("C2-20 alkenyl"). In certain embodiments, alkenyl does not contain any triple bonds. In some embodiments, an alkenyl group has 2 to 10 carbon atoms ("C2-10 alkenyl"). In some embodiments, an alkenyl group has 2 to 9 carbon atoms ("C2-9 alkenyl"). In some embodiments, an alkenyl group has 2 to 8 carbon atoms ("C2-8 alkenyl"). In some embodiments, an alkenyl group has 2 to 7 carbon atoms ("C2-7 alkenyl"). In some embodiments, an alkenyl group has 2 to 6 carbon atoms ("C2-6 alkenyl"). In some embodiments, an alkenyl group has 2 to 5 carbon atoms ("C2-5 alkenyl"). In some embodiments, an alkenyl group has 2 to 4 carbon atoms ("C2-4 alkenyl"). In some embodiments, an alkenyl group has 2 to 3 carbon atoms ("C2-3 alkenyl"). In some embodiments, an alkenyl group has 2 carbon atoms ("C2 alkenyl"). The one or more carbon-carbon double bonds can be internal (such as in 2-butenyl) or terminal (such as in 1-butenyl). Examples of C2-4 alkenyl groups include ethenyl (C2), 1-propenyl (C3), 2-propenyl (C3), 1-butenyl (C4), 2-butenyl (C4), butadienyl (C4), and the like. Examples of C2-6 alkenyl groups include the aforementioned C2-4 alkenyl groups as well as pentenyl (C5), pentadienyl (C5), hexenyl (C6), and the like. Additional examples of alkenyl include heptenyl (C7), octenyl (C8), octatrienyl (C8), and the like.

As used herein, "alkynyl" refers to a radical of a straight-chain or branched hydrocarbon group having from 2 to 20 carbon atoms, one or more carbon-carbon triple bonds (e.g., 1, 2, 3, or 4 carbon-carbon triple bonds), and optionally one or more carbon-carbon double bonds (e.g., 1, 2, 3, or 4 carbon-carbon double bonds) ("C2-20 alkynyl"). In certain embodiments, alkynyl does not contain any double bonds. In some embodiments, an alkynyl group has 2 to 10 carbon atoms ("C2-10 alkynyl"). In some embodiments, an alkynyl group has 2 to 9 carbon atoms ("C2-9 alkynyl"). In some embodiments, an alkynyl group has 2 to 8 carbon atoms ("C2-8 alkynyl"). In some embodiments, an alkynyl group has 2 to 7 carbon atoms ("C2-7 alkynyl"). In some embodiments, an alkynyl group has 2 to 6 carbon atoms ("C2-6 alkynyl"). In some embodiments, an alkynyl group has 2 to 5 carbon atoms ("C2-5 alkynyl"). In some embodiments, an alkynyl group has 2 to 4 carbon atoms ("C2-4 alkynyl"). In some embodiments, an alkynyl group has 2 to 3 carbon atoms ("C2-3 alkynyl"). In some embodiments, an alkynyl group has 2 carbon atoms ("C2 alkynyl"). The one or more carbon-carbon triple bonds can be internal (such as in 2-butyne) or terminal (such as in 1-butyne). Examples of C2-4 alkynyl groups include, without limitation, ethynyl (C2), 1-propynyl (C3), 2-propynyl (C3), 1-butyne (C4), 2-butyne (C4), and the like. Examples of C2-6 alkynyl groups include the aforementioned C2-4 alkynyl groups as well as pentynyl (C5), hexynyl (C6), and the like. Additional examples of alkynyl include heptynyl (C7), octynyl (C8), and the like.

As used herein, "alkylene," "alkenylene," and "alkynylene," refer to a divalent radical of an alkyl, alkenyl, and

alkynyl group respectively. When a range or number of carbons is provided for a particular "alkylene," "alkenylene," or "alkynylene," group, it is understood that the range or number refers to the range or number of carbons in the linear carbon divalent chain. "Alkylene," "alkenylene," and "alkynylene," groups may be substituted or unsubstituted with one or more substituents as described herein.

As used herein, the term "aryl" refers to aromatic groups (e.g., monocyclic, bicyclic and tricyclic structures) containing six to ten carbons in the ring portion. The aryl groups may be optionally substituted through available carbon atoms and in certain embodiments may include one or more heteroatoms such as oxygen, nitrogen or sulfur. In some embodiments, an aryl group has six ring carbon atoms ("C6 aryl"; e.g., phenyl). In some embodiments, an aryl group has ten ring carbon atoms ("C10 aryl"; e.g., naphthyl such as 1-naphthyl and 2-naphthyl).

The term "heteroalkyl" refers to a non-cyclic stable straight or branched chain, or combinations thereof, including at least one carbon atom and at least one heteroatom selected from the group consisting of O, N, P, Si. and S. and wherein the nitrogen and sulfur atoms may optionally be oxidized, and the nitrogen heteroatom may optionally be quaternized. The heteroatom(s) O, N, P, S. and Si may be placed at any interior position of the heteroalkyl group or at the position at which the alkyl group is attached to the remainder of the molecule. Exemplary heteroalkyl groups include, but are not limited to: $\text{—CH}_2\text{—CH}_2\text{—O—CH}_3$, $\text{—CH}_2\text{—CH}_2\text{—NH—CH}_3$, $\text{—CH}_2\text{—CH}_2\text{—N(CH}_3\text{)—CH}_3$, $\text{—CH}_2\text{—S—CH}_2\text{—CH}_3$, $\text{—CH}_2\text{—CH}_2\text{—S(O)}_2$, —S(O)—CH_3 , $\text{—S(O)}_2\text{—CH}_2\text{—}$, $\text{—CH}_2\text{—CH}_2\text{—S(O)}_2\text{—CH}_3$, —CH=CH—O—CH_3 , $\text{—Si(CH}_3\text{)}_3$, $\text{—CH}_2\text{—CH=N—OCH}_3$, $\text{—CH=CH—N(CH}_3\text{)—CH}_3$, —O—CH_3 , and $\text{—O—CH}_2\text{—CH}_3$. Up to two or three heteroatoms may be consecutive, such as, for example, $\text{—CH}_2\text{—NH—OCH}_3$ and $\text{—CH}_2\text{—O—Si(CH}_3\text{)}_3$. Where "heteroalkyl" is recited, followed by recitations of specific heteroalkyl groups, such as $\text{—CH}_2\text{O}$, $\text{—NR}^B\text{R}^C$, or the like, it will be understood that the terms heteroalkyl and $\text{—CH}_2\text{O}$ or $\text{—NR}^B\text{R}^C$ are not redundant or mutually exclusive. Rather, the specific heteroalkyl groups are recited to add clarity. Thus, the term "heteroalkyl" should not be interpreted herein as excluding specific heteroalkyl groups, such as $\text{—CH}_2\text{O}$, $\text{—NR}^B\text{R}^C$, or the like.

Similarly, the term "heteroalkylene," by itself or as part of another substituent, means, unless otherwise stated, a divalent radical derived from heteroalkyl, as exemplified, but not limited by, $\text{—CH}_2\text{O—}$ and $\text{—CH}_2\text{CH}_2\text{O—}$. A heteroalkylene group may be described as, e.g., a 2-7-membered heteroalkylene, wherein the term "membered" refers to the non-hydrogen atoms within the moiety. For heteroalkylene groups, heteroatoms can also occupy either or both of the chain termini (e.g., alkyleneoxy, alkylenedioxy, alkyleneamino, alkylenediamino, and the like). Still further, for alkylene and heteroalkylene linking groups, no orientation of the linking group is implied by the direction in which the formula of the linking group is written. For example, the formula $\text{—C(O)}_2\text{R'—}$ may represent both $\text{—C(O)}_2\text{R'—}$ and $\text{—R'C(O)}_2\text{—}$.

As used herein, "heteroaryl" refers to a radical of a 5-10 membered monocyclic or bicyclic $4n+2$ aromatic ring system (e.g., having 6 or 10 electrons shared in a cyclic array) having ring carbon atoms and 1-4 ring heteroatoms provided in the aromatic ring system, wherein each heteroatom is independently selected from nitrogen, oxygen and sulfur ("5-10 membered heteroaryl"). In heteroaryl groups that contain one or more nitrogen atoms, the point of attachment

can be a carbon or nitrogen atom, as valency permits. Heteroaryl bicyclic ring systems can include one or more heteroatoms in one or both rings. "Heteroaryl" includes ring systems wherein the heteroaryl ring, as defined above, is fused with one or more carbocyclyl or heterocyclyl groups wherein the point of attachment is on the heteroaryl ring, and in such instances, the number of ring members continue to designate the number of ring members in the heteroaryl ring system. "Heteroaryl" also includes ring systems wherein the heteroaryl ring, as defined above, is fused with one or more aryl groups wherein the point of attachment is either on the aryl or heteroaryl ring, and in such instances, the number of ring members designates the number of ring members in the fused (aryl/heteroaryl) ring system. Bicyclic heteroaryl groups wherein one ring does not contain a heteroatom (e.g., indolyl, quinoliny, carbazolyl, and the like) the point of attachment can be on either ring, i.e., either the ring bearing a heteroatom (e.g., 2-indolyl) or the ring that does not contain a heteroatom (e.g., 5-indolyl).

The term "cycloalkyl" refers to a monovalent saturated cyclic, bicyclic, or bridged cyclic (e.g., adamantyl) hydrocarbon group of 3-12, 3-8, 4-8, or 4-6 carbons, referred to herein, e.g., as "C₄-8cycloalkyl," derived from a cycloalkane. Exemplary cycloalkyl groups include, but are not limited to, cyclohexanes, cyclopentanes, cyclobutanes and cyclopropanes.

As used herein, "heterocyclyl" or "heterocyclic" refers to a radical of a 3- to 10-membered non-aromatic ring system having ring carbon atoms and 1 to 4 ring heteroatoms, wherein each heteroatom is independently selected from nitrogen, oxygen, sulfur, boron, phosphorus, and silicon ("3-10 membered heterocyclyl"). In heterocyclyl groups that contain one or more nitrogen atoms, the point of attachment can be a carbon or nitrogen atom, as valency permits. A heterocyclyl group can either be monocyclic ("monocyclic heterocyclyl") or a fused, bridged or spiro ring system such as a bicyclic system ("bicyclic heterocyclyl"), and can be saturated or can be partially unsaturated. Heterocyclyl bicyclic ring systems can include one or more heteroatoms in one or both rings. "Heterocyclyl" also includes ring systems wherein the heterocyclyl ring, as defined above, is fused with one or more carbocyclyl groups wherein the point of attachment is either on the carbocyclyl or heterocyclyl ring, or ring systems wherein the heterocyclyl ring, as defined above, is fused with one or more aryl or heteroaryl groups, w % herein the point of attachment is on the heterocyclyl ring, and in such instances, the number of ring members continue to designate the number of ring members in the heterocyclyl ring system. The terms "heterocycle," "heterocyclyl," "heterocyclyl ring," "heterocyclic group," "heterocyclic moiety," and "heterocyclic radical," may be used interchangeably.

As used herein, "cyano" refers to —CN .

The terms "halo" and "halogen" as used herein refer to an atom selected from fluorine (fluoro, F), chlorine (chloro, Cl), bromine (bromo, Br), and iodine (iodo, I). In certain embodiments, the halo group is either fluoro or chloro.

The term "alkoxy," as used herein, refers to an alkyl group which is attached to another moiety via an oxygen atom (—O(alkyl)). Non-limiting examples include e.g., methoxy, ethoxy, propoxy, and butoxy.

As used herein, "oxo" refers to —C=O .

In general, the term "substituted", whether preceded by the term "optionally" or not, means that at least one hydrogen present on a group (e.g., a hydrogen attached to a carbon or nitrogen atom of a group) is replaced with a permissible substituent, e.g., a substituent which upon substitution

results in a stable compound, e.g., a compound which does not spontaneously undergo transformation such as by rearrangement, cyclization, elimination, or other reaction. Unless otherwise indicated, a "substituted" group has a substituent at one or more substitutable positions of the group, and when more than one position in any given structure is substituted, the substituent is either the same or different at each position.

As used herein, "pharmaceutically acceptable salt" refers to those salts which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of humans and lower animals without undue toxicity, irritation, allergic response and the like, and are commensurate with a reasonable benefit/risk ratio. Pharmaceutically acceptable salts are well known in the art. For example, Berge et al., describes pharmaceutically acceptable salts in detail in *J. Pharmaceutical Sciences* (1977) 66:1-19. Pharmaceutically acceptable salts of the compounds of this invention include those derived from suitable inorganic and organic acids and bases. Examples of pharmaceutically acceptable, nontoxic acid addition salts are salts of an amino group formed with inorganic acids such as hydrochloric acid, hydrobromic acid, phosphoric acid, sulfuric acid and perchloric acid or with organic acids such as acetic acid, oxalic acid, maleic acid, tartaric acid, citric acid, succinic acid or malonic acid or by using other methods used in the art such as ion exchange. Other pharmaceutically acceptable salts include adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bisulfate, borate, butyrate, camphorate, camphorsulfonate, citrate, cyclopentanepropionate, digluconate, dodecylsulfate, ethanesulfonate, formate, fumarate, glucoheptonate, glycerophosphate, gluconate, hemisulfate, heptanoate, hexanoate, hydroiodide, 2-hydroxy-ethanesulfonate, lactobionate, lactate, laurate, lauryl sulfate, malate, maleate, malonate, methanesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, oleate, oxalate, palmitate, pamoate, pectinate, persulfate, 3-phenylpropionate, phosphate, picrate, pivalate, propionate, stearate, succinate, sulfate, tartrate, thiocyanate, p-toluenesulfonate, undecanoate, valerate salts, and the like. Pharmaceutically acceptable salts derived from appropriate bases include alkali metal, alkaline earth metal, ammonium and N+(C1-4alkyl)4 salts. Representative alkali or alkaline earth metal salts include sodium, lithium, potassium, calcium, magnesium, and the like. Further pharmaceutically acceptable salts include, when appropriate, nontoxic ammonium, quaternary ammonium, and amine cations formed using counterions such as halide, hydroxide, carboxylate, sulfate, phosphate, nitrate, lower alkyl sulfonate, and aryl sulfonate.

In typical embodiments, the present invention is intended to encompass the compounds disclosed herein, and the pharmaceutically acceptable salts, pharmaceutically acceptable esters, tautomeric forms, polymorphs, and prodrugs of such compounds. In some embodiments, the present invention includes a pharmaceutically acceptable addition salt, a pharmaceutically acceptable ester, a solvate (e.g., hydrate) of an addition salt, a tautomeric form, a polymorph, an enantiomer, a mixture of enantiomers, a stereoisomer or mixture of stereoisomers (pure or as a racemic or non-racemic mixture) of a compound described herein.

Compounds described herein can comprise one or more asymmetric centers, and thus can exist in various isomeric forms, e.g., enantiomers and/or diastereomers. For example, the compounds described herein can be in the form of an individual enantiomer, diastereomer or geometric isomer, or can be in the form of a mixture of stereoisomers, including racemic mixtures and mixtures enriched in one or more

stereoisomer. Isomers can be isolated from mixtures by methods known to those skilled in the art, including chiral high pressure liquid chromatography (HPLC) and the formation and crystallization of chiral salts, or preferred isomers can be prepared by asymmetric syntheses. See, for example, Jacques et al., *Enantiomers, Racemates and Resolutions* (Wiley Interscience, New York, 1981); Wilen et al., *Tetrahedron* 33:2725 (1977); Eliel, *Stereochemistry of Carbon Compounds* (McGraw-Hill, NY, 1962); and Wilen, *Tables of Resolving Agents and Optical Resolutions* p. 268 (E. L. Eliel, Ed., Univ. of Notre Dame Press, Notre Dame, IN 1972). The invention additionally encompasses compounds described herein as individual isomers substantially free of other isomers, and alternatively, as mixtures of various isomers.

In certain embodiments the compounds and the transfer vehicles of which such compounds are a component (e.g., lipid nanoparticles) exhibit an enhanced (e.g., increased) ability to transfect one or more target cells. Accordingly, also provided herein are methods of transfecting one or more target cells. Such methods generally comprise the step of contacting the one or more target cells with the compounds and/or pharmaceutical compositions disclosed herein such that the one or more target cells are transfected with the circular RNA encapsulated therein. As used herein, the terms "transfect" or "transfection" refer to the intracellular introduction of one or more encapsulated materials (e.g., nucleic acids and/or polynucleotides) into a cell, or preferably into a target cell. The term "transfection efficiency" refers to the relative amount of such encapsulated material (e.g., polynucleotides) up-taken by, introduced into and/or expressed by the target cell which is subject to transfection. In some embodiments, transfection efficiency may be estimated by the amount of a reporter polynucleotide product produced by the target cells following transfection. In some embodiments, a transfer vehicle has high transfection efficiency. In some embodiments, a transfer vehicle has at least about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, or 90% transfection efficiency.

As used herein, the term "liposome" generally refers to a vesicle composed of lipids (e.g., amphiphilic lipids) arranged in one or more spherical bilayer or bilayers. In certain embodiments, the liposome is a lipid nanoparticle (e.g., a lipid nanoparticle comprising one or more of the ionizable lipid compounds disclosed herein). Such liposomes may be unilamellar or multilamellar vesicles which have a membrane formed from a lipophilic material and an aqueous interior that contains the encapsulated circRNA to be delivered to one or more target cells, tissues and organs. In certain embodiments, the compositions described herein comprise one or more lipid nanoparticles. Examples of suitable lipids (e.g., ionizable lipids) that may be used to form the liposomes and lipid nanoparticles contemplated include one or more of the compounds disclosed herein (e.g., HGT4001, HGT4002, HGT4003, HGT4004 and/or HGT4005). Such liposomes and lipid nanoparticles may also comprise additional ionizable lipids such as C12-200, DLin-KC2-DMA, and/or HGT5001, helper lipids, structural lipids, PEG-modified lipids, MC3, DLinDMA, DLinkC2DMA, cKK-E12, ICE, HGT5000, DODAC, DDAB, DMRIE, DOSPA, DOGS, DODAP, DODMA, DMDMA, DODAC, DLenDMA, DMRIE, CLinDMA, CpLinDMA, DMOBA, DOcarbDAP, DLinDAP, DLincarbDAP, DLinCDAP, KLin-K-DMA, DLin-K-XTC2-DMA, HGT4003, and combinations thereof.

As used herein, the phrases “non-cationic lipid”, “non-cationic helper lipid”, and “helper lipid” are used interchangeably and refer to any neutral, zwitterionic or anionic lipid.

As used herein, the phrase “anionic lipid” refers to any of a number of lipid species that carry a net negative charge at a selected pH, such as physiological pH.

As used herein, the phrase “biodegradable lipid” or “degradable lipid” refers to any of a number of lipid species that are broken down in a host environment on the order of minutes, hours, or days ideally making them less toxic and unlikely to accumulate in a host over time. Common modifications to lipids include ester bonds, and disulfide bonds among others to increase the biodegradability of a lipid.

As used herein, the phrase “biodegradable PEG lipid” or “degradable PEG lipid” refers to any of a number of lipid species where the PEG molecules are cleaved from the lipid in a host environment on the order of minutes, hours, or days ideally making them less immunogenic. Common modifications to PEG lipids include ester bonds, and disulfide bonds among others to increase the biodegradability of a lipid.

In certain embodiments of the present invention, the transfer vehicles (e.g., lipid nanoparticles) are prepared to encapsulate one or more materials or therapeutic agents (e.g., circRNA). The process of incorporating a desired therapeutic agent (e.g., circRNA) into a transfer vehicle is referred to herein as or “loading” or “encapsulating” (Lasic, et al., FEBS Lett., 312: 255-258, 1992). The transfer vehicle-loaded or -encapsulated materials (e.g., circRNA) may be completely or partially located in the interior space of the transfer vehicle, within a bilayer membrane of the transfer vehicle, or associated with the exterior surface of the transfer vehicle.

As used herein, the term “structural lipid” refers to sterols and also to lipids containing sterol moieties.

As defined herein, “sterols” are a subgroup of steroids consisting of steroid alcohols.

As used herein, the term “PEG” means any polyethylene glycol or other polyalkylene ether polymer.

As generally defined herein, a “PEG-OH lipid” (also referred to herein as “hydroxy-PEGylated lipid”) is a PEGylated lipid having one or more hydroxyl (—OH) groups on the lipid.

As used herein, a “phospholipid” is a lipid that includes a phosphate moiety and one or more carbon chains, such as unsaturated fatty acid chains.

All nucleotide sequences disclosed herein can represent an RNA sequence or a corresponding DNA sequence. It is understood that deoxythymidine (dT or T) in a DNA is transcribed into a uridine (U) in an RNA. As such, “T” and “U” are used interchangeably herein in nucleotide sequences.

The recitations “sequence identity” or, for example, comprising a “sequence 50% identical to,” as used herein, refer to the extent that sequences are identical on a nucleotide-by-nucleotide basis or an amino acid-by-amino acid basis over a window of comparison. Thus, a “percentage of sequence identity” may be calculated by comparing two optimally aligned sequences over the window of comparison, determining the number of positions at which the identical nucleic acid base (e.g., A, T, C, G, I) or the identical amino acid residue (e.g., Ala, Pro, Ser, Thr, Gly, Val, Leu, Ile, Phe, Tyr, Trp, Lys, Arg, His, Asp, Glu, Asn, Gln, Cys and Met) occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of

comparison (i.e., the window size), and multiplying the result by 100 to yield the percentage of sequence identity. Included are nucleotides and polypeptides having at least about 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99% or 100% sequence identity to any of the reference sequences described herein, typically where the polypeptide variant maintains at least one biological activity of the reference polypeptide.

The expression sequences in the polynucleotide construct may be separated by a “cleavage site” sequence which enables polypeptides encoded by the expression sequences, once translated, to be expressed separately by the cell.

A “self-cleaving peptide” refers to a peptide which is translated without a peptide bond between two adjacent amino acids, or functions such that when the polypeptide comprising the proteins and the self-cleaving peptide is produced, it is immediately cleaved or separated into distinct and discrete first and second polypeptides without the need for any external cleavage activity.

The α and β chains of $\alpha\beta$ TCR’s are generally regarded as each having two domains or regions, namely variable and constant domains/regions. The variable domain consists of a concatenation of variable regions and joining regions. In the present specification and claims, the term “TCR alpha variable domain” therefore refers to the concatenation of TRAV and TRAJ regions, and the term TCR alpha constant domain refers to the extracellular TRAC region, or to a C-terminal truncated TRAC sequence. Likewise, the term “TCR beta variable domain” refers to the concatenation of TRBV and TRBD/TRBJ regions, and the term TCR beta constant domain refers to the extracellular TRBC region, or to a C-terminal truncated TRBC sequence.

The terms “duplexed,” “double-stranded,” or “hybridized” as used herein refer to nucleic acids formed by hybridization of two single strands of nucleic acids containing complementary sequences. In most cases, genomic DNA is double-stranded. Sequences can be fully complementary or partially complementary.

As used herein, “autoimmunity” is defined as persistent and progressive immune reactions to non-infectious self-antigens, as distinct from infectious non self-antigens from bacteria, viral, fungal, or parasitic organisms which invade and persist within mammals and humans. Autoimmune conditions include scleroderma, Grave’s disease, Crohn’s disease, Sjorgen’s disease, multiple sclerosis, Hashimoto’s disease, psoriasis, myasthenia gravis, autoimmune polyendocrinopathy syndromes, Type I diabetes mellitus (T1DM), autoimmune gastritis, autoimmune uveoretinitis, polymyositis, colitis, and thyroiditis, as well as in the generalized autoimmune diseases typified by human Lupus. “Autoantigen” or “self-antigen” as used herein refers to an antigen or epitope which is native to the mammal and which is immunogenic in said mammal.

As used herein, the phrase “cationic lipid” refers to any of a number of lipid species that carry a net positive charge at a selected pH, such as physiological pH.

The term “antibody” (Ab) includes, without limitation, a glycoprotein immunoglobulin which binds specifically to an antigen. In general, an antibody may comprise at least two heavy (H) chains and two light (L) chains interconnected by disulfide bonds, or an antigen-binding molecule thereof. Each H chain may comprise a heavy chain variable region (abbreviated herein as VH) and a heavy chain constant region. The heavy chain constant region can comprise three constant domains, CH1, CH2 and CH3. Each light chain can comprise a light chain variable region (abbreviated herein as VL) and a light chain constant region. The light chain

constant region can comprise one constant domain, CL. The VH and VL regions may be further subdivided into regions of hypervariability, termed complementarity determining regions (CDRs), interspersed with regions that are more conserved, termed framework regions (FR).

Each VH and VL may comprise three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4. The variable regions of the heavy and light chains contain a binding domain that interacts with an antigen. The constant regions of the Abs may mediate the binding of the immunoglobulin to host tissues or factors, including various cells of the immune system (e.g., effector cells) and the first component of the classical complement system. Antibodies may include, for example, monoclonal antibodies, recombinantly produced antibodies, monospecific antibodies, multispecific antibodies (including bispecific antibodies), human antibodies, engineered antibodies, humanized antibodies, chimeric antibodies, immunoglobulins, synthetic antibodies, tetrameric antibodies comprising two heavy chain and two light chain molecules, an antibody light chain monomer, an antibody heavy chain monomer, an antibody light chain dimer, an antibody heavy chain dimer, an antibody light chain-antibody heavy chain pair, intrabodies, antibody fusions (sometimes referred to herein as “antibody conjugates”), heteroconjugate antibodies, single domain antibodies, monovalent antibodies, single chain antibodies or single-chain variable fragments (scFv), camelized antibodies, affibodies, Fab fragments, F(ab')₂ fragments, disulfide-linked variable fragments (sdFv), anti-idiotypic (anti-id) antibodies (including, e.g., anti-anti-Id antibodies), minibodies, domain antibodies, synthetic antibodies (sometimes referred to herein as “antibody mimetics”), and antigen-binding fragments of any of the above. In some embodiments, antibodies described herein refer to polyclonal antibody populations.

An immunoglobulin may derive from any of the commonly known isotypes, including but not limited to IgA, secretory IgA, IgG and IgM. IgG subclasses are also well known to those in the art and include but are not limited to human IgG1, IgG2, IgG3 and IgG4. “Isotype” refers to the Ab class or subclass (e.g., IgM or IgG1) that is encoded by the heavy chain constant region genes. The term “antibody” includes, by way of example, both naturally occurring and non-naturally occurring Abs; monoclonal and polyclonal Abs; chimeric and humanized Abs; human or nonhuman Abs; wholly synthetic Abs; and single chain Abs. A nonhuman Ab may be humanized by recombinant methods to reduce its immunogenicity in humans. Where not expressly stated, and unless the context indicates otherwise, the term “antibody” also includes an antigen-binding fragment or an antigen-binding portion of any of the aforementioned immunoglobulins, and includes a monovalent and a divalent fragment or portion, and a single chain Ab.

An “antigen binding molecule,” “antigen binding portion,” or “antibody fragment” refers to any molecule that comprises the antigen binding parts (e.g., CDRs) of the antibody from which the molecule is derived. An antigen binding molecule may include the antigenic complementarity determining regions (CDRs). Examples of antibody fragments include, but are not limited to, Fab, Fab', F(ab')₂, Fv fragments, dAb, linear antibodies, scFv antibodies, and multispecific antibodies formed from antigen binding molecules. Peptibodies (i.e. Fc fusion molecules comprising peptide binding domains) are another example of suitable antigen binding molecules. In some embodiments, the antigen binding molecule binds to an antigen on a tumor cell. In

some embodiments, the antigen binding molecule binds to an antigen on a cell involved in a hyperproliferative disease or to a viral or bacterial antigen. In some embodiments, the antigen binding molecule binds to BCMA. In further embodiments, the antigen binding molecule is an antibody fragment, including one or more of the complementarity determining regions (CDRs) thereof, that specifically binds to the antigen. In further embodiments, the antigen binding molecule is a single chain variable fragment (scFv). In some embodiments, the antigen binding molecule comprises or consists of avimers.

As used herein, the term “variable region” or “variable domain” is used interchangeably and are common in the art. The variable region typically refers to a portion of an antibody, generally, a portion of a light or heavy chain, typically about the amino-terminal 110 to 120 amino acids in the mature heavy chain and about 90 to 115 amino acids in the mature light chain, which differ extensively in sequence among antibodies and are used in the binding and specificity of a particular antibody for its particular antigen. The variability in sequence is concentrated in those regions called complementarity determining regions (CDRs) while the more highly conserved regions in the variable domain are called framework regions (FR). Without wishing to be bound by any particular mechanism or theory, it is believed that the CDRs of the light and heavy chains are primarily responsible for the interaction and specificity of the antibody with antigen. In some embodiments, the variable region is a human variable region. In some embodiments, the variable region comprises rodent or murine CDRs and human framework regions (FRs). In particular embodiments, the variable region is a primate (e.g., non-human primate) variable region. In some embodiments, the variable region comprises rodent or murine CDRs and primate (e.g., non-human primate) framework regions (FRs).

The terms “VL” and “VL domain” are used interchangeably to refer to the light chain variable region of an antibody or an antigen-binding molecule thereof.

The terms “VH” and “VH domain” are used interchangeably to refer to the heavy chain variable region of an antibody or an antigen-binding molecule thereof.

A number of definitions of the CDRs are commonly in use: Kabat numbering, Chothia numbering, AbM numbering, or contact numbering. The AbM definition is a compromise between the two used by Oxford Molecular's AbM antibody modelling software. The contact definition is based on an analysis of the available complex crystal structures. The term “Kabat numbering” and like terms are recognized in the art and refer to a system of numbering amino acid residues in the heavy and light chain variable regions of an antibody, or an antigen-binding molecule thereof. In certain aspects, the CDRs of an antibody may be determined according to the Kabat numbering system (see, e.g., Kabat EA & Wu TT (1971) *Ann NY Acad Sci* 190: 382-391 and Kabat EA et al., (1991) *Sequences of Proteins of Immunological Interest*, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242). Using the Kabat numbering system, CDRs within an antibody heavy chain molecule are typically present at amino acid positions 31 to 35, which optionally may include one or two additional amino acids, following 35 (referred to in the Kabat numbering scheme as 35A and 35B) (CDR1), amino acid positions 50 to 65 (CDR2), and amino acid positions 95 to 102 (CDR3).

Using the Kabat numbering system, CDRs within an antibody light chain molecule are typically present at amino acid positions 24 to 34 (CDR1), amino acid positions 50 to

56 (CDR2), and amino acid positions 89 to 97 (CDR3). In a specific embodiment, the CDRs of the antibodies described herein have been determined according to the Kabat numbering scheme. In certain aspects, the CDRs of an antibody may be determined according to the Chothia numbering scheme, which refers to the location of immunoglobulin structural loops (see, e.g., Chothia C & Lesk AM, (1987), *J Mol Biol* 196: 901-917; A1-Lazikani B et al, (1997) *J Mol Biol* 273: 927-948; Chothia C et al., (1992) *J Mol Biol* 227: 799-817; Tramontano A et al, (1990) *J Mol Biol* 215(1): 175-82; and U.S. Pat. No. 7,709,226). Typically, when using the Kabat numbering convention, the Chothia CDR-H1 loop is present at heavy chain amino acids 26 to 32, 33, or 34, the Chothia CDR-H2 loop is present at heavy chain amino acids 52 to 56, and the Chothia CDR-H3 loop is present at heavy chain amino acids 95 to 102, while the Chothia CDR-L1 loop is present at light chain amino acids 24 to 34, the Chothia CDR-L2 loop is present at light chain amino acids 50 to 56, and the Chothia CDR-L3 loop is present at light chain amino acids 89 to 97. The end of the Chothia CDR-H1 loop when numbered using the Kabat numbering convention varies between H32 and H34 depending on the length of the loop (this is because the Kabat numbering scheme places the insertions at H35A and H35B: if neither 35A nor 35B is present, the loop ends at 32; if only 35A is present, the loop ends at 33; if both 35A and 35B are present, the loop ends at 34). In a specific embodiment, the CDRs of the antibodies described herein have been determined according to the Chothia numbering scheme.

As used herein, the terms "constant region" and "constant domain" are interchangeable and have a meaning common in the art. The constant region is an antibody portion, e.g., a carboxyl terminal portion of a light and/or heavy chain which is not directly involved in binding of an antibody to antigen but which may exhibit various effector functions, such as interaction with the Fc receptor. The constant region of an immunoglobulin molecule generally has a more conserved amino acid sequence relative to an immunoglobulin variable domain.

"Binding affinity" generally refers to the strength of the sum total of non-covalent interactions between a single binding site of a molecule (e.g., an antibody) and its binding partner (e.g., an antigen). Unless indicated otherwise, as used herein, "binding affinity" refers to intrinsic binding affinity which reflects a 1:1 interaction between members of a binding pair (e.g., antibody and antigen). The affinity of a molecule X for its partner Y may generally be represented by the dissociation constant (KD or Kd). Affinity may be measured and/or expressed in a number of ways known in the art, including, but not limited to, equilibrium dissociation constant (KD), and equilibrium association constant (KA or Ka). The KD is calculated from the quotient of koff/kon, whereas KA is calculated from the quotient of kon/koff. kon refers to the association rate constant of, e.g., an antibody to an antigen, and koff refers to the dissociation of, e.g., an antibody to an antigen. The kon and koff may be determined by techniques known to one of ordinary skill in the art, such as BIACORE® or KinExA.

As used herein, a "conservative amino acid substitution" is one in which the amino acid residue is replaced with an amino acid residue having a similar side chain. Families of amino acid residues having similar side chains have been defined in the art. These families include amino acids with basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polar side chains (e.g., glycine, asparagine, glutamine, serine, threonine, tyrosine, cysteine, tryptophan), nonpolar side

chains (e.g., alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine), beta-branched side chains (e.g., threonine, valine, isoleucine) and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine). In some embodiments, one or more amino acid residues within a CDR(s) or within a framework region(s) of an antibody or antigen-binding molecule thereof may be replaced with an amino acid residue with a similar side chain.

As used herein, the term "heterologous" means from any source other than naturally occurring sequences.

As used herein, an "epitope" is a term in the art and refers to a localized region of an antigen to which an antibody may specifically bind. An epitope may be, for example, contiguous amino acids of a polypeptide (linear or contiguous epitope) or an epitope can, for example, come together from two or more non-contiguous regions of a polypeptide or polypeptides (conformational, non-linear, discontinuous, or non-contiguous epitope). In some embodiments, the epitope to which an antibody binds may be determined by, e.g., NMR spectroscopy, X-ray diffraction crystallography studies, ELISA assays, hydrogen/deuterium exchange coupled with mass spectrometry (e.g., liquid chromatography electrospray mass spectrometry), array-based oligo-peptide scanning assays, and/or mutagenesis mapping (e.g., site-directed mutagenesis mapping). For X-ray crystallography, crystallization may be accomplished using any of the known methods in the art (e.g., Giese R et al., (1994) *Acta Crystallogr D Biol Crystallogr* 50(Pt 4): 339-350; McPherson A (1990) *Eur J Biochem* 189: 1-23; Chayen NE (1997) *Structure* 5: 1269-1274; McPherson A (1976) *J Biol Chem* 251: 6300-6303). Antibody: antigen crystals may be studied using well known X-ray diffraction techniques and may be refined using computer software such as X-PLOR (Yale University, 1992, distributed by Molecular Simulations, Inc.; see e.g. *Meth Enzymol* (1985) volumes 114 & 115, eds Wyckoff H W et al.; U.S. Patent Publication No. 2004/0014194), and BUSTER (Bricogne G (1993) *Acta Crystallogr D Biol Crystallogr* 49(Pt 1): 37-60; Bricogne G (1997) *Meth Enzymol* 276A: 361-423, ed Carter C W; Roversi P et al., (2000) *Acta Crystallogr D Biol Crystallogr* 56(Pt 10): 1316-1323).

As used herein, an antigen binding molecule, an antibody, or an antigen binding molecule thereof "cross-competes" with a reference antibody or an antigen binding molecule thereof if the interaction between an antigen and the first binding molecule, an antibody, or an antigen binding molecule thereof blocks, limits, inhibits, or otherwise reduces the ability of the reference binding molecule, reference antibody, or an antigen binding molecule thereof to interact with the antigen. Cross competition may be complete, e.g., binding of the binding molecule to the antigen completely blocks the ability of the reference binding molecule to bind the antigen, or it may be partial, e.g., binding of the binding molecule to the antigen reduces the ability of the reference binding molecule to bind the antigen. In some embodiments, an antigen binding molecule that cross-competes with a reference antigen binding molecule binds the same or an overlapping epitope as the reference antigen binding molecule. In other embodiments, the antigen binding molecule that cross-competes with a reference antigen binding molecule binds a different epitope as the reference antigen binding molecule.

Numerous types of competitive binding assays may be used to determine if one antigen binding molecule competes with another, for example: solid phase direct or indirect radioimmunoassay (RIA); solid phase direct or indirect enzyme immunoassay (EIA); sandwich competition assay

(Stahli et al., 1983, *Methods in Enzymology* 9:242-253); solid phase direct biotin-avidin EIA (Kirkland et al., 1986, *J. Immunol.* 137:3614-3619); solid phase direct labeled assay, solid phase direct labeled sandwich assay (Harlow and Lane, 1988, *Antibodies, A Laboratory Manual*, Cold Spring Harbor Press); solid phase direct label RIA using 1-125 label (Morel et al., 1988, *Molec. Immunol.* 25:7-15); solid phase direct biotin-avidin EIA (Cheung, et al., 1990, *Virology* 176:546-552); and direct labeled RIA (Moldenhauer et al., 1990, *Scand. J. Immunol.* 32:77-82).

As used herein, the terms “immunospecifically binds,” “immunospecifically recognizes,” “specifically binds,” and “specifically recognizes” are analogous terms in the context of antibodies and refer to molecules that bind to an antigen (e.g., epitope or immune complex) as such binding is understood by one skilled in the art. For example, a molecule that specifically binds to an antigen may bind to other peptides or polypeptides, generally with lower affinity as determined by, e.g., immunoassays, BIACORE®, KinExA 3000 instrument (Sapidyne Instruments, Boise, ID), or other assays known in the art. In a specific embodiment, molecules that specifically bind to an antigen bind to the antigen with a K_A that is at least 2 logs, 2.5 logs, 3 logs, 4 logs or greater than the K_A when the molecules bind to another antigen.

An “antigen” refers to any molecule that provokes an immune response or is capable of being bound by an antibody or an antigen binding molecule. The immune response may involve either antibody production, or the activation of specific immunologically-competent cells, or both. A person of skill in the art would readily understand that any macromolecule, including virtually all proteins or peptides, may serve as an antigen. An antigen may be endogenously expressed, i.e. expressed by genomic DNA, or may be recombinantly expressed.

An antigen may be specific to a certain tissue, such as a cancer cell, or it may be broadly expressed. In addition, fragments of larger molecules may act as antigens. In some embodiments, antigens are tumor antigens.

The term “autologous” refers to any material derived from the same individual to which it is later to be re-introduced. For example, the engineered autologous cell therapy (eACT™) method described herein involves collection of lymphocytes from a patient, which are then engineered to express, e.g., a CAR construct, and then administered back to the same patient.

The term “allogeneic” refers to any material derived from one individual which is then introduced to another individual of the same species, e.g., allogeneic T cell transplantation.

A “cancer” refers to a broad group of various diseases characterized by the uncontrolled growth of abnormal cells in the body. Unregulated cell division and growth results in the formation of malignant tumors that invade neighboring tissues and may also metastasize to distant parts of the body through the lymphatic system or bloodstream. A “cancer” or “cancer tissue” may include a tumor. The particular cancer may be responsive to chemo- or radiation therapy or the cancer may be refractory. A refractory cancer refers to a cancer that is not amenable to surgical intervention and the cancer is either initially unresponsive to chemo- or radiation therapy or the cancer becomes unresponsive over time.

An “anti-tumor effect” as used herein, refers to a biological effect that may present as a decrease in tumor volume, a decrease in the number of tumor cells, a decrease in tumor cell proliferation, a decrease in the number of metastases, an increase in overall or progression-free survival, an increase

in life expectancy, or amelioration of various physiological symptoms associated with the tumor. An anti-tumor effect may also refer to the prevention of the occurrence of a tumor, e.g., a vaccine.

A “cytokine,” as used herein, refers to a non-antibody protein that is released by one cell in response to contact with a specific antigen, wherein the cytokine interacts with a second cell to mediate a response in the second cell. “Cytokine” as used herein is meant to refer to proteins released by one cell population that act on another cell as intercellular mediators. A cytokine may be endogenously expressed by a cell or administered to a subject. Cytokines may be released by immune cells, including macrophages, B cells, T cells, neutrophils, dendritic cells, eosinophils and mast cells to propagate an immune response. Cytokines may induce various responses in the recipient cell. Cytokines may include homeostatic cytokines, chemokines, pro-inflammatory cytokines, effectors, and acute-phase proteins. For example, homeostatic cytokines, including interleukin (IL) 7 and IL-15, promote immune cell survival and proliferation, and pro-inflammatory cytokines may promote an inflammatory response.

Examples of homeostatic cytokines include, but are not limited to, IL-2, IL-4, IL-5, IL-7, IL-10, IL-12p40, IL-12p70, IL-15, and interferon (IFN) gamma. Examples of pro-inflammatory cytokines include, but are not limited to, IL-1a, IL-1b, IL-6, IL-13, IL-17a, IL-23, IL-27, tumor necrosis factor (TNF)-alpha, TNF-beta, fibroblast growth factor (FGF) 2, granulocyte macrophage colony-stimulating factor (GM-CSF), soluble intercellular adhesion molecule 1 (sICAM-1), soluble vascular adhesion molecule 1 (sVCAM-1), vascular endothelial growth factor (VEGF), VEGF-C, VEGF-D, and placental growth factor (PLGF). Examples of effectors include, but are not limited to, granzyme A, granzyme B, soluble Fas ligand (sFasL), TGF-beta, IL-35, and perforin. Examples of acute phase-proteins include, but are not limited to, C-reactive protein (CRP) and serum amyloid A (SAA).

The term “lymphocyte” as used herein includes natural killer (NK) cells, T cells, or B cells. NK cells are a type of cytotoxic (cell toxic) lymphocyte that represent a major component of the innate immune system. NK cells reject tumors and cells infected by viruses. It works through the process of apoptosis or programmed cell death. They were termed “natural killers” because they do not require activation in order to kill cells. T cells play a major role in cell-mediated-immunity (no antibody involvement). T cell receptors (TCR) differentiate T cells from other lymphocyte types. The thymus, a specialized organ of the immune system, is the primary site for T cell maturation. There are numerous types of T cells, including: helper T cells (e.g., CD4+ cells), cytotoxic T cells (also known as TC, cytotoxic T lymphocytes, CTL, T-killer cells, cytolytic T cells, CD8+ T cells or killer T cells), memory T cells ((i) stem memory cells (TSCM), like naive cells, are CD45RO-, CCR7+, CD45RA+, CD62L+(L-selectin), CD27+, CD28+ and IL-7R_a+, but also express large amounts of CD95, IL-2R, CXCR3, and LFA-1, and show numerous functional attributes distinctive of memory cells); (ii) central memory cells (TCM) express L-selectin and CCR7, they secrete IL-2, but not IFN γ or IL-4, and (iii) effector memory cells (TEM), however, do not express L-selectin or CCR7 but produce effector cytokines like IFN γ and IL-4), regulatory T cells (Tregs, suppressor T cells, or CD4+CD25+ or CD4+ FoxP3+ regulatory T cells), natural killer T cells (NKT) and gamma delta T cells. B-cells, on the other hand, play a principal role in humoral immunity (with antibody involvement). B-cells

make antibodies, are capable of acting as antigen-presenting cells (APCs) and turn into memory B-cells and plasma cells, both short-lived and long-lived, after activation by antigen interaction. In mammals, immature B-cells are formed in the bone marrow.

The term “genetically engineered” or “engineered” refers to a method of modifying the genome of a cell, including, but not limited to, deleting a coding or non-coding region or a portion thereof or inserting a coding region or a portion thereof. In some embodiments, the cell that is modified is a lymphocyte, e.g., a T cell, which may either be obtained from a patient or a donor. The cell may be modified to express an exogenous construct, such as, e.g., a chimeric antigen receptor (CAR) or a T cell receptor (TCR), which is incorporated into the cell’s genome.

An “immune response” refers to the action of a cell of the immune system (for example, T lymphocytes, B lymphocytes, natural killer (NK) cells, macrophages, eosinophils, mast cells, dendritic cells and neutrophils) and soluble macromolecules produced by any of these cells or the liver (including Abs, cytokines, and complement) that results in selective targeting, binding to, damage to, destruction of, and/or elimination from a vertebrate’s body of invading pathogens, cells or tissues infected with pathogens, cancerous or other abnormal cells, or, in cases of autoimmunity or pathological inflammation, normal human cells or tissues.

A “costimulatory signal,” as used herein, refers to a signal, which in combination with a primary signal, such as TCR/CD3 ligation, leads to a T cell response, such as, but not limited to, proliferation and/or upregulation or down regulation of key molecules.

A “costimulatory ligand,” as used herein, includes a molecule on an antigen presenting cell that specifically binds a cognate co-stimulatory molecule on a T cell. Binding of the costimulatory ligand provides a signal that mediates a T cell response, including, but not limited to, proliferation, activation, differentiation, and the like. A costimulatory ligand induces a signal that is in addition to the primary signal provided by a stimulatory molecule, for instance, by binding of a T cell receptor (TCR)/CD3 complex with a major histocompatibility complex (MHC) molecule loaded with peptide. A co-stimulatory ligand may include, but is not limited to, 3/TR6, 4-1BB ligand, agonist or antibody that binds Toll-like receptor, B7-1 (CD80), B7-2 (CD86), CD30 ligand, CD40, CD7, CD70, CD83, herpes virus entry mediator (HVEM), human leukocyte antigen G (HLA-G), ILT4, immunoglobulin-like transcript (ILT) 3, inducible costimulatory ligand (ICOS-L), intercellular adhesion molecule (ICAM), ligand that specifically binds with B7-H3, lymphotoxin beta receptor, MHC class I chain-related protein A (MICA), MHC class I chain-related protein B (MICB), OX40 ligand, PD-L2, or programmed death (PD) L1. A co-stimulatory ligand includes, without limitation, an antibody that specifically binds with a co-stimulatory molecule present on a T cell, such as, but not limited to, 4-1BB, B7-H3, CD2, CD27, CD28, CD30, CD40, CD7, ICOS, ligand that specifically binds with CD83, lymphocyte function-associated antigen-1 (LFA-1), natural killer cell receptor C (NKG2C), OX40, PD-1, or tumor necrosis factor superfamily member 14 (TNFSF14 or LIGHT).

A “costimulatory molecule” is a cognate binding partner on a T cell that specifically binds with a costimulatory ligand, thereby mediating a costimulatory response by the T cell, such as, but not limited to, proliferation. Costimulatory molecules include, but are not limited to, 4-1BB/CD137, B7-H3, BAFFR, BLAME (SLAMF8), BTLA, CD 33, CD 45, CD100 (SEMA4D), CD103, CD134, CD137, CD154,

CD16, CD160 (BY55), CD 18, CD19, CD19a, CD2, CD22, CD247, CD27, CD276 (B7-H3), CD28, CD29, CD3 (alpha; beta; delta; epsilon; gamma; zeta), CD30, CD37, CD4, CD4, CD40, CD49a, CD49D, CD49f, CD5, CD64, CD69, CD7, CD80, CD83 ligand, CD84, CD86, CD8alpha, CD8beta, CD9, CD96 (Tactile), CD1-1a, CD1-1b, CD1-1c, CD1-1d, CDS, CEACAM1, CRT AM, DAP-10, DNAM1 (CD226), Fc gamma receptor, GADS, GITR, HVEM (LIGHT), IA4, ICAM-1, ICAM-1, ICOS, Ig alpha (CD79a), IL2R beta, IL2R gamma, IL7R alpha, integrin, ITGA4, ITGA4, ITGA6, ITGAD, ITGAE, ITGAL, ITGAM, ITGAX, ITGB2, ITGB7, ITGB1, KIRDS2, LAT, LFA-1, LFA-1, LIGHT, LIGHT (tumor necrosis factor superfamily member 14: TNFSF14), LTBR, Ly9 (CD229), lymphocyte function-associated antigen-1 (LFA-1 (CD1 1a/CD18), MHC class I molecule, NKG2C, NKG2D, NKp30, NKp44, NKp46, NKp80 (KLRF1), OX40, PAG/Cbp, PD-1, PSGL1, SELPLG (CD162), signaling lymphocytic activation molecule, SLAM (SLAMF1; CD150: IPO-3), SLAMF4 (CD244; 2B4), SLAMF6 (NTB-A: Ly108), SLAMF7, SLP-76, TNF, TNF α , TNFR2, Toll ligand receptor, TRANCE/RANKL, VLA 1, or VLA-6, or fragments, truncations, or combinations thereof.

As used herein, a “vaccine” refers to a composition for generating immunity for the prophylaxis and/or treatment of diseases. Accordingly, vaccines are medicaments which comprise antigens and are intended to be used in humans or animals for generating specific defense and protective substances upon administration to the human or animal.

As used herein, a “neoantigen” refers to a class of tumor antigens which arises from tumor-specific mutations in an expressed protein.

As used herein, a “fusion protein” is a protein with at least two domains that are encoded by separate genes that have been joined to transcribe for a single peptide.

2. DNA Template, Precursor RNA & Circular RNA

According to the present invention, transcription of a DNA template provided herein (e.g., comprising a 3' enhanced intron element, 3' enhanced exon element, a core functional element, a 5' enhanced exon element, and a 5' enhanced intron element) results in formation of a precursor linear RNA polynucleotide capable of circularizing. In some embodiments, this DNA template comprises a vector, PCR product, plasmid, minicircle DNA, cosmid, artificial chromosome, complementary DNA (cDNA), extrachromosomal DNA (ecDNA), or a fragment therein. In certain embodiments, the minicircle DNA may be linearized or non-linearized. In certain embodiments, the plasmid may be linearized or non-linearized. In some embodiments, the DNA template may be single-stranded. In other embodiments, the DNA template may be double-stranded. In some embodiments, the DNA template comprises in whole or in part from a viral, bacterial or eukaryotic vector.

The present invention, as provided herein, comprises a DNA template that shares the same sequence as the precursor linear RNA polynucleotide prior to splicing of the precursor linear RNA polynucleotide (e.g., a 3' enhanced intron element, a 3' enhanced exon element, a core functional element, and a 5' enhanced exon element, a 5' enhanced intron element). In some embodiments, said linear precursor RNA polynucleotide undergoes splicing leading to the removal of the 3' enhanced intron element and 5' enhanced intron element during the process of circularization. In some embodiments, the resulting circular RNA polynucleotide lacks a 3' enhanced intron fragment and a 5'

enhanced intron fragment, but maintains a 3' enhanced exon fragment, a core functional element, and a 5' enhanced exon element.

In some embodiments, the precursor linear RNA polynucleotide circularizes when incubated in the presence of one or more guanosine nucleotides or nucleoside (e.g., GTP) and a divalent cation (e.g., Mg^{2+}). In some embodiments, the 3' enhanced exon element, 5' enhanced exon element, and/or core functional element in whole or in part promotes the circularization of the precursor linear RNA polynucleotide to form the circular RNA polynucleotide provided herein.

In certain embodiments circular RNA provided herein is produced inside a cell. In some embodiments, precursor RNA is transcribed using a DNA template (e.g., in some embodiments, using a vector provided herein) in the cytoplasm by a bacteriophage RNA polymerase, or in the nucleus by host RNA polymerase II and then circularized.

In certain embodiments, the circular RNA provided herein is injected into an animal (e.g., a human), such that a polypeptide encoded by the circular RNA molecule is expressed inside the animal.

In some embodiments, the DNA (e.g., vector), linear RNA (e.g., precursor RNA), and/or circular RNA polynucleotide provided herein is between 300 and 10000, 400 and 9000, 500 and 8000, 600 and 7000, 700 and 6000, 800 and 5000, 900 and 5000, 1000 and 5000, 1100 and 5000, 1200 and 5000, 1300 and 5000, 1400 and 5000, and/or 1500 and 5000 nucleotides in length. In some embodiments, the polynucleotide is at least 300 nt, 400 nt, 500 nt, 600 nt, 700 nt, 800 nt, 900 nt, 1000 nt, 1100 nt, 1200 nt, 1300 nt, 1400 nt, 1500 nt, 200 nt, 2500 nt, 3000 nt, 3500 nt, 4000 nt, 4500 nt, or 5000 nt in length. In some embodiments, the polynucleotide is no more than 3000 nt, 3500 nt, 4000 nt, 4500 nt, 5000 nt, 6000 nt, 7000 nt, 8000 nt, 9000 nt, or 10000 nt in length. In some embodiments, the length of a DNA, linear RNA, and/or circular RNA polynucleotide provided herein is about 300 nt, 400 nt, 500 nt, 600 nt, 700 nt, 800 nt, 900 nt, 1000 nt, 1100 nt, 1200 nt, 1300 nt, 1400 nt, 1500 nt, 2000 nt, 2500 nt, 3000 nt, 3500 nt, 4000 nt, 4500 nt, 5000 nt, 6000 nt, 7000 nt, 8000 nt, 9000 nt, or 10000 nt.

In some embodiments, the circular RNA provided herein has higher functional stability than mRNA comprising the same expression sequence. In some embodiments, the circular RNA provided herein has higher functional stability than mRNA comprising the same expression sequence, 5moU modifications, an optimized UTR, a cap, and/or a polyA tail.

In some embodiments, the circular RNA polynucleotide provided herein has a functional half-life of at least 5 hours, 10 hours, 15 hours, 20 hours, 30 hours, 40 hours, 50 hours, 60 hours, 70 hours or 80 hours. In some embodiments, the circular RNA polynucleotide provided herein has a functional half-life of 5-80, 10-70, 15-60, and/or 20-50 hours. In some embodiments, the circular RNA polynucleotide provided herein has a functional half-life greater than (e.g., at least 1.5-fold greater than, at least 2-fold greater than) that of an equivalent linear RNA polynucleotide encoding the same protein. In some embodiments, functional half-life can be assessed through the detection of functional protein synthesis.

In some embodiments, the circular RNA polynucleotide provided herein has a half-life of at least 5 hours, 10 hours, 15 hours, 20 hours, 30 hours, 40 hours, 50 hours, 60 hours, 70 hours or 80 hours. In some embodiments, the circular RNA polynucleotide provided herein has a half-life of 5-80, 10-70, 15-60, and/or 20-50 hours. In some embodiments, the circular RNA polynucleotide provided herein has a half-life

greater than (e.g., at least 1.5-fold greater than, at least 2-fold greater than) that of an equivalent linear RNA polynucleotide encoding the same protein. In some embodiments, the circular RNA polynucleotide, or pharmaceutical composition thereof, has a functional half-life in a human cell greater than or equal to that of a pre-determined threshold value. In some embodiments the functional half-life is determined by a functional protein assay. For example in some embodiments, the functional half-life is determined by an in vitro luciferase assay, wherein the activity of *Gaussia luciferase* (GLuc) is measured in the media of human cells (e.g. HepG2) expressing the circular RNA polynucleotide every 1, 2, 6, 12, or 24 hours over 1, 2, 3, 4, 5, 6, 7, or 14 days. In other embodiments, the functional half-life is determined by an in vivo assay, wherein levels of a protein encoded by the expression sequence of the circular RNA polynucleotide are measured in patient serum or tissue samples every 1, 2, 6, 12, or 24 hours over 1, 2, 3, 4, 5, 6, 7, or 14 days. In some embodiments, the pre-determined threshold value is the functional half-life of a reference linear RNA polynucleotide comprising the same expression sequence as the circular RNA polynucleotide.

In some embodiments, the circular RNA provided herein may have a higher magnitude of expression than equivalent linear mRNA, e.g., a higher magnitude of expression 24 hours after administration of RNA to cells. In some embodiments, the circular RNA provided herein has a higher magnitude of expression than mRNA comprising the same expression sequence. 5moU modifications, an optimized UTR, a cap, and/or a polyA tail.

In some embodiments, the circular RNA provided herein may be less immunogenic than an equivalent mRNA when exposed to an immune system of an organism or a certain type of immune cell. In some embodiments, the circular RNA provided herein is associated with modulated production of cytokines when exposed to an immune system of an organism or a certain type of immune cell. For example, in some embodiments, the circular RNA provided herein is associated with reduced production of IFN- β 1, RIG-I, IL-2, IL-6, IFN γ , and/or TNF α when exposed to an immune system of an organism or a certain type of immune cell as compared to mRNA comprising the same expression sequence. In some embodiments, the circular RNA provided herein is associated with less IFN- β 1, RIG-I, IL-2, IL-6, IFN γ , and/or TNF α transcript induction when exposed to an immune system of an organism or a certain type of immune cell as compared to mRNA comprising the same expression sequence. In some embodiments, the circular RNA provided herein is less immunogenic than mRNA comprising the same expression sequence. In some embodiments, the circular RNA provided herein is less immunogenic than mRNA comprising the same expression sequence, 5moU modifications, an optimized UTR, a cap, and/or a polyA tail.

In certain embodiments, the circular RNA provided herein can be transfected into a cell as is, or can be transfected in DNA vector form and transcribed in the cell. Transcription of circular RNA from a transfected DNA vector can be via added polymerases or polymerases encoded by nucleic acids transfected into the cell, or preferably via endogenous polymerases.

A. Enhanced Intron Elements & Enhanced Exon Elements

As present in the invention herein, the enhanced intron elements and enhanced exon elements may comprise spacers, duplex regions, affinity sequences, intron fragments, exon fragments and various untranslated elements. These

sequences within the enhanced intron elements or enhanced exon elements are arranged to optimize circularization or protein expression.

In certain embodiments, the DNA template, precursor linear RNA polynucleotide and circular RNA provided herein comprise a first (5') and/or a second (3') spacer. In some embodiments, the DNA template or precursor linear RNA polynucleotide comprises one or more spacers in the enhanced intron elements. In some embodiments, the DNA template, precursor linear RNA polynucleotide comprises one or more spacers in the enhanced exon elements. In certain embodiments, the DNA template or linear RNA polynucleotide comprises a spacer in the 3' enhanced intron fragment and a spacer in the 5' enhanced intron fragment. In certain embodiments, DNA template, precursor linear RNA polynucleotide, or circular RNA comprises a spacer in the 3' enhanced exon fragment and another spacer in the 5' enhanced exon fragment to aid with circularization or protein expression due to symmetry created in the overall sequence.

In some embodiments, including a spacer between the 3' group I intron fragment and the core functional element may conserve secondary structures in those regions by preventing them from interacting, thus increasing splicing efficiency. In some embodiments, the first (between 3' group I intron fragment and core functional element) and second (between the two expression sequences and core functional element) spacers comprise additional base pairing regions that are predicted to base pair with each other and not to the first and second duplex regions. In other embodiments, the first (between 3' group I intron fragment and core functional element) and second (between the one of the core functional element and 5' group I intron fragment) spacers comprise additional base pairing regions that are predicted to base pair with each other and not to the first and second duplex regions. In some embodiments, such spacer base pairing brings the group I intron fragments in close proximity to each other, further increasing splicing efficiency. Additionally, in some embodiments, the combination of base pairing between the first and second duplex regions, and separately, base pairing between the first and second spacers, promotes the formation of a splicing bubble containing the group I intron fragments flanked by adjacent regions of base pairing. Typical spacers are contiguous sequences with one or more of the following qualities: 1) predicted to avoid interfering with proximal structures, for example, the IRES, expression sequence, aptamer, or intron; 2) is at least 7 nt long and no longer than 100 nt; 3) is located after and adjacent to the 3' intron fragment and/or before and adjacent to the 5' intron fragment; and 4) contains one or more of the following: a) an unstructured region at least 5 nt long, b) a region of base pairing at least 5 nt long to a distal sequence, including another spacer, and c) a structured region at least 7 nt long limited in scope to the sequence of the spacer. Spacers may have several regions, including an unstructured region, a base pairing region, a hairpin/structured region, and combinations thereof. In an embodiment, the spacer has a structured region with high GC content. In an embodiment, a region within a spacer base pairs with another region within the same spacer. In an embodiment, a region within a spacer base pairs with a region within another spacer. In an embodiment, a spacer comprises one or more hairpin structures. In an embodiment, a spacer comprises one or more hairpin structures with a stem of 4 to 12 nucleotides and a loop of 2 to 10 nucleotides. In an embodiment, there is an additional spacer between the 3' group I intron fragment and the core functional element. In an embodiment, this additional spacer

prevents the structured regions of the IRES or aptamer of a TIE from interfering with the folding of the 3' group I intron fragment or reduces the extent to which this occurs. In some embodiments, the 5' spacer sequence is at least 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 25 or 30 nucleotides in length. In some embodiments, the 5' spacer sequence is no more than 100, 90, 80, 70, 60, 50, 45, 40, 35 or 30 nucleotides in length. In some embodiments the 5' spacer sequence is between 5 and 50, 10 and 50, 20 and 50, 20 and 40, and/or 25 and 35 nucleotides in length. In certain embodiments, the 5' spacer sequence is 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49 or 50 nucleotides in length. In one embodiment, the 5' spacer sequence is a polyA sequence. In another embodiment, the 5' spacer sequence is a polyAC sequence. In one embodiment, a spacer comprises about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 100% poly AC content. In one embodiment, a spacer comprises about 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 100% poly pyrimidine (C/T or C/U) content.

In some embodiments, the DNA template and precursor linear RNA polynucleotides and circular RNA polynucleotide provided herein comprise a first (5') duplex region and a second (3') duplex region. In certain embodiments, the DNA template and precursor linear RNA polynucleotide comprises a 5' external duplex region located within the 3' enhanced intron fragment and a 3' external duplex region located within the 5' enhanced intron fragment.

In some embodiments, the DNA template, precursor linear RNA polynucleotide and circular RNA polynucleotide comprise a 5' internal duplex region located within the 3' enhanced exon fragment and a 3' internal duplex region located within the 5' enhanced exon fragment. In some embodiments, the DNA polynucleotide and precursor linear RNA polynucleotide comprises a 5' external duplex region, 5' internal duplex region, a 3' internal duplex region, and a 3' external duplex region.

In certain embodiments, the first and second duplex regions may form perfect or imperfect duplexes. Thus, in certain embodiments at least 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% of the first and second duplex regions may be base paired with one another. In some embodiments, the duplex regions are predicted to have less than 50% (e.g., less than 45%, less than 40%, less than 35%, less than 30%, less than 25%) base pairing with unintended sequences in the RNA (e.g., non-duplex region sequences). In some embodiments, including such duplex regions on the ends of the precursor RNA strand, and adjacent or very close to the group I intron fragment, bring the group I intron fragments in close proximity to each other, increasing splicing efficiency. In some embodiments, the duplex regions are 3 to 100 nucleotides in length (e.g., 3-75 nucleotides in length, 3-50 nucleotides in length, 20-50 nucleotides in length, 35-50 nucleotides in length, 5-25 nucleotides in length, 9-19 nucleotides in length). In some embodiments, the duplex regions are about 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49 or 50 nucleotides in length. In some embodiments, the duplex regions have a length of about 9 to about 50 nucleotides. In one embodiment, the duplex regions have a length of about 9 to about 19 nucleotides. In some embodiments, the duplex regions have a length of about 20 to about 40 nucleotides. In certain embodiments, the duplex regions have a length of about 30 nucleotides.

In other embodiments, the DNA template, precursor linear RNA polynucleotide, or circular RNA polynucleotide does not comprise of any duplex regions to optimize translation or circularization.

As provided herein, the DNA template or precursor linear RNA polynucleotide may comprise an affinity tag. In some embodiments, the affinity tag is located in the 3' enhanced intron element. In some embodiments, the affinity tag is located in the 5' enhanced intron element. In some embodiments, both (3' and 5') enhanced intron elements each comprise an affinity tag. In one embodiment, an affinity tag of the 3' enhanced intron element is the length as an affinity tag in the 5' enhanced intron element. In some embodiments, an affinity tag of the 3' enhanced intron element is the same sequence as an affinity tag in the 5' enhanced intron element. In some embodiments, the affinity sequence is placed to optimize oligo-dT purification.

In some embodiments, an affinity tag comprises a polyA region. In some embodiments the polyA region is at least 15, 30, or 60 nucleotides long. In some embodiments, one or both polyA regions is 15-50 nucleotides long. In some embodiments, one or both polyA regions is 20-25 nucleotides long. The polyA sequence is removed upon circularization. Thus, an oligonucleotide hybridizing with the polyA sequence, such as a deoxythymine oligonucleotide (oligo (dT)) conjugated to a solid surface (e.g., a resin), can be used to separate circular RNA from its precursor RNA.

In certain embodiments, the 3' enhanced intron element comprises a leading untranslated sequence. In some embodiments, the leading untranslated sequence is at the 5' end of the 3' enhanced intron fragment. In some embodiments, the leading untranslated sequence comprises of the last nucleotide of a transcription start site (TSS). In some embodiments, the TSS is chosen from a viral, bacterial, or eukaryotic DNA template. In one embodiment, the leading untranslated sequence comprise the last nucleotide of a TSS and 0 to 100 additional nucleotides. In some embodiments, the TSS is a terminal spacer. In one embodiment, the leading untranslated sequence contains a guanosine at the 5' end upon translation of an RNA T7 polymerase.

In certain embodiments, the 5' enhanced intron element comprises a trailing untranslated sequence. In some embodiments, the 5' trailing untranslated sequence is located at the 3' end of the 5' enhanced intron element. In some embodiments, the trailing untranslated sequence is a partial restriction digest sequence. In one embodiment, the trailing untranslated sequence is in whole or in part a restriction digest site used to linearize the DNA template. In some embodiments, the restriction digest site is in whole or in part from a natural viral, bacterial or eukaryotic DNA template. In some embodiments, the trailing untranslated sequence is a terminal restriction site fragment.

A. Enhanced Intron Fragments

According to the present invention, the 3' enhanced intron element and 5' enhanced intron element each comprise an intron fragment. In certain embodiments, a 3' intron fragment is a contiguous sequence at least 75% homologous (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% homologous) to a 3' proximal fragment of a natural group I intron including the 3' splice site dinucleotide. Typically, a 5' intron fragment is a contiguous sequence at least 75% homologous (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% homologous) to a 5' proximal fragment of a natural group I intron including the 5' splice site dinucleotide. In some embodiments, the 3' intron fragment includes the first nucleotide of a 3' group I splice site

dinucleotide. In some embodiments, the 5' intron fragment includes the first nucleotide of a 5' group I splice site dinucleotide. In other embodiments, the 3' intron fragment includes the first and second nucleotides of a 3' group I intron fragment splice site dinucleotide; and the 5' intron fragment includes the first and second nucleotides of a 3' group I intron fragment splice site dinucleotide.

b. Enhanced Exon Fragments

In certain embodiments, as provided herein, the DNA template, linear precursor RNA polynucleotide, and circular RNA polynucleotide each comprise an enhanced exon fragment.

In some embodiments, following a 5' to 3' order, the 3' enhanced exon element is located upstream to core functional element. In some embodiments, following a 5' to 3' order, the 5' enhanced intron element is located downstream to the core functional element.

According to the present invention, the 3' enhanced exon element and 5' enhanced exon element each comprise an exon fragment. In some embodiments, the 3' enhanced exon element comprises a 3' exon fragment. In some embodiments, the 5' enhanced exon element comprises a 5' exon fragment. In certain embodiments, as provided herein, the 3' exon fragment and 5' exon fragment each comprises a group I intron fragment and 1 to 100 nucleotides of an exon sequence. In certain embodiments, a 3' intron fragment is a contiguous sequence at least 75% homologous (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% homologous) to a 3' proximal fragment of a natural group I intron including the 3' splice site dinucleotide. Typically, a 5' group I intron fragment is a contiguous sequence at least 75% homologous (e.g., at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% homologous) to a 5' proximal fragment of a natural group I intron including the 5' splice site dinucleotide. In some embodiments, the 3' exon fragment comprises a second nucleotide of a 3' group I intron splice site dinucleotide and 1 to 100 nucleotides of an exon sequence. In some embodiments, the 5' exon fragment comprises the first nucleotide of a 5' group I intron splice site dinucleotide and 1 to 100 nucleotides of an exon sequence. In some embodiments, the exon sequence comprises in part or in whole from a naturally occurring exon sequence from a virus, bacterium or eukaryotic DNA vector. In other embodiments, the exon sequence further comprises a synthetic, genetically modified (e.g., containing modified nucleotide), or other engineered exon sequence.

In one embodiment, where the 3' intron fragment comprises both nucleotides of a 3' group I splice site dinucleotide and the 5' intron fragment comprises both nucleotides of a 5' group I splice site dinucleotide, the exon fragments located within the 5' enhanced exon element and 3' enhanced exon element does not comprise of a group I splice site dinucleotide.

c. Exemplar Permutation of the Enhanced Intron Elements & Enhanced Exon Elements

For means of example and not intended to be limiting, in some embodiment, a 3' enhanced intron element comprises in the following 5' to 3' order: a leading untranslated sequence, a 5' affinity tag, an optional 5' external duplex region, a 5' external spacer, and a 3' intron fragment. In same embodiments, the 3' enhanced exon element comprises in the following 5' to 3' order: a 3' exon fragment, an optional 5' internal duplex region, an optional 5' internal duplex region, and a 5' internal spacer. In the same embodiments, the 5' enhanced exon element comprises in the following 5' to 3' order: a 3' internal spacer, an optional 3' internal duplex

region, and a 5' exon fragment. In still the same embodiments, the 3' enhanced intron element comprises in the following 5' to 3' order: a 5' intron fragment, a 3' external spacer, an optional 3' external duplex region, a 3' affinity tag, and a trailing untranslated sequence.

B. Core Functional Element

In some embodiments, the DNA template, linear precursor RNA polynucleotide, and circular RNA polynucleotide comprise a core functional element. In some embodiments, the core functional element comprises a coding or noncoding element. In certain embodiments, the core functional element may contain both a coding and noncoding element. In some embodiments, the core functional element further comprises translation initiation element (TIE) upstream to the coding or noncoding element. In some embodiments, the core functional element comprises a termination element. In some embodiments, the termination element is located downstream to the TIE and coding element. In some embodiments, the termination element is located downstream to the coding element but upstream to the TIE. In certain embodiments, where the coding element comprises a noncoding region, a core functional element lacks a TIE and/or a termination element.

A. Coding or Noncoding Element

In some embodiments, the polynucleotides herein comprise coding or noncoding element or a combination of both. In some embodiments, the coding element comprises an expression sequence. In some embodiments, the coding element encodes at least one therapeutic protein.

In some embodiments, the circular RNA encodes two or more polypeptides. In some embodiments, the circular RNA is a bicistronic RNA. The sequences encoding the two or more polypeptides can be separated by a ribosomal skipping element or a nucleotide sequence encoding a protease cleavage site. In certain embodiments, the ribosomal skipping element encodes thosea-asigna virus 2A peptide (T2A), porcine teschovirus-1 2A peptide (P2A), foot-and-mouth disease virus 2A peptide (F2A), equine rhinitis A virus 2A peptide (E2A), cytoplasmic polyhedrosis virus 2A peptide (BmCPV 2A), or flacherie virus of *B. mori* 2A peptide (BmIFV 2A).

b. TRANSLATION INITIATION ELEMENT (TIE)

As provided herein in some embodiments, the core functional element comprises at least one translation initiation element (TIE). TIEs are designed to allow translation efficiency of an encoded protein. Thus, optimal core functional elements comprising only of noncoding elements lack any TIEs. In some embodiments, core functional elements comprising one or more coding element will further comprise one or more TIEs.

In some embodiments, a TIE comprises an untranslated region (UTR). In certain embodiments, the TIE provided herein comprise an internal ribosome entry site (IRES). Inclusion of an IRES permits the translation of one or more open reading frames from a circular RNA (e.g., open reading frames that form the expression sequences). The IRES element attracts a eukaryotic ribosomal translation initiation complex and promotes translation initiation. See, e.g., Kaufman et al., *Nuc. Acids Res.* (1991) 19:4485-4490; Gurtu et al., *Biochem. Biophys. Res. Comm.* (1996) 229:295-298; Rees et al., *BioTechniques* (1996) 20: 102-110; Kobayashi et al., *BioTechniques* (1996) 21:399-402; and Mosser et al., *BioTechniques* 1997 22 150-161.

i. Natural Ties: Viral, & Eukaryotic/Cellular Internal Ribosome Entry Site (Ires)

A multitude of IRES sequences are available and include sequences derived from a wide variety of viruses, such as

from leader sequences of picornaviruses such as the encephalomyocarditis virus (EMCV) UTR (Jang et al., *J. Virol.* (1989) 63: 1651-1660), the polio leader sequence, the hepatitis A virus leader, the hepatitis C virus IRES, human rhinovirus type 2 IRES (Dobrikova et al., *Proc. Natl. Acad. Sci.* (2003) 100(25): 15125-15130), an IRES element from the foot and mouth disease virus (Ramesh et al., *Nucl. Acid Res.* (1996) 24:2697-2700), a giardiavirus IRES (Garlapati et al., *J. Biol. Chem.* (2004) 279(5):3389-3397), and the like.

For driving protein expression, the circular RNA comprises an IRES operably linked to a protein coding sequence. Modifications of IRES and accessory sequences are disclosed herein to increase or reduce IRES activities, for example, by truncating the 5' and/or 3' ends of the IRES, adding a spacer 5' to the IRES, modifying the 6 nucleotides 5' to the translation initiation site (Kozak sequence), modification of alternative translation initiation sites, and creating chimeric/hybrid IRES sequences. In some embodiments, the IRES sequence in the circular RNA disclosed herein comprises one or more of these modifications relative to a native IRES.

In some embodiments, the IRES is an IRES sequence of Taura syndrome virus, *Triatoma* virus, Theiler's encephalomyelitis virus, Simian Virus 40, *Solenopsis invicta* virus 1, *Rhopalosiphum padi* virus, Reticuloendotheliosis virus. Human poliovirus 1, *Plautia stali* intestine virus, Kashmir bee virus, Human rhinovirus 2, *Homalodisca coagulata* virus-1, Human Immunodeficiency Virus type 1, Himetobi P virus. Hepatitis C virus, Hepatitis A virus, Hepatitis GB virus, Foot and mouth disease virus, Human enterovirus 71, Equine rhinitis virus, *Ectropis obliqua* picoma-like virus, Encephalomyocarditis virus, *Drosophila* C Virus, Human coxsackievirus B3, Crucifer tobamovirus, Cricket paralysis virus, Bovine viral diarrhea virus 1, Black Queen Cell Virus, Aphid lethal paralysis virus. Avian encephalomyelitis virus, Acute bee paralysis virus. Hibiscus chlorotic ringspot virus, Classical swine fever virus, Human FGF2, Human SFTPA1, Human AML1/RUNX1, *Drosophila antennapedia*, Human AQP4, Human AT1R, Human BAG-1, Human BCL2, Human BiP, Human c-IAP1, Human c-myc, Human eIF4G, Mouse NDST4L, Human LEF1, Mouse HIF1 alpha, Human n.myc, Mouse Gtx, Human p27kipl, Human PDGF2/c-sis, Human p53, Human Pim-1, Mouse Rbm3, *Drosophila* reaper, Canine Scamper, *Drosophila* Ubx, Human UNR, Mouse UtrA, Human VEGF-A, Human X1AP, *Drosophila* hairless, *S. cerevisiae* TFIID, *S. cerevisiae* YAP1, tobacco etch virus, turnip crinkle virus, EMCV-A, EMCV-B, EMCV-Bf, EMCV-Cf, EMCV pEC9, Picobimavirus, HCV QC64, Human Cosavirus E/D, Human Cosavirus F. Human Cosavirus JMY, Rhinovirus NAT001, HRV14, HRV89, HRVC402, HRV-A21, Salivirus A SH1, Salivirus FHB, Salivirus NG-J1, Human Parechovirus 1, Crohivirus B, Yc-3, Rosavirus M-7, Shanbavirus A, Pasivirus A, Pasivirus A 2, Echovirus E14, Human Parechovirus 5, Aichi Virus, Hepatitis A Virus HA16, Phopivirus. CVA10, Enterovirus C, Enterovirus D, Enterovirus J. Human Pegivirus 2, GBV-C GT110, GBV-C K1737, GBV-C Iowa, Pegivirus A 1220, Pasivirus A 3, Sapelovirus, Rosavirus B, Bakunsa Virus, Tremovirus A, Swine Pasivirus 1, PLV-CHN, Pasivirus A, Sicinivirus, Hepacivirus K, Hepacivirus A, BVDV1, Border Disease Virus, BVDV2, CSFV-PK15C, SF573 Dicistovirus, Hubei Picoma-like Virus, CRPV, Salivirus A BN5, Salivirus A BN2, Salivirus A 02394, Salivirus A GUT, Salivirus A CH. Salivirus A SZ1, Salivirus FHB, CVB3, CVB1. Echovirus 7, CVB5, EVA71, CVA3, CVA12, EV24 or an aptamer to eIF4G.

In some embodiments, the IRES comprises in whole or in part from a eukaryotic or cellular IRES. In certain embodiments, the IRES is from a human gene, where the human gene is ABCF1, ABCG1, ACAD10, ACOT7, ACSS3, ACTG2, ADCYAP1, ADK, AGTR1, AHCYL2, AHIL, AKAP8L, AKR1A1, ALDH3A1, ALDOA, ALG13, AMMECR1L, ANGPTL4, ANK3, AOC3, AP4B1, AP4EL, APAF1, APBB1, APC, APH1A, APOBEC3D, APOM, APP, AQP4, ARHGAP36, ARL13B, ARMC8, ARMCX6, ARP-CIA, ARPC2, ARRDC3, ASAP1, ASB3, ASB5, ASCL1, ASMTL, ATF2, ATF3, ATG4A, ATP5B, ATP6V0A1, ATXN3, AURKA, AURKA, AURKA, AURKA, B3GALNT1, B3GNTL1, B4GALT3, BAAT, BAG1, BAIAP2, BAIAP2L2, BAZ2A, BBX, BCAR1, BCL2, BCS1L, BET1, BID, BIRC2, BPGM, BPIFA2, BRINP2, BSG, BTN3A2, C12orf43, C14orf93, C17orf62, Clorf226, C21orf62, C2orf15, C4BPB, C4orf22, C9orf84, CACNA1A, CALCOCO2, CAPN 11, CASP12, CASP8AP2, CAV1, CBX5, CCDIC120, CCDIC17, CCDIC186, CCDIC51, CCN1, CCND1, CCNT1, CD2BP2, CD9, CDC25C, CDC42, CDC7, CDCA7L, CDIP1, CDK1, CDK11A, CDKN1B, CEACAM7, CEP295NL, CFLAR, CHCHD7, CHIA, CHIC1, CHMP2A, CHRNA2, CLCN3, CLEC12A, CLEC7A, CLECL1, CLRN1, CMSS1, CN1H1, CNR1, CNTN5, COG4, COMMD1, COMMD5, CPEB1, CPS1, CRACR2B, CRBN, CREM, CRYBG1, CSDE1, CSF2RA, CSNK2A1, CSTF3, CTCFL, CTH, CTNNA3, CTNNB1, CTNNB1, CTNND1, CTSL, CUTA, CXCR5, CYB5R3, CYP24A1, CYP3A5, DAG1, DAP3, DAP5, DAXX, DCAF4, DCAF7, DCLRE1A, DCP1A, DCTN1, DCTN2, DDX19B, DDX46, DEFB123, DGKA, DGKD, DHRS4, DHX15, DIO3, DLG1, DLL4, DMD UTR, DMD ex5, DMKN, DNAH6, DNAL4, DUSP13, DUSP19, DYNC112, DYNLRB2, DYRK1A, ECI2, ECT2, EIF1AD, EIF2B4, EIF4G1, EIF4G2, EIF4G3, ELANE, ELOVL6, ELP5, EMCN, ENO1, EPB41, ERMN, ERVV-1, ESRRG, ETVB, ETVBKMT, ETV1, ETV4, EXD1, EXT1, EZH2, FAM111B, FAM157A, FAM213A, FBXO25, FBXO9, FBXW7, FCMR, FGF1, FGF1A, FGF2, FGF2, FGF-9, FHL5, FMR1, FN1, FOXP1, FTH1, FUBP1, G3BP1, GABBR1, GALT, GART, GAS7, gartin, GATA1, GATA4, GFM2, GHR, GJB2, GLI1, GLRA2, GMNN, GPAT3, GPATCH3, GPR137, GPR34, GPR55, GPR89A, GPRASP1, GRAP2, GSDMB, GSTO2, GTF2B, GTF2H4, GUCY1B2, HAX1, HCST, HIGD1A, HIGD1B, HIPK1, HIST1H1C, HIST1H3H, HK1, HLA-DRB4, HMBS, HMGA1, HNRNPC, HOPX, HOXA2, HOXA3, HPCAL1, HR, HSP90AB1, HSPA1A, HSPA4L, HSPA5, HYPK, IFFO1, IFT74, IFT81, IGF1, IGF1R, IGF1R, IGF2, IL 11, IL17RE, IL1RL1, IL1RN, IL32, IL6, ILF2, ILVBL, INSR, INTS13, IP6K1, ITGA4, ITGAE, KCNE4, KERA, KIAA0355, KIAA0895L, KIAA1324, KIAA1522, KIAA1683, KIF2C, KIZ, KLHL31, KLK7, KRR1, KRT14, KRT17, KRT33A, KRT6A, KRTAP10-2, KRTAP13-3, KRTAP13-4, KRTAP5-11, KRTCAP2, LACRT, LAMB1, LAMB3, LANCL1, LBX2, LCAT, LDHA, LDHAL6A, LEF1, LINC-PINT, LMO3, LRRC4C, LRRC7, LRTOMT, LSM5, LTB4R, LYRM1, LYRM2, MAGEA11, MAGEA8, MAGEB1, MAGEB16, MAGEB3, MAPT, MARS, MC1R, MCCCL, METTL12, METTL7A, MGC16025, MGC16025, MIA2, MIA2, MITE, MKLN1, MNT, MORF4L2, MPD6, MRAP1, MRPL21, MRPS12, MS12, MSLN, MSN, MT2A, MTFR1L, MTMR2, MTRR, MTUS1, MYB, MYC, MYCL, MYCN, MYL10, MYL3, MYLK, MYO1A, MYT2, MZB1, NAPIL1, NAV1, NBAS, NCF2, NDRG1, NDST2, NDUFA7, NDUFB11, NDUFC1, NDUFS1, NEDD4L, NFAT5, NFE2L2, NFE2L2, NFIA,

NHEJ1, NHP2, NIT1, NKRF, NME1-NME2, NPAT, NR3C1, NRBF2, NRF1, NTRK2, NUDCD1, NXF2, NXT2, ODC1, ODF2, OPTN, OR1OR2, OR11L1, OR2M2, OR2M3, OR2M5, OR2T10, OR4C15, OR4F17, OR4F5, OR5H1, OR5K1, OR6C3, OR6C75, OR6N1, OR7G2, p53, P2RY4, PAN2, PAQR6, PARP4, PARP9, PC, PCBP4, PCDHGC3, PCLAF, PDGFB, PDZRN4, PELO, PEMT, PEX2, PFKM, PGBD4, PGLYRP3, PHLDA2, PHTFI, P14 KB, PIGC, PIM1, PKD2L1, PKM, PLCB4, PLD3, PLEKHA1, PLEKHB1, PLS3, PML, PNMA5, PNN, POC1A, POC1B, POLD2, POLD4, POU5F1, PPIG, PQBP1, PRAME, PRPF4, PRR11, PRRT1, PRSS8, PSMA2, PSMA3, PSMA4, PSMD11, PSMD4, PSMD6, PSME3, PSMG3, PTBP3, PTCH1, PTHLH, PTPRD, PUS7L, PVRI, QPRT, RAB27A, RAB7B, RABGGTB, RAETIE, RALGDS, RALYL, RARB, RCVRN, REG3G, RFC5, RGL4, RGS19, RGS3, RHD, RINL, RIPOR2, RITA1, RMDN2, RNASE1, RNASE4, RNF4, RPA2, RPL17, RPL21, RPL26L1, RPL28, RPL29, RPL41, RPL9, RPSII, RPS13, RPS14, RRBPI, RSU1, RTP2, RUNX1, RUNX1T1, RUNX1T1, RUNX2, RUSC1, RXRG, S100A13, S100A4, SAT1, SCHIP1, SCMH1, SEC14L1, SEMA4A, SERPINA1, SERPINB4, SERTAD3, SFTPD, SH3D19, SHC1, SHMT1, SHPRH, SIM1, SIRT5, SLCH1A2, SLC12A4, SLC16A1, SLC25A3, SLC26A9, SLC5A1 1, SLC6A12, SLC6A19, SLC7A1, SLFN11, SL1RP, SMADS, SMARCA1, SMN1, SNCA, SNRNP200, SNRPB2, SNX12, SOD1, SOX13, SOX5, SPB, SPARCL1, SPATA12, SPATA31C2, SPN, SPOP, SQSTM1, SRBD1, SRC, SREBF1, SRPK2, SSB, SSB, SSBP1, ST3GAL6, STAB1, STAMPB, STAU1, STAU1, STAU1, STAU1, STAU1, STK16, STK24, STK38, STMN1, STX7, SULT2B1, SYK, SYNPR, TAF1C, TAGLN, TANK, TAS2R40, TBC1D15, TBXAS1, TCF4, TDGF1, TDP2, TDRD3, TDRD5, TESK2, THAP6, THBD, THTPA, TIAM2, TKFC, TKTL1, TLR10, TM9SF2, TMC6, TMCO2, TMED10, TMEM116, TMEM126A, TMEM159, TMEM208, TMEM230, TMEM67, TMPRSS13, TMUB2, TNFSF4, TNIP3, TP53, TP73, TRAF1, TRAK1, TRIM31, TRIM6, TRMT1, TRMT2B, TRPM7, TRPM8, TSPEAR, TTC39B, TLL11, TUBB6, TXLNB, TXNIP, TXNL1, TXNRD1, TYROBP, U2AF1, UBA1, UBE2D3, UBE21, UBE2L3, UBE2V1, UBE2V2, UMPS, UNG, UPP2, USMG5, USP18, UTP14A, UTRN, UTS2, VDR, VEGFA, VEGFA, VEPH1, VIPAS39, VPS29, VSIG10L, WDHD1, WDR12, WDR4, WDR45, WYDHV1, WRAP53, XIAP, XPNPEP3, YAP1, YWHAZ, YY1AP1, ZBTB32, ZNF146, ZNF250, ZNF385A, ZNF408, ZNF410, ZNF423, ZNF43, ZNF502, ZNF512, ZNF513, ZNF580, ZNF609, ZNF707, or ZNRD1.

ii. Synthetic Ties: Aptamer Complexes, Modified Nucleotides, Ires Variants & Other Engineered Ties

As contemplated herein, in certain embodiments, a translation initiation element (TIE) comprises a synthetic TIE. In some embodiments, a synthetic TIE comprises aptamer complexes, synthetic IRES or other engineered TIES capable of initiating translation of a linear RNA or circular RNA polynucleotide.

In some embodiments, one or more aptamer sequences is capable of binding to a component of a eukaryotic initiation factor to either enhance or initiate translation. In some embodiments, aptamer may be used to enhance translation in vivo and in vitro by promoting specific eukaryotic initiation factors (eIF) (e.g., aptamer in WO2019081383A1 is capable of binding to eukaryotic initiation factor 4F (eIF4F)).

In some embodiments, the aptamer or a complex of aptamers may be capable of binding to EIF4G, EIF4E, EIF4A, EIF4B, EIF3, EIF2, EIF5, EIF1, EIF1A, 40S ribosome, PCBP1

(polyC binding protein), PCBP2, PCBP3, PCBP4, PABP1 (polyA binding protein), PTB, Argonaute protein family, HNRNPK (heterogeneous nuclear ribonucleoprotein K), or La protein.

c. Termination Sequence

In certain embodiments, the core functional element comprises a termination sequence. In some embodiments, the termination sequence comprises a stop codon. In one embodiment, the termination sequence comprises a stop cassette. In some embodiments, the stop cassette comprises at least 2 stop codons. In some embodiments, the stop cassette comprises at least 2 frames of stop codons. In the same embodiment, the frames of the stop codons in a stop cassette each comprise 1, 2 or more stop codons. In some

C. Variants

In certain embodiments, a circular RNA polynucleotide provided herein comprises modified RNA nucleotides and/or modified nucleosides. In some embodiments, the modified nucleoside is m⁵C (5-methylcytidine). In another embodiment, the modified nucleoside is m⁵U (5-methyluridine). In another embodiment, the modified nucleoside is m⁶A (N⁶-methyladenosine). In another embodiment, the modified nucleoside is s²U (2-thiouridine). In another embodiment, the modified nucleoside is Ψ (pseudouridine). In another embodiment, the modified nucleoside is Um (2'-O-methyluridine). In other embodiments, the modified nucleoside is m¹A (1-methyladenosine); m²A (2-methyladenosine); Am (2'-O-methyladenosine); ms²m⁶A (2-methylthio-N⁶-methyladenosine); i⁶A (N⁶-isopentenyladenosine); ms²i⁶A (2-methylthio-N⁶-isopentenyladenosine); io⁶A (N⁶-(cis-hydroxyisopentenyl)adenosine); ms²io⁶A (2-methylthio-N⁶-(cis-hydroxyisopentenyl)adenosine); g⁶A (N⁶-glycylcarbamoyladenosine); t⁶A (N⁶-threonylcarbamoyladenosine); ms²t⁶A (2-methylthio-N⁶-threonylcarbamoyladenosine); m⁶t⁶A (N⁶-methyl-N⁶-threonylcarbamoyladenosine); hn⁶A (N⁶-hydroxynorvalylcarbamoyladenosine); ms²hn⁶A (2-methylthio-N⁶-hydroxynorvalylcarbamoyladenosine); Ar(p) (2'-O-ribosyladenosine (phosphate)); I (inosine); mil (1-methylinosine); m¹Im (1,2'-O-dimethylinosine); m³C (3-methylcytidine); Cm (2'-O-methylcytidine); s²C (2-thiocytidine); ac⁴C (N⁴-acetylcytidine); f⁵C (5-formylcytidine); m⁵Cm (5,2'-O-dimethylcytidine); ac⁴Cm (N⁴-acetyl-2'-O-methylcytidine); k²C (lysidine); m¹G (1-methylguanosine); m²G (N²-methylguanosine); m⁷G (7-methylguanosine); Gm (2'-O-methylguanosine); m²₂G (N²,N²-dimethylguanosine); m²Gm (N²,2'-O-dimethylguanosine); m²₂Gm (N²,N²,2'-O-trimethylguanosine); Gr(p) (2'-O-ribosylguanosine (phosphate)); yW (wybutosine); o₂yW (peroxywybutosine); OHyW (hydroxywybutosine); OHyW* (undermodified hydroxywybutosine); imG (wyosine); mimG (methylwyosine); Q (queuosine); oQ (epoxyqueuosine); galQ (galactosyl-queuosine); manQ (mannosyl-queuosine); preQ₀ (7-cyano-7-deazaguanosine); preQ₁ (7-aminomethyl-7-deazaguanosine); G⁺ (archaeosine); D (dihydrouridine); m⁵Um (5,2'-O-dimethyluridine); s⁴U (4-thiouridine); m⁵s²U (5-methyl-2-thiouridine); s²Um (2-thio-2'-O-methyluridine); acp³U (3-(3-amino-3-carboxypropyl)uridine); ho⁵U (5-hydroxyuridine); mo⁵U (5-methoxyuridine); cmo⁵U (uridine 5-oxyacetic acid); mcmo⁵U (uridine 5-oxyacetic acid methyl ester); chm⁵U (5-(carboxyhydroxymethyl)uridine); mchm⁵U (5-(carboxyhydroxymethyl)uridine methyl ester); mcm⁵U (5-methoxycarbonyl-

methyluridine); mcm⁵Um (5-methoxycarbonylmethyl-2'-O-methyluridine); mcm⁵s²U (5-methoxycarbonylmethyl-2-thiouridine); nm⁵S²U (5-aminomethyl-2-thiouridine); mnm⁵U (5-methylaminomethyluridine); mnm⁵s²U (5-methylaminomethyl-2-thiouridine); mnm⁵se²U (5-methylaminomethyl-2-selenouridine); ncm⁵U (5-carbamoylmethyluridine); ncm⁵Um (5-carbamoylmethyl-2'-O-methyluridine); cmnm⁵U (5-carboxymethylaminomethyluridine); cmnm⁵Um (5-carboxymethylaminomethyl-2'-O-methyluridine); cmnm⁵s²U (5-carboxymethylaminomethyl-2-thiouridine); m⁶₂A (N⁶,N⁶-dimethyladenosine); Im (2'-O-methylinosine); m⁴C (N⁴-methylcytidine); m⁴Cm (N⁴,2'-O-dimethylcytidine); hm⁵C (5-hydroxymethylcytidine); m³U (3-methyluridine); cm⁵U (5-carboxymethyluridine); m⁶Am (N⁶,2'-O-dimethyladenosine); m⁶₂Am (N⁶,N⁶,O-2'-trimethyladenosine); m^{2,7}G (N²,7-dimethylguanosine); m^{2,2,7}G (N²,N²,7-trimethylguanosine); m³Um (3,2'-O-dimethyluridine); m⁵D (5-methyldihydrouridine); f⁵Cm (5-formyl-2'-O-methylcytidine); m¹Gm (1,2'-O-dimethylguanosine); m¹Am (1,2'-O-dimethyladenosine); τM⁵U (5-taurinomethyluridine); τm⁵s²U (5-taurinomethyl-2-thiouridine); imG-14 (4-demethylwyosine); imG2 (isowyosine); or ac⁶A (N⁶-acetyladenosine).

In some embodiments, the modified nucleoside may include a compound selected from the group of: pyridin-4-one ribonucleoside, 5-aza-uridine, 2-thio-5-aza-uridine, 2-thiouridine, 4-thio-pseudouridine, 2-thio-pseudouridine, 5-hydroxyuridine, 3-methyluridine, 5-carboxymethyluridine, 1-carboxymethyl-pseudouridine, 5-propynyl-uridine, 1-propynyl-pseudouridine, 5-taurinomethyluridine, 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thio-uridine, 1-taurinomethyl-4-thio-uridine, 5-methyl-uridine, 1-methyl-pseudouridine, 4-thio-1-methyl-pseudouridine, 2-thio-1-methyl-pseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine, dihydrouridine, dihydropseudouridine, 2-thio-dihydrouridine, 2-thio-dihydropseudouridine, 2-methoxyuridine, 2-methoxy-4-thio-uridine, 4-methoxy-pseudouridine, 4-methoxy-2-thio-pseudouridine, 5-aza-cytidine, pseudoisocytidine, 3-methylcytidine, N⁴-acetylcytidine, 5-formylcytidine, N⁴-methylcytidine, 5-hydroxymethylcytidine, 1-methyl-pseudoisocytidine, pyrrolo-cytidine, pyrrolo-pseudoisocytidine, 2-thio-cytidine, 2-thio-5-methyl-cytidine, 4-thio-pseudoisocytidine, 4-thio-1-methyl-pseudoisocytidine, 4-thio-1-methyl-1-deaza-pseudoisocytidine, 1-methyl-1-deaza-pseudoisocytidine, zebularine, 5-aza-zebularine, 5-methyl-zebularine, 5-aza-2-thio-zebularine, 2-thio-zebularine, 2-methoxy-cytidine, 2-methoxy-5-methyl-cytidine, 4-methoxy-pseudoisocytidine, 4-methoxy-1-methyl-pseudoisocytidine, 2-aminopurine, 2, 6-diaminopurine, 7-deaza-adenine, 7-deaza-8-aza-adenine, 7-deaza-2-aminopurine, 7-deaza-8-aza-2-aminopurine, 7-deaza-2,6-diaminopurine, 7-deaza-8-aza-2,6-diaminopurine, 1-methyladenosine, N⁶-methyladenosine, N⁶-isopentenyladenosine, N⁶-(cis-hydroxyisopentenyl)adenosine, 2-methylthio-N⁶-(cis-hydroxyisopentenyl)adenosine, N⁶-glycylcarbamoyladenosine, N⁶-threonylcarbamoyladenosine, 2-methylthio-N⁶-threonylcarbamoyladenosine, N⁶,N⁶-dimethyladenosine, 7-methyladenine, 2-methylthio-adenine, 2-methoxy-adenine, inosine, 1-methyl-inosine, wyosine, wybutosine, 7-deaza-guanosine, 7-deaza-8-azaguanosine, 6-thio-guanosine, 6-thio-7-deaza-guanosine, 6-thio-7-deaza-8-azaguanosine, 7-methyl-guanosine, 6-thio-7-methyl-guanosine, 7-methylinosine, 6-methoxy-guanosine, 1-methylguanosine, N²-methylguanosine, N²,N²-dimethylguanosine, 8-oxo-guanosine, 7-methyl-8-

61

oxo-guanosine, 1-methyl-6-thio-guanosine, N2-methyl-6-thio-guanosine, and N2,N2-dimethyl-6-thio-guanosine. In another embodiment, the modifications are independently selected from the group consisting of 5-methylcytosine, pseudouridine and 1-methylpseudouridine.

In some embodiments, the modified ribonucleosides include 5-methylcytidine, 5-methoxyuridine, 1-methylpseudouridine, N6-methyladenosine, and/or pseudouridine. In some embodiments, such modified nucleosides provide additional stability and resistance to immune activation.

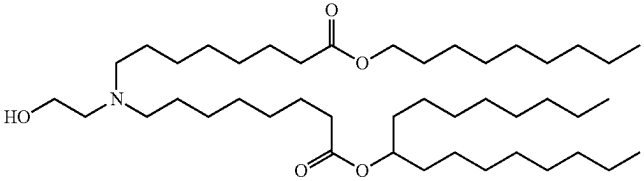
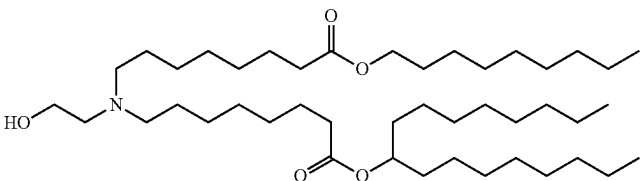
In particular embodiments, polynucleotides may be codon-optimized. A codon optimized sequence may be one in which codons in a polynucleotide encoding a polypeptide have been substituted in order to increase the expression, stability and/or activity of the polypeptide. Factors that influence codon optimization include, but are not limited to one or more of: (i) variation of codon biases between two or more organisms or genes or synthetically constructed bias tables, (ii) variation in the degree of codon bias within an organism, gene, or set of genes, (iii) systematic variation of

62

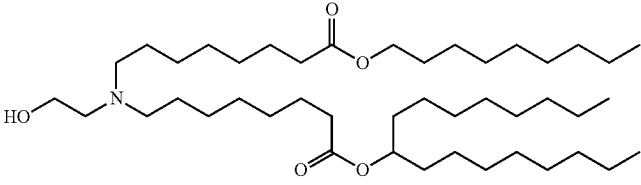
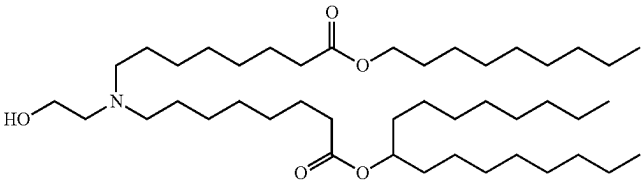
codons including context, (iv) variation of codons according to their decoding tRNAs, (v) variation of codons according to GC %, either overall or in one position of the triplet, (vi) variation in degree of similarity to a reference sequence for example a naturally occurring sequence, (vii) variation in the codon frequency cutoff, (viii) structural properties of mRNAs transcribed from the DNA sequence, (ix) prior knowledge about the function of the DNA sequences upon which design of the codon substitution set is to be based, and/or (x) systematic variation of codon sets for each amino acid. In some embodiments, a codon optimized polynucleotide may minimize ribozyme collisions and/or limit structural interference between the expression sequence and the core functional element.

3. Payloads

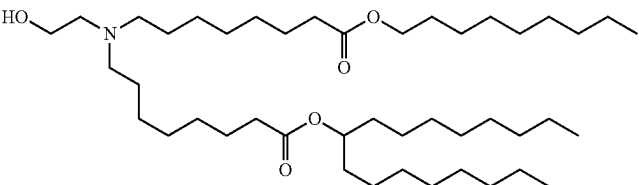
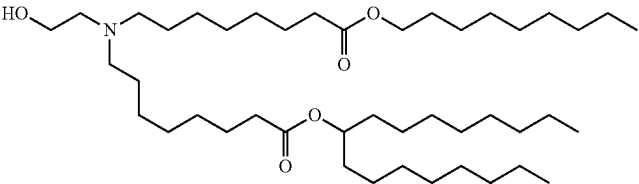
In some embodiments, the expression sequence encodes a therapeutic protein. In some embodiments, the therapeutic protein is selected from the proteins listed in the following table.

Payload	Sequence	Target cell/organ	Preferred delivery formulation
CD19 CAR	Any of sequences 309-314	T cells	 (50 mol %) DSPC (10 mol %) Beta-sitosterol (28.5% mol %) Cholesterol (10 mol %) PEG DMG (1.5 mol %)
BCMA CAR	MALPVTALLPLALLH AARPDIVLTQSPASLAVS LGERATINCRASESVSVI GAHLIHWHYQKPGQPPK LLIYLASNLETGVPARFS GSGSGTDFTLTISSLQAE DAAIYCLQSRIFPRTFG QGTLKLEIKGSTSGSGKPG SGEGSTKGQVQLVQSGS ELKKPGASVKVSCASG YFTTDYSINWVRQAPGQ GLEWMGWINTETREPA YAYDFRGRFVFSLDTSV STAYLQISSLKAEDTAVY YCARDYSYAMDYWGQ GTLVTVSSAAATTTPAP RPPTPAPTASQPLSLRPE ACRPAAGGAVHTRGLDF ACDIYIWAPLAGTCGVL LLSLVITLYCKRGRKKLL YIFKQPFMRPVQTQEE DGCCRFPEEEEGGCELR VKFSRSADAPAYQQGQN QLYNELNLGRREEYDVL DKRRGRDPGEMGKPRR KNPQEGLYNELQKDKM AEAYSEIGMKGERRRGK GHDGLYQGLSTATKDTY DALHMQALPPR	T cells	 (50 mol %) DSPC (10 mol %) Beta-sitosterol (28.5% mol %) Cholesterol (10 mol %) PEG DMG (1.5 mol %)

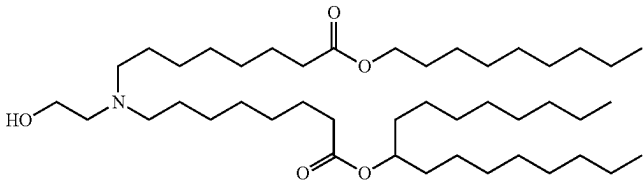
-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
MAGE-A4 TCR	TCR alpha chain: KNQVEQSPQSLILEGKN CTLQCNYTVSPFSNLRW YKQDTGRGPVSLTIMTF SENTKSNGRYTATLDAD TKQSSLHITASQLSDSAS YICVVNHSGGSYIPTFGR GTSLIVHPYIQKPDPAVY QL RDSKSSDKSVCLFTDF DSQTNVSQSKSDVYIT DKT VLDMRSMDFKSNS AVAWSNKSDFACANAF NNSHIPEDTFFPSPESS TCR beta chain: DVKVTQSSRYLVKRTGE KVFLECVQDMDHENMF WYRQDPGLGLRLIYFSY DVKMKEKGDIPGYSVS REKKERFSLILESASTNQ TSMYLCASSFLMTSGDP YEQYFGPGTRLTVTEDL KNVFPPEVAVFEPSEAEI SHTQKATLVCLATGFYP DHVELSWWVNGKEVHS GVSTDPQPLKEQPALND SRYCLSSRLRVSATFWQ NPRNHFRCQVQFYGLSE NDEWTQDRAKPVTQIVS AEAWGRAD	T cells	 (50 mol %) DSPC (10 mol %) Beta-sitosterol (28.5% mol %) Cholesterol (10 mol %) PEG DMG (1,5 mol %)
NY-ESO TCR	TCR alpha extracellular sequence MQEVTQIPAALSVPEGE NLVLNCSFTDSAIYNLQ WFRQDPGKGLTSLLLIQS SQREQTSGRLNASLDKS SGRSTLYIAASQPGDSAT YLC AVRPTSGGSYIPTFG RGTSLIVHPY TCR beta extracellular sequence MGVTTQTPKFQVLKTGQS MTLQCAQDMNHEYMS WYRQDPGMGLRLIHYS VGAGITDQGEVPNGYNV SRSTEDFPLRLLSAAPS QTSVYFCASSYVGNTGE LFFGEGSRLTVL	T cells	 (50 mol %) DSPC (10 mol %) Beta-sitosterol (28.5% mol %) Cholesterol (10 mol %) PEG DMG (1.5 mol %)
EPO	APPRLICDSRVLERYLLE AKEAENITGCAEHCSL NENITVPDTKVNIFYAWK RMEVGQQAWEVWQGLA LLSEAVLRGQALLVNSS QPWEPLQLHVDKAVSGL RSLTTLRLALGAQKEAIS PPDAASAAPLRTITADTF RKLFRVYSNFLRGKCLKL YTGEACRTGDR	Kidney or bone marrow	

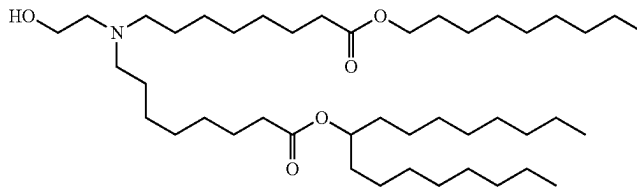
-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
PAH	MSTAVLENPGLGRKLSDFGQETSYIEDNCNQNGAISLIFSLKEEVGALAKVLR LFEENDVNLTHIESRPSRLCKQGDSIKAYGAGLLSLKKDEYEFTHLDRSL PALTNIIKILRHDIGATVHELSRDKKKDTVPWFPRTIQELDRFANQILSYGAEL DADHPGFKDPVYRARRKQFADIAYNYRHGQPIPRVEYMEEEEKKTWGTVF KTLKSLYKTHACYEYNHIFPLLEKYCGFHEDNIPQLEDVSQFLQTCTGFRLRP VAGLLSSRDFLGGLAFRVFHCTQYIRHGSKPMYTPEDICHELLGHVPLFSD RSFAQFSQEIGLASLGAPDEYIEKLATYWFTEFG LCKQGDSIKAYGAGLLS SFGELQYCLSEKPKLLPLELEKTAIQNYTVTEFQPL YYVAESFNDAKEKVRNFAATIPRPFVRYDPYTQRIEVLDTNTQQLKILADSIN SEIGILCSALQKIK	Hepatic cells	 (50 mol %) DSPC (10 mol %) Cholesterol (38.5% mol %) PEG-DMG (1.5%) OR MC3 (50 mol %) DSPC (10 mol %) Cholesterol (38.5% mol %) PEG-DMG (1.5%)
CPS1	LSVKAQTAHIVLEDGTMKGYSGFHPSSVAGEVVFNTGLGGYPEAITDPAY KGQILTMANPIIGNGGAPDTTALDELGLSKYLESN GIKVSGLLVLDYSKDYNHWLATKSLGQWLQEEK VPAIYGVDTRMLTKIIRDKGTMLGKIEFEGQPVDF VDPNKQNLIAEVSTKDV KVYGKGNPTKVVAVDC GIKNNVIRLLVKGAEVHLVPWNHDFTKMEYDGI LIAGGPGNPALAEPLIQ NVRKILESDRKEPLFGISTGNLITGLAAGAKTYKM SMANRGQNQPVLNITNKQAFITAQNHGYALDNTL PAGWKPLFVNVDQTN EGIMHESKPPFAVQFHPE VTPGPIDTEYLFDSFFSLIKKGGATTITSVLPKPALV ASRVEVSKVLILSGGGLSIGQAGEFDYSGSQAVKA MKEENVKTVLMNPNIASVQTNEVGLKQADTVYFL PITPQFVTEVIKAEQPDGLILGMGGQTALNCGVEL FKRGVLKEYGVKVLGTSVESIMATEDRQLFSDKL NEINEKIAPSFIVESIEDALKAADTIGYPMIRSAY ALGGLGSGICPNRETLM DLSTKAFAMTNQILVEK SVTGWKEIEYEVVRDAD DNCVTVCNMENVNDAMGVHTGDSVVVAPAQTLSEN AEFQMLRRTSINVVRHL GIVGECNIQFALHPTSM YCIIEVNARLSRSSALASKATGYPLAFIAKIALGI PLPEIKNVVSGKTSACFEP SLDYMTVTKIPRWDLDR FHGTSSRIGSSMKSVGEV MAIGRTFEESFQKALRM CHPSIEGFTPRLPMNKE WPSNLDLRKELSEPSSTR IYALAKAIDDNMSLDEIE KLTYIDKWFLYKMRDIL	Hepatic cells	 (50 mol %) DSPC (10 mol %) Cholesterol (38.5% mol %) PEG-DMG (1.5%) OR MC3 (50 mol %) DSPC (10 mol %) Cholesterol (38.5% mol %) PEG-DMG (1.5%)

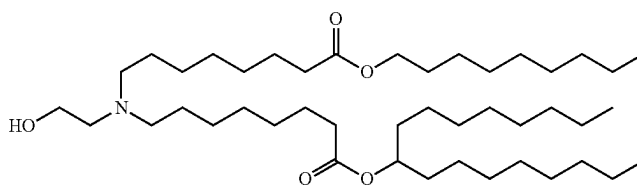
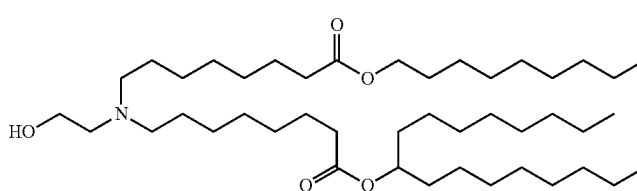
-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	NMEKTLKGLNSESMTTE TLKRAKEIGFSDKQISKC LGLTEAQTRELRKKNI HPWVKQIDTLAAEYPSV TNYLYVTYNGQEHDVN FDDHGMMLVLCGPGYHI GSSVEFDWC AVSSIRTLR QLGKKT VVVNCNPETVS TDFDECDKLYFEELSLE ILDIYHQBACGGCIISVG GQIPNNLAVPLYKNGVK IMGTSPLQIDRAEDRSIFS AVLDELKVAQAPWKAV NTLNEALEFAKSVDYPC LLRPSYVLSGSAMNVVF SEDEMKKFLEEATRVSQ EHPVVLTKFVEGAREVE MDAVGKDGRVISHAISE HVEDAGVHSGDATLML PTQTISQGAIEKVKDATR KIAKAF AISGPFNVQFLV KGNDVL VIECNLRASRS FPFVSKTLGVDFIDVATK VMIGENVDEKHLPTLDH PIIPADYVAIKAPMFSWP RLRDADPILRCEMASTG EVACFGEGIHTAFLKAM LSTGFKIPQKGILIGIQQS FRPRFLGVAEQLHNEGF KLFATEATSDWLNANN VPATPVAWPSQEGQNPS LSSIRKLIRDGSIDL VINL PNNNTKFVHDNYVIRRT AVDSGIPLLTNFQVTKLF AEAVQKSRKVDKSLFH YRQYSAGKAA		
Cas9	MKRNYILGLDIGITSVGY GIIDYETRDVIDAGVRLF KEANVENNEGRRSKRG ARRLKRRRRHRIQRVKK LLFDYNLLTDHSELSGIN PYEARVKGLS QKLSEEE FSAALLHLAKRRGVHNV NEVEEDTGNELSTKEQIS RNSKALEEKYVAELQLE RLKKDGEVRGSINRFTK SDYVKEAKQLKVQKA YHQLDQSFIDTYIDLLET RRTYYEGPGEGSPFGWK DIKEWYEMLMGHCTYF PEELRSVKYAYNADLYN ALNDLNNLVITRDENEK LEYYEKFQIENVFQKKK KPTLKQIAKEILVNEEDI KGYRVTS TGKPEFTNLK VYHDIKDITARKEIENA ELLDQIAKILTYQSSEDI QEELTNLNSLTQEEIEQI SNLKGYTGTHNLSLKAI NLILDELWHTNDNQIAIF NRLKLVPKKVDLSQOKE IPTTLVDDFILSPVVKRSF IQSIKVINAIHKYGLPND IHELAREKNSKDAQKMI NEMQKRNRQTNERIEEII RTTGKENAKYLIIEKIKLH DMQEGKCLYSLEAIPLE DLLNPNFNYEVDHIIPRS VSFDNSFNKVLVKQEE NSKKGNRTPFQYLSSSDS KISYETFKKHILNLAKGK GRISKTKEYLLEERDIN RFSVQKDFINRNLVDTR YATRGLMNLLRSYFRVN NLDVKVKSINGGFTSFLR RKWKFKKERNKGYKHH	Immune cells	 (50 mol %) DSPE (10 mol %) Beta-sitosterol (28.5% mol %) Cholesterol (10 mol %) PEG DMG (1.5 mol %)

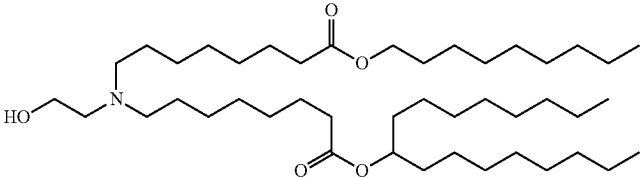
-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	AEDALIIANADFIFKEWK KLDKAKKVMENQMFEE KQAESMPEIETEQEYKEI FITPHQIKHIKDFKDYKY SHRVDKKPNRELINDTL YSTRKDDKGNTLIVNNL NGLYDKDNDKLIKLINK SPEKLLMYHHDPPQTYQK LKLIMEQYGDEKNPLYK YYEETGNYLTKYSKKDN GPVIKKIKYYGNKLNH LDITDDYPNSRNKVVKL SLKPYPYRFDVYLDNGVYK FVTVKNLDVIKKENYYE VNSKCYEEAKKLLKISN QAEFIASFYNNDLIKING ELYRVIGVNNDLLNRIEV NMIDITYREYLENMNDK RPPRIIKTLASKTQSIKKY STDILGNLYEVKSKKHP QIIKKG		
ADAMTS13	AAGGILHLELLVAVGPD VFQAHQEDTERYVLTNL NIGAEELLRDPGLGAQFRV HLVKMVLTEPEGAPNIT ANLTSSLLSVCWSQTIN PEDDTPDGHADLVLYIT RFDLELPDGNRQVRGVT QLGGACSPTWSCLTEDT GFDLGVTIAHEIGHSFGL EHDGAPGSGCGPSGHV MASDGAAPRAGLAWSP CSRRQLLSLLSAGRARC VWDPPRPQPGSAGHPPD AQPGLYYSANEQCRVAF GPKAVACTFAREHLDL CQALSCHTDPLDQSSCS RLLVPLLDGTECGVEKW CSKGRCRSLVELTPIAV HGRWSSWGPRSPCSRSC GGGVVTRRRQCNNPRPA FGGRACVGADLQAEMC NTQACEKTQLEFMSQQC ARTDQGQLRSPGGASF YHWGAAPHSQGDALC RHMCRAGESFIMKRGD SFLDGTRCMPSPGPREDG TSLSLCVSGSCRTFGCDG RMDSQQVWDRQCVCQG DNSTCSPRKGSFTAGRA REYVFTLTVTPNLTSVYI ANHRPLFTHLAVRIGGR YVAVGKMSISPNNTTYP LLEDGRVEYRVALTEDR LPRLEEIRIWGPLQEDAD IQVYRRYGEEYGNLTRP DITFTYFQPKPRQAWVW AAVRGPCSVSCGAGLR WVNYSCLDQARKELVE TVQCQGSQQPPAWPEAC VLEPCPPYWAVGDFGPC SASCGGGLRERPVRCVE AQGSLLKTLPPARCRA AQPPAVALTCNPQPCP ARWEVSEPSCTSAGGA GLALENETCVPGADGLE APVTEGPGSVDEKLPA EPCVGMSCPPGWGHLD ATSAGEKAPSPWGSIRT GAQAAHVWTPAAGSCS VSCGRGLMELRFLCMDS ALRVPVQEELCGLASKP GSRREVCAVPCPARW QYKLAACSVSCGRGVV RRILYCARAHGEDDGEI LLDTQCQGLPRPEPQEA	Hepatic cells	 (50 mol %) DSPC (10 mol %) Cholesterol (38.5 mol %) PEG-DMG (1.5%) OR MC3 (50 mol %) DSPC (10 mol %) Cholesterol (38.5 mol %) PEG-DMG (1.5%)

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	CSLEPCPPRWKVMSLGP CSASCLGTARRSVACV QLDQGQDVEVDEAACA ALVRPEASVPCLADCTY RWHVGTWMECSVSCGD GIQRRDTCCLGPAQAP VPADFCQHLPKPVTVRG CWAGPCVGQGTPLVPH EEAAPGRTTATPAGAS LEWSQARGLLFSPAPQP RRLPLPGQENSVQSSAC GRQHLEPTGTIDMRGPG QADCAVAIGRPLGEVVT LRVLESSLNCAGDMLL LWGRLTWRKMCRLLD MTFSSKTNTL VVRQRCG RPPGGVLLRYGSQLAPE TFYRECDMQLFGPWGEI VSPSLSPATSNAGGCRFL INVAPHARIAHALATNM GAGTEGANASYILIRDTH SLRTTAFHGQQVLYWES ESSQAEMEFSEGLKAQ ASLRGQYWTLSWVPE MQDPQSWKGKEGT		
FOXP3	MPNPRPGKPSAPSLALGP SPGASPSWRAAPKASDL LGARGPGGTFQGRDLRG GAHASSSSLNPMPPSQL QLPTLPLVMVAPSGARL GPLPHLQALLQDRPHFM HQLSTVDAHARTPVLQV HPLESPAMISLTPPTTAT GVFSLKARPLPPGINVA SLEWVSREPALLCTFPNP SAPRKDSTLSAVPQSSYP LLANGVCKWPGCEKVF EEPEDFLKHCQADHLLD EKGRAQCLLQREMVQSL EQQLVLEKEKLSAMQA HLAGKMALTKASSVASS DKGSCCIVAAGSQGPVV PAWSGPREADPSLFAVR RHLWGSHGNSTFPEFLH NMDYFKFHNMRPPFTY ATLIRWAILEAPEKQRTL NEIYHWFTRMFAFFRNH PATWKNAIRHNLSLHKC FVRVESEKGAVWTVDEL EFRKKRSQRPSRCSNPTP GP	Immune cells	 (50 mol %) DSPC (10 mol %) Beta-sitosterol (28.5 mol %) Cholesterol (10 mol %) PEG DMG (1.5 mol %)
IL-10	SPGQGTQSENSCTHFGP NLPNMLRDLRDAFSRVK TFFQMKGDLNLLKES LLEDFKGYLGQCALSEM IQFYLEEVMPQAENQDP DIKAHVNSLGENLKTLLR LRLRRCHRFLPCENKSK AVEQVKNAFNKLQEKGI YKAMSEFDIFINYIEAYM TMKIRN	Immune cells	 (50 mol %) DSPC (10 mol %) Beta-sitosterol (28.5 mol %) Cholesterol (10 mol %) PEG DMG (1.5 mol %)

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
IL-2	APTSSSTKKTQLQLEHLL LDLQMILNGINNYKNPK LTRMLTFKFYMPKKATE LKHLQCLEEELKPLEEVL NLAQSKNFHLRPRDLISN INVIVLELKGSETTFMCE YADETATIVEFLNRWITF CQSIISTLT	Immune cells	 (50 mol %) DSPC (10 mol %) Beta-sitosterol (28.5% mol %) Cholesterol (10 mol %) PEG DMG (1.5 mol %)
BCSP31 (BCSPBRUME)	MKFGSKIRRLAVAAVAG AIALGASFAVAQAPTFFR IGTGGTAGTYYPIGGLIA NAISGAGEKGVPLVAT AVSSNGSVANINAIKSGA LESGFTQSDVAYWAYN GTGLYDGGKGVEDLRL ATLYPETIHIVARKDANI KSVADLKGKRVSLDEPG SGTIVDARIVLEAYGLTE DDIKAE HLKPGPAGERLKDGALE AYFFVGGYPTGAISELAI SNGISLVPISGPEADKILE KYSFFSKDVVPAGAYKD VAETPTLAVAAQWVTS AKQPDDLIIYNITKVLWN EDTRKALDAGHAKGKLI KLDSATSSLGIPLHPGAE RFYKEAGVLIK	Immune cells	
MOMP (MOMP6_LP6)	MKKLLKSALLFAATGSA LSLQALPVGNPAPESLLI DGTMWEGASGDPCDPC ATWCDASIRAGYYGDY VFDRVLKVDVNKTFSG MAATPTQATGNASNTN QPEANGRPNIAYGRHMQ DAEWFSNAAFLALNIWD RFDIFCTLGASNGYFKAS SAAFNLVGLIGFSAASSIS TDLPMQLPNVGITQGVV EFYTDTSFSWSVGARGA LWECGCATLGAEFQYA QSNPKIEMLNVTSSPAQF VIHPRGYKGASSNFPLP ITAGTTEATDTKSATIKY HEWQVGLALSYRLNML VPYIGVNWSRATFDADT IRIAQPKLKSEILNITTWN PSLIGSTTALPNNSGKDV LSDVLQIASIQINKM KSRKACGVAVGATLIDA DKWSITGEARLINERAA HMNAQFRF	Immune cell	
FomA	MKKLALVLGLLLVVG VASAKEVMPAPTPAPEK VVEYVEKPVIVYRDREV APAWRPNGSVDVQYRW YGEVEKKNPKDDKDEN WATGKVNAGRLQTLTK VNFTEKQTLVTRNRHH TLNDT DANNKKSNGAADYRL RHFYNFGKLGSSKVNAT SRVEFKQKTNDGEKSLG ASVLEDFADYIYSNNFFK VDKLGLRPGYKYVWKG HGNGEETPTIVHNEYHL AFESDFTLPFNALNLEY	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	DLSYNRYREKFETTDGL KKAEWYGELTAVLSNY TPLYKAGAFELGENAEG GYDTYNMHQYKRIGGE DGTSVDRRDYELYLEPT LQVSYKPTDFVKLYAAA GADYRNRITGESEVKRW RWQP TASAGMKVTF		
MymA	MNQHFDVLIIGAGLSGIG TACHVTAEFDPKTIALLE RRERLGGTWDLFRYPGV RSDSDMFTFGYKFRPWR DVKVLADGASIRQYIAD TATEFGVDEKIHVGLKV NTAEWSSRQCRWTVAG VHEATGETRTYTCDYLLS CTGYNYDAGYLPDFPG VHRFGGRCVHPQHWPE DLDYSGKKVVVIGSGAT AVTLVPAMAGSNPGSAA HVTMLQRSPSYIFSLPAV DKISEVLGRFLPDRWVY EFGRRRNIAIQKLYQAC RRWPKLMRRLLLWEVR RRLGRSVDMSNFTPNYL PWDERLCAV PNGDLFKTLASGAASVV TDQIETFEKGILCKSGR EIEADIIIVTATGLNIQML GGMRLIVDGAEYQLPEK MTYKGVLLNAPNLAWI IGYTNASWTLKSDIAGA YLCRLLRHADNGYTV ATPRDAQDCALDVGMF DQLNSGYVKRGQDIMPR QGSKHPWR VLMHYEKD AKILLEDPIDDGVLFHFAA AAQDHAAA	Immune cell	
ESAT6	MTEQQWNFAGIEAAAS AIQGNVTSIHSLDEGKQ SLTKLAAWGGSGSEAY QGVQQKWDATATELNN ALQNLARTISEAGQAMA STEGNVTGMFA	Immune cell	
PorB	MKKSLIALTLAALPVAA MADVTLYGTIKAGVETY RFVAHNGAQASGVETAT EIADLGSKIGFKGQEDLG NGLKAIWQLEQKAYVS GTNTGWGNRQSFILKG GFGKVRVGRVNSVLKDT GGFNPWEGKSEYLSLSNI ARPEERPISVRYDSPEFA GFSGSVQYVPNDNSGEN KSESYHAGFNYKNSGFF VQYAGSYKRHNYYTTEK HQIHRLVGGYDHDALY ASVAVQQQDAKLAWPD DNSHNSQTEVATTVAYR FGNVTPRVSYAHGFKGS VYEANHDNTYDQVVVG AEYDFSKRTSALVSAGW LQEGKGA	Immune cell	
PVL (Panton Valentine- leukocidin)	FVGYPYSQNPRDYFVP DNELPPLVHSGFNPSFIA TVSHEKSGGDTSEFEITY GRNMDVTHATRRTTTHY GNSYLEGSRIHNAFVNR NYTVKYEVDNWKTHEIK VKGHN	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
Porin	EVKLSGDARMGVMYNG DDWNFSSRSRVLFMSG TTDSGLEFGASFKAHES VGAETGEDGTVFLSGAF GKIEMGDALGASEALFG DLYEVGYTDLDDRGGN DIPYLTGDERLTAEDNPV LLYTSAGAFSVAASMS DGKVGETSEDDAQEMA VAAAYTFGNYTVGLGY EKIDSPDTALMADMEQL ELAAIAKFGATNVKAYY ADGELDRDFARAVFDLT PVAAAATAVDHKAYGL SVDSTFGATTGGYVQV LDIDTIDDVITYYGLGAS YDLGGGASIVGGIADND LPNSDMVADLGVKFKF	Immune cell	
OmpA	MKKTAAIAVALAGFAT VAQAAPKDNTWYTGA LGWSQYHDTGFINNNGP THENQLGAGAFGGYQV NPYVGFEMGYDWLGRM PYKGSVENGAYKAQGV QLTAKLGYPTDDLDIYT RLGGMVWRADTKSNVY GKNHDTGVSPVFAGGVE YAITPEIATRLEYQWTNN IGDAHTIGTRPDNGMLS GVSYRFGQGEAAPVVAP APAPAPEVQTKHFTLKS DVLNFNFKATLKPEGQA ALDQLYSQSLNLDPKDG SVVVLGYTDRIGSDAYN QGLSERRAQSVVDYLIS KGIPADKISARGM GESNPVTGNTCDNVKQR AALIDCLAPDRRVEIEVK GIKDVVTQPQA	Immune cell	
MOMP	AGVATATGTKSATINYH EWQVGASLSYRLNSLVP YIGVQWSRATEDADNIRI AQPKLPTAVLNLTAWNP SLLG NATALSTTDSFSDF	Immune cell	
PepO	MTTYQDDFYQAVNGKW AETAVIPDDKPRTGGFSD LADEIEALMLDITDAWL AGENIPDDAILKNFVKFH RLVADYAKRDEVGVSP LPLIEEYQSLKSFSEFVA NIAKYELAGLPNEFPFSV APDFMNAQLNVLWAEA PSILLPDTTTYEEGNEKA EELRGHWRSQEKLLPQF GFSTEEIKDLLDKVIELD KQLAKYVLSREEGSEYA KLYHPYVWADFKKLAP ELPLDSIFEKILGQVPDK VIVPEERFWTEFAATYY EANWDLLKANLIVDAA NAYNAYLTDDIRVESGA YSRALSGTPQAMDKQK AAFYLAQGPFQALGLW YAGQKFSPEAKADVESK VARMIEVYKSRLTADW LAPATREKAITKLNVTTP HIGYPEKLPETY AKKVID ESLSLVENAQNLAKITIA HTWSKWNKPVDRSEWH MPAHLVNAYYDPQQNQ IVFPAILQEPFYSLDQSS SANYGGIGAVIAHEISHA FDTNGASFDEHGSLNDW	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	WTQEDY AAFKERTDKIV AQFDGLES HGAKVNGK LTVSENVADLGGVACAL EAAQSEEDFSARDFFINF ATIWRMKAREEYMQML ASIDVHAPGELRTNVTLT NFD AFHETFDIKEGDAM WRAPKDRV IIW		
OmpU	MNKTLIALAVSAAAVAT GAYADGINQSGDKAGST VYS AKGTSLEVGGRAEA RLSLKD GKAQDN S RVL NFLGKAEINDSLYGVGF YEGEFTTNDQ GKNASNN SLDNRYTYAGIGGTYGE VTYGKNDGALGVITDFT DIMS YHGNTAAEKIAVA DRVDNMLAYKGQFGDL GVKASYRFADRNAVDA MGNVVTETNA AKYS DN GEDGYSLSAIYTFGDTGF NVGAGYADQDDQNEY MLAASYRMENLYFAGL FTDGELAKDVDTGYEL AAGYKLGQAAFTATYN NAETAKETSADNF AIDA TYYFKPNFRSYISYQFNL LDS DKVGVK VASEDE LAI GLRYDF	Immune cell	
Lamazine synthase	MKGGAGVPDLPSLDASG VRLAIVASSWHGKICDA LLDGARKVAAGCGLDD PTVVRVLGAIEIPVVAQE LARNHDAVVALGVVIRG QTPHFDYVCDAVTQGLT RVSLDSS TPIANGVLT TN TEEQALDRAGLPTS AED KGAQATVAALATALT LR ELRAHS	Immune cell	
Omp16	MKKLT KVLLVAGSVAV LAACGSSKKDESAGQMF GGYSVQDLQQRYNTVY FGFDKYNIEGEYVQILDA HAAFLNATPATKVVVEG NTDERGTPEYNIALGQR RADAVKH YLSAKGVQA GOVSTVSYGEEKPAVLG HDEAAYS KNRRAVLAY	Immune cell	
Omp19	MGISKASLLSLAAAGIVL AGCQSSRLGNLDNVSP PPAPVNAVPA GTVQKG NLDSPTQFPNAPSTDMS AQSGTQVASLPPASAPD LTPGAVAGVWNASLGG QSCKIATPQTKYGQGYR AGPLRCPGELANLASWA VNGKQLVLYDANGGT V ASLYSSGQGRFDGQTTG GQAVTL SR	Immune cell	
CobT	MQILADLLNTIP AIDSTA MSRAQRHIDGLLKP VGS LGKLEVLAIQLAGMPGL NGIPHVGGKAVLVMCA DHGVWEEGVAISPKEVT AIQAENMTRGTTGVCVL AEQAGANVHVIVDVGIDT AEPPIGLINMRVARGSGN IASAPAMSRRQAEKLLL DVICYTQELAKNGVTLF GVGELGMANTTPAAAI V STITGRDPEEVVGIGANL	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	PTDKLANKIDVVRRAITL NQPNPQDGVVDVLAKVG GFDLVGIAGVMLGAASC GLPVLLDGFLSYAAALA ACQMSPAIPYLPISHLS AEKGARIALSHLGLEPYL NMEMRLGEGSGAALAM PIEAACAIYNNMGELAA SNIVLPGNTTSDLNS		
RpfE	MKNARTTLIAAAIAGTL VTTSPAGIANADDAGLD PNAAAGPDVAVGFDPNLP PAPDAAPVDTPPAPEDA GFDPNLPPPLAPDFLSPP AEEAPPVPVAVSVNWD AIAQCESGGNWSINTGN GYYGGLRFTAGTWRAN GGSGSAANASREEQIRV AENVLRSQGIRAWPVCG RRG	Immune cell	
Rv0652	MAKLSTDELLDAFKEMT LLELSDFVKKFEETFEVT AAAPVAVAAAGAAPAG AAVEAAEEQSEFDVILE AAGDKKIGVIKVVREIVS GLGLKEAKDLVDGAPKP LLEKVAKEAADEAKAK LEAAGATVTVK	Immune cell	
HBHA	MAENSNIDDIKAPLLAA LGAADLALATVNELITN LRERAETRTDTRSRVEE SRARLTKLQEDLPEQLTE LREKFTAELRKAAEGY LEAATSRYNELVERGEA ALERLSQQSFEEVSAR AEGYVDQAVELTQEAL GTVASQTRAVGERAAKL VGIELPKKAAPAKKAAP AKKAAPAKKAAAKKAP AKKAAAKKVTQK	Immune cell	
NhhA	MNKIYRIIWNALNAWV AVSELTRNHTKRASATV ATAVLATLLFATVQAST TDDDDLYLEPVQRTAVV LSFRSDKEGTGEKEVTE DSNWGVYFDKKGVLT GTITLKAGDNLKIKQNT NENTNASSFTYSLKKDL TDLTSVGTEKLSFSANSN KVNITSDTKGLNFAKKT AETNGDTTVHLNGIGST LTDILLNTGATTNVIND NVTDDDEKKRAASVKDV LNAGWNIKGVPKGTAS DNVDFVRTYDTVEFLSA DTKTITVNVESKDNGKR TEVKIGAKTSVIKEKDG KLVTGKDKGENDSSTDK GEGLVTAKEVIDAVNKA GWRMKTTTANGQTGQA DKFETVTSGTNVTFASG KGTATVSKDDQGNITV MYDVNVGDALNVNQLQ NSGWNLDSKAVAGSSG KVISGNVSPSKGKMDT VNINAGNNIEITRNGKNI DIATSMTPQFSSVSLGAG ADAPTLSVDDEGALNVG SKDANKPVRIITNVAPGV KEGDVTNVAQLKGVAQ NLNNHIDNVNARAGI AQAIATAGLVQAYLPGK	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	SMMAIGGGTYRGEAGY AIGYSSISDGGNWIIGT ASGNSRGHFGASASVGY QW		
DnaJ	MAKQDYIEILGVSKTAE EREIRKAYKRLAMKYHP DRNQGDKEAEAKFKEIK EAYEVLTDSSQKRAAYD QYGHAAFEQGGMGGGG FGGGADFSDFGDFVFDI FGGGRGRQRAARGADL RYNMELTLEEAVRGVTK EIRIPTLEECDVCHGSGA KPGTQPQTCPTCHGSGQ VQMRQGFFAVQQTCPH CQGRGTLIKDPCKNCHG HGRVERSCTLVSKIPAG VDTGDRIRLAGEGEAGE HGAPAGDLYVQVQVKQ HPIFEREGNNLYCEVPIN FAMAALGGEIEVPTLDG RVKLKVPGETQTGKLFR MRGKGKSVRGGAAQGD LLCRVVVETPVGLNERQ KQLLQELQESFGGPTGE HNSPRSKSFFDGVKKFF DDLTR	Immune cell	
Pneumolysin	MANKA VNDFILAMNYD KKKLLTHQGESIENRFIK EGNQLPDEFVIERKKRS LSTNTSDISVTATNDSRL YPGALL VVDETLLENNP TLLAVDRAPMTYSIDL GLASSDSFLQVEDPSNSS VRGAVNDLLAKWHQDY GOVNNVPARMQYELIT AHSMEQLKVKFGSDFEK TGNSLDIDFNSVHSGEK QIQIVNFKQIYYTVSVDA VKNPGDVFQDVTVTEDL KQRGISAERPLVYISSVA YGRQVYLKLETTSKSDE VEAAFEALIKGVKAPQ TEWKQILDNTEVKAVIL GGDPSSGARVVTGKVD MVEDLIQEGSRFTADHP GLPISYTTSLRDNVAT FQNSTDYVETKVTAYRN GDLLLDHSGAYVAQYYI TWDELSYDHQGEVLTP KAWDRNGQDLTAHFTT SIPLKGNVRNLSVKIREC TGLAWEWVRTVYEKTD LPLVRKRTISIWGTTLYP QVEDK VEND	Immune cell	
Flagellin (FLIC_ECOLI Flagellin OS = <i>Escherichia</i> <i>coli</i> (strain K12))	MAQVINTNSLSLITQNNI NKNQSALSSSIERLSSGL RINSAKDDAAGQAIANR FTSNIKGLTQAARNAND GISVAQTTEGALSEINNN LQVRRELTVQATTGINS ESDLSSIQDEIKSRLDEID RVSGQTQFNGVNVLA NGSMKIQVGANDNQTTT IDLKQIDAKTLGLDGFVS KNNDTVTTSAPVTAFGA TTTNNIKLTGTLSTEAA TDTGGTNPASIEGVYTD NGNDYYAKITGGDNDG KYYAVTVANDGTVTMA TGATANATVTDANTTK ATTITSGGTPVQIDNTAG SATANLGAVSLVKLQDS	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	KGNDTDTYALKDTNGN LYAADVNETTGAVSVKT ITYTDSSGAASSPTAVKL GGDDGKTEVVDIDGKTY DSADLNGGNIQTGLTAG GEALTAVANGKTTDPLK ALDDAIAVVDKFRSSLG AVQNRLDSAVINLNNTT TNLSEAQSRIQDADYAT EVSNNMSKAQIIQQAGNS VLAKANQVPQQVLSLLQ G		
IFN-alpha (IFNAI_HUMAN Interferon alpha- 1/13)	MASPFALLMVLVLSCK SSCSLGCDLPETHSLDNR RTLMLLAQMSRISPSCL MDRHDGFGPQEEFDGNQ FQKAPAISVLELIIQQIFN LFTTKDSSAAWDEDLDD KFCTELYQQLNDLEACV MQEERVGETPLMNADSI LAVKKYFRRTLYLTEK KYSPCAWEVVRAEIMRS LSLSTNLQERLRRKE	Immune cell	
IFN-gamma (IFNG_HUMAN Interferon gamma)	MKYTSYILAFQLCIVLGS LGICYQDPYVKEAENLK KYFNAGHSDVADNGTLF LGILKNWKEESDRKIMQ SQIVSFYFKLFKNFKDDQ SIQKSVEITIKEDMNVKFF NSNKKKRDDFEKLTNYS VTDLNVQRKAIHELIQV MAELSPAAGTKRKRISQ MLFRGRRASQ		
IL-2 (IL2_HUMAN Interleukin-2)	MYRMQLLSICIALSLALV TNSAPTSSSTKKTQLQLE HLLDLQMLNGINNYK NPKLTRMLTFKFMPPKK ATELKHLQCLEEELKPLE EVLNLAQSKNFHLRPRD LISNINVIVLELKGSETTF MCEYADETATIVEFLNR WITFCQSIISTLT	Immune cell	
Interleukin- 12p35 subunit	MWPPGSASQPPSPAAA TGLHPAARPVSLQCRLS MCPAR	Immune cell	
p40	MGKKQNRKTGNSKTQS ASPPPKERSSSPATEQSW MENDFDELREEGFRRSN YSELREDIQTGKEVENF EKNLEECITRISNTEKCL KELMELKTKTRELREEC RSLRSRCDQLEERVSA EDEMNMKREGKFREK RIKRNEQTLQEIWDYVK RPNLRLIGVPESDVENG KLENTLQDIIQENFPNLA RQANVQIQEIQRTPQRYS SRRATPRHIIVRFKIVEM KEKMLRAAREKGRVTL KGKPIRLTADLLAETLQ ARREWGPIFNILKGKNF QPRISYPAKLSFISEGEIK YFIDKQMLRDFVTTRPA LKELLKEALNMERNNRY QLLQNHAKM	Immune cell	
IL-15 (IL15_HUMAN Interleukin-15)	MRISKPHLRSISIQCYLCL LLNSHFLTEAGIHVFILG CFSAGLPKTEANWVNV SDLKKIEDLIQSMHIDAT LYTESDVHPSCKVTAMK	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
IL-18 (IL18_HUMAN Interleukin-18)	CFLLELQVISLESGDASIH DTVENLIILANNSLSSNG NVTESGCKECEEELEEKNI KEFLQSFVHIVQMFINIS MAAEPVEDNCINFVAM KFIDNTLYFLAEDDENLE SDYFGKLESKLSVIRNLN DQVLFIDQGNRPLFEDM TSDSCRDNAPRTIFHSM YKDSQPRGMAVTISVKC EKISTLSCENKIISFKEMN PPDNIKDTKSDIIFQRSV PGHDNKMQFESSSYEGY FLACEKERDLFKLILKKE DELGDRSIMFTVQNED		
IL-21	MRSSPGNMERIVICLMVI FLGTL VHKSSSQGQDRH MIRMRLIDIVDQLKNY VNDLVPEFLPAPEDVET NCEWSAFSCFQKAQLKS ANTGNNERIINVSIKCLK RKPPSTNAGRRQKHRLT CPSCDSEYKPPKEFLER FKSLQKMIHQHLSRT HGSEDS	Immune cell	
GM-CSF	MWLQSLLLGTVACSIS APARSPSPSTQPWEHVN AIQEARRLLNLSRDTAA EMNETVEVISEMFDLQE PTCLQTRLELYKQGLRG SLTKLKGPLTMMASHYK QHCPPTPETSCATQIITFE FKENLKDFLL VIPFDCW EPVQE	Immune cell	
IL -1beta	MAEVPPELASEMMAYYS GNEDDLFFEADGPKQM KCSFQDL DLCPLDGGIQL RISDHHYSKGFRAASV VVAMDCLRKMLVPCPQ TFQENDLSTFFPFIFEEPI FFDTWDNEAYVHDAPV RSLNCTLRDSQQKSLVM SGPYELKALHLQGQDME QQVVFMSFVQGEESND KIPVALGLKEKNLYLSC VLKDDKPTLQLESVDPK NYPKKKMEKRFVFNKIE INNKEFESAQFPNWIIS TSQAENMPVFLGGTKGG QDITDFTMQFVSS	Immune cell	
IL-6	MNSFSTSAFGPVAFLGL LLVLPAAFPAPVPPGEDS KDVAAPHRQPLTSSERID KQIRYILDGISALRKETC NKSNMCESSKEALAENN LNLPKMAEKDGCFCQSGF NEETCLVKIITGLLEFEV YLEYLQNRFESSEEQAR AVQMSTKVLILQLKKA KNLDAITPDPTTNASLL TKLQAQNQWLQDMTTH LILRSFKEFLQSSLRALR QM	Immune cell	
TNF-a	MSTESMIRDVELAEEL PKKTGGPQGSRRCLFLSL FSFLIVAGATTFLCLLHF GVIGPQREEFPRDLSLSP LAQAVRSSRTPSDKPV AHVVANPQAEGLQWL NRRANALLANGVELRD	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
	NQLVVPSEGLYLIYSQVL FKGQGCPSTHVLLHTIS RIAVSYQTKVNLLSAIKS PCQRETPEGAEAKPWYE PIYLGGVFQLEKGDRLS AEINRPDYLDFAESGQV YFGIHAL		
IL-7	MFHVSFRYIFGLPPLILV LLPVASSDCDIEGKDGK QYESVLMVSIQLLDSM KEIGSNCLNNEFNFFKRH ICDANKEGMFLFRAARK LRQFLKMNSTGDFDLHL LKVSEGTILLNCTGQV KGRKPAALGEAQPTKSL EENKSLKEQKKLNDLCF LKRLLQEIKTCWNKILM GTKEH	Immune cell	
IL-17a	MTPGKTSLVSLLLLSLE AIVKAGITIPRNPGPCNSE DKNFPRTVMVNLNIHNR NTNTNPKRSSDYYNRST SPWNLHRNEDPERYPSVI WEAKCRHLGCINADGN VDYHMNSVPIQEIIVL RREPPHCPNSFRLEKILV SVGCTCVTPIVHHVA	Immune cell	
Flt3-ligand	MTVLAPAWSPTTYLLLL LLSSGLSGTQDCSFQHS PISSDFAVKIRELSDYLL QDYPVTVASNLDDEELC GGLWRLVLAQRWMERL KTVAGSKMQGLLERVN TEIHFVTKCAFPQPPSCL RFVQTNISRLLQETSEQL VALKWPWTRQNFSRCLE LQCQPDSSSTLPPWSPRP LEATAPTAPQPPLLLLL LPVGLLLAAAWCLHW QRTRRRTPRPGEQVPPVP SPQDLLLVEH	Immune cell	
anti-CTLA4 (ipilumimab)	QVQLVESGGGVVQPGRS LRLSCAASGFTFSSYTM HWVRQAPGKGLEWVTF ISYDGNKYYADSVKGR FTISRDNSKNTLYLQMN SLRAEDTAIYYCARTGW LGPFDYWGQGTLVTVSS AS TKGPSVFPLAPSSKSTSG GTAALGCLVKDYFPEPV TVSWNSGALTSGVHTFP AVLQSSGLYSLSSVVTVP SSSLGTQTYICNVNHKPS NTKVDKRVEPKSCDKTH TCPPCPAPELLGGPSVFL FPPKPKDTLMISRTPEVT CVVVDVSHEDPEVKEN WYVDGVEVHNAKTKPR EEQYNST YRVVSVLTVLHQDWLN GKEYKCKVSNKALPAPI EKTISKAKGQPREPQVY TLPPSRDELTKNQVSLTC LVKGFYPSDIAVEWESN GQPENNYKTTTPVLDSQ GSFFLYSKLTVDKSRWQ QGNVFSCSVMHEALHN HYTQKSLSLSPGK	Immune cell	

-continued

Payload	Sequence	Target cell/organ	Preferred delivery formulation
anti-PDI (nivo)	QVQLVESGGGVVQPGRS LRLDCKASGITFSNSGM HWVRQAPGKGLEWVAV IWYDGSKRYADSVKG RFTISRDN SKNTLFLQMN SLRAEDTAVYYCATNDD YWGQGTILTVSSASTKG PSVFPLAPCSRSTSESTA ALGCLVKDYFPEPVTVS WNSGALTSGVHTFPAVL QSSGLYSLSSVTVPSSS LGTKITYTCNVDHKPSNT KVDKRVESKYGPCCPPC PAPEFLGGPSVFLFPPKP KDTLMISRTPETVCVVV DVSQEDPEVQFNWYVD GVEVHNAKTKPREEQFN STYRVVSVLTVLHQDWL NGKEYKCKVSNKGLPSS IEKTISKAKGQPREPQVY TLPSPQEEMTKNQVSLT CLVKGFYPSDIAVEWES NGQPENNYKTTPVLDSD DGSFFLYSRLTVDKSRW QEGNVFSCSVMHEALHN HYTQKSLSLSGK	Immune cell	
anti-41BB (utomilumab)	EVQLVQSGAEVKKPGES LRISCKGSGYSFSTYWIS WVRQMPGKGLEWMGKI YPGDSYTNYSFQGGQV TISADKSISTAYLQWSSL KASDTAMYYCARGYGIF DYWGQGTILTVSSASTK GPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTV SWNSGALTSGVHTFPAV LQSSGLYSLSSVTVPSSS NFGTQITYTCNVDHKPSN TKVDKTVKCCVECPP CPAPPVAGPSVFLFPPKP KDTLMISRTPETVCVVV DVSHEDPEVQFNWYVD GVEVHNAKTKPREEQFN STFRVSVLTVVHQDWL NGKEYKCKVSNKGLPAP IEKTISKTKGQPREPQVY TLPSPREEMTKNQVSLT CLVKGFYPSDIAVEWES NGQPENNYKTTPMLDS DGSFFLYSKLTVDKSRW QQGNVFSCSVMHEALH NHYTQKSLSLSPGK	Immune cell	

50

In some embodiments, the expression sequence encodes a therapeutic protein. In some embodiments, the expression sequence encodes a cytokine, e.g., IL-12p70, IL-15, IL-2, IL-18, IL-21, IFN- α , IFN- β , IL-10, TGF-beta, IL-4, or IL-35, or a functional fragment thereof. In some embodiments, the expression sequence encodes an immune check- point inhibitor. In some embodiments, the expression sequence encodes an agonist (e.g., a TNFR family member such as CD137L, OX40L, ICOSL, LIGHT, or CD70). In some embodiments, the expression sequence encodes a chimeric antigen receptor. In some embodiments, the expression sequence encodes an inhibitory receptor agonist (e.g., PDL1, PDL2, Galectin-9, VISTA, B7H4, or MHCII) or inhibitory receptor (e.g., PD1, CTLA4, TIGIT, LAG3, or TIM3). In some embodiments, the expression sequence encodes an inhibitory receptor antagonist. In some embodiments, the expression sequence encodes one or more TCR

55

60

65

chains (alpha and beta chains or gamma and delta chains). In some embodiments, the expression sequence encodes a secreted T cell or immune cell engager (e.g., a bispecific antibody such as BiTE, targeting, e.g., CD3, CD137, or CD28 and a tumor-expressed protein e.g., CD19, CD20, or BCMA etc.). In some embodiments, the expression sequence encodes a transcription factor (e.g., FOXP3, HELIOS, TOX1, or TOX2). In some embodiments, the expression sequence encodes an immunosuppressive enzyme (e.g., IDO or CD39/CD73). In some embodiments, the expression sequence encodes a GvHD (e.g., anti-HLA-A2 CAR-Tregs).

In some embodiments, a polynucleotide encodes a protein that is made up of subunits that are encoded by more than one gene. For example, the protein may be a heterodimer, wherein each chain or subunit of the protein is encoded by a separate gene. It is possible that more than one circRNA

molecule is delivered in the transfer vehicle and each circRNA encodes a separate subunit of the protein. Alternatively, a single circRNA may be engineered to encode more than one subunit. In certain embodiments, separate circRNA molecules encoding the individual subunits may be administered in separate transfer vehicles.

A. Antigen-Recognition Receptors

A. Chimeric Antigen Receptors (CARs)

Chimeric antigen receptors (CARs or CAR-Ts) are genetically-engineered receptors.

These engineered receptors may be inserted into and expressed by immune cells, including T cells via circular RNA as described herein. With a CAR, a single receptor may be programmed to both recognize a specific antigen and, when bound to that antigen, activate the immune cell to attack and destroy the cell bearing that antigen. When these antigens exist on tumor cells, an immune cell that expresses the CAR may target and kill the tumor cell. In some embodiments, the CAR encoded by the polynucleotide comprises (i) an antigen-binding molecule that specifically binds to a target antigen, (ii) a hinge domain, a transmembrane domain, and an intracellular domain, and (iii) an activating domain.

In some embodiments, an orientation of the CARs in accordance with the disclosure comprises an antigen binding domain (such as an scFv) in tandem with a costimulatory domain and an activating domain. The costimulatory domain may comprise one or more of an extracellular portion, a transmembrane portion, and an intracellular portion. In other embodiments, multiple costimulatory domains may be utilized in tandem.

i. Antigen Binding Domain

CARs may be engineered to bind to an antigen (such as a cell-surface antigen) by incorporating an antigen binding molecule that interacts with that targeted antigen. In some embodiments, the antigen binding molecule is an antibody fragment thereof, e.g., one or more single chain antibody fragment (scFv). An scFv is a single chain antibody fragment having the variable regions of the heavy and light chains of an antibody linked together. See U.S. Pat. Nos. 7,741,465, and 6,319,494 as well as Eshhar et al., *Cancer Immunol Immunotherapy* (1997) 45: 131-136. An scFv retains the parent antibody's ability to specifically interact with target antigen. scFvs are useful in chimeric antigen receptors because they may be engineered to be expressed as part of a single chain along with the other CAR components. Id. See also Krause et al., *J. Exp. Med.*, Volume 188, No. 4, 1998 (619-626); Finney et al., *Journal of Immunology*, 1998, 161: 2791-2797. It will be appreciated that the antigen binding molecule is typically contained within the extracellular portion of the CAR such that it is capable of recognizing and binding to the antigen of interest. Bispecific and multispecific CARs are contemplated within the scope of the invention, with specificity to more than one target of interest.

In some embodiments, the antigen binding molecule comprises a single chain, wherein the heavy chain variable region and the light chain variable region are connected by a linker. In some embodiments, the VH is located at the N terminus of the linker and the VL is located at the C terminus of the linker. In other embodiments, the VL is located at the N terminus of the linker and the VH is located at the C terminus of the linker. In some embodiments, the linker comprises at least about 5, at least about 8, at least about 10, at least about 13, at least about 15, at least about 18, at least about 20, at least about 25, at least about 30, at least about 35, at least about 40, at least about 45, at least about 50, at

least about 60, at least about 70, at least about 80, at least about 90, or at least about 100 amino acids.

In some embodiments, the antigen binding molecule comprises a nanobody. In some embodiments, the antigen binding molecule comprises a DARPIn. In some embodiments, the antigen binding molecule comprises an anticalin or other synthetic protein capable of specific binding to target protein.

In some embodiments, the CAR comprises an antigen binding domain specific for an antigen selected from the group CD19, CD123, CD22, CD30, CD171, CS-1, C-type lectin-like molecule-1, CD33, epidermal growth factor receptor variant III (EGFRvIII), ganglioside G2 (GD2), ganglioside GD3, TNF receptor family member B cell maturation (BCMA), Tn antigen ((Tn Ag) or (GalNAc-Ser/Thr)), prostate-specific membrane antigen (PSMA), Receptor tyrosine kinase-like orphan receptor 1 (ROR 1), Fms-Like Tyrosine Kinase 3 (FLT3), Tumor-associated glycoprotein 72 (TAG72), CD38, CD44v6, Carcinoembryonic antigen (CEA), Epithelial cell adhesion molecule (EP-CAM), B7H3 (CD276), KIT (CD117), Interleukin-13 receptor subunit alpha-2, mesothelin, Interleukin II receptor alpha (IL-1 IRa), prostate stem cell antigen (PSCA), Protease Serine 21, vascular endothelial growth factor receptor 2 (VEGFR2), Lewis(Y) antigen, CD24, Platelet-derived growth factor receptor beta (PDGFR-beta), Stage-specific embryonic antigen-4 (SSEA-4), CD20, Folate receptor alpha, HER2, HER3, Mucin 1, cell surface associated (MUC1), epidermal growth factor receptor (EGFR), neural cell adhesion molecule (NCAM), Prostase, prostatic acid phosphatase (PAP), elongation factor 2 mutated (ELF2M), Ephrin B2, fibroblast activation protein alpha (FAP), insulin-like growth factor 1 receptor (IGF-I receptor), carbonic anhydrase IX (CAIX), Proteasome (Prosome, Macropain) Subunit, Beta Type, 9 (LMP2), glycoprotein 100 (gp100), oncogene fusion protein consisting of breakpoint cluster region (BCR) and Abelson murine leukemia viral oncogene homolog 1 (Abl) (bcr-abl), tyrosinase, ephrin type-A receptor 2 (EphA2), Fucosyl GM1, sialyl Lewis adhesion molecule (sLe), ganglioside GM3, transglutaminase 5 (TGS5), high molecular weight-melanoma-associated antigen (HMWMAA), o-acetyl-GD2 ganglioside (OAcGD2), Folate receptor beta, tumor endothelial marker 1 (TEM1/CD248), tumor endothelial marker 7-related (TEM7R), claudin 6 (CLDN6), thyroid stimulating hormone receptor (TSHR), G protein-coupled receptor class C group 5, member D (GPRC5D), chromosome X open reading frame 61 (CX-ORF61), CD97, CD179a, anaplastic lymphoma kinase (ALK), Polysialic acid, placenta-specific 1 (PLAC1), hexa-saccharide portion of globoH glycosphingolipid (GloboH), mammary gland differentiation antigen (NY-BR-1), uroplakin 2 (UPK2), Hepatitis A virus cellular receptor 1 (HAVCR1), adrenoceptor beta 3 (ADRB3), pannexin 3 (PANX3), G protein-coupled receptor 20 (GPR20), lymphocyte antigen 6 complex, locus K 9 (LY6K), Olfactory receptor 51 E2(OR51E2), TCR Gamma Alternate Reading Frame Protein (TARP), Wilms tumor protein (WT1), Cancer/testis antigen 1 (NY-ESO-1), Cancer/testis antigen 2 (LAGE-1a), MAGE family members (including MAGE-A1, MAGE-A3 and MAGE-A4), ETS translocation-variant gene 6, located on chromosome 12p (ETV6-AML), sperm protein 17 (SPA17), X Antigen Family, Member 1A (XAGE1), angiopoietin-binding cell surface receptor 2 (Tie 2), melanoma cancer testis antigen-I (MAD-CT-1), melanoma cancer testis antigen-2 (MAD-CT-2), Fos-related antigen 1, tumor protein p53 (p53), p53 mutant, prostein, surviving, telomerase, prostate carcinoma tumor antigen-1, melanoma

antigen recognized by T cells 1, Rat sarcoma (Ras) mutant, human Telomerase reverse transcriptase (hTERT), sarcoma translocation breakpoints, melanoma inhibitor of apoptosis (ML-IAP), ERG (transmembrane protease, serine 2 (TMPRSS2) ETS fusion gene), N-Acetyl glucosaminyl-transferase V (NA17), paired box protein Pax-3 (PAX3), Androgen receptor, Cyclin B1, v-myc avian myelocytomatosis viral oncogene neuroblastoma derived homolog (MYCN), Ras Homolog Family Member C (RhoC), Tyrosinase-related protein 2 (TRP-2), Cytochrome P450 1B1 (CYP1B1), CCCTC-Binding Factor (Zinc Finger Protein)-Like, Squamous Cell Carcinoma Antigen Recognized By T Cells 3 (SART3), Paired box protein Pax-5 (PAX5), proacrosin binding protein sp32 (OY-TES1), lymphocyte-specific protein tyrosine kinase (LCK), A kinase anchor protein 4 (AKAP-4), synovial sarcoma, X breakpoint 2 (SSX2), Receptor for Advanced Glycation Endproducts (RAGE-I), renal ubiquitous 1 (RU1), renal ubiquitous 2 (RU2), legumain, human papilloma virus E6 (HPV E6), human papilloma virus E7 (HPV E7), intestinal carboxyl esterase, heat shock protein 70-2 mutated (mut hsp70-2), CD79a, CD79b, CD72, Leukocyte-associated immunoglobulin-like receptor 1 (LAIR1), Fc fragment of IgA receptor (FCAR or CD89), Leukocyte immunoglobulin-like receptor subfamily A member 2 (LILRA2), CD300 molecule-like family member f (CD300LF), C-type lectin domain family 12 member A (CLEC12A), bone marrow stromal cell antigen 2 (BST2), EGF-like module-containing mucin-like hormone receptor-like 2 (EMR2), lymphocyte antigen 75 (LY75), Glypican-3 (GPC3), Fc receptor-like 5 (FCRL5), MUC16, 5T4, 8H9, α v β 0 integrin, α v β 6 integrin, alphafetoprotein (AFP), B7-H6, ca-125, CA9, CD44, CD44v7/8, CD52, E-cadherin, EMA (epithelial membrane antigen), epithelial glycoprotein-2 (EGP-2), epithelial glycoprotein-40 (EGP-40), ErbB4, epithelial tumor antigen (ETA), folate binding protein (FBP), kinase insert domain receptor (KDR), k-light chain, L1 cell adhesion molecule, MUC18, NKG2D, oncofetal antigen (h5T4), tumor/testis-antigen 1B, GAGE, GAGE-1, BAGE, SCP-1, CTZ9, SAGE, CAGE, CT10, MART-1, immunoglobulin lambda-like polypeptide I (IG-LLI), Hepatitis B Surface Antigen Binding Protein (HBsAg), viral capsid antigen (VCA), early antigen (EA), EBV nuclear antigen (EBNA), HHV-6 p41 early antigen, HHV-6B U94 latent antigen, HHV-6B p98 late antigen, cytomegalovirus (CMV) antigen, large T antigen, small T antigen, adenovirus antigen, respiratory syncytial virus (RSV) antigen, haemagglutinin (HA), neuraminidase (NA), parainfluenza type 1 antigen, parainfluenza type 2 antigen, parainfluenza type 3 antigen, parainfluenza type 4 antigen, Human Metapneumovirus (HMPV) antigen, hepatitis C virus (HCV) core antigen, HIV p24 antigen, human T-cell lymphotropic virus (HTLV-1) antigen, Merkel cell polyoma virus small T antigen, Merkel cell polyoma virus large T antigen, Kaposi sarcoma-associated herpesvirus (KSHV) lytic nuclear antigen and KSHV latent nuclear antigen.

ii. Hinge/Spacer Domain

In some embodiments, a CAR of the instant disclosure comprises a hinge or spacer domain. In some embodiments, the hinge/spacer domain may comprise a truncated hinge/spacer domain (THD) the THD domain is a truncated version of a complete hinge/spacer domain ("CHD"). In some embodiments, an extracellular domain is from or derived from (e.g., comprises all or a fragment of) ErbB2, glycoporin A (GpA), CD2, CD3 delta, CD3 epsilon, CD3 gamma, CD4, CD7, CD8a, CD8/T CD1 1a (IT GAL), CD1 1b (IT GAM), CD1 1c (ITGAX), CD1 1d (IT GAD), CD18 (ITGB2), CD1 9 (B4), CD27 (TNFRSF7), CD28, CD28T,

CD29 (ITGB1), CD30 (TNFRSF8), CD40 (TNFRSF5), CD48 (SLAMF2), CD49a (ITGA1), CD49d (ITGA4), CD49f (ITGA6), CD66a (CEACAM1), CD66b (CEACAM8), CD66c (CEACAM6), CD66d (CEACAM3), CD66e (CEACAM5), CD69 (CLEC2), CD79A (B-cell antigen receptor complex-associated alpha chain), CD79B (B-cell antigen receptor complex-associated beta chain), CD84 (SLAMF5), CD96 (Tactile), CD100 (SEMA4D), CD103 (ITGAE), CD134 (OX40), CD137 (4-1BB), CD150 (SLAMF1), CD158A (KIR2DL1), CD158B1 (KIR2DL2), CD158B2 (KIR2DL3), CD158C (KIR3 DPI), CD158D (KIRDL4), CD158F1 (KIR2DL5A), CD158F2 (KIR2DL5B), CD158K (KIR3DL2), CD160 (BY55), CD162 (SELPLG), CD226 (DNAM1), CD229 (SLAMF3), CD244 (SLAMF4), CD247 (CD3-zeta), CD258 (LIGHT), CD268 (BAFFR), CD270 (TNFSF14), CD272 (BTLA), CD276 (B7-H3), CD279 (PD-1), CD314 (NKG2D), CD319 (SLAMF7), CD335 (NK-p46), CD336 (NK-p44), CD337 (NK-p30), CD352 (SLAMF6), CD353 (SLAMF8), CD355 (CRT AM), CD357 (TNFRSF18), inducible T cell co-stimulator (ICOS), LFA-I (CD1 1a/CD18), NKG2C, DAP-10, ICAM-1, NKp80 (KLRf1), IL-2R beta, IL-2R gamma, IL-7R alpha, LFA-1, SLAMF9, LAT, GADS (GrpL), SLP-76 (LCP2), PAG1/CBP, a CD83 ligand, Fc gamma receptor, MHC class 1 molecule, MHC class 2 molecule, a TNF receptor protein, an immunoglobulin protein, a cytokine receptor, an integrin, activating NK cell receptors, a Toll ligand receptor, and fragments or combinations thereof. A hinge or spacer domain may be derived either from a natural or from a synthetic source.

In some embodiments, a hinge or spacer domain is positioned between an antigen binding molecule (e.g., an scFv) and a transmembrane domain. In this orientation, the hinge/spacer domain provides distance between the antigen binding molecule and the surface of a cell membrane on which the CAR is expressed. In some embodiments, a hinge or spacer domain is from or derived from an immunoglobulin. In some embodiments, a hinge or spacer domain is selected from the hinge/spacer regions of IgG1, IgG2, IgG3, IgG4, IgA, IgD, IgE, IgM, or a fragment thereof. In some embodiments, a hinge or spacer domain comprises, is from, or is derived from the hinge/spacer region of CD8 alpha. In some embodiments, a hinge or spacer domain comprises, is from, or is derived from the hinge/spacer region of CD28. In some embodiments, a hinge or spacer domain comprises a fragment of the hinge/spacer region of CD8 alpha or a fragment of the hinge/spacer region of CD28, wherein the fragment is anything less than the whole hinge/spacer region. In some embodiments, the fragment of the CD8 alpha hinge/spacer region or the fragment of the CD28 hinge/spacer region comprises an amino acid sequence that excludes at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 amino acids at the N-terminus or C-Terminus, or both, of the CD8 alpha hinge/spacer region, or of the CD28 hinge/spacer region.

iii. Transmembrane Domain

The CAR of the present disclosure may further comprise a transmembrane domain and/or an intracellular signaling domain. The transmembrane domain may be designed to be fused to the extracellular domain of the CAR. It may similarly be fused to the intracellular domain of the CAR. In some embodiments, the transmembrane domain that naturally is associated with one of the domains in a CAR is used. In some instances, the transmembrane domain may be selected or modified (e.g., by an amino acid substitution) to

avoid binding of such domains to the transmembrane domains of the same or different surface membrane proteins to minimize interactions with other members of the receptor complex. The transmembrane domain may be derived either from a natural or from a synthetic source. Where the source is natural, the domain may be derived from any membrane-bound or transmembrane protein.

Transmembrane regions may be derived from (i.e. comprise) a receptor tyrosine kinase (e.g., ErbB2), glycoprotein A (GpA), 4-1BB/CD137, activating NK cell receptors, an immunoglobulin protein. B7-H3, BAFFR, BFAME (SEAMF8), BTEA, CD100 (SEMA4D), CD103, CD160 (BY55), CD18, CD19, CD19a, CD2, CD247, CD27, CD276 (B7-H3), CD28, CD29, CD3 delta, CD3 epsilon, CD3 gamma, CD30, CD4, CD40, CD49a, CD49D, CD49f, CD69, CD7, CD84, CD8alpha, CD8beta, CD96 (Tactile), CD1 1a, CD1 1b, CD1 1c, CD1 1d, CDS, CEACAM1, CRT AM, cytokine receptor, DAP-10, DNAM1 (CD226), Fc gamma receptor, GADS, GITR, HVEM (EIGHTR), IA4, ICAM-1, ICAM-1, Ig alpha (CD79a), IE-2R beta, IE-2R gamma, IE-7R alpha, inducible T cell costimulator (ICOS), integrins, ITGA4, ITGA4, ITGA6, ITGAD, ITGAE, ITGAE, ITGAM, ITGAX, ITGB2, ITGB7, ITGB1, KIRDS2, EAT, LFA-1, LFA-1, a ligand that specifically binds with CD83, LIGHT, LIGHT, LTBR, Ly9 (CD229), lymphocyte function-associated antigen-1 (LFA-1; CD1-1a/CD18), MHC class 1 molecule, NKG2C, NKG2D, NKp30, NKp44, NKp46, NKp80 (KLR1), OX-40, PAG/Cbp, programmed death-1 (PD-1), PSGL1, SELPLG (CD162), Signaling Lymphocytic Activation Molecules (SLAM proteins), SLAM (SLAMF1; CD150; IPO-3), SLAMF4 (CD244; 2B4), SLAMF6 (NTB-A; Lyl08), SLAMF7, SLP-76, TNF receptor proteins, TNFR2, TNFSF14, a Toll ligand receptor, TRANCE/RANKL, VLA1, or VLA-6, or a fragment, truncation, or a combination thereof.

In some embodiments, suitable intracellular signaling domain include, but are not limited to, activating Macrophage/Myeloid cell receptors CSFR1, MYD88, CD14, TIE2, TLR4, CR3, CD64, TREM2, DAP10, DAP12, CD169, DECTIN1, CD206, CD47, CD163, CD36, MARCO, TIM4, MERTK, F4/80, CD91, C1QR, LOX-1, CD68, SRA, BAI-1, ABCA7, CD36, CD31, Lactoferrin, or a fragment, truncation, or combination thereof.

In some embodiments, a receptor tyrosine kinase may be derived from (e.g., comprise) Insulin receptor (InsR), Insulin-like growth factor 1 receptor (IGF1R), Insulin receptor-related receptor (IRR), platelet derived growth factor receptor alpha (PDGFRa), platelet derived growth factor receptor beta (PDGFRb), KIT proto-oncogene receptor tyrosine kinase (Kit), colony stimulating factor 1 receptor (CSFR), fms related tyrosine kinase 3 (FLT3), fms related tyrosine kinase 1 (VEGFR-1), kinase insert domain receptor (VEGFR-2), fms related tyrosine kinase 4 (VEGFR-3), fibroblast growth factor receptor 1 (FGFR1), fibroblast growth factor receptor 2 (FGFR2), fibroblast growth factor receptor 3 (FGFR3), fibroblast growth factor receptor 4 (FGFR4), protein tyrosine kinase 7 (CCK4), neurotrophic receptor tyrosine kinase 1 (trkA), neurotrophic receptor tyrosine kinase 2 (trkB), neurotrophic receptor tyrosine kinase 3 (trkC), receptor tyrosine kinase like orphan receptor 1 (ROR1), receptor tyrosine kinase like orphan receptor 2 (ROR2), muscle associated receptor tyrosine kinase (MuSK), MET proto-oncogene, receptor tyrosine kinase (MET), macrophage stimulating 1 receptor (Ron), AXL receptor tyrosine kinase (Ax1), TYRO3 protein tyrosine kinase (Tyro3), MER proto-oncogene, tyrosine kinase (Mer), tyrosine kinase with immunoglobulin like and EGF

like domains 1 (TIE1), TEK receptor tyrosine kinase (TIE2), EPH receptor A1 (EphA1), EPH receptor A2 (EphA2), (EPH receptor A3) EphA3, EPH receptor A4 (EphA4), EPH receptor A5 (EphA5), EPH receptor A6 (EphA6), EPH receptor A7 (EphA7), EPH receptor A8 (EphA8), EPH receptor A10 (EphA10), EPH receptor B1 (EphB1), EPH receptor B2 (EphB2), EPH receptor B3 (EphB3), EPH receptor B4 (EphB4), EPH receptor B6 (EphB6), ret proto oncogene (Ret), receptor-like tyrosine kinase (RYK), discoidin domain receptor tyrosine kinase 1 (DDR1), discoidin domain receptor tyrosine kinase 2 (DDR2), c-ros oncogene 1, receptor tyrosine kinase (ROS), apoptosis associated tyrosine kinase (Lmr1), lemur tyrosine kinase 2 (Lmr2), lemur tyrosine kinase 3 (Lmr3), leukocyte receptor tyrosine kinase (LTK), ALK receptor tyrosine kinase (ALK), or serine/threonine/tyrosine kinase 1 (STYK1).

iv. Costimulatory Domain

In certain embodiments, the CAR comprises a costimulatory domain. In some embodiments, the costimulatory domain comprises 4-1BB (CD137), CD28, or both, and/or an intracellular T cell signaling domain. In a preferred embodiment, the costimulatory domain is human CD28, human 4-1BB, or both, and the intracellular T cell signaling domain is human CD3 zeta (ζ). 4-1BB, CD28, CD3 zeta may comprise less than the whole 4-1BB, CD28 or CD3 zeta, respectively. Chimeric antigen receptors may incorporate costimulatory (signaling) domains to increase their potency. See U.S. Pat. Nos. 7,741,465, and 6,319,494, as well as Krause et al. and Finney et al. (supra), Song et al., Blood 119:696-706 (2012); Kalos et al., Sci Transl. Med. 3:95 (2011); Porter et al., N. Engl. J. Med. 365:725-33 (2011), and Gross et al., Amur. Rev. Pharmacol. Toxicol. 56:59-83 (2016).

v. Intracellular Signalling Domain

The intracellular (signaling) domain of the engineered T cells disclosed herein may provide signaling to an activating domain, which then activates at least one of the normal effector functions of the immune cell. Effector function of a T cell, for example, may be cytolytic activity or helper activity including the secretion of cytokines.

In some embodiments, suitable intracellular signaling domain include (e.g., comprise), but are not limited to 4-1BB/CD137, activating NK cell receptors, an Immunoglobulin protein, B7-H3, BAFFR, BLAME (SLAMF8), BTLA, CD100 (SEMA4D), CD103, CD160 (BY55), CD18, CD19, CD 19a, CD2, CD247, CD27, CD276 (B7-H3), CD28, CD29, CD3 delta, CD3 epsilon, CD3 gamma, CD30, CD4, CD40, CD49a, CD49D, CD49f, CD69, CD7, CD84, CD8alpha, CD8beta, CD96 (Tactile), CD1 1a, CD1 1b, CD1 1c, CD1 1d, CDS, CEACAM1, CRT AM, cytokine receptor, DAP-10, DNAM1 (CD226), Fc gamma receptor, GADS, GITR, HVEM (LIGHTR), IA4, ICAM-1, Ig alpha (CD79a), IL-2R beta, IL-2R gamma, IL-7R alpha, inducible T cell costimulator (ICOS), integrins, ITGA4, ITGA6, ITGAD, ITGAE, ITGAL, ITGAM, ITGAX, ITGB2, ITGB7, ITGB1, KIRDS2, LAT, LFA-1, ligand that specifically binds with CD83, LIGHT, LTBR, Ly9 (CD229), Lyl08, lymphocyte function-associated antigen-1 (LFA-1; CD1-1a/CD18), MHC class 1 molecule, NKG2C, NKG2D, NKp30, NKp44, NKp46, NKp80 (KLR1), OX-40, PAG/Cbp, programmed death-1 (PD-1), PSGL1, SELPLG (CD162), Signaling Lymphocytic Activation Molecules (SLAM proteins), SLAM (SLAMF1; CD150; IPO-3), SLAMF4 (CD244; 2B4), SLAMF6 (NTB-A), SLAMF7, SLP-76, TNF receptor proteins, TNFR2, TNFSF14, a Toll ligand receptor, TRANCE/RANKL, VLA1, or VLA-6, or a fragment, truncation, or a combination thereof.

CD3 is an element of the T cell receptor on native T cells, and has been shown to be an important intracellular activating element in CARs. In some embodiments, the CD3 is CD3 zeta.

b. T-Cell Receptors (TCR)

TCRs are described using the International Immunogenetics (IMGT) TCR nomenclature, and links to the IMGT public database of TCR sequences. Native alpha-beta heterodimeric TCRs have an alpha chain and a beta chain. Broadly, each chain may comprise variable, joining and constant regions, and the beta chain also usually contains a short diversity region between the variable and joining regions, but this diversity region is often considered as part of the joining region. Each variable region may comprise three CDRs (Complementarity Determining Regions) embedded in a framework sequence, one being the hypervariable region named CDR3. There are several types of alpha chain variable ($V\alpha$) regions and several types of beta chain variable ($V\beta$) regions distinguished by their framework, CDR1 and CDR2 sequences, and by a partly defined CDR3 sequence. The $V\alpha$ types are referred to in IMGT nomenclature by a unique TRAV number. Thus "TRAV21" defines a TCR $V\alpha$ region having unique framework and CDR1 and CDR2 sequences, and a CDR3 sequence which is partly defined by an amino acid sequence which is preserved from TCR to TCR but which also includes an amino acid sequence which varies from TCR to TCR. In the same way, "TRBV5-1" defines a TCR $V\beta$ region having unique framework and CDR1 and CDR2 sequences, but with only a partly defined CDR3 sequence.

The joining regions of the TCR are similarly defined by the unique IMGT TRAJ and TRBJ nomenclature, and the constant regions by the IMGT TRAC and TRBC nomenclature.

The beta chain diversity region is referred to in IMGT nomenclature by the abbreviation TRBD, and, as mentioned, the concatenated TRBD/TRBJ regions are often considered together as the joining region.

The unique sequences defined by the IMGT nomenclature are widely known and accessible to those working in the TCR field. For example, they can be found in the IMGT public database. The "T cell Receptor Factsbook", (2001) LeFranc and LeFranc, Academic Press, ISBN 0-12-441352-8 also discloses sequences defined by the IMGT nomenclature, but because of its publication date and consequent time-lag, the information therein sometimes needs to be confirmed by reference to the IMGT database.

Native TCRs exist in heterodimeric $\alpha\beta$ or $\gamma\delta$ forms. However, recombinant TCRs consisting of $\alpha\alpha$ or $\beta\beta$ homodimers have previously been shown to bind to peptide MHC molecules. Therefore, the TCR of the invention may be a heterodimeric $\alpha\beta$ TCR or may be an $\alpha\alpha$ or $\beta\beta$ homodimeric TCR.

For use in adoptive therapy, an $\alpha\beta$ heterodimeric TCR may, for example, be transfected as full length chains having both cytoplasmic and transmembrane domains. In certain embodiments TCRs of the invention may have an introduced disulfide bond between residues of the respective constant domains, as described, for example, in WO 2006/000830.

TCRs of the invention, particularly alpha-beta heterodimeric TCRs, may comprise an alpha chain TRAC constant domain sequence and/or a beta chain TRBC1 or TRBC2 constant domain sequence. The alpha and beta chain constant domain sequences may be modified by truncation or substitution to delete the native disulfide bond between Cys4 of exon 2 of TRAC and Cys2 of exon 2 of TRBC1 or TRBC2. The alpha and/or beta chain constant domain

sequence(s) may also be modified by substitution of cysteine residues for Thr 48 of TRAC and Ser 57 of TRBC1 or TRBC2, the said cysteines forming a disulfide bond between the alpha and beta constant domains of the TCR.

Binding affinity (inversely proportional to the equilibrium constant KD) and binding half-life (expressed as TW) can be determined by any appropriate method. It will be appreciated that doubling the affinity of a TCR results in halving the KD. Th is calculated as $\ln 2$ divided by the off-rate (koff). So doubling of Th results in a halving in koff. Kn and koff values for TCRs are usually measured for soluble forms of the TCR, i.e. those forms which are truncated to remove cytoplasmic and transmembrane domain residues. Therefore it is to be understood that a given TCR has an improved binding affinity for, and/or a binding half-life for the parental TCR if a soluble form of that TCR has the said characteristics. Preferably the binding affinity or binding half-life of a given TCR is measured several times, for example 3 or more times, using the same assay protocol, and an average of the results is taken.

Since the TCRs of the invention have utility in adoptive therapy, the invention includes a non-naturally occurring and/or purified and/or engineered cell, especially a T-cell, presenting a TCR of the invention. There are a number of methods suitable for the transfection of T cells with nucleic acid (such as DNA, cDNA or RNA) encoding the TCRs of the invention (see for example Robbins et al., (2008) J Immunol. 180: 6116-6131). T cells expressing the TCRs of the invention will be suitable for use in adoptive therapy-based treatment of cancers such as those of the pancreas and liver. As will be known to those skilled in the art, there are a number of suitable methods by which adoptive therapy can be carried out (see for example Rosenberg et al., (2008) Nat Rev Cancer 8(4): 299-308).

As is well-known in the art TCRs of the invention may be subject to post-translational modifications when expressed by transfected cells. Glycosylation is one such modification, which may comprise the covalent attachment of oligosaccharide moieties to defined amino acids in the TCR chain. For example, asparagine residues, or serine/threonine residues are well-known locations for oligosaccharide attachment. The glycosylation status of a particular protein depends on a number of factors, including protein sequence, protein conformation and the availability of certain enzymes. Furthermore, glycosylation status (i.e. oligosaccharide type, covalent linkage and total number of attachments) can influence protein function. Therefore, when producing recombinant proteins, controlling glycosylation is often desirable.

Glycosylation of transfected TCRs may be controlled by mutations of the transfected gene (Kuball J et al. (2009), J Exp Med 206(2):463-475). Such mutations are also encompassed in this invention.

A TCR may be specific for an antigen in the group MAGE-A1, MAGE-A2, MAGE-A3, MAGE-A4, MAGE-A5, MAGE-A6, MAGE-A7, MAGE-A8, MAGE-A9, MAGE-A10, MAGE-A11, MAGE-A12, MAGE-A13, GAGE-1, GAGE-2, GAGE-3, GAGE-4, GAGE-5, GAGE-6, GAGE-7, GAGE-8, BAGE-1, RAGE-1, LB33/MUM-1, PRAME, NAG, MAGE-Xp2 (MAGE-B2), MAGE-Xp3 (MAGE-B3), MAGE-Xp4 (AGE-B4), tyrosinase, brain glycogen phosphorylase, Melan-A, MAGE-C1, MAGE-C2, NY-ESO-1, LAGE-1, SSX-1, SSX-2(HOM-MEL-40), SSX-1, SSX-4, SSX-5, SCP-1, CT-7, alpha-actinin-4, Bcr-Abl fusion protein, Casp-8, beta-catenin, cdc27, cdk4, cdkn2a, coa-1, dek-can fusion protein, EF2, ETV6-AML1 fusion protein, LDLR-fucosyltransferaseAS fusion protein, HLA-

A2, HLA-All, hsp70-2, KIAA0205, Mart2, Mum-2, and 3, neo-PAP, myosin class I. OS-9, pml-RARa fusion protein, PTPRK, K-ras, N-ras, Triosephosphate isomeras, GnTV, Herv-K-mel, Lage-1, Mage-C2, NA-88, Lage-2, SP17, and TRP2-Int2, (MART-1), gp100 (Pmel 17), TRP-1, TRP-2, MAGE-1, MAGE-3, p15(58), CEA, NY-ESO (LAGE), SCP-1, Hom/Mel-40, p53, H-Ras, HER-2/neu, BCR-ABL, E2A-PRL, H4-RET, IGH-IGK, MYL-RAR, Epstein Barr virus antigens, EBNA, human papillomavirus (HPV) antigens E6 and E7, TSP-180, MAGE-4, MAGE-5, MAGE-6, p185erbB2, p180erbB-3, c-met, nm-23H1, PSA, TAG-72-4, CA 19-9, CA 72-4, CAM 17.1, NuMa, K-ras, beta-catenin, CDK4, Mum-1, p16, TAGE, PSMA, PSCA, CT7, telomerase, 43-9F, 5T4, 791Tgp72, a-fetoprotein, 13HCG, BCA225, BTAA, CA 125, CA 15-3 (CA 27.29BCAA), CA 195, CA 242, CA-50, CAM43, CD68KP1, CO-029, FGF-5, G250, Ga733 (EpCAM), HTgp-175, M344, MA-50, MG7-Ag, MOV18, NB170K, NY-CO-1, RCAS1, SDCCAGI6, TA-90 (Mac-2 binding protein/cyclophilin C-associated protein), TAAI6, TAG72, TLP, and TPS.

c. B-CELL RECEPTORS (BCR)

B-cell receptors (BCRs) or B-cell antigen receptors are immunoglobulin molecules that form a type I transmembrane protein on the surface of a B cell. A BCR is capable of transmitting activatory signal into a B cell following recognition of a specific antigen. Prior to binding of a B cell to an antigen, the BCR will remain in an unstimulated or "resting" stage. Binding of an antigen to a BCR leads to signaling that initiates a humoral immune response.

A BCR is expressed by mature B cells. These B cells work with immunoglobulins (Igs) in recognizing and tagging pathogens. The typical BCR comprises a membrane-bound immunoglobulin (e.g., mIgA, mIgD, mIgE, mIgG, and mIgM), along with associated and Ig α /Ig β (CD79a/CD79b) heterodimers (α/β). These membrane-bound immunoglobulins are tetramers consisting of two identical heavy and two light chains. Within the BCR, the membrane bound immunoglobulins is capable of responding to antigen binding by signal transmission across the plasma membrane leading to B cell activation and consequently clonal expansion and specific antibody production (Friess M et al. (2018), *Front. Immunol.* 2947(9)). The Ig α /Ig β heterodimers is responsible for transducing signals to the cell interior.

A Ig α /Ig β heterodimer signaling relies on the presence of immunoreceptor tyrosine-based activation motifs (ITAMs) located on each of the cytosolic tails of the heterodimers. ITAMs comprise two tyrosine residues separated by 9-12 amino acids (e.g., tyrosine, leucine, and/or valine). Upon binding of an antigen, the tyrosine of the BCR's ITAMs become phosphorylated by Src-family tyrosine kinases Blk, Fyn, or Lyn (Janeway C et al., *Immunobiology: The Immune System in Health and Disease* (Garland Science, 5th ed. 2001)).

d. Other Chimeric Proteins

In addition to the chimeric proteins provided above, the circular RNA polynucleotide may encode for a various number of other chimeric proteins available in the art. The chimeric proteins may include recombinant fusion proteins, chimeric mutant protein, or other fusion proteins.

B. Immune Modulatory Ligands

In some embodiments, the circular RNA polynucleotide encodes for an immune modulatory ligand. In certain embodiments, the immune modulatory ligand may be immunostimulatory; while in other embodiments, the immune modulatory ligand may be immunosuppressive.

1. Cytokines: Interferon, Chemokines, Interleukins, Growth Factor & Others

In some embodiments, the circular RNA polynucleotide encodes for a cytokine. In some embodiments, the cytokine comprises a chemokine, interferon, interleukin, lymphokine, and tumor necrosis factor. Chemokines are chemotactic cytokine produced by a variety of cell types in acute and chronic inflammation that mobilizes and activates white blood cells. An interferon comprises a family of secreted α -helical cytokines induced in response to specific extracellular molecules through stimulation of TLRs (Borden, *Molecular Basis of Cancer* (Fourth Edition) 2015). Interleukins are cytokines expressed by leukocytes.

Descriptions and/or amino acid sequences of IL-2, IL-7, IL-10, IL-12, IL-15, IL-18, IL-270, IFN γ , and/or TGF β 1 are provided herein and at the www.uniprot.org database at accession numbers: P60568 (IL-2), P29459 (IL-12A), P29460 (IL-12B), P13232 (IL-7), P22301 (IL-10), P40933 (IL-15), Q14116 (IL-18), Q14213 (IL-270), P01579 (IFN γ), and/or P01137 (TGF β 1).

20 C. Transcription Factors

Regulatory T cells (Treg) are important in maintaining homeostasis, controlling the magnitude and duration of the inflammatory response, and in preventing autoimmune and allergic responses.

In general, Tregs are thought to be mainly involved in suppressing immune responses, functioning in part as a "self-check" for the immune system to prevent excessive reactions. In particular, Tregs are involved in maintaining tolerance to self-antigens, harmless agents such as pollen or food, and abrogating autoimmune disease.

Tregs are found throughout the body including, without limitation, the gut, skin, lung, and liver. Additionally, Treg cells may also be found in certain compartments of the body that are not directly exposed to the external environment such as the spleen, lymph nodes, and even adipose tissue. Each of these Treg cell populations is known or suspected to have one or more unique features and additional information may be found in Lehtimäki and Laheesmaa, *Regulatory T cells control immune responses through their non-redundant tissue specific features*, 2013, *FRONTIERS IN IMMUNOL.*, 4(294): 1-10, the disclosure of which is hereby incorporated in its entirety.

Typically, Tregs are known to require TGF-0 and IL-2 for proper activation and development. Tregs, expressing abundant amounts of the IL-2 receptor (IL-2R), are reliant on IL-2 produced by activated T cells. Tregs are known to produce both IL-10 and TGF-0, both potent immune suppressive cytokines. Additionally, Tregs are known to inhibit the ability of antigen presenting cells (APCs) to stimulate T cells. One proposed mechanism for APC inhibition is via CTLA-4, which is expressed by Foxp3+ Tregs. It is thought that CTLA-4 may bind to B7 molecules on APCs and either block these molecules or remove them by causing internalization resulting in reduced availability of B7 and an inability to provide adequate co-stimulation for immune responses. Additional discussion regarding the origin, differentiation and function of Tregs may be found in Dhamne et al., *Peripheral and thymic Foxp3+ regulatory T cells in search of origin, distinction, and function*, 2013, *Frontiers in Immunol.*, 4 (253): 1-11, the disclosure of which is hereby incorporated in its entirety.

D. Checkpoint Inhibitors & Agonists

As provided herein, in certain embodiments, the coding element of the circular RNA encodes for one or more checkpoint inhibitors or agonists.

In some embodiments, the immune checkpoint inhibitor is an inhibitor of Programmed Death-Ligand 1 (PD-L1, also known as B7-H1, CD274), Programmed Death 1 (PD-1),

CTLA-4, PD-L2 (B7-DC, CD273), LAG3, TIM3, 2B4, A2aR, B7H1, B7H3, B7H4, BTLA, CD2, CD27, CD28, CD30, CD40, CD70, CD80, CD86, CD137, CD160, CD226, CD276, DR3, GAL9, GITR, HAVCR2, HVEM, IDO1, IDO2, ICOS (inducible T cell costimulator), KIR, LAIR1, LIGHT, MARCO (macrophage receptor with collagenous structure), PS (phosphatidylserine), OX-40, SLAM, TIGIT, VISTA, VTCN1, or any combinations thereof.

In some embodiments, the immune checkpoint inhibitor is an inhibitor of IDO1, CTLA4, PD-1, LAG3, PD-L1, TIM3, or combinations thereof. In some embodiments, the immune checkpoint inhibitor is an inhibitor of PD-L1. In some embodiments, the immune checkpoint inhibitor is an inhibitor of PD-1. In some embodiments, the immune checkpoint inhibitor is an inhibitor of CTLA-4. In some embodiments, the immune checkpoint inhibitor is an inhibitor of LAG3. In some embodiments, the immune checkpoint inhibitor is an inhibitor of TIM3. In some embodiments, the immune checkpoint inhibitor is an inhibitor of IDO1.

As described herein, at least in one aspect, the invention encompasses the use of immune checkpoint antagonists. Such immune checkpoint antagonists include antagonists of immune checkpoint molecules such as Cytotoxic T-Lymphocyte Antigen 4 (CTLA-4), Programmed Cell Death Protein 1 (PD-1), Programmed Death-Ligand 1 (PDL-1), Lymphocyte-activation gene 3 (LAG-3), and T-cell immunoglobulin and mucin domain 3 (TIM-3). An antagonist of CTLA-4, PD-1, PDL-1, LAG-3, or TIM-3 interferes with CTLA-4, PD-1, PDL-1, LAG-3, or TIM-3 function, respectively. Such antagonists of CTLA-4, PD-1, PDL-1, LAG-3, and TIM-3 can include antibodies which specifically bind to CTLA-4, PD-1, PDL-1, LAG-3, and TIM-3, respectively and inhibit and/or block biological activity and function.

E. Others

In some embodiments, the payload encoded within one or more of the coding elements is a hormone, FC fusion protein, anticoagulant, blood clotting factor, protein associated with deficiencies and genetic disease, a chaperone protein, an antimicrobial protein, an enzyme (e.g., metabolic enzyme), a structural protein (e.g., a channel or nuclear pore protein), protein variant, small molecule, antibody, nanobody, an engineered non-body antibody, or a combination thereof.

4. Additional Accessory Elements (Sequence Elements)

As described in this invention, the circular RNA polynucleotide, linear RNA polynucleotide, and/or DNA template may further comprise of accessory elements. In certain embodiments, these accessory elements may be included within the sequences of the circular RNA, linear RNA polynucleotide and/or DNA template for enhancing circularization, translation or both. Accessory elements are sequences, in certain embodiments that are located with specificity between or within the enhanced intron elements, enhanced exon elements, or core functional element of the respective polynucleotide. As an example, but not intended to be limiting, an accessory element includes, a IRES transacting factor region, a miRNA binding site, a restriction site, an RNA editing region, a structural or sequence element, a granule site, a zip code element, an RNA trafficking element or another specialized sequence as found in the art that enhances promotes circularization and/or translation of the protein encoded within the circular RNA polynucleotide.

A. Ires Transacting Factors

In certain embodiments, the accessory element comprises an IRES transacting factor (ITAF) region. In some embodiments, the IRES transacting factor region modulates the initiation of translation through binding to PCBP1-PCBP4 (polyC binding protein), PABP1 (polyA binding protein), PTB (polyprimidine tract binding). Argonaute protein family, HNRNPK (Heterogeneous nuclear ribonucleoprotein K protein), or La protein. In some embodiments, the IRES transacting factor region comprises a polyA, polyC, polyAC, or polyprimidine track.

In some embodiments, the ITAF region is located within the core functional element. In some embodiments, the ITAF region is located within the TIE.

B. MiRNA Binding Sites

In certain embodiments, the accessory element comprises a miRNA binding site. In some embodiments the miRNA binding site is located within the 5' enhanced intron element, 5' enhanced exon element, core functional element, 3' enhanced exon element, and/or 3' enhanced intron element.

In some embodiments, wherein the miRNA binding site is located within the spacer within the enhanced intron element or enhanced exon element. In certain embodiments, the miRNA binding site comprises the entire spacer regions.

In some embodiments, the 5' enhanced intron element and 3' enhanced intron elements each comprise identical miRNA binding sites. In another embodiment, the miRNA binding site of the 5' enhanced intron element comprises a different, in length or nucleotides, miRNA binding site than the 3' enhanced intron element. In one embodiment, the 5' enhanced exon element and 3' enhanced exon element comprise identical miRNA binding sites. In other embodiments, the 5' enhanced exon element and 3' enhanced exon element comprises different, in length or nucleotides, miRNA binding sites.

In some embodiments, the miRNA binding sites are located adjacent to each other within the circular RNA polynucleotide, linear RNA polynucleotide precursor, and/or DNA template. In certain embodiments, the first nucleotide of one of the miRNA binding sites follows the first nucleotide last nucleotide of the second miRNA binding site.

In some embodiments, the miRNA binding site is located within a translation initiation element (TIE) of a core functional element. In one embodiment, the miRNA binding site is located before, trailing or within an internal ribosome entry site (IRES). In another embodiment, the miRNA binding site is located before, trailing, or within an aptamer complex.

The unique sequences defined by the miRNA nomenclature are widely known and accessible to those working in the microRNA field. For example, they can be found in the miRDB public database.

5. Production of Polynucleotides

The DNA templates provided herein can be made using standard techniques of molecular biology. For example, the various elements of the vectors provided herein can be obtained using recombinant methods, such as by screening cDNA and genomic libraries from cells, or by deriving the polynucleotides from a DNA template known to include the same.

The various elements of the DNA template provided herein can also be produced synthetically, rather than cloned, based on the known sequences. The complete sequence can be assembled from overlapping oligonucleotides prepared by standard methods and assembled into the complete sequence. See, e.g., Edge, Nature (1981) 292:756;

Nambair et al., Science (1984) 223: 1299; and Jay et al., J. Biol. Chem. (1984) 259:631 1.

Thus, particular nucleotide sequences can be obtained from DNA template harboring the desired sequences or synthesized completely, or in part, using various oligonucleotide synthesis techniques known in the art, such as site-directed mutagenesis and polymerase chain reaction (PCR) techniques where appropriate. One method of obtaining nucleotide sequences encoding the desired DNA template elements is by annealing complementary sets of overlapping synthetic oligonucleotides produced in a conventional, automated polynucleotide synthesizer, followed by ligation with an appropriate DNA ligase and amplification of the ligated nucleotide sequence via PCR. See, e.g., Jayaraman et al., Proc. Natl. Acad. Sci. USA (1991) 88:4084-4088. Additionally, oligonucleotide-directed synthesis (Jones et al., Nature (1986) 54:75-82), oligonucleotide directed mutagenesis of preexisting nucleotide regions (Riechmann et al., Nature (1988) 332:323-327 and Verhoeyen et al., Science (1988) 239: 1534-1536), and enzymatic filling-in of gapped oligonucleotides using T4 DNA polymerase (Queen et al., Proc. Natl. Acad. Sci. USA (1989) 86: 10029-10033) can be used.

The precursor RNA provided herein can be generated by incubating a DNA template provided herein under conditions permissive of transcription of the precursor RNA encoded by the DNA template. For example, in some embodiments a precursor RNA is synthesized by incubating a DNA template provided herein that comprises an RNA polymerase promoter upstream of its 5' duplex sequence and/or expression sequences with a compatible RNA polymerase enzyme under conditions permissive of in vitro transcription. In some embodiments, the DNA template is incubated inside of a cell by a bacteriophage RNA polymerase or in the nucleus of a cell by host RNA polymerase II.

In certain embodiments, provided herein is a method of generating precursor RNA by performing in vitro transcription using a DNA template provided herein as a template (e.g., a vector provided herein with an RNA polymerase promoter positioned upstream of the 5' duplex region).

In certain embodiments, the resulting precursor RNA can be used to generate circular RNA (e.g., a circular RNA polynucleotide provided herein) by incubating it in the presence of magnesium ions and guanosine nucleotide or nucleoside at a temperature at which RNA circularization occurs (e.g., between 20° C. and 60° C.).

Thus, in certain embodiments provided herein is a method of making circular RNA. In certain embodiments, the method comprises synthesizing precursor RNA by transcription (e.g., run-off transcription) using a vector provided herein (e.g., a 5' enhanced intron element, a 5' enhanced exon element, a core functional element, a 3' enhanced exon element, and a 3' enhanced intron element) as a template, and incubating the resulting precursor RNA in the presence of divalent cations (e.g., magnesium ions) and GTP such that it circularizes to form circular RNA. In some embodiments, the precursor RNA disclosed herein is capable of circularizing in the absence of magnesium ions and GTP and/or without the step of incubation with magnesium ions and GTP. It has been discovered that circular RNA has reduced immunogenicity relative to a corresponding mRNA, at least partially because the mRNA contains an immunogenic 5' cap. When transcribing a DNA vector from certain promoters (e.g., a T7 promoter) to produce a precursor RNA, it is understood that the 5' end of the precursor RNA is G. To reduce the immunogenicity of a circular RNA composition that contains a low level of contaminant linear mRNA, an

excess of GMP relative to GTP can be provided during transcription such that most transcripts contain a 5' GMP, which cannot be capped. Therefore, in some embodiments, transcription is carried out in the presence of an excess of GMP. In some embodiments, transcription is carried out where the ratio of GMP concentration to GTP concentration is within the range of about 3:1 to about 15:1, for example, about 3:1 to about 10:1, about 3:1 to about 5:1, about 3:1, about 4:1, or about 5:1.

In some embodiments, a composition comprising circular RNA has been purified. Circular RNA may be purified by any known method commonly used in the art, such as column chromatography, gel filtration chromatography, and size exclusion chromatography. In some embodiments, purification comprises one or more of the following steps: phosphatase treatment, HPLC size exclusion purification, and RNase R digestion. In some embodiments, purification comprises the following steps in order: RNase R digestion, phosphatase treatment, and HPLC size exclusion purification. In some embodiments, purification comprises reverse phase HPLC. In some embodiments, a purified composition contains less double stranded RNA, DNA splints, triphosphorylated RNA, phosphatase proteins, protein ligases, capping enzymes and/or nicked RNA than unpurified RNA. In some embodiments, a purified composition is less immunogenic than an unpurified composition. In some embodiments, immune cells exposed to a purified composition produce less TNF α , RIG-I, IL-2, IL-6, IFN γ , and/or a type 1 interferon, e.g., IFN-p1, than immune cells exposed to an unpurified composition.

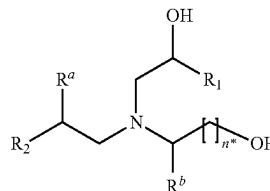
6. Overview of Transfer Vehicle & Other Delivery Mechanisms

A. Ionizable Lipids

In certain embodiments disclosed herein are ionizable lipids that may be used as a component of a transfer vehicle to facilitate or enhance the delivery and release of circular RNA to one or more target cells (e.g., by permeating or fusing with the lipid membranes of such target cells). In certain embodiments, an ionizable lipid comprises one or more cleavable functional groups (e.g., a disulfide) that allow, for example, a hydrophilic functional head-group to dissociate from a lipophilic functional tail-group of the compound (e.g., upon exposure to oxidative, reducing or acidic conditions), thereby facilitating a phase transition in the lipid bilayer of the one or more target cells.

In various embodiments, an ionizable lipid of the disclosure is a compound of Formula (13*):

Formula (13*)



wherein:

n^* is an integer between 1 to 7.

R^a is hydrogen or hydroxyl,

R^b is hydrogen or C_1 - C_6 alkyl,

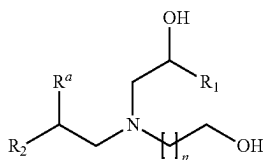
R_1 and R_2 are each independently a linear or branched C_1 - C_{30} alkyl, C_2 - C_{30} alkenyl, or C_1 - C_{30} heteroalkyl,

107

optionally substituted by one or more substituents selected from oxo, halo, hydroxy, cyano, alkyl, alkenyl, aldehyde, heterocyclylalkyl, hydroxyalkyl, dihydroxyalkyl, hydroxyalkylaminoalkyl, aminoalkyl, alkylaminoalkyl, dialkylaminoalkyl, (heterocyclyl)(alkyl)aminoalkyl, heterocyclyl, heteroaryl, alkylheteroaryl, alkynyl, alkoxy, amino, dialkylamino, aminoalkylcarbonylamino, aminocarbonylalkylamino, (aminocarbonylalkyl)(alkyl)amino, alkenylcarbonylamino, hydroxycarbonyl, alkylloxycarbonyl, alkylcarbonyloxy, alkylcarbonate, alkenyloxycarbonyl, alkenylcarbonyloxy, alkenylcarbonate, alkynylloxycarbonyl, alkynylcarbonyloxy, alkynylcarbonate, aminocarbonyl, aminoalkylaminocarbonyl, alkylaminoalkylaminocarbonyl, dialkylaminoalkylaminocarbonyl, heterocyclylalkylaminocarbonyl, (alkylaminoalkyl)(alkyl)aminocarbonyl, alkylaminoalkylcarbonyl, dialkylaminoalkylcarbonyl, heterocyclylcarbonyl, alkenylcarbonyl, alkynylcarbonyl, alkylsulfoxide, alkylsulfoxidealkyl, alkylsulfonyl, and alkylsulfonealkyl.

In some embodiments of Formula (13*), R^b is C_1 - C_6 alkyl. In some embodiments of Formula (13*), R^b is methyl. In some embodiments of Formula (13*), R^b is ethyl.

In some embodiments of Formula (13*), R^b is H and the ionizable lipid is of Formula (13):



Formula (13)

wherein n is an integer between 1 to 7. In some embodiments of Formula (13), n is an integer between 1 to 4.

In some embodiments of Formula (13*), and Formula (13), R_1 and R_2 are the same. In some embodiments of Formula (13*) and Formula (13), R_1 and R_2 are different.

In some embodiments of Formula (13*) and Formula (13), R_1 and R_2 are each independently an optionally substituted linear or branched alkyl, alkenyl, or heteroalkyl, where the total number of carbon atoms present in the optionally substituted linear or branched group is 30 carbons or less, such as 6-30 carbon atoms, or 6-20 carbon atoms.

In some embodiments of Formula (13*) and Formula (13), at least one of R_1 and R_2 is an unsubstituted, linear or branched C_6 - C_{30} alkyl, C_6 - C_{30} alkenyl, or C_6 - C_{30} heteroalkyl.

In some embodiments of Formula (13*) and Formula (13), R_1 and R_2 are each independently a linear or branched C_6 - C_{30} alkyl, C_6 - C_{30} alkenyl, or C_9 - C_{20} heteroalkyl, optionally substituted by one or more substituents (e.g., as described above). In some embodiments of Formula (13*) and Formula (13), R_1 and R_2 are independently selected from a linear or branched C_6 - C_{30} alkyl, C_6 - C_{30} alkenyl, or C_1 - C_{20} heteroalkyl, substituted with alkyloxycarbonyl, alkylcarbonyloxy, alkylcarbonate, alkenyloxycarbonyl, alkenylcarbonyloxy, alkenylcarbonate.

In some embodiments of Formula (13*) and Formula (13), R_1 and R_2 are each independently a linear or branched C_1 - C_{20} alkyl, C_2 - C_{20} alkenyl, or C_1 - C_{20} heteroalkyl, optionally substituted by one or more substituents each independently selected from linear or branched C_1 - C_{20} alkoxy,

108

linear or branched C_1 - C_{20} alkyloxycarbonyl, linear or branched C_1 - C_{20} alkylcarbonyloxy, linear or branched C_1 - C_{20} alkylcarbonate, linear or branched C_2 - C_{20} alkenyloxycarbonyl, linear or branched C_2 - C_{20} alkenylcarbonyloxy, linear or branched C_2 - C_{20} alkenylcarbonate, linear or branched C_2 - C_{20} alkynylloxycarbonyl, linear or branched C_2 - C_{20} alkynylcarbonyloxy, and linear or branched C_2 - C_{20} alkynylcarbonate.

In some embodiments of Formula (13*) and Formula (13), at least one of R_1 and R_2 is a linear C_1 - C_{12} alkyl substituted by $-\text{O}(\text{CO})\text{R}^6$, $-\text{C}(\text{O})\text{OR}^6$, or $-\text{O}(\text{CO})\text{OR}^6$, wherein each R^6 is independently linear or branched C_1 - C_{20} alkyl or C_2 - C_{20} alkenyl. In some embodiments of Formula (13*) and Formula (13), R_1 and R_2 are each independently a linear C_1 - C_{12} alkyl substituted by $-\text{O}(\text{CO})\text{R}^6$, $-\text{C}(\text{O})\text{OR}^6$, or $-\text{O}(\text{CO})\text{OR}^6$, wherein each R^6 is independently linear or branched C_1 - C_{20} alkyl or C_2 - C_{20} alkenyl.

In some embodiments, at least one of R_1 and R_2 is substituted with an alkyloxycarbonyl. In some embodiments, the alkyloxycarbonyl is of the formula $-\text{C}(\text{O})\text{OR}'$, wherein R' is unsubstituted C_6 - C_{30} alkyl or C_6 - C_{30} alkenyl.

In some embodiments, at least one of R_1 and R_2 is substituted with an alkylcarbonyloxy. In some embodiments, the alkylcarbonyloxy is of the formula $-\text{OC}(\text{O})\text{R}^{6*}$, wherein R^{6*} is unsubstituted C_6 - C_{30} alkyl or C_6 - C_{30} alkenyl.

In some embodiments, at least one of R_1 and R_2 is substituted with an alkylcarbonate. In some embodiments, the alkylcarbonate is of the formula $-\text{O}(\text{CO})\text{OR}^{6*}$, wherein R^{6*} is unsubstituted C_6 - C_{30} alkyl or C_6 - C_{30} alkenyl.

In some embodiments, R_1 and R_2 are each independently C_1 - C_{12} alkyl substituted by $-\text{O}(\text{CO})\text{R}'$, $-\text{C}(\text{O})\text{OR}^{6*}$, or $-\text{O}(\text{CO})\text{OR}^{6*}$, wherein R^{6*} is unsubstituted C_6 - C_{30} alkyl or C_6 - C_{30} alkenyl. In some embodiments, R_1 and R_2 are each C_1 - C_{12} alkyl substituted by $-\text{O}(\text{CO})\text{R}^{6*}$. In some embodiments, R_1 and R_2 are each C_1 - C_{12} alkyl substituted by $-\text{C}(\text{O})\text{OR}^{6*}$. In some embodiments, R_1 and R_2 are each C_1 - C_{12} alkyl substituted by $-\text{O}(\text{CO})\text{OR}^{6*}$. In some embodiments, R_1 is $-\text{C}(\text{O})\text{OR}^{6*}$ or $-\text{O}(\text{CO})\text{R}^{6*}$ and R_2 is $-\text{O}(\text{CO})\text{OR}^{6*}$. In some embodiments, R_1 is $-\text{O}(\text{CO})\text{OR}^{6*}$ and R_2 is $-\text{C}(\text{O})\text{OR}^{6*}$ or $-\text{O}(\text{CO})\text{R}^{6*}$.

In some embodiments, at least one of R_1 and R_2 is selected from the following formulae:

- (i) $-(\text{CH}_2)_q\text{C}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$,
- (ii) $-(\text{CH}_2)_q\text{OC}(\text{O})(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$, and
- (iii) $-(\text{CH}_2)_q\text{OC}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$,

wherein:

q is an integer between 0 to 12,

r is an integer between 0 to 6,

R^8 is H or R^{10} , and

R^9 and R^{10} are independently unsubstituted linear C_1 - C_2 alkyl or unsubstituted linear C_2 - C_{12} -alkenyl.

In some embodiments, each of R_1 and R_2 is independently selected from one of the following formulae:

- (i) $-(\text{CH}_2)_q\text{C}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$,
- (ii) $-(\text{CH}_2)_q\text{OC}(\text{O})(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$, and
- (iii) $-(\text{CH}_2)_q\text{OC}(\text{O})\text{O}(\text{CH}_2)_r\text{CH}(\text{R}^8)(\text{R}^9)$,

wherein:

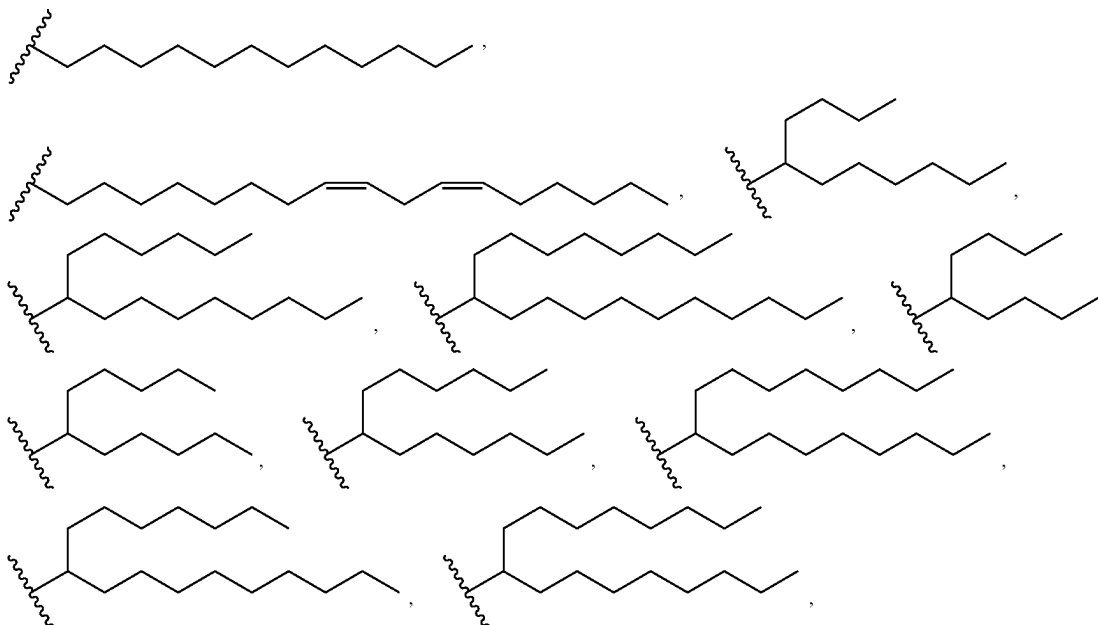
q is an integer between 0 to 12,

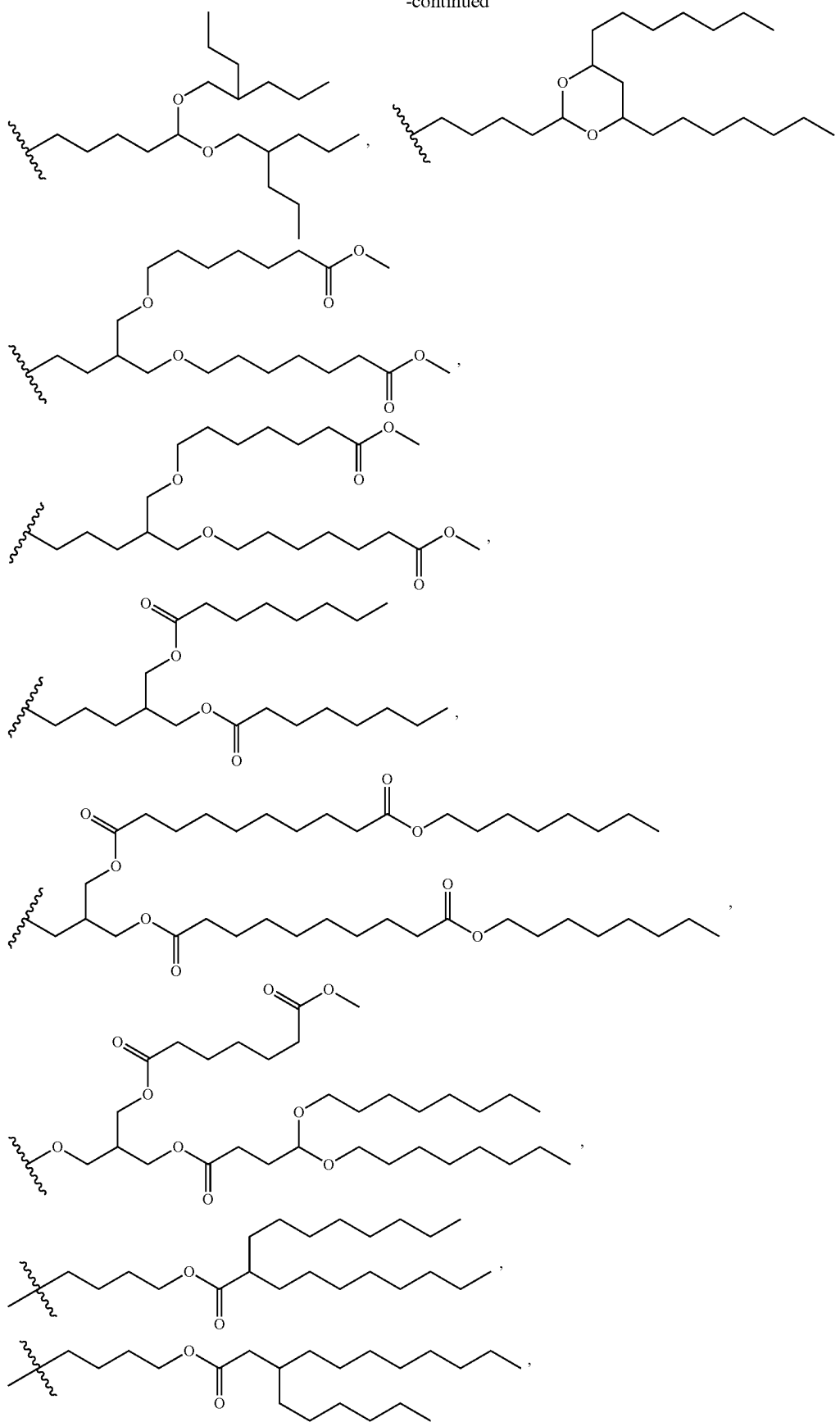
r is an integer between 0 to 6,

R^8 is H or R^{10} , and

R^9 and R^{10} are independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_2 - C_{12} -alkenyl.

In some embodiments, of any one of formulae (i)-(iii), q is an integer between 1 to 6. In some embodiments, of any one of formulae (i)-(iii), q is 0. In some embodiments, of any one of formulae (i)-(iii), q is 1. In some embodiments, of any

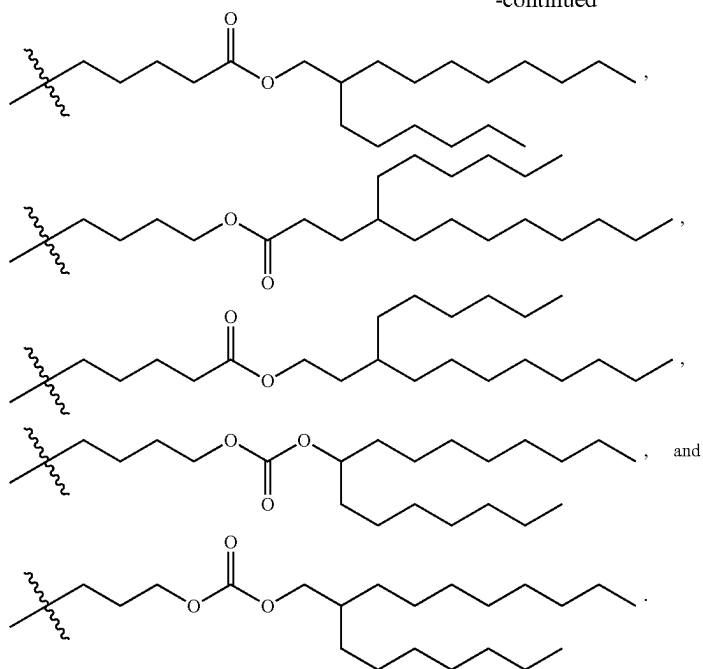




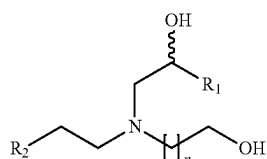
115

116

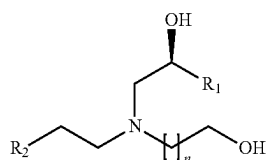
-continued



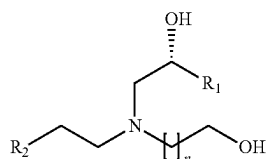
In some embodiments, the ionizable lipid of Formula (13) is represented by Formula (13a-1), Formula (13a-2), or Formula (13a-3):



Formula (13a-1)

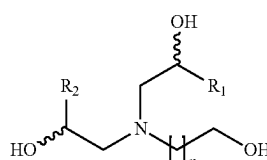


Formula (13a-2)



Formula (13a-3)

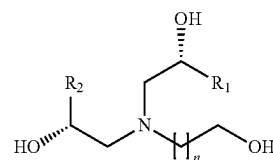
In some embodiments, the ionizable lipid is represented by Formula (13b-1), Formula (13b-2), or Formula (13b-3):



Formula (13b-1)

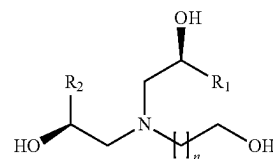
-continued

Formula (13b-2)



35

40



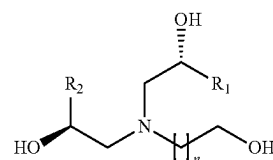
Formula (13b-3)

45

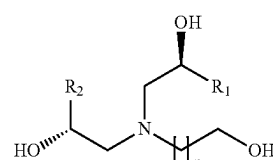
In some embodiments, the ionizable lipid is represented by Formula (13b-4), Formula (13b-5), Formula (13b-6), Formula (13b-7), Formula (13b-8), or Formula (13b-9):

50

Formula (13b-4)



55

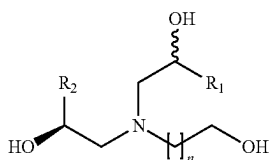


Formula (13b-5)

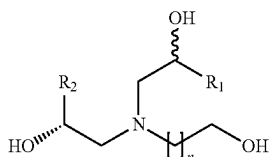
65

117

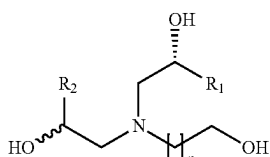
-continued



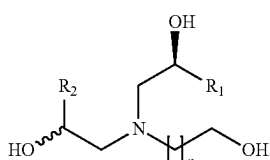
Formula (13b-6)



Formula (13b-7)



Formula (13b-8)



Formula (13b-9)

In some embodiments of Formula (13a-1) to (13b-9), R^1 and R^2 are independently C_1 - C_{12} alkyl optionally substituted by $-O(CO)R^6$, $-C(O)OR^6$, or $-O(CO)OR^6$, wherein R^6 is unsubstituted linear or branched C_1 - C_{20} alkyl or C_2 - C_{20} alkenyl. In some embodiments, R^6 is unsubstituted linear C_1 - C_{20} alkyl. In some embodiments, R^6 is unsubstituted branched C_6 - C_{20} alkyl. In some embodiments, R^6 is unsubstituted linear C_6 - C_{20} alkyl. In some embodiments, R^6 is unsubstituted branched C_6 - C_{20} alkyl.

In some embodiments of Formula (13a-1) to (13b-9), R^1 and R^2 are independently selected from linear or branched C_6 - C_{30} alkyl, linear or branched C_6 - C_{30} alkenyl, linear or branched C_6 - C_{30} heteroalkyl, $-(CH_2)_qC(O)O(CH_2CH(R^8)(R^9))$, $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, and $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, wherein q is 0 to 12, r is 0 to 6, R^8 is H or R^{10} , and R^9 and R^{10} are independently unsubstituted linear C_1 - C_{20} alkyl or unsubstituted linear C_2 - C_{20} alkenyl. In some embodiments, q is 1 to 8, such as 1 to 6, or 2 to 6. In some embodiments, r is 0. In some embodiments, r is 1 to 6, such as 1 to 3. In some embodiments, r is 1. In some embodiments, r is 2.

In some embodiments of Formula (13a-1) to (13b-9), R^1 and R^2 are different groups. In some embodiments of Formula (13a-1) to (13b-9), R^1 and R^2 are the same. In some embodiments of Formula (13a-1) to (13b-9), one of R^1 and R^2 is a linear group, and the other of R^1 and R^2 includes a branched group.

In some embodiments of Formula (13a-1) to (13b-9), at least one of R^1 and R^2 is selected from the following formulae:

- (i) $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$,
- (ii) $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, and
- (iii) $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$,

wherein:

- q is an integer between 0 to 12,
- r is an integer between 0 to 6,
- R^8 is H or R^{10} , and

118

R^9 and R^{10} are independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_2 - C_{12} alkenyl.

In some embodiments of Formula (13a-1) to (13b-9), each of R^1 and R^2 is independently selected from one of the following formulae:

- (i) $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$,
- (ii) $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, and
- (iii) $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$,

wherein:

q is an integer between 0 to 12,

r is an integer between 0 to 6,

R^8 is H or R^{10} , and

R^9 and R^{10} are independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_2 - C_{12} alkenyl.

In some embodiments, of any one of formulae (i)-(iii), q is an integer between 1 to 6. In some embodiments of any one of formulae (i)-(iii), q is 1. In some embodiments of any one of formulae (i)-(iii), q is 2. In some embodiments, of any one of formulae (i)-(iii), q is an integer between 3 to 12. In some embodiments, of any one of formulae (i)-(iii), q is an integer between 3 to 6.

In some embodiments of any one of formulae (i)-(iii), r is 0. In some embodiments of any one of formulae (i)-(iii), r is an integer between 1 to 6. In some embodiments of any one of formulae (i)-(iii), r is 1. In some embodiments of any one of formulae (i)-(iii), r is 2.

In some embodiments of formulae (i)-(iii), R^8 is H. In some embodiments of formulae (i)-(iii), R^8 is R^{10} . In some embodiments of formulae (i)-(iii), R^8 and R^{10} are different. In some embodiments of formulae (i)-(iii), R^9 and R^{10} are the same.

In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_1 - C_{12} alkenyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_2 - C_{12} alkyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_2 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_4 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_6 - C_8 alkyl.

In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_1 - C_{12} alkenyl. In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_2 - C_{12} alkyl. In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_2 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_4 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_6 - C_8 alkyl.

In some embodiments of Formula (13a-1) to (13b-9), at least one of R^1 and R^2 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$ (R^9), where q , r , R^8 and R^9 are as defined above. In some embodiments of Formula (13a-1) to (13b-9), at least one of R^1 and R^2 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, where q , r , R^8 and R^9 are as defined above. In some embodiments of Formula (13a-1) to (13b-9), at least one of R^1 and R^2 is $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, where q , r , R^8 and R^9 are as defined above.

In certain embodiments of Formula (13a-1) to (13b-9), at least one of R^1 and R^2 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$ (R^9) or $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$ (R^9), where q , r , R^8 and R^9 are as defined above. In some embodiments of Formula (13a-1) to (13b-9), at least one of R^1 and R^2 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$ (R^9) or $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$ (R^9), where q , r , R^8 and R^9 are as defined above. In some

In some embodiments of Formula (13d-2), R^a is hydrogen, Z^A is selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, and $-\text{OC}(\text{O})\text{O}-$, and Z^B is absent. In some embodiments of 65 Formula (13d-2), R^a is hydrogen, Z^B is a linking group selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, and $-\text{OC}(\text{O})\text{O}-$, and Z^A is absent.

121

In some embodiments of Formula (13d-2), R^a is hydroxyl, Z^A is selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, and $-\text{OC}(\text{O})\text{O}-$, and Z^B is absent. In some embodiments of Formula (13d-2), R^a is hydroxyl, Z^B is selected from $-\text{C}(\text{O})\text{O}-$, $-\text{OC}(\text{O})-$, and $-\text{OC}(\text{O})\text{O}-$, and Z^A is absent.

122

In some embodiments, the ionizable lipid of the disclosure is of Formula (13d-2) as described in the compounds of the Table 1 below, where any undefined variables are as described above.

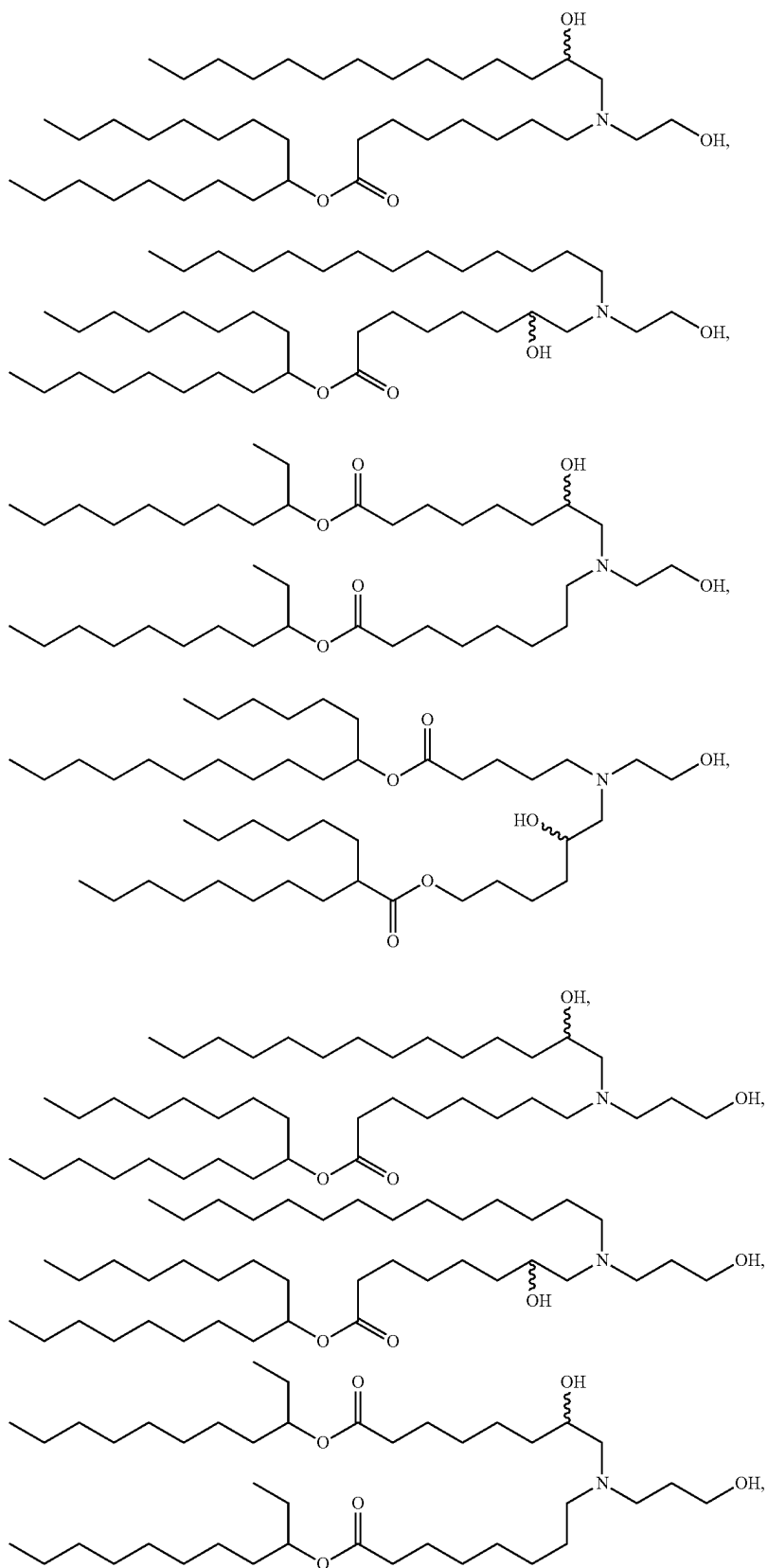
TABLE 1

Exemplary ionizable lipids of Formula (13d-2).											
Formula (13d-2)											
Cmpd #	R^{9A}	R^{8A}	r	Z^A	q	R^a	q^1	Z^B	r^1	R^{8B}	R^{9B}
1	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{C}(\text{O})-$	3-6	H	3-6	$-\text{OC}(\text{O})-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
2	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	1	$-\text{C}(\text{O})-$	3-6	H	3-6	$-\text{OC}(\text{O})-$		C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
3	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	2	$-\text{C}(\text{O})-$	3-6	H	3-6	$-\text{OC}(\text{O})-$	2	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
4	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})-$	3-6	H	3-6	$-\text{C}(\text{O})-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
5	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	1	$-\text{OC}(\text{O})-$	3-6	H	3-6	$-\text{C}(\text{O})-$	1	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
6	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	2	$-\text{OC}(\text{O})-$	3-6	H	3-6	$-\text{C}(\text{O})-$	2	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
7	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})\text{O}-$	3-6	H	3-6	$-\text{OC}(\text{O})\text{O}-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
8	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	1	$-\text{OC}(\text{O})\text{O}-$	3-6	H	3-6	$-\text{OC}(\text{O})\text{O}-$	1	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
9	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})-$	3-6	H	1-12	absent	0	H	C ₁ -C ₁₂ alkyl
10	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})-$	3-6	H	3-6	$-\text{C}(\text{O})-$	0	H	C ₄ -C ₁₂ alkyl
11	C ₁ -C ₁₂ alkyl	H	0	absent	3-6	H	3-6	$-\text{C}(\text{O})-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
12	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})-$	3-6	OH	1-12	absent	0	H	C ₁ -C ₁₂ alkyl
13	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})-$	3-6	OH	3-6	$-\text{C}(\text{O})-$	0	H	C ₄ -C ₁₂ alkyl
14	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})\text{O}-$	3-6	H	3-6	$-\text{C}(\text{O})-$	0	H	C ₄ -C ₁₂ alkyl
15	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})-$	3-6	H	3-6	$-\text{OC}(\text{O})\text{O}-$	0	H	C ₄ -C ₁₂ alkyl
16	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})\text{O}-$	3-6	H	3-6	$-\text{OC}(\text{O})\text{O}-$	0	H	C ₄ -C ₁₂ alkyl
17	C ₄ -C ₈ alkyl	H	0	$-\text{OC}(\text{O})-$	3-6	H	3-6	$-\text{OC}(\text{O})\text{O}-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
18	C ₄ -C ₈ alkyl	H	0	$-\text{OC}(\text{O})\text{O}-$	3-6	H	3-6	$-\text{C}(\text{O})\text{O}-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
19	C ₄ -C ₈ alkyl	H	0	$-\text{OC}(\text{O})\text{O}-$	3-6	H	3-6	$-\text{OC}(\text{O})\text{O}-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
20	C ₄ -C ₈ alkyl	H	0	$-\text{OC}(\text{O})-$	3-6	OH	3-6	$-\text{OC}(\text{O})\text{O}-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
21	C ₄ -C ₈ alkyl	H	0	$-\text{OC}(\text{O})\text{O}-$	3-6	OH	3-6	$-\text{C}(\text{O})\text{O}-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
22	C ₄ -C ₈ alkyl	H	0	$-\text{OC}(\text{O})\text{O}-$	3-6	OH	3-6	$-\text{OC}(\text{O})\text{O}-$	0	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl
23	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})\text{O}-$	3-6	H	3-6	$-\text{C}(\text{O})-$	0	C ₁ -C ₄ alkyl	C ₄ -C ₁₂ alkyl
24	C ₄ -C ₈ alkyl	C ₄ -C ₈ alkyl	0	$-\text{OC}(\text{O})\text{O}-$	3-6	H	3-6	$-\text{C}(\text{O})-$	0	C ₁ -C ₅ alkyl	C ₁ -C ₁₂ alkyl

In some embodiments of the exemplary lipids of Table 1, n is 1 or 2.

123

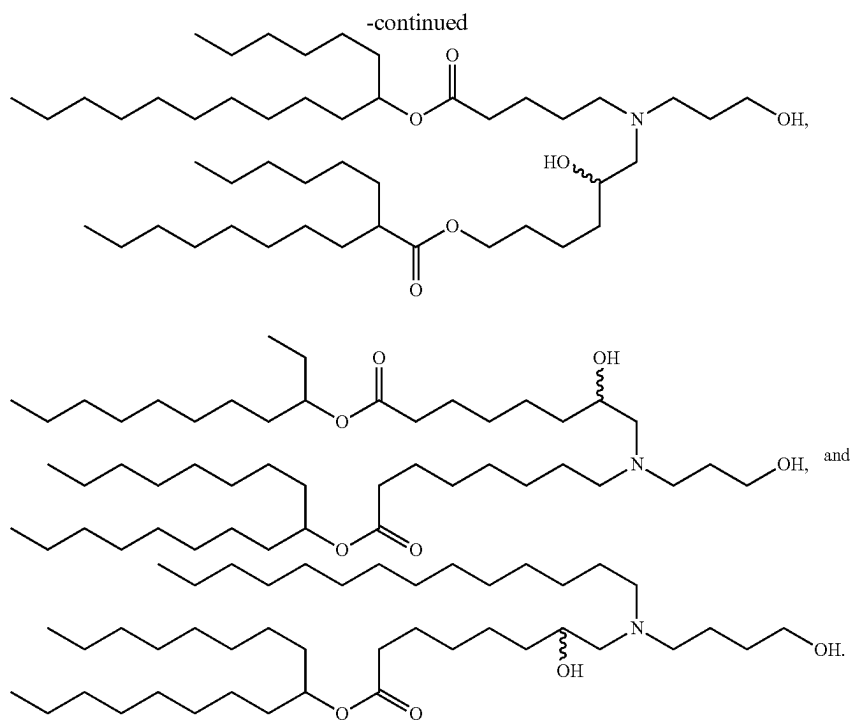
In some embodiments, the ionizable lipid of the disclosure is selected from the group consisting of:



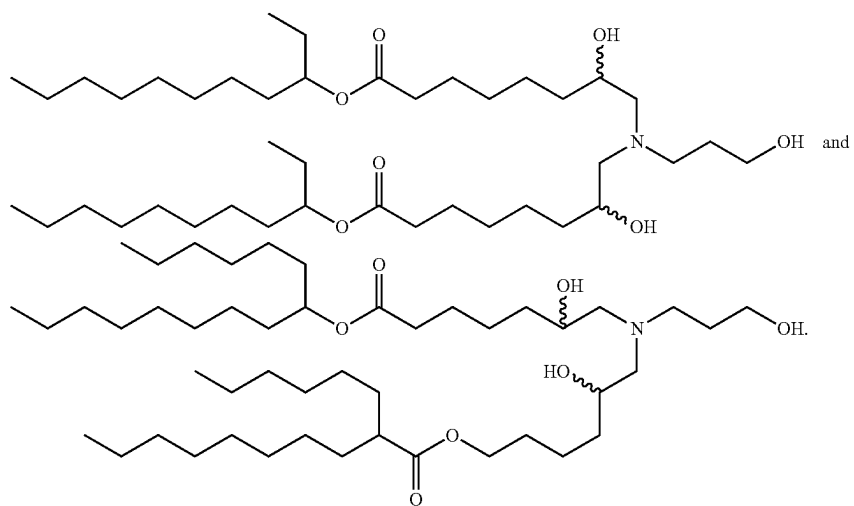
124

125

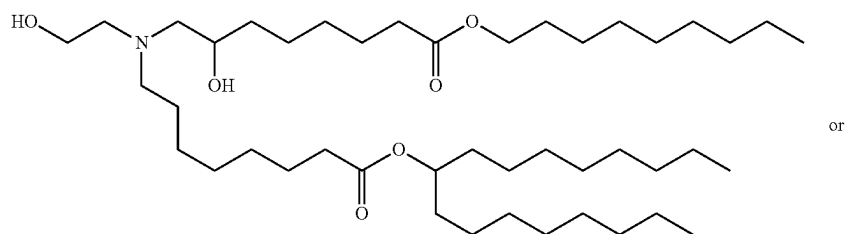
126



In some embodiments, the ionizable lipid is selected from ³⁰ the group consisting of

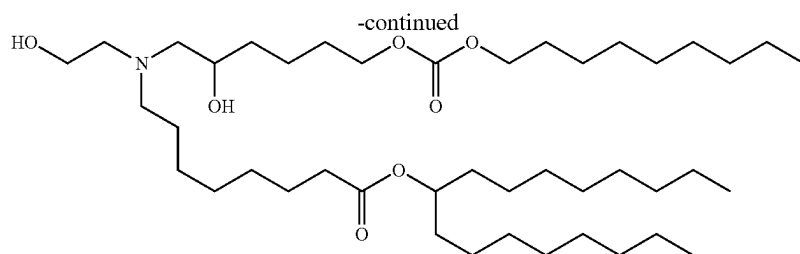


In some embodiments, the ionizable lipid is not



127

128



In some embodiments of Formula (13c-1) and/or (13c-2), each R^b is hydrogen.

In some embodiments of Formula (13c-1) and/or to (13c-2), one and only one R^b is C_1 - C_6 alkyl, and the other R^b group(s), if present, are hydrogen. In some embodiments of Formula (13c-1) and/or (13c-2), one and only one R^b is methyl or ethyl. In some embodiments, the one and only one R^b that is C_1 - C_6 alkyl is attached to the carbon atom adjacent to the nitrogen atom of the ionizable lipid. In some embodiments of Formula (13c-1) and/or (13c-2), n is 2 to 7, one and only one R^b is C_1 - C_6 alkyl, and the other R^b group(s), if present, are hydrogen.

In some embodiments, an ionizable lipid of the disclosure is a lipid selected from Table 10e-Table 10h.

In some embodiments, an ionizable lipid of the disclosure has a beta-hydroxyl amine head group. In some embodiments, the ionizable lipid has a gamma-hydroxyl amine head group.

In an embodiment, the ionizable lipid is described in US patent publication number US20170210697A1. In an embodiment, the ionizable lipid is described in US patent publication number US20170119904A1.

TABLE 10e

Ionizable lipid number	Structure
1	
2	
3	
4	

TABLE 10e-continued

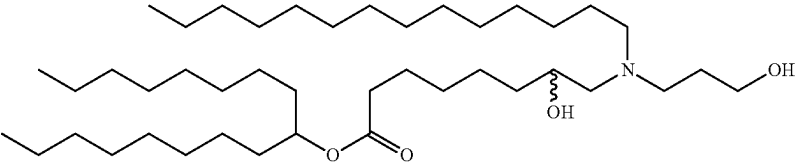
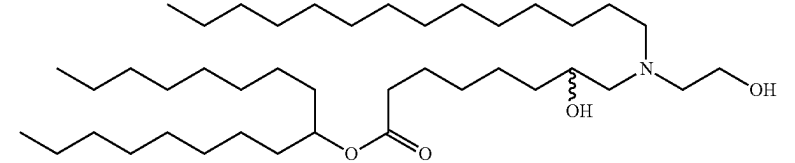
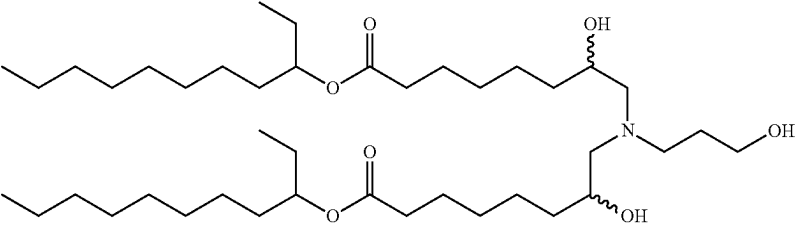
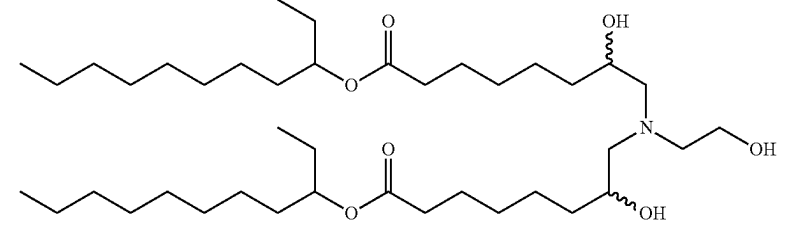
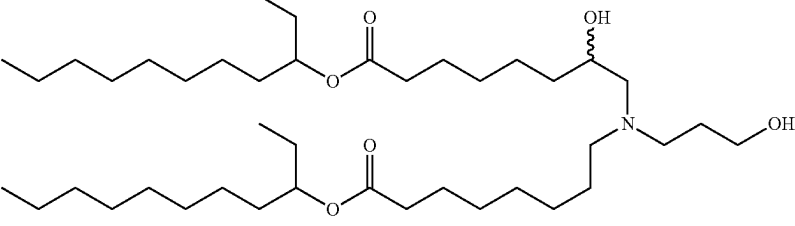
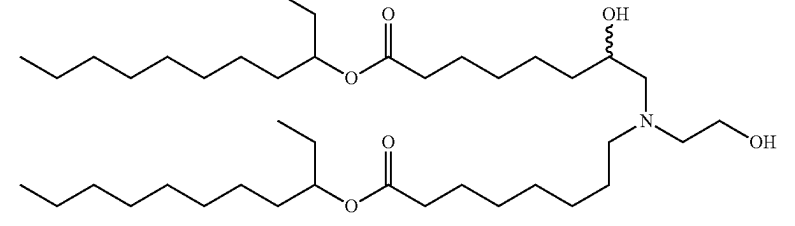
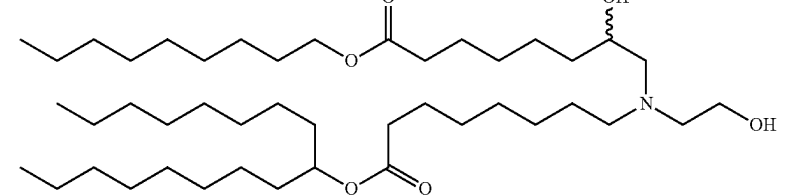
Ionizable lipid number	Structure
5	
6	
7	
8	
9	
10	
11	

TABLE 10e-continued

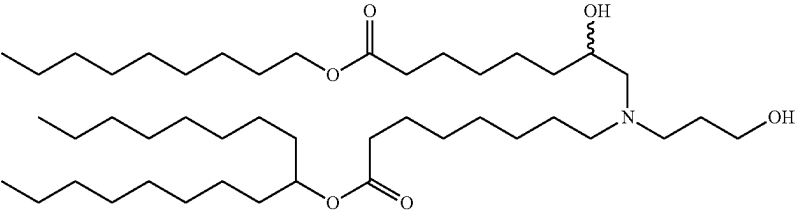
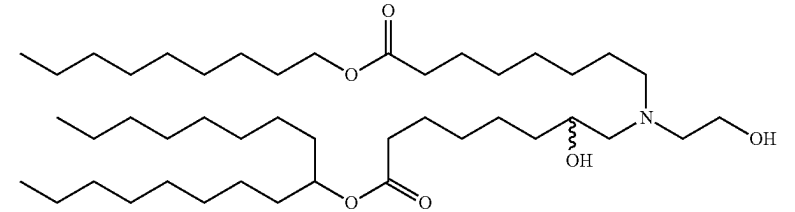
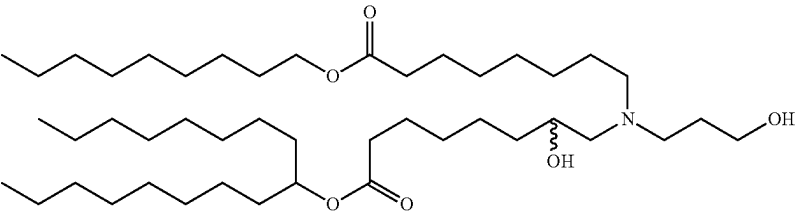
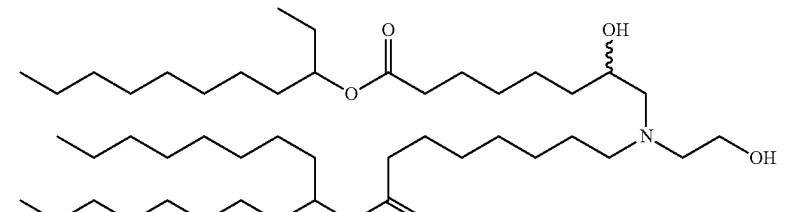
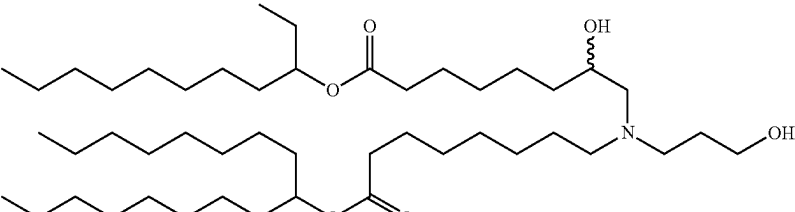
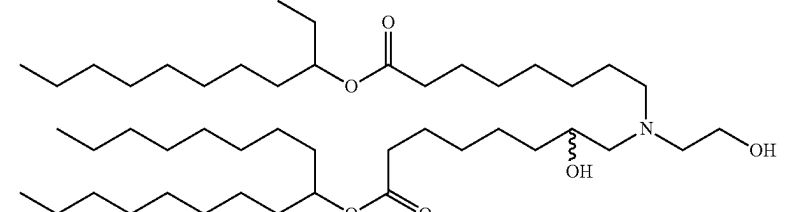
Ionizable lipid number	Structure
12	
13	
14	
15	
16	
17	

TABLE 10e-continued

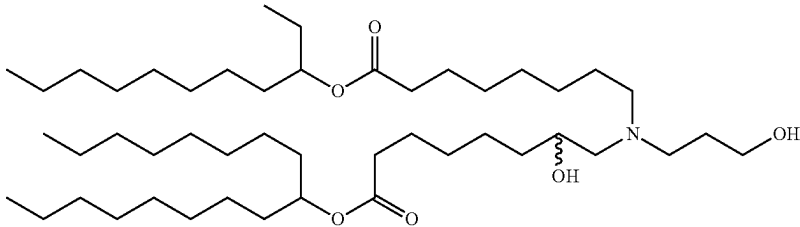
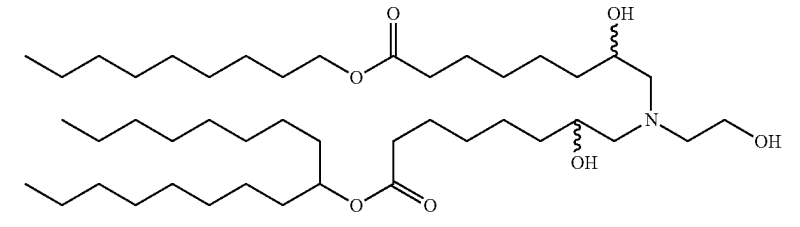
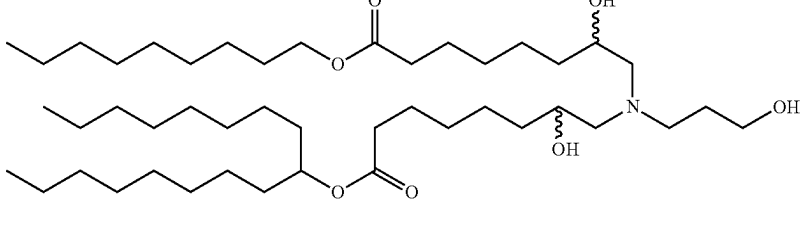
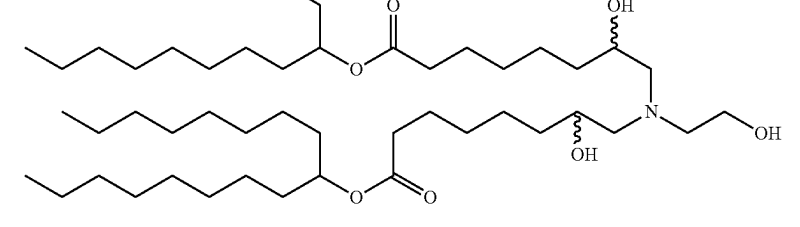
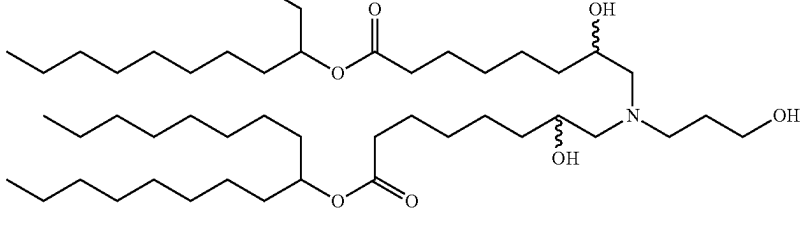
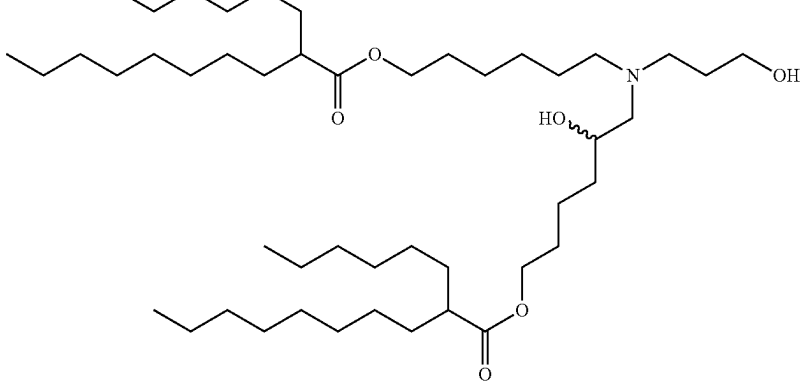
Ionizable lipid number	Structure
18	
19	
20	
21	
22	
23	

TABLE 10e-continued

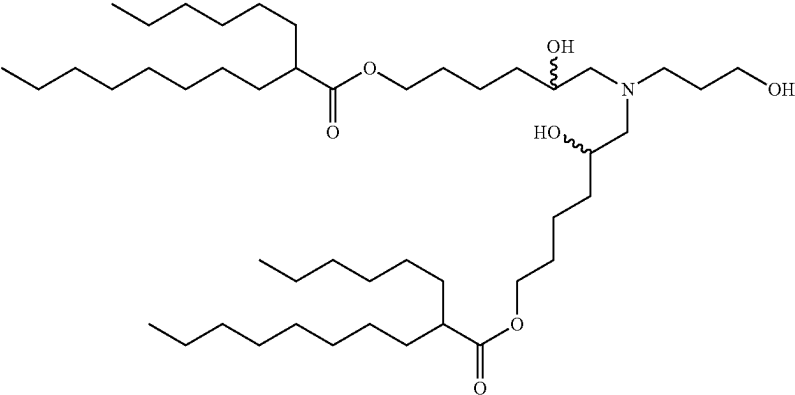
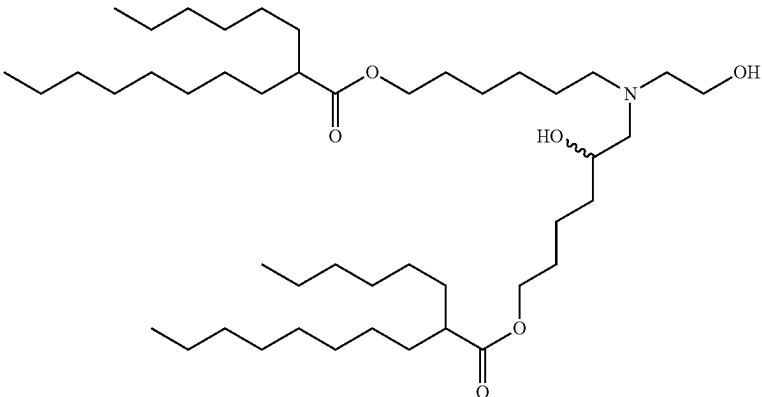
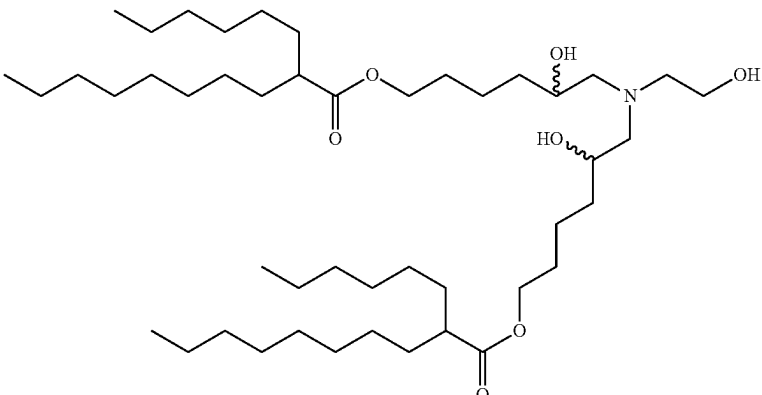
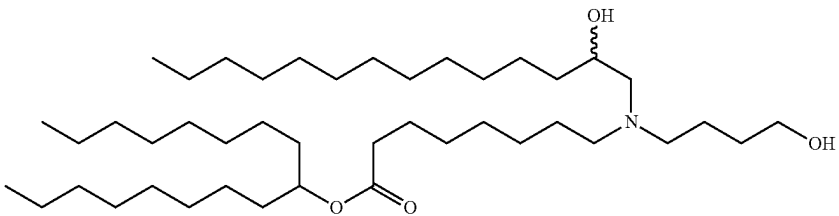
Ionizable lipid number	Structure
24	
25	
26	
27	

TABLE 10e-continued

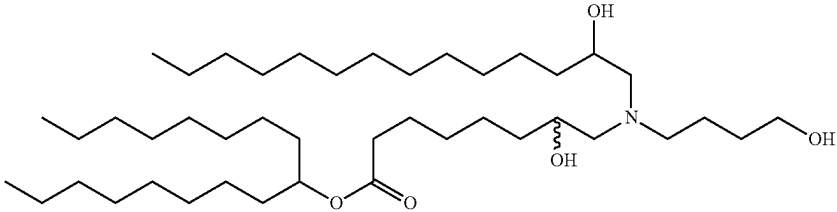
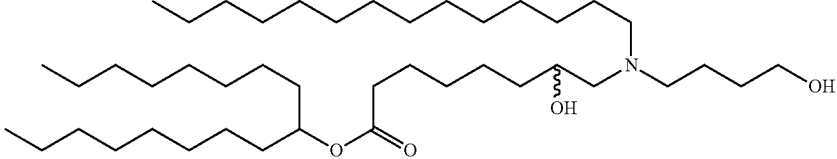
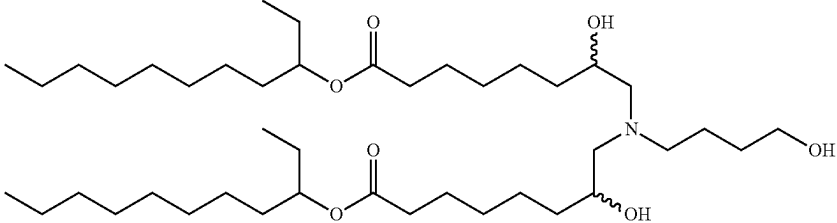
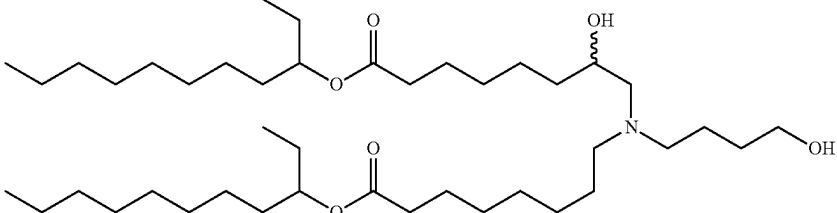
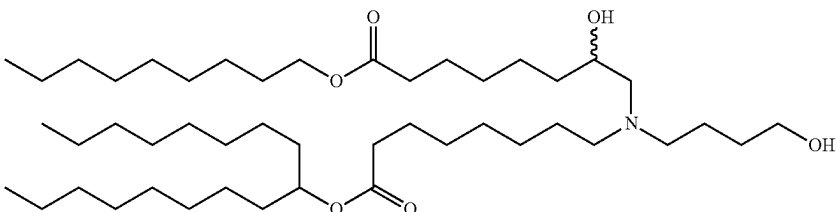
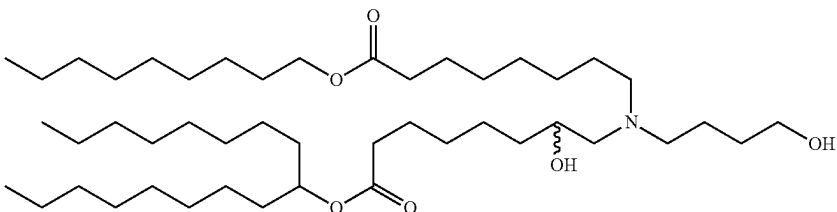
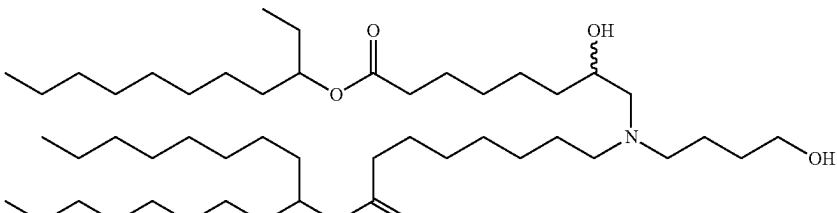
Ionizable lipid number	Structure
28	
29	
30	
31	
32	
33	
34	

TABLE 10e-continued

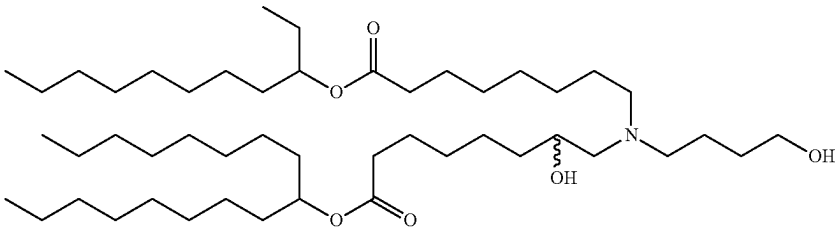
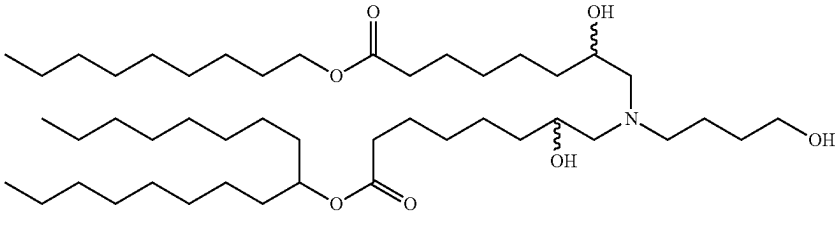
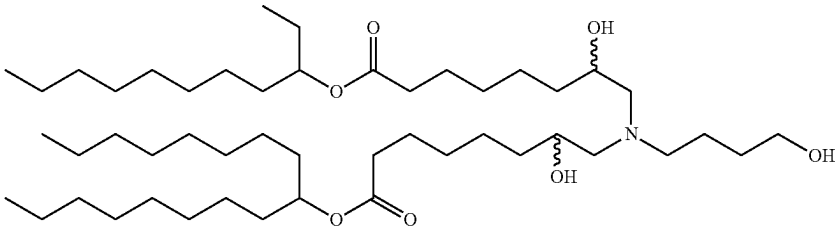
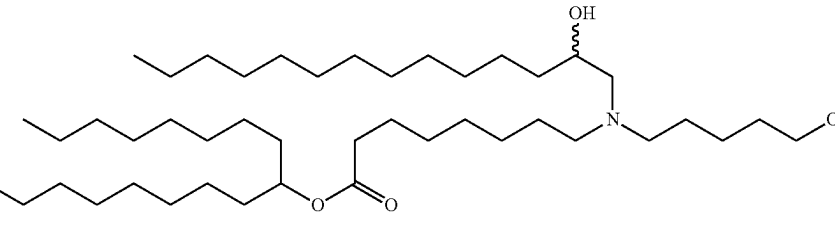
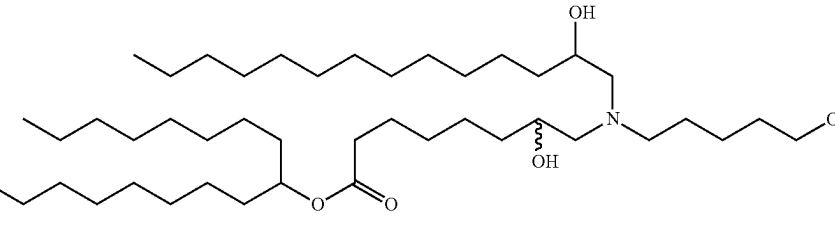
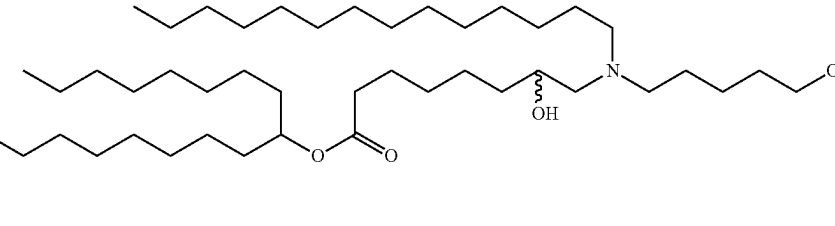
Ionizable lipid number	Structure
35	
36	
37	
38	
39	
40	

TABLE 10e-continued

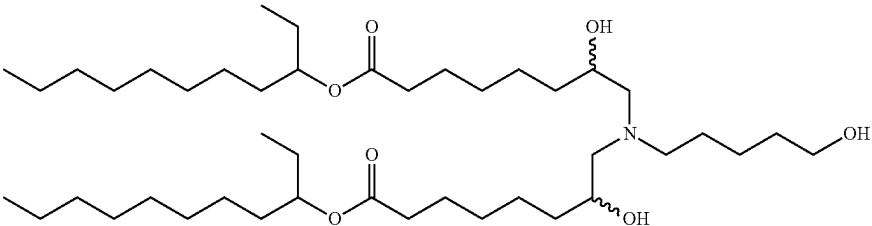
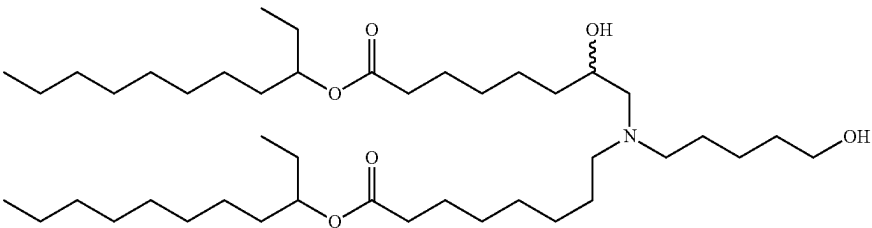
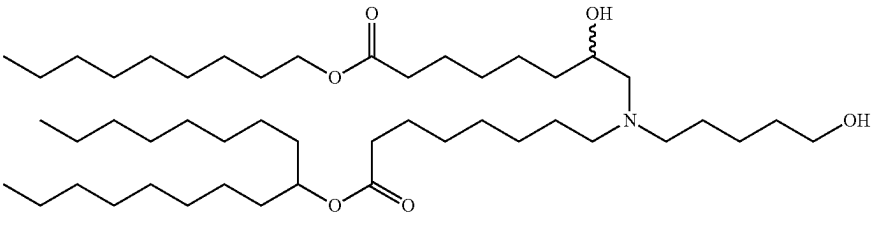
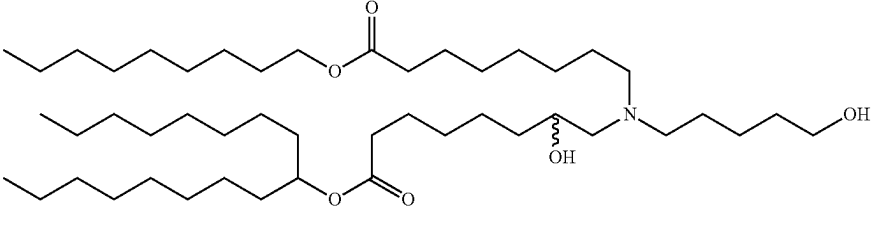
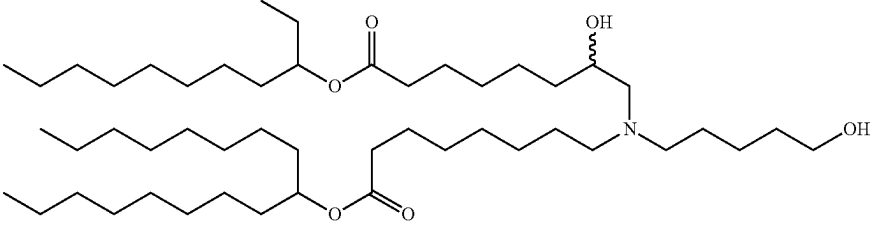
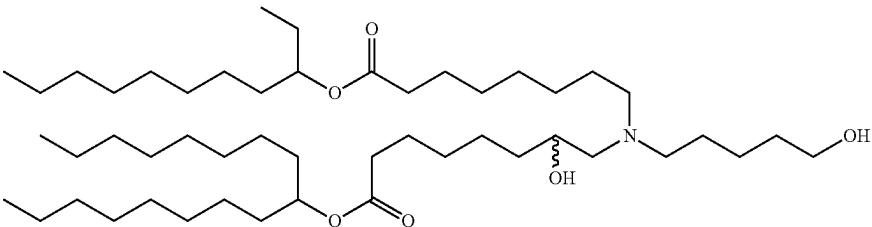
Ionizable lipid number	Structure
41	
42	
43	
44	
45	
46	

TABLE 10e-continued

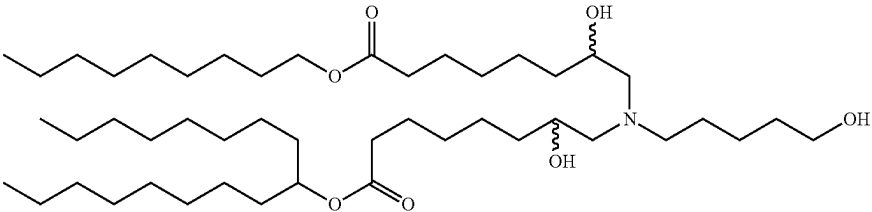
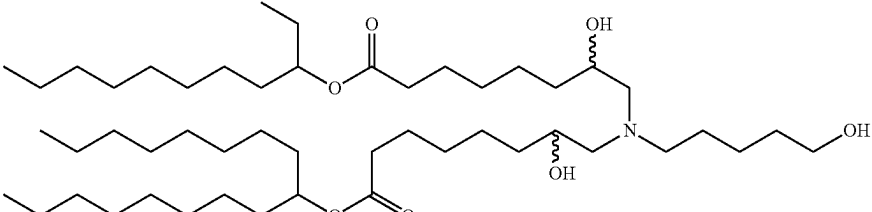
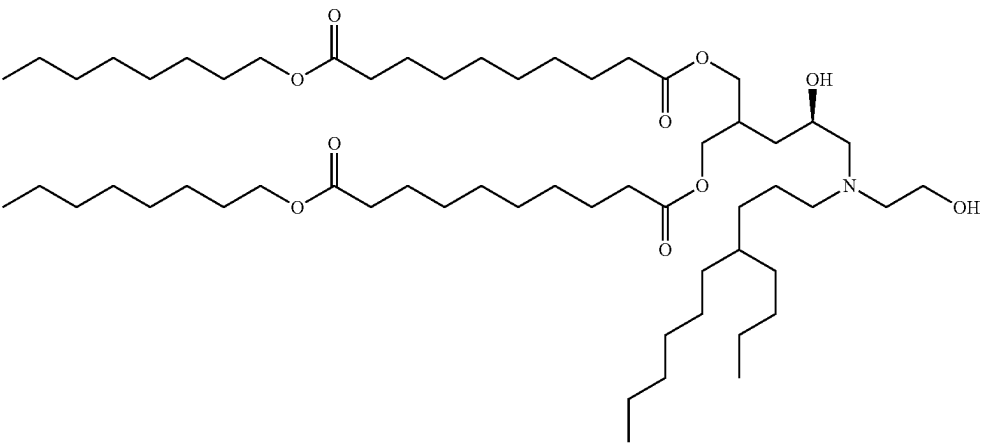
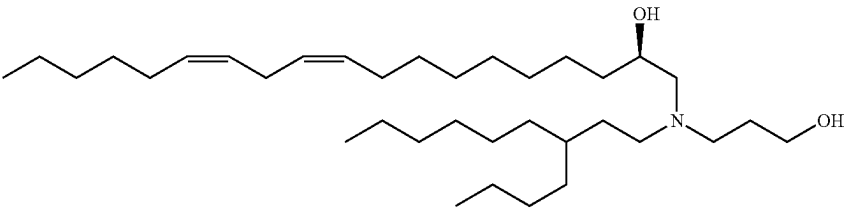
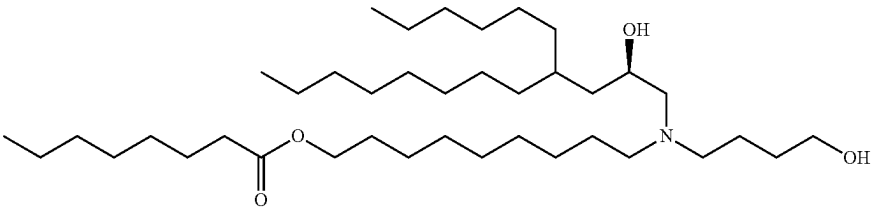
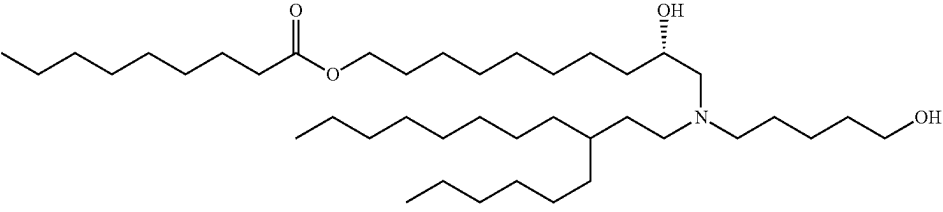
Ionizable lipid number	Structure
47	
48	
49	
50	
51	
52	

TABLE 10e-continued

Ionizable lipid number	Structure
53	
54	
55	
56	

TABLE 10e-continued

Ionizable lipid number	Structure
57	
58	
59	
60	

TABLE 10e-continued

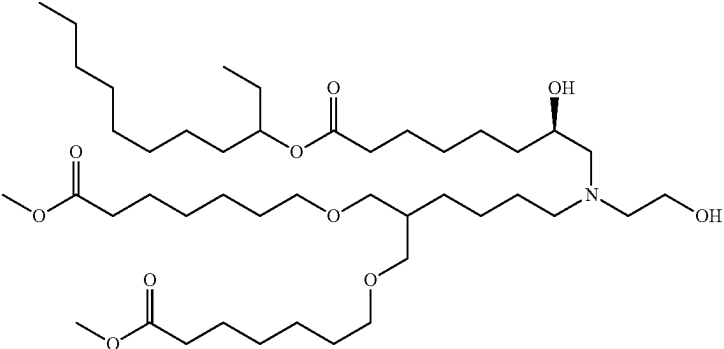
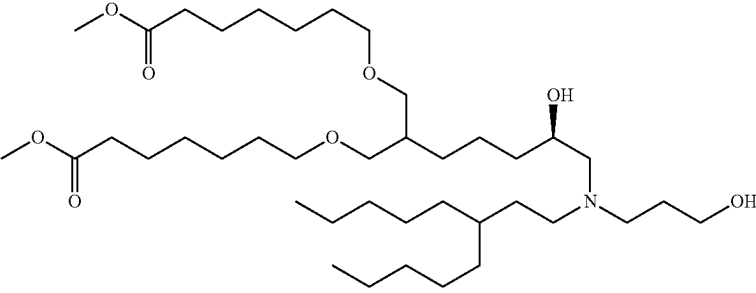
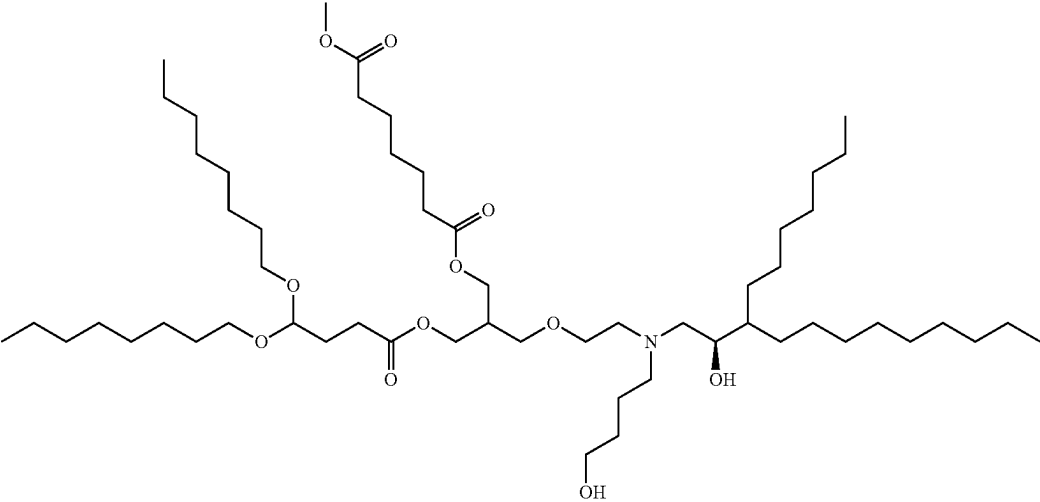
Ionizable lipid number	Structure
61	
62	
63	

TABLE 10e-continued

Ionizable lipid number	Structure
64	
65	
66	
67	

TABLE 10e-continued

Ionizable lipid number	Structure
68	
69	
70	
71	
72	

TABLE 10e-continued

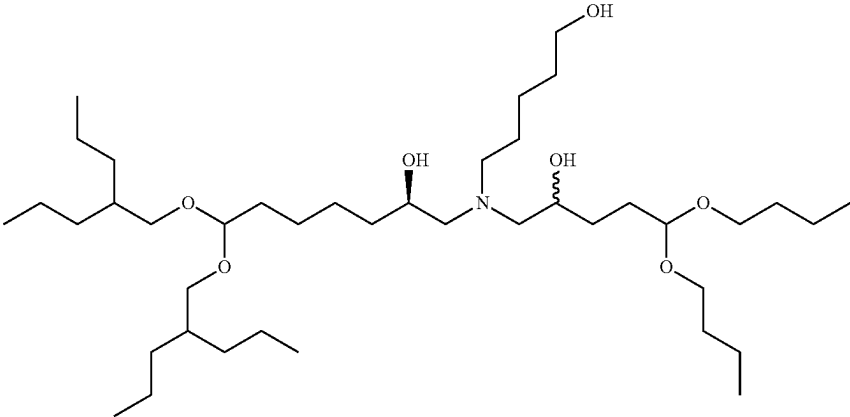
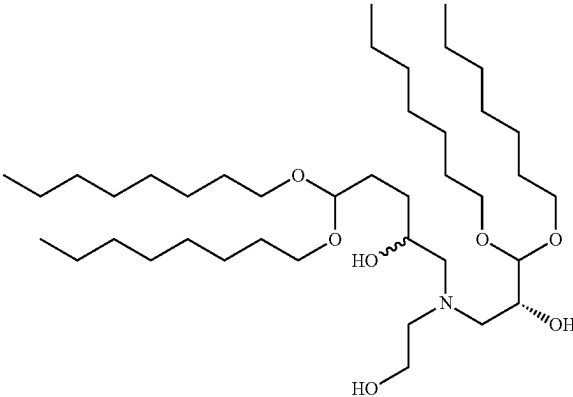
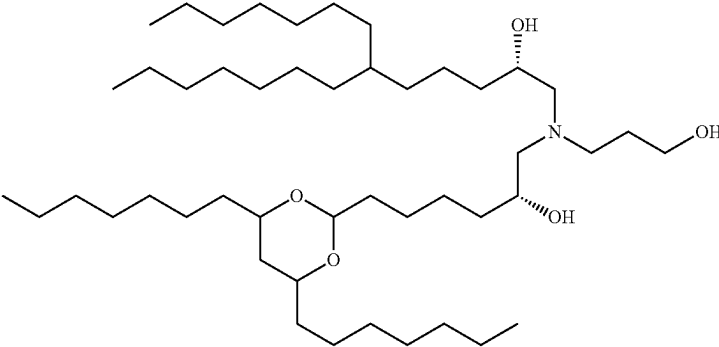
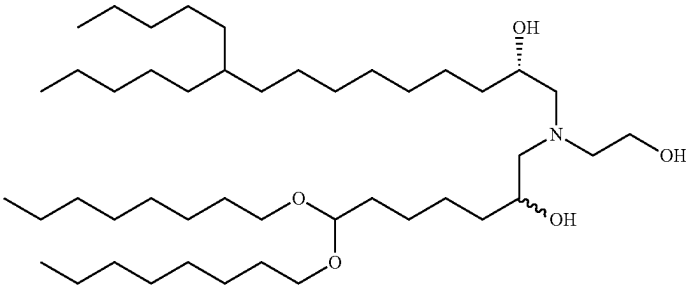
Ionizable lipid number	Structure
73	
74	
75	
76	

TABLE 10e-continued

Ionizable lipid number	Structure
77	
78	
79	

TABLE 10e-continued

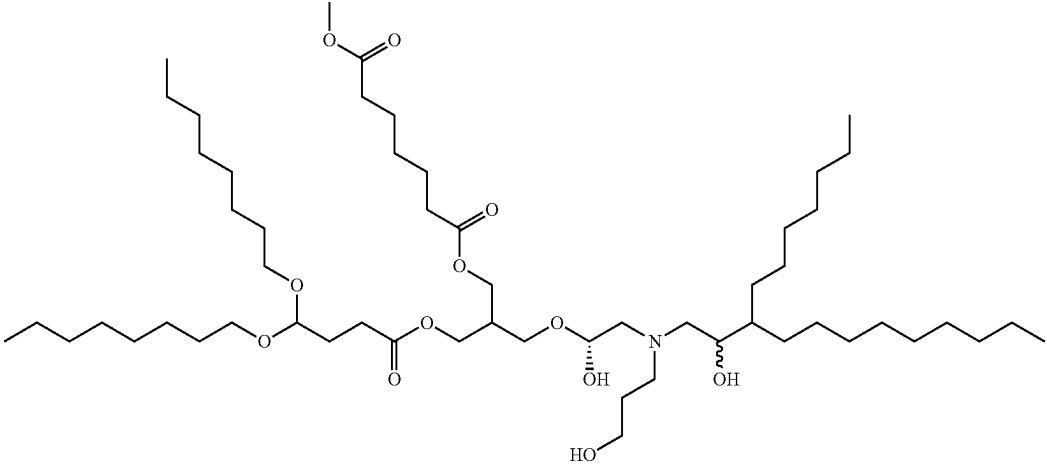
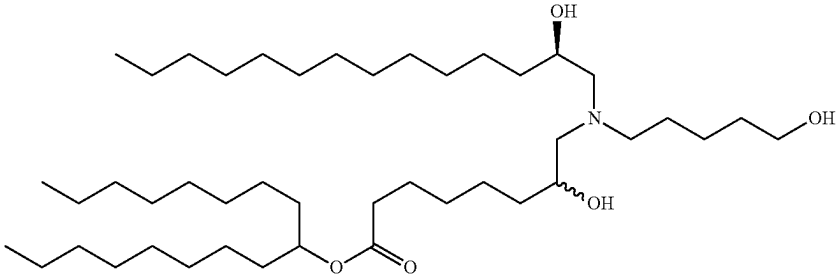
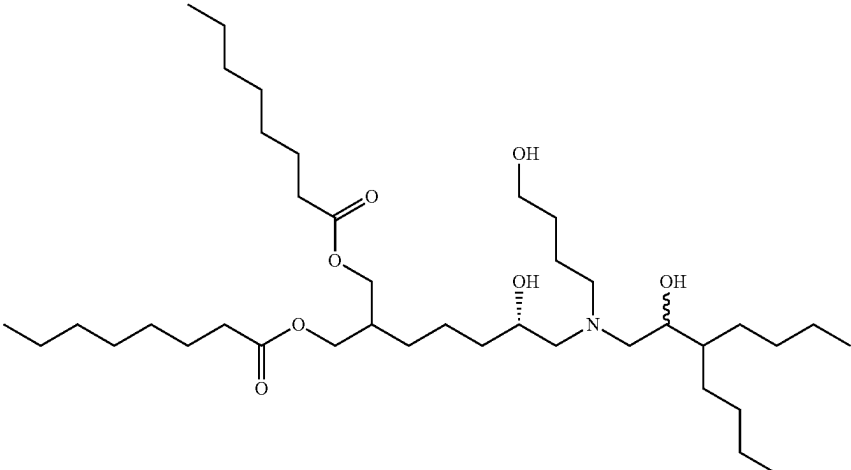
Ionizable lipid number	Structure
80	
81	
82	

TABLE 10e-continued

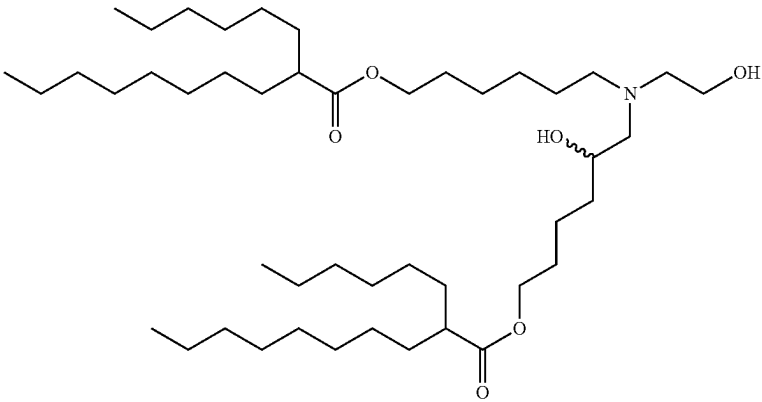
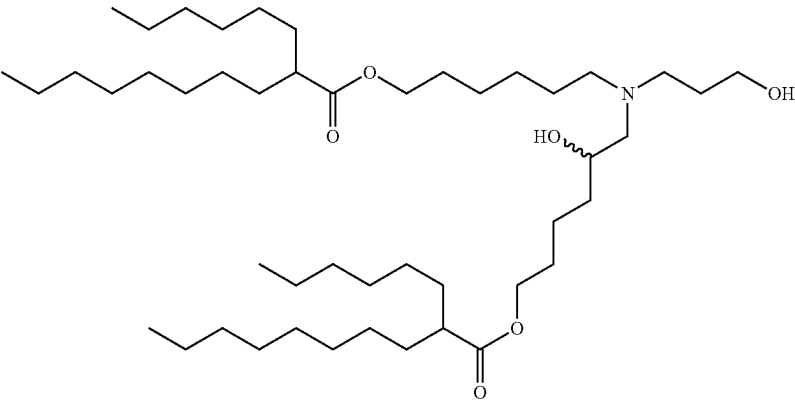
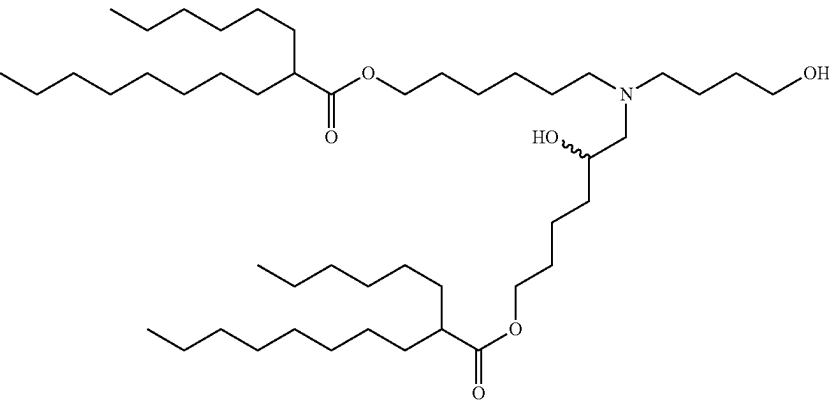
Ionizable lipid number	Structure
83	
84	
85	

TABLE 10e-continued

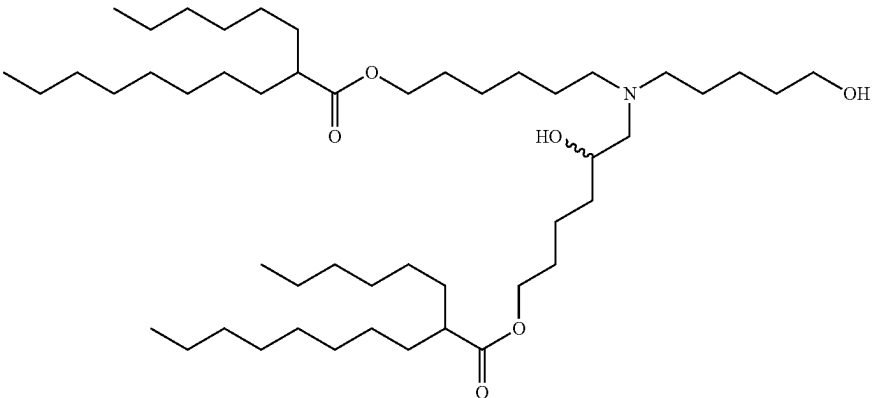
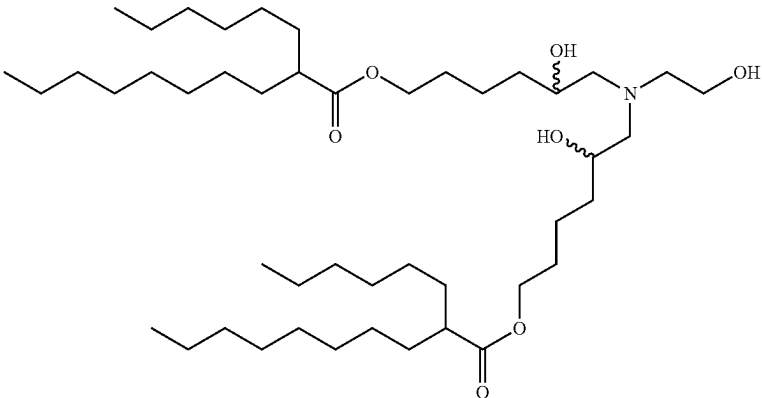
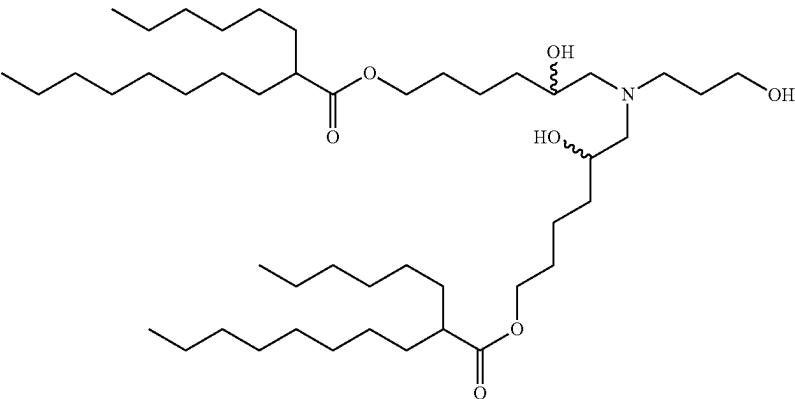
Ionizable lipid number	Structure
86	
87	
88	

TABLE 10e-continued

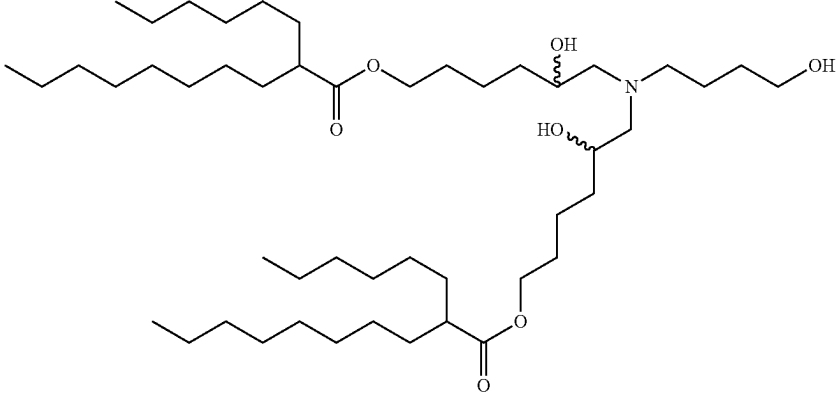
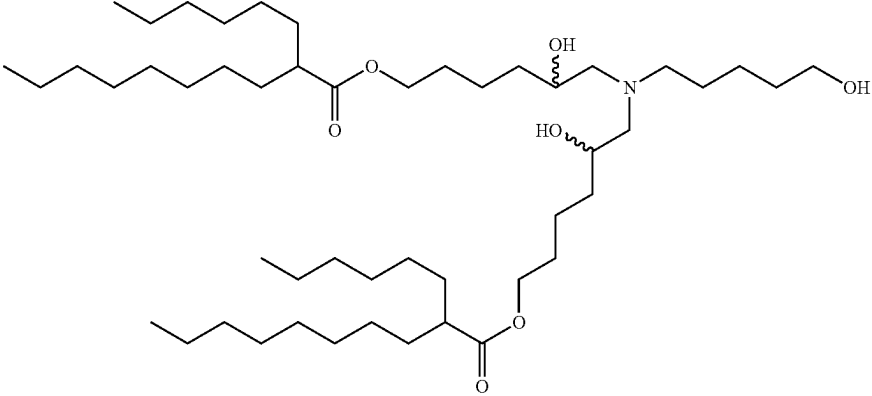
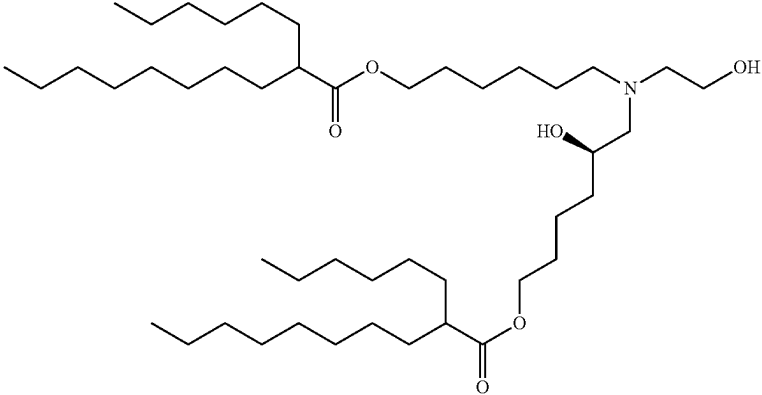
Ionizable lipid number	Structure
89	
90	
91	

TABLE 10e-continued

Ionizable lipid number	Structure
92	<chem>CCCCCCCCCCCC(=O)OCCN(CCCO)CCOC(=O)CCCCCCCCCCC</chem>
93	<chem>CCCCCCCCCCCC(=O)OCCN(CCCO)CCOC(=O)CCCCCCCCCCC</chem>
94	<chem>CCCCCCCCCCCC(=O)OCCN(CCCO)CCOC(=O)CCCCCCCCCCC</chem>

TABLE 10e-continued

Ionizable lipid number	Structure
95	<chem>CCCCCCCCCCCC(=O)OCC[C@H](O)CNCCOCCCC(=O)OCC[C@H](O)CO</chem>
96	<chem>CCCCCCCCCCCC(=O)OCC[C@H](O)CNCCOCCCC(=O)OCC[C@@H](O)CO</chem>
97	<chem>CCCCCCCCCCCC(=O)OCC[C@@H](O)CNCCOCCCC(=O)OCC[C@H](O)CO</chem>

TABLE 10e-continued

Ionizable lipid number	Structure
98	<chem>CCCCCCCCCCCC(=O)OCC(O)CN(CCO)CCO</chem>
99	<chem>CCCCCCCCCCCC(=O)OCC(O)CN(CCO)CCO</chem>
100	<chem>CCCCCCCCCCCC(=O)OCC(O)CN(CCO)CCO</chem>

TABLE 10e-continued

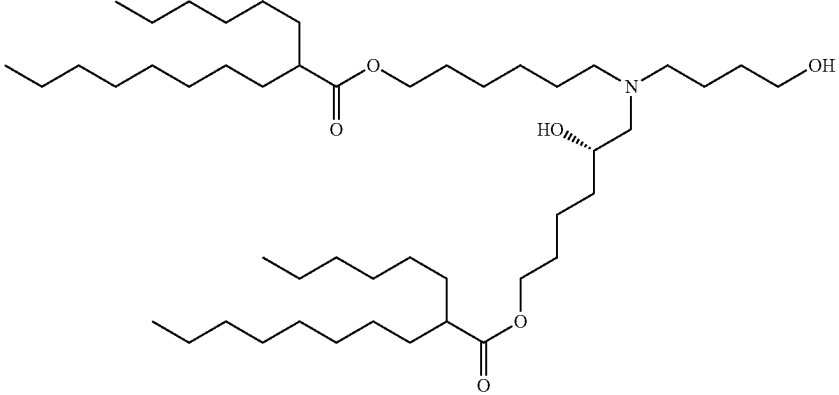
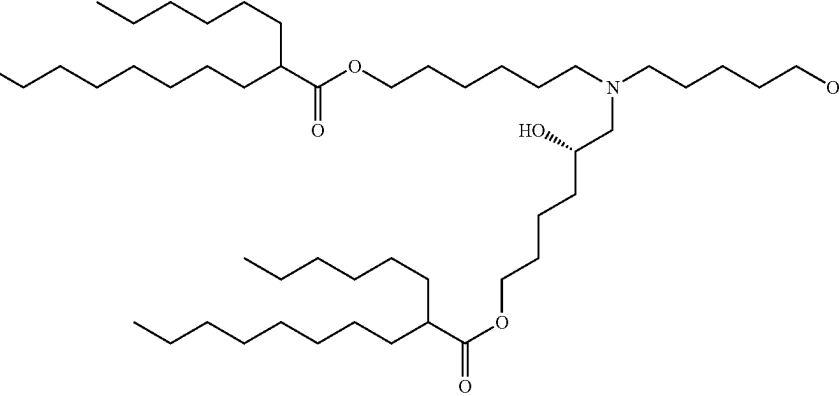
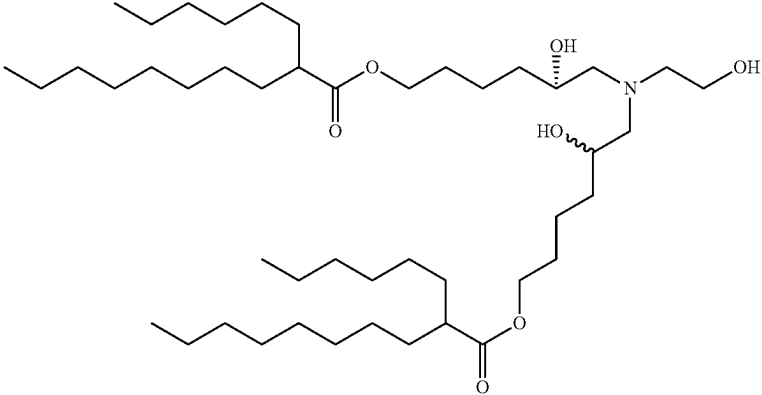
Ionizable lipid number	Structure
101	
102	
103	

TABLE 10e-continued

Ionizable lipid number	Structure
104	
105	
106	

TABLE 10e-continued

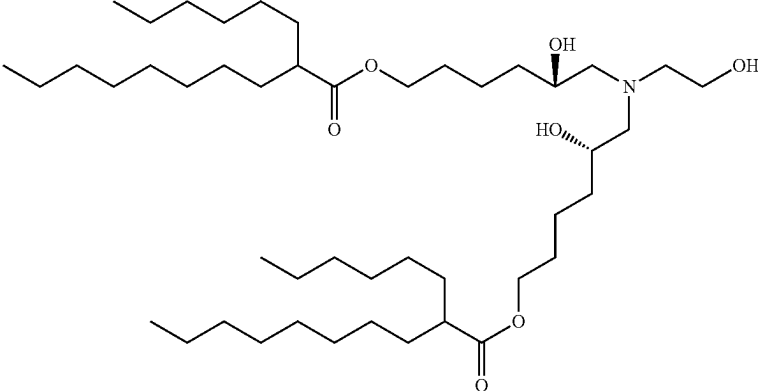
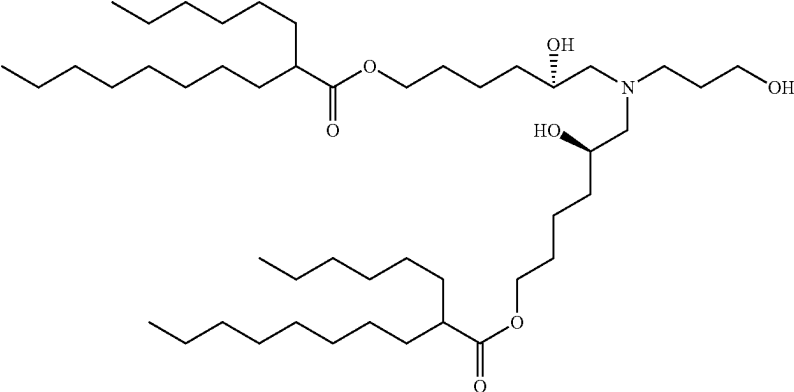
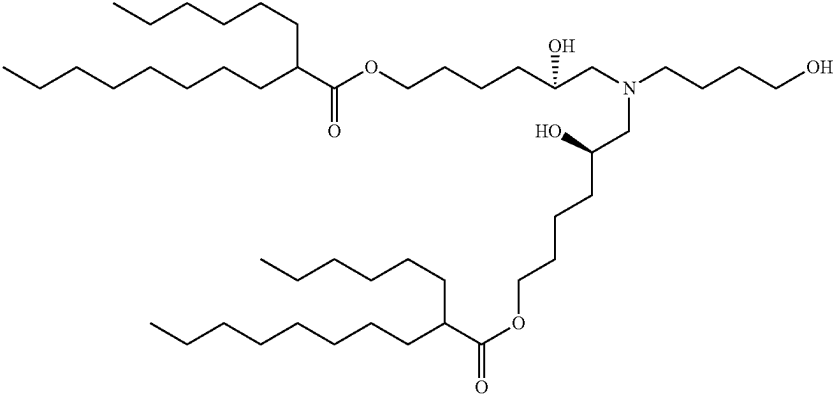
Ionizable lipid number	Structure
107	
108	
109	

TABLE 10e-continued

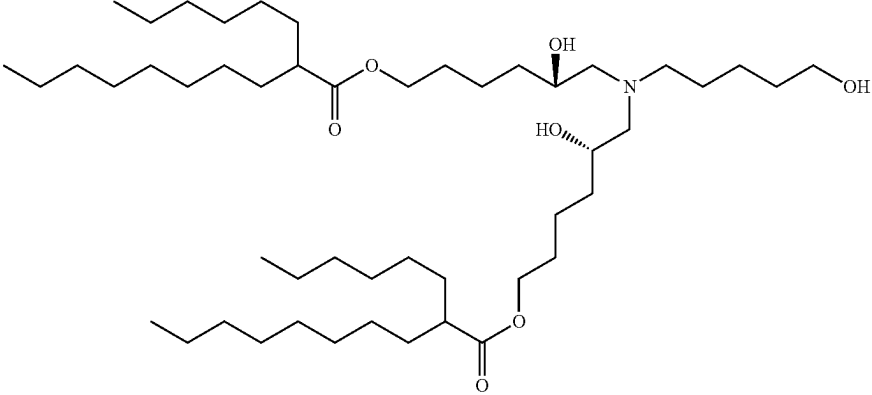
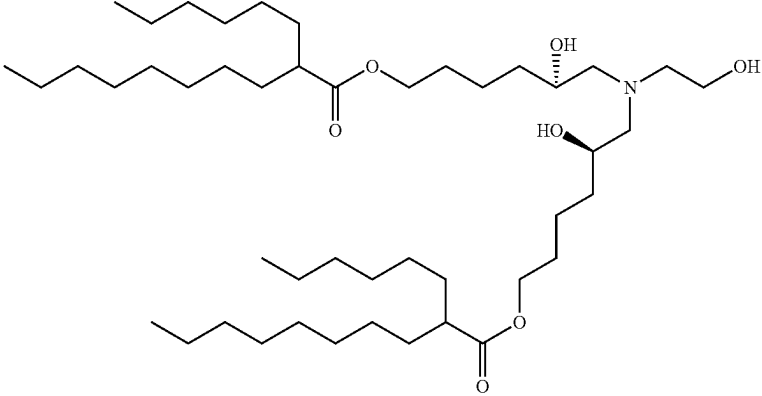
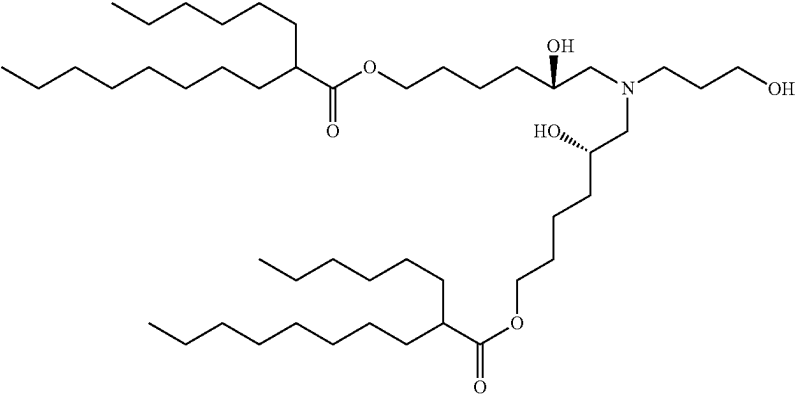
Ionizable lipid number	Structure
110	
111	
112	

TABLE 10e-continued

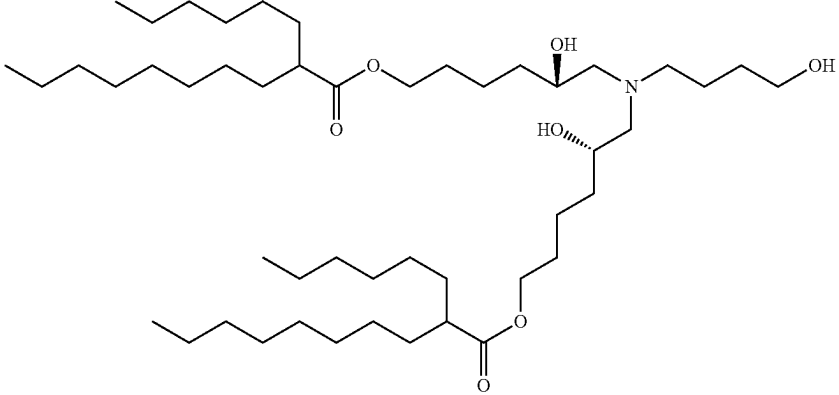
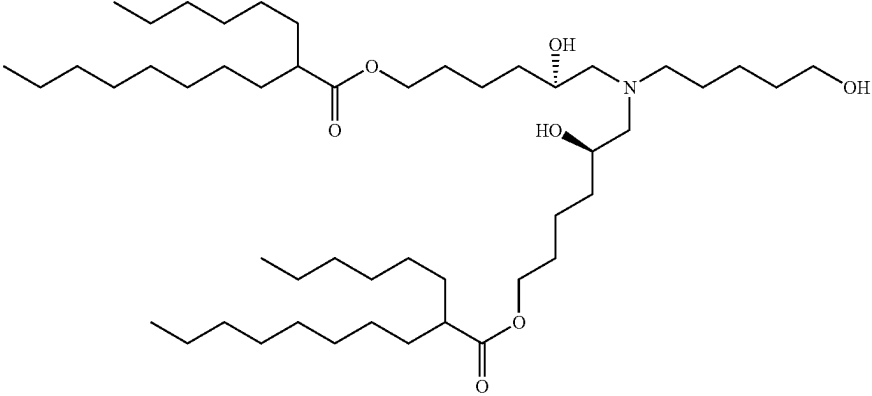
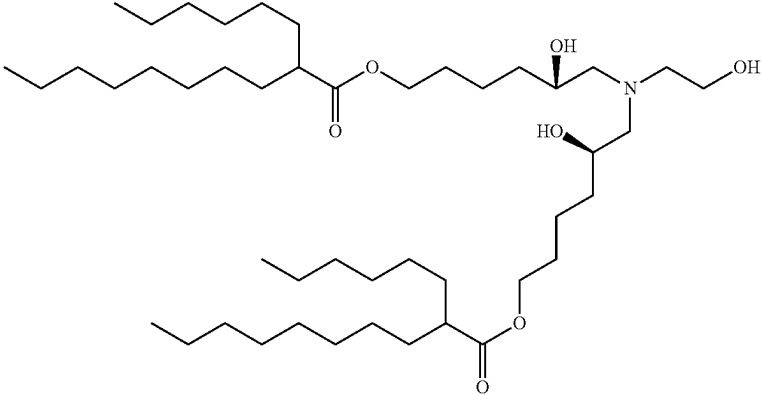
Ionizable lipid number	Structure
113	
114	
115	

TABLE 10e-continued

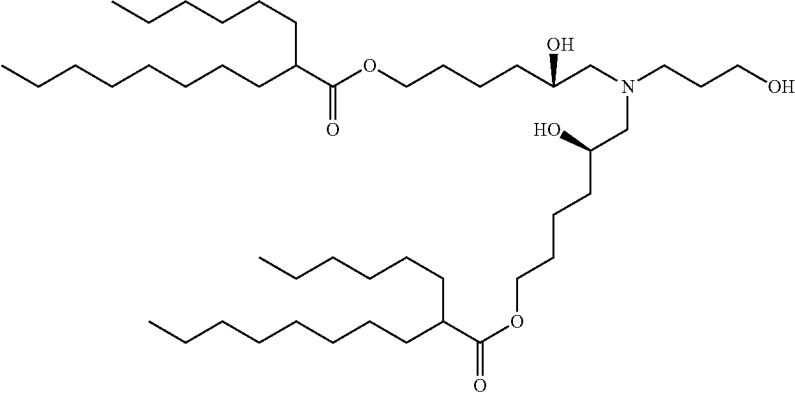
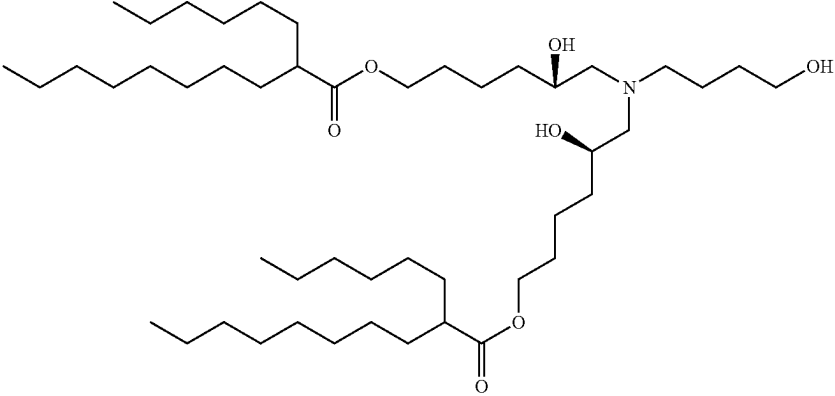
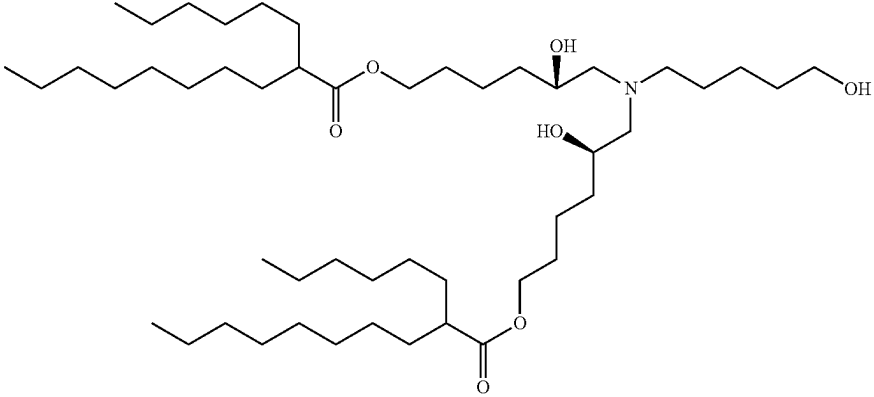
Ionizable lipid number	Structure
116	
117	
118	

TABLE 10e-continued

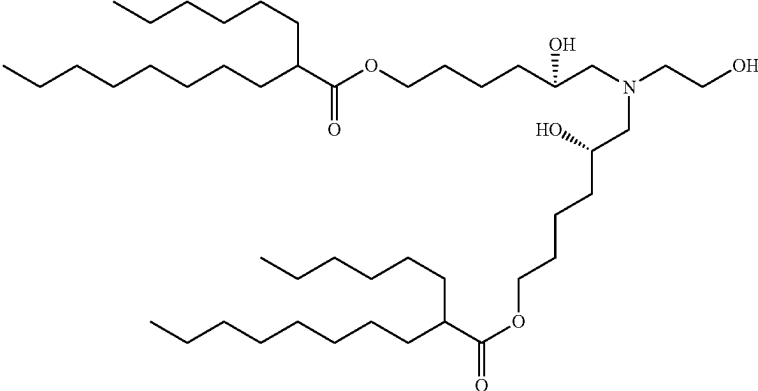
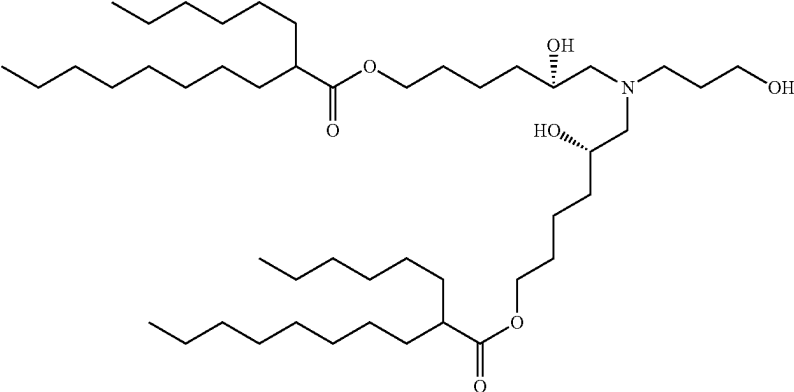
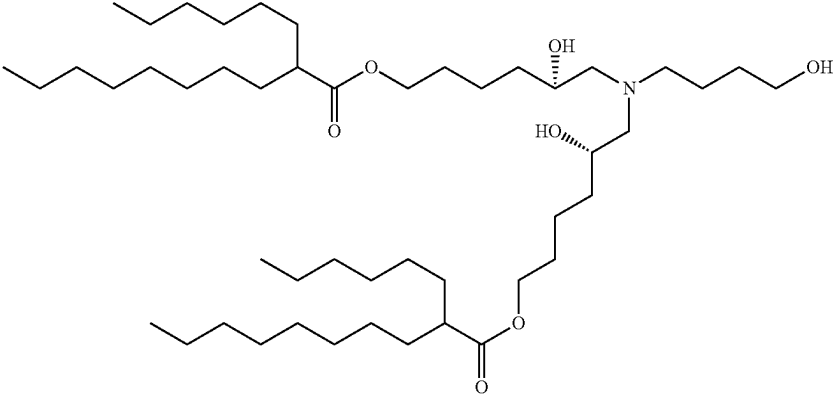
Ionizable lipid number	Structure
119	
120	
121	

TABLE 10e-continued

Ionizable lipid number	Structure
122	

TABLE 10f

Ionizable lipid number	Structure
1	
2	

TABLE 10f-continued

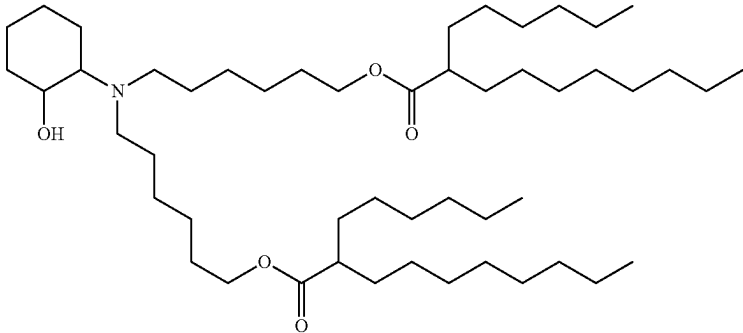
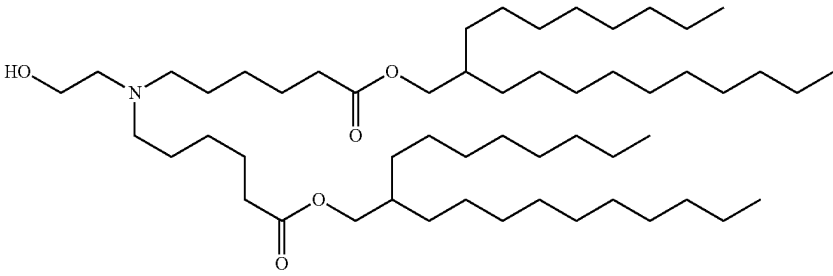
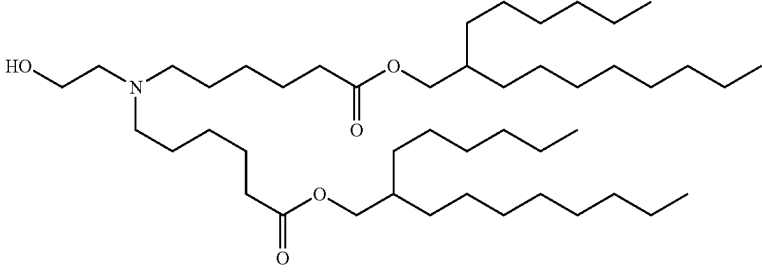
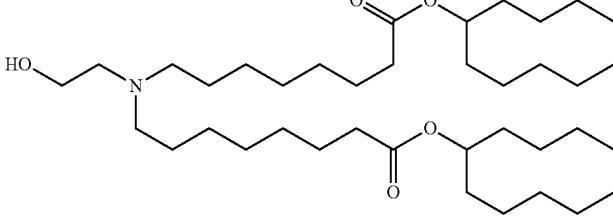
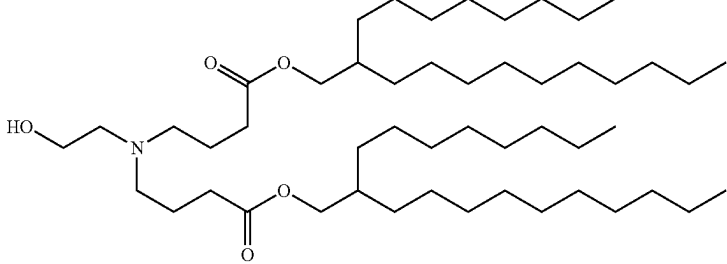
Ionizable lipid number	Structure
3	
4	
5	
6	
7	

TABLE 10f-continued

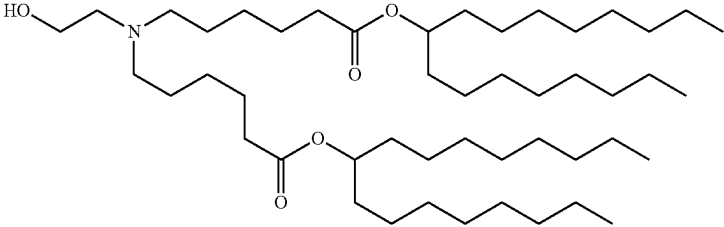
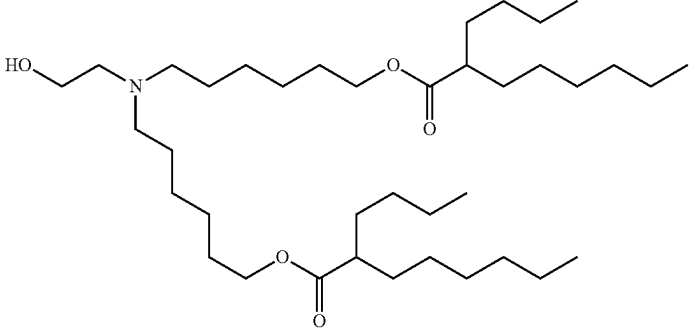
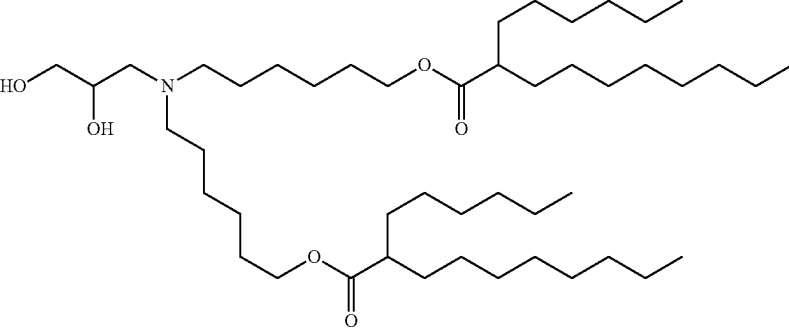
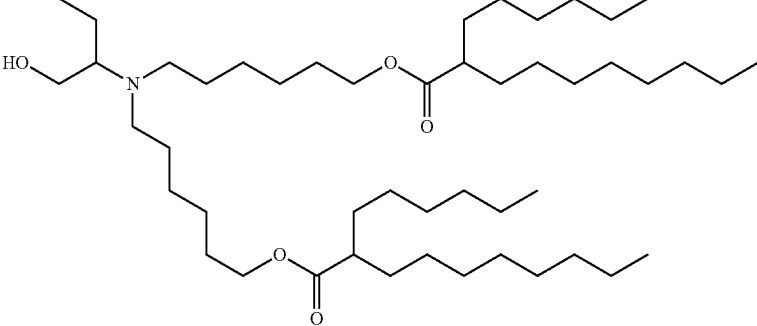
Ionizable lipid number	Structure
8	
9	
10	
11	

TABLE 10f-continued

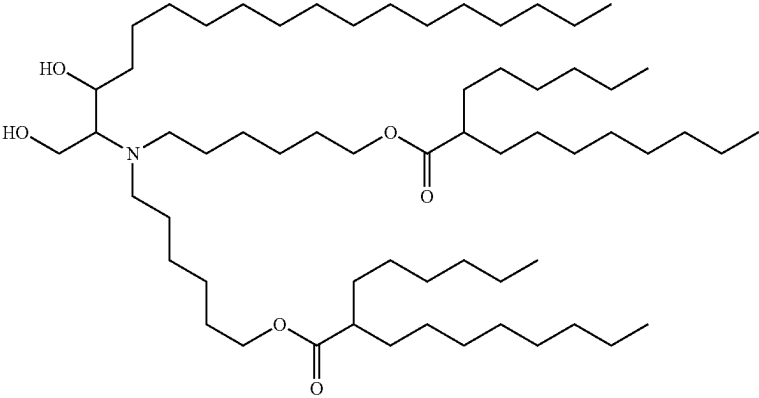
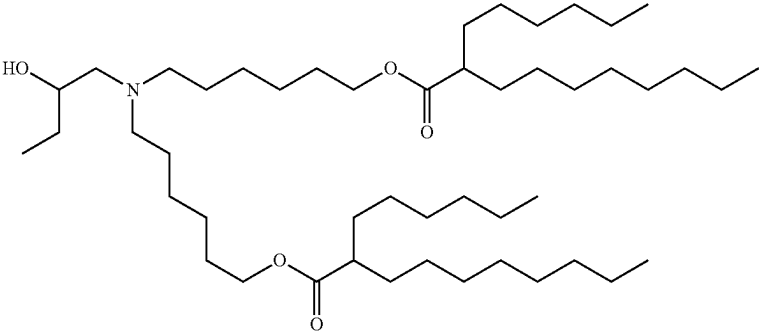
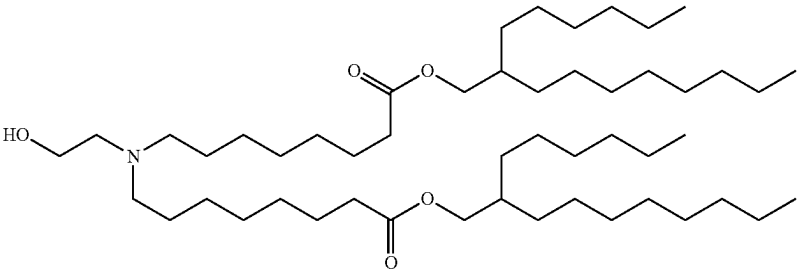
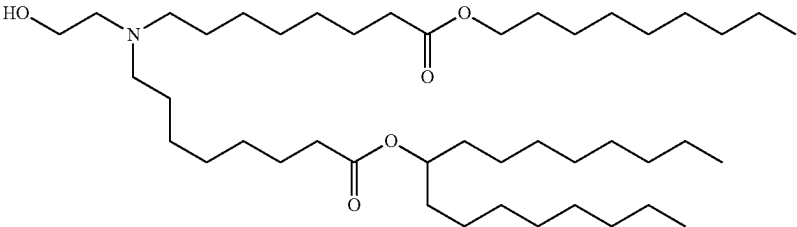
Ionizable lipid number	Structure
12	
13	
14	
15	

TABLE 10f-continued

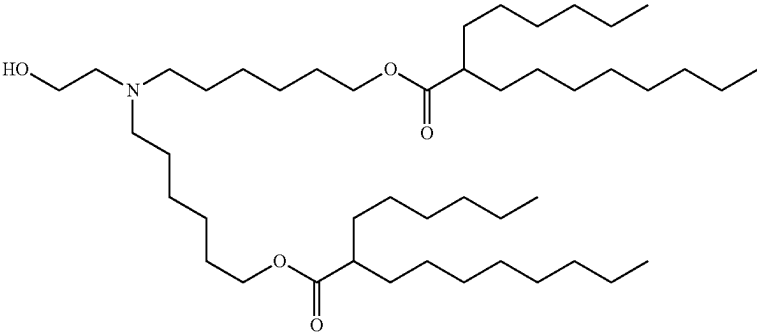
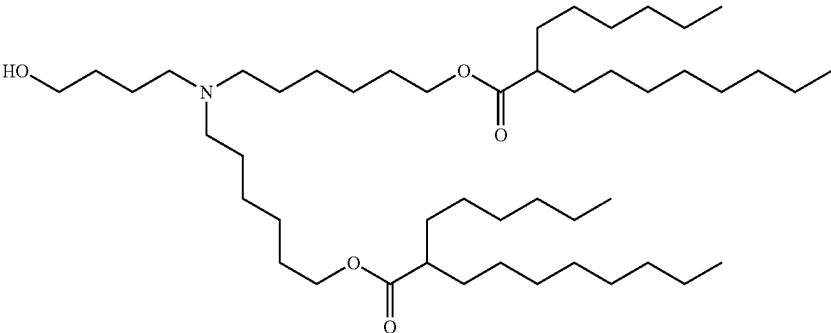
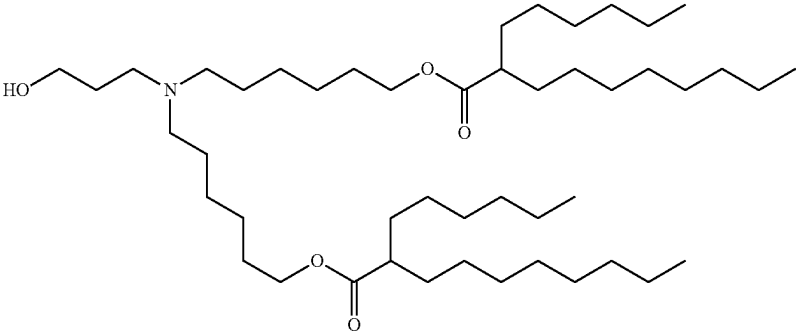
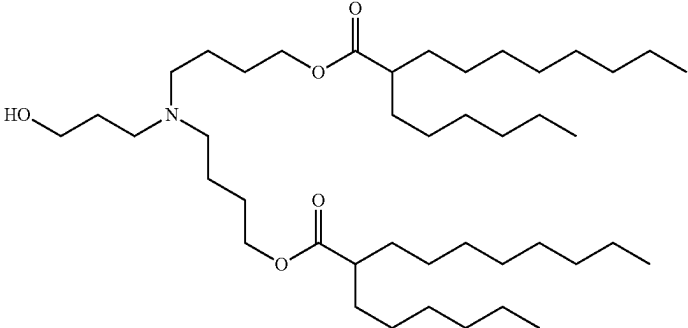
Ionizable lipid number	Structure
16	
17	
18	
19	

TABLE 10f-continued

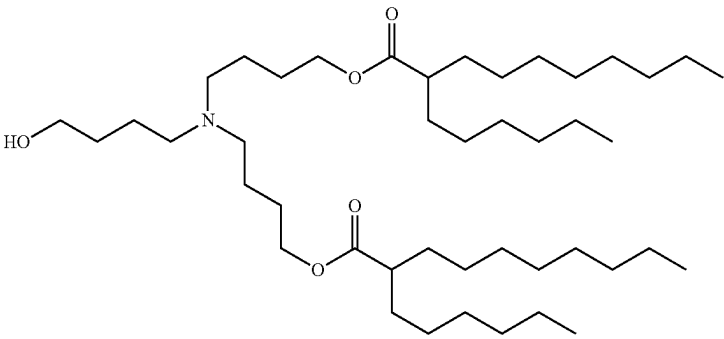
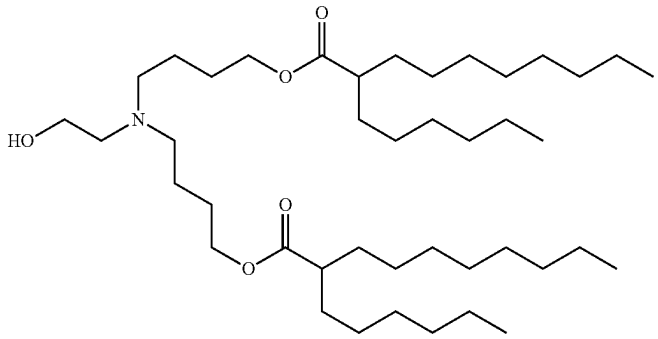
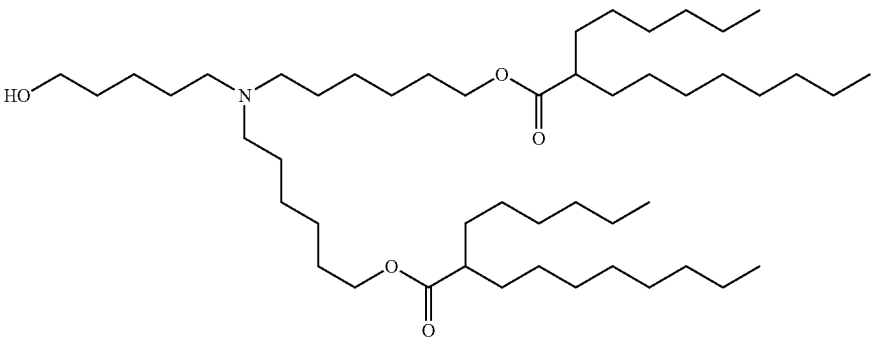
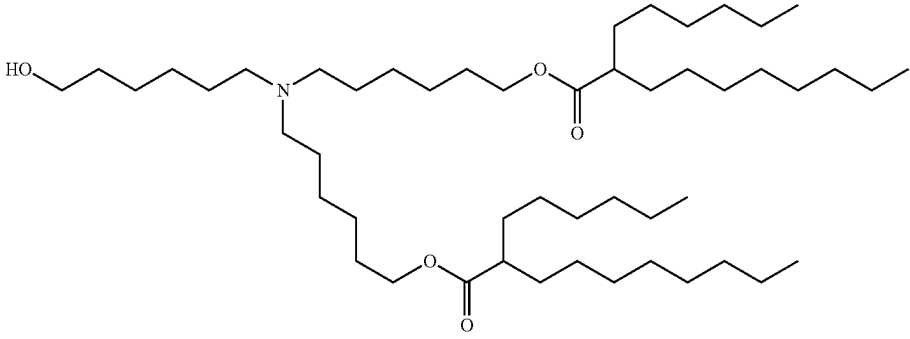
Ionizable lipid number	Structure
20	
21	
22	
23	

TABLE 10f-continued

Ionizable lipid number	Structure
24	
25	
26	
27	
28	

TABLE 10f-continued

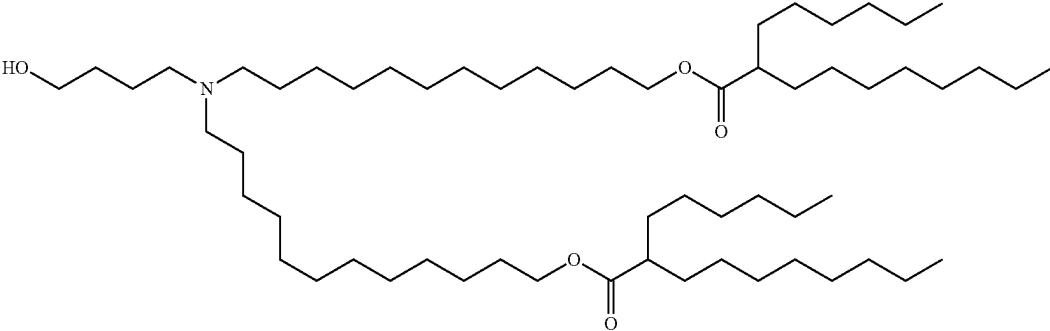
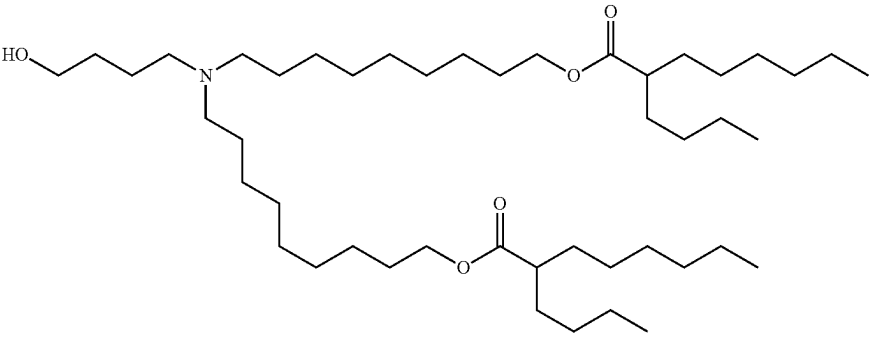
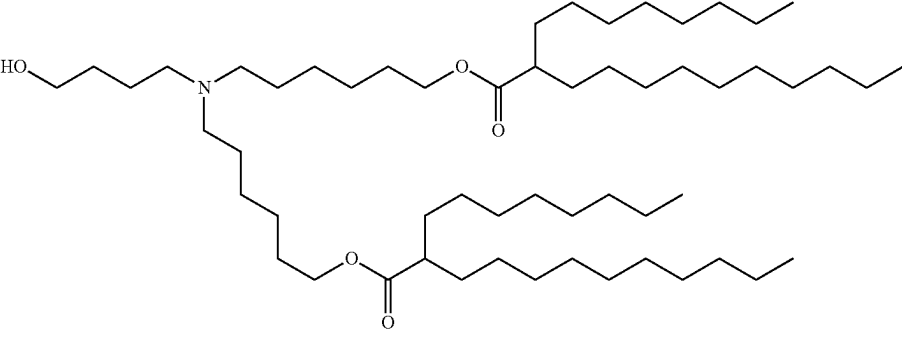
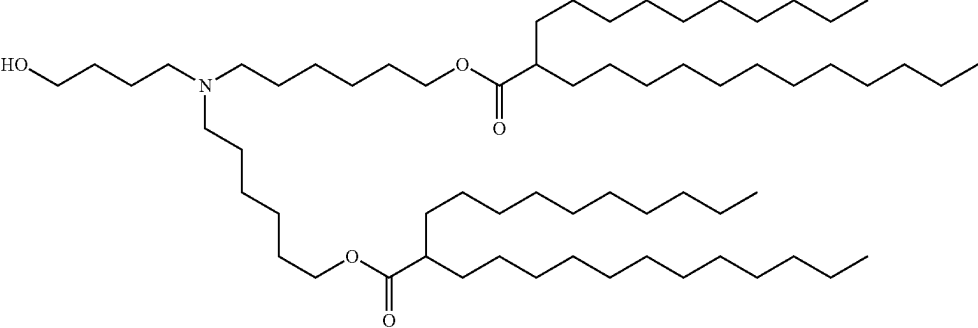
Ionizable lipid number	Structure
29	
30	
31	
32	

TABLE 10f-continued

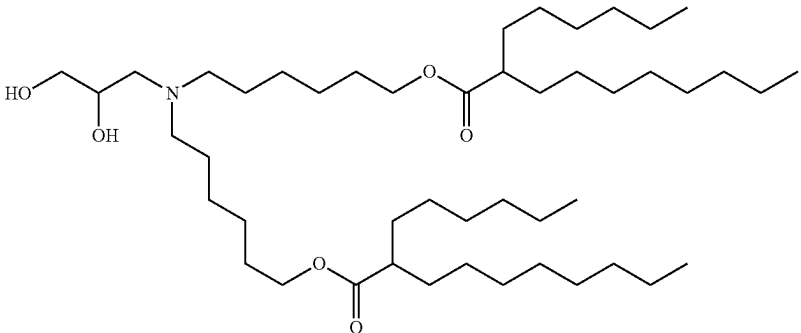
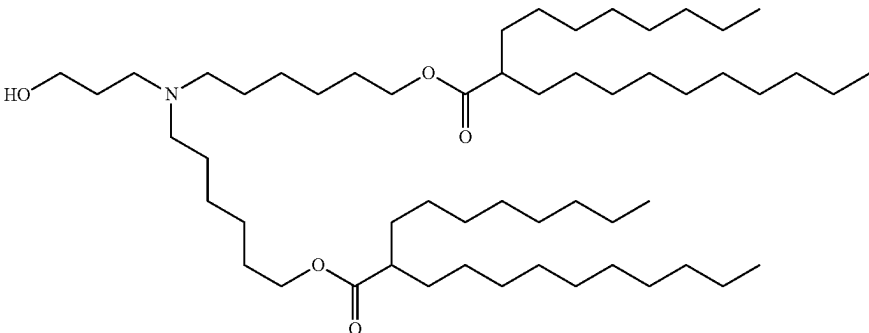
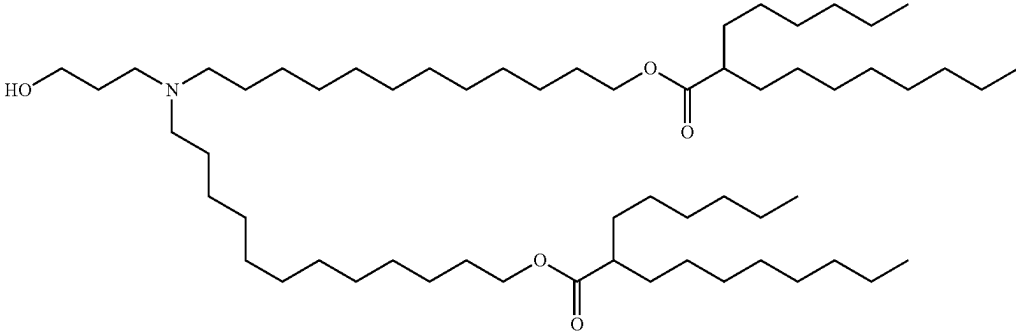
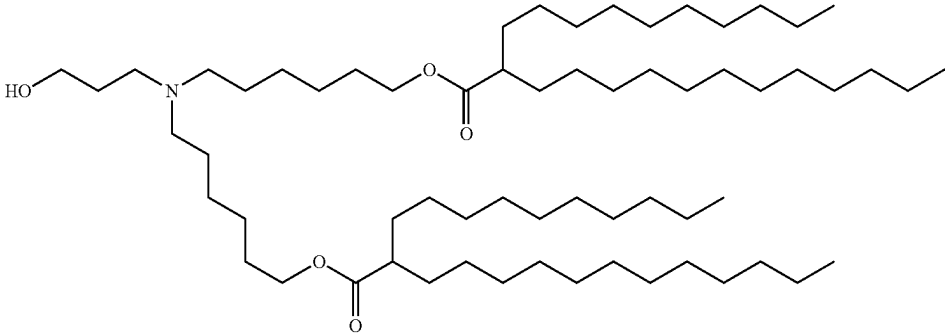
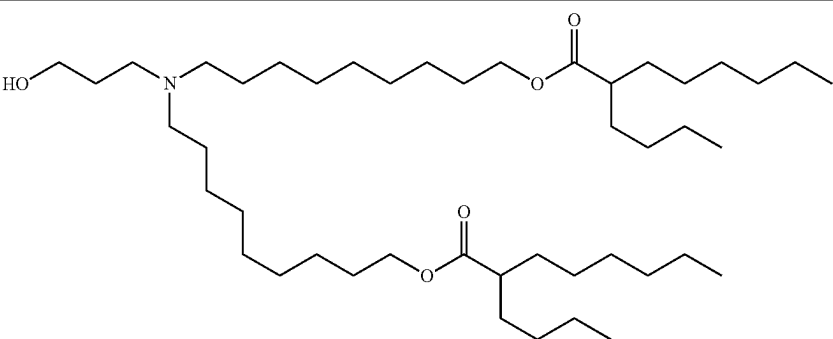
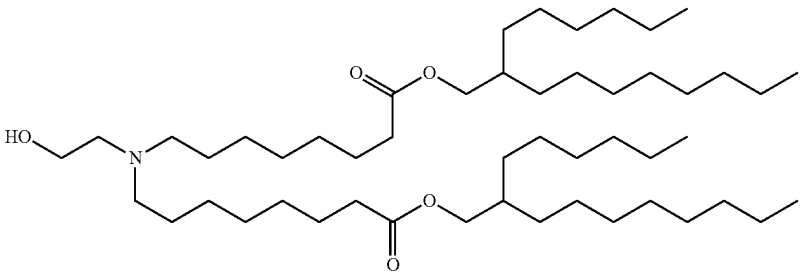
Ionizable lipid number	Structure
33	
34	
35	
36	

TABLE 10f-continued

Ionizable lipid number	Structure
37	
38	

In some embodiments, an ionizable lipid has one of the structures set forth in Table 10 below.

TABLE 10g

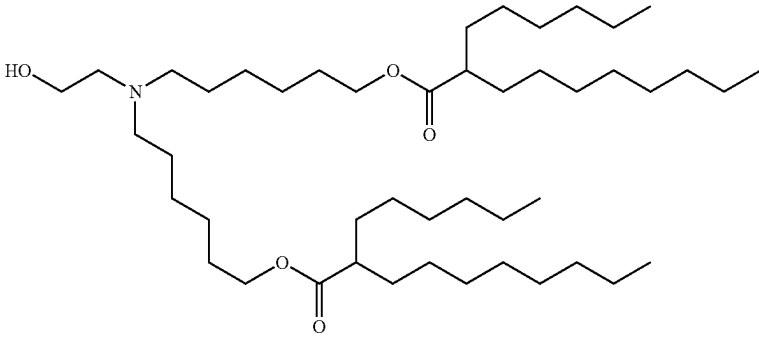
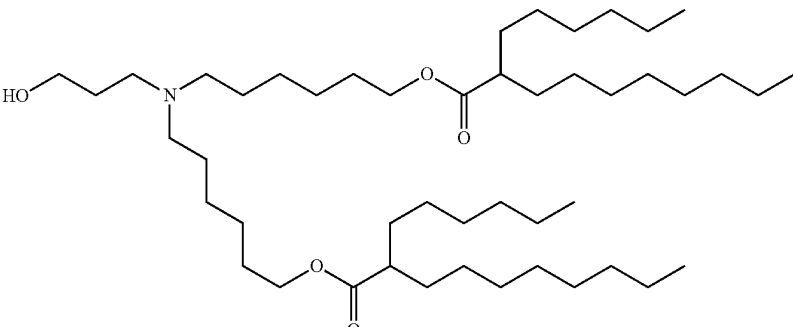
Number	Structure
1	
2	

TABLE 10g-continued

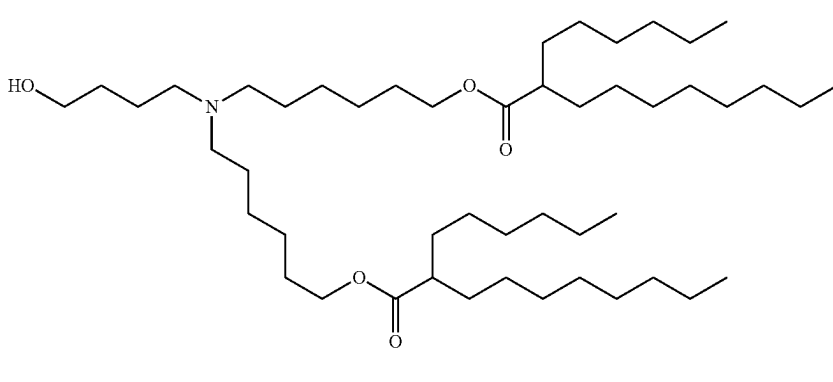
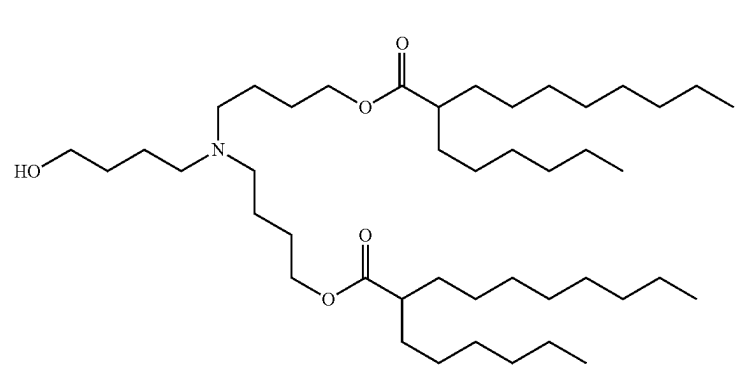
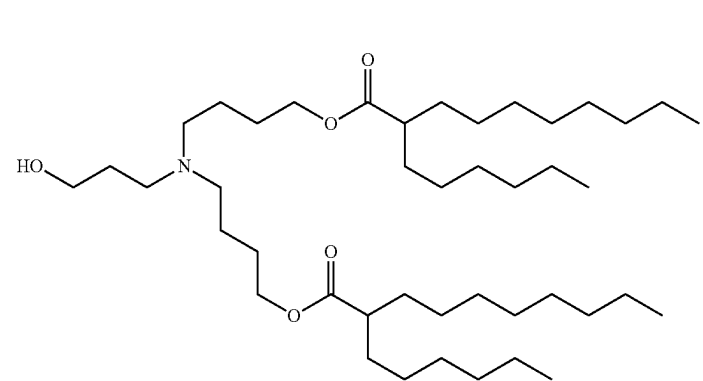
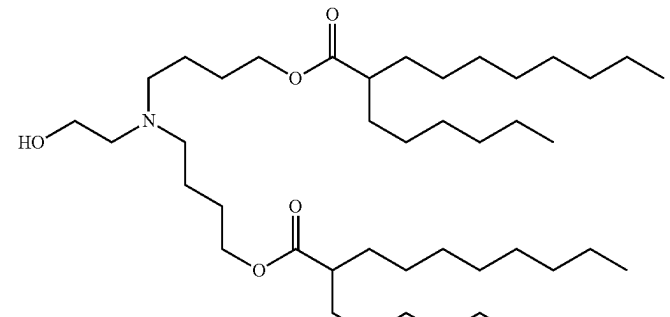
Number	Structure
3	
4	
5	
6	

TABLE 10g-continued

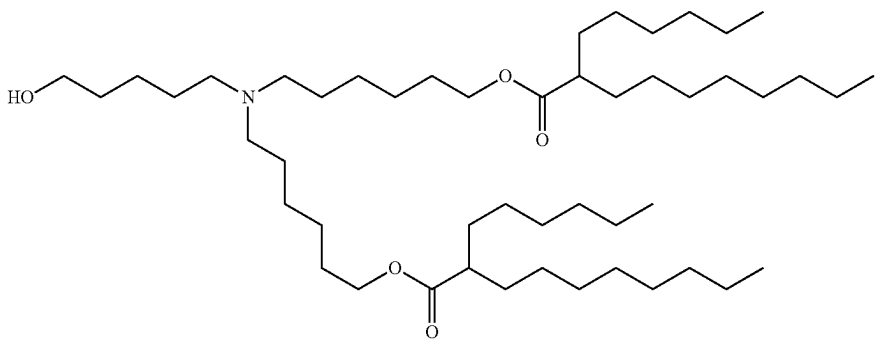
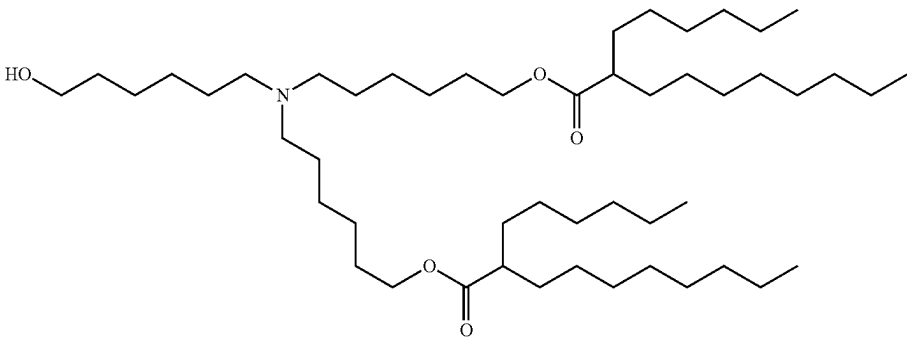
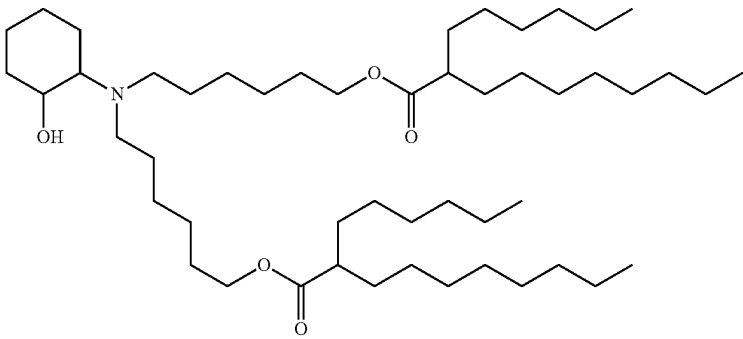
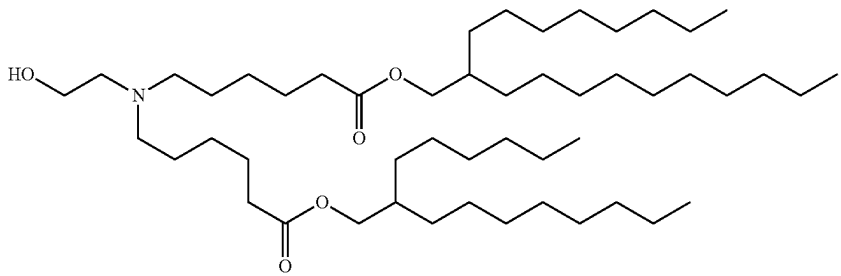
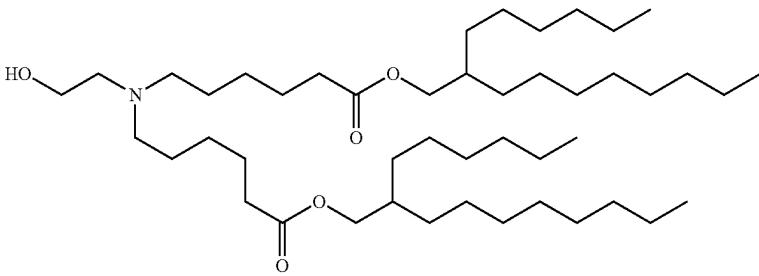
Number	Structure
7	
8	
9	
10	
11	

TABLE 10g-continued

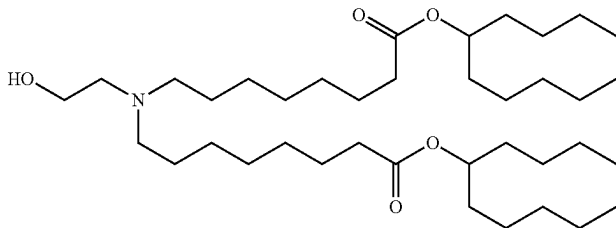
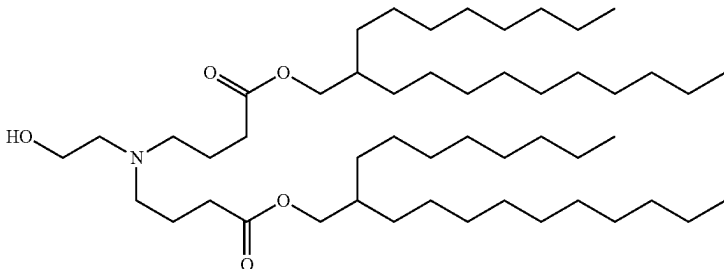
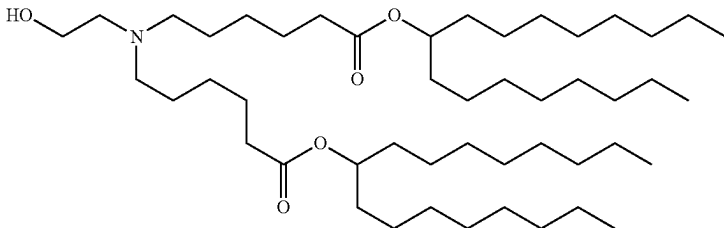
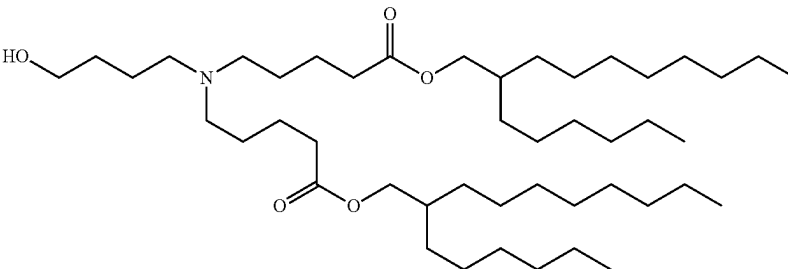
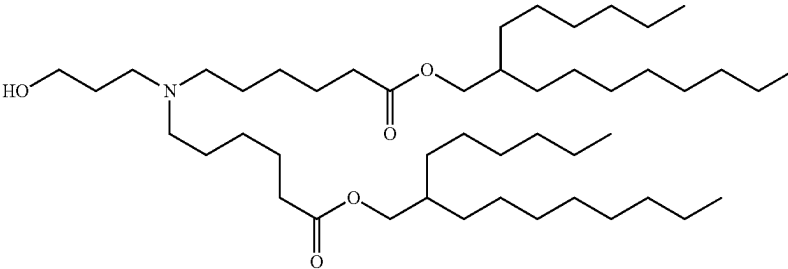
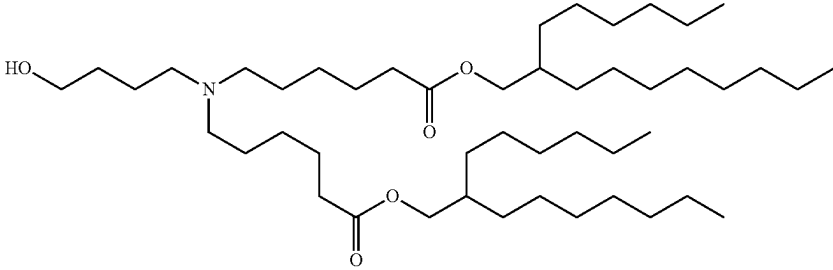
Number	Structure
12	
13	
14	
15	
16	
17	

TABLE 10g-continued

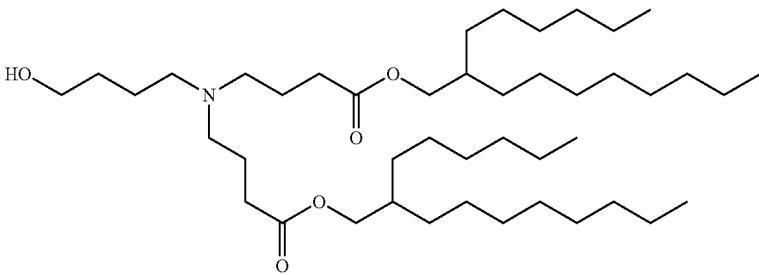
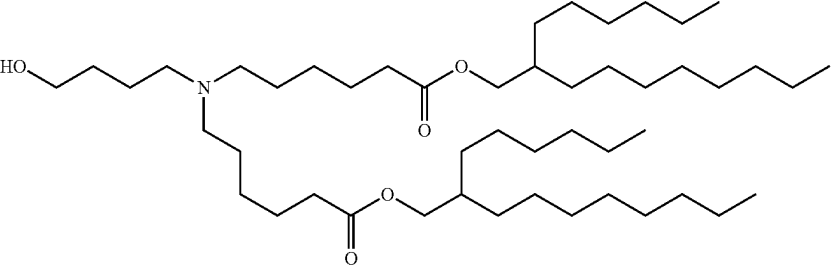
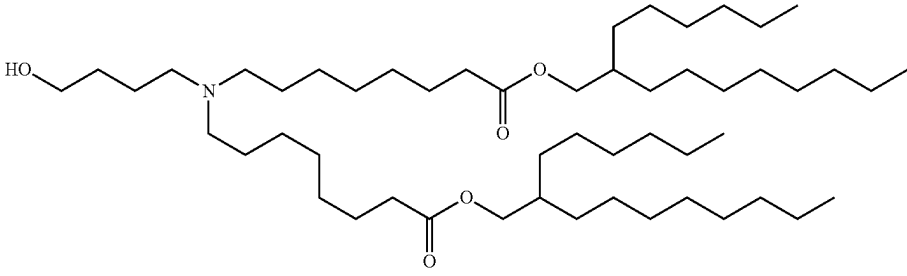
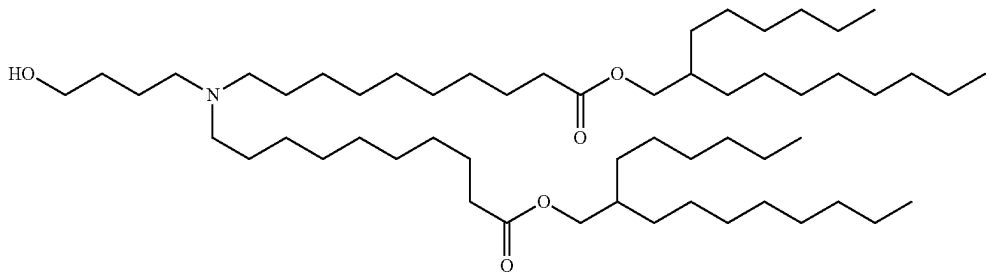
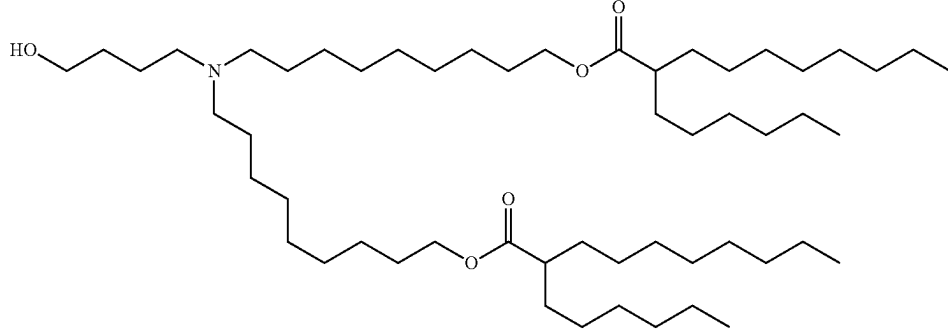
Number	Structure
18	
19	
20	
21	
22	

TABLE 10g-continued

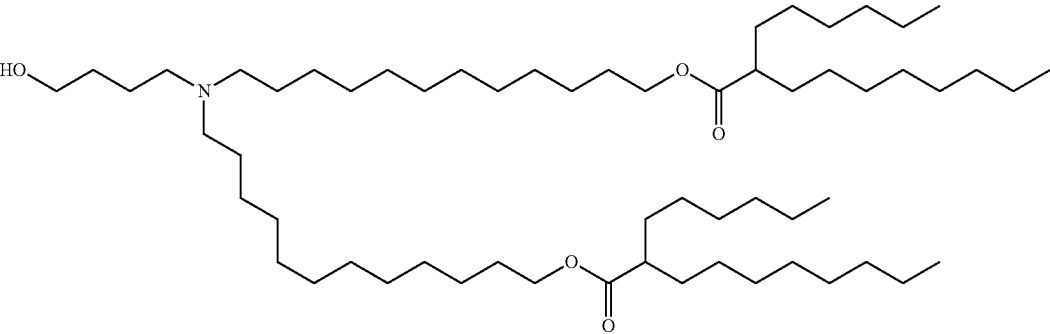
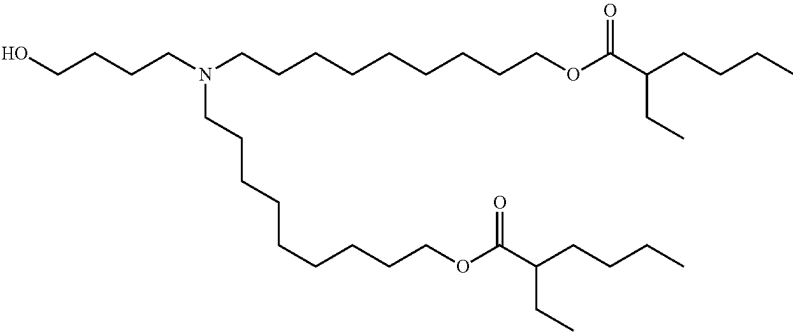
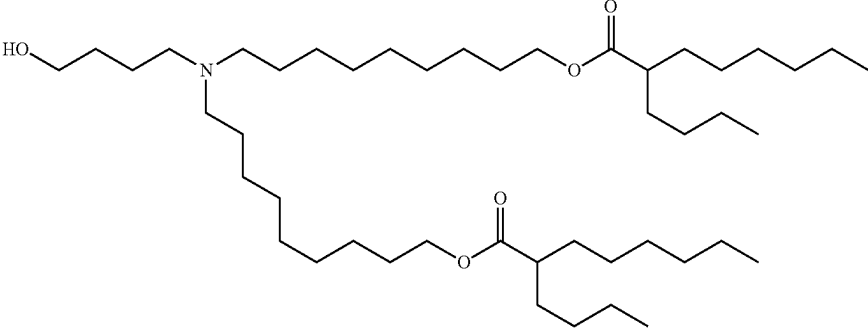
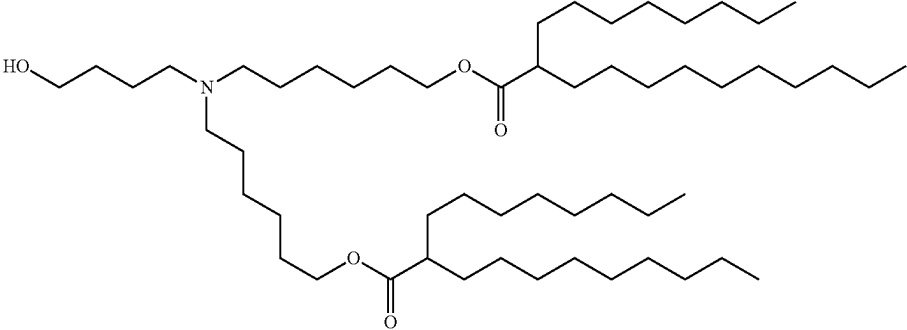
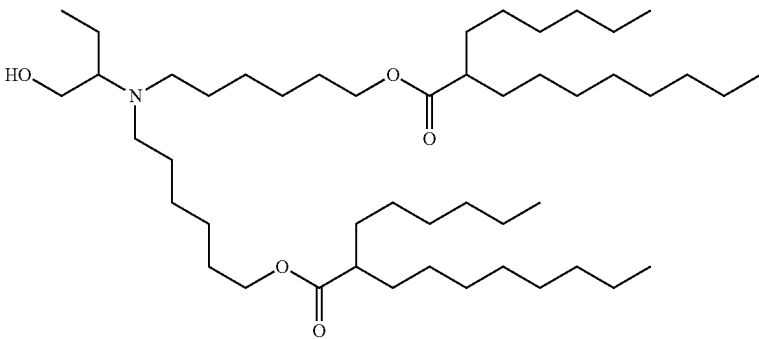
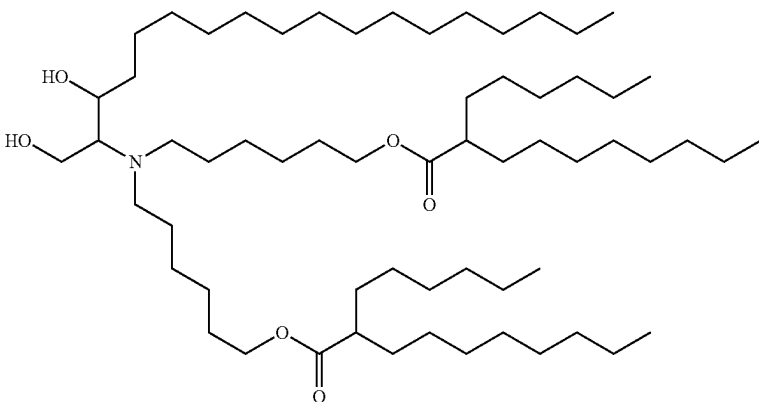
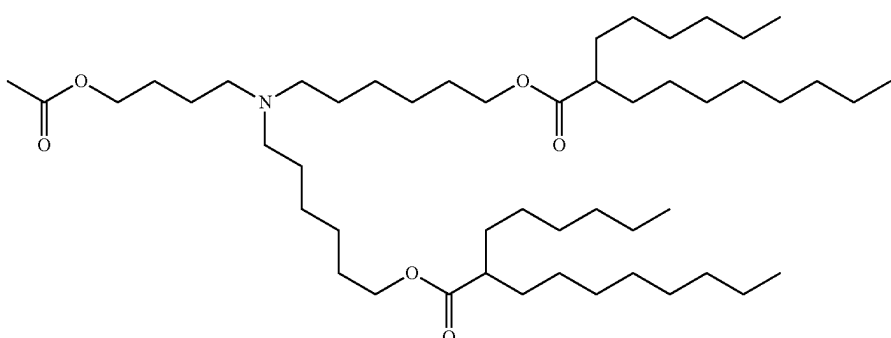
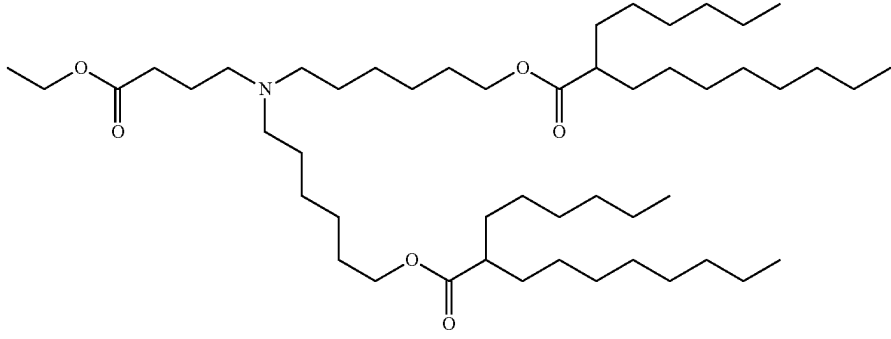
Number	Structure
23	
24	
25	
26	

TABLE 10g-continued

Number	Structure
27	<chem>CCCCOCCN(CCCCCCCCCOC(=O)CCCCCCCC)CCCCCCCCOC(=O)CCCCCCCC</chem>
28	<chem>CCCCOCCN(CCCCCCCCCOC(=O)CCCCCCCC)CCCCCCCCOC(=O)CCCCCCCC</chem>
29	<chem>CCCCOCCN(CCCCCCCCCOC(=O)CCCCCCCC)CCCCCCCCOC(=O)CCCCCCCC</chem>
30	<chem>CC(C)(O)CCN(CCCCCCCCCOC(=O)CCCCCCCC)CCCCCCCCOC(=O)CCCCCCCC</chem>

TABLE 10g-continued

Number	Structure
31	
32	
33	
34	

Number	Structure
--------	-----------

Structure

CCCCCCCCCCCCCCCC(=O)OCCN(CCCCCC#N)CCCCCCCCCCCCCCCC(=O)OCCCCCCN(CCCCOC(=O)CCCCCCCCC(C)CCCC)CCCC(=O)COCCCN(CCCCCCCCOC(=O)CCCCCCCCC)CCCCCCCCOC(=O)CCCCCCCCCOCCCN(CCCCCCCCCCCCCCCCCCCOC(=O)C(CC)(CC)CCCCCCCCCCCCCCC)CCCCCCCCCCCCCCCCCOC(=O)C(CC)(CC)CCCCCCCCCCCCCCC

TABLE 10g-continued

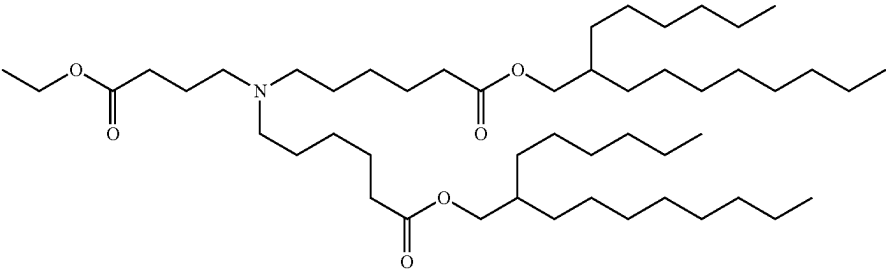
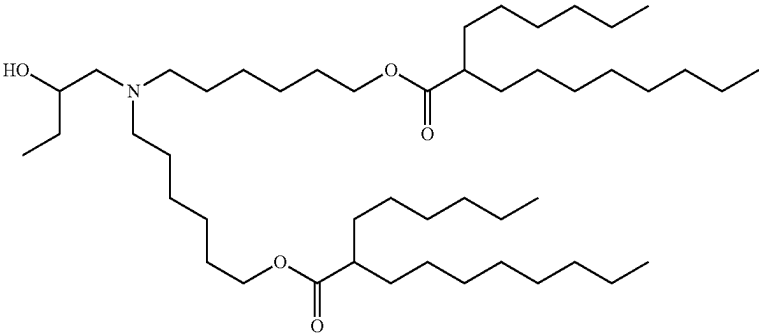
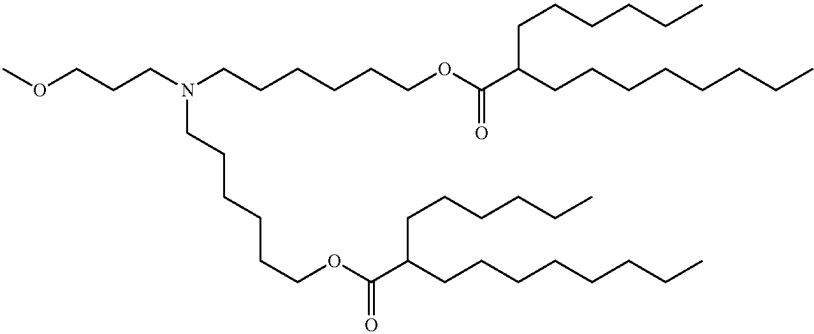
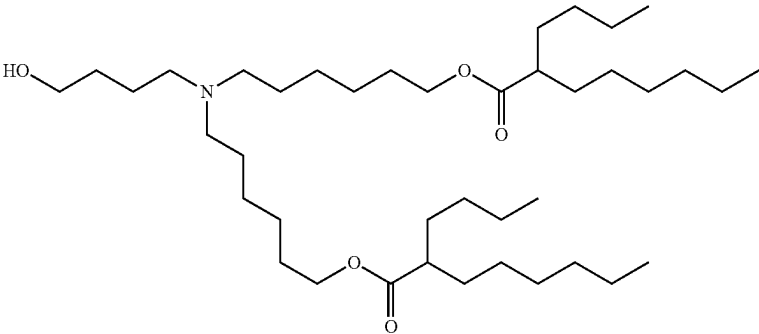
Number	Structure
39	
40	
41	
42	

TABLE 10g-continued

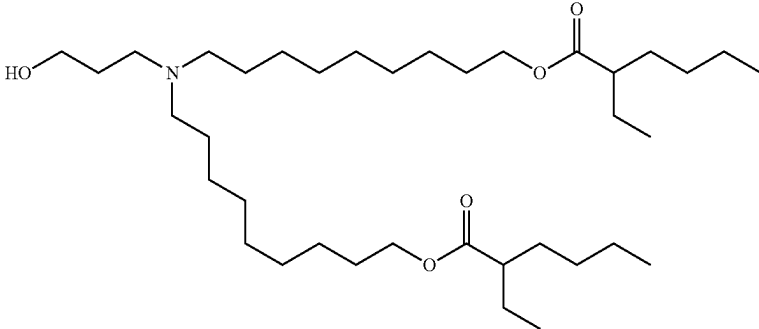
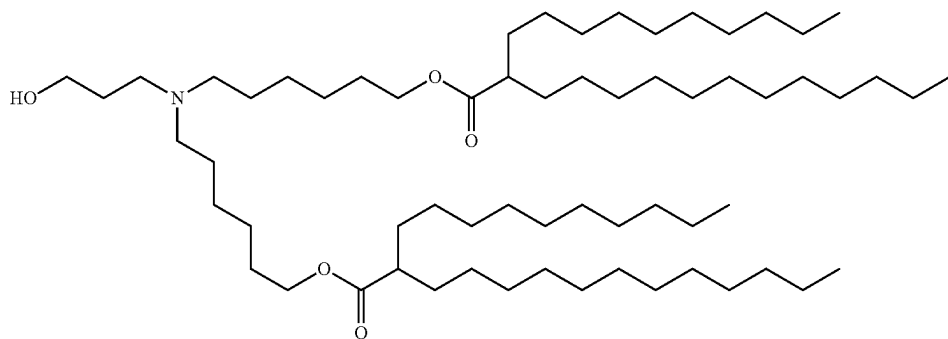
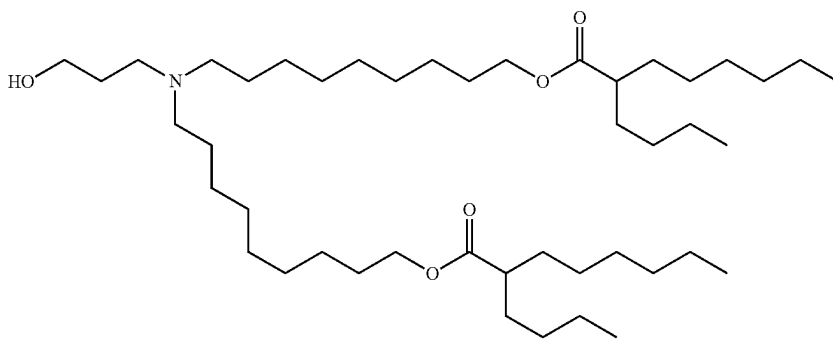
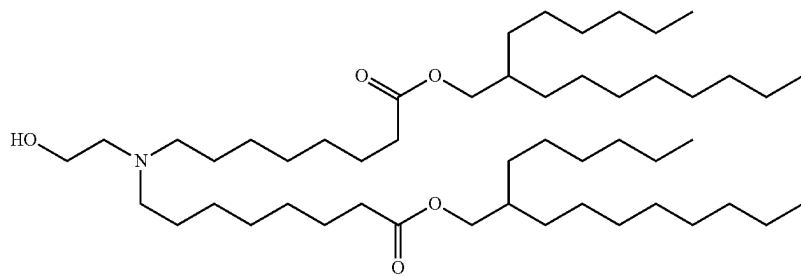
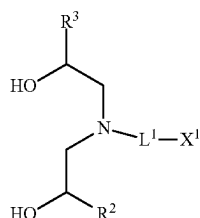
Number	Structure
43	
44	
45	
46	

TABLE 10g-continued

Number	Structure
47	
48	
49	

In some embodiments, an ionizable lipid is as described in international patent application PCT/US2020/038678.

In some embodiments, the ionizable lipid is represented by Formula (14*):



Formula (14*)

or a pharmaceutically acceptable salt thereof, wherein L¹ is C₂-C₁₁ alkylene, C₄-C₁₀-alkenylene, or C₄-C₁₀-alkynylene:

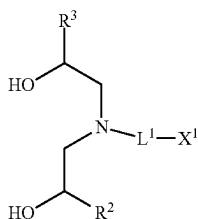
X¹ is OR¹, SR¹, or N(R¹)₂, where R¹ is independently H or unsubstituted C₁-C₆ alkyl; and

R₂ and R₃ are each independently a linear or branched C₁-C₃₀ alkyl, C₂-C₃₀ alkenyl, or C₁-C₃₀ heteroalkyl, optionally substituted by one or more substituents

selected from oxo, halo, hydroxy, cyano, alkyl, alkenyl, aldehyde, heterocyclalkyl, hydroxyalkyl, dihydroxyalkyl, hydroxyalkylaminoalkyl, aminoalkyl, alkylaminoalkyl, dialkylaminoalkyl, (heterocycl)(alkyl)aminoalkyl, heterocycl, heteroaryl, alkylheteroaryl, alkynyl, alkoxy, amino, dialkylamino, aminoalkylcarbonylamino, aminocarbonylalkylamino, (aminocarbonylalkyl)(alkyl)amino, alkenylcarbonylamino, hydroxycarbonyl, alkyloxy-carbonyl, alkylcarbonyloxy, alkylcarbonate, alkenyloxy-carbonyl, alkenylcarbonyloxy, alkenylcarbonate, alkynylloxy-carbonyl, alkynylcarbonyloxy, alkynylcarbonate, aminocarbonyl, aminoalkylaminocarbonyl, alkylaminoalkylaminocarbonyl, dialkylaminoalkylaminocarbonyl, heterocyclalkylaminocarbonyl, (alkylaminoalkyl)(alkyl)aminocarbonyl, alkylaminoalkylcarbonyl, dialkylaminoalkylcarbonyl, heterocyclcarbonyl, alkenylcarbonyl, alkynylcarbonyl, alkylsulfoxide, alkylsulfoxide-alkyl, alkylsulfonyl, and alkylsulfonealkyl.

229

In some embodiments, the ionizable lipid is represented by Formula (14*):



or a pharmaceutically acceptable salt thereof, wherein L^1 is C_2 - C_{11} alkylene, C_4 - C_{10} -alkenylene, or C_4 - C_{10} -alkynylene:

X^1 is OR^1 , SR^1 , or $N(R^1)_2$, where R^1 is independently H or unsubstituted C_1 - C_6 alkyl; and

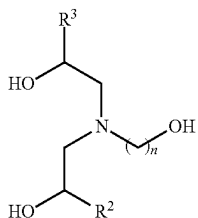
R^2 and R^3 are each independently C_6 - C_{30} -alkyl, C_6 - C_{30} -alkenyl, or C_6 - C_{30} -alkynyl.

In some embodiments, X^1 is OR^1 . In some embodiments, X^1 is SR^1 . In some embodiments, X^1 is SH . In some embodiments, X^1 is $N(R^1)_2$.

In some embodiments, X^1 is NH_2 .

In some embodiments, L^1 is C_2 - C_{10} alkylene. In some embodiments, L^1 is unsubstituted C_2 - C_{10} alkylene. In some embodiments, L^1 is C_4 - C_{10} alkenylene. In some embodiments, L^1 is unsubstituted C_4 - C_{10} alkenylene. In some embodiments, L^1 is C_4 - C_{10} alkynylene. In some embodiments, L^1 is unsubstituted C_4 - C_{10} alkynylene.

In some embodiments of Formula (14), a lipid has a structure according to Formula (14-2),



or a pharmaceutically acceptable salt thereof, wherein n is an integer of 2-10.

In some embodiments, n is 2, 3, 4, or 5. In some embodiments, n is 2. In some embodiments, n is 3. In some embodiments, n is 4. In some embodiments, n is 5. In some embodiments, n is 6. In some embodiments, n is 7. In some embodiments, n is 8. In some embodiments, n is 9. In some embodiments, n is 10.

In some embodiments of Formula (14*) or Formula (14-2), R_2 and R_3 are independently a linear or branched C_1 - C_{20} alkyl, C_2 - C_{20} alkenyl, or C_1 - C_{20} heteroalkyl, optionally substituted by one or more substituents each independently selected from linear or branched C_1 - C_{20} alkoxy, linear or branched C_1 - C_{20} alkyloxycarbonyl, linear or branched C_1 - C_{20} alkylcarbonyloxy, linear or branched C_1 - C_{20} alkylcarbonate, linear or branched C_2 - C_{20} alkenyloxycarbonyl, linear or branched C_2 - C_{20} alkenylcarbonyloxy, linear or branched C_2 - C_{20} alkenylcarbonate, linear or branched C_2 - C_{20} alkynylloxycarbonyl, linear or branched C_2 - C_{20} alkynylcarbonyloxy, and linear or branched C_2 - C_{20} alkynylcarbonate.

230

In certain embodiments of Formula (14*) or Formula (14-2), one or each of R^2 and R^3 is unsubstituted C_6 - C_{30} -alkyl, unsubstituted C_6 - C_{30} -alkenyl, or unsubstituted C_6 - C_{30} -alkynyl. In certain embodiments, each of R^2 and R^3 is unsubstituted C_6 - C_{30} -alkyl. In certain embodiments, each of R^2 and R^3 is unsubstituted C_6 - C_{30} -alkenyl. In certain embodiments, each of R^2 and R^3 is unsubstituted C_6 - C_{30} -alkynyl.

In some embodiments of Formula (14*), the alkyloxycarbonyl substituent is of the formula $-C(O)OR^6$, wherein R^6 is unsubstituted C_6 - C_{30} alkyl or C_6 - C_{30} alkenyl. In some embodiments of Formula (14*) or Formula (14-2), at least one of R_2 and R_3 is substituted with an alkylcarbonyloxy. In some embodiments, the alkylcarbonyloxy is of the formula $-OC(O)R^6$, wherein R^6 is unsubstituted C_6 - C_{30} alkyl or C_6 - C_{30} alkenyl. In some embodiments, at least one of R_2 and R_3 is substituted with an alkylcarbonate. In some embodiments, the alkylcarbonate is of the formula $-O(CO)OR^6$, wherein R^6 is unsubstituted C_6 - C_{30} alkyl or C_6 - C_{30} alkenyl. In some embodiments, R_2 and R_3 are independently C_1 - C_{12} alkyl substituted by $-O(CO)R^6$, $-C(O)OR^6$, or $-O(CO)OR^6$, wherein R^6 is unsubstituted C_6 - C_{30} alkyl or C_6 - C_{30} alkenyl. In some embodiments, R_2 and R_3 are each C_1 - C_{12} alkyl substituted by $-O(CO)R^6$. In some embodiments, R_2 and R_3 are each C_1 - C_{12} alkyl substituted by $-C(O)OR^6$. In some embodiments, R_2 is $-C(O)OR^6$ or $-O(CO)R^6$ and R_3 is $-O(CO)OR^6$. In some embodiments, R_2 is $-O(CO)OR^6$ and R_3 is $-C(O)OR^6$ or $-O(CO)R^6$.

In some embodiments of Formula (14*) or Formula (14-2), at least one of R_2 and R_3 is selected from the following formulae:

- (i) $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$,
- (ii) $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, and
- (iii) $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$,

wherein:

q is 0 to 12,

r is 0 to 6,

R^8 is H or R^{10} , and

R^9 and R^{10} are independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_2 - C_{12} -alkenyl.

In some embodiments of Formula (14*) or Formula (14-2), each of R_2 and R_3 is independently selected from one of the following formulae:

- (i) $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$,
- (ii) $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, and
- (iii) $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$,

wherein:

q is 0 to 12,

r is 0 to 6.

R^8 is H or R^{10} , and

R^9 and R^{10} are independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_2 - C_{12} -alkenyl.

In some embodiments, of any one of formulae (i)-(iii), q is 1 to 6. In some embodiments of any one of formulae (i)-(iii), q is 0. In some embodiments, of any one of formulae (i)-(iii), q is 1. In some embodiments of any one of formulae (i)-(iii), q is 2. In some embodiments, of any one of formulae (i)-(iii), q is 3 to 12. In some embodiments, of any one of formulae (i)-(iii), q is 3 to 6.

In some embodiments of any one of formulae (i)-(iii), r is 0. In some embodiments of any one of formulae (i)-(iii), r is 1 to 6. In some embodiments of any one of formulae (i)-(iii), r is 1. In some embodiments of any one of formulae (i)-(iii),

r is 2. In some embodiments of any one of formulae (i)-(iii), r is 3. In some embodiments of any one of formulae (i)-(iii), r is 4.

In some embodiments of formulae (i)-(iii), R^8 is H. In some embodiments of formulae (i)-(iii), R^8 is R^{10} . In some embodiments of formulae (i)-(iii), R^9 and R^{10} are different. In some embodiments of formulae (i)-(iii), R^9 and R^{10} are the same.

In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_1 - C_{12} -alkenyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_2 - C_{12} alkyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_2 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_4 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_5 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 is H, and R^9 is unsubstituted linear C_6 - C_8 alkyl.

In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_1 - C_{12} -alkenyl. In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_2 - C_{12} alkyl. In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_2 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_4 - C_8 alkyl. In some embodiments of formulae (i)-(iii), R^8 and R^9 are each independently unsubstituted linear C_6 - C_8 alkyl.

In some embodiments of Formula (14*) or Formula (14-2), at least one of R_2 and R_3 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above. In some embodiments at least one of R_2 and R_3 is $-(CH_2)_qOC(O)(CH_2CH(R^8)(R^9))$, where q, r, R^8 and R^9 are as defined above. In some embodiments at least one of R_2 and R_3 is $-(CH_2)_qOC(O)O(CH_2CH(R^8)(R^9))$, where q, r, R^8 and R^9 are as defined above. In some embodiments of Formula (14*) or Formula (14-2), at least one of R_2 and R_3 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$, where q is 3 to 12 (e.g., 6 to 12), r is 1 to 6 (e.g., 1, 2 or 3), and R^8 and R^9 are each independently unsubstituted linear C_4 - C_8 alkyl.

In certain embodiments of Formula (14*) or Formula (14-2), at least one of R_2 and R_3 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$ or $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above. In other embodiments, at least one of R_2 and R_3 is $-(CH_2)_qOC(O)(CH_2CH(R^8)(R^9))$ or $-(CH_2)_qOC(O)O(CH_2CH(R^8)(R^9))$, where q, r, R^8 and R^9 are as defined above. In some embodiments, at least one of R_2 and R_3 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$ or $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above.

In other embodiments of Formula (14*) or Formula (14-2), at least one of R_2 and R_3 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$ where q, r, R^8 and R^9 are as defined above. In some embodiments, at least one of R_2 and R_3 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$ or $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above.

In certain embodiments of Formula (14*) or Formula (14-2), R_2 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above. In certain embodiments, R_2 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above. In certain embodiments, R is

$-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above.

In certain embodiments of Formula (14*) or Formula (14-2), R_2 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above. In certain embodiments, R_2 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above. In certain embodiments, R_2 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above.

In certain embodiments of Formula (14*) or Formula (14-2), R_2 is $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above. In certain embodiments, R_2 is $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above. In certain embodiments, R_2 is $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, and R_3 is $-(CH_2)_qOC(O)O(CH_2)_rCH(R^8)(R^9)$, where q, r, R^8 and R^9 are as defined above.

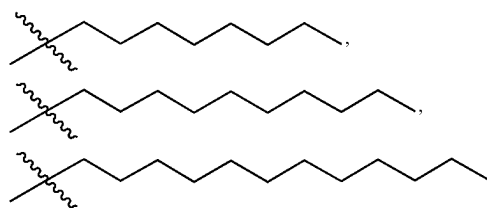
In certain embodiments of Formula (14*) or Formula (14-2), one or each of R_2 and R_3 is unsubstituted C_6 - C_{22} -alkyl, or one or each of R_2 and R_3 is unsubstituted C_6 - C_{22} -alkenyl. In certain embodiments, each of R_2 and R_3 is unsubstituted C_6 - C_{22} -alkyl. In certain embodiments, each of R_2 and R_3 is unsubstituted C_6 - C_{22} -alkenyl.

In certain embodiments, one or each of R_2 and R_3 is $-C_6H_{13}$, $-C_7H_{15}$, $-C_8H_{17}$, $-C_9H_{19}$, $-C_{10}H_{21}$, $-C_{11}H_{23}$, $-C_{12}H_{25}$, $-C_{13}H_{27}$, $-C_{14}H_{29}$, $-C_{15}H_{31}$, $-C_{16}H_{33}$, $-C_{17}H_{35}$, $-C_{18}H_{37}$, $-C_{19}H_{39}$, $-C_{20}H_{41}$, $-C_{21}H_{43}$, $-C_{22}H_{45}$, $-C_{23}H_{47}$, $-C_{24}H_{49}$, $-C_{25}H_{51}$.

In certain embodiments, one or each of R_2 and R_3 is $-(CH_2)_4CH=CH_2$, $-(CH_2)_5CH=CH_2$, $-(CH_2)_6CH=CH_2$, $-(CH_2)_7CH=CH_2$, $-(CH_2)_8SCH=CH_2$, $-(CH_2)_9CH=CH_2$, $-(CH_2)_{10}CH=CH_2$, $-(CH_2)_{11}CH=CH_2$, $-(CH_2)_{12}CH=CH_2$, $-(CH_2)_{13}CH=CH_2$, $-(CH_2)_{14}CH=CH_2$, $-(CH_2)_{15}CH=CH_2$, $-(CH_2)_{16}CH=CH_2$, $-(CH_2)_{17}CH=CH_2$, $-(CH_2)_{18}CH=CH_2$, $-(CH_2)_7CH=CH(CH_2)_3CH_3$, $-(CH_2)_7CH=CH(CH_2)_5CH_3$, $-(CH_2)_4CH=CH(CH_2)_3SCH_3$, $-(CH_2)_7CH=CH(CH_2)_7CH_3$, $-(CH_2)_6CH=CHCH_2CH=CH(CH_2)_4CH_3$, $-(CH_2)_2CH=CHCH_2CH=CH(CH_2)_4CH_3$, $-(CH_2)_7CH=CHCH_2CH=CHCH_2CH=CHCH_2CH_3$, $-(CH_2)_3CH=CHCH_2CH=CHCH_2CH=CHCH_2CH=CH(CH_2)_4CH_3$, $4CH_2)_3CH=CHCH_2CH=CHCH_2CH=CHCH_2CH=CHCH_2CH=CHCH_2CH_3$, $-(CH_2)_{11}CH=CH(CH_2)_7CH_3$, or $-(CH_2)_2CH=CHCH_2CH=CHCH_2CH=CHCH_2CH=CHCH_2CH=CHCH_2CH=CHCH_2CH_3$.

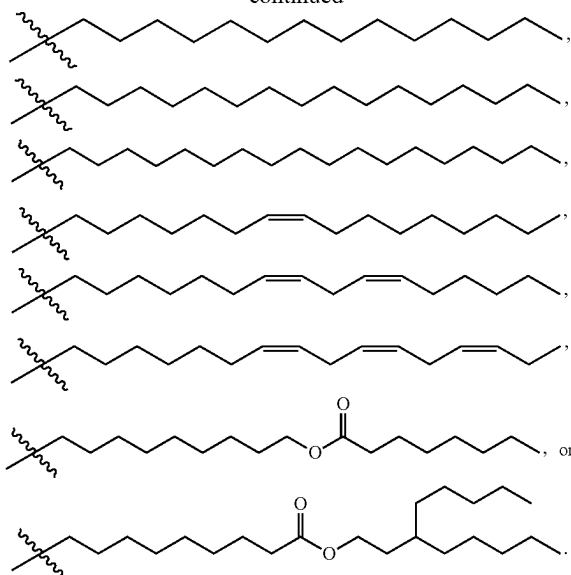
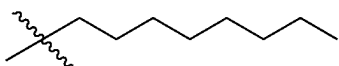
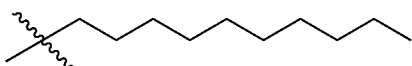
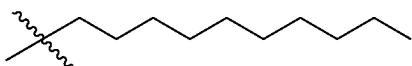
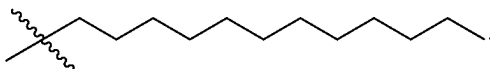
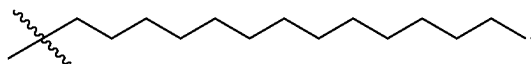
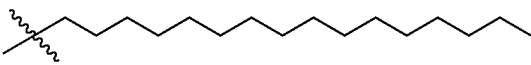
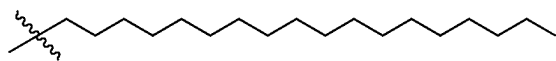
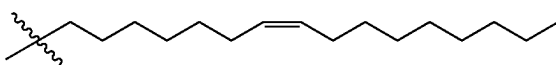
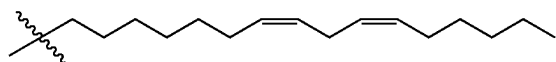
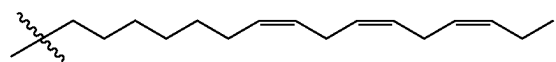
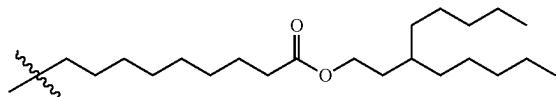
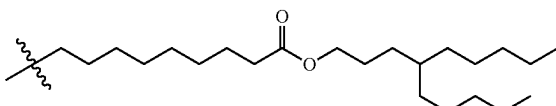
In certain embodiments, one or each of R_2 and R_3 is C_6 - C_{12} alkyl substituted by $-O(CO)R^6$ or $-C(O)OR^6$, wherein R^6 is unsubstituted C_6 - C_{14} alkyl. In certain embodiments, R^6 is unsubstituted linear C_6 - C_{14} alkyl. In certain embodiments, R^6 is unsubstituted branched C_6 - C_{14} alkyl.

In certain embodiments, one or each of R_2 and R_3 is $(CH_2)_7C(O)O(CH_2)_2CH(C_5H_{11})_2$ or $(CH_2)_8C(O)O(CH_2)_2CH(C_5H_{11})_2$. In certain embodiments, one or each of R^2 and R^3 is



233

-continued

In certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R₂ and R₃ isIn certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ is**234**In certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ isIn certain embodiments, one or each of R² and R³ is

In some embodiments, an ionizable lipid is described in Table 10h.

TABLE 10h

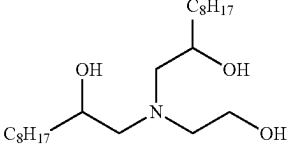
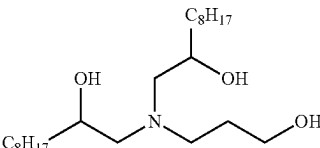
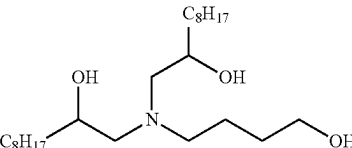
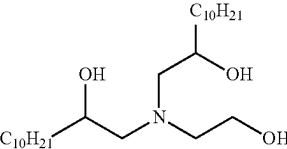
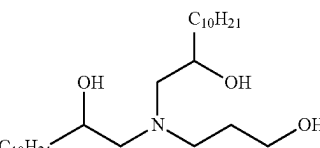
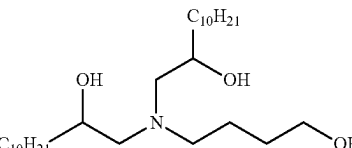
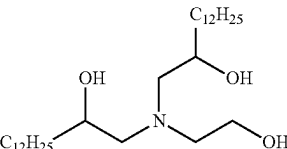
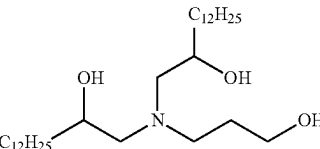
Compound n		R	Structure
1	1	C ₈ H ₁₇	
2	2	C ₈ H ₁₇	
3	3	C ₈ H ₁₇	
4	1	C ₁₀ H ₂₁	
5	2	C ₁₀ H ₂₁	
6	3	C ₁₀ H ₂₁	
7	1	C ₁₂ H ₂₅	
8	2	C ₁₂ H ₂₅	

TABLE 10h-continued

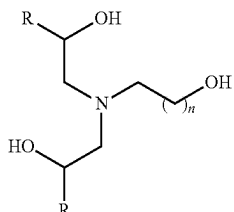
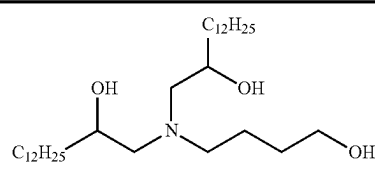
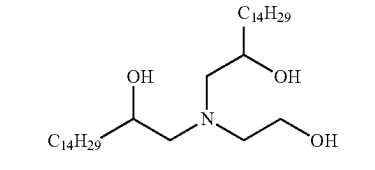
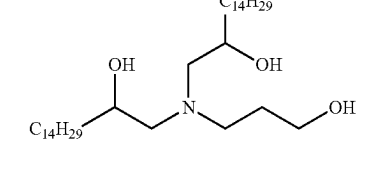
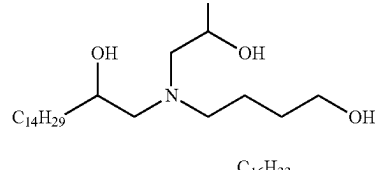
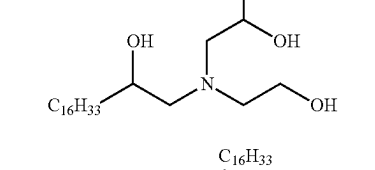
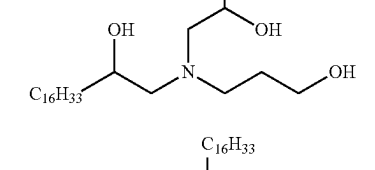
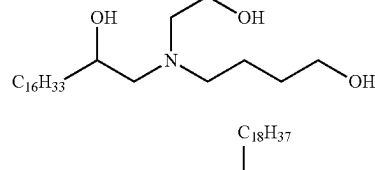
Compound n		R	Structure
9	3	C ₁₂ H ₂₅	
10	1	C ₁₄ H ₂₉	
11	2	C ₁₄ H ₂₉	
12	3	C ₁₄ H ₂₉	
13	1	C ₁₆ H ₃₃	
14	2	C ₁₆ H ₃₃	
15	3	C ₁₆ H ₃₃	
16	1	C ₁₈ H ₃₇	

TABLE 10h-continued

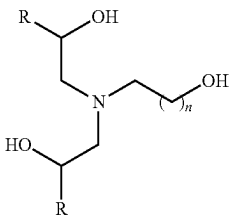
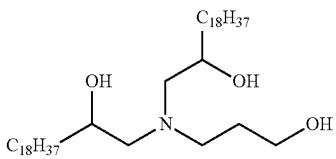
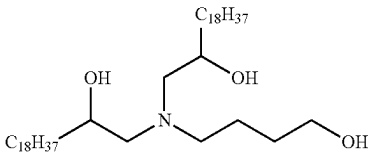
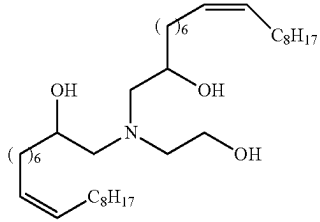
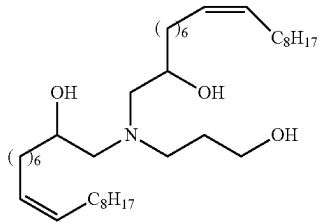
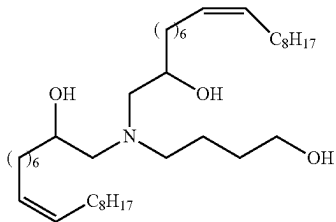
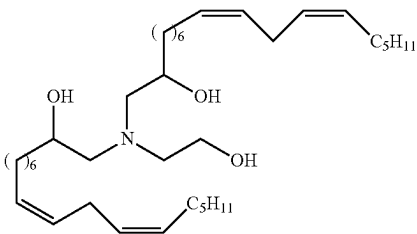
			
Compound	n	R	Structure
17	2	C ₁₈ H ₃₇	
18	3	C ₁₈ H ₃₇	
19	1	C ₆ H ₁₂ CH=CHC ₈ H ₁₇	
20	2	C ₆ H ₁₂ CH=CHC ₈ H ₁₇	
21	3	C ₆ H ₁₂ CH=CHC ₈ H ₁₇	
22	1	C ₆ H ₁₂ CH=CHCH ₂ CH=CHC ₅ H ₁₁	

TABLE 10h-continued

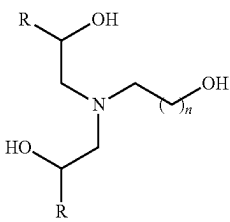
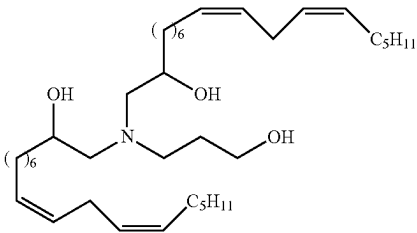
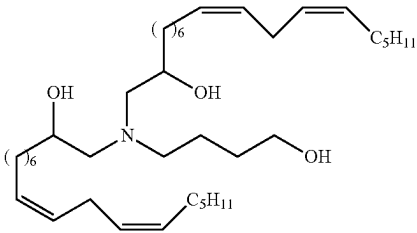
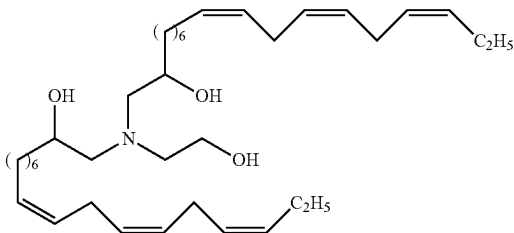
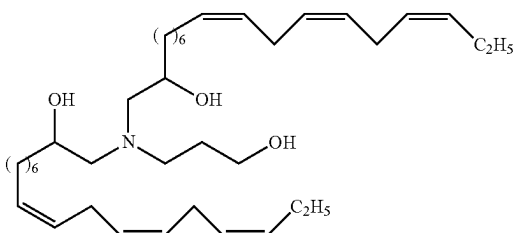
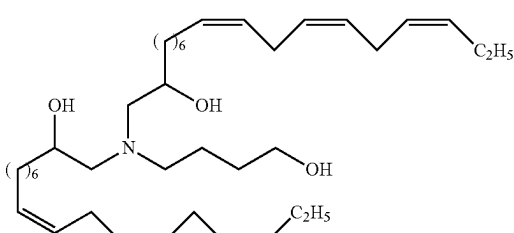
			
Compound	n	R	Structure
23	2	$C_6H_{12}CH=CHCH_2CH=CHC_5H_{11}$	
24	3	$C_6H_{12}CH=CHCH_2CH=CHC_5H_{11}$	
25	1	$(CH_2)_6CH=CH(CH_2)CH=CH(CH_2)CH=CHC_2H_5$	
26	2	$(CH_2)_6CH=CH(CH_2)CH=CH(CH_2)CH=CHC_2H_5$	
27	3	$(CH_2)_6CH=CH(CH_2)CH=CH(CH_2)CH=CHC_2H_5$	

TABLE 10h-continued

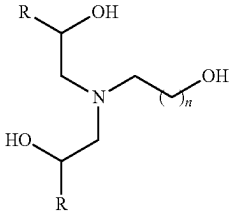
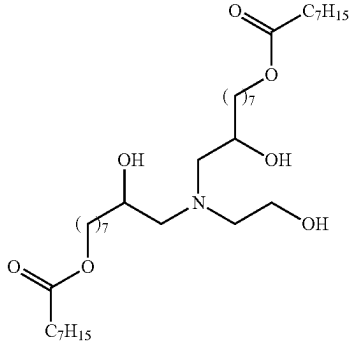
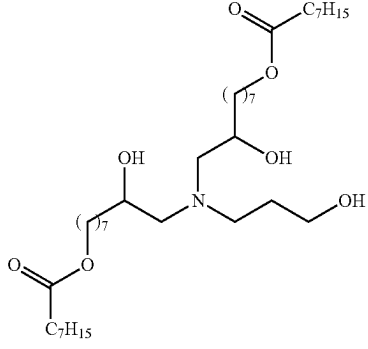
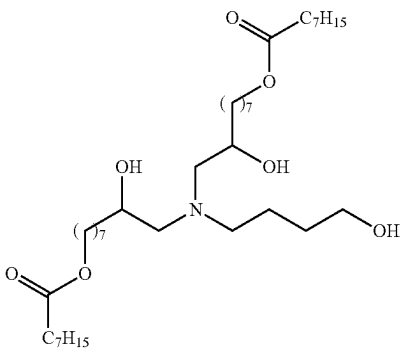
			
Compound	n	R	Structure
28	1	$C_7H_{14}O(CO)C_7H_{15}$	
29	2	$C_7H_{14}O(CO)C_7H_{15}$	
30	3	$C_7H_{14}O(CO)C_7H_{15}$	

TABLE 10h-continued

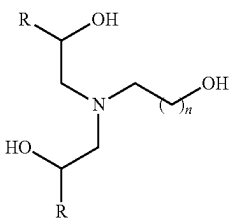
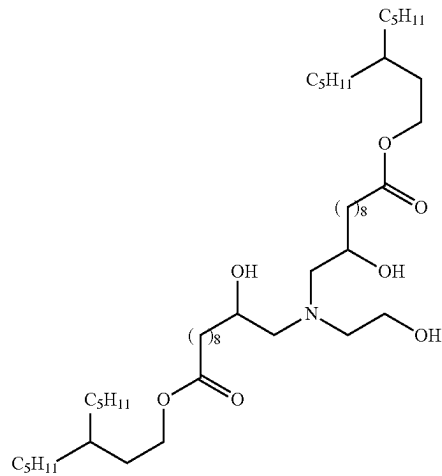
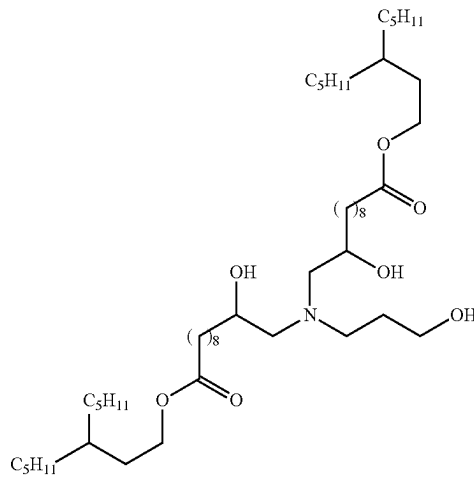
			
Compound	n	R	Structure
31	1	$\text{C}_8\text{H}_{16}\text{C}(\text{O})\text{O}(\text{CH}_2)_2\text{CH}(\text{C}_5\text{H}_{11})_2$	
32	2	$\text{C}_8\text{H}_{16}\text{C}(\text{O})\text{O}(\text{CH}_2)_2\text{CH}(\text{C}_5\text{H}_{11})_2$	

TABLE 10h-continued

Compound n		R	Structure
33	3	$C_8H_{16}C(O)O(CH_2)_2CH(C_5H_{11})_2$	

5.1 Other Ionizable Lipids

In some embodiments, one or more (e.g., two or more, or three or more) ionizable lipids are utilized in the transfer vehicles of this disclosure. In some embodiments, the transfer vehicle includes a first ionizable lipid (e.g., as described herein, such as a lipid of Formula (13*) or (14*)), and one or more additional ionizable lipids.

Lipids of interest, including ionizable lipids that can be used in combination with a first ionizable lipid as described herein, such as by being incorporated into the transfer vehicles of this disclosure, include, but are not limited to, lipids as described in: international application PCT/US2018/058555, international application PCT/US2020/038678, US publication US2019/0314524, WO2019/152848, international application PCT/US2010/001058, international application PCT/US2017/028981, WO2015/095340, WO2014/136086, US2019/0321489, WO2010/053572, U.S. provisional patent application 61/617,468, international patent application PCT/US2019/025246. US patent publications 2017/0190661 and 2017/0114010, US publication 20190314284, WO2015/095340, WO2019/152557, WO2019/152848, international application PCT/US2019/015913, U.S. Pat. Nos. 9,708,628, 9,765,022; Wang et al., ACS Synthetic Biology, 1, 403-07 (2012); WO 2008/042973, U.S. Pat. No. 8,071,082, the disclosures of which are incorporated herein by reference in their entirety.

In some embodiments, tail groups as used in the lipids may be as described in. WO2015/095340, WO2019/152557, and WO2019/152848, the disclosures of which are incorporated herein by reference in their entirety.

The lipid-like compounds can be prepared by methods well known the art. See

In some embodiments, the ionizable lipid N-[1-(2,3-dioleoyloxy)propyl]-N,N,N-trimethylammonium chloride

or "DOTMA" is used. (Felgner et al. Proc. Nat'l Acad. Sci. 84, 7413 (1987); U.S. Pat. No. 4,897,355). DOTMA can be formulated with an ionizable lipid (e.g., as described herein), and/or can be combined with a neutral lipid, dioleoylphosphatidylethanolamine or "DOPE" or other cationic or non-cationic lipids into a lipid nanoparticle.

Other suitable lipids include, for example, ionizable cationic lipids, such as, e.g., (15Z,18Z)-N,N-dimethyl-6-(9Z,12Z)-octadeca-9,12-dien-1-yl)tetracos-15,18-dien-1-amine (-HGT5000), (15Z,18Z)-N,N-dimethyl-6-((9Z,12Z)-octadeca-9,12-dien-1-yl)tetracos-4,15,18-trien-1-amine (HGT5001), and (15Z,18Z)-N,N-dimethyl-6-((9Z,12Z)-octadeca-9,12-dien-1-yl)tetracos-5,15,18-trien-1-amine (HGT5002), C12-200 (described in WO 2010/053572), 2-(2,2-di((9Z,12Z)-octadeca-9,12-dien-1-yl)-1,3-dioxolan-4-yl)-N,N-dimethylethanamine (DLinKC2-DMA)) (See, WO 2010/042877; Semple et al., Nature Biotech, 28:172-176 (2010)), 2-(2,2-di((9Z,2Z)-octadeca-9,12-dien-1-yl)-1,3-dioxolan-4-yl)-N,N-dimethylethanamine (DLin-KC2-DMA), (3S,10R,13R,17R)-10,13-dimethyl-17-((R)-6-methylheptan-2-yl)-2,3,4,7,8,9,10,11,12,13,14,15,16,17-tetradecahydro-1H-cyclopenta[a]phenanthren-3-yl 3-(1H-imidazol-4-yl)propanoate (ICE), (15Z,18Z)-N,N-dimethyl-6-(9Z,12Z)-octadeca-9,12-dien-1-yl)tetracos-15,18-dien-1-amine (HGT5000), (15Z,18Z)-N,N-dimethyl-6-((9Z,12Z)-octadeca-9,12-dien-1-yl)tetracos-4,15,18-trien-1-amine (HGT5001), (15Z,18 Z)-N,N-dimethyl-6-((9Z,12Z)-octadeca-9,12-dien-1-yl)tetracos-5,15,18-trien-1-amine (HGT5002), 5-carboxyspermylglycine-diactadecylamide (DOGS), 2,3-dioleoyloxy-N-[2(spermine-carboxamido)ethyl]-N,N-dimethyl-1-propanaminium (DOSPA) (Behr et al. Proc. Nat'l Acad. Sci. 86, 6982 (1989); U.S. Pat. No. 5,171,678: 5.334,761), 1,2-Dioleoyl-3-Dimethylammonium-Propane (DODAP), 1,2-Dioleoyl-3-

Trimethylammonium-Propane or (DOTAP). Contemplated ionizable lipids also include 1,2-distearoyloxy-N,N-dimethyl-3-aminopropane (DSDMA), 1,2-dioleoyloxy-N,N-dimethyl-3-aminopropane (DODMA), 1,2-dilinoleoyloxy-N,N-dimethyl-3-aminopropane (DLinDMA), 1,2-dilinolenyloxy-N,N-dimethyl-3-aminopropane (DLenDMA), N-dioleoyl-N,N-dimethylammonium chloride (DODAC), N,N-distearyl-N,N-dimethylammonium bromide (DDAB), N-(1,2-dimyristyloxyprop-3-yl)-N,N-dimethyl-N-hydroxyethyl ammonium bromide (DMRIE), 3-dimethylamino-2-(cholest-5-en-3-beta-oxybutan-4-oxy)-1-(cis,cis-9,12-octadecadienoxy)propane (CLinDMA), 2-[5'-(cholest-5-en-3-beta-oxy)-3'-oxapentoxyl]-3-dimethyl-1-(cis,cis-9',1'-octadecadienoxy)propane (CpLinDMA), N,N-dimethyl-3,4-dioleoyloxybenzylamine (DMOBA), 1,2-N,N'-dioleoylcarbaryl-3-dimethylaminopropane (DOcarbDAP), 2,3-Dilinoleoyloxy-N,N-dimethylpropylamine (DLinDAP), 1,2-N,N'-Dilinoleoylcarbaryl-3-dimethylaminopropane (DLincarbDAP), 1,2-Dilinoleoylcarbaryl-3-dimethylaminopropane (DLinCDAP), 2,2-dilinoleyl-4-dimethylaminomethyl-[1,3]-dioxolane (DLin-K-DMA), 2,2-dilinoleyl-4-dimethylaminomethyl-[1,3]-dioxolane (DLin-K-XTC2-DMA) or GL67, or mixtures thereof. (Heyes, J., et al., *J Controlled Release* 107: 276-287 (2005), Morrissey, D V., et al., *Nat. Biotechnol.* 23(8): 1003-1007 (2005); PCT Publication WO2005/121348A1). The use of cholesterol-based ionizable lipids to formulate the transfer vehicles (e.g. lipid nanoparticles) is also contemplated by the present invention. Such cholesterol-based ionizable lipids can be used, either alone or in combination with other lipids. Suitable cholesterol-based ionizable lipids include, for example, DC-Cholesterol (N,N-dimethyl-N-ethylcarboxamidcholesterol), and 1,4-bis(3-N-oleylamino-propyl)piperazine (Gao, et al., *Biochem. Biophys. Res. Comm.* 179, 280 (1991); Wolf et al., *BioTechniques* 23, 139 (1997); U.S. Pat. No. 5,744,335).

Also contemplated are cationic lipids such as dialkylamino-based, imidazole-based, and guanidinium-based lipids. For example, also contemplated is the use of the ionizable lipid (3S, 10R, 13R, 17R)-10,13-dimethyl-17-((R)-6-methylheptan-2-yl)-2,3,4,7,8,9,10,11,12,13,14,15,16,17-tetradecahydro-1H-cyclopenta[a]phenanthren-3-yl 3-(1H-imidazol-4-yl)propanoate (ICE), as disclosed in International Application No. PCT/US2010/058457, incorporated herein by reference.

Also contemplated are ionizable lipids such as the dialkylamino-based, imidazole-based, and guanidinium-based lipids. For example, certain embodiments are directed to a composition comprising one or more imidazole-based ionizable lipids, for example, the imidazole cholesterol ester or "ICE" lipid, (3S, 10R, 13R, 17R)-10,13-dimethyl-17-((R)-6-methylheptan-2-yl)-2,3,4,7,8,9,10,11,12,13,14,15,16,17-tetradecahydro-1H-cyclopenta[a]phenanthren-3-yl 3-(1H-imidazol-4-yl)propanoate.

Without wishing to be bound by a particular theory, it is believed that the fusogenicity of the imidazole-based cationic lipid ICE is related to the endosomal disruption which is facilitated by the imidazole group, which has a lower pKa relative to traditional ionizable lipids.

The endosomal disruption in turn promotes osmotic swelling and the disruption of the liposomal membrane, followed by the transfection or intracellular release of the nucleic acid(s) contents loaded therein into the target cell.

The imidazole-based ionizable lipids are also characterized by their reduced toxicity relative to other ionizable lipids.

In certain embodiments, transfer vehicle compositions for the delivery of circular RNA comprise an amine lipid. In

certain embodiments, an ionizable lipid is an amine lipid. In some embodiments, an amine lipid is described in international patent application PCT/US2018/053569.

In some embodiments, the amine lipid is Lipid E, which is (9Z, 12Z)-3-((4,4-bis(octyloxy)butanoyloxy)-2-(((3-(diethylamino)propoxy)carbonyloxy)methyl)propyl octadeca-9, 12-dienoate.

In certain embodiments, an amine lipid is an analog of Lipid E. In certain embodiments, a Lipid E analog is an acetal analog of Lipid E. In particular transfer vehicle compositions, the acetal analog is a C4-C12 acetal analog. In some embodiments, the acetal analog is a C5-C12 acetal analog. In additional embodiments, the acetal analog is a C5-C10 acetal analog. In further embodiments, the acetal analog is chosen from a C4, C5, C6, C7, C9, C10, C11 and C12 acetal analog.

Amine lipids and other biodegradable lipids suitable for use in the transfer vehicles, e.g., lipid nanoparticles, described herein are biodegradable in vivo. The amine lipids described herein have low toxicity (e.g., are tolerated in animal models without adverse effect in amounts of greater than or equal to 10 mg/kg). In certain embodiments, transfer vehicles composing an amine lipid include those where at least 75% of the amine lipid is cleared from the plasma within 8, 10, 12, 24, or 48 hours, or 3, 4, 5, 6, 7, or 10 days.

Biodegradable lipids include, for example, the biodegradable lipids of WO2017/173054, WO2015/095340, and WO2014/136086.

Lipid clearance may be measured by methods known by persons of skill in the art. See, for example. Maier, M. A., et al. *Biodegradable Lipids Enabling Rapidly Eliminated Lipid Nanoparticles for Systemic Delivery of RNAi Therapeutics*. *Mol. Ther.* 2013, 21(8), 1570-78.

Transfer vehicle compositions comprising an amine lipid can lead to an increased clearance rate. In some embodiments, the clearance rate is a lipid clearance rate, for example the rate at which a lipid is cleared from the blood, serum, or plasma. In some embodiments, the clearance rate is an RNA clearance rate, for example the rate at which an circRNA is cleared from the blood, serum, or plasma. In some embodiments, the clearance rate is the rate at which transfer vehicles are cleared from the blood, serum, or plasma. In some embodiments, the clearance rate is the rate at which transfer vehicles are cleared from a tissue, such as liver tissue or spleen tissue. In certain embodiments, a high rate of clearance leads to a safety profile with no substantial adverse effects. The amine lipids and biodegradable lipids may reduce transfer vehicle accumulation in circulation and in tissues. In some embodiments, a reduction in transfer vehicle accumulation in circulation and in tissues leads to a safety profile with no substantial adverse effects.

Lipids may be ionizable depending upon the pH of the medium they are in. For example, in a slightly acidic medium, the lipid, such as an amine lipid, may be protonated and thus bear a positive charge. Conversely, in a slightly basic medium, such as, for example, blood, where pH is approximately 7.35, the lipid, such as an amine lipid, may not be protonated and thus bear no charge.

The ability of a lipid to bear a charge is related to its intrinsic pKa. In some embodiments, the amine lipids of the present disclosure may each, independently, have a pKa in the range of from about 5.1 to about 7.4. In some embodiments, the bioavailable lipids of the present disclosure may each, independently, have a pKa in the range of from about 5.1 to about 7.4. For example, the amine lipids of the present disclosure may each, independently, have a pKa in the range of from about 5.8 to about 6.5. Lipids with a pKa ranging

from about 5, 1 to about 7.4 are effective for delivery of cargo in vivo, e.g., to the liver. Further, it has been found that lipids with a pKa ranging from about 5.3 to about 6.4 are effective for delivery in vivo, e.g., into tumors. See, e.g., WO2014/136086.

A lipid of the present disclosure may have an —S—S— (disulfide) bond.

Lipid-like compounds of this disclosure can be prepared using suitable starting materials through synthetic route known in the art. The method can include an additional step(s) to add or remove suitable protecting groups in order to ultimately allow synthesis of the lipid-like compounds. In addition, various synthetic steps can be performed in an alternate sequence or order to give the desired material. Synthetic chemistry transformations and protecting group methodologies (protection and deprotection) useful in synthesizing applicable lipid-like compounds are known in the art, including, for example, R. Larock, *Comprehensive Organic Transformations* (2nd Ed., VCH Publishers 1999); P. G. M. Wuts and T. W. Greene, *Greene's Protective Groups in Organic Synthesis* (4th Ed., John Wiley and Sons 2007); L. Fieser and M. Fieser, *Fieser and Fieser's Reagents for Organic Synthesis* (John Wiley and Sons 1994); and L. Paquette, ed., *Encyclopedia of Reagents for Organic Synthesis* (2nd ed., John Wiley and Sons 2009) and subsequent editions thereof. Certain lipid-like compounds may contain a non-aromatic double bond and one or more asymmetric centers. Thus, they can occur as racemates and racemic mixtures, single enantiomers, individual diastereomers, diastereomeric mixtures, and cis- or trans-isomeric forms. All such isomeric forms are contemplated.

Preparation methods for the above compounds and compositions are described herein below and/or known in the art.

It will be appreciated by those skilled in the art that in the process described herein the functional groups of intermediate compounds may need to be protected by suitable protecting groups. Such functional groups include, e.g., hydroxyl, amino, mercapto, and carboxylic acid. Suitable protecting groups for hydroxyl include, e.g., trialkylsilyl or diarylalkylsilyl (for example, t-butyltrimethylsilyl, t-butyl-diphenylsilyl or trimethylsilyl), tetrahydropyranyl, benzyl, and the like. Suitable protecting groups for amino, amidino, and guanidino include, e.g., t-butoxycarbonyl, benzyloxycarbonyl, and the like. Suitable protecting groups for mercapto include, e.g., —C(O)—R" (where R" is alkyl, aryl, or arylalkyl), p-methoxybenzyl, trityl, and the like. Suitable protecting groups for carboxylic acid include, e.g., alkyl, aryl, or arylalkyl esters. Protecting groups may be added or removed in accordance with standard techniques, which are known to one skilled in the art and as described herein. The use of protecting groups is described in detail in, e.g., Green, T. W. and P. G. M. Wuts, *Protective Groups in Organic Synthesis* (1999), 3rd Ed., Wiley. As one of skill in the art would appreciate, the protecting group may also be a polymer resin such as a Wang resin, Rink resin, or a 2-chlorotrityl-chloride resin.

It will also be appreciated by those skilled in the art, although such protected derivatives of compounds of this invention may not possess pharmacological activity as such, they may be administered to a mammal and thereafter metabolized in the body to form compounds of the invention which are pharmacologically active. Such derivatives may therefore be described as prodrugs. All prodrugs of compounds of this invention are included within the scope of the invention.

Furthermore, all compounds of the invention which exist in free base or acid form can be converted to their pharma-

ceutically acceptable salts by treatment with the appropriate inorganic or organic base or acid by methods known to one skilled in the art. Salts of the compounds of the invention can also be converted to their free base or acid form by standard techniques.

It is understood that one skilled in the art may be able to make these compounds by similar methods or by combining other methods known to one skilled in the art. It is also understood that one skilled in the art would be able to make other compounds of Formula (1) not specifically illustrated herein by using the appropriate starting materials and modifying the parameters of the synthesis. In general, starting materials may be obtained from sources such as Sigma Aldrich, Lancaster Synthesis, Inc., Maybridge, Matrix Scientific, TCI, and Fluorochem USA, etc. or synthesized according to sources known to those skilled in the art (see, e.g., *Advanced Organic Chemistry: Reactions, Mechanisms, and Structure*, 5th edition (Wiley, December 2000)) or prepared as described in this invention.

As mentioned above, these lipid-like compounds are useful for delivery of pharmaceutical agents. They can be preliminarily screened for their efficacy in delivering pharmaceutical agents by an in vitro assay and then confirmed by animal experiments and clinic trials. Other methods will also be apparent to those of ordinary skill in the art.

Not to be bound by any theory, the lipid-like compounds of this disclosure can facilitate delivery of pharmaceutical agents by forming complexes, e.g., nanocomplexes and microparticles. The hydrophilic head of such a lipid-like compound, positively or negatively charged, binds to a moiety of a pharmaceutical agent that is oppositely charged and its hydrophobic moiety binds to a hydrophobic moiety of the pharmaceutical agent. Either binding can be covalent or non-covalent.

The above described complexes can be prepared using procedures described in publications such as Wang et al., *ACS Synthetic Biology*, 1, 403-07 (2012). Generally, they are obtained by incubating a lipid-like compound and a pharmaceutical agent in a buffer such as a sodium acetate buffer or a phosphate buffered saline ("PBS").

5.2 Hydrophilic Groups

In certain embodiments, the selected hydrophilic functional group or moiety may alter or otherwise impart properties to the compound or to the transfer vehicle of which such compound is a component (e.g., by improving the transfection efficiencies of a lipid nanoparticle of which the compound is a component). For example, the incorporation of guanidinium as a hydrophilic head-group in the compounds disclosed herein may promote the fusogenicity of such compounds (or of the transfer vehicle of which such compounds are a component) with the cell membrane of one or more target cells, thereby enhancing, for example, the transfection efficiencies of such compounds. It has been hypothesized that the nitrogen from the hydrophilic guanidinium moiety forms a six-membered ring transition state which grants stability to the interaction and thus allows for cellular uptake of encapsulated materials. (Wender, et al., *Adv. Drug Del. Rev.* (2008) 60: 452-472.) Similarly, the incorporation of one or more amino groups or moieties into the disclosed compounds (e.g., as a head-group) may further promote disruption of the endosomal/lysosomal membrane of the target cell by exploiting the fusogenicity of such amino groups. This is based not only on the pKa of the amino group of the composition, but also on the ability of the amino group to undergo a hexagonal phase transition and fuse with the target cell surface, i.e. the vesicle membrane. (Koltover, et al. *Science* (1998) 281: 78-81.) The result is

believed to promote the disruption of the vesicle membrane and release of the lipid nanoparticle contents into the target cell.

Similarly, in certain embodiments the incorporation of, for example, imidazole as a hydrophilic head-group in the compounds disclosed herein may serve to promote endosomal or lysosomal release of, for example, contents that are encapsulated in a transfer vehicle (e.g., lipid nanoparticle) of the invention. Such enhanced release may be achieved by one or both of a proton-sponge mediated disruption mechanism and/or an enhanced fusogenicity mechanism. The proton-sponge mechanism is based on the ability of a compound, and in particular a functional moiety or group of the compound, to buffer the acidification of the endosome. This may be manipulated or otherwise controlled by the pKa of the compound or of one or more of the functional groups comprising such compound (e.g., imidazole). Accordingly, in certain embodiments the fusogenicity of, for example, the imidazole-based compounds disclosed herein (e.g., HGT4001 and HGT4004) are related to the endosomal disruption properties, which are facilitated by such imidazole groups, which have a lower pKa relative to other traditional ionizable lipids. Such endosomal disruption properties in turn promote osmotic swelling and the disruption of the liposomal membrane, followed by the transfection or intracellular release of the polynucleotide materials loaded or encapsulated therein into the target cell. This phenomenon can be applicable to a variety of compounds with desirable pKa profiles in addition to an imidazole moiety. Such embodiments also include multi-nitrogen based functionalities such as polyamines, poly-peptide (histidine), and nitrogen-based dendritic structures.

Exemplary ionizable and/or cationic lipids are described in International PCT patent publications WO2015/095340, WO2015/199952, WO2018/011633, WO2017/049245, WO2015/061467, WO2012/040184, WO2012/000104, WO2015/074085, WO2016/081029, WO2017/004 143, WO2017/075531, WO2017/117528, WO2011/022460, WO2013/148541, WO2013/116126, WO2011/153120, WO2012/044638, WO2012/054365, WO2011/090965, WO2013/016058, WO2012/162210, WO2008/042973, WO2010/129709, WO2010/144740, WO20 12/099755, WO2013/049328, WO2013/086322, WO2013/086373, WO2011/071860, WO2009/132131, WO2010/048536, WO2010/088537, WO2010/054401, WO2010/054406, WO2010/054405, WO2010/054384, WO2012/016184, WO2009/086558, WO2010/042877, WO2011/000106, WO2011/000107, WO2005/120152, WO2011/141705, WO2013/126803, WO2006/007712, WO2011/038160, WO2005/121348, WO2011/066651, WO2009/127060, WO2011/141704, WO2006/069782, WO2012/031043, WO2013/006825, WO2013/033563, WO2013/089151, WO2017/099823, WO2015/095346, and WO2013/086354, and US patent publications US2016/0311759, US2015/0376115, US2016/0151284, US2017/0210697, US2015/0140070, US2013/0178541, US2013/0303587, US2015/

0141678, US2015/0239926, US2016/0376224, US2017/0119904, US2012/0149894, US2015/0057373, US2013/0090372, US2013/0274523, US2013/0274504, US2013/0274504, US2009/0023673, US2012/0128760, US2010/0324120, US2014/0200257, US2015/0203446, US2018/0005363, US2014/0308304, US2013/0338210, US2012/0101148, US2012/0027796, US2012/0058144, US2013/0323269, US2011/0117125, US2011/0256175, US2012/0202871, US2011/0076335, US2006/0083780, US2013/0123338, US2015/0064242, US2006/0051405, US2013/0065939, US2006/0008910, US2003/0022649, US2010/0130588, US2013/0116307, US2010/0062967, US2013/0202684, US2014/0141070, US2014/0255472, US2014/0039032, US2018/0028664, US2016/0317458, and US2013/0195920, the contents of all of which are incorporated herein by reference in their entirety. International patent application WO 2019/131770 is also incorporated herein by reference in its entirety.

B. Peg Lipids

The use and inclusion of polyethylene glycol (PEG)-modified phospholipids and derivatized lipids such as derivatized ceramides (PEG-CER), including N-Octanoyl-Sphingosine-1-[Succinyl(Methoxy Polyethylene Glycol)-2000] (C8 PEG-2000 ceramide) in the liposomal and pharmaceutical compositions described herein is contemplated, preferably in combination with one or more of the compounds and lipids disclosed herein. Contemplated PEG-modified lipids include, but are not limited to, a polyethylene glycol chain of up to 5 kDa in length covalently attached to a lipid with alkyl chain(s) of C6-C20 length. In some embodiments, the PEG-modified lipid employed in the compositions and methods of the invention is 1,2-dimyristoyl-sn-glycerol, methoxypolyethylene Glycol (2000 MW PEG) "DMG-PEG2000." The addition of PEG-modified lipids to the lipid delivery vehicle may prevent complex aggregation and may also provide a means for increasing circulation lifetime and increasing the delivery of the lipid-polynucleotide composition to the target tissues. (Klibanov et al. (1990) FEBS Letters, 268 (1): 235-237), or they may be selected to rapidly exchange out of the formulation in vivo (see U.S. Pat. No. 5,885,613). Particularly useful exchangeable lipids are PEG-ceramides having shorter acyl chains (e.g., C14 or C18). The PEG-modified phospholipid and derivatized lipids of the present invention may comprise a molar ratio from about 0% to about 20%, about 0.5% to about 20%, about 1% to about 15%, about 4% to about 10%, or about 2% of the total lipid present in a liposomal lipid nanoparticle.

In an embodiment, a PEG-modified lipid is described in International Pat. Appl. No. PCT/US2019/015913 or PCT/US2020/046407, which are incorporated herein by reference in their entirety. In an embodiment, a transfer vehicle comprises one or more PEG-modified lipids.

Non-limiting examples of PEG-modified lipids include PEG-modified phosphatidylethanolamines and phosphatidic acids, PEG-ceramide conjugates (e.g., PEG-CerC14 or PEG-CerC20). PEG-modified dialkylamines and PEG-

255

modified 1,2-diacyloxypropan-3-amines. In some further embodiments, a PEG-modified lipid may be, e.g., PEG-DOMG, PEG-DMG, PEG-DLPE, PEG-DMPE, PEG-DPPC, or a PEG-DSPE.

In some still further embodiments, the PEG-modified lipid includes, but is not limited to 1,2-dimyristoyl-sn-glycerol methoxypolyethylene glycol (PEG-DMG), 1,2-distearoyl-sn-glycerol-3-phosphoethanolamine-N-[amino(polyethylene glycol)](PEG-DSPE), PEG-disteryl glycerol (PEG-DSG), PEG-dipalmitoyl, PEG-dioleoyl, PEG-distearyl, PEG-diacylglyceramide (PEG-DAG), PEG-dipalmitoyl phosphatidylethanolamine (PEG-DPPE), PEG-1,2-dimyristyloxylpropyl-3-amine (PEG-c-DMA).

In some still further embodiments, the PEG-modified lipid is DSPE-PEG, DMG-PEG, PEG-DAG, PEG-S-DAG, PEG-PE, PEG-S-DMG, PEG-cer, PEG-dialkoxypropylcarbamate, PEG-OR, PEG-OH, PEG-c-DOMG, or PEG-1. In some embodiments, the PEG-modified lipid is DSPE-PEG (2000).

In some embodiments, the PEG-modified lipid comprises a PEG moiety comprising 10-70 (e.g., 30-60) oxyethylene ($-\text{O}-\text{CH}_2-\text{CH}_2-$) units or portions thereof. In some embodiments, the PEG-modified lipid comprises ($-\text{O}-\text{CH}_2-\text{CH}_2-$) OR_w , and v is an integer between 0 and 70 (inclusive) (e.g., an integer between 30 and 60), w is 30

hydrogen or alkyl.

In various embodiments, a PEG-modified lipid may also be referred to as “PEGylated lipid” or “PEG-lipid.”

In one embodiment, the PEG-lipid is selected from the group consisting of a PEG-modified phosphatidylethanolamine, a PEG-modified phosphatidic acid, a PEG-modified ceramide, a PEG-modified dialkylamine, a PEG-modified diacylglycerol, a PEG-modified dialkylglycerol, and mixtures thereof.

In some embodiments, the lipid moiety of the PEG-lipids includes those having lengths of from about C_{14} to about

256

C_{22} , such as from about C_{14} to about C_{16} . In some embodiments, a PEG moiety, for example a mPEG- NH_2 , has a size of about 1000, about 2000, about 5000, about 10,000, about 15,000 or about 20,000 daltons. In one embodiment, the PEG-lipid is PEG2k-DMG.

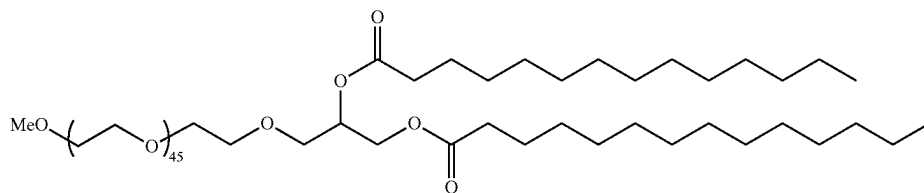
In one embodiment, the lipid nanoparticles described herein can comprise a lipid modified with a non-diffusible PEG. Non-limiting examples of non-diffusible PEGs include PEG-DSG and PEG-DSPE.

PEG-lipids are known in the art, such as those described in U.S. Pat. No. 8,158,601 and International Pat. Publ. No. WO2015/130584 A2, which are incorporated herein by reference in their entirety.

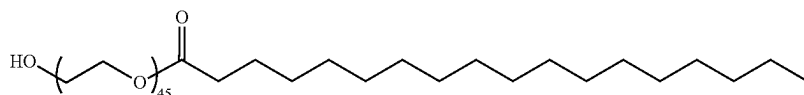
In various embodiments, lipids (e.g., PEG-lipids), described herein may be synthesized as described International Pat. Publ. No. PCT/US2016/000129, which is incorporated by reference in its entirety.

The lipid component of a lipid nanoparticle composition may include one or more molecules comprising polyethylene glycol, such as PEG or PEG-modified lipids. Such species may be alternately referred to as PEGylated lipids. A PEG lipid is a lipid modified with polyethylene glycol. A PEG lipid may be selected from the non-limiting group including PEG-modified phosphatidylethanolamines, PEG-modified phosphatidic acids, PEG-modified ceramides, PEG-modified dialkylamines, PEG-modified diacylglycerols, PEG-modified dialkylglycerols, and mixtures thereof. For example, a PEG lipid may be PEG-c-DOMG, PEG-DMG, PEG-DLPE, PEG-DMPE, PEG-DPPC, or a PEG-DSPE lipid.

In some embodiments the PEG-modified lipids are a modified form of PEG-DMG. PEG-DMG has the following structure:



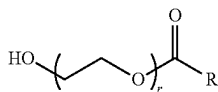
In some embodiments the PEG-modified lipids are a modified form of PEG-C18, or PEG-1. PEG-1 has the following structure



257

In one embodiment, PEG lipids useful in the present invention can be PEGylated lipids described in International Publication No. WO2012099755, the contents of which is herein incorporated by reference in its entirety. Any of these exemplary PEG lipids described herein may be modified to comprise a hydroxyl group on the PEG chain. In certain embodiments, the PEG lipid is a PEG-OH lipid. In certain embodiments, the PEG-OH lipid includes one or more hydroxyl groups on the PEG chain. In certain embodiments, a PEG-OH or hydroxy-PEGylated lipid comprises an —OH group at the terminus of the PEG chain. Each possibility represents a separate embodiment of the present invention.

In some embodiments, the PEG lipid is a compound of Formula (P1):

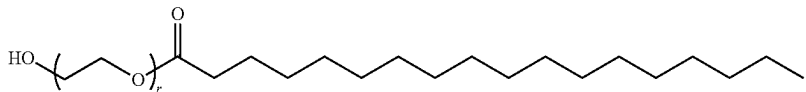


or a salt or isomer thereof, wherein:

r is an integer between 1 and 100;

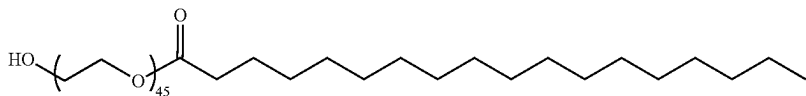
R is C₁₀₋₄₀ alkyl, C₁₀₋₄₀ alkenyl, or C₁₀₋₄₀ alkynyl; and optionally one or more methylene groups of R are independently replaced with C₃₋₁₀ carbocyclylene, 4 to 10 membered heterocyclylene, C₆₋₁₀ arylene, 4 to 10 membered heteroarylene, —N(R^N)—, —O—, —S—, —C(O)—, —C(O)N(R^N)—, —NR^NC(O)—, —NR^NC(O)N(R^N)—, —C(O)O—, —OC(O)—, —OC(O)O—, —OC(O)N(R^N)—, —NR^NC(O)O—, —C(O)S—, —SC(O)—, —C(=NR^N)—, —C(=NR^N)N(R^N)—, —NR^NC(=NR^N)—, —NR^NC(=NR^N)N(R^N)—, —C(S)—, —C(S)N(R^N)—, —NR^NC(S)—, —NR^NC(S)N(R^N)—, —S(O)—, —OS(O)—, —S(O)O—, —OS(O)O—, —OS(O)₂—, —S(O)₂O—, —OS(O)₂O—, —N(R^N)S(O)—, —S(O)N(R^N)—, —N(R^N)S(O)N(R^N)—, —OS(O)N(RN)—, —N(R^N)S(O)O—, —S(O)₂—, —N(R^N)S(O)₂—, —S(O)₂N(R^N)—, —N(R^N)S(O)₂N(R^N)—, —OS(O)₂N(R^N)—, or —N(R^N)S(O)₂O—; and each instance of R^N is independently hydrogen, C₁₋₆ alkyl, or a nitrogen protecting group.

For example, R is C17 alkyl. For example, the PEG lipid is a compound of Formula (P1-a):



or a salt or isomer thereof, wherein r is an integer between 1 and 100.

For example, the PEG lipid is a compound of the following formula:



C. Helper Lipids

In some embodiments, the transfer vehicle (e.g., LNP) described herein comprises one or more non-cationic helper

258

lipids. In some embodiments, the helper lipid is a phospholipid. In some embodiments, the helper lipid is a phospholipid substitute or replacement. In some embodiments, the phospholipid or phospholipid substitute can be, for example, one or more saturated or (poly)unsaturated phospholipids, or phospholipid substitutes, or a combination thereof. In general, phospholipids comprise a phospholipid moiety and one or more fatty acid moieties.

A phospholipid moiety can be selected, for example, from the non-limiting group consisting of phosphatidyl choline, phosphatidyl ethanolamine, phosphatidyl glycerol, phosphatidyl serine, phosphatidic acid, 2-lysophosphatidyl choline, and a sphingomyelin.

A fatty acid moiety can be selected, for example, from the non-limiting group consisting of lauric acid, myristic acid, myristoleic acid, palmitic acid, palmitoleic acid, stearic acid, oleic acid, linoleic acid, alpha-linolenic acid, erucic acid, phytanoic acid, arachidic acid, arachidonic acid, eicosapentaenoic acid, behenic acid, docosapentaenoic acid, and docosahexaenoic acid.

Phospholipids include, but are not limited to, glycerophospholipids such as phosphatidylcholines, phosphatidylethanolamines, phosphatidylserines, phosphatidylinositols, phosphatidyl glycerols, and phosphatidic acids. Phospholipids also include phosphosphingolipid, such as sphingomyelin.

In some embodiments, the helper lipid is a 1,2-distearoyl-177-glycero-3-phosphocholine (DSPC) analog, a DSPC substitute, oleic acid, or an oleic acid analog.

In some embodiments, a helper lipid is a non-phosphatidyl choline (PC) zwitterionic lipid, a DSPC analog, oleic acid, an oleic acid analog, or a DSPC substitute.

In some embodiments, a helper lipid is described in PCT/US2018.053569. Helper lipids suitable for use in a lipid composition of the disclosure include, for example, a variety of neutral, uncharged or zwitterionic lipids. Such helper lipids are preferably used in combination with one or more of the compounds and lipids disclosed herein. Examples of helper lipids include, but are not limited to, 5-heptadecylbenzene-1,3-diol (resorcinol), dipalmitoylphosphatidylcholine (DPPC), distearoylphosphatidylcholine (DSPC), dioleoylphosphatidylcholine (DOPC), dimyristoylphosphatidylcholine (DMPC), phosphatidylcholine (PLPC), 1,2-distearoylsn-glycero-3-phosphocholine

(DAPC), phosphatidylethanolamine (PE), egg phosphatidylcholine (EPC), dilaurylphosphatidylcholine (DLPC), dimyristoylphosphatidylcholine (DMPC), 1-myristoyl-2-palmitoyl phosphatidylcholine (MPPC), 1-palmitoyl-2-

myristoyl phosphatidylcholine (PMPC), 1-palmitoyl-2-stearoyl phosphatidylcholine (PSPC), 1,2-diarachidoyl-sn-glycero-3-phosphocholine (DBPC), 1-stearoyl-2-palmitoyl

259

phosphatidylcholine (SPPC), 1,2-dieicosenoyl-sn-glycero-3-phosphocholine (DEPC), palmitoyioieoyl phosphatidylcholine (POPC), lysophosphatidyl choline, dioleoyl phosphatidylethanol amine (DOPE) dilinoleoylphosphatidylcholine distearoylphosphatidylethanolamine (DSPE), dimyristoyl phosphatidylethanolamine (DMPE), dipalmitoyl phosphatidylethanolamine (DPPE), palmitoyloleoyl phosphatidylethanolamine (POPE), lysophosphatidylethanolamine and combinations thereof. In one embodiment, the helper lipid may be distearoylphosphatidylcholine (DSPC) or dimyristoyl phosphatidyl ethanolamine (DMPE). In another embodiment, the helper lipid may be distearoylphosphatidylcholine (DSPC). Helper lipids function to stabilize and improve processing of the transfer vehicles. Such helper lipids are preferably used in combination with other excipients, for example, one or more of the ionizable lipids disclosed herein. In some embodiments, when used in combination with an ionizable lipid, the helper lipid may comprise a molar ratio of 5% to about 90%, or about 10% to about 70% of the total lipid present in the lipid nanoparticle.

D. Structural Lipids

In an embodiment, a structural lipid is described in international patent application PCT/US2019/015913.

The transfer vehicles described herein comprise one or more structural lipids. Incorporation of structural lipids in the lipid nanoparticle may help mitigate aggregation of other lipids in the particle. Structural lipids can include, but are not limited to, cholesterol, fecosterol, ergosterol, bassicasterol, tomatidine, tomatine, ursolic, alpha-tocopherol, and mixtures thereof. In certain embodiments, the structural lipid is cholesterol. In certain embodiments, the structural lipid includes cholesterol and a corticosteroid (such as, for example, prednisolone, dexamethasone, prednisone, and hydrocortisone), or a combination thereof.

In some embodiments, the structural lipid is a sterol. In certain embodiments, the structural lipid is a steroid. In certain embodiments, the structural lipid is cholesterol. In certain embodiments, the structural lipid is an analog of cholesterol. In certain embodiments, the structural lipid is alpha-tocopherol.

260

The transfer vehicles described herein comprise one or more structural lipids. Incorporation of structural lipids in a transfer vehicle, e.g., a lipid nanoparticle, may help mitigate aggregation of other lipids in the particle. In certain embodiments, the structural lipid includes cholesterol and a corticosteroid (such as, for example, prednisolone, dexamethasone, prednisone, and hydrocortisone), or a combination thereof.

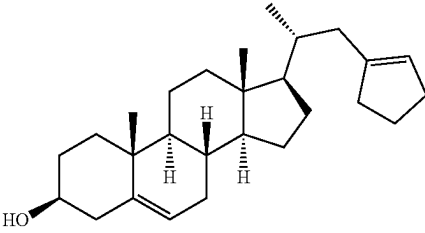
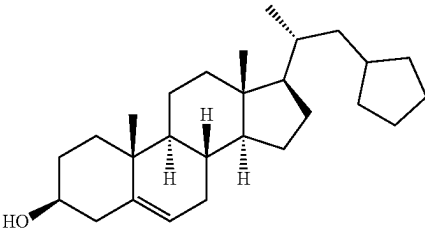
In some embodiments, the structural lipid is a sterol. Structural lipids can include, but are not limited to, sterols (e.g., phytosterols or zoosterols).

In certain embodiments, the structural lipid is a steroid. For example, sterols can include, but are not limited to, cholesterol, 0-sitosterol, fecosterol, ergosterol, sitosterol, campesterol, stigmasterol, brassicasterol, ergosterol, tomatidine, tomatine, ursolic acid, or alpha-tocopherol.

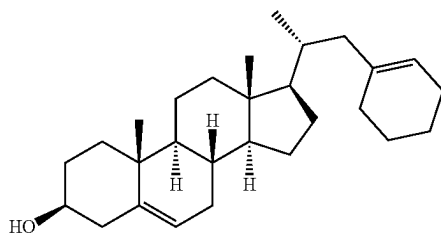
In some embodiments, a transfer vehicle includes an effective amount of an immune cell delivery potentiating lipid, e.g., a cholesterol analog or an amino lipid or combination thereof, that, when present in a transfer vehicle, e.g., a lipid nanoparticle, may function by enhancing cellular association and/or uptake, internalization, intracellular trafficking and/or processing, and/or endosomal escape and/or may enhance recognition by and/or binding to immune cells, relative to a transfer vehicle lacking the immune cell delivery potentiating lipid. Accordingly, while not intending to be bound by any particular mechanism or theory, in one embodiment, a structural lipid or other immune cell delivery potentiating lipid of the disclosure binds to C1q or promotes the binding of a transfer vehicle comprising such lipid to C1q. Thus, for in vitro use of the transfer vehicles of the disclosure for delivery of a nucleic acid molecule to an immune cell, culture conditions that include C1q are used (e.g., use of culture media that includes serum or addition of exogenous C1q to serum-free media). For in vivo use of the transfer vehicles of the disclosure, the requirement for C1q is supplied by endogenous C1q.

In certain embodiments, the structural lipid is cholesterol. In certain embodiments, the structural lipid is an analog of cholesterol. In some embodiments, the structural lipid is a lipid in Table 16:

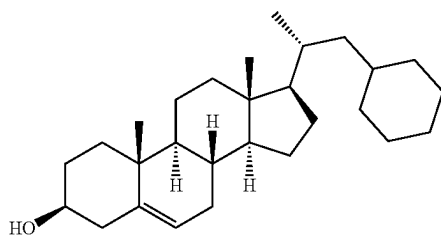
TABLE 16

CMPD No. S-	Structure
1	
2	

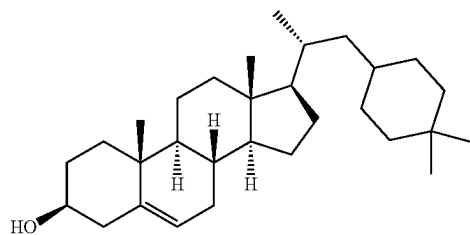
3



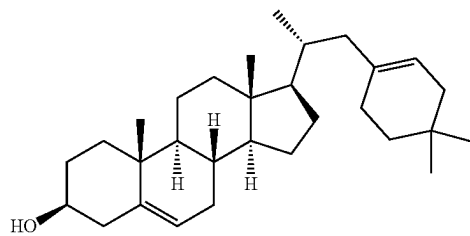
4



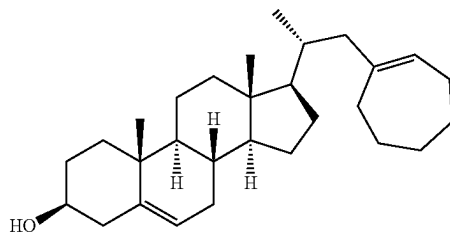
5



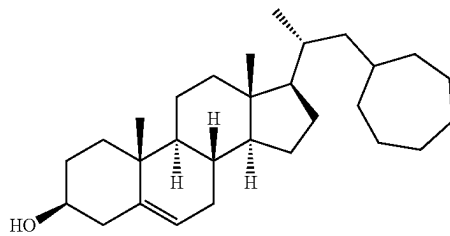
6



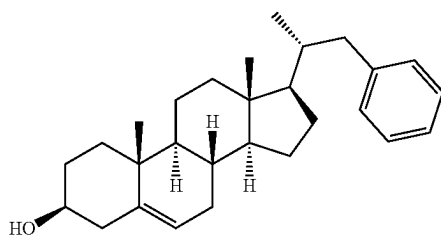
7



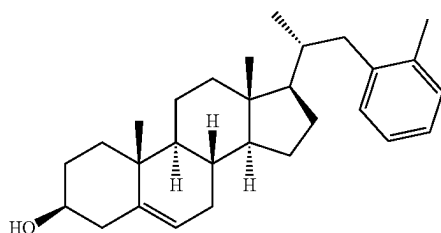
8



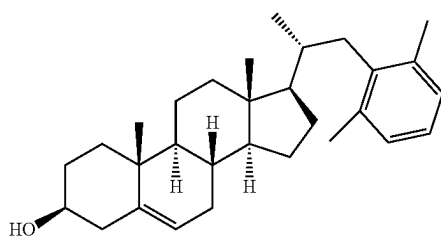
9



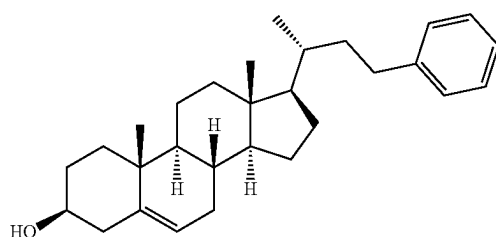
10



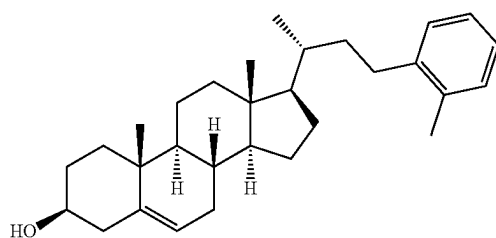
11



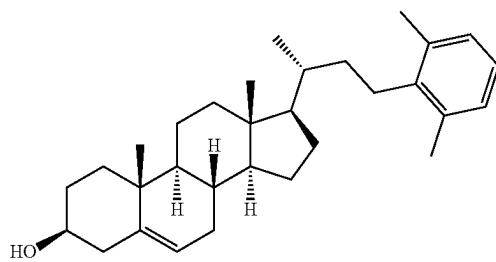
12



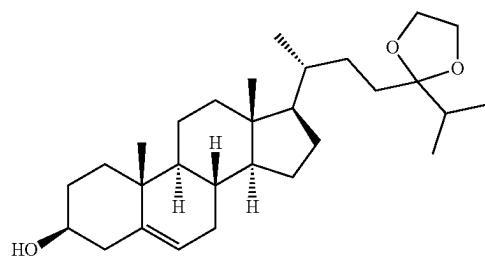
13



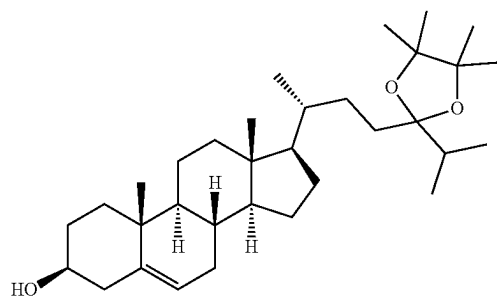
14



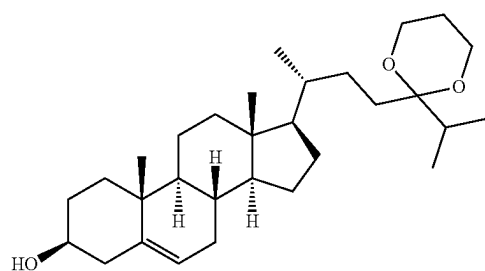
15



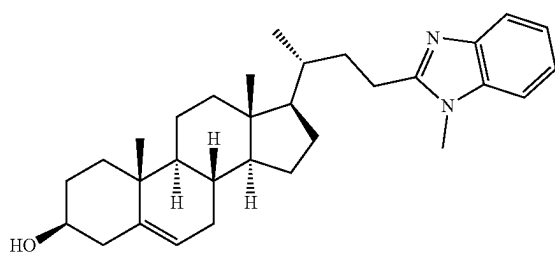
16



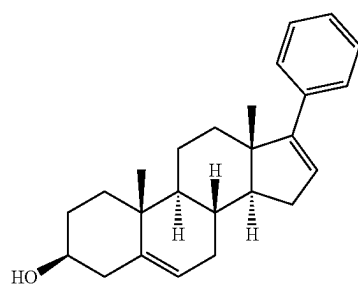
17



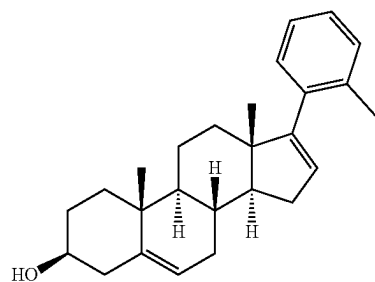
18



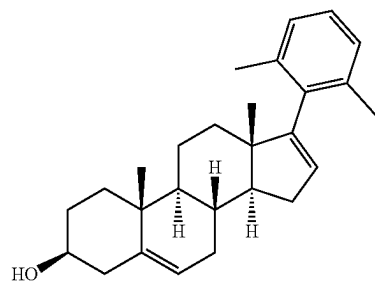
19



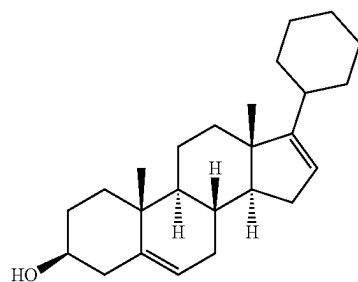
20



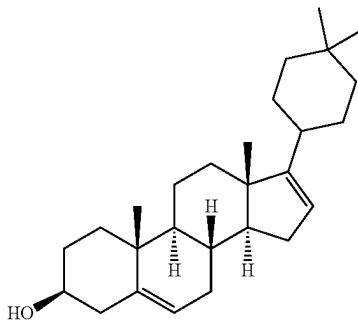
21



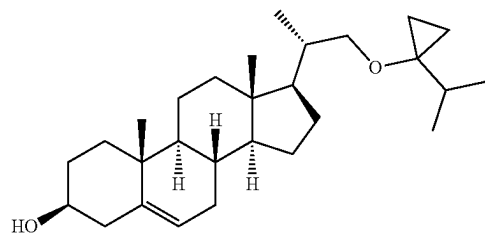
22



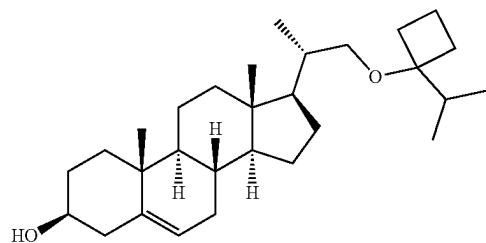
23



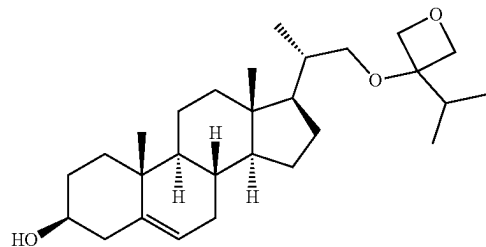
24



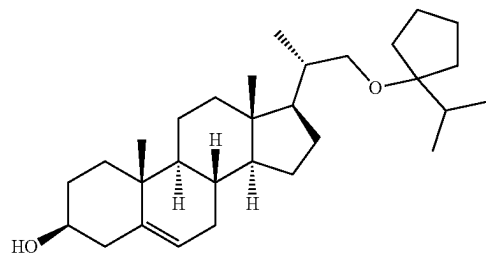
25



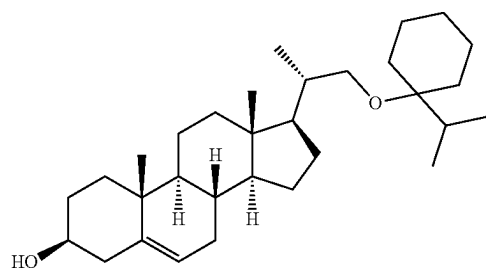
26



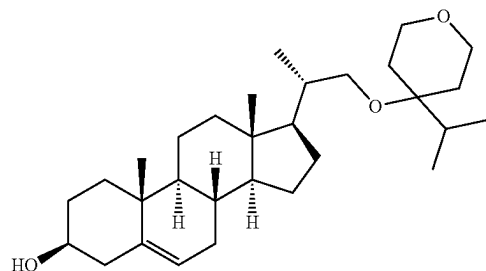
27



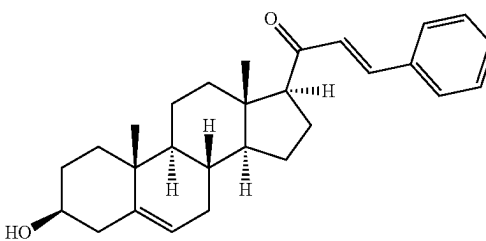
28



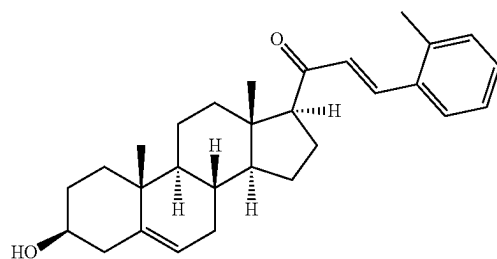
29



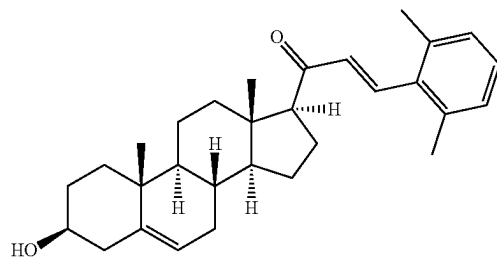
30



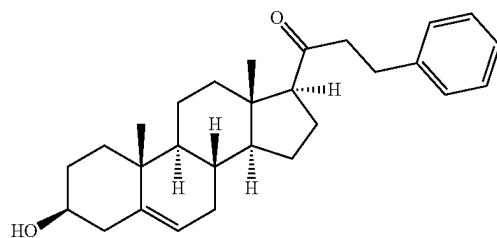
31



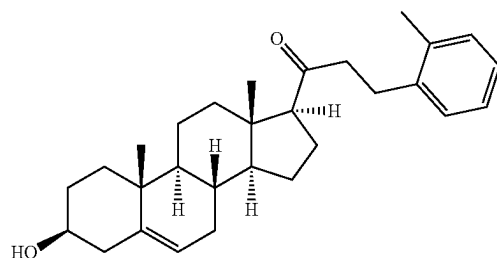
32



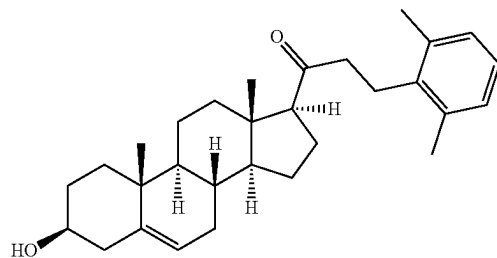
33



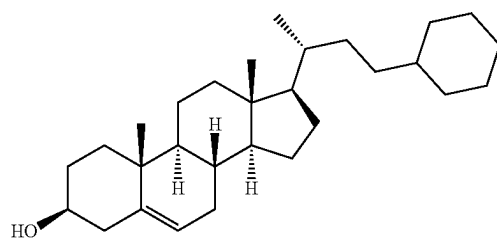
34



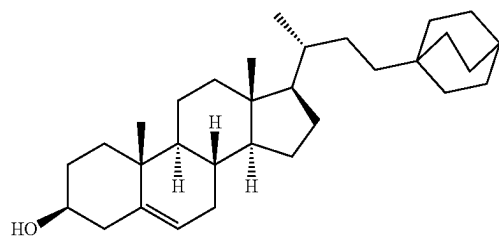
35



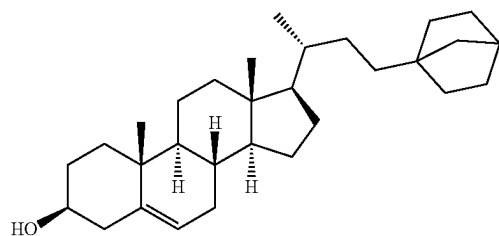
36



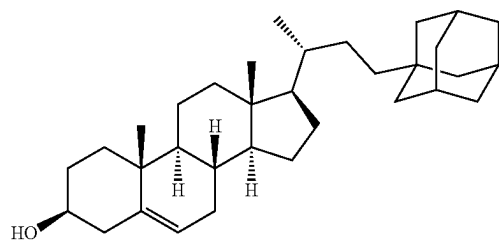
37



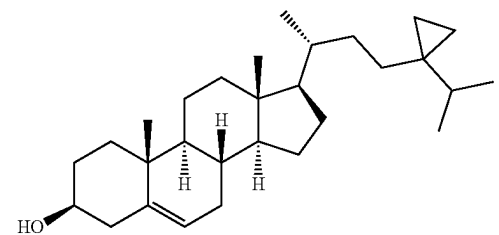
38



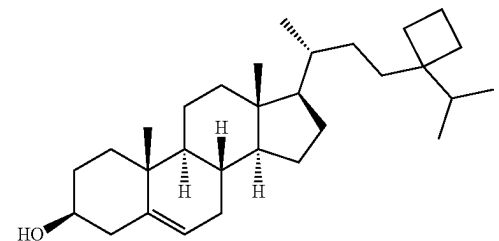
39



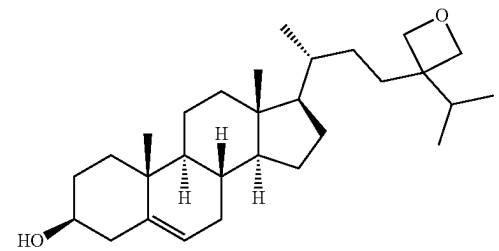
40



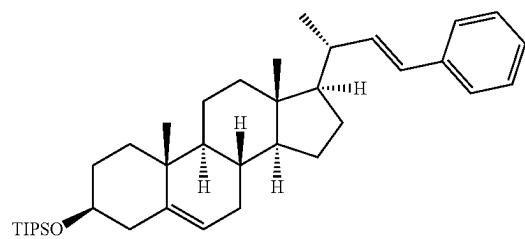
41



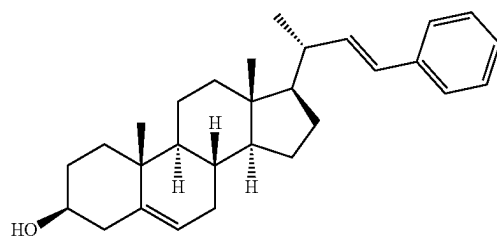
42



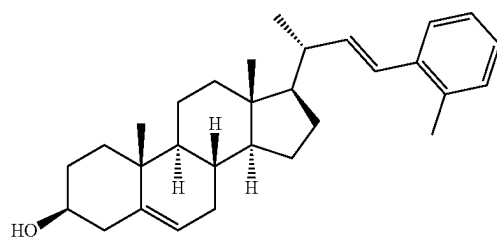
150



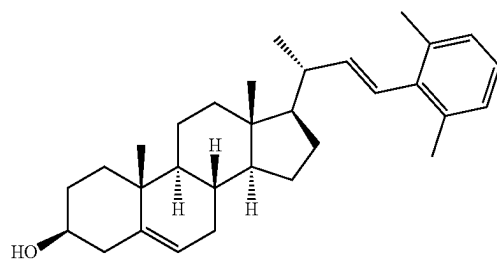
154



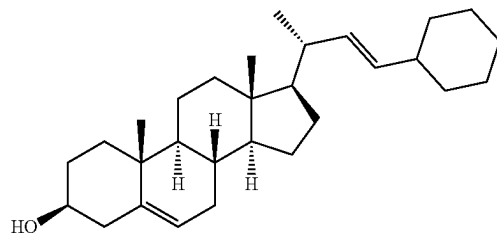
162



163



164



165

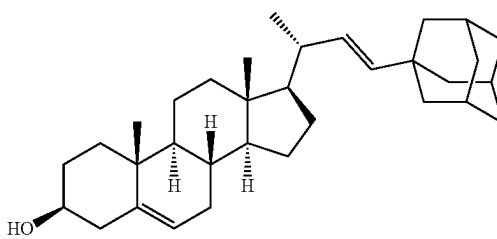
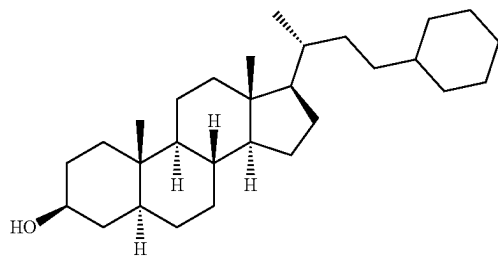
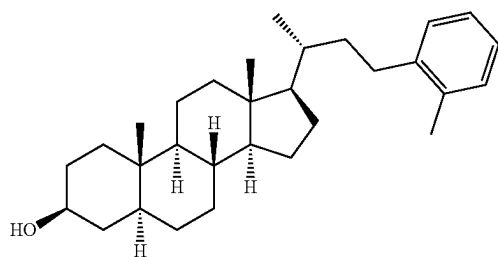


TABLE 16-continued

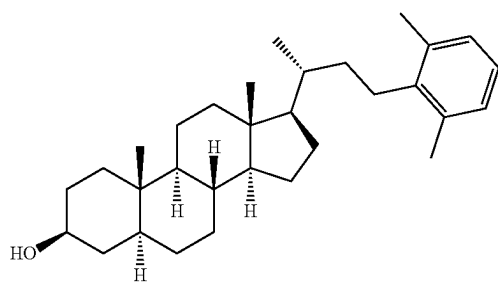
169



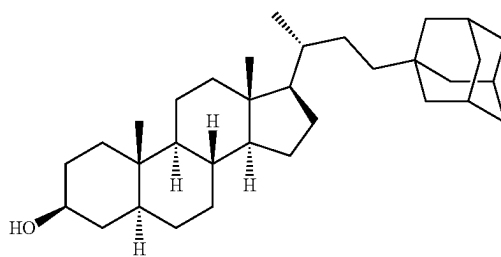
170



171



172



184

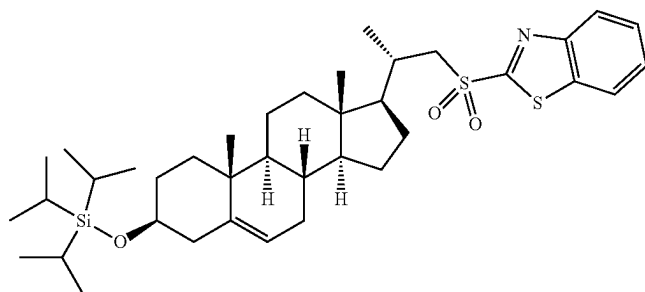
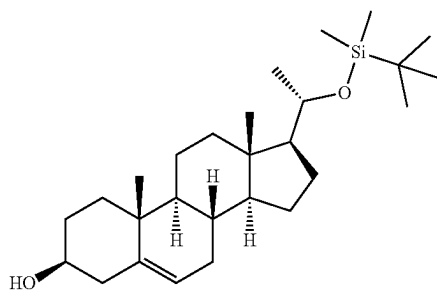
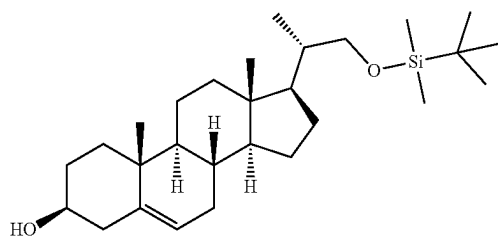


TABLE 16-continued

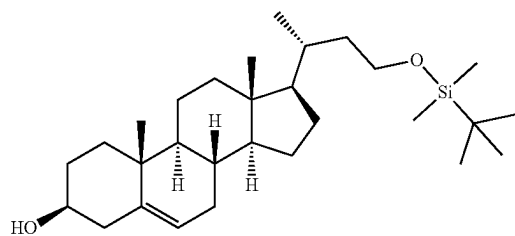
43



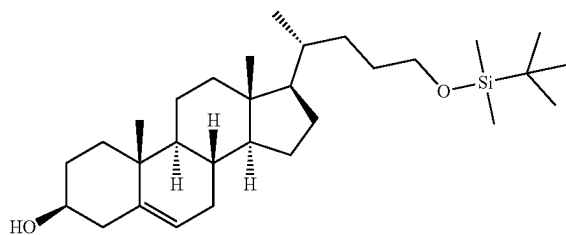
44



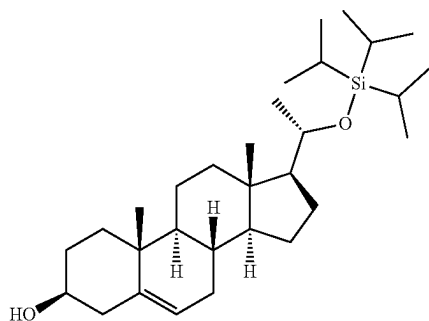
45



46



47



48

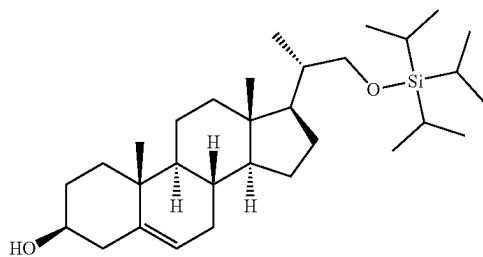
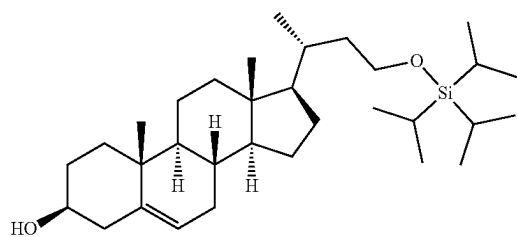
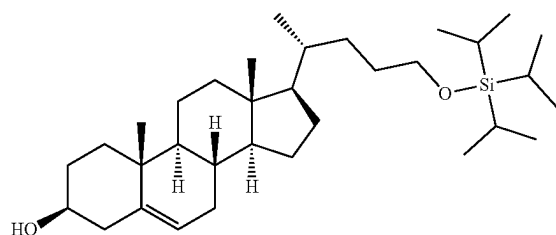


TABLE 16-continued

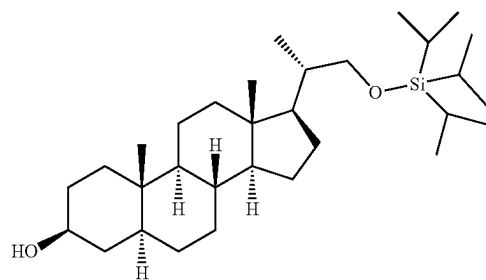
49



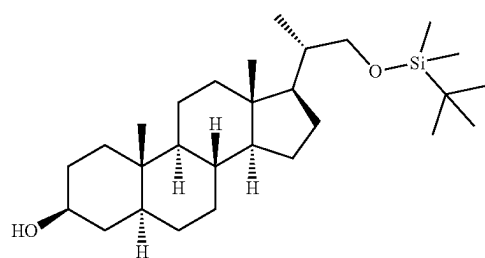
50



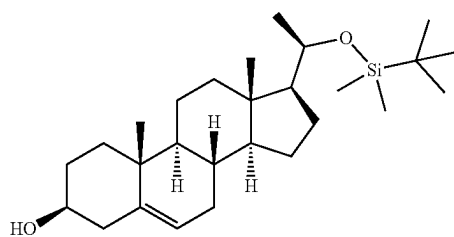
175



176



177



178

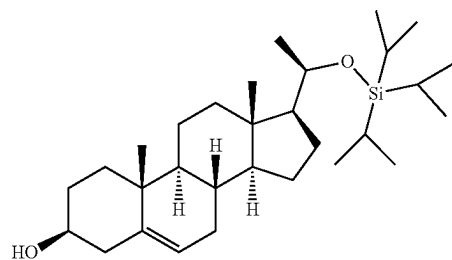
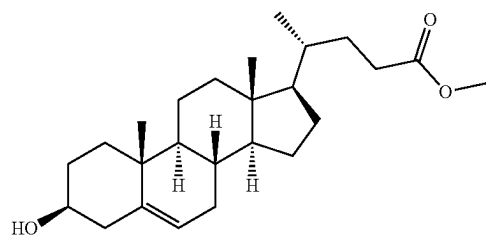
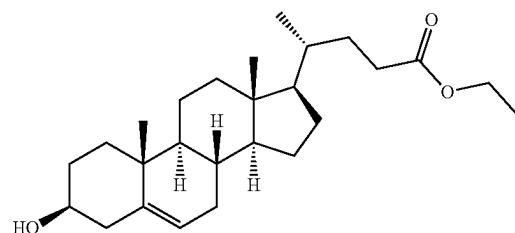


TABLE 16-continued

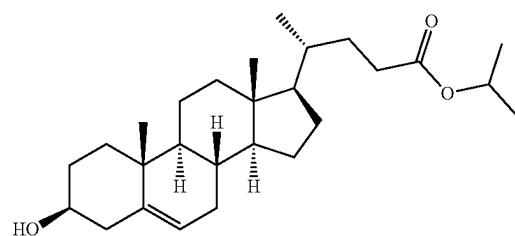
51



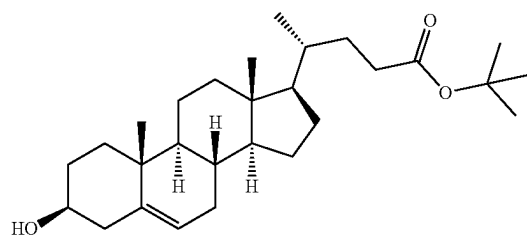
52



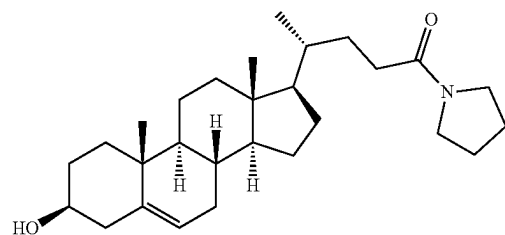
53



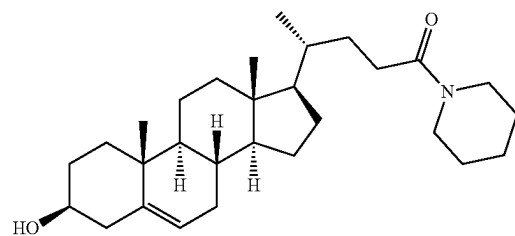
54



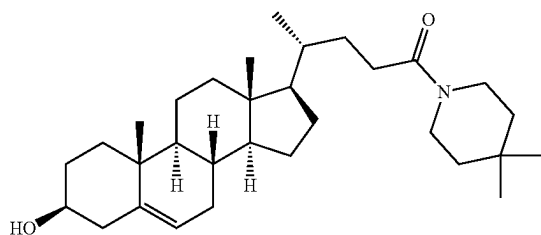
55



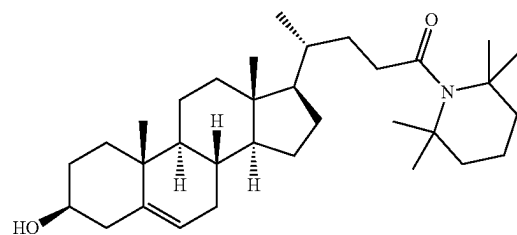
56



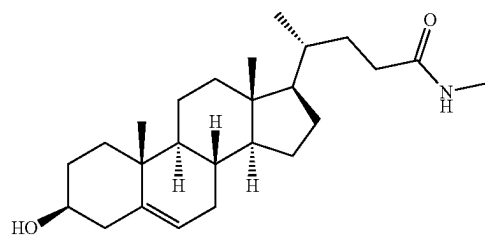
57



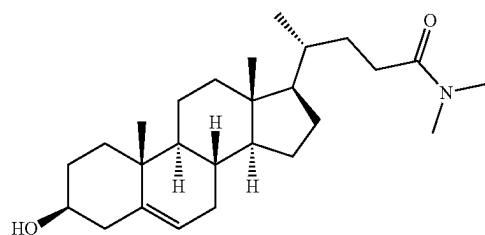
58



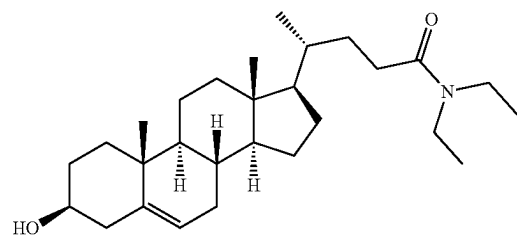
59



60



61



62

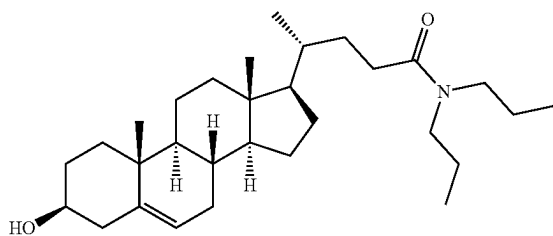
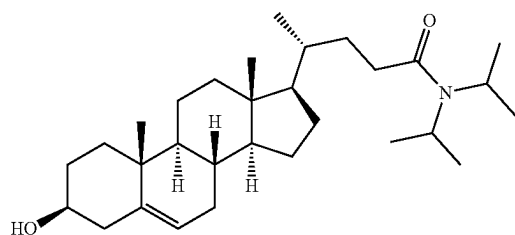
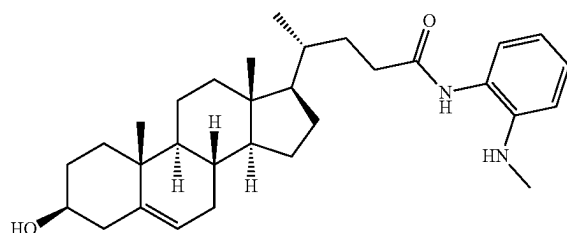


TABLE 16-continued

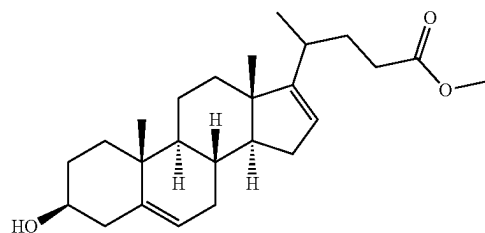
63



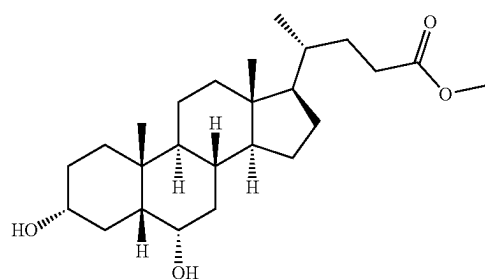
64



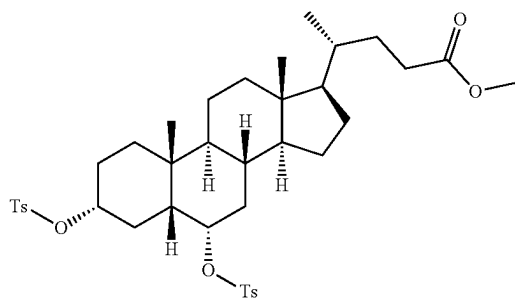
65



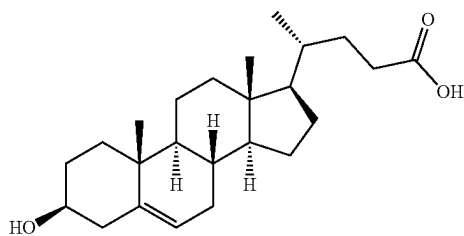
66



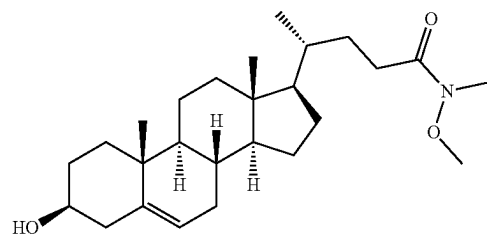
67



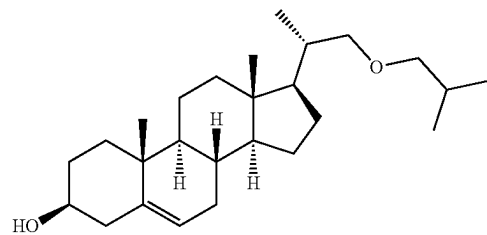
149



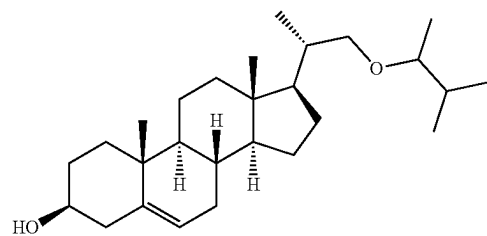
153



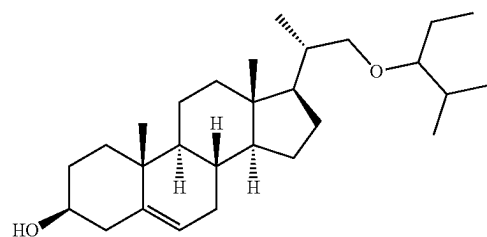
68



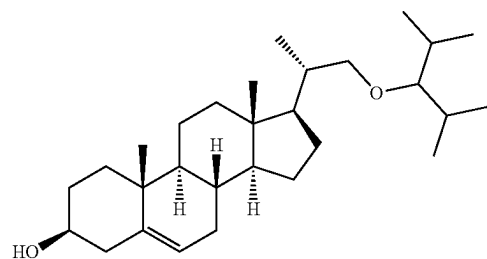
69



70



71



72

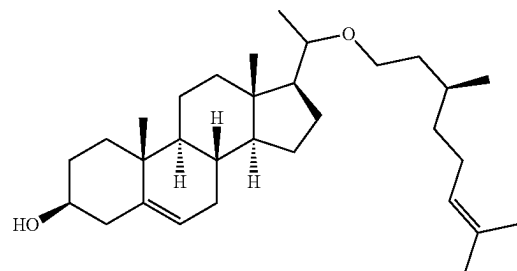
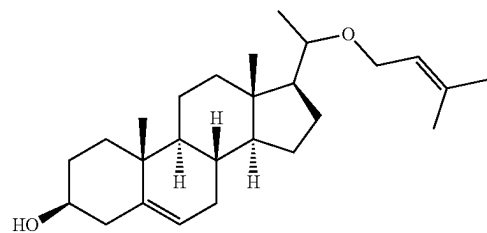
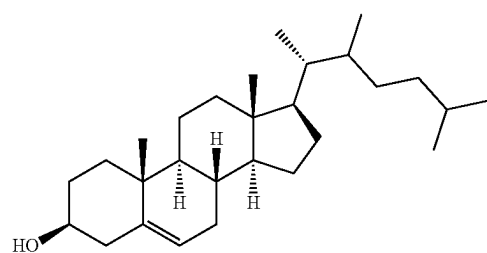


TABLE 16-continued

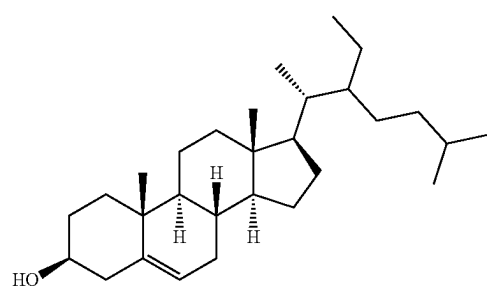
73



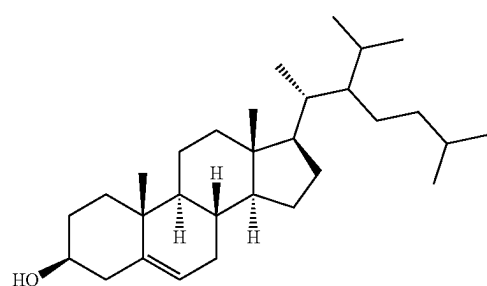
74



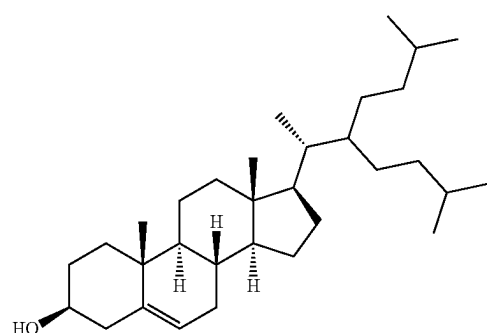
75



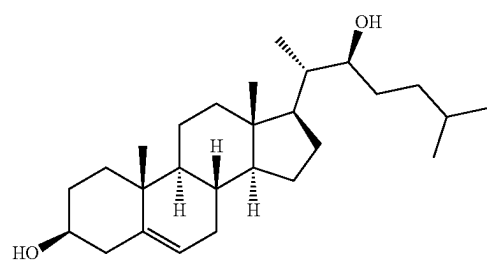
76



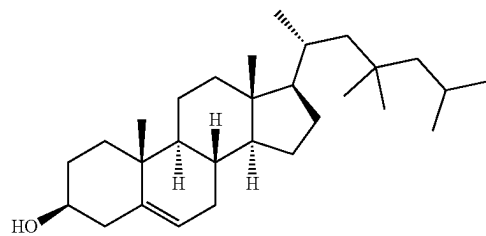
77



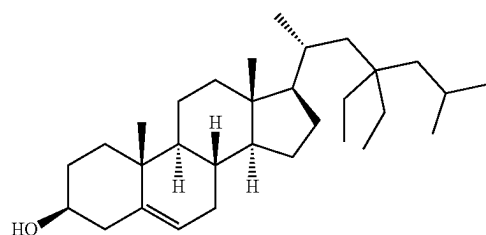
78



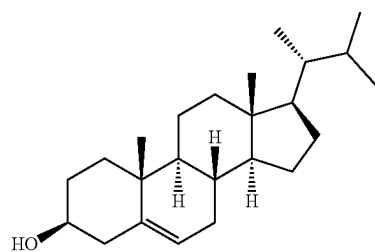
79



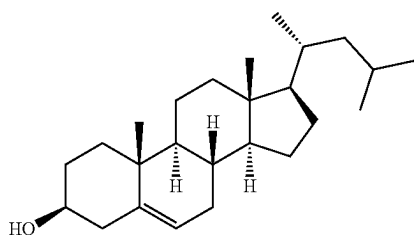
80



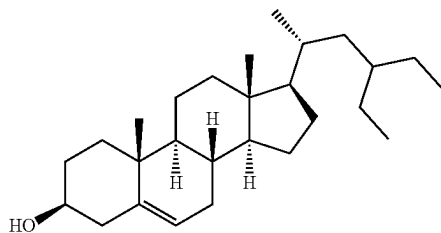
81



82



83



84

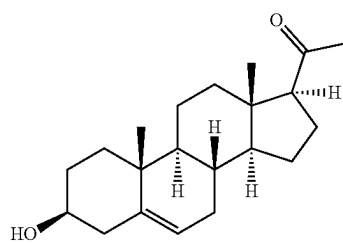
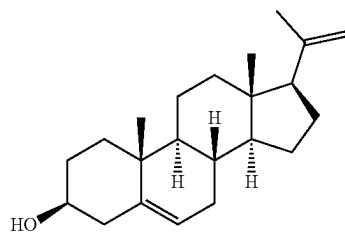
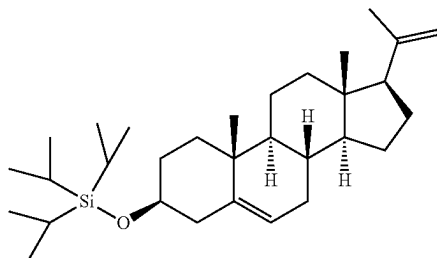


TABLE 16-continued

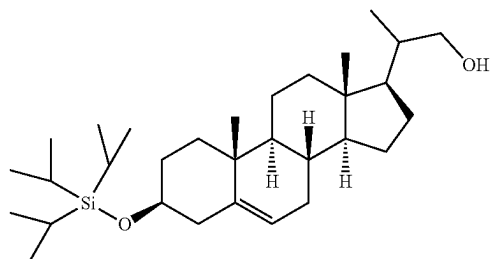
85



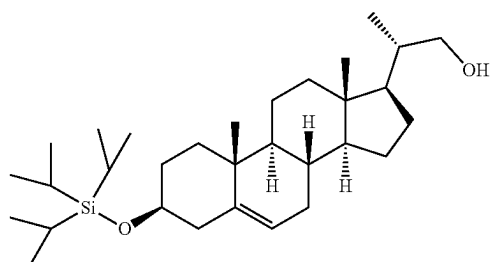
86



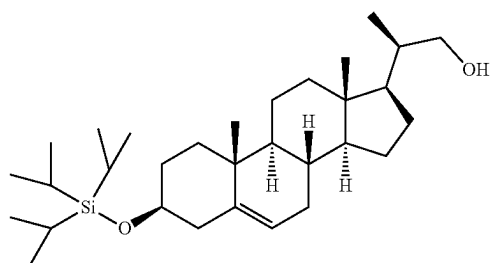
87



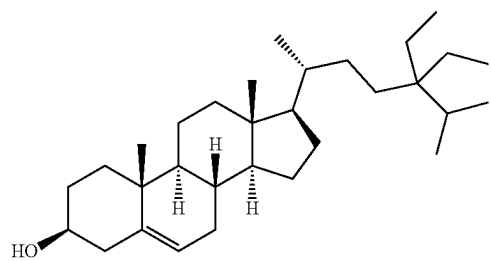
152



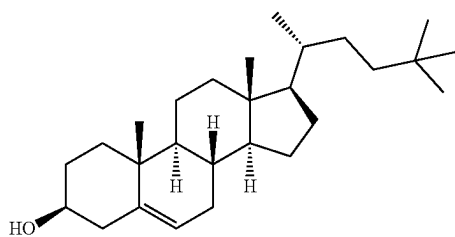
157



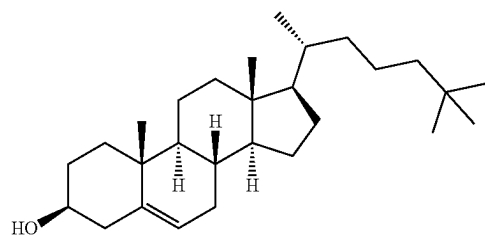
88



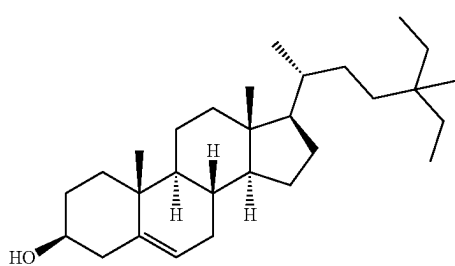
89



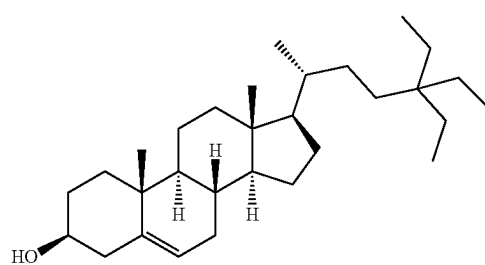
90



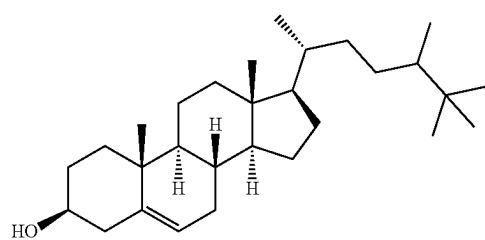
91



92



93



94

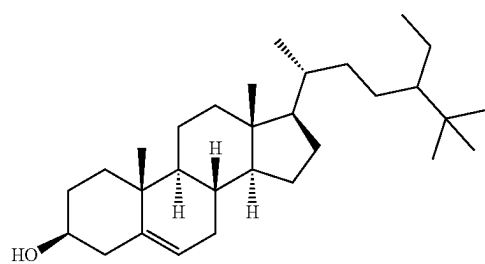
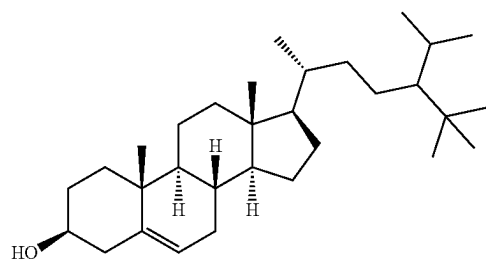
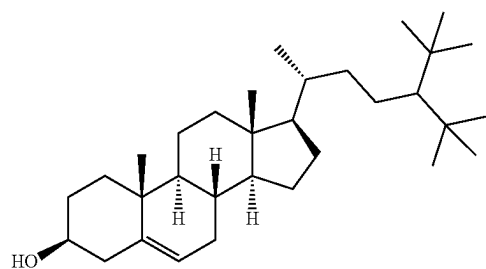


TABLE 16-continued

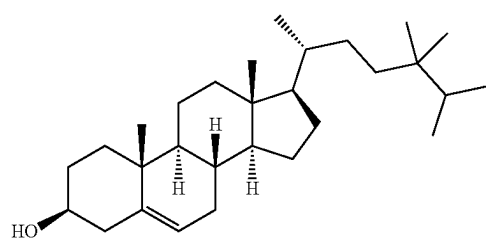
95



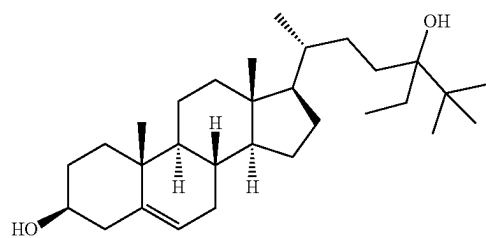
96



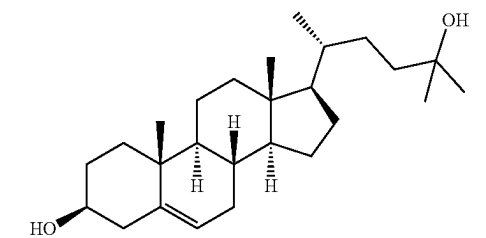
97



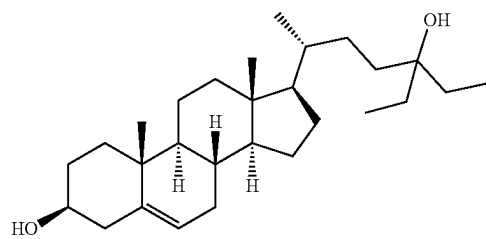
98



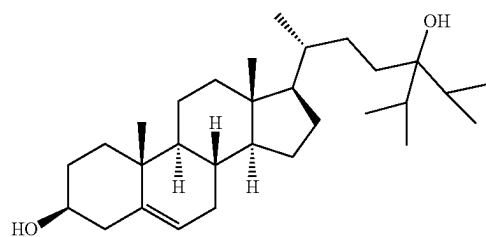
99



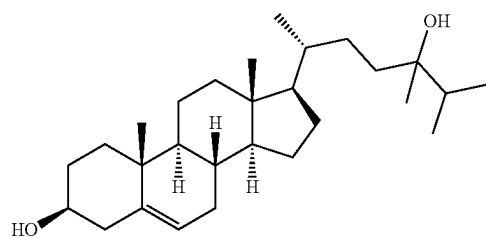
100



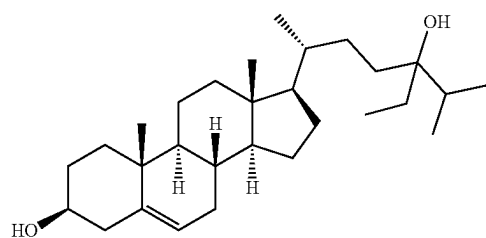
101



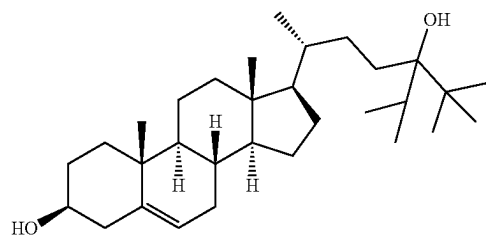
102



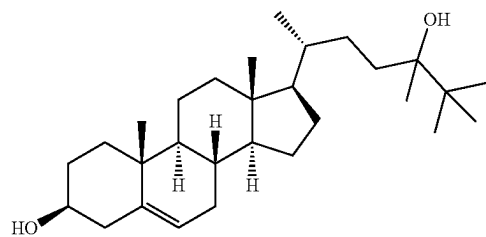
103



104



105



180

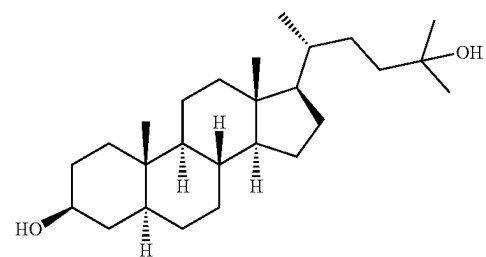
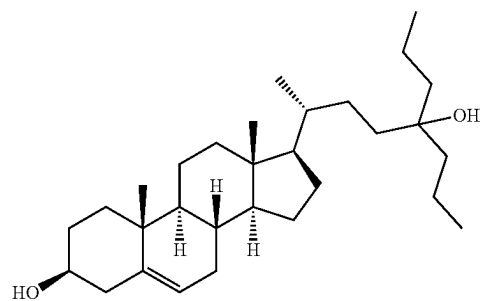
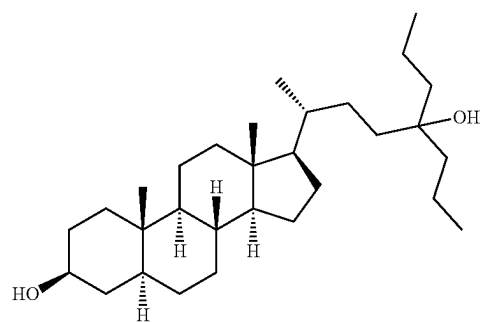


TABLE 16-continued

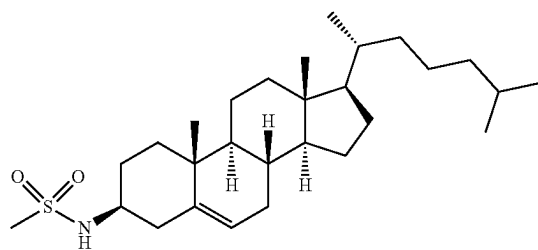
181



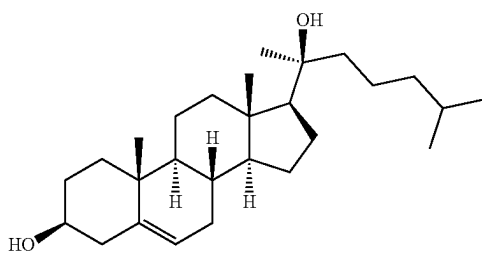
182



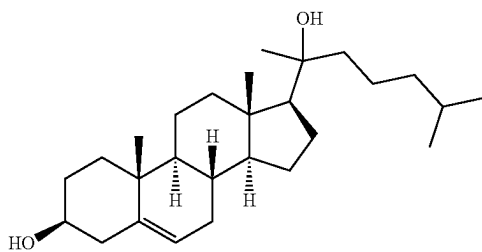
106



107



108



109

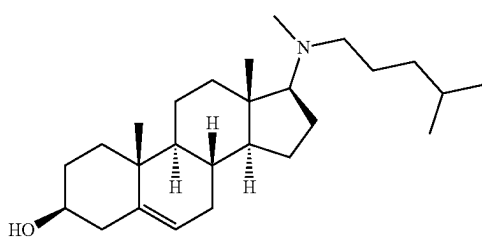
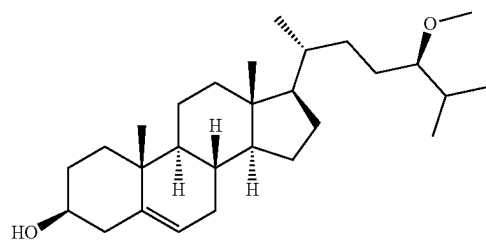
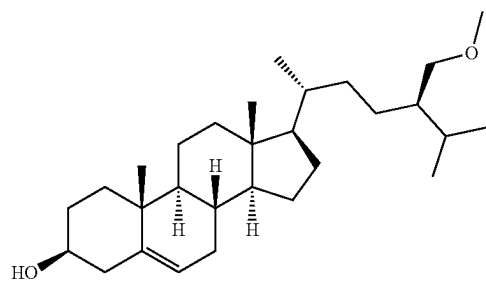


TABLE 16-continued

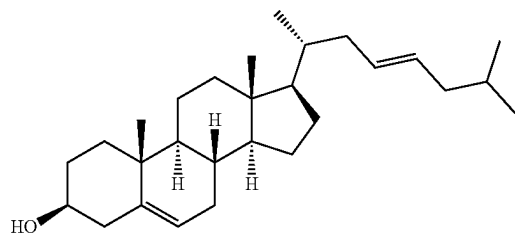
110



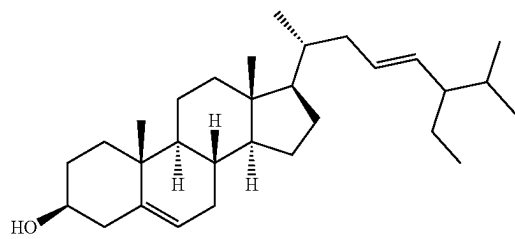
111



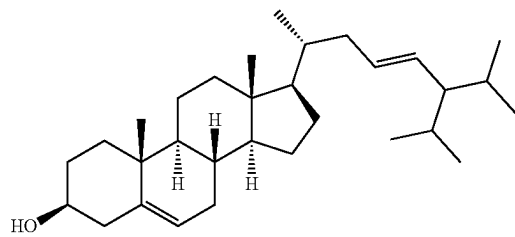
112



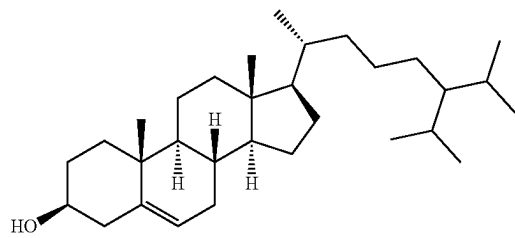
113



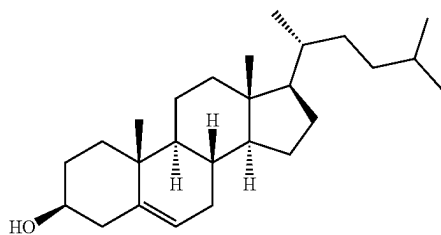
114



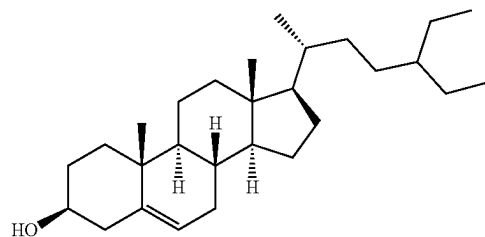
115



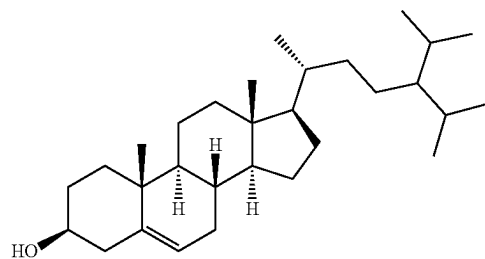
116



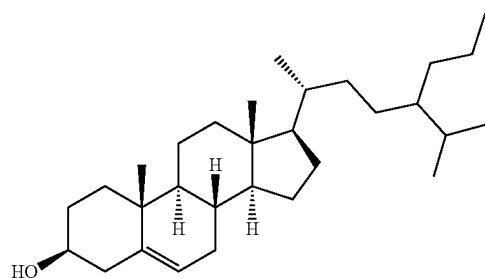
117



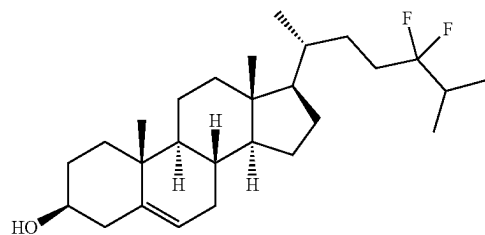
118



119



120



121

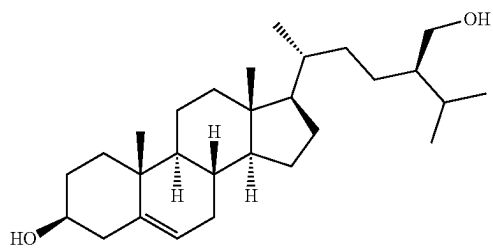
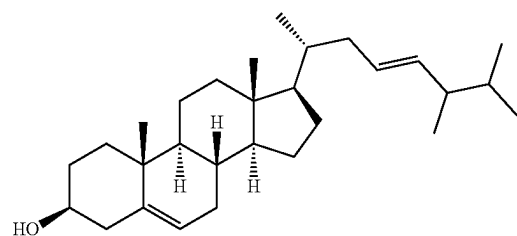
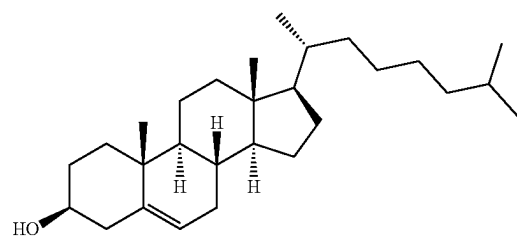


TABLE 16-continued

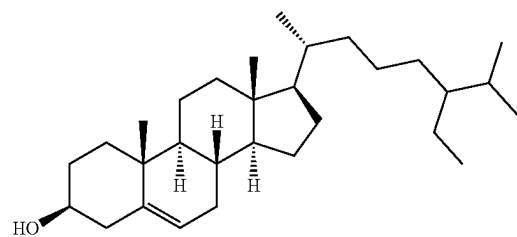
122



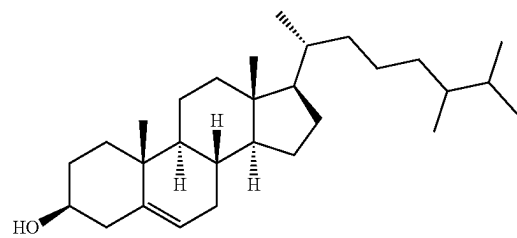
123



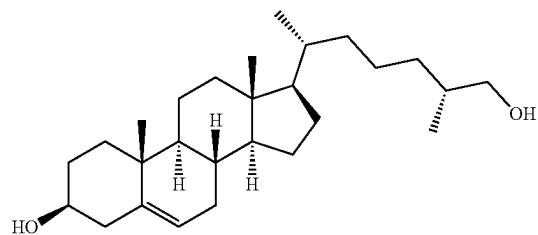
124



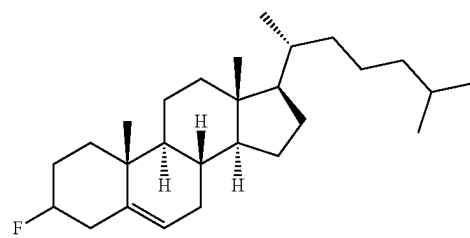
125



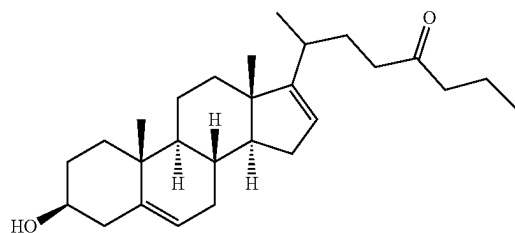
126



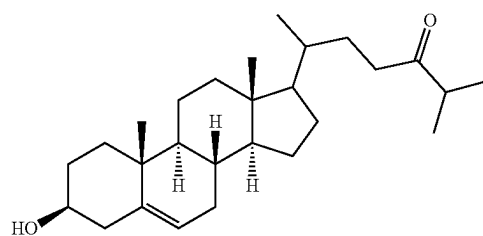
127



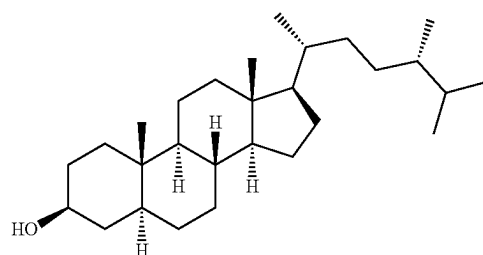
128



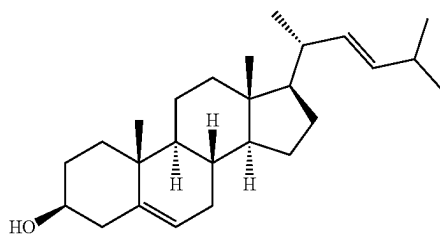
129



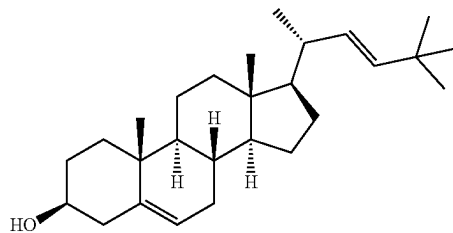
130



155



156



158

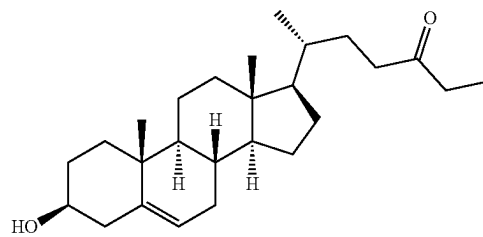
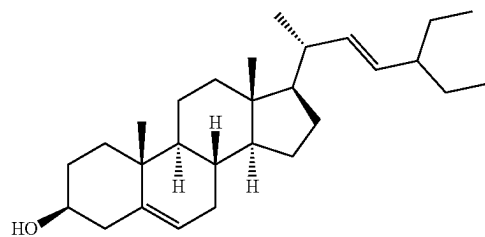
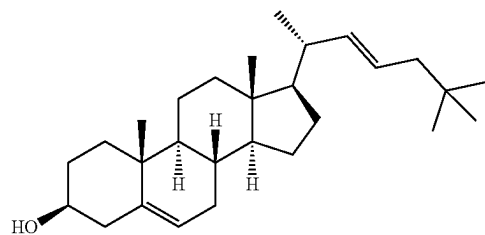


TABLE 16-continued

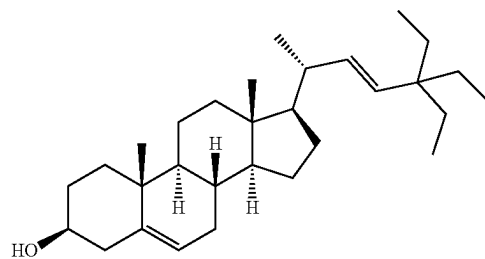
160



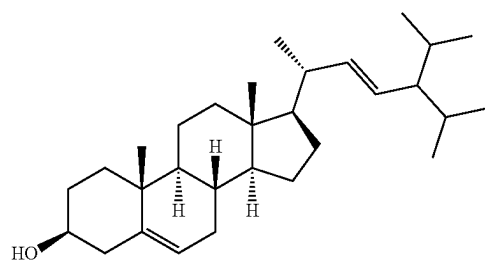
161



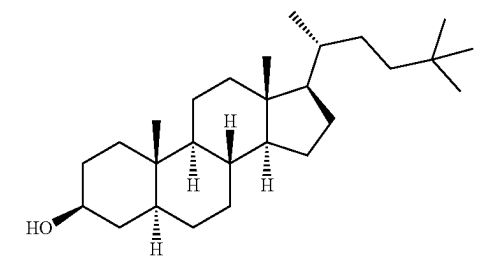
166



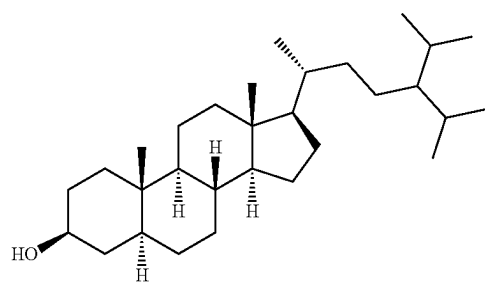
167



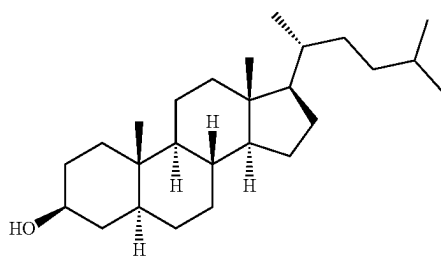
168



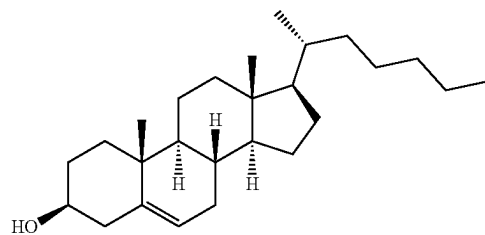
173



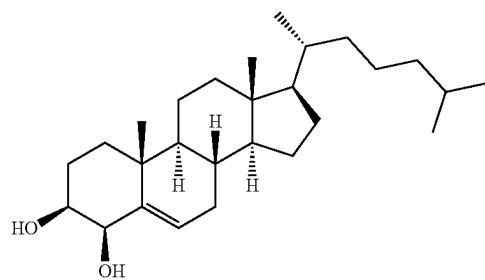
174



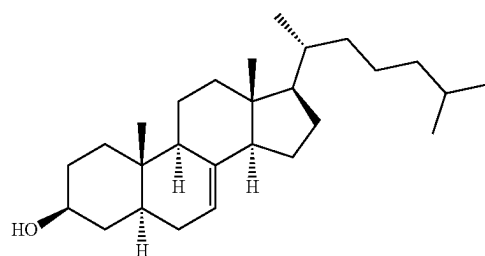
179



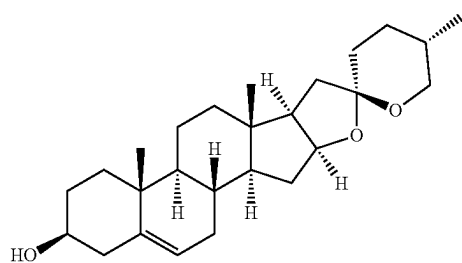
131



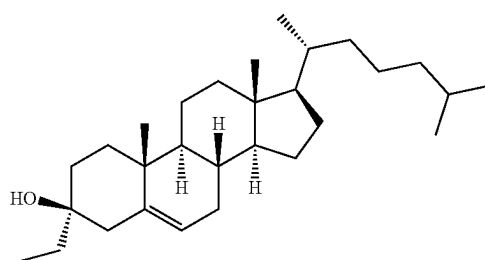
132



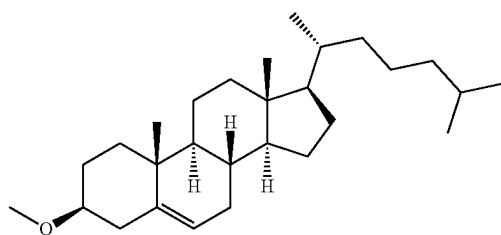
133



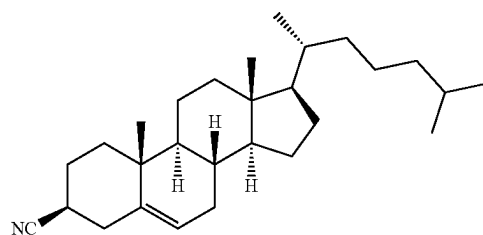
134



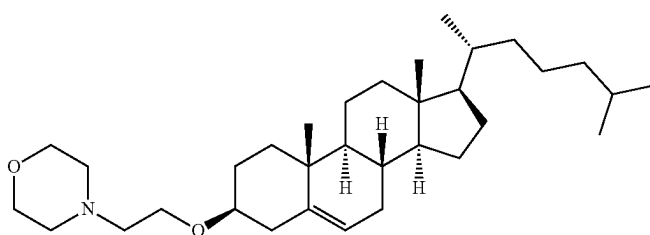
135



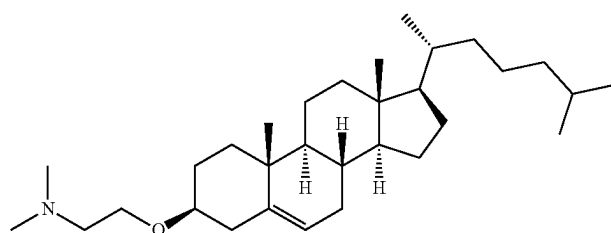
136



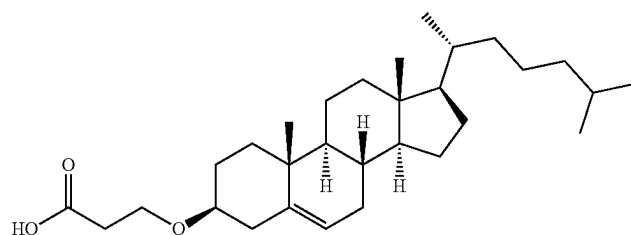
137



138



139



140

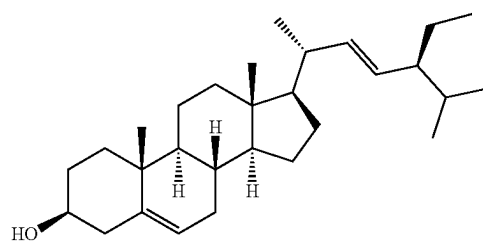
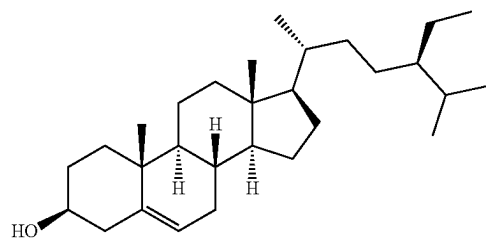
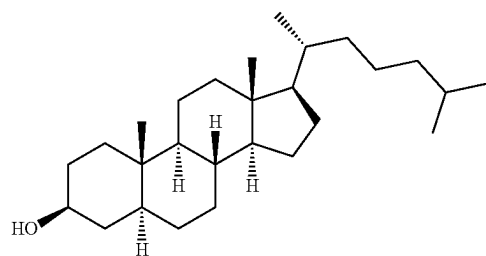


TABLE 16-continued

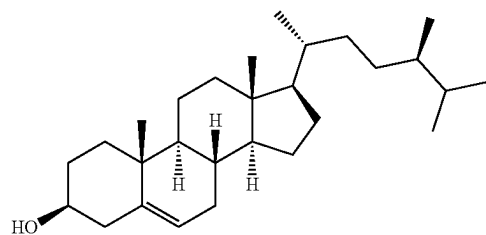
141



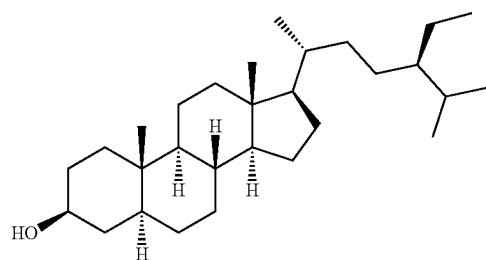
142



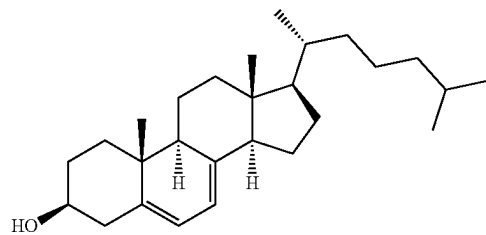
143



144



145



146

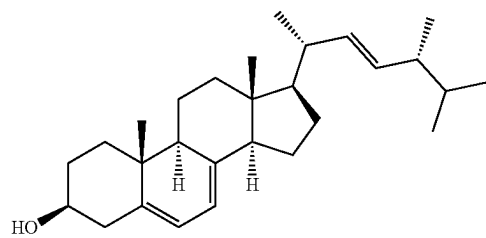
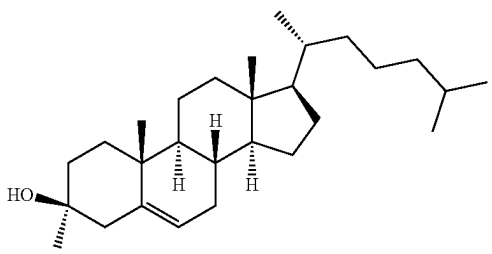
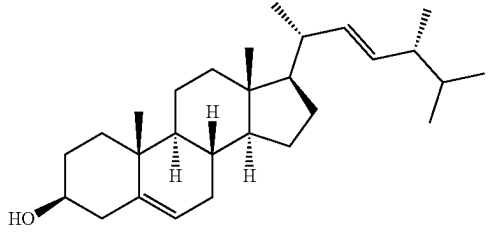
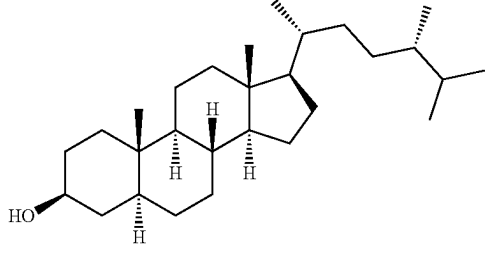
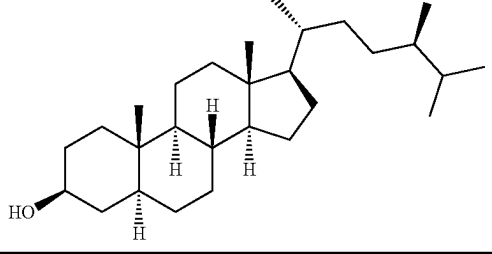
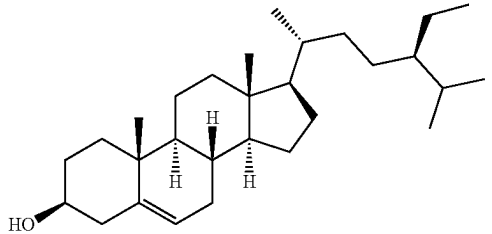
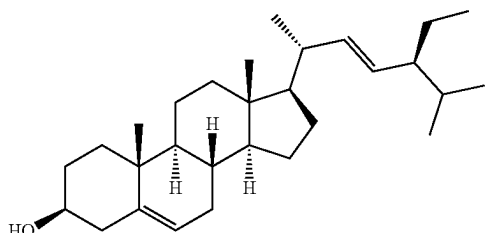
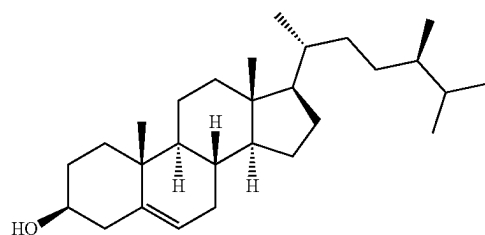
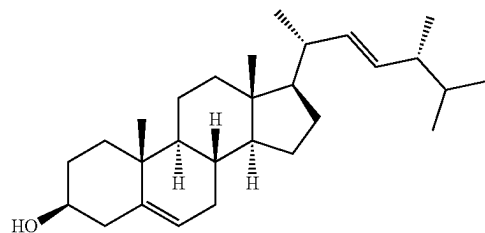


TABLE 16-continued

147	
148	
151	
159	
Composition S- No.	Structure
183	
	Compound 141
	
	Compound 140



Compound 143



Compound 148

E. Lipid Nanoparticle (LNP) Formulations

The formation of a lipid nanoparticle (LNP) described herein may be accomplished by any methods known in the art. For example, as described in U.S. Pat. Pub. No. US2012/0178702 A1, which is incorporated herein by reference in its entirety. Non-limiting examples of lipid nanoparticle compositions and methods of making them are described, for example, in Semple et al. (2010) *Nat. Biotechnol.* 28:172-176, Jayarama et al. (2012), *Angew. Chem. Int. Ed.*, 51:8529-8533; and Maier et al. (2013) *Molecular Therapy* 21, 1570-1578 (the contents of each of which are incorporated herein by reference in their entirety).

In one embodiment, the LNP formulation may be prepared by, e.g., the methods described in International Pat. Pub. No. WO 2011/127255 or WO 2008/103276, the contents of each of which are herein incorporated by reference in their entirety.

In one embodiment, LNP formulations described herein may comprise a polycationic composition. As a non-limiting example, the polycationic composition may be a composition selected from Formulae 1-60 of U.S. Pat. Pub. No. US2005/0222064 A1, the content of which is herein incorporated by reference in its entirety.

In one embodiment, the lipid nanoparticle may be formulated by the methods described in U.S. Pat. Pub. No. US2013/0156845 A1, and International Pat. Pub. No. WO2013/093648 A2 or WO2012/024526 A2, each of which is herein incorporated by reference in its entirety.

In one embodiment, the lipid nanoparticles described herein may be made in a sterile environment by the system and/or methods described in U.S. Pat. Pub. No. US2013/0164400 A1, which is incorporated herein by reference in its entirety.

In one embodiment, the LNP formulation may be formulated in a nanoparticle such as a nucleic acid-lipid particle described in U.S. Pat. No. 8,492,359, which is incorporated herein by reference in its entirety.

A nanoparticle composition may optionally comprise one or more coatings. For example, a nanoparticle composition may be formulated in a capsule, film, or tablet having a coating. A capsule, film, or tablet including a composition described herein may have any useful size, tensile strength, hardness, or density.

25

In some embodiments, the lipid nanoparticles described herein may be synthesized using methods comprising microfluidic mixers. Exemplary microfluidic mixers may include, but are not limited to, a slit interdigital micromixer including, but not limited to, those manufactured by Precision Nanosystems (Vancouver, BC, Canada). Microinova (Allerheiligen bei Wildon, Austria) and/or a staggered herringbone micromixer (SHM) (Zhigaltsev, I. V. et al. (2012) *Langmuir*. 28:3633-40; Belliveau, N. M. et al. *Mol. Ther. Nucleic. Acids.* (2012) 1:e37; Chen, D. et al. *J. Am. Chem. Soc.* (2012) 134(16):6948-51; each of which is herein incorporated by reference in its entirety).

In some embodiments, methods of LNP generation comprising SHM, further comprise the mixing of at least two input streams wherein mixing occurs by microstructure-induced chaotic advection (MICA). According to this method, fluid streams flow through channels present in a herringbone pattern causing rotational flow and folding the fluids around each other. This method may also comprise a surface for fluid mixing wherein the surface changes orientations during fluid cycling. Methods of generating LNPs using SHM include those disclosed in U.S. Pat. Pub. Nos. US2004/0262223 A1 and US2012/0276209 A1, each of which is incorporated herein by reference in their entirety.

In one embodiment, the lipid nanoparticles may be formulated using a micromixer such as, but not limited to, a Slit Interdigital Microstructured Mixer (SIMM-V2) or a Standard Slit Interdigital Micro Mixer (SSIMM) or Caterpillar (CPMM) or Impinging-jet (IJMM) from the Institut für Mikrotechnik Mainz GmbH, Mainz Germany). In one embodiment, the lipid nanoparticles are created using microfluidic technology (see, Whitesides (2006) *Nature*. 442: 368-373; and Abraham et al. (2002) *Science*. 295: 647-651; each of which is herein incorporated by reference in its entirety). As a non-limiting example, controlled microfluidic formulation includes a passive method for mixing streams of steady pressure-driven flows in micro channels at a low Reynolds number (see, e.g., Abraham et al. (2002) *Science*. 295: 647-651; which is herein incorporated by reference in its entirety).

In one embodiment, the circRNA of the present invention may be formulated in lipid nanoparticles created using a

micromixer chip such as, but not limited to, those from Harvard Apparatus (Holliston, MA), Dolomite Microfluidics (Royston, UK), or Precision Nanosystems (Vanouver, BC, Canada). A micromixer chip can be used for rapid mixing of two or more fluid streams with a split and recombine mechanism.

In some embodiments, the LNP of the present disclosure comprises a molar ratio of between about 40% and about 60% ionizable lipid, a molar ratio of between about 3.5% and about 14% helper lipid, a molar ratio of between about 28% and about 50% structural lipid, and a molar ratio of between about 0.5% and about 5% PEG-lipid, inclusive of all endpoints. In some embodiments, the total molar percentage of the ionizable lipid, the helper lipid, the structural lipid, and the PEG-lipid is 100% in the LNP.

In some embodiments, the molar ratio of the ionizable lipid in the LNP is from about 40 to about 60% of the total lipid present in the LNP. In some embodiments, the molar ratio of the ionizable lipid in the LNP is about 40%, about 41%, about 42%, about 43%, about 44%, about 45%, about 46%, about 47%, about 48%, about 49%, about 50%, about 51%, about 52%, about 53%, about 54%, about 55%, about 56%, about 57%, about 58%, about 59%, or about 60% of the total lipid present in the LNP. In some embodiments, the ionizable lipid is represented by Formula (7). In some embodiments, the ionizable lipid is represented by Formula (8). All values are inclusive of all endpoints.

In some embodiments, the molar ratio of the helper lipid in the LNP is from about 3.5% to about 14% of the total lipid present in the LNP. In some embodiments, the molar ratio of the helper lipid in the LNP is about 3, about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, about 10%, about 11%, about 12%, about 13%, or about 14% of the total lipid present in the LNP. In some embodiments, the helper lipid is DSPC. In some embodiments, the helper lipid is DOPE. All values are inclusive of all endpoints.

In some embodiments, the molar ratio of the structural lipid in the LNP is from about 28% to about 50% of the total lipid present in the LNP. In some embodiments, the molar ratio of the structural lipid in the LNP is about 28%, about 29%, about 30%, about 31%, about 32%, about 33%, about 34%, about 35%, about 36%, about 37%, about 38%, about 39%, about 40%, about 41%, about 42%, about 43%, about 44%, about 45%, about 46%, about 47%, about 48%, about 49%, or about 50% of the total lipid present in the LNP. In some embodiments, the structural lipid is cholesterol. All values are inclusive of all endpoints.

In some embodiments, the molar ratio of the PEG-lipid in the LNP is from about 0.5% to about 5% of the total lipid present in the LNP. In some embodiments, the molar ratio of the PEG-lipid in the LNP is about 0.5%, about 0.6%, about 0.7%, about 0.8%, about 0.9%, about 1.0%, about 1.1%, about 1.2%, about 1.3%, about 1.4%, about 1.5%, about 1.6%, about 1.7%, about 1.8%, about 1.9%, about 2.0%, about 2.1%, about 2.2%, about 2.3%, about 2.4%, about 2.5%, about 2.6%, about 2.7%, about 2.8%, about 2.9%, about 3.0%, about 3.1%, about 3.2%, about 3.3%, 3.4%, about 3.5%, about 4.0%, about 4.5%, or about 5% of the total lipid present in the LNP. In some embodiments, the PEG-lipid is DSPE-PEG(2000). In some embodiments, the PEG-lipid is DMG-PEG(2000). All values are inclusive of all endpoints.

In some embodiments, the molar ratio of ionizable lipid: helper lipid: structural lipid: PEG-lipid in the LNP is about 45:9:44:2. In some embodiments, the molar ratio of ionizable lipid: helper lipid: structural lipid: PEG-lipid in the LNP is about 50:10:38.5:1.5. In some embodiments, the molar

ratio of ionizable lipid: helper lipid: structural lipid: PEG-lipid in the LNP is about 41:12:45:2. In some embodiments, the molar ratio of ionizable lipid: helper lipid: structural lipid: PEG-lipid in the LNP is about 62:4:33:1. In some embodiments, the molar ratio of ionizable lipid: helper lipid: structural lipid: PEG-lipid in the LNP is about 53:5:41:1. In some embodiments, the molar ratio of each of the ionizable lipid, helper lipid, structural lipid, and PEG-lipid is within 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.9%, 0.8%, 0.7%, 0.6%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.09%, 0.08%, 0.07%, 0.06%, 0.05%, 0.04%, 0.03%, 0.02%, or 0.01% of the stated value.

In one embodiment, the lipid nanoparticles may have a diameter from about 10 to about 100 nm such as, but not limited to, about 10 to about 20 nm, about 10 to about 30 nm, about 10 to about 40 nm, about 10 to about 50 nm, about 10 to about 60 nm, about 10 to about 70 nm, about 10 to about 80 nm, about 10 to about 90 nm, about 20 to about 30 nm, about 20 to about 40 nm, about 20 to about 50 nm, about 20 to about 60 nm, about 20 to about 70 nm, about 20 to about 80 nm, about 20 to about 90 nm, about 20 to about 100 nm, about 30 to about 40 nm, about 30 to about 50 nm, about 30 to about 60 nm, about 30 to about 70 nm, about 30 to about 80 nm, about 30 to about 90 nm, about 30 to about 100 nm, about 40 to about 50 nm, about 40 to about 60 nm, about 40 to about 70 nm, about 40 to about 80 nm, about 40 to about 90 nm, about 40 to about 100 nm, about 50 to about 60 nm, about 50 to about 70 nm, about 50 to about 80 nm, about 50 to about 90 nm, about 50 to about 100 nm, about 60 to about 70 nm, about 60 to about 80 nm, about 60 to about 90 nm, about 60 to about 100 nm, about 70 to about 80 nm, about 70 to about 90 nm, about 70 to about 100 nm, about 80 to about 90 nm, about 80 to about 100 nm and/or about 90 to about 100 nm. In one embodiment, the lipid nanoparticles may have a diameter from about 10 to 500 nm. In one embodiment, the lipid nanoparticle may have a diameter greater than 100 nm, greater than 150 nm, greater than 200 nm, greater than 250 nm, greater than 300 nm, greater than 350 nm, greater than 400 nm, greater than 450 nm, greater than 500 nm, greater than 550 nm, greater than 600 nm, greater than 650 nm, greater than 700 nm, greater than 750 nm, greater than 800 nm, greater than 850 nm, greater than 900 nm, greater than 950 nm or greater than 1000 nm. Each possibility represents a separate embodiment of the present invention.

In some embodiments, a nanoparticle (e.g., a lipid nanoparticle) has a mean diameter of 10-500 nm, 20-400 nm, 30-300 nm, or 40-200 nm. In some embodiments, a nanoparticle (e.g., a lipid nanoparticle) has a mean diameter of 50-150 nm, 50-200 nm, 80-100 nm, or 80-200 nm.

In some embodiments, the lipid nanoparticles described herein can have a diameter from below 0.1 μm to up to 1 mm such as, but not limited to, less than 0.1 μm , less than 1.0 μm , less than 5 μm , less than 10 μm , less than 15 μm , less than 20 μm , less than 25 μm , less than 30 μm , less than 35 μm , less than 40 μm , less than 50 μm , less than 55 μm , less than 60 μm , less than 65 μm , less than 70 μm , less than 75 μm , less than 80 μm , less than 85 μm , less than 90 μm , less than 95 μm , less than 100 μm , less than 125 μm , less than 150 μm , less than 175 μm , less than 200 μm , less than 225 μm , less than 250 μm , less than 275 μm , less than 300 μm , less than 325 μm , less than 350 μm , less than 375 μm , less than 400 μm , less than 425 μm , less than 450 μm , less than 475 μm , less than 500 μm , less than 525 μm , less than 550 μm , less than 575 μm , less than 600 μm , less than 625 μm , less than 650 μm , less than 675 μm , less than 700 μm , less than 725 μm , less than 750 μm , less than 775 μm , less than 800 μm ,

less than 825 μm , less than 850 μm , less than 875 μm , less than 900 μm , less than 925 μm , less than 950 μm , less than 975 μm .

In another embodiment, LNPs may have a diameter from about 1 nm to about 100 nm, from about 1 nm to about 10 nm, about 1 nm to about 20 nm, from about 1 nm to about 30 nm, from about 1 nm to about 40 nm, from about 1 nm to about 50 nm, from about 1 nm to about 60 nm, from about 1 nm to about 70 nm, from about 1 nm to about 80 nm, from about 1 nm to about 90 nm, about 10 to about 50 nm, from about 20 to about 100 nm, from about 5 nm to about 10 nm, about 5 nm to about 20 nm, from about 5 nm to about 30 nm, from about 5 nm to about 40 nm, from about 5 nm to about 50 nm, from about 5 nm to about 60 nm, from about 5 nm to about 70 nm, from about 5 nm to about 80 nm, from about 5 nm to about 90 nm, about 10 to about 50 nm, from about 20 to about 50 nm, from about 30 to about 50 nm, from about 40 to about 50 nm, from about 20 to about 60 nm, from about 30 to about 60 nm, from about 20 to about 70 nm, from about 30 to about 70 nm, from about 40 to about 70 nm, from about 50 to about 70 nm, from about 60 to about 70 nm, from about 20 to about 80 nm, from about 30 to about 80 nm, from about 40 to about 80 nm, from about 50 to about 80 nm, from about 60 to about 80 nm, from about 20 to about 90 nm, from about 30 to about 90 nm, from about 40 to about 90 nm, from about 50 to about 90 nm, from about 60 to about 90 nm and/or from about 70 to about 90 nm. Each possibility represents a separate embodiment of the present invention.

A nanoparticle composition may be relatively homogeneous. A polydispersity index may be used to indicate the homogeneity of a nanoparticle composition, e.g., the particle size distribution of the nanoparticle compositions. A small (e.g., less than 0.3) polydispersity index generally indicates a narrow particle size distribution. A nanoparticle composition may have a polydispersity index from about 0 to about 0.25, such as 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.10, 0.11, 0.12, 0.13, 0.14, 0.15, 0.16, 0.17, 0.18, 0.19, 0.20, 0.21, 0.22, 0.23, 0.24, or 0.25. In some embodiments, the polydispersity index of a nanoparticle composition may be from about 0.10 to about 0.20. Each possibility represents a separate embodiment of the present invention.

The zeta potential of a nanoparticle composition may be used to indicate the electrokinetic potential of the composition. For example, the zeta potential may describe the surface charge of a nanoparticle composition. Nanoparticle compositions with relatively low charges, positive or negative, are generally desirable, as more highly charged species may interact undesirably with cells, tissues, and other elements in the body. In some embodiments, the zeta potential of a nanoparticle composition may be from about -20 mV to about +20 mV, from about -20 mV to about +15 mV, from about -20 mV to about +10 mV, from about -20 mV to about +5 mV, from about -20 mV to about 0 mV, from about -20 mV to about -5 mV, from about -20 mV to about -10 mV, from about -20 mV to about -15 mV from about -20 mV to about +20 mV, from about -20 mV to about +15 mV, from about -20 mV to about +10 mV, from about -20 mV to about +5 mV, from about -20 mV to about 0 mV, from about 0 mV to about +20 mV, from about 0 mV to about +15 mV, from about 0 mV to about +10 mV, from about 0 mV to about +5 mV, from about +5 mV to about +20 mV, from about +5 mV to about +15 mV, or from about +5 mV to about +10 mV. Each possibility represents a separate embodiment of the present invention.

The efficiency of encapsulation of a therapeutic agent describes the amount of therapeutic agent that is encapsu-

lated or otherwise associated with a nanoparticle composition after preparation, relative to the initial amount provided. The encapsulation efficiency is desirably high (e.g., close to 100%). The encapsulation efficiency may be measured, for example, by comparing the amount of therapeutic agent in a solution containing the nanoparticle composition before and after breaking up the nanoparticle composition with one or more organic solvents or detergents. Fluorescence may be used to measure the amount of free therapeutic agent (e.g., nucleic acids) in a solution. For the nanoparticle compositions described herein, the encapsulation efficiency of a therapeutic agent may be at least 50%, for example 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%. In some embodiments, the encapsulation efficiency may be at least 80%. In certain embodiments, the encapsulation efficiency may be at least 90%. Each possibility represents a separate embodiment of the present invention. In some embodiments, the lipid nanoparticle has a polydispersity value of less than 0.4. In some embodiments, the lipid nanoparticle has a net neutral charge at a neutral pH. In some embodiments, the lipid nanoparticle has a mean diameter of 50-200 nm.

The properties of a lipid nanoparticle formulation may be influenced by factors including, but not limited to, the selection of the cationic lipid component, the degree of cationic lipid saturation, the selection of the non-cationic lipid component, the degree of noncationic lipid saturation, the selection of the structural lipid component, the nature of the PEGylation, ratio of all components and biophysical parameters such as size. As described herein, the purity of a PEG lipid component is also important to an LNP's properties and performance.

F. Methods for Lipid Nanoparticles (LNP)

In one embodiment, a lipid nanoparticle formulation may be prepared by the methods described in International Publication Nos. WO2011127255 or WO2008103276, each of which is herein incorporated by reference in their entirety. In some embodiments, lipid nanoparticle formulations may be as described in International Publication No. WO2019131770, which is herein incorporated by reference in its entirety.

In some embodiments, circular RNA is formulated according to a process described in U.S. patent application Ser. No. 15/809,680. In some embodiments, the present invention provides a process of encapsulating circular RNA in transfer vehicles comprising the steps of forming lipids into pre-formed transfer vehicles (i.e. formed in the absence of RNA) and then combining the pre-formed transfer vehicles with RNA. In some embodiments, the novel formulation process results in an RNA formulation with higher potency (peptide or protein expression) and higher efficacy (improvement of a biologically relevant endpoint) both in vitro and in vivo with potentially better tolerability as compared to the same RNA formulation prepared without the step of preforming the lipid nanoparticles (e.g., combining the lipids directly with the RNA).

For certain cationic lipid nanoparticle formulations of RNA, in order to achieve high encapsulation of RNA, the RNA in buffer (e.g., citrate buffer) has to be heated. In those processes or methods, the heating is required to occur before the formulation process (i.e. heating the separate components) as heating post-formulation (post-formation of nanoparticles) does not increase the encapsulation efficiency of the RNA in the lipid nanoparticles. In contrast, in some embodiments of the novel processes of the present invention, the order of heating of RNA does not appear to affect the RNA encapsulation percentage. In some embodiments,

no heating (i.e. maintaining at ambient temperature) of one or more of the solutions comprising the pre-formed lipid nanoparticles, the solution comprising the RNA and the mixed solution comprising the lipid nanoparticle encapsulated RNA is required to occur before or after the formulation process.

RNA may be provided in a solution to be mixed with a lipid solution such that the RNA may be encapsulated in lipid nanoparticles. A suitable RNA solution may be any aqueous solution containing RNA to be encapsulated at various concentrations. For example, a suitable RNA solution may contain an RNA at a concentration of or greater than about 0.01 mg/ml, 0.05 mg/ml, 0.06 mg/ml, 0.07 mg/ml, 0.08 mg/ml, 0.09 mg/ml, 0.1 mg/ml, 0.15 mg/ml, 0.2 mg/ml, 0.3 mg/ml, 0.4 mg/ml, 0.5 mg/ml, 0.6 mg/ml, 0.7 mg/ml, 0.8 mg/ml, 0.9 mg/ml, or 1.0 mg/ml. In some embodiments, a suitable RNA solution may contain an RNA at a concentration in a range from about 0.01-1.0 mg/ml, 0.01-0.9 mg/ml, 0.01-0.8 mg/ml, 0.01-0.7 mg/ml, 0.01-0.6 mg/ml, 0.01-0.5 mg/ml, 0.01-0.4 mg/ml, 0.01-0.3 mg/ml, 0.01-0.2 mg/ml, 0.01-0.1 mg/ml, 0.05-1.0 mg/ml, 0.05-0.9 mg/ml, 0.05-0.8 mg/ml, 0.05-0.7 mg/ml, 0.05-0.6 mg/ml, 0.05-0.5 mg/ml, 0.05-0.4 mg/ml, 0.05-0.3 mg/ml, 0.05-0.2 mg/ml, 0.05-0.1 mg/ml, 0.1-1.0 mg/ml, 0.2-0.9 mg/ml, 0.3-0.8 mg/ml, 0.4-0.7 mg/ml, or 0.5-0.6 mg/ml.

Typically, a suitable RNA solution may also contain a buffering agent and/or salt. Generally, buffering agents can include HEPES, Tris, ammonium sulfate, sodium bicarbonate, sodium citrate, sodium acetate, potassium phosphate or sodium phosphate. In some embodiments, suitable concentration of the buffering agent may be in a range from about 0.1 mM to 100 mM, 0.5 mM to 90 mM, 1.0 mM to 80 mM, 2 mM to 70 mM, 3 mM to 60 mM, 4 mM to 50 mM, 5 mM to 40 mM, 6 mM to 30 mM, 7 mM to 20 mM, 8 mM to 15 mM, or 9 to 12 mM.

Exemplary salts can include sodium chloride, magnesium chloride, and potassium chloride. In some embodiments, suitable concentration of salts in an RNA solution may be in a range from about 1 mM to 500 mM, 5 mM to 400 mM, 10 mM to 350 mM, 15 mM to 300 mM, 20 mM to 250 mM, 30 mM to 200 mM, 40 mM to 190 mM, 50 mM to 180 mM, 50 mM to 170 mM, 50 mM to 160 mM, 50 mM to 150 mM, or 50 mM to 100 mM.

In some embodiments, a suitable RNA solution may have a pH in a range from about 3.5-6.5, 3.5-6.0, 3.5-5.5, 3.5-5.0, 3.5, 4.5, 4.0-5.5, 4.0-5.0, 4.0-4.9, 4.0-4.8, 4.0-4.7, 4.0-4.6, or 4.0-4.5.

Various methods may be used to prepare an RNA solution suitable for the present invention. In some embodiments, RNA may be directly dissolved in a buffer solution described herein. In some embodiments, an RNA solution may be generated by mixing an RNA stock solution with a buffer solution prior to mixing with a lipid solution for encapsulation. In some embodiments, an RNA solution may be generated by mixing an RNA stock solution with a buffer solution immediately before mixing with a lipid solution for encapsulation.

According to the present invention, a lipid solution contains a mixture of lipids suitable to form transfer vehicles for encapsulation of RNA. In some embodiments, a suitable lipid solution is ethanol based. For example, a suitable lipid solution may contain a mixture of desired lipids dissolved in pure ethanol (i.e. 100% ethanol). In another embodiment, a suitable lipid solution is isopropyl alcohol based. In another embodiment, a suitable lipid solution is dimethylsulfoxide-based. In another embodiment, a suitable lipid solution is a

mixture of suitable solvents including, but not limited to, ethanol, isopropyl alcohol and dimethylsulfoxide.

A suitable lipid solution may contain a mixture of desired lipids at various concentrations. In some embodiments, a suitable lipid solution may contain a mixture of desired lipids at a total concentration in a range from about 0.1-100 mg/ml, 0.5-90 mg/ml, 1.0-80 mg/ml, 1.0-70 mg/ml, 1.0-60 mg/ml, 1.0-50 mg/ml, 1.0-40 mg/ml, 1.0-30 mg/ml, 1.0-20 mg/ml, 1.0-15 mg/ml, 1.0-10 mg/ml, 1.0-9 mg/ml, 1.0-8 mg/ml, 1.0-7 mg/ml, 1.0-6 mg/ml, or 1.0-5 mg/ml.

G. Liposomes

In certain embodiments, liposomes or other lipid bilayer vesicles are disclosed herein and may be used as a component or as the whole transfer vehicle to facilitate or enhance the delivery and release of circular RNA to one or more target cells. Liposomes are usually characterized by having an interior space sequestered from an outer medium by a membrane of one or more bilayers forming a microscopic sack or vesicle. Bilayer membranes of liposomes are typically formed by lipids, i.e. amphiphilic molecules of synthetic or natural origin that comprise spatially separated hydrophobic or hydrophilic domains (Lasic, D. and Papahadjopoulos, D., eds. *Medical Applications of Liposomes*. Elsevier, Amsterdam, 1998).

In some embodiments, the circular RNA is encapsulated, or the liposome can be prepared using various methods, including but not limited to mechanical dispersion, solvent dispersion, and or detergent removal. Each of these methods include the steps of drying the lipids from organic solvents, dispersing the lipid in aqueous media, resizing the liposomes and purifying the/liposome suspension (Gomez et al., *ACS Omega*. 2019, 4(6): 10866-10876). Various other methods of liposome preparation can be found in Akbarzadeh et al., *Nanoscale Res Lett*. 2013; 8(1): 102. In some embodiments, the circular RNA may be loaded passively (i.e. the circular RNA is encapsulated during liposome formation) or actively (i.e. after liposome formation).

In some embodiments, the liposome disclosed herein may comprise one or more bilayers. In certain embodiments, the liposome may comprise a multilamellar vesicle or a unilamellar vesicle.

In certain embodiments, the liposome as described herein comprises of naturally derived or engineered phospholipids. In some embodiments, the liposomes may further comprise PEG-lipids that aid with stability of the overall liposome structure. Other improvements, including but not limited to corticosteroid and other steroids may be used to help with maintaining structure and stability of the liposome.

H. Dendrimer

In certain embodiments, the transfer vehicle for transporting the circular RNA comprises a dendrimer. Use of "dendrimer" describes the architectural motif of the transfer vehicle. In some embodiments, the dendrimer includes but is not limited to containing an interior core and one or more layers (i.e. generations) that extend or attach out from the interior core. In some of the embodiments, the generations may contain one or more branching points and an exterior surface of terminal groups that attach to the outermost generation. The branching points, in certain embodiments, may be mostly monodispersed and contain symmetric branching units built around the interior core.

Synthesis of the dendrimer may comprise the divergent method, convergent growth, hypercore and branched monomer growth, double exponential growth, lego chemistry, click chemistry and other methods as available in the art (Mendes L. et al., *Molecules*. 2017. 22 (9): 1401 further describes these methods).

I. Polymer-Based Delivery

In certain embodiments, as described herein, the transfer vehicle for the circular RNA polynucleotide comprises a polymer nanoparticle. In some embodiments, the polymer nanoparticle includes nanocapsules and nanospheres. Nanocapsules, in some embodiments, are composed of an oily core surrounded by a polymeric shell. In some embodiments, the circular RNA is contained within the core and the polymeric shell controls the release of the circular RNA. On the other hand, nanospheres comprise a continuous polymeric network in which the circular RNA is retained or absorbed onto the surface. In some embodiments, cationic polymers is used to encapsulate the circular RNA due to the favorable electrostatic interaction of the cations to the negatively charged nucleic acids and cell membrane.

The polymer nanoparticle may be prepared by various methods. In some embodiments, the polymer nanoparticle may be prepared by nanoprecipitation, emulsion techniques, solvent evaporation, solvent diffusion, reverse salting-out or other methods available in the art.

J. Polymer-Lipid Hybrids

In certain embodiments, as described herein, the transfer vehicle for the circular RNA polynucleotide comprises a polymer-lipid hybrid nanoparticle (LPHNP). In some embodiments, the LPHNP comprises a polymer core enveloped within a lipid bilayer. In some embodiments, the polymer core encapsulates the circular RNA polynucleotide. In some embodiments, the LPHNP further comprises an outer lipid bilayer. In certain embodiments this outer lipid bilayer comprises a PEG-lipid, helper lipid, cholesterol or other molecule as known in the art to help with stability in a lipid-based nanoparticle. The lipid bilayer closest to the polymer core mitigates the loss of the entrapped circular RNA during LPHNP formation and protects from degradation of the polymer core by preventing diffusion of water from outside of the transfer vehicle into the polymer core (Mukherjee et al., *In J. Nanomedicine*. 2019; 14: 1937-1952).

There are various methods of developing and formulating a LPHNP. In certain embodiments, the LPHNP is developed using a one-step or a two-step method available in the art. In some embodiments, the one-step method for forming an LPHNP is through nanoprecipitation or emulsification-solvent evaporation. In certain embodiments, the two-step method includes nanoprecipitation, emulsification-solvent evaporation, high-pressure homogenization, or other method available in the art.

K. Peptide-Based Delivery

In certain embodiments, the circular RNA can be transported using a peptide-based delivery mechanism. In some embodiments, the peptide-based delivery mechanism comprises a lipoprotein. Based on the size of the drug to be delivered, the lipoprotein may be either a low-density (LDL) or high-density lipoprotein (HDL). As seen in U.S. Pat. No. 8,734,853B2, high-density lipoproteins are capable of transporting a nucleic acid *in vivo* and *in vitro*.

In particular embodiments, the lipid component includes cholesterol. In more particular embodiments, the lipid component includes a combination of cholesterol and cholesterol oleate.

The HDL-nucleic acid particle can be of any size, but in particular embodiments the particle has a molecular size of from about 100 Angstroms to about 500 Angstroms. In more particular embodiments, the particle has a molecular size of from about 100 Angstroms to about 300 Angstroms. The size may be dependent on the size of the nucleic acid component incorporated into the particle.

The HDL-nucleic acid particle can have a broad range in molecular weight. The weight is dependent on the size of the nucleic acid incorporated into the particle. For example, in some embodiments, the particle has a molecular weight of between about 100,000 Daltons to about 1,000,000 Daltons. In more particular embodiments, the particle has a molecular weight of between about 100,000 Daltons to about 500,000 Daltons. In specific embodiments, the particle has a molecular weight of between about 100,000 Daltons to about 300,000 Daltons.

The HDL-nucleic acid particles of the present invention can be made by different methods. For example, a nucleic acid (e.g., siRNA) may be neutralized by combining the nucleic acid with peptides or polypeptides composed of contiguous positively-charged amino acids. For example, as discussed above, amino acid sequences may include 2 or more contiguous lysine residues. The positive charge of the amino acid sequences neutralizes the negatively charged nucleic acid molecule. The nucleic acid can then be encapsulated in an HDL particle using a method as described in Lacko et al. (2002).

L. Carbohydrate Carrier

In certain embodiments, the circular RNA polynucleotide can be transported using a carbohydrate carrier or a sugar-nanocapsule. In certain embodiments, the carbohydrate carrier comprises a sugar-decorated nanoparticle, peptide- and saccharide-conjugated dendrimer, nanoparticles based on polysaccharides, and other carbohydrate-based carriers available in the art. As described herein, the incorporation of carbohydrate molecules may be through synthetic means.

In some embodiments, the carbohydrate carrier comprises polysaccharides. These polysaccharides may be made from the microbial cell wall of the target cell. For example, carbohydrate carriers comprise of mannan carbohydrates have been shown to successfully deliver mRNA (Son et al., *Nano Lett.* 2020, 20(3): 1499-1509).

M. Glycan-Decorated Nanoparticles/Glyconanoparticles

In certain embodiments, as provided herein, the transfer vehicle for the circular RNA is a glyconanoparticle (GlycoNP). As known in the art, glyconanoparticles comprise a core comprising gold, iron oxide, semiconductor nanoparticles or a combination thereof. In some embodiments, the glyconanoparticle is functionalized using carbohydrates. In certain embodiments, the glyconanoparticle comprises a carbon nanotube or graphene. In one embodiment the glyconanoparticle comprises a polysaccharide-based GlycoNP (e.g., chitosan-based GlycoNP). In certain embodiments, the glyconanoparticle is a glycodendrimer.

7. Combinations of Proteins and Ires

In certain embodiments, as provided herein, the payload encoded by the circular RNA polynucleotide may be optimized through use of a specific internal ribosome entry sites (IRES) within the translation initiation element (TIE). In some embodiments, IRES specificity within a circular RNA can significantly enhance expression of specific proteins encoded within the coding element.

8. Targeting

A. Targeting Methods

The present invention also contemplates the discriminatory targeting of target cells and tissues by both passive and active targeting means. The phenomenon of passive targeting exploits the natural distributions patterns of a transfer vehicle *in vivo* without relying upon the use of additional

excipients or means to enhance recognition of the transfer vehicle by target cells. For example, transfer vehicles which are subject to phagocytosis by the cells of the reticulo-endothelial system are likely to accumulate in the liver or spleen, and accordingly may provide a means to passively direct the delivery of the compositions to such target cells.

Alternatively, the present invention contemplates active targeting, which involves the use of targeting moieties that may be bound (either covalently or non-covalently) to the transfer vehicle to encourage localization of such transfer vehicle at certain target cells or target tissues. For example, targeting may be mediated by the inclusion of one or more endogenous targeting moieties in or on the transfer vehicle to encourage distribution to the target cells or tissues. Recognition of the targeting moiety by the target tissues actively facilitates tissue distribution and cellular uptake of the transfer vehicle and/or its contents in the target cells and tissues (e.g., the inclusion of an apolipoprotein-E targeting ligand in or on the transfer vehicle encourages recognition and binding of the transfer vehicle to endogenous low density lipoprotein receptors expressed by hepatocytes). As provided herein, the composition can comprise a moiety capable of enhancing affinity of the composition to the target cell. Targeting moieties may be linked to the outer bilayer of the lipid particle during formulation or post-formulation. These methods are well known in the art. In addition, some lipid particle formulations may employ fusogenic polymers such as PEAA, hemagglutinin, other lipopeptides (see U.S. patent application Ser. No. 08/835,281, and 60/083,294, which are incorporated herein by reference) and other features useful for in vivo and/or intracellular delivery. In other some embodiments, the compositions of the present invention demonstrate improved transfection efficacies, and/or demonstrate enhanced selectivity towards target cells or tissues of interest. Contemplated therefore are compositions which comprise one or more moieties (e.g., peptides, aptamers, oligonucleotides, a vitamin or other molecules) that are capable of enhancing the affinity of the compositions and their nucleic acid contents for the target cells or tissues. Suitable moieties may optionally be bound or linked to the surface of the transfer vehicle. In some embodiments, the targeting moiety may span the surface of a transfer vehicle or be encapsulated within the transfer vehicle. Suitable moieties are selected based upon their physical, chemical or biological properties (e.g., selective affinity and/or recognition of target cell surface markers or features). Cell-specific target sites and their corresponding targeting ligand can vary widely. Suitable targeting moieties are selected such that the unique characteristics of a target cell are exploited, thus allowing the composition to discriminate between target and non-target cells. For example, compositions of the invention may include surface markers (e.g., apolipoprotein-B or apolipoprotein-E) that selectively enhance recognition of, or affinity to hepatocytes (e.g., by receptor-mediated recognition of and binding to such surface markers). As an example, the use of galactose as a targeting moiety would be expected to direct the compositions of the present invention to parenchymal hepatocytes, or alternatively the use of mannose containing sugar residues as a targeting ligand would be expected to direct the compositions of the present invention to liver endothelial cells (e.g., mannose containing sugar residues that may bind preferentially to the asialoglycoprotein receptor present in hepatocytes). (See Hillery A M. et al. "Drug Delivery and Targeting: For Pharmacists and Pharmaceutical Scientists" (2002) Taylor & Francis, Inc.) The presentation of such targeting moieties that have been conjugated to moieties

present in the transfer vehicle (e.g., a lipid nanoparticle) therefore facilitate recognition and uptake of the compositions of the present invention in target cells and tissues. Examples of suitable targeting moieties include one or more peptides, proteins, aptamers, vitamins and oligonucleotides.

In particular embodiments, a transfer vehicle comprises a targeting moiety. In some embodiments, the targeting moiety mediates receptor-mediated endocytosis selectively into a specific population of cells. In some embodiments, the targeting moiety is capable of binding to a T cell antigen. In some embodiments, the targeting moiety is capable of binding to a NK, NKT, or macrophage antigen. In some embodiments, the targeting moiety is capable of binding to a protein selected from the group CD3, CD4, CD8, PD-1, 4-1BB, and CD2. In some embodiments, the targeting moiety is an single chain Fv (scFv) fragment, nanobody, peptide, peptide-based macrocycle, minibody, heavy chain variable region, light chain variable region or fragment thereof. In some embodiments, the targeting moiety is selected from T-cell receptor motif antibodies, T-cell α chain antibodies, T-cell β chain antibodies, T-cell γ chain antibodies, T-cell δ chain antibodies, CCR7 antibodies, CD3 antibodies, CD4 antibodies, CD5 antibodies, CD7 antibodies, CD8 antibodies, CD11b antibodies, CD11c antibodies, CD16 antibodies, CD19 antibodies, CD20 antibodies, CD21 antibodies, CD22 antibodies, CD25 antibodies, CD28 antibodies, CD34 antibodies, CD35 antibodies, CD40 antibodies, CD45RA antibodies, CD45RO antibodies, CD52 antibodies, CD56 antibodies, CD62L antibodies, CD68 antibodies, CD80 antibodies, CD95 antibodies, CD117 antibodies, CD127 antibodies, CD133 antibodies, CD137 (4-1BB) antibodies, CD163 antibodies, F4/80 antibodies, IL-4R α antibodies, Sca-1 antibodies, CTLA-4 antibodies, GTR antibodies GARP antibodies, LAP antibodies, granulocyte B antibodies, LFA-1 antibodies, transferrin receptor antibodies, and fragments thereof. In some embodiments, the targeting moiety is a small molecule binder of an ectoenzyme on lymphocytes. Small molecule binders of ectoenzymes include A2A inhibitors CD73 inhibitors, CD39 or adenosine receptors A2aR and A2bR. Potential small molecules include AB928.

In some embodiments, transfer vehicles are formulated and/or targeted as described in Shobaki N, Sato Y, Harashima H. Mixing lipids to manipulate the ionization status of lipid nanoparticles for specific tissue targeting. *Int J Nanomedicine*. 2018;13:8395-8410. Published 2018 Dec 10. In some embodiments, a transfer vehicle is made up of 3 lipid types. In some embodiments, a transfer vehicle is made up of 4 lipid types. In some embodiments, a transfer vehicle is made up of 5 lipid types. In some embodiments, a transfer vehicle is made up of 6 lipid types.

B. Target Cells

Where it is desired to deliver a nucleic acid to an immune cell, the immune cell represents the target cell. In some embodiments, the compositions of the invention transfect the target cells on a discriminatory basis (i.e., do not transfect non-target cells). The compositions of the invention may also be prepared to preferentially target a variety of target cells, which include, but are not limited to, T cells, B cells, macrophages, and dendritic cells.

In some embodiments, the target cells are deficient in a protein or enzyme of interest. For example, where it is desired to deliver a nucleic acid to a hepatocyte, the hepatocyte represents the target cell. In some embodiments, the compositions of the invention transfect the target cells on a discriminatory basis (i.e., do not transfect non-target cells). The compositions of the invention may also be prepared to

335

preferentially target a variety of target cells, which include, but are not limited to, hepatocytes, epithelial cells, hematopoietic cells, epithelial cells, endothelial cells, lung cells, bone cells, stem cells, mesenchymal cells, neural cells (e.g., meninges, astrocytes, motor neurons, cells of the dorsal root ganglia and anterior horn motor neurons), photoreceptor cells (e.g., rods and cones), retinal pigmented epithelial cells, secretory cells, cardiac cells, adipocytes, vascular smooth muscle cells, cardiomyocytes, skeletal muscle cells, beta cells, pituitary cells, synovial lining cells, ovarian cells, testicular cells, fibroblasts, B cells, T cells, reticulocytes, leukocytes, granulocytes and tumor cells.

The compositions of the invention may be prepared to preferentially distribute to target cells such as in the heart, lungs, kidneys, liver, and spleen. In some embodiments, the compositions of the invention distribute into the cells of the liver or spleen to facilitate the delivery and the subsequent expression of the circRNA comprised therein by the cells of the liver (e.g., hepatocytes) or the cells of spleen (e.g., immune cells). The targeted cells may function as a biological “reservoir” or “depot” capable of producing, and systemically excreting a functional protein or enzyme. Accordingly, in one embodiment of the invention the transfer vehicle may target hepatocytes or immune cells and/or preferentially distribute to the cells of the liver or spleen upon delivery. In an embodiment, following transfection of the target hepatocytes or immune cells, the circRNA loaded in the vehicle are translated and a functional protein product is produced, excreted and systemically distributed. In other embodiments, cells other than hepatocytes (e.g., lung, spleen, heart, ocular, or cells of the central nervous system) can serve as a depot location for protein production.

In one embodiment, the compositions of the invention facilitate a subject's endogenous production of one or more functional proteins and/or enzymes. In an embodiment of the present invention, the transfer vehicles comprise circRNA which encode a deficient protein or enzyme. Upon distribution of such compositions to the target tissues and the subsequent transfection of such target cells, the exogenous circRNA loaded into the transfer vehicle (e.g., a lipid nanoparticle) may be translated in vivo to produce a functional protein or enzyme encoded by the exogenously administered circRNA (e.g., a protein or enzyme in which the subject is deficient). Accordingly, the compositions of the present invention exploit a subject's ability to translate exogenously- or recombinantly-prepared circRNA to produce an endogenously-translated protein or enzyme, and thereby produce (and where applicable excrete) a functional protein or enzyme. The expressed or translated proteins or enzymes may also be characterized by the in vivo inclusion of native post-translational modifications which may often be absent in recombinantly-prepared proteins or enzymes, thereby further reducing the immunogenicity of the translated protein or enzyme.

The administration of circRNA encoding a deficient protein or enzyme avoids the need to deliver the nucleic acids to specific organelles within a target cell. Rather, upon transfection of a target cell and delivery of the nucleic acids to the cytoplasm of the target cell, the circRNA contents of a transfer vehicle may be translated and a functional protein or enzyme expressed.

In some embodiments, a circular RNA comprises one or more miRNA binding sites. In some embodiments, a circular RNA comprises one or more miRNA binding sites recognized by miRNA present in one or more non-target cells or non-target cell types (e.g., Kupffer cells or hepatic cells) and not present in one or more target cells or target cell types

336

(e.g., hepatocytes or T cells). In some embodiments, a circular RNA comprises one or more miRNA binding sites recognized by miRNA present in an increased concentration in one or more non-target cells or non-target cell types (e.g., Kupffer cells or hepatic cells) compared to one or more target cells or target cell types (e.g., hepatocytes or T cells). miRNAs are thought to function by pairing with complementary sequences within RNA molecules, resulting in gene silencing.

In some embodiments, the compositions of the invention transfect or distribute to target cells on a discriminatory basis (i.e. do not transfect non-target cells). The compositions of the invention may also be prepared to preferentially target a variety of target cells, which include, but are not limited to, hepatocytes, epithelial cells, hematopoietic cells, epithelial cells, endothelial cells, lung cells, bone cells, stem cells, mesenchymal cells, neural cells (e.g., meninges, astrocytes, motor neurons, cells of the dorsal root ganglia and anterior horn motor neurons), photoreceptor cells (e.g., rods and cones), retinal pigmented epithelial cells, secretory cells, cardiac cells, adipocytes, vascular smooth muscle cells, cardiomyocytes, skeletal muscle cells, beta cells, pituitary cells, synovial lining cells, ovarian cells, testicular cells, fibroblasts, B cells, T cells, reticulocytes, leukocytes, granulocytes and tumor cells.

9. Pharmaceutical Compositions

In certain embodiments, provided herein are compositions (e.g., pharmaceutical compositions) comprising a therapeutic agent provided herein. In some embodiments, the therapeutic agent is a circular RNA polynucleotide provided herein. In some embodiments the therapeutic agent is a vector provided herein. In some embodiments, the therapeutic agent is a cell comprising a circular RNA or vector provided herein (e.g., a human cell, such as a human T cell). In certain embodiments, the composition further comprises a pharmaceutically acceptable carrier. In some embodiments, the compositions provided herein comprise a therapeutic agent provided herein in combination with other pharmaceutically active agents or drugs, such as anti-inflammatory drugs or antibodies capable of targeting B cell antigens, e.g., anti-CD20 antibodies, e.g., rituximab.

With respect to pharmaceutical compositions, the pharmaceutically acceptable carrier can be any of those conventionally used and is limited only by chemico-physical considerations, such as solubility and lack of reactivity with the active agent(s), and by the route of administration. The pharmaceutically acceptable carriers described herein, for example, vehicles, adjuvants, excipients, and diluents, are well-known to those skilled in the art and are readily available to the public. It is preferred that the pharmaceutically acceptable carrier be one which is chemically inert to the therapeutic agent(s) and one which has no detrimental side effects or toxicity under the conditions of use.

The choice of carrier will be determined in part by the particular therapeutic agent, as well as by the particular method used to administer the therapeutic agent. Accordingly, there are a variety of suitable formulations of the pharmaceutical compositions provided herein.

In certain embodiments, the pharmaceutical composition comprises a preservative. In certain embodiments, suitable preservatives may include, for example, methylparaben, propylparaben, sodium benzoate, and benzalkonium chloride. Optionally, a mixture of two or more preservatives may be used. The preservative or mixtures thereof are typically

present in an amount of about 0.0001% to about 2% by weight of the total composition.

In some embodiments, the pharmaceutical composition comprises a buffering agent. In some embodiments, suitable buffering agents may include, for example, citric acid, sodium citrate, phosphoric acid, potassium phosphate, and various other acids and salts. A mixture of two or more buffering agents optionally may be used. The buffering agent or mixtures thereof are typically present in an amount of about 0.001% to about 4% by weight of the total composition.

In some embodiments, the concentration of therapeutic agent in the pharmaceutical composition can vary, e.g., less than about 1%, or at least about 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, or about 50% or more by weight, and can be selected primarily by fluid volumes, and viscosities, in accordance with the particular mode of administration selected.

The following formulations for oral, aerosol, parenteral (e.g., subcutaneous, intravenous, intraarterial, intramuscular, intradermal, intraperitoneal, and intrathecal), and topical administration are merely exemplary and are in no way limiting. More than one route can be used to administer the therapeutic agents provided herein, and in certain instances, a particular route can provide a more immediate and more effective response than another route.

Formulations suitable for oral administration can comprise or consist of (a) liquid solutions, such as an effective amount of the therapeutic agent dissolved in diluents, such as water, saline, or orange juice; (b) capsules, sachets, tablets, lozenges, and troches, each containing a predetermined amount of the active ingredient, as solids or granules, (c) powders; (d) suspensions in an appropriate liquid; and (e) suitable emulsions. Liquid formulations may include diluents, such as water and alcohols, for example, ethanol, benzyl alcohol and the polyethylene alcohols, either with or without the addition of a pharmaceutically acceptable surfactant. Capsule forms can be of the ordinary hard or soft shelled gelatin type containing, for example, surfactants, lubricants, and inert fillers, such as lactose, sucrose, calcium phosphate, and corn starch. Tablet forms can include one or more of lactose, sucrose, mannitol, corn starch, potato starch, alginic acid, microcrystalline cellulose, acacia, gelatin, guar gum, colloidal silicon dioxide, croscarmellose sodium, talc, magnesium stearate, calcium stearate, zinc stearate, stearic acid, and other excipients, colorants, diluents, buffering agents, disintegrating agents, moistening agents, preservatives, flavoring agents, and other pharmacologically compatible excipients. Lozenge forms can comprise the therapeutic agent with a flavorant, usually sucrose, acacia or tragacanth. Pastilles can comprise the therapeutic agent with an inert base, such as gelatin and glycerin, or sucrose and acacia, emulsions, gels, and the like containing, in addition to, such excipients as are known in the art.

Formulations suitable for parenteral administration include aqueous and nonaqueous isotonic sterile injection solutions, which can contain antioxidants, buffers, bacteriostats, and solutes that render the formulation isotonic with the blood of the intended recipient, and aqueous and nonaqueous sterile suspensions that can include suspending agents, solubilizers, thickening agents, stabilizers, and preservatives. In some embodiments, the therapeutic agents provided herein can be administered in a physiologically acceptable diluent in a pharmaceutical carrier, such as a sterile liquid or mixture of liquids including water, saline, aqueous dextrose and related sugar solutions, an alcohol such as ethanol or hexadecyl alcohol, a glycol such as

propylene glycol or polyethylene glycol, dimethylsulfoxide, glycerol, ketals such as 2,2-dimethyl-1,3-dioxolane-4-methanol, ethers, poly(ethyleneglycol) 400, oils, fatty acids, fatty acid esters or glycerides, or acetylated fatty acid glycerides with or without the addition of a pharmaceutically acceptable surfactant such as a soap or a detergent, suspending agent such as pectin, carbomers, methylcellulose, hydroxypropylmethylcellulose, or carboxymethylcellulose, or emulsifying agents and other pharmaceutical adjuvants.

Oils, which can be used in parenteral formulations in some embodiments, include petroleum, animal oils, vegetable oils, or synthetic oils. Specific examples of oils include peanut, soybean, sesame, cottonseed, com, olive, petrolatum, and mineral oil. Suitable fatty acids for use in parenteral formulations include oleic acid, stearic acid, and isostearic acid. Ethyl oleate and isopropyl myristate are examples of suitable fatty acid esters.

Suitable soaps for use in certain embodiments of parenteral formulations include fatty alkali metal, ammonium, and triethanolamine salts, and suitable detergents include (a) cationic detergents such as, for example, dimethyl dialkyl ammonium halides, and alkyl pyridinium halides, (b) anionic detergents such as, for example, alkyl, aryl, and olefin sulfonates, alky, olefin, ether, and monoglyceride sulfates, and sulfosuccinates, (c) nonionic detergents such as, for example, fatty amine oxides, fatty acid alkanol-amides, and polyoxyethylenepolypropylene copolymers, (d) amphoteric detergents such as, for example, alkyl- β -amino-propionates, and 2-alkyl-imidazoline quaternary ammonium salts, and (e) mixtures thereof.

In some embodiments, the parenteral formulations will contain, for example, from about 0.5% to about 25% by weight of the therapeutic agent in solution. Preservatives and buffers may be used. In order to minimize or eliminate irritation at the site of injection, such compositions may contain one or more nonionic surfactants having, for example, a hydrophile-lipophile balance (HLB) of from about 12 to about 17. The quantity of surfactant in such formulations will typically range, for example, from about 5% to about 15% by weight. Suitable surfactants include polyethylene glycol, sorbitan, fatty acid esters such as sorbitan monooleate, and high molecular weight adducts of ethylene oxide with a hydrophobic base formed by the condensation of propylene oxide with propylene glycol. The parenteral formulations can be presented in unit-dose or multi-dose sealed containers, such as ampoules or vials, and can be stored in a freeze-dried (lyophilized) condition requiring only the addition of a sterile liquid excipient, for example, water for injections, immediately prior to use. Extemporaneous injection solutions and suspensions can be prepared from sterile powders, granules, and tablets of the kind previously described.

In certain embodiments, injectable formulations are provided herein. The requirements for effective pharmaceutical carriers for injectable compositions are well-known to those of ordinary skill in the art (see, e.g., *Pharmaceutics and Pharmacy Practice*, J. B. Lippincott Company, Philadelphia, PA, Banker and Chalmers, eds., pages 238-250 (1982), and *ASHP Handbook on Injectable Drugs*, Toissel, 4th ed, pages 622-630 (1986)).

In some embodiments, topical formulations are provided herein. Topical formulations, including those that are useful for transdermal drug release, are suitable in the context of certain embodiments provided herein for application to skin. In some embodiments, the therapeutic agent alone or in combination with other suitable components, can be made

into aerosol formulations to be administered via inhalation. These aerosol formulations can be placed into pressurized acceptable propellants, such as dichlorodifluoromethane, propane, nitrogen, and the like. They also may be formulated as pharmaceuticals for non-pressured preparations, such as in a nebulizer or an atomizer. Such spray formulations also may be used to spray mucosa.

In certain embodiments, the therapeutic agents provided herein can be formulated as inclusion complexes, such as cyclodextrin inclusion complexes, or liposomes. Liposomes can serve to target the therapeutic agents to a particular tissue. Liposomes also can be used to increase the half-life of the therapeutic agents. Many methods are available for preparing liposomes, as described in, for example, Szoka et al., *Ann. Rev. Biophys. Bioeng.*, 9, 467 (1980) and U.S. Pat. Nos. 4,235,871, 4,501,728, 4,837,028, and 5,019,369.

In some embodiments, the therapeutic agents provided herein are formulated in time-released, delayed release, or sustained release delivery systems such that the delivery of the composition occurs prior to, and with sufficient time to, cause sensitization of the site to be treated. Such systems can avoid repeated administrations of the therapeutic agent, thereby increasing convenience to the subject and the physician, and may be particularly suitable for certain composition embodiments provided herein. In one embodiment, the compositions of the invention are formulated such that they are suitable for extended-release of the circRNA contained therein. Such extended-release compositions may be conveniently administered to a subject at extended dosing intervals. For example, in one embodiment, the compositions of the present invention are administered to a subject twice a day, daily or every other day. In an embodiment, the compositions of the present invention are administered to a subject twice a week, once a week, every ten days, every two weeks, every three weeks, every four weeks, once a month, every six weeks, every eight weeks, every three months, every four months, every six months, every eight months, every nine months or annually.

In some embodiments, a protein encoded by an inventive polynucleotide is produced by a target cell for sustained amounts of time. For example, the protein may be produced for more than one hour, more than four, more than six, more than 12, more than 24, more than 48 hours, or more than 72 hours after administration. In some embodiments the polypeptide is expressed at a peak level about six hours after administration. In some embodiments the expression of the polypeptide is sustained at least at a therapeutic level. In some embodiments, the polypeptide is expressed at least at a therapeutic level for more than one, more than four, more than six, more than 12, more than 24, more than 48, or more than 72 hours after administration. In some embodiments, the polypeptide is detectable at a therapeutic level in patient tissue (e.g., liver or lung). In some embodiments, the level of detectable polypeptide is from continuous expression from the circRNA composition over periods of time of more than one, more than four, more than six, more than 12, more than 24, more than 48, or more than 72 hours after administration.

In certain embodiments, a protein encoded by an inventive polynucleotide is produced at levels above normal physiological levels. The level of protein may be increased as compared to a control. In some embodiments, the control is the baseline physiological level of the polypeptide in a normal individual or in a population of normal individuals. In other embodiments, the control is the baseline physiological level of the polypeptide in an individual having a deficiency in the relevant protein or polypeptide or in a

population of individuals having a deficiency in the relevant protein or polypeptide. In some embodiments, the control can be the normal level of the relevant protein or polypeptide in the individual to whom the composition is administered. In other embodiments, the control is the expression level of the polypeptide upon other therapeutic intervention, e.g., upon direct injection of the corresponding polypeptide, at one or more comparable time points.

In certain embodiments, the levels of a protein encoded by an inventive polynucleotide are detectable at 3 days, 4 days, 5 days, or 1 week or more after administration. Increased levels of protein may be observed in a tissue (e.g., liver or lung).

In some embodiments, the method yields a sustained circulation half-life of a protein encoded by an inventive polynucleotide. For example, the protein may be detected for hours or days longer than the half-life observed via subcutaneous injection of the protein or mRNA encoding the protein. In some embodiments, the half-life of the protein is 1 day, 2 days, 3 days, 4 days, 5 days, or 1 week or more.

Many types of release delivery systems are available and known to those of ordinary skill in the art. They include polymer based systems such as poly(lactide-glycolide), copolyoxalates, polycaprolactones, polyesteramides, polyorthoesters, polyhydroxybutyric acid, and polyanhydrides. Microcapsules of the foregoing polymers containing drugs are described in, for example, U.S. Pat. No. 5,075,109. Delivery systems also include non-polymer systems that are lipids including sterols such as cholesterol, cholesterol esters, and fatty acids or neutral fats such as mono-di- and tri-glycerides; hydrogel release systems; systatic systems; peptide based systems; wax coatings; compressed tablets using conventional binders and excipients; partially fused implants; and the like. Specific examples include, but are not limited to: (a) erosional systems in which the active composition is contained in a form within a matrix such as those described in U.S. Patents 4,452,775, 4,667,014, 4,748,034, and 5,239,660 and (b) diffusional systems in which an active component permeates at a controlled rate from a polymer such as described in U.S. Pat. Nos. 3,832,253 and 3,854,480. In addition, pump-based hardware delivery systems can be used, some of which are adapted for implantation.

In some embodiments, the therapeutic agent can be conjugated either directly or indirectly through a linking moiety to a targeting moiety. Methods for conjugating therapeutic agents to targeting moieties is known in the art. See, for instance, Wadwa et al., *J. Drug Targeting* 3:111 (1995) and U.S. Pat. No. 5,087,616.

In some embodiments, the therapeutic agents provided herein are formulated into a depot form, such that the manner in which the therapeutic agent is released into the body to which it is administered is controlled with respect to time and location within the body (see, for example, U.S. Pat. No. 4,450,150). Depot forms of therapeutic agents can be, for example, an implantable composition comprising the therapeutic agents and a porous or non-porous material, such as a polymer, wherein the therapeutic agents are encapsulated by or diffused throughout the material and/or degradation of the non-porous material. The depot is then implanted into the desired location within the body and the

341

therapeutic agents are released from the implant at a predetermined rate.

10. Therapeutic Methods

A. Areas for Treatment & Relevant Diseases/Disorders

In certain aspects, provided herein is a method of treating and/or preventing a condition, e.g., an autoimmune disorder or cancer.

In certain embodiments, the therapeutic agents provided herein are co-administered with one or more additional therapeutic agents (e.g., in the same pharmaceutical composition or in separate pharmaceutical compositions). In some embodiments, the therapeutic agent provided herein can be administered first and the one or more additional therapeutic agents can be administered second, or vice versa. Alternatively, the therapeutic agent provided herein and the one or more additional therapeutic agents can be administered simultaneously.

In some embodiments, the subject is a mammal. In some embodiments, the mammal referred to herein can be any mammal, including, but not limited to, mammals of the order Rodentia, such as mice and hamsters, or mammals of the order Logomorpha, such as rabbits. The mammals may be from the order Carnivora, including Felines (cats) and Canines (dogs). The mammals may be from the order Artiodactyla, including Bovines (cows) and Swines (pigs), or of the order Perssodactyla, including Equines (horses). The mammals may be of the order Primates, Ceboids, or Simoids (monkeys) or of the order Anthropoids (humans and apes). Preferably, the mammal is a human.

342

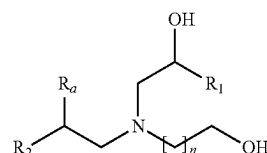
11. Additional Embodiments

The invention is further described by the following non-limiting exemplary embodiments:

Embodiment 1. A pharmaceutical composition comprising:

- an RNA polynucleotide, and
- a transfer vehicle comprising an ionizable lipid represented by Formula (13):

Formula (13)

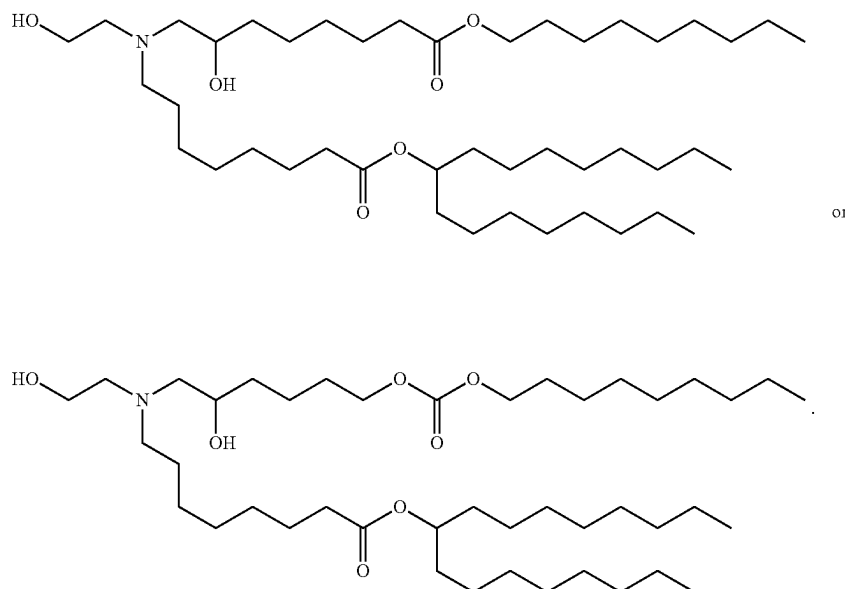


wherein:

n is an integer between 1 and 4;

R_a is hydrogen or hydroxyl;

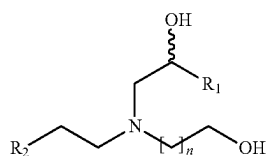
R₁ and R₂ are each independently a linear or branched C₆-C₃₀ alkyl, C₆-C₃₀ alkenyl, or C₆-C₃₀ heteroalkyl, optionally substituted by one or more substituents selected from a group consisting of oxo, halo, hydroxy, cyano, alkyl, alkenyl, aldehyde, heterocyclalkyl, hydroxyalkyl, dihydroxyalkyl, hydroxyalkylaminoalkyl, aminoalkyl, alkylaminoalkyl, dialkylaminoalkyl, (heterocyclalkyl)(alkyl)aminoalkyl, heterocyclalkyl, heteroaryl, alkylheteroaryl, alkynyl, alkoxy, amino, dialkylamino, aminoalkylcarbonylamino, aminocarbonylalkylamino, (aminocarbonylalkyl)(alkyl)amino, alkenylcarbonylamino, hydroxycarbonyl, alkylloxycarbonyl, aminocarbonyl, aminoalkylaminocarbonyl, alkylaminoalkylaminocarbonyl, dialkylaminoalkylaminocarbonyl, heterocyclalkylaminocarbonyl, (alkylaminoalkyl)(alkyl)aminocarbonyl, alkylaminoalkylcarbonyl, dialkylaminoalkylcarbonyl, heterocyclalkylcarbonyl, alkenylcarbonyl, alkynylcarbonyl, alkylsulfoxide, alkylsulfoxidealkyl, alkylsulfonyl, and alkylsulfonealkyl; with the proviso that the ionizable lipid is not



343

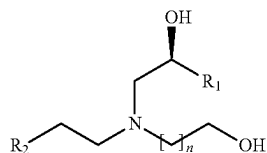
Embodiment 2. The pharmaceutical composition of embodiment 1, wherein R_a is hydrogen.

Embodiment 3. The pharmaceutical composition of embodiment 2, wherein the ionizable lipid is represented by Formula (13a-1), Formula (13a-2), or Formula (13a-3):



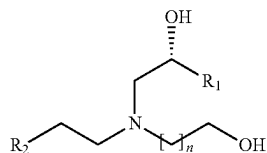
Formula (13a-1)

10



Formula (13a-2)

15



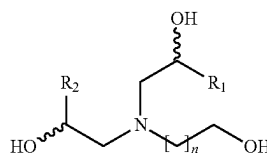
Formula (13a-3)

20

25

Embodiment 4. The pharmaceutical composition of embodiment 1, wherein R_a is hydroxyl.

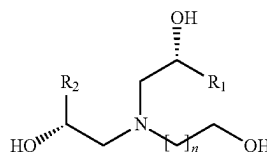
Embodiment 5. The pharmaceutical composition of embodiment 4, wherein the ionizable lipid is represented by Formula (13b-1), Formula (13b-2), or Formula (13b-3):



Formula (13b-1)

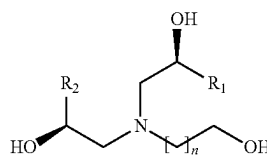
30

35



Formula (13b-2)

40



Formula (13b-3)

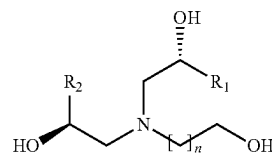
45

50

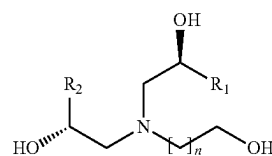
55

344

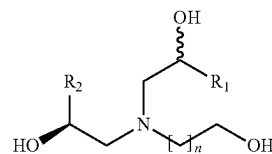
Embodiment 6. The pharmaceutical composition of embodiment 4, wherein the ionizable lipid is represented by Formula (13b-4), Formula (13b-5), Formula (13b-6), Formula (13b-7), Formula (13b-8), or Formula (13b-9):



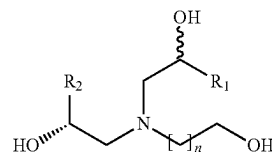
Formula (13b-4)



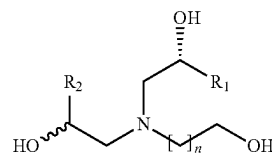
Formula (13b-5)



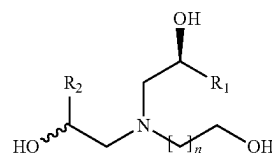
Formula (13b-6)



Formula (13b-7)

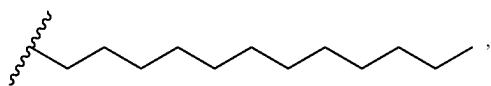


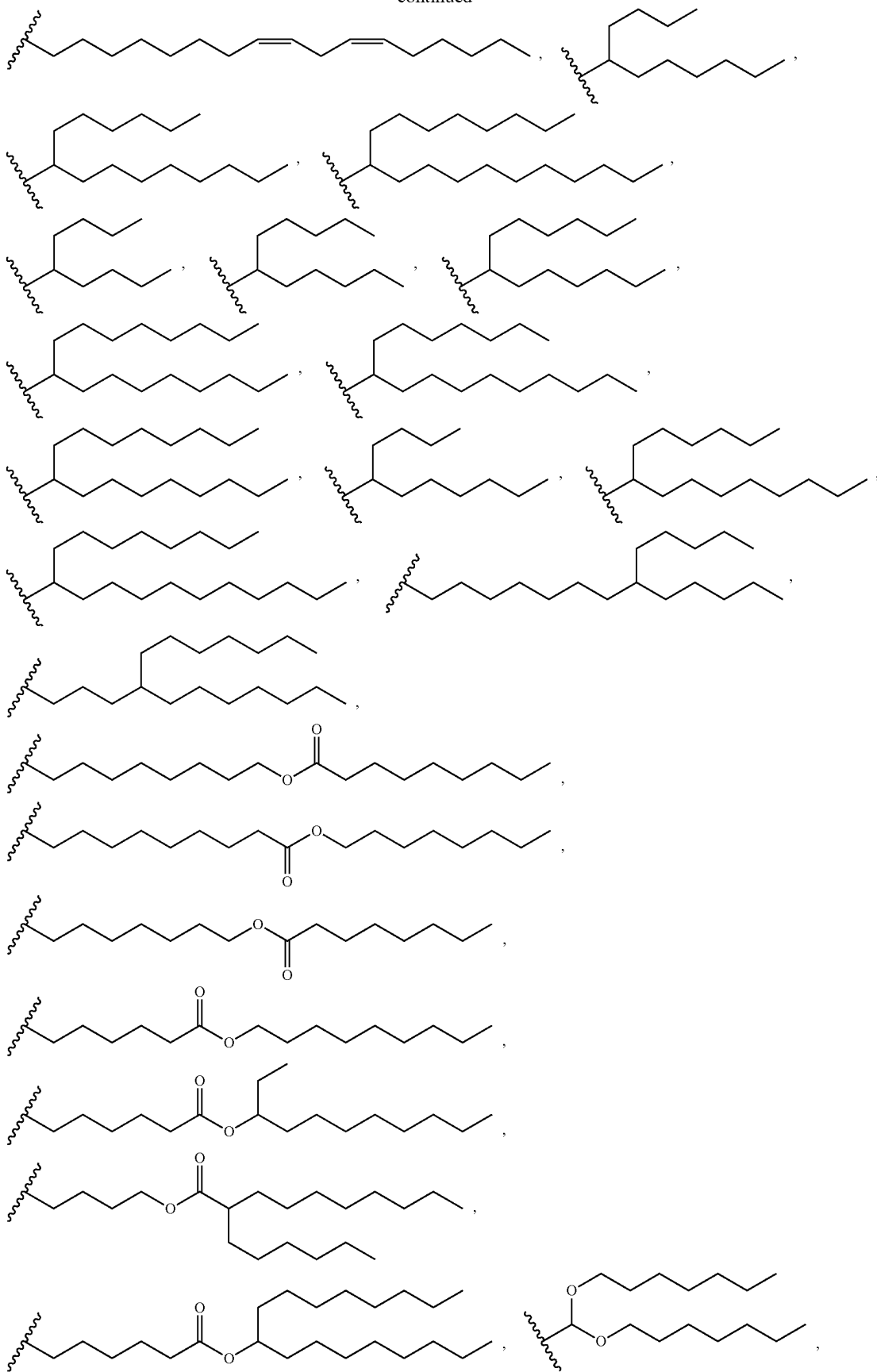
Formula (13b-8)



Formula (13b-9)

Embodiment 7. The pharmaceutical composition of any one of embodiments 1-6, wherein R_1 and R_2 are each independently selected from the group consisting of:

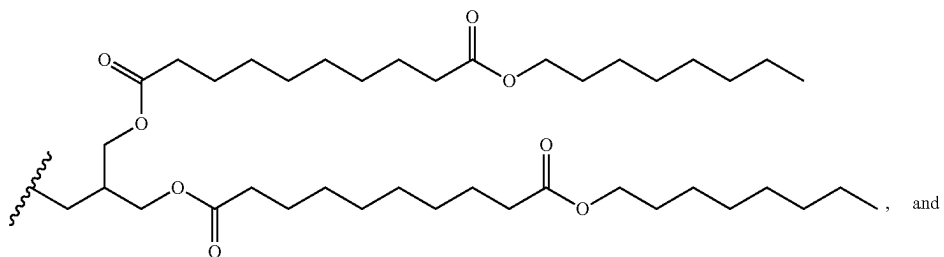
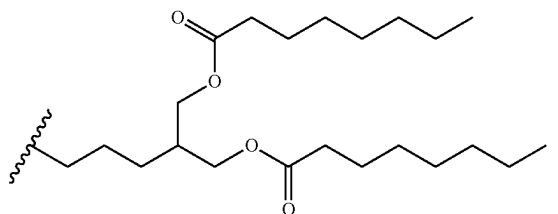
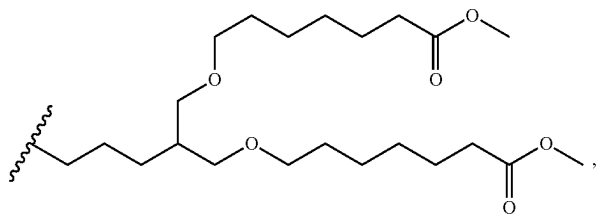
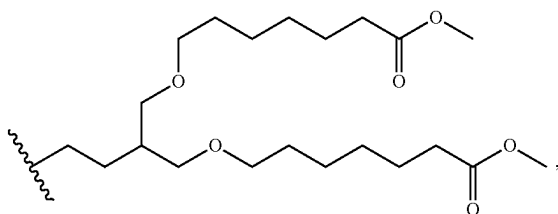
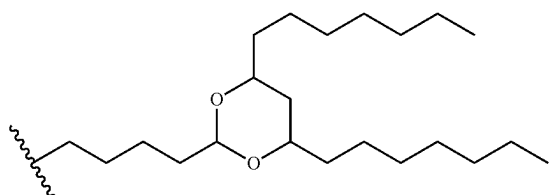
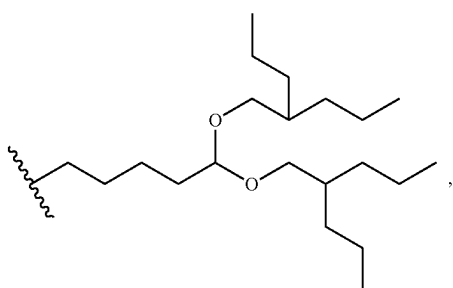
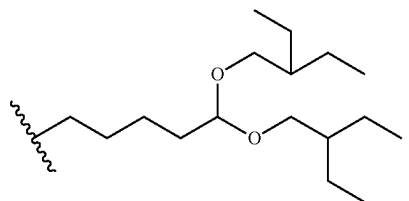
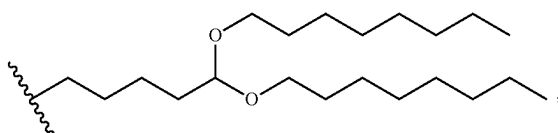
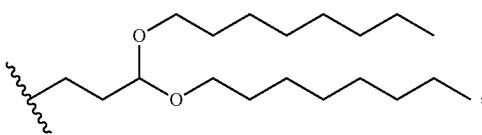
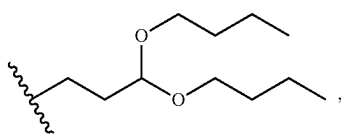




347

348

-continued

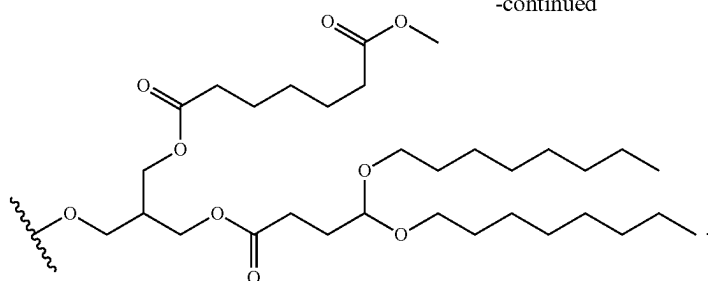


and

349

350

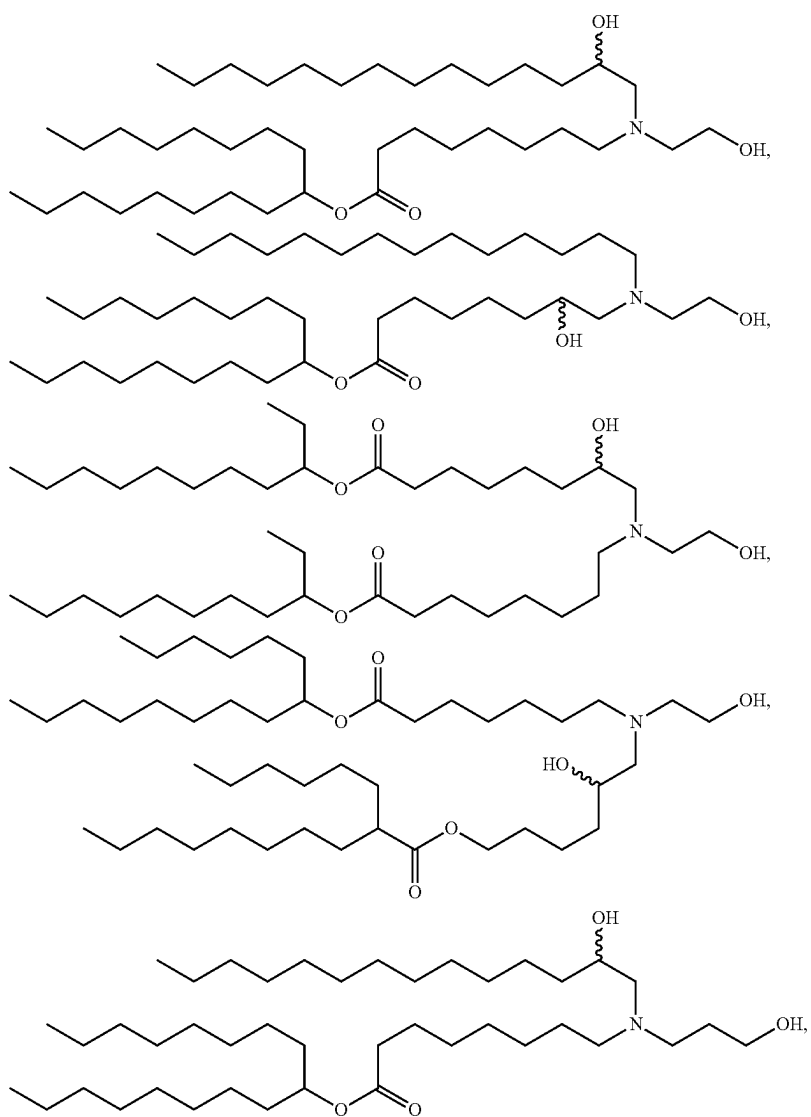
-continued

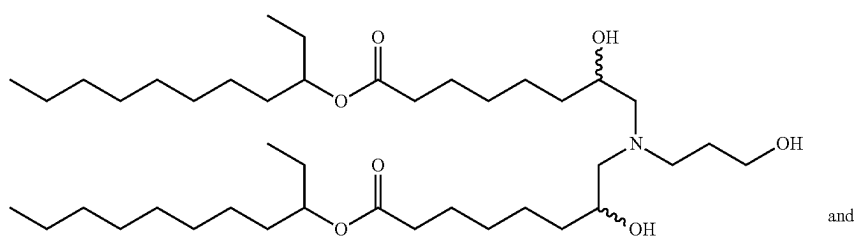
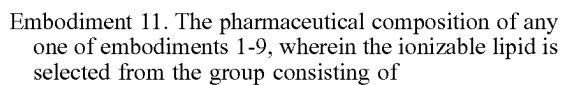


Embodiment 8. The pharmaceutical composition of any
one of embodiments 1-7, wherein R_1 and R_2 are the
same.

Embodiment 9. The pharmaceutical composition of any
one of embodiments 1-7, wherein R_1 and R_2 are dif-
ferent.

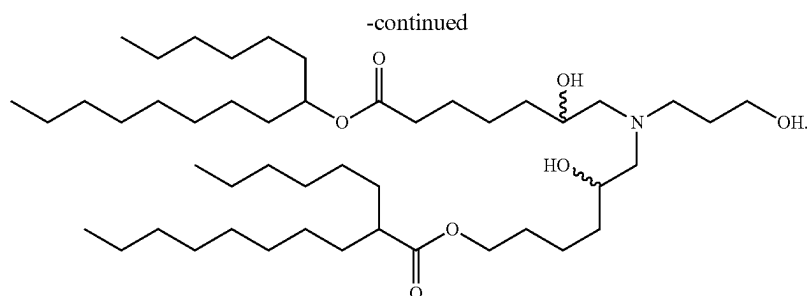
Embodiment 10. The pharmaceutical composition of any
one of embodiments 1-9, wherein the ionizable lipid is
selected from the group consisting of





353

354



Embodiment 12. The pharmaceutical composition of any one of embodiments 1-9, wherein the ionizable lipid is selected from Table 10e.

Embodiment 13. The pharmaceutical composition of any one of embodiments 1-12, wherein the RNA polynucleotide is a linear or circular RNA polynucleotide.

Embodiment 14. The pharmaceutical composition of any one of embodiments 1-13, wherein the RNA polynucleotide is a circular RNA polynucleotide.

Embodiment 15. A pharmaceutical composition comprising:

- a. an RNA polynucleotide, wherein the RNA polynucleotide is a circular RNA polynucleotide, and
- b. a transfer vehicle comprising an ionizable lipid selected from

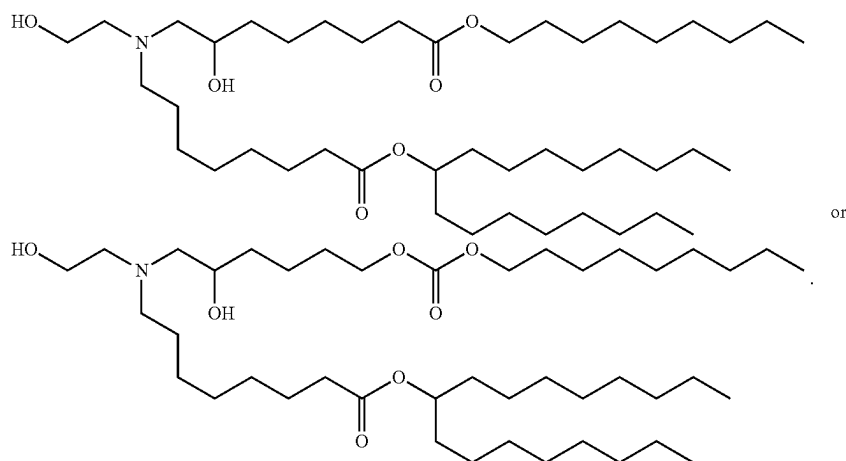
Embodiment 21. The pharmaceutical composition of embodiment 19, wherein the first expression sequence encodes a transcription factor.

Embodiment 22. The pharmaceutical composition of embodiment 19, wherein the first expression sequence encodes an immune checkpoint inhibitor.

Embodiment 23. The pharmaceutical composition of embodiment 19, wherein the first expression sequence encodes a chimeric antigen receptor (CAR).

Embodiment 24. The pharmaceutical composition of any one of embodiments 1-23, wherein the RNA polynucleotide further comprises a second expression sequence.

Embodiment 25. The pharmaceutical composition of embodiment 24, wherein the RNA polynucleotide further comprises an internal ribosome entry site (IRES).



Embodiment 16. The pharmaceutical composition of any one of embodiments 1-15, wherein the RNA polynucleotide is encapsulated in the transfer vehicle.

Embodiment 17. The pharmaceutical composition of any one of embodiments 1-16, wherein the RNA polynucleotide is encapsulated in the transfer vehicle with an encapsulation efficiency of at least 80%.

Embodiment 18. The pharmaceutical composition of any one of embodiments 1-14, wherein the RNA comprises a first expression sequence.

Embodiment 19. The pharmaceutical composition of embodiment 18, wherein the first expression sequence encodes a therapeutic protein.

Embodiment 20. The pharmaceutical composition of embodiment 19, wherein the first expression sequence encodes a cytokine or a functional fragment thereof.

Embodiment 26. The pharmaceutical composition of embodiment 25, wherein the first and second expression sequences are separated by a ribosomal skipping element or a nucleotide sequence encoding a protease cleavage site.

Embodiment 27. The pharmaceutical composition of any one of embodiments 24 or 26, wherein the first expression sequence encodes a first T-cell receptor (TCR) chain, and the second expression sequence encodes a second TCR chain.

Embodiment 28. The pharmaceutical composition of any one of embodiments 1-27, wherein the RNA polynucleotide comprises one or more microRNA binding sites.

Embodiment 29. The pharmaceutical composition of embodiment 28, wherein the microRNA binding site is recognized by a microRNA expressed in the liver.

355

Embodiment 30. The pharmaceutical composition of embodiment 28 or 29, wherein the microRNA binding site is recognized by miR-122.

Embodiment 31. The pharmaceutical composition of any one of embodiments 1-30, wherein the RNA polynucleotide comprises a first IRES associated with greater protein expression in a human immune cell than in a reference human cell.

Embodiment 32. The pharmaceutical composition of embodiment 31, wherein the human immune cell is a T cell, an NK cell, an NKT cell, a macrophage, or a neutrophil.

Embodiment 33. The pharmaceutical composition of embodiment 31 or 32, wherein the reference human cell is a hepatic cell.

Embodiment 34. The pharmaceutical composition of any one of embodiments 1-33, wherein the RNA polynucleotide comprises, in the following order:

- a 5' enhanced exon element,
- a core functional element, and
- a 3' enhanced exon element.

Embodiment 35. The pharmaceutical composition of any one of embodiments 1-34, further comprising a post-splicing intron fragment.

Embodiment 36. The pharmaceutical composition of embodiment 34 or 35, wherein the 5' enhanced exon element comprises a 3' exon fragment.

Embodiment 37. The pharmaceutical composition of any one of embodiments 34-36, wherein the 5' enhanced exon element comprises a 5' internal duplex region located downstream to the 3' exon fragment.

Embodiment 38. The pharmaceutical composition of any one of embodiments 34-37, wherein the 5' enhanced exon element comprises a 5' internal spacer located downstream to the 3' exon fragment.

Embodiment 39. The pharmaceutical composition of embodiment 38, wherein the 5' internal spacer has a length of about 10 to about 60 nucleotides.

Embodiment 40. The pharmaceutical composition of embodiment 38 or 39, wherein the 5' internal spacer comprises a polyA or polyA-C sequence.

Embodiment 41. The pharmaceutical composition of embodiment 40, wherein the polyA or polyA-C sequence comprises a length of about 10-50 nucleotides.

Embodiment 42. The pharmaceutical composition of any one of embodiments 34-41, wherein the core functional element comprises a translation initiation element (TIE).

Embodiment 43. The pharmaceutical composition of any one of embodiments 42, wherein the translation initiation element (TIE) comprises an untranslated region (UTR) or fragment thereof.

Embodiment 44. The pharmaceutical composition of embodiment 43, wherein the UTR or fragment thereof comprises a viral internal ribosome entry site (IRES) or eukaryotic IRES.

Embodiment 45. The pharmaceutical composition of embodiment 44, wherein the IRES is selected from Table 17, or is a functional fragment or variant thereof.

Embodiment 46. The pharmaceutical composition of embodiment 44 or 45, wherein the IRES has a sequence in whole or in part from a Taura syndrome virus, *Triatoma* virus, Theiler's encephalomyelitis virus, Simian Virus 40, *Solenopsis invicta* virus 1, *Rhopalosiphum padi* virus, Reticuloendotheliosis virus, Human poliovirus 1, *Plautia stali* intestine virus, Kashmir bee

356

virus, Human rhinovirus 2, *Homalodisca coagulata* virus- 1, Human Immunodeficiency Virus type 1, *Homalodisca coagulata* virus- 1, Himetobi P virus, Hepatitis C virus, Hepatitis A virus, Hepatitis GB virus, Foot and mouth disease virus.

Human enterovirus 71, Equine rhinitis virus, Ectropis obliqua picoma-like virus, Encephalomyocarditis virus, *Drosophila* C Virus, Human coxsackievirus B3, Cruxifer tobamovirus, Cricket paralysis virus, Bovine viral diarrhea virus 1. Black Queen Cell Virus, Aphid lethal paralysis virus, Avian encephalomyelitis virus, Acute bee paralysis virus, Hibiscus chlorotic ringspot virus, Classical swine fever virus, Human FGF2, Human SFTPA1, Human AML1/RUNX1, *Drosophila antennapedia*, Human AQP4, Human AT1R, Human BAG-1, Human BCL2, Human BiP, Human c-IAP1, Human c-myc, Human eIF4G, Mouse NDST4L, Human LEF1, Mouse HIF1 alpha, Human n.myc, Mouse Gtx, Human p27kipl, Human PDGF2/c-sis, Human p53, Human Pim-1, Mouse Rbm3, *Drosophila* reaper, Canine Scamper. *Drosophila* Ubx, Human UNR, Mouse UtrA, Human VEGF-A, Human XIAP, *Drosophila* hairless, *S. cerevisiae* TFIID, *S. cerevisiae* YAP1, tobacco etch virus, turnip crinkle virus, EMCV-A, EMCV-B, EMCV-Bf, EMCV-Cf, EMCV pEC9, Picobimavirus, HCV QC64, Human Cosavirus E/D, Human Cosavirus F, Human Cosavirus JMY, Rhinovirus NAT001, HRV14, HRV89, HRVC-02, HRV-A21, Salivirus A SH1, Salivirus FHB, Salivirus NG-J1, Human Parechovirus 1, Crohivirus B, Yc-3, Rosavirus M-7, Shanbavirus A, Pasivirus A, Pasivirus A 2, Echovirus E14, Human Parechovirus 5, Aichi Virus, Hepatitis A Virus HA16, Phopivirus, CVA10, Enterovirus C, Enterovirus D, Enterovirus J, Human Pegivirus 2, GBV-C GT110, GBV-C K1737, GBV-C Iowa, Pegivirus A 1220, Pasivirus A 3, Sapelovirus, Rosavirus B, Bakunsa Virus, Tremovirus A, Swine Pasivirus 1, PLV-CHN, Pasivirus A, Sicinivirus, Hepacivirus K, Hepacivirus A, BVDV1, Border Disease Virus, BVDV2, CSFV-PK15C, SF573 Dicistrovirus, Hubei Picoma-like Virus, CRPV, Apodemus Agrarius Picomavirus, Caprine Kobuvirus, Parabovirus, Salivirus A BN5, Salivirus A BN2, Salivirus A 02394, Salivirus A GUT, Salivirus A CH, Salivirus A SZ1, Salivirus FHB, CVB3, CVB1, Echovirus 7, CVB5, EVA71, CVA3, CVA12, EV24, or an aptamer to eIF4G.

Embodiment 47. The pharmaceutical composition of any one of embodiments 42-46, wherein the translation initiation element (TIE) comprises an aptamer complex.

Embodiment 48. The pharmaceutical composition of embodiment 42, wherein the aptamer complex comprises at least two aptamers.

Embodiment 49. The pharmaceutical composition of any one of embodiments 34-48, wherein the core functional element comprises a coding region.

Embodiment 50. The pharmaceutical composition of embodiment 49, wherein the coding region encodes for a therapeutic protein.

Embodiment 51. The pharmaceutical composition of embodiment 50, wherein the therapeutic protein is a chimeric antigen receptor (CAR), a cytokine, a transcription factor, a T cell receptor (TCR), B-cell receptor (BCR), ligand, immune cell activation or inhibitory receptor, recombinant fusion protein, chimeric mutant protein, or fusion protein or a functional fragment thereof.

Embodiment 52. The pharmaceutical composition of embodiment 51, wherein the therapeutic protein is an antigen.

357

Embodiment 53. The pharmaceutical composition of embodiment 52, wherein the antigen is a viral polypeptide from an adenovirus; Herpes simplex, type 1: Herpes simplex, type 2; encephalitis virus, papillomavirus. Varicella-zoster virus; Epstein-barr virus, Human cytomegalovirus; Human herpes virus, type 8. Human papillomavirus; BK virus; JC virus; Smallpox; polio virus; Hepatitis B virus; Human bocavirus; Parvovirus B19; Human astrovirus; Norwalk virus; coxsackievirus; hepatitis A virus; poliovirus; rhinovirus; Severe acute respiratory syndrome virus; Hepatitis C virus; Yellow Fever virus; Dengue virus; West Nile virus; Rubella virus; Hepatitis E virus; Human Immunodeficiency virus (HIV); Influenza virus; Guanarito virus; Junin virus; Lassa virus; Machupo virus, Sabia virus, Crimean-Congo hemorrhagic fever virus; Ebola virus; Marburg virus, Measles virus; Mumps virus; Parainfluenza virus; Respiratory syncytial virus; Human metapneumo virus; Hendra virus; Nipah virus; Rabies virus; Hepatitis D; Rotavirus; Orbivirus, Coltivirus; Banna virus; Human Enterovirus; Hanta virus; West Nile virus; Middle East Respiratory Syndrome Corona Virus; Japanese encephalitis virus; Vesicular exanthnavirus; SARS-CoV-2; Eastern equine encephalitis, or a combination of any two or more of the foregoing.

Embodiment 54. The pharmaceutical composition of any one of embodiments 34-53, wherein the core functional element comprises a stop codon or a stop cassette.

Embodiment 55. The pharmaceutical composition of any one of embodiment 34-53, wherein the core functional element comprises a noncoding region.

Embodiment 56. The pharmaceutical composition of any one of embodiment 34-53, wherein the core functional element comprises an accessory or modulatory element.

Embodiment 57. The pharmaceutical composition of embodiment 56, wherein the accessory or modulatory element comprises a miRNA binding site or a fragment thereof, a restriction site or a fragment thereof, a RNA editing motif or a fragment thereof, a zip code element or a fragment thereof, a RNA trafficking element or fragment thereof, or a combination thereof.

Embodiment 58. The pharmaceutical composition of embodiment 56, wherein the accessory or modulatory element comprises a binding domain to an IRES transacting factor (ITAF).

Embodiment 59. The pharmaceutical composition of any one of embodiments 34-58, wherein the 3' enhanced exon element comprises a 5' exon fragment.

Embodiment 60. The pharmaceutical composition of embodiments 59, wherein the 3' enhanced exon element comprises a 3' internal spacer located upstream to the 5' exon fragment.

Embodiment 61. The pharmaceutical composition of embodiment 60, wherein the 3' internal spacer is a polyA or polyA-C sequence.

Embodiment 62. The pharmaceutical composition of embodiment 60 or 61, wherein the 3' internal spacer has a length of about 10 to about 60 nucleotides.

Embodiment 63. The pharmaceutical composition of any one of embodiments 59-62, wherein the 3' enhanced exon element comprises a 3' internal duplex element located upstream to the 5' exon fragment.

Embodiment 64. The pharmaceutical composition of any one of embodiments 1-63, wherein the RNA polynucleotide is made via circularization of a RNA polynucleotide comprising, in the following order:

358

a. a 5' enhanced intron element,
 b. a 5' enhanced exon element,
 c. a core functional element,
 d. a 3' enhanced exon element, and
 e. a 3' enhanced intron element.

Embodiment 65. The pharmaceutical composition of embodiment 64, wherein the 5' enhanced intron element comprises a 3' intron fragment.

Embodiment 66. The pharmaceutical composition of embodiment 65, wherein the 3' intron fragment comprises a first or a first and second nucleotide of a 3' group I intron splice site dinucleotide.

Embodiment 67. The pharmaceutical composition of embodiment 64 or 65, wherein the 5' enhanced intron element comprises a 5' affinity tag located upstream to the 3' intron fragment.

Embodiment 68. The pharmaceutical composition of any one of embodiments 65-67 wherein the 5' enhanced intron element comprises a 5' external spacer located upstream to the 3' intron fragment.

Embodiment 69. The pharmaceutical composition of any one of embodiments 64-68, wherein the 5' enhanced intron element comprises a leading untranslated sequence located at the 5' end of said 5' enhanced intron element.

Embodiment 70. The pharmaceutical composition of any one of embodiments 64-69, wherein the 3' enhanced intron element comprises a 5' intron fragment.

Embodiment 71. The pharmaceutical composition of any one of embodiments 64-70, wherein the 3' enhanced intron element comprises a 3' external spacer located downstream to the 5' intron fragment.

Embodiment 72. The pharmaceutical composition of any one of embodiments 64-71, wherein the 3' enhanced intron element comprises a 3' affinity tag located downstream to the 5' intron fragment.

Embodiment 73. The pharmaceutical composition of any one of embodiments 64-72, wherein the 3' enhanced intron element comprises a 3' terminal untranslated sequence at the 3' end of the said 5' enhanced intron element.

Embodiment 74. The pharmaceutical composition of any one of embodiments 64-73, wherein the 5' enhanced intron element comprises a 5' external duplex region upstream to the 3' intron fragment, and the 3' enhanced intron element comprises a 3' external duplex region downstream to the 5' intron fragment.

Embodiment 75. The pharmaceutical composition of embodiment 74, wherein the 5' external duplex region and the 3' external duplex region are the same.

Embodiment 76. The pharmaceutical composition of embodiment 74, wherein the 5' external duplex region and the 3' external duplex region are different.

Embodiment 77. The pharmaceutical composition of any one of embodiments 66-76, wherein the group I intron comprises in part or in whole from a bacterial phage, viral vector, organelle genome, or a nuclear rDNA gene.

Embodiment 78. The pharmaceutical composition of embodiment 77, wherein the nuclear rDNA gene comprises a nuclear rDNA gene derived from a fungi, plant, or algae, or a fragment thereof.

Embodiment 79. The pharmaceutical composition of any one of embodiments 1-78, wherein the RNA polynucleotide contains at least about 80%, at least about 90%, at least about 95%, or at least about 99% naturally occurring nucleotides.

359

Embodiment 80. The pharmaceutical composition of any one of embodiments 1-79, wherein the RNA polynucleotide consists of naturally occurring nucleotides.

Embodiment 81. The pharmaceutical composition of any one of embodiments 34-80, wherein the expression sequence is codon optimized. 5

Embodiment 82. The pharmaceutical composition of any one of embodiments 1-81, wherein the RNA polynucleotide is optimized to lack at least one microRNA binding site present in an equivalent pre-optimized polynucleotide. 10

Embodiment 83. The pharmaceutical composition of any one of embodiments 1-82, wherein the RNA polynucleotide is optimized to lack at least one microRNA binding site capable of binding to a microRNA present in a cell within which the RNA polynucleotide is expressed. 15

Embodiment 84. The pharmaceutical composition of any one of embodiments 1-83, wherein the RNA polynucleotide is optimized to lack at least one endonuclease susceptible site present in an equivalent pre-optimized polynucleotide. 20

Embodiment 85. The pharmaceutical composition of any one of embodiments 1-84, wherein the RNA polynucleotide is optimized to lack at least one endonuclease susceptible site capable of being cleaved by an endonuclease present in a cell within which the endonuclease is expressed. 25

Embodiment 86. The pharmaceutical composition of any one of embodiments 1-85, wherein the RNA polynucleotide is optimized to lack at least one RNA editing susceptible site present in an equivalent pre-optimized polynucleotide. 30

Embodiment 87. The pharmaceutical composition of any one of embodiments 1-86, wherein the RNA polynucleotide is from about 100 nt to about 10,000 nt in length. 35

Embodiment 88. The pharmaceutical composition of any one of embodiments 1-87, wherein the RNA polynucleotide is from about 100 nt to about 15,000 nt in length.

Embodiment 89. The pharmaceutical composition of any one of embodiments 1-88, wherein the RNA polynucleotide is a circular RNA polynucleotide, and wherein the circular RNA polynucleotide is more compact than a reference linear RNA polynucleotide having the same expression sequence as the circular RNA polynucleotide. 40 45

Embodiment 90. The pharmaceutical composition of any one of embodiments 1-89, wherein the RNA polynucleotide is a circular RNA polynucleotide, and wherein the composition has a duration of therapeutic effect in a human cell greater than or equal to that of a composition comprising a reference linear RNA polynucleotide having the same expression sequence as the circular RNA polynucleotide. 50

Embodiment 91. The pharmaceutical composition of embodiment 90, wherein the reference linear RNA polynucleotide is a linear, unmodified or nucleoside-modified, fully-processed mRNA comprising a cap1 structure and a polyA tail at least 80 nt in length. 55

Embodiment 92. The pharmaceutical composition of any one of embodiments 1-91, wherein the RNA polynucleotide is a circular RNA polynucleotide, and wherein the composition has a duration of therapeutic effect in vivo in humans greater than that of a composition comprising a reference linear RNA polynucleotide having the same expression sequence as the circular RNA polynucleotide. 60 65

360

Embodiment 93. The pharmaceutical composition of any one of embodiments 1-92, wherein the composition has a duration of therapeutic effect in vivo in humans of at least about 10, at least about 20, at least about 30, at least about 40, at least about 50, at least about 60, at least about 70, at least about 80, at least about 90, or at least about 100 hours.

Embodiment 94. The pharmaceutical composition of any one of embodiments 1-93, wherein the composition has a functional half-life in a human cell greater than or equal to that of a pre-determined threshold value.

Embodiment 95. The pharmaceutical composition of any one of embodiments 1-94, wherein the composition has a functional half-life in vivo in humans greater than that of a pre-determined threshold value.

Embodiment 96. The pharmaceutical composition of embodiment 94 or 95, wherein the functional half-life is determined by a functional protein assay.

Embodiment 97. The pharmaceutical composition of embodiment 96, wherein the functional protein assay is an in vitro luciferase assay.

Embodiment 98. The pharmaceutical composition of embodiment %, wherein the functional protein assay comprises measuring levels of protein encoded by the expression sequence of the RNA polynucleotide in a patient serum or tissue sample.

Embodiment 99. The pharmaceutical composition of any one of embodiments 94-98, wherein the pre-determined threshold value is the functional half-life of a reference linear RNA polynucleotide comprising the same expression sequence as the RNA polynucleotide.

Embodiment 100. The pharmaceutical composition of any one of embodiments 1-99, wherein the composition has a functional half-life of at least about 20 hours.

Embodiment 101. The pharmaceutical composition of any one of embodiments 1-100, further comprising a structural lipid and a PEG-modified lipid.

Embodiment 102. The pharmaceutical composition of any one of embodiment 101, wherein the structural lipid binds to C1q and/or promotes the binding of the transfer vehicle comprising said lipid to C1q compared to a control transfer vehicle lacking the structural lipid and/or increases uptake of C1q-bound transfer vehicle into an immune cell compared to a control transfer vehicle lacking the structural lipid.

Embodiment 103. The pharmaceutical composition of any one of embodiment 97-102, wherein the immune cell is a T cell, an NK cell, an NKT cell, a macrophage, or a neutrophil.

Embodiment 104. The pharmaceutical composition of any one of embodiments 101-103, wherein the structural lipid is cholesterol.

Embodiment 105. The pharmaceutical composition of embodiment 102, wherein the structural lipid is beta-sitosterol.

Embodiment 106. The pharmaceutical composition of embodiment 102, wherein the structural lipid is not beta-sitosterol.

Embodiment 107. The pharmaceutical composition of any one of embodiments 101-106, wherein the PEG-modified lipid is DSPE-PEG, DMG-PEG, PEG-DAG, PEG-S-DAG, PEG-PE, PEG-S-DMG, PEG-cer, PEG-dialkoxypyrrolcarbamate, PEG-OR, PEG-OH, PEG-c-DOMG, or PEG-1.

Embodiment 108. The pharmaceutical composition of embodiment 107, wherein the PEG-modified lipid is DSPE-PEG(2000).

Embodiment 117. The pharmaceutical composition of embodiment 117, wherein the transfer vehicle comprises the helper lipid of DOPE and the PEG-lipid of DSPE-PEG(2000), and wherein the molar ratio of ionizable lipid:DOPE:cholesterol:DSPE-PEG(2000) is 45:9:44:2, 50:10:38.5:1.5, 41:12:45:2, 62:4:33:1, or 53:5:41:1.

Embodiment 118. The pharmaceutical composition of embodiment 117, wherein the transfer vehicle comprises the helper lipid of DOPE and the PEG-lipid of DSPE-PEG(2000), and wherein the molar ratio of ionizable lipid:DOPE:cholesterol:DSPE-PEG(2000) is 62:4:33:1.

Embodiment 119. The pharmaceutical composition of embodiment 114, wherein the transfer vehicle comprises the helper lipid of DOPE and the PEG-lipid of DSPE-PEG(2000), and wherein the molar ratio of ionizable lipid:DOPE:cholesterol:DSPE-PEG(2000) is 53:5:41:1.

Embodiment 120. The pharmaceutical composition of embodiment 114, wherein the transfer vehicle com-

Embodiment 114. The pharmaceutical composition of any one of embodiments 1-113, wherein the transfer vehicle comprises:

a an ionizable lipid selected from:

The chemical structures are as follows:

- 1**: 1,3-bis(sn-3'-phosphatidyl)-2-glycerol. It consists of a central glycerol backbone with two 3'-phosphatidyl groups attached at the 1 and 3 positions. The phosphatidyl groups are linked via their 3' phosphate groups to the 1 and 3 positions of the central glycerol.
- 2**: 1,3-bis(sn-3'-phosphatidyl)-2-glycerol. It consists of a central glycerol backbone with two 3'-phosphatidyl groups attached at the 1 and 3 positions. The phosphatidyl groups are linked via their 3' phosphate groups to the 1 and 3 positions of the central glycerol.
- 3**: 1,3-bis(sn-3'-phosphatidyl)-2-glycerol. It consists of a central glycerol backbone with two 3'-phosphatidyl groups attached at the 1 and 3 positions. The phosphatidyl groups are linked via their 3' phosphate groups to the 1 and 3 positions of the central glycerol.
- 4**: 1,3-bis(sn-3'-phosphatidyl)-2-glycerol. It consists of a central glycerol backbone with two 3'-phosphatidyl groups attached at the 1 and 3 positions. The phosphatidyl groups are linked via their 3' phosphate groups to the 1 and 3 positions of the central glycerol.
- 5**: 1,3-bis(sn-3'-phosphatidyl)-2-glycerol. It consists of a central glycerol backbone with two 3'-phosphatidyl groups attached at the 1 and 3 positions. The phosphatidyl groups are linked via their 3' phosphate groups to the 1 and 3 positions of the central glycerol.
- 6**: 1,3-bis(sn-3'-phosphatidyl)-2-glycerol. It consists of a central glycerol backbone with two 3'-phosphatidyl groups attached at the 1 and 3 positions. The phosphatidyl groups are linked via their 3' phosphate groups to the 1 and 3 positions of the central glycerol.

- b. a helper lipid selected from DOPE or DSPC,
- c. cholesterol, and

prises the helper lipid of DSPC and the PEG-lipid of DMG-PEG(2000), and wherein the molar ratio of ionizable lipid:DSPC:cholesterol:DMG-PEG(2000) is 45:9:44:2, 50:10:38.5:1.5, 41:12:45:2, 62:4:33:1, or 53:5:41:1.

Embodiment 121. The pharmaceutical composition of embodiment 120, wherein the transfer vehicle comprises the helper lipid of DSPC and the PEG-lipid of DMG-PEG(2000), and wherein the molar ratio of ionizable lipid:DSPC:cholesterol:DMG-PEG(2000) is 50:10:38.5:1.5.

Embodiment 122. The pharmaceutical composition of embodiment 120, wherein the transfer vehicle comprises the helper lipid of DSPC and the PEG-lipid of DMG-PEG(2000), and wherein the molar ratio of ionizable lipid:DSPC:cholesterol:DGM-PEG(2000) is 41:12:45:2.

363

Embodiment 123. The pharmaceutical composition of embodiment 120, wherein the transfer vehicle comprises the helper lipid of DSPC and the PEG-lipid of DMG-PEG(2000), and wherein the molar ratio of ionizable lipid:DSPC:cholesterol:DMG-PEG(2000) is 45:9:44:2. Embodiment 124. The pharmaceutical composition of embodiment 114, wherein the transfer vehicle comprises the helper lipid of DSPC and the PEG-lipid of DSPE-PEG(2000), and wherein the molar ratio of ionizable lipid: DSPC:cholesterol:DSPE-PEG(2000) is 45:9:44:2, 50:10:38.5:1.5, 41:12:45:2, 62:4:33:1, or 53:5:41:1.

Embodiment 125. The pharmaceutical composition of embodiment 114, wherein the transfer vehicle comprises the helper lipid of DOPE and the PEG-lipid is C14-PEG(2000), and wherein the molar ratio of ionizable lipid:DOPE:cholesterol:C14-PEG(2000) is 45:9:44:2, 50:10:38.5:1.5, 41:12:45:2, 62:4:33:1, or 53:5:41:1.

Embodiment 126. The pharmaceutical composition of embodiment 114, wherein the transfer vehicle comprises the helper lipid of DOPE and the PEG-lipid of DMG-PEG(2000), wherein the molar ratio of ionizable lipid:DOPE:cholesterol:DMG-PEG(2000) is 45:9:44:2, 50:10:38.5:1.5, 41:12:45:2, 62:4:33:1, or 53:5:41:1.

Embodiment 127. The pharmaceutical composition of any one of embodiments 1-126, having a lipid to phosphate (IL:P) molar ratio of about 3 to about 9.

Embodiment 128. The pharmaceutical composition of any one of embodiments 1-127, having a lipid to phosphate (IL:P) molar ratio of about 3, about 4, about 4.5, about 5, about 5.5, about 5.7, about 6, about 6.2, about 6.5, or about 7.

Embodiment 129. The pharmaceutical composition of any one of embodiments 1-128, wherein the transfer vehicle is formulated for endosomal release of the RNA polynucleotide.

Embodiment 130. The pharmaceutical composition of any one of embodiments 1-129, wherein the transfer vehicle is capable of binding to APOE.

Embodiment 131. The pharmaceutical composition of any one of embodiments 1-130, wherein the transfer vehicle interacts with apolipoprotein E (APOE) less than an equivalent transfer vehicle loaded with a reference linear RNA having the same expression sequence as the RNA polynucleotide.

Embodiment 132. The pharmaceutical composition of any one of embodiments 1-131, wherein the exterior surface of the transfer vehicle is substantially free of APOE binding sites.

Embodiment 133. The pharmaceutical composition of any one of embodiments 1-132, wherein the transfer vehicle has a diameter of less than about 120 nm.

Embodiment 134. The pharmaceutical composition of any one of embodiments 1-133, wherein the transfer vehicle does not form aggregates with a diameter of more than 300 nm. Embodiment 135. The pharmaceutical composition of any one of embodiments 1-134, wherein the transfer vehicle has an in vivo half-life of less than about 30 hours.

Embodiment 136. The pharmaceutical composition of any one of embodiments 1-135, wherein the transfer vehicle is capable of low density lipoprotein receptor (LDLR) dependent uptake into a cell.

364

Embodiment 137. The pharmaceutical composition of any one of embodiments 1-136, wherein the transfer vehicle is capable of LDLR independent uptake into a cell.

Embodiment 138. The pharmaceutical composition of any one of embodiments 1-137, wherein the pharmaceutical composition is substantially free of linear RNA.

Embodiment 139. The pharmaceutical composition of any one of embodiments 1-138, further comprising a targeting moiety operably connected to the transfer vehicle.

Embodiment 140. The pharmaceutical composition of embodiment 139, wherein the targeting moiety specifically binds an immune cell antigen or indirectly.

Embodiment 141. The pharmaceutical composition of embodiment 140, wherein the immune cell antigen is a T cell antigen.

Embodiment 142. The pharmaceutical composition of embodiment 141, wherein the T cell antigen is selected from the group consisting of CD2, CD3, CD5, CD7, CD8, CD4, beta7 integrin, beta2 integrin, and C1qR.

Embodiment 143. The pharmaceutical composition of embodiment 142, further comprising an adapter molecule comprising a transfer vehicle binding moiety and a cell binding moiety, wherein the targeting moiety specifically binds the transfer vehicle binding moiety and the cell binding moiety specifically binds a target cell antigen.

Embodiment 144. The pharmaceutical composition of embodiment 143, wherein the target cell antigen is an immune cell antigen.

Embodiment 145. The pharmaceutical composition of embodiment 144, wherein the immune cell antigen is a T cell antigen, an NK cell, an NKT cell, a macrophage, or a neutrophil.

Embodiment 146. The pharmaceutical composition of embodiment 145, wherein the T cell antigen is selected from the group consisting of CD2, CD3, CD5, CD7, CD8, CD4, beta7 integrin, beta2 integrin, CD25, CD39, CD73, A2a Receptor, A2b Receptor, and C1qR.

Embodiment 147. The pharmaceutical composition of embodiment 140 or 143, wherein the immune cell antigen is a macrophage antigen.

Embodiment 148. The pharmaceutical composition of embodiment 147, wherein the macrophage antigen is selected from the group consisting of mannose receptor, CD206, and C1q.

Embodiment 149. The pharmaceutical composition of any one of embodiments 139-148, wherein the targeting moiety is a small molecule.

Embodiment 150. The pharmaceutical composition of embodiment 149, wherein the small molecule is mannose, a lectin, acivicin, biotin, or digoxigenin.

Embodiment 151. The pharmaceutical composition of embodiment 149, wherein the small molecule binds to an ectoenzyme on an immune cell, wherein the ectoenzyme is selected from the group consisting of CD38, CD73, adenosine 2a receptor, and adenosine 2b receptor.

Embodiment 152. The pharmaceutical composition of any one of embodiments 139-148, wherein the targeting moiety is a single chain Fv (scFv) fragment, nanobody, peptide, peptide-based macrocycle, minibody, small molecule ligand such as folate, arginylglycylaspartic acid (RGD), or phenol-soluble modulin alpha 1 peptide (PSMA1), heavy chain variable region, light chain variable region or fragment thereof.

365

Embodiment 153. The pharmaceutical composition of any one of embodiments 1-152, wherein the ionizable lipid has a half-life in a cell membrane less than about 2 weeks.

Embodiment 154. The pharmaceutical composition of any one of embodiments 1-153, wherein the ionizable lipid has a half-life in a cell membrane less than about 1 week.

Embodiment 155. The pharmaceutical composition of any one of embodiments 1-154, wherein the ionizable lipid has a half-life in a cell membrane less than about 30 hours.

Embodiment 156. The pharmaceutical composition of any one of embodiments 1-155, wherein the ionizable lipid has a half-life in a cell membrane less than the functional half-life of the RNA polynucleotide.

Embodiment 157. A method of treating or preventing a disease, disorder, or condition, comprising administering an effective amount of a pharmaceutical composition of any one of embodiments 1-156.

Embodiment 158. The method of embodiment 157, wherein the disease, disorder, or condition is associated with aberrant expression, activity, or localization of a polypeptide selected from ASCII Tables L and M.

Embodiment 159. The method of embodiment 157 or 158, wherein the RNA polynucleotide encodes a therapeutic protein.

Embodiment 160. The method of embodiment 159, wherein therapeutic protein expression in the spleen is higher than therapeutic protein expression in the liver.

Embodiment 161. The method of embodiment 160, wherein therapeutic protein expression in the spleen is at least about 2.9× therapeutic protein expression in the liver.

Embodiment 162. The method of embodiment 160, wherein the therapeutic protein is not expressed at functional levels in the liver.

Embodiment 163. The method of embodiment 160, wherein the therapeutic protein is not expressed at detectable levels in the liver.

Embodiment 164. The method of embodiment 160, wherein therapeutic protein expression in the spleen is at least about 50% of total therapeutic protein expression.

Embodiment 165. The method of embodiment 160, wherein therapeutic protein expression in the spleen is at least about 63% of total therapeutic protein expression.

Embodiment 166. A pharmaceutical composition of any one of embodiments 1-156, wherein the transfer vehicle comprises a nanoparticle, and optionally, a targeting moiety operably connected to the nanoparticle.

Embodiment 167. The pharmaceutical composition of embodiment 166, wherein the nanoparticle is a lipid nanoparticle, a core-shell nanoparticle, a biodegradable nanoparticle, a biodegradable lipid nanoparticle, a polymer nanoparticle, or a biodegradable polymer nanoparticle.

Embodiment 168. The pharmaceutical composition of embodiment 166 or 167, comprising a targeting moiety, wherein the targeting moiety mediates receptor-mediated endocytosis or direct fusion selectively into cells of a selected cell population or tissue in the absence of cell isolation or purification.

Embodiment 169. The pharmaceutical composition of any one of embodiments 166-168, wherein the targeting

366

moiety is a scfv, nanobody, peptide, minibody, polynucleotide aptamer, heavy chain variable region, light chain variable region or fragment thereof.

Embodiment 170. The pharmaceutical composition of any one of embodiments 166-169, wherein less than 1%, by weight, of the polynucleotides in the composition are double stranded RNA, DNA splints, or triphosphorylated RNA.

Embodiment 171. The pharmaceutical composition of any one of embodiments 166-170, wherein less than 1%, by weight, of the polynucleotides and proteins in the pharmaceutical composition are double stranded RNA, DNA splints, triphosphorylated RNA, phosphatase proteins, protein ligases, and capping enzymes.

Embodiment 172. A method of treating a subject in need thereof comprising administering a therapeutically effective amount of the pharmaceutical composition of any one of embodiments 166-171.

Embodiment 173. The method of embodiment 172, wherein the targeting moiety is a scfv, nanobody, peptide, minibody, heavy chain variable region, light chain variable region, an extracellular domain of a TCR, or a fragment thereof.

Embodiment 174. The method of embodiment 172 or 173, wherein the nanoparticle comprises one or more cationic lipids, ionizable lipids, or poly β -amino esters.

Embodiment 175. The method of any one of embodiments 172-174, wherein the nanoparticle comprises one or more non-cationic lipids.

Embodiment 176. The method of any one of embodiments 172-175, wherein the nanoparticle comprises one or more PEG-modified lipids, polyglutamic acid lipids, or Hyaluronic acid lipids.

Embodiment 177. The method of any one of embodiments 172-176, wherein the nanoparticle comprises cholesterol.

Embodiment 178. The method of any one of embodiments 172-177, wherein the nanoparticle comprises arachidonic acid or oleic acid.

Embodiment 179. The method of any one of embodiments 172-178, wherein the pharmaceutical composition comprises a targeting moiety, wherein the targeting moiety mediates receptor-mediated endocytosis selectively into cells of a selected cell population in the absence of cell selection or purification.

Embodiment 180. The method of any one of embodiments 172-179, wherein the nanoparticle comprises more than one circular RNA polynucleotide.

Embodiment 181. A DNA vector encoding the RNA polynucleotide of any one of embodiments 64-78.

Embodiment 182. The DNA vector of embodiment 181, further comprising a transcription regulatory sequence.

Embodiment 183. The DNA vector of embodiment 182, wherein the transcription regulatory sequence comprises a promoter and/or an enhancer.

Embodiment 184. The DNA vector of embodiment 183, wherein the promoter comprises a T7 promoter.

Embodiment 185. The DNA vector of any one of embodiments 181-184, wherein the DNA vector comprises a circular DNA.

Embodiment 186. The DNA vector of any one of embodiments 181-185, wherein the DNA vector comprises a linear DNA.

Embodiment 187. A prokaryotic cell comprising the DNA vector according to any one of embodiments 181-186.

367

Embodiment 188. A eukaryotic cell comprising the RNA polynucleotide according to any one of embodiments 1-187.

Embodiment 189. The eukaryotic cell of embodiment 189, wherein the eukaryotic cell is a human cell.

Embodiment 190. A method of producing a circular RNA polynucleotide, the method comprising incubating the RNA polynucleotide of any one of embodiments 64-78 under suitable conditions for circularization.

Embodiment 191. A method of producing a circular RNA polynucleotide, the method comprising incubating DNA of any one of embodiments 181-186 under suitable conditions for transcription.

Embodiment 192. The method of embodiment 191, wherein the DNA is transcribed in vitro.

Embodiment 193. The method of embodiment 191, wherein the suitable conditions comprises adenosine triphosphate (ATP), guanine triphosphate (GTP), cytosine triphosphate (CTP), uridine triphosphate (UTP), and an RNA polymerase.

Embodiment 194. The method of embodiment 191, wherein the suitable conditions further comprises guanine monophosphate (GMP).

Embodiment 195. The method of embodiment 194, wherein the ratio of GMP concentration to GTP concentration is within the range of about 3:1 to about 15:1, optionally about 4:1, 5:1, or 6:1.

Embodiment 196. A method of producing a circular RNA polynucleotide, the method comprising culturing the prokaryotic cell of embodiment 187 under suitable conditions for transcribing the DNA in the cell.

Embodiment 197. The method of any one of embodiments 190-196, further comprising purifying a circular RNA polynucleotide.

Embodiment 198. The method of embodiment 197, wherein the circular RNA polynucleotide is purified by negative selection using an affinity oligonucleotide that hybridizes with the first or second spacer conjugated to a solid surface.

Embodiment 199. The method of embodiment 198, wherein the first or second spacer comprises a polyA sequence, and wherein the affinity oligonucleotide is a deoxythymine oligonucleotide.

EXAMPLES

Wesselhoeft et al., (2019) RNA Circularization Diminishes Immunogenicity and Can Extend Translation Duration In vivo. *Molecular Cell*. 74(3), 508-520 and Wesselhoeft et al., (2018) Engineering circular RNA for Potent and Stable Translation in Eukaryotic Cells. *Nature Communications*. 9, 2629 are incorporated by reference in their entirety.

The invention is further described in detail by reference to the following examples but are not intended to be limited to the following examples. These examples encompass any and all variations of the illustrations with the intention of providing those of ordinary skill in the art with complete disclosure and description of how to make and use the subject invention and are not intended to limit the scope of what is regarded as the invention.

Example 1

Example 1A: External Duplex Regions Allow for Circularization of Long Precursor RNA Using the Permuted Intron Exon (PIE) Circularization Strategy

A 1.1 kb sequence containing a full-length encephalomyocarditis virus (EMCV) IRES, a *Gaussia luciferase*

368

(GLuc) expression sequence, and two short exon fragments of the permuted intron-exon (PIE) construct were inserted between the 3' and 5' introns of the permuted group 1 catalytic intron in the thymidylate synthase (Td) gene of the T4 phage.

Precursor RNA was synthesized by run-off transcription. Circularization was attempted by heating the precursor RNA in the presence of magnesium ions and GTP, but splicing products were not obtained.

Perfectly complementary 9 nucleotide and 19 nucleotide long duplex regions were designed and added at the 5' and 3' ends of the precursor RNA. Addition of these homology arms increased splicing efficiency from 0 to 16% for 9 nucleotide duplex regions and to 48% for 19 nucleotide duplex regions as assessed by disappearance of the precursor RNA band.

The splicing product was treated with RNase R. Sequencing across the putative splice junction of RNase R-treated splicing reactions revealed ligated exons, and digestion of the RNase R-treated splicing reaction with oligonucleotide-targeted RNase H produced a single band in contrast to two bands yielded by RNase H-digested linear precursor. This shows that circular RNA is a major product of the splicing reactions of precursor RNA containing the 9 or 19 nucleotide long external duplex regions

Example 1B: Spacers that Conserve Secondary Structures of IRES and PIE Splice Sites Increase Circularization Efficiency

A series of spacers was designed and inserted between the 3' PIE splice site and the IRES. These spacers were designed to either conserve or disrupt secondary structures within intron sequences in the IRES, 3' PIE splice site, and/or 5' splice site. The addition of spacer sequences designed to conserve secondary structures resulted in 87% splicing efficiency, while the addition of a disruptive spacer sequences resulted in no detectable splicing.

Example 2

Example 2A: Internal Duplex Regions in Addition to External Duplex Regions Create a Splicing Bubble and Allows for Translation of Several Expression Sequences

Spacers were designed to be unstructured, non-homologous to the intron and IRES sequences, and to contain spacer-spacer duplex regions. These were inserted between the 5' exon and IRES and between the 3' exon and expression sequence in constructs containing external duplex regions, EMCV IRES, and expression sequences for *Gaussia luciferase* (total length: 1289 nt). Firefly luciferase (2384 nt), eGFP (1451 nt), human erythropoietin (1313 nt), and Cas9 endonuclease (4934 nt). Circularization of all 5 constructs was achieved. Circularization of constructs utilizing T4 phage and *Anabaena* introns were roughly equal. Circularization efficiency was higher for shorter sequences. To measure translation, each construct was transfected into HEK293 cells. *Gaussia* and Firefly luciferase transfected cells produced a robust response as measured by luminescence, human erythropoietin was detectable in the media of cells transfected with erythropoietin circRNA, and EGFP fluorescence was observed from cells transfected with EGFP circRNA. Co-transfection of Cas9 circRNA with sgRNA directed against GFP into cells constitutively expressing

369

GFP resulted in ablated fluorescence in up to 97% of cells in comparison to an sgRNA-only control.

Example 2B: Use of CVB3 IRES Increases Protein Production

Constructs with internal and external duplex regions and differing IRES containing either *Gaussia luciferase* or Firefly luciferase expression sequences were made. Protein production was measured by luminescence in the supernatant of HEK293 cells 24 hours after transfection. The Coxsackievirus B3 (CVB3) IRES construct produced the most protein in both cases.

Example 2C: Use of polyA or polyAC Spacers Increases Protein Production

Thirty nucleotide long polyA or polyAC spacers were added between the IRES and splice junction in a construct with each IRES that produced protein in example 2B. *Gaussia luciferase* activity was measured by luminescence in the supernatant of HEK293 cells 24 hours after transfection. Both spacers improved expression in every construct over control constructs without spacers.

Example 3

HEK293 or HeL Cells Transfected with Circular RNA Produce More Protein than Those Transfected with Comparable Unmodified or Modified Linear RNA.

HPLC-purified *Gaussia luciferase*-coding circRNA (CVB3-GLuc-pAC) was compared with a canonical unmodified 5' methylguanosine-capped and 3' polyA-tailed linear GLuc mRNA, and a commercially available nucleoside-modified (pseudouridine, 5-methylcytosine) linear GLuc mRNA (from Trilink). Luminescence was measured 24 h post-transfection, revealing that circRNA produced 811.2% more protein than the unmodified linear mRNA in HEK293 cells and 54.5% more protein than the modified mRNA. Similar results were obtained in HeLa cells and a comparison of optimized circRNA coding for human erythropoietin with linear mRNA modified with 5-methoxyuridine.

Luminescence data was collected over 6 days. In HEK293 cells, circRNA transfection resulted in a protein production half-life of 80 hours, in comparison with the 43 hours of unmodified linear mRNA and 45 hours of modified linear mRNA. In HeLa cells, circRNA transfection resulted in a protein production half-life of 116 hours, in comparison with the 44 hours of unmodified linear mRNA and 49 hours of modified linear mRNA. CircRNA produced substantially more protein than both the unmodified and modified linear mRNAs over its lifetime in both cell types.

Example 4

Example 4A: Purification of circRNA by RNase Digestion, HPLC Purification, and Phosphatase Treatment Decreases Immunogenicity. Completely Purified Circular RNA is Significantly Less Immunogenic than Unpurified or Partially Purified Circular RNA. Protein Expression Stability and Cell Viability are Dependent on Cell Type and Circular RNA Purity

Human embryonic kidney 293 (HEK293) and human lung carcinoma A549 cells were transfected with:

370

- a. products of an unpurified GLuc circular RNA splicing reaction,
- b. products of RNase R digestion of the splicing reaction,
- c. products of RNase R digestion and HPLC purification of the splicing reaction, or
- d. products of RNase digestion, HPLC purification, and phosphatase treatment of the splicing reaction.

RNase R digestion of splicing reactions was insufficient to prevent cytokine release in A549 cells in comparison to untransfected controls.

The addition of HPLC purification was also insufficient to prevent cytokine release, although there was a significant reduction in interleukin-6 (IL-6) and a significant increase in interferon- α 1 (IFN α 1) compared to the unpurified splicing reaction.

The addition of a phosphatase treatment after HPLC purification and before RNase R digestion dramatically reduced the expression of all upregulated cytokines assessed in A549 cells. Secreted monocyte chemoattractant protein 1 (MCP1), IL-6, IFN α 1, tumor necrosis factor α (TNF α), and IFN γ inducible protein-10 (IP-10) fell to undetectable or untransfected baseline levels.

There was no substantial cytokine release in HEK293 cells. A549 cells had increased GLuc expression stability and cell viability when transfected with higher purity circular RNA. Completely purified circular RNA had a stability phenotype similar to that of transfected 293 cells.

Example 4B: Circular RNA does not Cause Significant Immunogenicity and is not a RIG-I Ligand

A549 cells were transfected with:

- a. unpurified circular RNA,
- b. high molecular weight (linear and circular concatenations) RNA,
- c. circular (nicked) RNA,
- d. an early fraction of purified circular RNA (more overlap with nicked RNA peak),
- e. a late fraction of purified circular RNA (less overlap with nicked RNA peak),
- f. introns excised during circularization, or
- g. vehicle (i.e. untransfected control).

Precursor RNA was separately synthesized and purified in the form of the splice site deletion mutant (DS) due to difficulties in obtaining suitably pure linear precursor RNA from the splicing reaction. Cytokine release and cell viability was measured in each case.

Robust IL-6, RANTES, and IP-10 release was observed in response to most of the species present within the splicing reaction, as well as precursor RNA. Early circRNA fractions elicited cytokine responses comparable to other non-circRNA fractions, indicating that even relatively small quantities of linear RNA contaminants are able to induce a substantial cellular immune response in A549 cells. Late circRNA fractions elicited no cytokine response in excess of that from untransfected controls. A549 cell viability 36 hours post-transfection was significantly greater for late circRNA fractions compared with all of the other fractions.

RIG-I and IFN- β 1 transcript induction upon transfection of A549 cells with late circRNA HPLC fractions, precursor RNA or unpurified splicing reactions were analyzed. Induction of both RIG-I and IFN- β 1 transcripts were weaker for late circRNA fractions than precursor RNA and unpurified splicing reactions. RNase R treatment of splicing reactions alone was not sufficient to ablate this effect. Addition of very small quantities of the RIG-I ligand 3p-hpRNA to circular

371

RNA induced substantial RIG-I transcription. In HeLa cells, transfection of RNase R-digested splicing reactions induced RIG-I and IFN- β 1, but purified circRNA did not. Overall, HeLa cells were less sensitive to contaminating RNA species than A549 cells.

A time course experiment monitoring RIG-I, IFN- β 2, IL-6, and RANTES transcript induction within the first 8 hours after transfection of A549 cells with splicing reactions or fully purified circRNA did not reveal a transient response to circRNA. Purified circRNA similarly failed to induce pro-inflammatory transcripts in RAW264.7 murine macrophages.

A549 cells were transfected with purified circRNA containing an EMCV IRES and EGFP expression sequence. This failed to produce substantial induction of pro-inflammatory transcripts. These data demonstrate that non-circular components of the splicing reaction are responsible for the immunogenicity observed in previous studies and that circRNA is not a natural ligand for RIG-I.

Example 5

Circular RNA Avoids Detection by TLRs.

TLR 3, 7, and 8 reporter cell lines were transfected with multiple linear or circular RNA constructs and secreted embryonic alkaline phosphatase (SEAP) was measured.

Linearized RNA was constructed by deleting the intron and homology arm sequences.

The linear RNA constructs were then treated with phosphatase (in the case of capped RNAs, after capping) and purified by HPLC.

None of the attempted transfections produced a response in TLR7 reporter cells. TLR3 and TLR8 reporter cells were activated by capped linearized RNA, polyadenylated linearized RNA, the nicked circRNA HPLC fraction, and the early circRNA fraction. The late circRNA fraction and m1 Ψ -mRNA did not provoke TLR-mediated response in any cell line.

In a second experiment, circRNA was linearized using two methods: treatment of circRNA with heat in the presence of magnesium ions and DNA oligonucleotide-guided RNase H digestion. Both methods yielded a majority of full-length linear RNA with small amounts of intact circRNA. TLR3, 7, and 8 reporter cells were transfected with circular RNA, circular RNA degraded by heat, or circular RNA degraded by RNase H, and SEAP secretion was measured 36 hours after transfection. TLR8 reporter cells secreted SEAP in response to both forms of degraded circular RNA, but did not produce a greater response to circular RNA transfection than mock transfection. No activation was observed in TLR3 and TLR7 reporter cells for degraded or intact conditions, despite the activation of TLR3 by in vitro transcribed linearized RNA.

Example 6

Unmodified Circular RNA Produces Increased Sustained in Vivo Protein Expression than Linear RNA.

Mice were injected and HEK293 cells were transfected with unmodified and m1 Ψ -modified human erythropoietin (hEpo) linear mRNAs and circRNAs. Equimolar transfection of m1 Ψ -mRNA and unmodified circRNA resulted in robust protein expression in HEK293 cells. hEpo linear mRNA and circRNA displayed similar relative protein expression patterns and cell viabilities in comparison to GLuc linear mRNA and circRNA upon equal weight transfection of HEK293 and A549 cells.

372

In mice, hEpo was detected in serum after the injection of hEpo circRNA or linear mRNA into visceral adipose. hEpo detected after the injection of unmodified circRNA decayed more slowly than that from unmodified or m1 Ψ -mRNA and was still present 42 hours post-injection. Serum hEpo rapidly declined upon the injection of unpurified circRNA splicing reactions or unmodified linear mRNA. Injection of unpurified splicing reactions produced a cytokine response detectable in serum that was not observed for the other RNAs, including purified circRNA.

Example 7

Circular RNA can be Effectively Delivered In Vivo or In Vitro Via Lipid Nanoparticles.

Purified circular RNA was formulated into lipid nanoparticles (LNPs) with the ionizable lipidoid cKK-E12 (Dong et al., 2014; Kauffman et al., 2015). The particles formed uniform multilamellar structures with an average size, polydispersity index, and encapsulation efficiency similar to that of particles containing commercially available control linear mRNA modified with 5moU.

Purified hEpo circRNA displayed greater expression than 5moU-mRNA when encapsulated in LNPs and added to HEK293 cells. Expression stability from LNP-RNA in HEK293 cells was similar to that of RNA delivered by transfection reagent, with the exception of a slight delay in decay for both 5moU-mRNA and circRNA. Both unmodified circRNA and 5moU-mRNA failed to activate RIG-I/IFN- β 1 in vitro.

In mice, LNP-RNA was delivered by local injection into visceral adipose tissue or intravenous delivery to the liver. Serum hEpo expression from circRNA was lower but comparable with that from 5moU-mRNA 6 hours after delivery in both cases. Serum hEpo detected after adipose injection of unmodified LNP-circRNA decayed more slowly than that from LNP-5moU-mRNA, with a delay in expression decay present in serum that was similar to that noted in vitro, but serum hEpo after intravenous injection of LNP-circRNA or LNP-5moU-mRNA decayed at approximately the same rate. There was no increase in serum cytokines or local RIG-I, TNF α , or IL-6 transcript induction in any of these cases.

Example 8

Expression and Functional Stability by IRES in HEK293, HepG2, and 1C1C7 Cells.

Constructs including *Anabaena* intron/exon regions, a *Gussia luciferase* expression sequence, and varying IRES were circularized. 100 ng of each circularization reaction was separately transfected into 20,000 HEK293 cells, HepG2 cells, and 1C1C7 cells using Lipofectamine MessengerMax. Luminescence in each supernatant was assessed after 24 hours as a measure of protein expression. In HEK293 cells, constructs including Crohivirus B. Salivirus FHB, Aichi Virus, Salivirus HG-J1, and Enterovirus J IRES produced the most luminescence at 24 hours (FIG. 1A). In HepG2 cells, constructs including Aichi Virus, Salivirus FHB, EMCV-Cf, and CVA3 IRES produced high luminescence at 24 hours (FIG. 1B). In 1C1C7 cells, constructs including Salivirus FHB, Aichi Virus, Salivirus NG-J1, and Salivirus A SZ-1 IRES produced high luminescence at 24 hours (FIG. 1C).

A trend of larger IRES producing greater luminescence at 24 hours was observed. Shorter total sequence length tends to increase circularization efficiency, so selecting a high expression and relatively short IRES may result in an

improved construct. In HEK293 cells, a construct using the Crohivirus B IRES produced the highest luminescence, especially in comparison to other IRES of similar length (FIG. 2A). Expression from IRES constructs in HepG2 and 1C1C7 cells plotted against IRES size are in FIGS. 2B and 2C.

Functional stability of select IRES constructs in HepG2 and 1C1C7 cells were measured over 3 days. Luminescence from secreted *Gaussia luciferase* in supernatant was measured every 24 hours after transfection of 20,000 cells with 100 ng of each circularization reaction, followed by complete media replacement. Salivirus A GUT and Salivirus FHB exhibited the highest functional stability in HepG2 cells, and Salivirus N-J1 and Salivirus FHB produced the most stable expression in 1C1C7 cells (FIGS. 3A and 3B).

Example 9

Expression and Functional Stability by IRES in Jurkat Cells.

2 sets of constructs including *Anabaena* intron/exon regions, a *Gaussia luciferase* expression sequence, and a subset of previously tested IRES were circularized. 60,000 Jurkat cells were electroporated with 1 µg of each circularization reaction. Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 hours after electroporation. A CVB3 IRES construct was included in both sets for comparison between sets and to previously defined IRES efficacy. CVB1 and Salivirus A SZ1 IRES constructs produced the most expression at 24 h. Data can be found in FIGS. 4A and 4B.

Functional stability of the IRES constructs in each round of electroporated Jurkat cells was measured over 3 days. Luminescence from secreted *Gaussia luciferase* in supernatant was measured every 24 hours after electroporation of 60,000 cells with 1 µg of each circularization reaction, followed by complete media replacement (FIGS. 5A and 5B).

Salivirus A SZ1 and Salivirus A BN2 IRES constructs had high functional stability compared to other constructs.

Example 10

Expression, Functional Stability, and Cytokine Release of Circular and Linear RNA in Jurkat Cells.

A construct including *Anabaena* intron/exon regions, a *Gaussia luciferase* expression sequence, and a Salivirus FHB IRES was circularized. mRNA including a *Gaussia luciferase* expression sequence and a ~150 nt polyA tail, and modified to replace 100% of uridine with 5-methoxy uridine (5moU) is commercially available and was purchased from Trilink. 5moU nucleotide modifications have been shown to improve mRNA stability and expression (Bioconjug Chem. 2016 Mar. 16; 27(3):849-53). Expression of modified mRNA, circularization reactions (unpure), and circRNA purified by size exclusion HPLC (pure) in Jurkat cells were measured and compared (FIG. 6A). Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 hours after electroporation of 60000 cells with 1 µg of each RNA species.

Luminescence from secreted *Gaussia luciferase* in supernatant was measured every 24 hours after electroporation of 60,000 cells with 1 µg of each RNA species, followed by complete media replacement. A comparison of functional stability data of modified mRNA and circRNA in Jurkat cells over 3 days is in FIG. 6B.

IFN γ (FIG. 7A), IL-6 (FIG. 7B), IL-2 (FIG. 7C), RIG-I (FIG. 7D), IFN- β 1 (FIG. 7E), and TNF α (FIG. 7F) transcript

induction was measured 18 hours after electroporation of 60,000 Jurkat cells with 1 µg of each RNA species described above and 3p-hpRNA (5' triphosphate hairpin RNA, which is a known RIG-I agonist).

Example 11

Expression of Circular and Linear RNA in Monocytes and Macrophages.

A construct including *Anabaena* intron/exon regions, a *Gaussia luciferase* expression sequence, and a Salivirus FHB IRES was circularized. mRNA including a *Gaussia luciferase* expression sequence and a ~150 nt polyA tail, and modified to replace 100% of uridine with 5-methoxy uridine (5moU) was purchased from Trilink. Expression of circular and modified mRNA was measured in human primary monocytes (FIG. 8A) and human primary macrophages (FIG. 8B). Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 hours after electroporation of 60,000 cells with 1 µg of each RNA species. Luminescence was also measured 4 days after electroporation of human primary macrophages with media changes every 24 hours (FIG. 8C). The difference in luminescence was statistically significant in each case ($p < 0.05$).

Example 12

Expression and Functional Stability by IRES in Primary T Cells.

Constructs including *Anabaena* intron/exon regions, a *Gaussia luciferase* expression sequence, and a subset of previously tested IRES were circularized and reaction products were purified by size exclusion HPLC. 150,000 primary human CD3+ T cells were electroporated with 1 µg of each circRNA. Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 hours after electroporation (FIG. 9A). Aichi Virus and CVB3 IRES constructs had the most expression at 24 hours.

Luminescence was also measured every 24 hours after electroporation for 3 days in order to compare functional stability of each construct (FIG. 9B). The construct with a Salivirus A SZ1 IRES was the most stable.

Example 13

Expression and Functional Stability of Circular and Linear RNA in Primary T Cells and PBMCs.

Constructs including *Anabaena* intron/exon regions, a *Gaussia luciferase* expression sequence, and a Salivirus A SZ1 IRES or Salivirus FHB IRES were circularized. mRNA including a *Gaussia luciferase* expression sequence and a ~150 nt polyA tail, and modified to replace 100% of uridine with 5-methoxy uridine (5moU) and was purchased from Trilink. Expression of Salivirus A SZ1 IRES HPLC purified circular and modified mRNA was measured in human primary CD3+ T cells. Expression of Salivirus FHB HPLC purified circular, unpurified circular and modified mRNA was measured in human PBMCs. Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 hours after electroporation of 150,000 cells with 1 µg of each RNA species. Data for primary human T cells is in FIGS. 10A and 10B, and data for PBMCs is in FIG. 10C. The difference in expression between the purified circular RNA and unpurified circular RNA or linear RNA was significant in each case ($p < 0.05$).

Luminescence from secreted *Gaussia luciferase* in primary T cell supernatant was measured every 24 hours after

375

electroporation over 3 days in order to compare construct functional stability. Data is shown in FIG. 10B. The difference in relative luminescence from the day 1 measurement between purified circular RNA and linear RNA was significant at both day 2 and day 3 for primary T cells.

Example 14

Circularization Efficiency by Permutation Site in *Anabaena* Intron.

RNA constructs including a CVB3 IRES, a *Gaussia luciferase* expression sequence, *Anabaena* intron/exon regions, spacers, internal duplex regions, and homology arms were produced. Circularization efficiency of constructs using the traditional *Anabaena* intron permutation site and 5 consecutive permutations sites in P9 was measured by HPLC. HPLC chromatograms for the 5 consecutive permutation sites in P9 are shown in FIG. 11A.

Circularization efficiency was measured at a variety of permutation sites. Circularization efficiency is defined as the area under the HPLC chromatogram curve for each of: circRNA/(circRNA+precursor RNA). Ranked quantification of circularization efficiency at each permutation site is in FIG. 11B. 3 permutation sites (indicated in FIG. 11B) were selected for further investigation.

Circular RNA in this example was circularized by in vitro transcription (IVT) then purified via spin column. Circularization efficiency for all constructs would likely be higher if the additional step of incubation with Mg^{2+} and guanosine nucleotide were included; however, removing this step allowed for comparison between, and optimization of, circular RNA constructs. This level of optimization is especially useful for maintaining high circularization efficiency with large RNA constructs, such as those encoding chimeric antigen receptors.

Example 15

Circularization Efficiency of Alternative Introns.

Precursor RNA containing a permuted group 1 intron of variable species origin or permutation site and several constant elements including: a CVB3 IRES, a *Gaussia luciferase* expression sequence, spacers, internal duplex regions, and homology arms were created. Circularization data can be found in FIG. 12. FIG. 12A shows chromatograms resolving precursor, CircRNA and introns. FIG. 12B provides ranked quantification of circularization efficiency, based on the chromatograms shown in FIG. 12A, as a function of intron construct.

Circular RNA in this example was circularized by in vitro transcription (IVT) then spin column purification. Circularization efficiency for all constructs would likely be higher if the additional step of incubation with Mg^{2+} and guanosine nucleotide were included; however, removing this step allows for comparison between, and optimization of, circular RNA constructs. This level of optimization is especially useful for maintaining high circularization efficiency with large RNA constructs, such as those encoding chimeric antigen receptors.

Example 16

Circularization Efficiency by Homology Arm Presence or Length.

RNA constructs including a CVB3 IRES, a *Gaussia luciferase* expression sequence, *Anabaena* intron/exon regions, spacers, and internal duplex regions were produced.

376

Constructs representing 3 *Anabaena* intron permutation sites were tested with 30 nt, 25% GC homology arms or without homology arms ("NA"). These constructs were allowed to circularize without the step of incubation with Mg^{2+} . Circularization efficiency was measured and compared. Data can be found in FIG. 13. Circularization efficiency was higher for each construct lacking homology arms. FIG. 13A provides ranked quantification of circularization efficiency; FIG. 25 13B provides chromatograms resolving precursor, circRNA and introns.

For each of the 3 permutation sites, constructs were created with 10 nt, 20 nt, and 30 nt arm length and 25%, 50%, and 75% GC. Splicing efficiency of these constructs was measured and compared to constructs without homology arms (FIG. 14). Splicing efficiency is defined as the proportion of free introns relative to the total RNA in the splicing reaction.

FIG. 15 A (left) contains HPLC chromatograms showing the contribution of strong homology arms to improved splicing efficiency. Top left: 75% GC content, 10 nt homology arms. Center left: 75% GC content, 20 nt homology arms. Bottom left: 75% GC content, 30 nt homology arms.

FIG. 15A (right) shows HPLC chromatograms indicating increased splicing efficiency paired with increased nicking, appearing as a shoulder on the circRNA peak. Top right: 75% GC content, 10 nt homology arms. Center right: 75% GC content, 20 nt homology arms. Bottom right: 75% GC content, 30 nt homology arms.

FIG. 15B (left) shows select combinations of permutation sites and homology arms hypothesized to demonstrate improved circularization efficiency.

FIG. 15B (right) shows select combinations of permutation sites and homology arms hypothesized to demonstrate improved circularization efficiency, treated with *E coli* polyA polymerase.

Circular RNA in this example was circularized by in vitro transcription (IVT) then spin-column purified. Circularization efficiency for all constructs would likely be higher if an additional Mg^{2+} incubation step with guanosine nucleotide were included; however, removing this step allowed for comparison between, and optimization of, circular RNA constructs. This level of optimization is especially useful for maintaining high circularization efficiency with large RNA constructs, such as those encoding chimeric antigen receptors.

Example 17

Circular RNA Encoding Chimeric Antigen Receptors.

Constructs including *Anabaena* intron/exon regions, a Kymriah chimeric antigen receptor (CAR) expression sequence, and a CVB3 IRES were circularized. 100,000 human primary CD3+ T cells were electroporated with 500 ng of circRNA and co-cultured for 24 hours with Raji cells stably expressing GFP and firefly luciferase. Effector to target ratio (E:T ratio) 0.75:1. 100,000 human primary CD3+ T cells were mock electroporated and co-cultured as a control (FIG. 16).

Sets of 100,000 human primary CD3+ T cells were mock electroporated or electroporated with 1 μ g of circRNA then co-cultured for 48 hours with Raji cells stably expressing GFP and firefly luciferase. E:T ratio 10:1 (FIG. 17).

Quantification of specific lysis of Raji target cells was determined by detection of firefly luminescence (FIG. 18). 100,000 human primary CD3+ T cells either mock electroporated or electroporated with circRNA encoding different CAR sequences were co-cultured for 48 hours with Raji

377

cells stably expressing GFP and firefly luciferase. % Specific lysis defined as 1-[CAR condition luminescence]/[mock condition luminescence]. E:T ratio 10:1.

Example 18

Expression and Functional Stability of Circular and Linear RNA in Jurkat Cells and Resting Human T Cells.

Constructs including *Anabaena* intron/exon regions, a *Gaussia luciferase* expression sequence, and a subset of previously tested IRES were circularized and reaction products were purified by size exclusion HPLC. 150,000 Jurkat cells were electroporated with 1 µg of circular RNA or 5moU-mRNA. Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 hours after electroporation (FIG. 19A left). 150,000 resting primary human CD3+T cells (10 days post-stimulation) were electroporated with 1 µg of circular RNA or 5moU-mRNA. Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 hours after electroporation (FIG. 19A right).

Luminescence from secreted *Gaussia luciferase* in supernatant was measured every 24 hours after electroporation, followed by complete media replacement. Functional stability data is shown in FIG. 19B. Circular RNA had more functional stability than linear RNA in each case, with a more pronounced difference in Jurkat cells.

Example 19

IFN- γ 1, RIG-I, IL-2, IL-6, IFN γ , and TNF α transcript induction of cells electroporated with linear RNA or varying circular RNA constructs.

Constructs including *Anabaena* intron/exon regions, a *Gaussia luciferase* expression sequence, and a subset of previously tested IRES were circularized and reaction products were purified by size exclusion HPLC. 150,000 CD3+ human T cells were electroporated with 1 µg of circular RNA, 5moU-mRNA, or immunostimulatory positive control poly inosine:cytosine. IFN- β 1 (FIG. 20A), RIG-I (FIG. 20B), IL-2 (FIG. 20C), IL-6 (FIG. 20D), IFN- γ (FIG. 20E), and TNF- α (FIG. 20F) transcript induction was measured 18 hours after electroporation.

Example 20

Specific Lysis of Target Cells and IFN γ Transcript Induction by CAR Expressing Cells Electroporated with Different Amounts of Circular or Linear RNA: Specific Lysis of Target and Non-Target Cells by CAR Expressing Cells at Different E:T Ratios.

Constructs including *Anabaena* intron/exon regions, an anti-CD19 CAR expression sequence, and a CVB3 IRES were circularized and reaction products were purified by size exclusion HPLC. 150,000 human primary CD3+ T cells either mock electroporated or electroporated with different quantities of circRNA encoding an anti-CD19 CAR sequence were co-cultured for 12 hours with Raji cells stably expressing GFP and firefly luciferase at an E:T ratio of 2:1. Specific lysis of Raji target cells was determined by detection of firefly luminescence (FIG. 21A). % Specific lysis was defined as 1-[CAR condition luminescence]/[mock

378

condition luminescence]. IFN γ transcript induction was measured 24 hours after electroporation (FIG. 21B).

150,000 human primary CD3+ T cells were either mock electroporated or electroporated with 500 ng circRNA or m1 Ψ -mRNA encoding an anti-CD19 CAR sequence, then co-cultured for 24 hours with Raji cells stably expressing firefly luciferase at different E:T ratios. Specific lysis of Raji target cells was determined by detection of firefly luminescence (FIG. 22A). Specific lysis was defined as 1-[CAR condition luminescence]/[mock condition luminescence].

CAR expressing T cells were also co-cultured for 24 hours with Raji or K562 cells stably expressing firefly luciferase at different E:T ratios. Specific lysis of Raji target cells or K562 non-target cells was determined by detection of firefly luminescence (FIG. 22B). % Specific lysis was defined as 1-[CAR condition luminescence]/[mock condition luminescence].

Example 21

Specific Lysis of Target Cells by T Cells Electroporated with Circular RNA or Linear RNA Encoding a CAR.

Constructs including *Anabaena* intron/exon regions, an anti-CD19 CAR expression sequence, and a CVB3 IRES were circularized and reaction products were purified by size exclusion HPLC. Human primary CD3+ T cells were electroporated with 500 ng of circular RNA or an equimolar quantity of m1 Ψ -mRNA, each encoding a CD19-targeted CAR. Raji cells were added to CAR-T cell cultures over 7 days at an E:T ratio of 10:1. % Specific lysis was measured for both constructs at 1, 3, 5, and 7 days (FIG. 23).

Example 22

Specific Lysis of Raji Cells by T Cells Expressing an Anti-CD19 CAR or an Anti-BCMA CAR.

Constructs including *Anabaena* intron/exon regions, anti-CD19 or anti-BCMA CAR expression sequence, and a CVB3 IRES were circularized and reaction products were purified by size exclusion HPLC. 150,000 primary human CD3+ T cells were electroporated with 500 ng of circRNA, then were co-cultured with Raji cells at an E:T ratio of 2:1. % Specific lysis was measured 12 hours after electroporation (FIG. 24).

Example 23

Expression, Functional Stability, and Cytokine Transcript Induction of Circular and Linear RNA Expressing Antigens.

Constructs including one or more antigen expression sequences are circularized and reaction products are purified by size exclusion HPLC. Antigen presenting cells are electroporated with circular RNA or mRNA.

In vitro antigen production is measured via ELISA. Optionally, antigen production is measured every 24 hours after electroporation. Cytokine transcript induction or release is measured 18 hours after electroporation of antigen presenting cells with circular or linear RNA encoding antigens. The tested cytokines may include IFN- β 1, RIG-I, IL-2, IL-6, IFN γ , RANTES, and TNF α .

379

In vitro antigen production and cytokine induction are measured using purified circRNA, purified circRNA plus antisense circRNA, and unpurified circRNA in order to find the ratio that best preserves expression and immune stimulation.

Example 24

In Vivo Antigen and Antibody Expression in Animal Models.

To assess the ability of antigen encoding circRNAs to facilitate antigen expression and antibody production in vivo, escalating doses of RNA encoding one or more antigens is introduced into mice via intramuscular injection.

Mice are injected once, blood collected after 28 days, then injected again, with blood collected 14 days thereafter. Neutralizing antibodies against antigen of interest is measured via ELISA.

Example 25

Protection Against Infection.

To assess the ability of antigen encoding circRNAs to protect against or cure an infection, RNA encoding one or

more antigens of a virus (such as influenza) is introduced into mice via intramuscular injection.

Mice receive an initial injection and boost injections of circRNA encoding one or more antigens. Protection from a virus such as influenza is determined by weight loss and mortality over 2 weeks.

Example 26

Example 26A: Synthesis of Compounds

Synthesis of representative ionizable lipids of the invention are described in PCT applications PCT/US2016/

380

applications with publication numbers 20190314524, 20190321489, and 20190314284, the contents of each of which are incorporated herein by reference in their entireties.

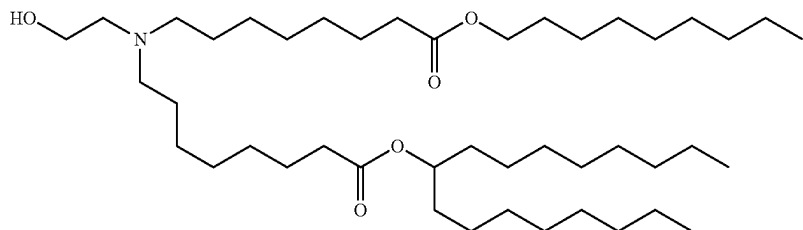
Example 26B: Synthesis of Compounds

Synthesis of representative ionizable lipids of the invention are described in US patent publication number US20170210697A1, the contents of which is incorporated herein by reference in its entirety.

Example 27

Protein Expression by Organ

Circular or linear RNA encoding FLuc was generated and loaded into transfer vehicles with the following formulation: 50% ionizable lipid represented by

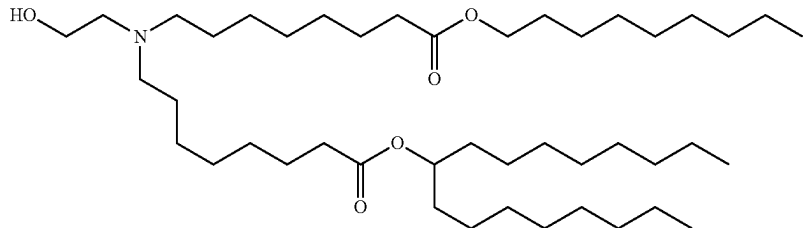


10% DSPC, 1.5% PEG-DMG, 38.5% cholesterol. CD-1 mice were dosed at 0.2 mg/kg and luminescence was measured at 6 hours (live IVIS) and 24 hours (live IVIS and ex vivo IVIS). Total Flux (photons/second over a region of interest) of the liver, spleen, kidney, lung, and heart was measured.

Example 28

Distribution of Expression in the Spleen

Circular or linear RNA encoding GFP is generated and loaded into transfer vehicles with the following formulation: 50% ionizable Lipid represented by



052352, PCT/US2016/068300, PCT/US2010/061058, PCT/US2018/058555, PCT/US2018/053569, PCT/US2017/028981, PCT/US2019/025246, PCT/US2018/035419, PCT/US2019/015913, PCT/US2020/063494, and US

10% DSPC, 1.5% PEG-DMG, 38.5% cholesterol. The formulation is administered to CD-1 mice. Flow cytometry is run on spleen cells to determine the distribution of expression across cell types.

Example 29A: Production of Nanoparticle Compositions

In order to investigate safe and efficacious nanoparticle compositions for use in the delivery of circular RNA to cells, a range of formulations are prepared and tested. Specifically, the particular elements and ratios thereof in the lipid component of nanoparticle compositions are optimized.

Nanoparticles can be made in a 1 fluid stream or with mixing processes such as microfluidics and T-junction mixing of two fluid streams, one of which contains the circular RNA and the other has the lipid components.

Lipid compositions are prepared by combining an ionizable lipid, optionally a helper lipid (such as DOPE, DSPC, or oleic acid obtainable from Avanti Polar Lipids, Alabaster, AL), a PEG lipid (such as 1,2-dimyristoyl-sn-glycerol methoxypolyethylene glycol, also known as PEG-DMG, obtainable from Avanti Polar Lipids, Alabaster, AL), and a structural lipid such as cholesterol at concentrations of about, e.g., 40 or 50 mM in a solvent, e.g., ethanol. Solutions should be refrigerated for storage at, for example, -20° C. Lipids are combined to yield desired molar ratios (see, for example, Tables 17a and 17b below) and diluted with water and ethanol to a final lipid concentration of e.g., between about 5.5 mM and about 25 mM.

TABLE 17a

Formulation number	Description
1	Aliquots of 50 mg/mL ethanolic solutions of C12-200, DOPE, Chol and DMG-PEG2K (40:30:25:5) are mixed and diluted with ethanol to 3 mL final volume. Separately, an aqueous buffered solution (10 mM citrate/150 mM NaCl, pH 4.5) of circRNA is prepared from a 1 mg/mL stock. The lipid solution is injected rapidly into the aqueous circRNA solution and shaken to yield a final suspension in 20% ethanol. The resulting nanoparticle suspension is filtered, diafiltrated with 1 × PBS (pH 7.4), concentrated and stored at 2-8° C.
2	Aliquots of 50 mg/mL ethanolic solutions of DODAP, DOPE, cholesterol and DMG-PEG2K (18:56:20:6) are mixed and diluted with ethanol to 3 mL final volume. Separately, an aqueous buffered solution (10 mM citrate/150 mM NaCl, pH 4.5) of EPO circRNA is prepared from a 1 mg/mL stock. The lipid solution is injected rapidly into the aqueous circRNA solution and shaken to yield a final suspension in 20% ethanol. The resulting nanoparticle suspension is filtered, diafiltrated with 1 × PBS (pH 7.4), concentrated and stored at 2-8° C. Final concentration = 1.35 mg/mL EPO circRNA (encapsulated). Zave = 75.9 nm (Dv(50) = 57.3 nm; Dv(90) = 92.1 nm).
3	Aliquots of 50 mg/mL ethanolic solutions of HGT4003, DOPE, cholesterol and DMG-PEG2K (50:25:20:5) are mixed and diluted with ethanol to 3 mL final volume. Separately, an aqueous buffered solution (10 mM citrate/150 mM NaCl, pH 4.5) of circRNA is prepared from a 1 mg/mL stock. The lipid solution is injected rapidly into the aqueous circRNA solution and shaken to yield a final suspension in 20% ethanol. The resulting nanoparticle suspension is filtered, diafiltrated with 1 × PBS (pH 7.4), concentrated and stored at 2-8° C.
4	Aliquots of 50 mg/mL ethanolic solutions of ICE, DOPE and DMG-PEG2K (70:25:5) are mixed and diluted with ethanol to 3 mL final volume. Separately, an aqueous buffered solution (10 mM citrate/150 mM NaCl, pH 4.5) of circRNA is prepared from a 1 mg/mL stock. The lipid solution is injected rapidly into the aqueous circRNA solution and shaken to yield a final suspension in 20% ethanol. The resulting nanoparticle suspension is filtered, diafiltrated with 1 × PBS (pH 7.4), concentrated and stored at 2-8° C.

TABLE 17a-continued

Formulation number	Description
5	Aliquots of 50 mg/mL ethanolic solutions of HGT5000, DOPE, cholesterol and DMG-PEG2K (40:20:35:5) are mixed and diluted with ethanol to 3 mL final volume. Separately, an aqueous buffered solution (10 mM citrate/150 mM NaCl, pH 4.5) of EPO circRNA is prepared from a 1 mg/mL stock. The lipid solution is injected rapidly into the aqueous circRNA solution and shaken to yield a final suspension in 20% ethanol. The resulting nanoparticle suspension is filtered, diafiltrated with 1 × PBS (pH 7.4), concentrated and stored at 2-8° C. Final concentration = 1.82 mg/mL EPO mRNA (encapsulated). Zave = 105.6 nm (Dv(50) = 53.7 nm; Dv(90) = 157 nm).
6	Aliquots of 50 mg/mL ethanolic solutions of HGT5001, DOPE, cholesterol and DMG-PEG2K (40:20:35:5) are mixed and diluted with ethanol to 3 mL final volume. Separately, an aqueous buffered solution (10 mM citrate/150 mM NaCl, pH 4.5) of EPO circRNA is prepared from a 1 mg/mL stock. The lipid solution is injected rapidly into the aqueous circRNA solution and shaken to yield a final suspension in 20% ethanol. The resulting nanoparticle suspension is filtered, diafiltrated with 1 × PBS (pH 7.4), concentrated and stored at 2-8° C.
7	Aliquots of 50 mg/mL ethanolic solutions of HGT5001, DOPE, cholesterol and DMG-PEG2K (35:16:46.5:2.5) are mixed and diluted with ethanol to 3 mL final volume. Separately, an aqueous buffered solution (10 mM citrate/150 mM NaCl, pH 4.5) of EPO circRNA is prepared from a 1 mg/mL stock. The lipid solution is injected rapidly into the aqueous circRNA solution and shaken to yield a final suspension in 20% ethanol. The resulting nanoparticle suspension is filtered, diafiltrated with 1 × PBS (pH 7.4), concentrated and stored at 2-8° C.
8	Aliquots of 50 mg/mL ethanolic solutions of HGT5001, DOPE, cholesterol and DMG-PEG2K (40:10:40:10) are mixed and diluted with ethanol to 3 mL final volume. Separately, an aqueous buffered solution (10 mM citrate/150 mM NaCl, pH 4.5) of EPO circRNA is prepared from a 1 mg/mL stock. The lipid solution is injected rapidly into the aqueous circRNA solution and shaken to yield a final suspension in 20% ethanol. The resulting nanoparticle suspension is filtered, diafiltrated with 1 × PBS (pH 7.4), concentrated and stored at 2-8° C.

In some embodiments, transfer vehicle has a formulation as described in Table 17a.

TABLE 17b

Composition (mol %)	Components
40:20:38.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
45:15:38.5:1.5	Compound:Phospholipid:Phytosterol*:PRO-DMG
50:10:38.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
55:5:38.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
60:5:33.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
45:20:33.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
50:20:28.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
55:20:23.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
60:20:18.5:1.5	Compound:Phospholipid:Phytosterol*:PBG-DMG
40:15:43.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
50:15:33.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
55:15:28.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
60:15:23.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
40:10:48.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
45:10:43.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
55:10:33.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
60:10:28.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
40:5:53.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
45:5:48.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
50:5:43.5:1.5	Compound:Phospholipid:Phytosterol*:PEG-DMG
40:20:40:0	Compound:Phospholipid:Phytosterol*:PEG-DMG
45:20:35:0	Compound:Phospholipid:Phytosterol*:PEG-DMG
50:20:30:0	Compound:Phospholipid:Phytosterol*:PEG-DMG

TABLE 17b-continued

Composition (mol %)	Components
55:20:25:0	Compound:Phospholipid:Phytosterol*:PEG-DMG
60:20:20:0	Compound:Phospholipid:Phytosterol*:PEG-DMG
40:15:45:0	Compound:Phospholipid:Phytosterol*:PEG-DMG

In some embodiments, transfer vehicle has a formulation as described in Table 17b.

For nanoparticle compositions including circRNA, solutions of the circRNA at concentrations of 0.1 mg/ml in deionized water are diluted in a buffer, e.g., 50 mM sodium citrate buffer at a pH between 3 and 4 to form a stock solution. Alternatively, solutions of the circRNA at concentrations of 0.15 mg/ml in deionized water are diluted in a buffer, e.g., 6.25 mM sodium acetate buffer at a pH between 3 and 4.5 to form a stock solution.

Nanoparticle compositions including a circular RNA and a lipid component are prepared by combining the lipid solution with a solution including the circular RNA at lipid component to circRNA wt:wt ratios between about 5:1 and about 50:1. The lipid solution is rapidly injected using, e.g., a NanoAssemblr microfluidic based system at flow rates between about 10 ml/min and about 18 ml/min or between about 5 ml/min and about 18 ml/min into the circRNA solution, to produce a suspension with a water to ethanol ratio between about 1:1 and about 4:1.

Nanoparticle compositions can be processed by dialysis to remove ethanol and achieve buffer exchange. Formulations are dialyzed twice against phosphate buffered saline (PBS), pH 7.4, at volumes 200 times that of the primary product using Slide-A-Lyzer cassettes (Thermo Fisher Scientific Inc., Rockford, IL) with a molecular weight cutoff of 10 kDa or 20 kDa. The formulations are then dialyzed overnight at 4° C. The resulting nanoparticle suspension is filtered through 0.2 µm sterile filters (Sarstedt, Numbrecht, Germany) into glass vials and sealed with crimp closures. Nanoparticle composition solutions of 0.01 mg/ml to 0.15 mg/ml are generally obtained.

The method described above induces nano-precipitation and particle formation.

Alternative processes including, but not limited to, T-junction and direct injection, may be used to achieve the same nano-precipitation. B. Characterization of nanoparticle compositions

A Zetasizer Nano ZS (Malvern Instruments Ltd, Malvern, Worcestershire, UK) can be used to determine the particle size, the polydispersity index (PDI) and the zeta potential of the nanoparticle compositions in 1×PBS in determining particle size and 15 mM PBS in determining zeta potential.

Ultraviolet-visible spectroscopy can be used to determine the concentration of circRNA in nanoparticle compositions. 100 µL of the diluted formulation in 1-PBS is added to 900 µL of a 4:1 (v/v) mixture of methanol and chloroform. After mixing, the absorbance spectrum of the solution is recorded, for example, between 230 nm and 330 nm on a DU 800 spectrophotometer (Beckman Coulter, Beckman Coulter, Inc., Brea, CA). The concentration of circRNA in the nanoparticle composition can be calculated based on the extinction coefficient of the circRNA used in the composition and on the difference between the absorbance at a wavelength of, for example, 260 nm and the baseline value at a wavelength of, for example, 330 nm.

A QUANT-IT™ RIBOGREEN®RNA assay (Invitrogen Corporation Carlsbad, CA) can be used to evaluate the

encapsulation of circRNA by the nanoparticle composition. The samples are diluted to a concentration of approximately 5 µg/mL or 1 µg/mL in a TE buffer solution (10 mM Tris-HCl, 1 mM EDTA, pH 7.5). 50 µL of the diluted samples are transferred to a polystyrene 96 well plate and either 50 µL of TE buffer or 50 µL of a 2-4% Triton X-100 solution is added to the wells. The plate is incubated at a temperature of 37° C. for 15 minutes. The RIBOGREEN® reagent is diluted 1:100 or 1:200 in TE buffer, and 100 µL of this solution is added to each well. The fluorescence intensity can be measured using a fluorescence plate reader (Wallac Victor 1420 Multilabel Counter; Perkin Elmer, Waltham, MA) at an excitation wavelength of, for example, about 480 nm and an emission wavelength of, for example, about 520 nm. The fluorescence values of the reagent blank are subtracted from that of each of the samples and the percentage of free circRNA is determined by dividing the fluorescence intensity of the intact sample (without addition of Triton X-100) by the fluorescence value of the disrupted sample (caused by the addition of Triton X-100).

Example 29B: In Vivo Formulation Studies

In order to monitor how effectively various nanoparticle compositions deliver circRNA to targeted cells, different nanoparticle compositions including circRNA are prepared and administered to rodent populations. Mice are intravenously, intramuscularly, intraarterially, or intratumorally administered a single dose including a nanoparticle composition with a lipid nanoparticle formulation. In some instances, mice may be made to inhale doses. Dose sizes may range from 0.001 mg/kg to 10 mg/kg, where 10 mg/kg describes a dose including 10 mg of a circRNA in a nanoparticle composition for each 1 kg of body mass of the mouse. A control composition including PBS may also be employed.

Upon administration of nanoparticle compositions to mice, dose delivery profiles, dose responses, and toxicity of particular formulations and doses thereof can be measured by enzyme-linked immunosorbent assays (ELISA), bioluminescent imaging, or other methods. Time courses of protein expression can also be evaluated. Samples collected from the rodents for evaluation may include blood and tissue (for example, muscle tissue from the site of an intramuscular injection and internal tissue): sample collection may involve sacrifice of the animals.

Higher levels of protein expression induced by administration of a composition including a circRNA will be indicative of higher circRNA translation and/or nanoparticle composition circRNA delivery efficiencies. As the non-RNA components are not thought to affect translational machineries themselves, a higher level of protein expression is likely indicative of a higher efficiency of delivery of the circRNA by a given nanoparticle composition relative to other nanoparticle compositions or the absence thereof.

Example 30

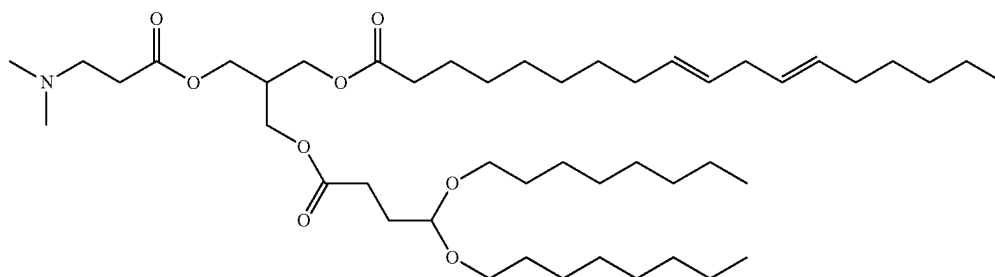
Characterization of Nanoparticle Compositions

A Zetasizer Nano ZS (Malvern Instruments Ltd, Malvern, Worcestershire, UK) can be used to determine the particle size, the polydispersity index (PDI) and the zeta potential of the transfer vehicle compositions in 1×PBS in determining particle size and 15 mM PBS in determining zeta potential.

Ultraviolet-visible spectroscopy can be used to determine the concentration of a therapeutic and/or prophylactic (e.g., RNA) in transfer vehicle compositions. 100 µL of the diluted

385

formulation in 1×PBS is added to 900 μL of a 4:1 (v/v) mixture of methanol and chloroform. After mixing, the absorbance spectrum of the solution is recorded, for example, between 230 nm and 330 nm on a DU 800 spectrophotometer (Beckman Coulter, Beckman Coulter, Inc., Brea, CA). The concentration of therapeutic and/or prophylactic in the transfer vehicle composition can be calculated based on the extinction coefficient of the therapeutic and/or prophylactic used in the composition and on the difference between the absorbance at a wavelength of, for example, 260 nm and the baseline value at a wavelength of, for example, 330 nm.



RV88

386

of SM(PEG)24 and centrifuged to remove unreacted cross-linker. Activated transfer vehicles are then incubated with an excess of reduced anti-T cell antigen antibody. Unbound antibody is removed using a centrifugal filtration device.

Example 32

RNA Containing Transfer Vehicle Using RV88.

In this example RNA containing transfer vehicles are synthesized using the 2-D vortex microfluidic chip with the cationic lipid RV88 for delivery of circRNA.

For transfer vehicle compositions including RNA, a QUANT-IT™ RIBOGREEN®, RNA assay (Invitrogen Corporation Carlsbad, CA) can be used to evaluate the encapsulation of RNA by the transfer vehicle composition. The samples are diluted to a concentration of approximately 5 μg/mL or 1 μg/mL in a TE buffer solution (10 mM Tris-HCl, 1 mM EDTA, pH 7.5). 50 μL of the diluted samples are transferred to a polystyrene 96 well plate and either 50 μL of TE buffer or 50 μL of a 2-4% Triton X-100 solution is added to the wells. The plate is incubated at a temperature of 37° C. for 15 minutes. The RIBOGREEN® reagent is diluted 1:100 or 1:200 in TE buffer, and 100 μL of this solution is added to each well. The fluorescence intensity can be measured using a fluorescence plate reader (Wallac Victor 1420 Multilabel Counter; Perkin Elmer, Waltham, MA) at an excitation wavelength of, for example, about 480 nm and an emission wavelength of, for example, about 520 nm. The fluorescence values of the reagent blank are subtracted from that of each of the samples and the percentage of free RNA is determined by dividing the fluorescence intensity of the intact sample (without addition of Triton X-100) by the fluorescence value of the disrupted sample (caused by the addition of Triton X-100).

Example 31

T Cell Targeting

To target transfer vehicles to T-cells, T cell antigen binders, e.g., anti-CD8 antibodies, are coupled to the surface of the transfer vehicle. Anti-T cell antigen antibodies are mildly reduced with an excess of DTT in the presence of EDTA in PBS to expose free hinge region thiols. To remove DTT, antibodies are passed through a desalting column. The heterobifunctional cross-linker SM(PEG)24 is used to anchor antibodies to the surface of circRNA-loaded transfer vehicles (Amine groups are present in the head groups of PEG lipids, free thiol groups on antibodies were created by DTT, SM(PEG)24 cross-links between amines and thiol groups). Transfer vehicles are first incubated with an excess

TABLE 18a

Materials and Instrument	Vendor	Cat.#
1M Tris-HCl, pH 8.0, Sterile	Teknova	T1080
5M Sodium Chloride solution	Teknova	S0250
QB Citrate buffer, pH 6.0 (100 mM)	Teknova	Q2446
Nuclease-free water	Ambion	AM9937
Triton X-100	Sigma-Aldrich	T8787-100ML
RV88	GVK bio	
DSPC	Lipoid	556500
Cholesterol	Sigma	C3045-5G
PEG2K	Avanti Polar Lipids	880150
Ethanol	Acros Organic	615090010
5 mL Borosilicate glass vials	Thermo Scientific	ST5-20
PD MiniTrap G-25 Desalting Columns	GE Healthcare	VWR Cat. #95055-984
Quant-iT RiboGreen RNA Assay kit	Molecular Probes/Life Technologies	R11490
Black 96-well microplates	Greiner	655900

RV88, DSPC, and cholesterol all being prepared in ethanol at a concentration of 10 mg/ml in borosilica vials. The lipid 14:0-PEG2K PE is prepared at a concentration of 4 mg/ml also in a borosilica glass vial. Dissolution of lipids at stock concentrations is attained by sonication of the lipids in ethanol for 2 min. The solutions are then heated on an orbital tilting shaker set at 170 rpm at 37° C. for 10 min. Vials are then equilibrated at 26° C. for a minimum of 45 min. The lipids are then mixed by adding volumes of stock lipid as shown in Table 18b. The solution is then adjusted with ethanol such that the final lipid concentration was 7.92 mg/ml.

TABLE 18b

Composition	MW	%	nmoles	mg	Stock (mg/ml)	ul	Ethanol (ul)
RV88	794.2	40%	7200	5.72	10	571.8	155.3
DSPC	790.15	10%	1800	1.42	10	142.2	

387

TABLE 18b-continued

Composition	MW	%	nmoles	mg	Stock (mg/ml)	Ethanol (ul)
Cholesterol	386.67	48%	8640	3.34	10	334.1
PEG2K	2693.3	2%	380	0.97	4	242.4

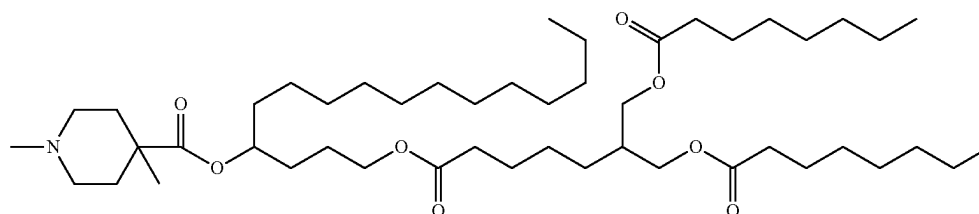
RNA is prepared as a stock solution with 75 mM Citrate buffer at pH 6.0 and a concentration of RNA at 1.250 mg/ml. The concentration of the RNA is then adjusted to 0.1037 mg/ml with 75 mM citrate buffer at pH 6.0, equilibrated to 26° C. The solution is then incubated at 26° C. for a minimum of 25 min.

The microfluidic chamber is cleaned with ethanol and neMYSIS syringe pumps are prepared by loading a syringe with the RNA solution and another syringe with the ethanol lipid. Both syringes are loaded and under the control of neMESYS software. The solutions are then applied to the mixing chip at an aqueous to organic phase ratio of 2 and a total flow rate of 22 ml/min (14.67 ml/min for RNA and 7.33 ml/min for the lipid solution). Both pumps are started synchronously. The mixer solution that flowed from the microfluidic chip is collected in 4x1 ml fractions with the first fraction being discarded as waste. The remaining solution containing the RNA-liposomes is exchanged by using G-25 mini desalting columns to 10 mM Tris-HCl, 1 mM EDTA, at pH 7.5. Following buffer exchange, the materials are characterized for size, and RNA entrapment through DLS analysis and Ribogreen assays, respectively.

Example 33

RNA Containing Transfer Vehicle Using RV94.

In this example, RNA containing liposome are synthesized using the 2-D) vortex microfluidic chip with the cationic lipid RV94 for delivery of circRNA.



RV94

TABLE 19

Materials and Instrument	Vendor	Cat#
1M Tris-HCl, pH 8.0, Sterile	Teknova	T1080
5M Sodium Chloride solution	Teknova	S0250
QB Citrate buffer, pH 6.0 (100 mM)	Teknova	Q2446
Nuclease-free water	Ambion	AM9937
Triton X-100	Sigma-Aldrich	T8787-100ML
RV94	GvKbio	
DSPC	Lipoid	556500
Cholesterol	Sigma	C3045-5G
PEG2K	Avanti Polar Lipids	880150
Ethanol	Acros Organic	615090010
5 mL Borosilicate glass vials	Thermo Scientific	ST5-20

388

TABLE 19-continued

Materials and Instrument	Vendor	Cat#
PD MiniTrap G-25 Desalting Columns	GE Healthcare	VWR Cat. #95055-984
Quant-iT RiboGreen RNA Assay kit	Molecular Probes/Life Technologies	R11490
Black 96-well microplates	Greiner	655900

The lipids were prepared as in Example 29 using the material amounts named in Table 20 to a final lipid concentration of 7.92 mg/ml.

TABLE 20

Composition	MW	%	nmoles	mg	Stock (mg/ml)	Ethanol (ul)
RV94	808.22	40%	2880	2.33	10	232.8
DSPC	790.15	10%	720	0.57	10	56.9
Cholesterol	386.67	48%	3456	1.34	10	133.6
PEG2K	2683.3	2%	144	0.39	4	97.0

The aqueous solution of circRNA is prepared as a stock solution with 75 mM Citrate buffer at pH 6.0 the circRNA at 1 0.250 mg/ml. The concentration of the RNA is then adjusted to 0.1037 mg/ml with 75 mM citrate buffer at pH 6.0, equilibrated to 26° C. The solution is then incubated at 26° C. for a minimum of 25 min.

The microfluidic chamber is cleaned with ethanol and neMYSIS syringe pumps are prepared by loading a syringe with the RNA solution and another syringe with the ethanol lipid. Both syringes are loaded and under the control of neMESYS software. The solutions are then applied to the

mixing chip at an aqueous to organic phase ratio of 2 and a total flow rate of 22 ml/min (14.67 ml/min for RNA and 7.33 ml/min for the lipid solution). Both pumps are started synchronously. The mixer solution that flowed from the microfluidic chip is collected in 4x1 ml fractions with the first fraction being discarded as waste. The remaining solution containing the circRNA-transfer vehicles is exchanged by using G-25 mini desalting columns to 10 mM Tris-HCl, 1 mM EDTA, at pH 7.5, as described above. Following buffer exchange, the materials are characterized for size, and RNA entrapment through DLS analysis and Ribogreen assays, respectively. The biophysical analysis of the liposomes is shown in Table

TABLE 21

Sample	N:P	TFR	Ratio (aqueous/ org phase)	RNA encapsulation amount ($\mu\text{g}/\text{mL}$)	RNA encapsulation yield %	size d · nm	PDI
SAM- RV94	8	22	2	31.46	86.9	113.1	0.12

Example 34

General Protocol for in Line Mixing.

Individual and separate stock solutions are prepared—one containing lipid and the other circRNA. Lipid stock containing a desired lipid or lipid mixture, DSPC, cholesterol and PEG lipid is prepared by solubilized in 90% ethanol. The remaining 10% is low pH citrate buffer. The concentration of the lipid stock is 4 mg/mL. The pH of this citrate buffer can range between pH 3 and pH 5, depending on the type of lipid employed. The circRNA is also solubilized in citrate buffer at a concentration of 4 mg/mL. 5 mL of each stock solution is prepared.

Stock solutions are completely clear and lipids are ensured to be completely solubilized before combining with circRNA. Stock solutions may be heated to completely solubilize the lipids. The circRNAs used in the process may be unmodified or modified oligonucleotides and may be conjugated with lipophilic moieties such as cholesterol.

The individual stocks are combined by pumping each solution to a T-junction. A dual-head Watson-Marlow pump was used to simultaneously control the start and stop of the two streams. A 1.6 mm polypropylene tubing is further downsized to 0.8 mm tubing in order to increase the linear flow rate. The polypropylene line (ID=0.8 mm) are attached to either side of a T-junction. The polypropylene T has a linear edge of 1.6 mm for a resultant volume of 4.1 mm³. Each of the large ends (1.6 mm) of polypropylene line is placed into test tubes containing either solubilized lipid stock or solubilized circRNA. After the T-junction, a single tubing is placed where the combined stream exited. The tubing is then extended into a container with 2× volume of PBS, which is rapidly stirred. The flow rate for the pump is at a setting of 300 rpm or 110 mL/min. Ethanol is removed and exchanged for PBS by dialysis. The lipid formulations are then concentrated using centrifugation or diafiltration to an appropriate working concentration.

C57BL/6 mice (Charles River Labs, MA) receive either saline or formulated circRNA via tail vein injection. At various time points after administration, serum samples are collected by retroorbital bleed. Serum levels of Factor VII protein are determined in samples using a chromogenic assay (Biophen FVTI, Aniera Corporation, OH). To determine liver RNA levels of Factor VII, animals are sacrificed and livers are harvested and snap frozen in liquid nitrogen. Tissue lysates are prepared from the frozen tissues and liver RNA levels of Factor VII are quantified using a branched DNA assay (QuantiGene Assay, Panomics, CA).

FVII activity is evaluated in FVTI siRNA-treated animals at 48 hours after intravenous (bolus) injection in C57BL/6 mice. FVII is measured using a commercially available kit for determining protein levels in serum or tissue, following the manufacturer's instructions at a microplate scale. FVII reduction is determined against untreated control mice, and the results are expressed as % Residual FVII. Two dose

levels (0.05 and 0.005 mg/kg FVII siRNA) are used in the screen of each novel liposome composition.

Example 36

circRNA Formulation Using Preformed Vesicles.

Cationic lipid containing transfer vehicles are made using the preformed vesicle method. Cationic lipid, DSPC, cholesterol and PEG-lipid are solubilized in ethanol at a molar ratio of 40/10/40/10, respectively. The lipid mixture is added to an aqueous buffer (50 mM citrate, pH 4) with mixing to a final ethanol and lipid concentration of 30% (vol/vol) and 6.1 mg/mL respectively and allowed to equilibrate at room temperature for 2 min before extrusion. The hydrated lipids are extruded through two stacked 80 nm pore-sized filters (Nuclepore) at 22° C. using a Lipex Extruder (Northern Lipids, Vancouver, BC) until a vesicle diameter of 70-90 nm, as determined by Nicomp analysis, is obtained. For cationic lipid mixtures which do not form small vesicles, hydrating the lipid mixture with a lower pH buffer (50 mM citrate, pH 3) to protonate the phosphate group on the DSPC headgroup helps form stable 70-90 nm vesicles.

The FVII circRNA (solubilized in a 50 mM citrate, pH 4 aqueous solution containing 30% ethanol) is added to the vesicles, pre-equilibrated to 35° C., at a rate of ~5 mL/min with mixing. After a final target circRNA/lipid ratio of 0.06 (wt wt) is achieved, the mixture is incubated for a further 30 min at 35° C. to allow vesicle re-organization and encapsulation of the FVII RNA. The ethanol is then removed and the external buffer replaced with PBS (155 mM NaCl, 3 mM Na₂HPO₄, 1mM KH₂PO₄, pH 7.5) by either dialysis or tangential flow diafiltration. The final encapsulated circRNA-to-lipid ratio is determined after removal of unencapsulated RNA using size-exclusion spin columns or ion exchange spin columns.

Example 37

Example 37A: Expression of Trispecific Antigen Binding Proteins from Engineered Circular RNA

Circular RNAs are designed to include: (1) a 3' post splicing group I intron fragment; (2) an Internal Ribosome Entry Site (IRES); (3) a trispecific antigen-binding protein coding region; and (4) a 3' duplex region. The trispecific antigen-binding protein regions are constructed to produce an exemplary trispecific antigen-binding protein that will bind to a target antigen, e.g., GPC3.

Example 37B: Generation of a scFv CD3 Binding Domain

The human CD3epsilon chain canonical sequence is Uniprot Accession No. P07766. The human CD3gamma chain canonical sequence is Uniprot Accession No. P09693. The human CD3delta chain canonical sequence is Uniprot

391

Accession No. P043234. Antibodies against CD3epsilon, CD3gamma or CD3delta are generated via known technologies such as affinity maturation. Where murine anti-CD3 antibodies are used as a starting material, humanization of murine anti-CD3 antibodies is desired for the clinical setting, where the mouse-specific residues may induce a human-anti-mouse antigen (HAMA) response in subjects who receive treatment of a trispecific antigen-binding protein described herein. Humanization is accomplished by grafting CDR regions from murine anti-CD3 antibody onto appropriate human germline acceptor frameworks, optionally including other modifications to CDR and/or framework regions.

Human or humanized anti-CD3 antibodies are therefore used to generate scFv sequences for CD3 binding domains of a trispecific antigen-binding protein. DNA sequences coding for human or humanized VL and VH domains are obtained, and the codons for the constructs are, optionally, optimized for expression in cells from *Homo sapiens*. The order in which the VL and VH domains appear in the scFv is varied (i.e. VL-VH, or VH-VL orientation), and three copies of the "G4S" or "G₄S" (SEQ ID NO: 124) subunit (G₄S)₃ (SEQ ID NO: 125) connect the variable domains to create the scFv domain. Anti-CD3 scFv plasmid constructs can have optional Flag, His or other affinity tags, and are electroporated into HEK293 or other suitable human or mammalian cell lines and purified. Validation assays include binding analysis by FACS, kinetic analysis using Proteon, and staining of CD3-expressing cells.

Example 37C: Generation of a scFv Glypican-3 (GPC3) Binding Domain

Glypican-3 (GPC3) is one of the cell surface proteins present on Hepatocellular Carcinoma but not on healthy normal liver tissue. It is frequently observed to be elevated in hepatocellular carcinoma and is associated with poor prognosis for HCC patients. It is known to activate Wnt signalling. GPC3 antibodies have been generated including MDX-1414, HN3, GC33, and YP7.

A scFv binding to GPC-3 or another target antigen is generated similarly to the above method for generation of a scFv binding domain to CD3.

Example 37D: Expression of Trispecific Antigen-Binding Proteins In Vitro

A CHO cell expression system (Flp-In®, Life Technologies), a derivative of CHO-K1 Chinese Hamster ovary cells (ATCC, CCL-61) (Kao and Puck, Proc. Natl. Acad. Sci. USA 1968; 60(4):1275-81), is used. Adherent cells are subcultured according to standard cell culture protocols provided by Life Technologies.

For adaption to growth in suspension, cells are detached from tissue culture flasks and placed in serum-free medium. Suspension-adapted cells are cryopreserved in medium with 10% DMSO.

Recombinant CHO cell lines stably expressing secreted trispecific antigen-binding proteins are generated by transfection of suspension-adapted cells. During selection with the antibiotic Hygromycin B viable cell densities are measured twice a week, and cells are centrifuged and resuspended in fresh selection medium at a maximal density of 0.1×10^6 viable cells/mL. Cell pools stably expressing trispecific antigen-binding proteins are recovered after 2-3 weeks of selection at which point cells are transferred to standard culture medium in shake flasks. Expression of recombinant

392

secreted proteins is confirmed by performing protein gel electrophoresis or flow cytometry. Stable cell pools are cryopreserved in DMSO containing medium.

Trispecific antigen-binding proteins are produced in 10-day fed-batch cultures of stably transfected CHO cell lines by secretion into the cell culture supernatant. Cell culture supernatants are harvested after 10 days at culture viabilities of typically >75%. Samples are collected from the production cultures every other day and cell density and viability are assessed. On day of harvest, cell culture supernatants are cleared by centrifugation and vacuum filtration before further use.

Protein expression titers and product integrity in cell culture supernatants are analyzed by SDS-PAGE.

Example 37E: Purification of Trispecific Antigen-Binding Proteins

Trispecific antigen-binding proteins are purified from CHO cell culture supernatants in a two-step procedure. The constructs are subjected to affinity chromatography in a first step followed by preparative size exclusion chromatography (SEC) on Superdex 200 in a second step. Samples are buffer-exchanged and concentrated by ultrafiltration to a typical concentration of >1 mg/mL. Purity and homogeneity (typically >90%) of final samples are assessed by SDS PAGE under reducing and non-reducing conditions, followed by immunoblotting using an anti-(half-life extension domain) or anti idiotype antibody as well as by analytical SEC, respectively. Purified proteins are stored at aliquots at -80° C. until use.

Example 38

Expression of Engineered Circular RNA with a Half-Life Extension Domain has Improved Pharmacokinetic Parameters than without a Half-Life Extension Domain

The trispecific antigen-binding protein encoded on a circRNA molecule of Example 37 is administered to cynomolgus monkeys as a 0.5 mg/kg bolus injection intramuscularly. Another cynomolgus monkey group receives a comparable protein encoded on a circRNA molecule in size with binding domains to CD3 and GPC-3 but lacking a half-life extension domain. A third and fourth group receive a protein encoded on a circRNA molecule with CD3 and half-life extension domain binding domains and a protein with GPC-3 and half-life extension domains, respectively. Both proteins encoded by circRNA are comparable in size to the trispecific antigen-binding protein. Each test group consists of 5 monkeys. Serum samples are taken at indicated time points, serially diluted, and the concentration of the proteins is determined using a binding ELISA to CD3 and/or GPC-3.

Pharmacokinetic analysis is performed using the test article plasma concentrations. Group mean plasma data for each test article conforms to a multi-exponential profile when plotted against the time post-dosing. The data are fit by a standard two-compartment model with bolus input and first-order rate constants for distribution and elimination phases. The general equation for the best fit of the data for i.v. administration is: $c(t) = Ae^{-\alpha t} + Be^{-\beta t}$, where $c(t)$ is the plasma concentration at time t , A and B are intercepts on the Y-axis, and α and β are the apparent first-order rate constants for the distribution and elimination phases, respectively. The α -phase is the initial phase of the clearance and reflects distribution of the protein into all extracellular fluid of the animal, whereas the second or β -phase portion of the decay curve represents true plasma clearance. Methods for fitting

393

such equations are well known in the art. For example, $A=D/V(a-k_{21})/(a-p)$, $B=DN(p-k_{21})/(a-p)$, and a and β (for $\alpha>\beta$) are roots of the quadratic equation: $r^2+(k_{12}+k_{21}+k_{10})r+k_{21}k_{10}=0$ using estimated parameters of V =volume of distribution, k_{10} =elimination rate, k_{12} =transfer rate from compartment 1 to compartment 2 and k_{21} =transfer rate from compartment 2 to compartment 1, and D =the administered dose.

Data analysis: Graphs of concentration versus time profiles are made using KaleidaGraph (KaleidaGraph™ V. 3.09 Copyright 1986-1997. Synergy Software. Reading, Pa.). Values reported as less than reportable (LTR) are not included in the PK analysis and are not represented graphically. Pharmacokinetic parameters are determined by compartmental analysis using WinNonlin software (WinNonlin Professional V. 3.1 WinNonlin™ Copyright 1998-1999. Pharsight Corporation. Mountain View, Calif). Pharmacokinetic parameters are computed as described in Ritschel W A and Kearns G L, 1999, EST: Handbook Of Basic Pharmacokinetics Including Clinical Applications, 5th edition, American Pharmaceutical Assoc., Washington, D C.

It is expected that the trisppecific antigen-binding protein encoded on a circRNA molecule of Example 37 has improved pharmacokinetic parameters such as an increase in elimination half-time as compared to proteins lacking a half-life extension domain.

Example 39

Cytotoxicity of the Trisppecific Antigen-Binding Protein

The trisppecific antigen-binding protein encoded on a circRNA molecule of Example 37 is evaluated in vitro on its mediation of T cell dependent cytotoxicity to GPC-3+ target cells.

Fluorescence labeled GPC3 target cells are incubated with isolated PBMC of random donors or T-cells as effector cells in the presence of the trisppecific antigen-binding protein of Example 37. After incubation for 4 h at 37° C. in a humidified incubator, the release of the fluorescent dye from the target cells into the supernatant is determined in a spectrofluorimeter. Target cells incubated without the trisppecific antigen-binding protein of Example 37 and target cells totally lysed by the addition of saponin at the end of the incubation serve as negative and positive controls, respectively.

Based on the measured remaining living target cells, the percentage of specific cell lysis is calculated according to the following formula: $[1-(\text{number of living targets}(\text{sample})/\text{number of living targets}(\text{spontaneous}))]\times 100\%$. Sigmoidal dose response curves and EC50 values are calculated by non-linear regression/4-parameter logistic fit using the GraphPad Software. The lysis values obtained for a given antibody concentration are used to calculate sigmoidal dose-response curves by 4 parameter logistic fit analysis using the Prism software.

Example 40

Lipid Nanoparticle Formulation with Circular RNA

Lipid Nanoparticles (LNPs) were formed using a Precision Nanosystems Ignite instrument with a 'NextGen' mixing chamber. Ethanol phase contained ionizable Lipid 10c-7, DSPC, Cholesterol, and DSPE-PEG 2000 (Avanti Polar Lipids Inc.) at a weight ratio of 16:1:4:1 or 62:4:33:1 molar ratio was combined with an aqueous phase containing circular RNA and 25 mM sodium acetate buffer at pH 5.2. A 3:1 aqueous to ethanol mixing ratio was used. The

394

formulated LNP then were dialyzed in 1 L of water and exchanged 2 times over 18 hours. Dialyzed LNPs were filtered using 0.2 μ m filter. Prior to in vivo dosing, LNPs were diluted in PBS. LNP sizes were determined by dynamic light scattering. A cuvette with 1 mL of 20 μ g/mL LNPs in PBS (pH 7.4) was measured for Z-average using the Malvern Panalytical Zetasizer Pro. The Z-average and polydispersity index were recorded.

40.1 Formulation of Lipids 10c-7 and 10c-8

Lipid Nanoparticles (LNPs) were formed using a Precision Nanosystems Ignite instrument with a 'NextGen' mixing chamber. Ethanol phase contained ionizable Lipid 10c-7 or Lipid 10c-28, DOPE, Cholesterol, and DSPE-PEG 2000 (Avanti Polar Lipids Inc.) at a weight ratio of 16:1:4:1 or 62:4:33:1 molar ratio was combined with an aqueous phase containing circular RNA and 25 mM sodium acetate buffer at pH 5.2. A 3:1 aqueous to ethanol mixing ratio was used. The formulated LNPs were then dialyzed in 1 L of water and exchanged 2 times over 18 hours. Dialyzed LNPs were filtered using 0.2 μ m filter. Prior to in vivo dosing, LNPs were diluted in PBS. LNP sizes were determined by dynamic light scattering. A cuvette with 1 mL of 20 μ g/mL LNPs in PBS (pH 7.4) was measured for Z-average using the Malvern Panalytical Zetasizer Pro. The Z-average and polydispersity index were recorded.

Example 41

In Vitro Delivery of Green Fluorescent Protein (GFP) or Chimeric Antigen Receptor (CAR)

Human PBMCs (Stemcell Technologies) were transfected with LNP encapsulating GFP and examined by flow cytometry. PBMCs from five different donors (PBMC A-E) were incubated in vitro with one LNP composition, containing circular RNA encoding either GFP or CD19-CAR (200 ng), at 37° C. in RPMI. 2% human serum, IL-2 (10 ng/mL), and 50 uM BME. PBMCs incubated without LNP were used as a negative control. After 24, 48, or 72 hours post-LNP incubation, cells were analyzed for CD3, CD19, CD56, CD14, CD11b, CD45, fixable live dead, and payload (GFP or CD19-CAR).

Representative data are presented in FIGS. 27A and 27B, showing that the tested LNP is capable of delivering circular RNA into primary human immune cells resulting in protein expression.

Example 42

Multiple IRES Variants can Mediate Expression of Murine CD19 CAR In Vitro

Multiple circular RNA constructs, encoding anti-murine CD19 CAR, contains unique IRES sequences and were lipotransfected into 1C1C7 cell lines. Prior to lipotransfection, 1C1C7 cells are expanded for several days in complete RPMI. Once the cells expanded to appropriate numbers, 1C1C7 cells were lipotransfected (Invitrogen RNAiMAX) with four different circular RNA constructs. After 24 hours, 1C1C7 cells were incubated with His-tagged recombinant murine CD19 (Sino Biological) protein, then stained with a secondary anti-His antibody. Afterwards, the cells were analyzed via flow cytometry.

Representative data are presented in FIGS. 26, showing that IRES sourced from the indicated virus (apodemus agrarius picomavirus, caprine kobuvirus, parbovirus, and salivirus) are capable of driving expression of an anti-mouse CD19 CAR in murine T cells.

Murine CD19 CAR Mediates Tumor Cell Killing In Vitro

Circular RNA encoding anti-mouse CD19 CAR were electroporated into murine T cells to evaluate CAR-mediated cytotoxicity. For electroporation, T cells were electroporated with circular RNA encoding anti-mouse CD19 CAR using ThermoFisher's Neon Transfection System then rested overnight. For the cytotoxicity assay, electroporated T cells were co-cultured with Fluc+ target and non-target cells at 1:1 ratio in complete RPMI containing 10% FBS, IL-2 (10 ng/mL), and 50 uM BME and incubated overnight at 37° C. Cytotoxicity was measured using a luciferase assay system 24 hours post-co-culture (Promega Brightglo Luciferase System) to detect lysis of Fluc+ target and non-target cells. Values shown are calculated relative to the untransfected mock signal.

Representative data are presented in FIG. 27, showing that an anti-mouse CD19 CAR expressed from circular RNA is functional in murine T cells in vitro.

Example 44

CD9 CAR Expressed from Circular RNA has Higher Yield and Greater Cytotoxic Effect Compared to that Expressed from mRNA

Circular RNA encoding encoding anti-CD19 chimeric antigen antigen receptor, which includes, from N-terminus to C-terminus, a FMC63-derived scFv, a CD8 transmembrane domain, a 4-1BB costimulatory domain, and a CD3 (intracellular domain, were electroporated into human peripheral T cells to evaluate surface expression and CAR-mediated cytotoxicity. For comparison, circular RNA-electroporated T cells were compared to mRNA-electroporated T cells in this experiment. For electroporation, CD3+ T cells were isolated from human PBMCs using commercially available T cell isolation kits (Miltenyi Biotec) from donor human PBMCs. After isolation, T cells were stimulated with anti-CD3/anti-CD28 (Stemcell Technologies) and expanded over 5 days at 37° C. in complete RPMI containing 10% FBS, IL-2 (10 ng/mL), and 50 uM BME. Five days post stimulation, T cells were electroporated with circular RNA encoding anti-human CD19 CAR using ThermoFisher's Neon Transfection System and then rested overnight. For the cytotoxicity assay, electroporated T cells were co-cultured with Fluc+ target and non-target cells at 1:1 ratio in complete RPMI containing 10% FBS, IL-2 (10 ng/mL), and 50 uM BME and incubated overnight at 37° C. Cytotoxicity was measured using a luciferase assay system 24 hours post-co-culture (Promega Brightglo Luciferase System) to detect lysis of Fluc+ target and non-target cells. Furthermore, an aliquot of electroporated T cells were taken and stained for live dead fixable staining, CD3, CD45, and chimeric antigen receptors (FMC63) at the day of analysis.

Representative data are presented in FIGS. 28 and 29. FIGS. 28A and 28B show that an anti-human CD19 CAR expressed from circular RNA is expressed at higher levels and longer than an anti-human CD19 CAR expressed from linear mRNA. FIGS. 29A and 29B show that an anti-human CD19 CAR expressed from circular RNA is exerts a greater cytotoxic effect relative to anti-human CD19 CAR expressed from linear mRNA.

Functional Expression of Two CARs from a Single Circular RNA

Circular RNA encoding chimeric antigen receptors were electroporated into human peripheral T cells to evaluate surface expression and CAR-mediated cytotoxicity. The purpose of this study is to evaluate if circular RNA encoding for two CARs can be stochastically expressed with a 2A (P2A) or an IRES sequence. For electroporation, CD3+ T cells were commercially purchased (Cellero) and stimulated with anti-CD3/anti-CD28 (Stemcell Technologies) and expanded over 5 days at 37° C. in complete RPMI containing 10% FBS, IL-2 (10 ng/mL), and 50 uM BME. Four days post stimulation, T cells were electroporated with circular RNA encoding anti-human CD19 CAR, anti-human CD19 CAR-2A-anti-human BCMA CAR, and anti-human CD19 CAR-IRES-anti-human BCMA CAR using ThermoFisher's Neon Transfection System then rested overnight. For the cytotoxicity assay, electroporated T cells were co-cultured with Fluc+ K562 cells expressing human CD19 or BCMA antigens at 1:1 ratio in complete RPMI containing 10% FBS, IL-2 (10 ng/mL), and 50 uM BME and incubated overnight at 37° C. Cytotoxicity was measured using a luciferase assay system 24 hours post-co-culture (Promega BrightGlo Luciferase System) to detect lysis of Fluc+ target cells.

Representative data are presented in FIG. 30, showing that two CARs can be functionally expressed from the same circular RNA construct and exert cytotoxic effector function.

Example 46

Example 46A: Built-In polyA Sequences and Affinity-Purification to Produce Immue-Silent Circular RNA

PolyA sequences (20-30 nt) were inserted into the 5' and 3' ends of the RNA construct (precursor RNA with built-in polyA sequences in the introns). Precursor RNA and introns can alternatively be polyadenylated post-transcriptionally using, e.g., *E. coli*. polyA polymerase or yeast polyA polymerase, which requires the use of an additional enzyme.

Circular RNA in this example was circularized by in vitro transcription (IVT) and affinity-purified by washing over a commercially available oligo-dT resin to selectively remove polyA-tagged sequences (including free introns and precursor RNA) from the splicing reaction. The IVT was performed with a commercial IVT kit (New England Biolabs) or a customerized IVT mix (Orna Therapeutics), containing guanosine monophosphate (GMP) and guanosine triphosphate (GTP) at different ratios (GMP:GTP=8, 12.5, or 13.75). In some embodiments, GMP at a high GMP:GTP ratio may be preferentially included as the first nucleotide, yielding a majority of monophosphate-capped precursor RNAs. As a comparison, the circular RNA product was alternatively purified by the treatment with Xmi1, Rnase R, and Dnase 1 (enzyme purification).

Immunogenicity of the circular RNAs prepared using the affinity purification or enzyme purification process were then assessed. Briefly, the prepared circular RNAs were transfected into A549 cells. After 24 hours, the cells were lysed and interferon beta-I induction relative to mock samples was measured by qPCR. 3p-hpRNA, a triphosphorylated RNA, was used as a positive control.

FIGS. 31B and 31C show that the negative selection affinity purification removes non-circular products from splicing reactions when polyA sequences are included on

397

elements that are removed during splicing and present in unspliced precursor molecules. FIG. 31D shows circular RNAs prepared with tested IVT conditions and purification methods are all immunoquiescent. These results suggest the negative selection affinity purification is equivalent or superior to enzyme purification for circular RNA purification and that customized circular RNA synthesis conditions (IVT conditions) may reduce the reliance on GMP excess to achieve maximal immunoquiescence.

Example 46B: Dedicated Binding Site and Affinity-Purification for Circular RNA Production

Instead of polyA tags, one can include specifically design sequences (DBS, dedicated binding site).

Instead of a polyA tag, a dedicated binding site (DBS), such as a specifically designed complementary oligonucleotide that can bind to a resin, may be used to selectively deplete precursor RNA and free introns. In this example, DBS sequences (30 nt) were inserted into the 5' and 3' ends of the precursor RNA. RNA was transcribed and the transcribed product was washed over a custom complementary oligonucleotide linked to a resin.

FIGS. 32B and 32C demonstrates that including the designed DBS sequence in elements that are removed during splicing enables the removal of unspliced precursor RNA and free intron components in a splicing reaction, via negative affinity purification.

Example 46C: Production of a Circular RNA Encoding Dystrophin

A 12kb12,000 nt circular RNA encoding dystrophin was produced by in vitro transcription of RNA precursors followed by enzyme purification using a mixture of Xnl. DNase 1, and RNase R to degrade remaining linear components. FIG. 33 shows that the circular RNA encoding dystrophin was successfully produced.

Example 47

S spacer between 3' intron fragment and the IRES improves circular RNA expression

Expression level of purified circRNAs with different 5' spacers between the 3' intron fragment and the IRES in Jurkat cells were compared. Briefly, luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 hours after electroporation of 60,000 cells with 250 ng of each RNA.

Additionally, stability of purified circRNAs with different 5' spacers between the 3' intron fragment and the IRES in Jurkat cells were compared. Briefly, luminescence from secreted *Gaussia luciferase* in supernatant was measured over 2 days after electroporation of 60,000 cells with 250 ng of each RNA and normalized to day 1 expression.

The results are shown in FIGS. 34A and 34B, indicating that adding a spacer can enhance IRES function and the importance of sequence identity and length of the added spacer. A potential explanation is that the spacer is added right before the IRES and likely functions by allowing the IRES to fold in isolation from other structured elements such as the intron fragments.

Example 48

This example describes deletion scanning from 5' or 3' end of the caprine kobuvirus IRES. IRES borders are

398

generally poorly characterized and require empirical analysis, and this example can be used for locating the core functional sequences required for driving translation. Briefly, circular RNA constructs were generated with truncated IRES elements operably linked to a *Gaussia luciferase* coding sequence. The truncated IRES elements had nucleotide sequences of the indicated lengths removed from the 5' or 3' end. Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 and 48 hours after electroporation of primary human T cells with RNA. Stability of expression was calculated as the ratio of the expression level at the 48-hour time point relative to that at the 24-hour time point.

As shown in FIG. 35, deletion of more than 40 nucleotides from the 5' end of the IRES reduced expression and disrupted IRES function. Stability of expression was relatively unaffected by the truncation of the IRES element but expression level was substantially reduced by deletion of 141 nucleotides from the 3' end of the IRES, whereas deletion of 57 or 122 nucleotides from the 3' end had a positive impact on the expression level.

It was also observed that deletion of the 6-nucleotide pre-start sequence reduced the expression level of the luciferase reporter. Replacement of the 6-nucleotide sequence with a classical kozak sequence (GCCACC) did not have a significant impact but at least maintained expression.

Example 49

This example describes modifications (e.g. truncations) of selected selected IRES sequences, including Caprine Kobuvirus (CKV) IRES, Parabovirus IRES, Apodemus Picomavirus (AP) IRES, Kobuvirus SZAL6 IRES, Crohivirus B (CrVB) IRES, CVB3 IRES, and SAFV IRES. The sequences of the IRES elements are provided in SEQ ID NOs: 55-96. Briefly, circular RNA constructs were generated with truncated IRES elements operably linked to a *Gaussia luciferase* coding sequence. HepG2 cells were transfected with the circular RNAs. Luminescence in the supernatant was assessed 24 and 48 hours after transfection. Stability of expression was calculated as the ratio of the expression level at the 48-hour time point relative to that at the 24-hour time point.

As shown in FIG. 36, truncations had variable effects depending on the identity of the IRES, which may depend on the initiation mechanism and protein factors used for translation, which often differs between IRESs. 5' and 3' deletions can be effectively combined, for example, in the context of CKV IRES. Addition of a canonical Kozak sequence in some cases significantly improved expression (as in SAFV. Full vs Full+K) or diminished expression (as in CKV, 5d40/3d122 vs 5d40/3d122+K).

Example 50

This example describes modifications of CK-739, AP-748, and PV-743 IRES sequences, including mutations at the translation initiation elements. Briefly, circular RNA constructs were generated with modified IRES elements operably linked to a *Gaussia luciferase* coding sequence. Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 and 48 hours after transfection of IC1C₇ cells with RNA.

CUG was the most commonly found alternative start site but many others were also characterized. These triplets can be present in the IRES scanning tract prior to the start codon

and can affect translation of correct polypeptides. Four alternative start site mutations were created, with the IRES sequences provided in SEQ ID NOs: 97-99. As shown in FIG. 37, mutations of alternative translation initiation sites in the CK-739 IRES affected translation of correct polypeptides, positively in some instances and negatively in other instances. Mutation of all the alternative translation initiation sites reduced the level of translation.

Alternative Kozak sequences, 6 nucleotides before start codon, can also affect expression levels. The 6-nucleotide sequence upstream of the start codon were gTcacG, aaagtc, gTcacG, gtcacg, gcaaac, and acaacc, respectively, in CK-739 IRES and Sample Nos. 1-5 in the "6 nt Pre-Start" group. As shown in FIG. 37, substitution of certain 6-nucleotide sequences prior to the start codon affected translation.

It was also observed that 5' and 3' terminal deletions in AP-748 and PV-743 IRES sequences reduced expression. However, in the CK-739 IRES, which had a long scanning tract, translation was relatively unaffected by deletions in the scanning tract.

Example 51

This example describes modifications of selected IRES sequences by inserting 5' and/or 3' untranslated regions (UTRs) and creating IRES hybrids. Briefly, circular RNA constructs were generated with modified IRES elements operably linked to a *Gaussia luciferase* coding sequence. Luminescence from secreted *Gaussia luciferase* in supernatant was measured 24 and 48 hours after transfection of HepG2 cells with RNA.

IRES sequences with UTRs inserted are provided in SEQ ID NOs: 100-104, 106-110 and GTCACG. As shown in FIG. 53, insertion of 5' UTR right after the 3' end of the IRES and before the start codon slightly increased the translation from Caprine Kobuvirus (CK) IRES but in some instances abrogated translation from Salivirus SZ1 IRES. Insertion of 3' UTR right after the stop cassette had no impact on both IRES sequences.

Hybrid CK IRES sequences are provided in SEQ ID NOs: 100-104, 106-110 and GTCACG. CK IRES was used as a base, and specific regions of the CK IRES were replaced with similar-looking structures from other IRES sequences, for example, SZ1 and AV (Aichivirus). As shown in FIG. 38, certain hybrid synthetic IRES sequences were functional, indicating that hybrid IRES can be constructed using parts from distinct IRES sequences that show similar predicted structures while deleting these structures completely abrogates IRES function.

Example 52

This example describes modifications of circular RNAs by introducing stop codon or cassette variants. Briefly, circular RNA constructs were generated with IRES elements operably linked to a *Gaussia luciferase* coding sequence followed by variable stop codon cassettes, which included a stop codon in each frame and two stop codons in the reading frame of the *Gaussia luciferase* coding sequence. 1C1C7 cells were transfected with the circular RNAs. Luminescence in supernatant was assessed 24 and 48 hours after transfection.

The sequences of the stop codon cassettes are set forth in SEQ ID NOs: 112-115, 117-118 and TAA. As shown in FIG. 39, certain stop codon cassettes improved expression levels, although they had little impact on expression stability. In particular, a stop cassette with two frame 1 (the reading

frame of the *Gaussia luciferase* coding sequence) stop codons, the first being TAA, followed by a frame 2 stop codon and a frame 3 stop codon, is effective for promoting functional translation.

Example 53

This example describes modifications of circular RNAs by inserting 5' UTR variants. Briefly, circular RNA constructs were generated with IRES elements with 5' UTR variants inserted between the 3' end of the IRES and the start codon, the IRES being operably linked to a *Gaussia luciferase* coding sequence. 1C1C7 cells were transfected with the circular RNAs. Luminescence in supernatant was assessed 24 and 48 hours after transfection.

The sequences of the 5' UTR variants are set forth in SEQ ID NOs: 120-121, GTCACG, and AGCCACC. As shown in FIG. 40, a CK IRES with a canonical Kozak sequence (UTR4) was more effective when a 36-nucleotide unstructured/low GC spacer sequence was added (UTR2), suggesting that the GC-rich Kozak sequences may interfere with core IRES folding. Using a higher-GC/structured spacer with a kozak sequence did not show the same benefit (UTR3), possibly due to interference with IRES folding by the spacer itself. Mutating the kozak sequence to gTcacG (UTR1) enhanced translation to the same level as the Kozak+spacer alternative without the need for a spacer.

Example 54

This example describes the impact of miRNA target sites in circular RNAs on expression levels. Briefly, circular RNA constructs were generated with IRES elements operably linked to a human erythropoietin (hEPO) coding sequence, where 2 tandem miR-122 target sites were inserted into the construct. miR-122-expressing Huh7 cells were transfected with the circular RNAs. hEPO expression in supernatant was assessed 24 and 48 hours after transfection by sandwich ELISA.

As shown in FIG. 41, the hEPO expression level was obrogated where the miR-122 target sites were inserted into the circular RNA. This result demonstrates that expression from circular RNA can be regulated by miRNA. As such, cell type- or tissue-specific expression can be achieved by incorporating target sites of the miRNAs expressed in the cell types in which expression of the recombinant protein is undesirable.

Example 55

LNP and Circular RNA Construct Containing Anti-CD19 CAR Reduces B Cells in the Blood and Spleen In Vivo.

Circular RNA constructs encoding an anti-CD19 CAR expression were encapsulated within lipid nanoparticles as described above. For comparison, circular RNAs encoding luciferase expression were encapsulated within separate lipid nanoparticle.

C57BL/6 mice at 6 to 8 weeks old were injected with either LNP solution every other day for a total of 4 LNP injections within each mouse. 24 hours after the last LNP injection, the mice's spleen and blood were harvested, stained, and analyzed via flow cytometry. As shown in FIG. 42A and FIG. 42B, mice containing LNP-circular RNA constructs encoding an anti-CD19 CAR led to a statistically significant reduction in CD19+ B cells in the peripheral

401

blood and spleen compared to mice treated with LNP-circular RNA encoding a luciferase.

Example 56

IRES Sequences Contained within Circular RNA Encoding CARs Improves CAR Expressions and Cytotoxicity of T-Cells.

Activated murine T-cells were electroporated with 200 ng of circular RNA constructs containing a unique IRES and a murine anti-CD19 1D3CAR expression sequence. The IRES contained in these constructs were derived either in whole or in part from a Caprine Kobuvirus, Apodemus Picomavirus, Parabovirus, or Salivirus. A Caprine Kobuvirus derived IRES was additionally codon optimized. As a control, a circular RNA containing a wild-type zeta mouse CAR with no IRES was used for comparison. The T-cells were stained for the CD-19 CAR 24 hours post electroporation to evaluate for surface expression and then co-cultured with A20 Fluc target cells. The assay was then evaluated for cytotoxic killing of the Fluc+ A20 cells 24 hours after co-culture of the T-cells with the target cells.

As seen in FIGS. 43A, 43B, 43C, and 44, the unique IRES were able to increase the frequency that the T-cells expressed the CAR protein and level of CAR expression on the surface of the cells. The increase frequency of expression of the CAR protein and level of CAR expression on the surface of cells lead to an improved anti-tumor response.

Example 57

Cytosolic and Surface Proteins Expressed from Circular RNA Construct in Primary Human T-Cells.

Circular RNA construct contained either a sequence encoding for a fluorescent cytosolic reporter or a surface antigen reporter. Fluorescent reporters included green fluorescent protein, mCitrine, mWasabi, Tsapphire. Surface reporters included CD52 and Thy1.1^{bio}. Primary human T-cells were activated with an anti-CD3/anti-CD28 antibody and electroporated 6 days post activation of the circular RNA containing a reporter sequence. T-cells were harvested and analyzed via flow cytometry 24 hours post electroporation. Surface antigens were stained with commercially available antibodies (e.g., Biolegend, Miltenyi, and BD).

As seen in FIG. 45A and FIG. 45B, cytosolic and surface proteins can be expressed from circular RNA encoding the proteins in primary human T-cells.

Example 58

Circular RNAs Containing Unique IRES' Sequences have Improved Translation Expression Over Linear mRNA.

Circular RNA constructs contained a unique IRES along with an expression sequence for Firefly luciferase (FLuc).

Human T-cells from 2 donors were enriched and stimulated with anti-CD3/anti-CD28 antibodies. After several days of proliferation, activated T cells were harvested and electroporated with equal molar of either mRNA or circular RNA expressing FLuc payloads. Various IRES sequences, including those derived from Caprine Kobuvirus, Apodemus Picomavirus, and Parabovirus, were studied to evaluate expression level and durability of the payload expression across 7 days. Across the 7 days, the T-cells were lysed with Promega Brightglo to evaluate for bioluminescence.

As shown in FIGS. 46C, 46D, 46E, 46F, and 46G, the presence of an IRES within a circular RNA can increase translation and expression of a cytosolic protein by orders of

402

magnitude and can improve expression compared to linear mRNA. This was found consistent across multiple human T-cell donors.

Example 59

Example 59A: LNP-Circular RNA Encoding Anti-CD19 Mediates Human T-Cell Killing of K562 Cells

Circular RNA constructs contained a sequence encoding for anti-CD19 antibodies. Circular RNA constructs were then encapsulated within a lipid nanoparticle (LNP).

Human T-cells were stimulated with anti-CD3/anti-CD28 and left to proliferate up to 6 says. At day 6, LNP-circular RNA and ApoE3 (1 µg/mL) were co-cultured with the T-cells to mediate transfection. 24 hours later, Fluc+ K562 cells were electroporated with 200 ng of circular RNA encoding anti-CD19 antibodies and were later co-cultured at day 7. 48 hours post co-culture, the assay was assessed for CAR expression and cytotoxic killing of K562 cells through Fluc expression.

As shown in FIG. 47A and FIG. 47B, there is T-cell expression of anti-CD19 CAR from the LNP-mediated delivery of a CAR in vitro to T-cells and its capability to lyse tumor cells in a specific, antigen dependent manner in engineered K562 cells.

Example 59B: LNP-Circular RNA Encoding Anti-BCMA Antibody Mediates Human T-Cell Killing of K562 Cells

Circular RNA constructs contained a sequence encoding for anti-BCMA antibodies. Circular RNA constructs were then encapsulated within a lipid nanoparticle (LNP).

Human T-cells were stimulated with anti-CD3/anti-CD28 and left to proliferate up to 6 says. At day 6, LNP-circular RNA and ApoE3 (1 µg/mL) were co-cultured with the T-cells to mediate transfection. 24 hours later, Fluc+ K562 cells were electroporated with 200 ng of circular RNA encoding anti-BCMA antibodies and were later co-cultured at day 7. 48 hours post co-culture, the assay was assessed for CAR expression and cytotoxic killing of K562 cells through Fluc expression.

As shown in FIG. 47B, there is T-cell expression of BCMA CAR from the LNP-mediated delivery of a CAR in vitro to T-cells and its capability to lyse tumor cells in a specific, antigen dependent manner in engineered K562 cells.

Example 60

Anti-CD19 CAR T-Cells Exhibit Anti-Tumor Activity In Vitro.

Human T-cells were activated with anti-CD3/anti-CD28 and electroporated once with 200 ng of anti-CD19 CAR-expressing circular RNA. Electroporated T-cells were co-cultured with FLuc+ Nalm6 target cells and non-target Fluc+K562 cells to evaluate CAR-mediated killing. After 24 hours post co-culture, the T-cells were lysed and examined for remanent FLuc expression by target and non-target cells to evaluate expression and stability of expression across 8 days total.

As shown in FIGS. 48A and 48B, T-cells express circular RNA CAR constructs in specific, antigen-dependent manner. Results also shows improved cytotoxicity of circular

403

RNAs encoding CARs compared to linear mRNA encoding CARs and delivery of a functional surface receptor.

Example 61

Effective LNP transfection of circular RNA mediated with ApoE3

Human T-cells were stimulated with anti-CD3/anti-CD28 and left to proliferate up to 6 days. At day 6, lipid nanoparticle (LNP) was and circular RNA expressing green fluorescence protein solution with or without ApoE3 (1 µg/mL) were co-cultured with the T-cells. 24 hours later, the T-cells were stained for live/dead T-cells and the live T-cells were analyzed for GFP expression on a flow cytometer.

As shown by FIGS. 49A, 49B, 49C, 49D, and 49E, efficient LNP transfection can be mediated by ApoE3 on activated T-cells, followed by significant payload expression. These results were exhibited across multiple donors.

Example 62

This example illustrates expression of SARS-CoV2 spike protein expression in vitro. Circular RNA encoding SARS-CoV2 stabilized spike protein was transfected into 293 cells using MessengerMax Transfection Reagent. 24 hours after transfection, the 293 cells were stained with a CR3022 anti-spike primary antibody and APC-labeled secondary antibody.

FIG. 50A shows circularization efficiency of roughly 4.5 kb SARS-Cov2 stabilized spike protein-encoding RNA resulting from an in vitro transcription reaction. FIG. 50B and FIG. 73C show SARS-CoV2 stabilized spike protein expression on 293 cells after the circular RNA transfection with MessengerMax Transfection Reagent relative to mock transfected cells.

FIG. 54A and FIG. 54B show SARS-CoV2 stabilized spike protein expression by percentage of cells and gMFI on 293 cells after transfection of a panel of circular RNAs, containing variable IRES sequences, codon optimized coding regions, and stabilized SARS-CoV2 spike proteins, using MessengerMax Transfection Reagent. FIG. 54C shows the relationship between MFI and percentage.

Example 63

This example shows in vivo cytokine response after IV injection of 0.2 mg/kg circRNA preparations encapsulated in a lipid nanoparticle formulation. circRNA splicing reactions synthesized with GTP as a precursor RNA initiator and splicing nucleotide incited greater cytokine responses than purified circRNA and linear m1Ψ-mRNA due to the presence of triphosphorylated 5' termini in the splicing reaction. Levels of IL-10, IL-6, IL-10, IL-12p70, RANTES, TNFα were measured from blood drawn 6 hours following intravenous injection of the LNP-formulated circRNA preparation. Mice injected with PBS were used as a control.

As seen in FIG. 51, circRNA splicing reactions synthesized with GTP as a precursor RNA initiator and splicing nucleotide incite greater cytokine responses than purified circRNA and linear m1Ψ-mRNA due to the presence of triphosphorylated 5' termini in the splicing reaction.

Example 64

This example illustrates intramuscular delivery of varying doses of lipid nanoparticle containing circular RNAs. Mice were dosed with either 0.1 µg, 1 µg, or 10 µg of circRNA

404

formulated in lipid nanoparticles. Whole body IVIS imaging was conducted at 6 hours following an injection of luciferin (FIG. 52A and FIG. 52B). Ex vivo IVIS imaging was conducted at 24-hour. Flux values for liver, quad, and calf are shown in FIG. 52C. FIG. 53B and FIG. 53C show that the expression of the circular RNA is present in the muscle tissue of the mice.

Example 65

This example illustrates expression of multiple circular RNAs in LNP formulations. Circular RNA constructs encoding either hEPO or fLuc were dosed in a single and mixed set of LNPs. hEPO concentration in the serum (FIG. 53A) and total flux by IVIS imaging (FIG. 53B) was determined. The results show that the circular RNA hEPO or fLuc constructs individually formulated or co-formulated expressed protein efficiently.

Example 66

Example 66A: Hepatocyte Plating and Culture

Primary human hepatocytes (PHH), primary mouse hepatocytes (PMH), primary cynomolgus monkey hepatocytes (PCH) were thawed and resuspended in hepatocyte thawing medium (Xenotech, cat #K8600/K8650) followed by centrifugation. The supernatant was discarded, and the pelleted cells were resuspended in hepatocyte plating medium (Xenotech, cat #K8200). Cells were counted via hemocytometer and plated on Bio-coat collagen-I coated 96-well plates at a density of 25,000 cells/well for PHH, 25,000 cells/well for PMH, and 50,000 cells/well for PCH in 100 µL of plating media. Plated cells were allowed to settle and adhere for 6 hours in a tissue culture incubator at 37° C. and 5% CO₂ atmosphere. After incubation cells were checked for monolayer formation after which the plating media was aspirated and replaced with 100 ul of culture media (Xenotech, cat #K8300). Media was replaced every 24 hours for the duration of the experiment.

Example 66B: In Vitro Screening of LNP Formulated Circular RNA Encoding Firefly Luciferase in Primary Human, Mouse, And Cynomolgus Monkey Hepatocytes

A circular RNA construct comprising a TIE and a coding element encoding for firefly luciferase was produced and transfected into LNPs. Various concentrations of LNPs formulated with the circularized RNA (oRNA) were diluted in hepatocyte media supplemented with 3% fetal bovine serum (FBS) (ThermoFisher, cat #A3160401). Media was aspirated from the cells prior to addition of 100 µL of LNP/FBS/media mixture to the cells.

Luciferase activity was detected in primary human (FIG. 67A), mouse (FIG. 67B), and cynomolgus monkey (FIG. 67C) hepatocyte. 24 hours post-transfection plates were removed from the incubator and allowed to equilibrate to room temperature for 15 mins. A volume of 100 µL of Firefly Luciferase one-step glow assay working solution (Pierce, cat #16196) was added to each well. The plate was placed on a microplate shaker (ThermoFisher, cat #S72050) and mixed at 300 rpm for 3 min. Post-mixing, the plate was allowed to incubate at room temperature for 10 min. Luminescence was read using a Varioskan or Bio-Tek Cytation5 instrument.

As seen in FIG. 67A, FIG. 67B, and FIG. 67C, TIE containing circular RNAs are capable of driving firefly

luciferase protein expression in primary hepatocytes from multiple species in a dose-dependent manner when transfected in vitro with an LNP.

Example 66C: In Vitro Screening of LNP
Formulated Circular RNA Encoding Firefly
Luciferase in Multiple Primary Human Hepatocyte
Donors

A circular RNA construct comprising a TIE and a coding element encoding for firefly luciferase was produced and transfected into LNPs. Various concentrations of LNPs formulated with the circularized RNA (oRNA) were diluted in hepatocyte media supplemented with 3% fetal bovine serum (FBS) (ThermoFisher, cat #A3160401). Media was aspirated from the cells prior to addition of 100 μ l of LNP/FBS/media mixture to the cells.

Luciferase activity was detected in primary human (FIG. 68A), mouse (FIG. 68B), and cynomolgus monkey (FIG. 68C) hepatocyte. 24 hours post-transfection plates were removed from the incubator and allowed to equilibrate to room temperature for 15 mins. A volume of 100 μ l of Firefly Luciferase one-step glow assay working solution (Pierce, cat #16196) was added to each well. The plate was placed on a microplate shaker (ThermoFisher, cat #S72050) and mixed at 300 rpm for 3 min. Post-mixing, the plate was allowed to incubate at room temperature for 10 min. Luminescence was read using a Varioskan or Bio-Tek Cytation5 instrument.

As seen in FIG. 68A, FIG. 68B, and FIG. 68C, TIE containing circular RNAs are capable of driving firefly luciferase protein expression in primary hepatocytes from multiple human donors in a dose-dependent manner when transfected in vitro with an LNP.

Example 67

In vitro expression of LNP formulated with circular RNA encoding for GFP in multiple human cell models.

A circular RNA construct was produced comprising a TIE and coding element encoding for a GFP protein. LNP were formulated with the circular RNA construct. Then various concentrations of the LNP containing the circular RNA construct were diluted in hepatocyte media supplemented with 3% fetal bovine serum (FBS) (ThermoFisher, cat #A3160401). Media was aspirated from the cells prior to addition of 100 μ l of LNP/FBS/media mixture to the cells.

HeLa (human cervical adenocarcinoma: ATCC, cat #CCL-2), HEK293 (human embryonic kidney; ATCC, cat #CRL-1573), and HUH7 (human liver hepatocellular carcinoma; JCRB, cat #JCRB0403) were transfected as previously described with LNP formulated oRNAs. Twenty-four hours post-transfection, the media was removed and the cells were trypsinized. The trypsinized cells were neutralized with PBS supplemented with 10% FBS, harvested, and transferred to a tube. The tube was centrifuged to pellet the cells and the supernatant was aspirated. The pellet was stored at -80° C. prior to lysis. For lysis the cells were thawed on ice and were lysed with 100 μ l/well RIPA buffer (Boston Bio Products, Cat. BP-115) plus freshly added 1 mM DTT, and 250 U/mL Benzonase (EMD Millipore, cat #71206-3), and protease inhibitor mixture consisting of complete protease inhibitor cocktail (Sigma, cat #11697498001). Cells were kept on ice for 30 minutes at which time NaCl (IM final concentration) was added. Cell lysates were thoroughly mixed and retained on ice for 30 min. The whole cell extracts (WCE) were centrifuged to pellet debris. A Bradford assay (Bio-Rad, cat #500-0001)

was used to assess protein content of the lysates. The Bradford assay procedure was completed according to the manufacturer's protocol. Extracts were stored at -20° C. prior to use. Western blots were performed to assess GFP protein levels. Whole cell extract lysates were mixed with Laemmli buffer and denatured at 95° C. for 10 min. Western blots were run using the NuPage system on 4-12% Bis-Tris gels (ThermoFisher, cat #NPO335BOX) according to the manufacturer's protocol followed by wet transfer onto 0.45 μ m nitrocellulose membrane (ThermoFisher, cat #LC2001). After transfer membranes were rinsed thoroughly with water and stained with Ponceau S solution (Boston Bio Products, cat #ST-180) to confirm complete and even transfer. Blots were blocked using 5% Dry Milk in TBS for 30 minutes on a lab rocker at room temperature. Blots were rinsed with 1 $\times$ TBST (Boston BioProducts, cat #IBB-180) and probed with mouse dylight 680-tagged anti-GFP monoclonal antibody (ThermoFisher, cat #MA515256D680) at 1:1,000 in 1 $\times$ TBST. Anti-p-actin or GAPDH was used as a loading control (ThermoFisher, cat #AM4302/AM4300) at 1:4,000 in 1 $\times$ TBST and incubated simultaneously with the GFP primary antibody. Blots were sealed in a bag and kept overnight at 4° C. on a lab rocker. After incubation, blots were rinsed 3 times for 5 minutes each in 1 $\times$ TBST and probed with mouse secondary antibodies (ThermoFisher, cat #P135519) at 1:25,000 each in 1 $\times$ TBST for 30 minutes at room temperature. After incubation, blots were rinsed 3 times for 5 minutes each in 1 $\times$ TBST. Blots were visualized and analyzed using a Licor Odyssey system.

As shown in FIG. 69, TIE-containing circular RNA is capable of expressing GFP protein in diverse human cell lines (e.g. HeLa, HEK293, and HUH7 cells) in a dose dependent manner when transfected in vitro with an LNP.

Example 68

In Vitro Expression of LNP Formulated with Circular RNA Encoding for GFP in Primary Human Hepatocytes.

A circular RNA construct was produced comprising a TIE and coding element encoding for a GFP protein. Various concentrations of LNP containing circularized RNA (oRNA) were diluted in hepatocyte media supplemented with 3% fetal bovine serum (FBS) (ThermoFisher, cat #A3160401). Media was aspirated from the cells prior to addition of 100 μ l of LNP/FBS/media mixture to the cells.

Primary human hepatocytes (PHH) were thawed and resuspended in hepatocyte thawing medium (Xenotech, cat #K8600/K8650) followed by centrifugation. The supernatant was discarded, and the pelleted cells were resuspended in hepatocyte plating medium (Xenotech, cat #K8200). Cells were counted via hemocytometer and plated on Bio-coat collagen-I coated 96-well plates at a density of 25,000 cells/well for PHH, 25,000 cells/well for PMH, and 50,000 cells/well for PCH in 100 μ l of plating media. Plated cells were allowed to settle and adhere for 6 hours in a tissue culture incubator at 37° C. and 5% CO_2 atmosphere.

After incubation cells were checked for monolayer formation after which the plating media was aspirated and replaced with 100 μ l of culture media (Xenotech, cat #K8300). Media was replaced every 24 hours for the duration of the experiment.

Primary human hepatocytes were transfected as previously described with LNP formulated oRNAs. Twenty-four hours post-transfection, the media was removed and the cells were trypsinized. The trypsinized cells were neutralized with PBS supplemented with 10% FBS, harvested, and transferred to a tube. The tube was centrifuged to pellet the

407

cells and the supernatant was aspirated. The pellet was stored at -80°C . prior to lysis. For lysis the cells were thawed on ice and were lysed with 100 μL /well RIPA buffer (Boston Bio Products, Cat. BP-115) plus freshly added 1 mM DTT, and 250 U/ml Benzonase (EMD Millipore, cat #71206-3), and protease inhibitor mixture consisting of complete protease inhibitor cocktail (Sigma, cat #11697498001). Cells were kept on ice for 30 minutes at which time NaCl (1 M final concentration) was added. Cell lysates were thoroughly mixed and retained on ice for 30 min. The whole cell extracts (WCE) were centrifuged to pellet debris. A Bradford assay (Bio-Rad, cat #500-0001) was used to assess protein content of the lysates. The Bradford assay procedure was completed according to the manufacturer's protocol. Extracts were stored at -20°C . prior to use. Western blots were performed to assess GFP protein levels. Whole cell extract lysates were mixed with Laemmli buffer and denatured at 95°C . for 10 min. Western blots were run using the NuPage system on 4-12% Bis-Tris gels (ThermoFisher, cat #NPO335BOX) according to the manufacturer's protocol followed by wet transfer onto 0.45 μm nitrocellulose membrane (ThermoFisher, cat #LC2001). After transfer membranes were rinsed thoroughly with water and stained with Ponceau S solution (Boston Bio Products, cat #ST-180) to confirm complete and even transfer. Blots were blocked using 5% Dry Milk in TBS for 30 minutes on a lab rocker at room temperature. Blots were rinsed with 1 $\times$ TBST (Boston BioProducts, cat #IBB-180) and probed with mouse dylight 680-tagged anti-GFP monoclonal antibody (ThermoFisher, cat #MA515256D680) at 1:1,000 in 1 $\times$ TBST. Anti- β -actin or GAPDH was used as a loading control (ThermoFisher, cat #AM4302/AM4300) at 1:4,000 in 1 $\times$ TBST and incubated simultaneously with the GFP primary antibody. Blots were sealed in a bag and kept overnight at 4°C . on a lab rocker. After incubation, blots were rinsed 3 times for 5 minutes each in 1 $\times$ TBST and probed with mouse secondary antibodies (ThermoFisher, cat #PI35519) at 1:25,000 each in 1 $\times$ TBST for 30 minutes at room temperature. After incubation, blots were rinsed 3 times for 5 minutes each in 1 $\times$ TBST. Blots were visualized and analyzed using a Licor Odyssey system.

As shown in the western blot in FIG. 70, circular RNAs containing a TIE is capable of successfully encoding a GFP protein in primary human hepatocytes when transfected in vivo with an LNP.

Example 69

In Vitro Expression of Firefly Luciferase in Circular RNA Encoding Firefly Luciferase in Mouse Myoblast and Primary Human Skeletal Muscle Myoblast Cells Using Lipofectamine.

A circular RNA construct comprising a TIE and coding element encoding firefly luciferase protein.

Primary human skeletal muscle (HSkM) cells (Lonza, cat #20TL356514) were thawed in a 37°C . water bath and plated at recommended seeding density (3,000 to 5,000 per cm^2) in SkGM-2 BulletKit growth media (Lonza, cat #CC-3245) and allowed to grow overnight. Cells were detached using ReagentPack subculture reagents (Lonza, cat #CC-5034) and plated on tissue culture grade 96-well plates at recommended seeding density and allowed to grow overnight in a tissue culture incubator at 37°C . and 5% CO_2 atmosphere or to 70-80% confluency with growth media changed every 2 days.

For one 96-well plate reaction, 0.3 μL of Lipofectamine-3000 transfection reagent (Lipo3K) (ThermoFisher, cat

408

#L3000015) was mixed with 5 μL Opti-MEM reduced serum media (ThermoFisher, cat #51985091). In a separate tube, per reaction, firefly luciferase (fluc) oRNA (at 10-200 ng) was combined with 5 μL Opti-MEM and 0.2 μL P3000TM enhancer reagent (ThermoFisher, cat #L3000015). Equal volumes of Lipo3K/Opti-MEM mix was combined with oRNA/Opti-MEM mix and incubated at room temperature for 15 min. The Lipo3K/oRNA mixture was added to each well to be transfected and placed in a tissue culture incubator at 37°C . and 5% CO_2 atmosphere for 24 hours.

After 24 hours, the transfection plates were removed from the incubator and allowed to equilibrate to room temperature for 15 minutes. A volume of 100 μL of firefly luciferase one-step glow assay working solution (Pierce, cat #16196) was added to each well. The plate was placed on a microplate shaker (ThermoFisher, cat #S72050) and mixed at 300 rpm for 3 minutes. Post-mixing, the plate was allowed to incubate at room temperature for 10 minutes. Luminescence was read using a Varioskan or Bio-Tek Cytation5 instrument.

As shown in FIG. 71A and FIG. 71B, circular RNAs comprising a TIE is capable of driving firefly luciferase protein expression in myoblasts from different species in a dose-dependent manner when transfected in vitro with lipofectamine.

Example 70

In Vitro Expression of Firefly Luciferase in Circular RNA Encoding Firefly Luciferase in Differentiated Primary Human Skeletal Muscles Myotubes

A circular RNA construct comprising a TIE and coding element encoding firefly luciferase protein.

Primary human skeletal muscle (HSkM) cells (Lonza, cat #20TL356514) were thawed in a 37°C . water bath and plated at recommended seeding density (3,000 to 5,000 per cm^2) in SkGM-2 BulletKit growth media (Lonza, cat #CC-3245) and allowed to grow overnight. Cells were detached using ReagentPack subculture reagents (Lonza, cat #CC-5034) and plated on tissue culture grade 96-well plates at recommended seeding density and allowed to grow overnight in a tissue culture incubator at 37°C . and 5% CO_2 atmosphere or to 70-80% confluency with growth media changed every 2 days. Once cells reached 70-80% confluency, growth media was removed, cells were washed twice in 1 $\times$ PBS (Gibco, cat #10010023) and changed to differentiation media consisting of F-10 (1 $\times$) (Gibco, cat #11550-043) supplemented with 2% Horse Serum (Gibco, cat #26050088) and 1% Pen-Strep (Gibco, cat #15140-122). Media was changed daily for 5 to 6 days until nearly all myoblasts had fused to form myotubes.

For one 96-well plate reaction, 0.3 μL of Lipofectamine-3000 transfection reagent (Lipo3K) (ThermoFisher, cat #L3000015) was mixed with 5 μL Opti-MEM reduced serum media (ThermoFisher, cat #51985091). In a separate tube, per reaction, firefly luciferase (f.luc) oRNA (at 10-200 ng) was combined with 5 μL Opti-MEM and 0.2 μL P3000TM enhancer reagent (ThermoFisher, cat #L3000015). Equal volumes of Lipo3K/Opti-MEM mix was combined with oRNA/Opti-MEM mix and incubated at room temperature for 15 min. The Lipo3K/oRNA mixture was added to each well to be transfected and placed in a tissue culture incubator at 37°C . and 5% CO_2 atmosphere for 24 hours.

After 24 hours, transfection plates were removed from the incubator and allowed to equilibrate to room temperature for 15 minutes. A volume of 100 μL of Firefly Luc one-step

409

glow assay working solution (Pierce, cat #16196) was added to each well. The plate was placed on a microplate shaker (ThermoFisher, cat #S72050) and mixed at 300 rpm for 3 minutes. Post-mixing, the plate was allowed to incubate at room temperature for 10 minutes. Luminescence was read using a Varioskan or Bio-Tek Cytation5 instrument.

As shown in FIG. 72A and FIG. 72B, a circular RNA comprising a TIE is capable of driving firefly luciferase protein expression in primary muscles cells through differentiated states (e.g., in myoblast and differentiated myotubes) in multiple human donors in a dose-dependent manner when transfected in vitro with lipofectamine.

Example 71

Cell-Free In Vitro Translation of Circular RNAs Containing TIEs

A cell-free rabbit reticulocyte in vitro translation assay (Promega, cat #L4540) was completed to characterize protein products from various RNA templates. Both linear mRNA and circular oRNA templates were used in the assay and reaction components were assembled according to the manufacturer's protocol. Prior to assay, RNA templates were denatured at 65° C. for 3 minutes and immediately cooled on ice. All reaction components were assembled on ice. Flexi Rabbit Reticulocyte kit components, complete amino acid mixture (Promega, cat #L5061), RNasin TIEuclease inhibitor (Promega, cat #N2111), and transcend tRNA (Promega, cat #L5061) were added to denatured RNA templates. The reaction was vortexed to mix and incubated at 30° C. for 60 minutes and then placed on ice.

The reaction mixture was added to 1× sample buffer (ThermoFisher, cat #NP(007) and heated at 70° C. for 15 minutes. The denatured protein sample was loaded onto 4-12% Bis-Tris gels (ThermoFisher, cat #NPO335BOX). Gel electrophoresis was completed, and the gel was wet transferred onto 0.45 μm nitrocellulose membrane (ThermoFisher, cat #LC2001). Post-transfer, the membrane was blocked for 1 hour with rocking in freshly made TBS with 0.5% Tween-20 (Boston Bioproducts Inc Cat #IBB-180). The membrane was incubated for 45 min with rocking with streptavidin-AP (Promega, cat #V5591) at a 1:2,500 dilution. The membrane was rinsed with four cycles of: twice with TBST and twice with deionized water for 1 min per rinse. The membrane was incubated with Western Blue substrate (Promega, cat #53841) for 45 minutes and the membrane was rinsed in water and scanned on a LICOR Odyssey CLx imaging system.

As shown in FIG. 73A and FIG. 73B, a circular RNA comprising a TIE is capable of driving protein expression in a cell-free lysate, independent of any cell type. FIG. 73A illustrates expression of firefly luciferase from a linear or circular RNA input. FIG. 73B illustrates expression of human and mouse ATP7B proteins with different codon optimization (co) approaches compared to wild-type native sequence (WT). The codon optimized-circular RNAs expressing ARP7B protein and the circular RNA expressing firefly luciferase showed protein full-length protein expression.

Example 72

TIE Selection Methodology

Putative TIEs were identified for activity assessment from sequences in GenBank. Briefly, Riboviria and Unclassified Virus sequences greater than 1 kb in length were identified. 5' and intergenic UTRs were extracted based on putative

410

CDS start and end sites with a minimum length cutoff of 250 nt. Reverse sequences were also collected for negative sense CDS annotations. For genres not expected to contain TIE sequences, a few noncoding regions per genus were selected at random. Duplicates, >10 nt repeat-containing sequences, sequences with both XbaI and BamHI sites, and low-quality sequences (non acgt) were culled, then sequences were clustered through CDHit with an 80% sequence similarity cutoff for clustering; representative sequences from each cluster were selected for further study. For unclassified sequences or sequences expected to contain a TIE, all 5' and intergenic UTRs were selected; duplicates, >10 nt repeat-containing sequences, sequences with both XbaI and BamHI sites, and low-quality sequences (non acgt) were culled, then sequences were clustered through CDHit with a 95% (low risk, known IREs) sequence similarity cutoff for clustering; representative sequences from each cluster were selected for further study. For both strategies, sequences shorter than 300 nt or unable to be synthesized due to sequence complexity were eliminated.

Example 73

Example 73A: TIE Activity in Primary Human T Cells

Nucleic acid sequences containing putative TIEs were inserted into a circular RNA (oRNA) construct prior to the start codon of a *Gaussia luciferase* reporter sequence. oRNA containing the TIE was synthesized and purified. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNA was transfected into T cells in vitro. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates higher TIE function. Higher luminescence at 48 hours relative to 24 hours indicates higher oRNA stability due to TIE function.

Example 73B: TIE Activity in Primary Human Hepatocytes

Nucleic acid sequences containing putative TIEs were inserted into a circular RNA (oRNA) construct prior to the start codon of a *Gaussia luciferase* reporter sequence. oRNA containing the TIE was synthesized and purified. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNA was transfected into hepatocytes in vitro. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates higher TIE function. Higher luminescence at 48 hours relative to 24 hours indicates higher oRNA stability due to TIE function.

Example 73C: TIE Activity in Primary Human Myotubes

Nucleic acid sequences containing putative TIEs were inserted into a circular RNA (oRNA) construct prior to the start codon of a *Gaussia luciferase* reporter sequence. oRNA containing the TIE was synthesized and purified. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNA was transfected into human myotubes in vitro. Supernatant was harvested and replaced 24 and 48 hours after transfection.

411

tion, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates higher TIE function. Higher luminescence at 48 hours relative to 24 hours indicates higher oRNA stability due to TIE function.

Example 74

TIE Tissue Tropism

Select TIE-containing oRNAs were formulated into LNPs. LNP-oRNAs were transfected into T cells, hepatocytes, and myotubes. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates higher TIE function. Higher luminescence at 48 hours relative to 24 hours may indicate higher oRNA stability due to TIE function. TIE activity was compared between cell types and differences resulting from TIE tissue preference were noted. Differences may be a result of the TIE engaging proteins that show tissue-specific expression and promoting enhanced translation initiation, degradation, or stability.

Example 75

Example 75A: TIE Deletion Scanning

Select TIE sequences with progressive deletions from either the 5' end or 3' end of the TIE were inserted into a circular RNA (oRNA) construct prior to the start codon of a *Gaussia luciferase* reporter sequence. oRNA containing the TIE variant was synthesized and purified. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNA was transfected into human primary T cells. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates higher TIE function. Higher luminescence at 48 hours relative to 24 hours may indicate higher oRNA stability due to TIE function. Expression or stability impairment due to progressive deletions identifies the core functional unit of the TIE.

Example 75B: TIE Variant Generation and Identification

Select TIE-containing oRNA synthesis plasmids were subjected to error-prone PCR to introduce random mutations into the PCR product. PCR product was used as a template for oRNA synthesis. Purified oRNA was formulated into LNPs and transfected into primary human T cells. Polysome fractions were harvested from T cells at 6, 24, 48, and 72 hours post-transfection by HPLC. RNA associated with each polysome fraction was extracted from polysome fractions and sequenced by NGS. TIE mutation enrichment in each polysome fraction at each time point was analyzed to identify mutations that 1) maintain or improve translation activity from the TIE and/or 2) improve stability of the oRNA.

Example 75C: TIE Single and Multi-Variant Validation

Nucleic acid sequences containing putative beneficial TIE mutations from example 6 alone or in combination were

412

inserted into a circular RNA (oRNA) construct prior to the start codon of a *Gaussia luciferase* reporter sequence. oRNA containing the TIE variant was synthesized and purified. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNA was transfected into human primary T cells. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates higher TIE function. Higher luminescence at 48 hours relative to 24 hours indicates higher oRNA stability due to TIE function.

Example 76

Example 76A: Selection of Eukaryotic TIEs

Selection of eukaryotic TIEs. Putative eukaryotic TIEs were identified using several databases. TIEs selected include sequences 40-1578 nucleotides in length and may or may not contain identified modification (m6A) sites.

Example 76B: TIEs Containing Modified Nucleotides (m6A)

Nucleic acid sequences containing putative TIEs were inserted into a circular RNA (oRNA) construct preceding the coding region of a *Gaussia luciferase* reporter sequence. oRNAs were synthesized with a titration of modified nucleotide. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNAs were transfected into T cells, hepatocytes, and myotubes. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence of modified nucleotide containing TIEs at 24 hours indicates necessity for modification for enhanced function. Higher luminescence at 48 hours relative to 24 hours indicates modified nucleotide containing TIEs enhance stability of oRNA.

Example 77

Expression of TIEs in Cells Undergoing Oxidative and/or Hypoxic Stress

Nucleic acid sequences containing putative TIEs were inserted into a circular RNA (oRNA) construct preceding the coding region of a *Gaussia luciferase* reporter sequence. Purified oRNA was formulated into lipid nanoparticles. Hepatocytes were treated with hydrogen peroxide to induce oxidative stress or CoCl₂ to induce hypoxic stress. LNP-oRNA was transfected into hepatocytes (under hypoxic stress, oxidative stress, or untreated) in vitro. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates higher TIE function. Higher luminescence at 48 hours relative to 24 hours may indicate higher oRNA stability due to TIE function.

Example 78

Aptamer as a TIE

Nucleic acid sequences containing aptamers against translation initiation factors (ie eIF4E, eIF4G, eIF4a) were inserted into a circular RNA (oRNA) construct preceding the

413

coding region of a *Gaussia luciferase* reporter sequence. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNAs were transfected into hepatocytes. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates higher TIE function. Higher luminescence at 48 hours relative to 24 hours may indicate higher oRNA stability due to TIE function.

Example 79

Tandem TIEs

Select combinations of viral, eukaryotic, and/or aptamer TIEs were inserted into a circular RNA (oRNA) construct preceding the coding region of a *Gaussia luciferase* reporter sequence. oRNAs were synthesized with a titration of modified nucleotide. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNAs were transfected into T cells, hepatocytes, and myotubes. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence of constructs containing multiple TIEs at 24 hours indicates a synergy of TIEs. Higher luminescence at 48 hours relative to 24 hours may indicate having multiple TIEs in one construct enhance stability of oRNA.

Example 80

Example 80A: Coding Aptamers to Enhance Cap-Independent Translation

Certain aptamers that bind to eIF4E, eIF4a, and other translation initiators are known to inhibit translation by forcing the proteins to adopt a non-functional conformation. Nucleic acid sequences containing aptamers against trans-

414

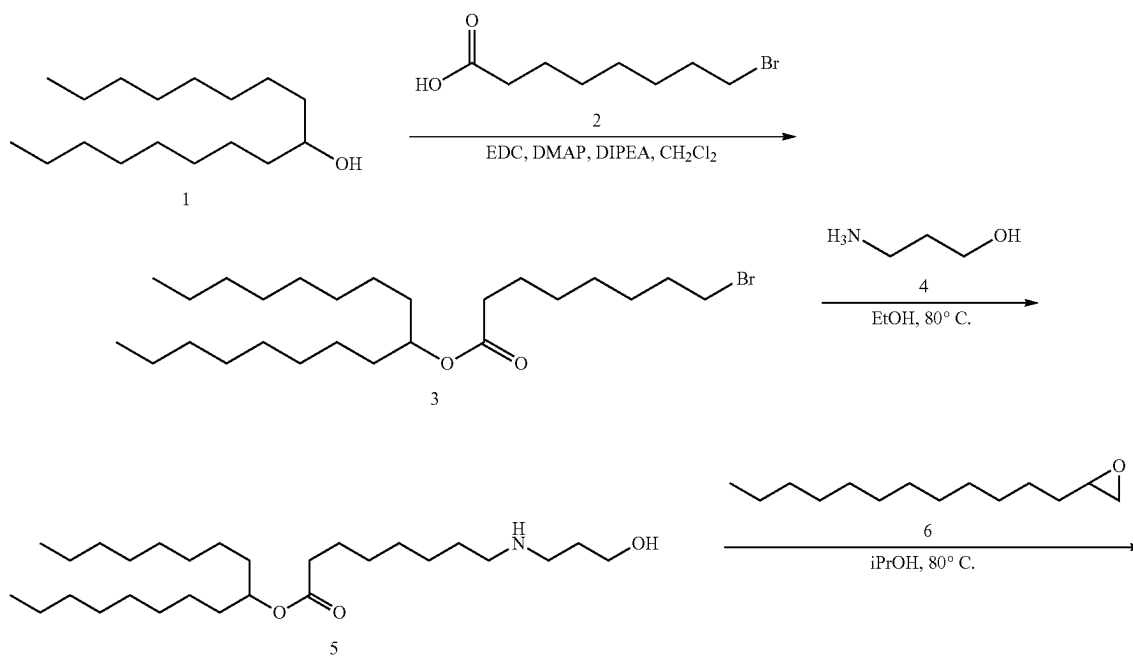
lation initiation factors (ie eIF4E, eIF4a) were inserted into a circular RNA (oRNA) construct preceding a functional TIE and the coding region of a *Gaussia luciferase* reporter sequence. Purified oRNA was formulated into lipid nanoparticles. LNP-oRNAs were transfected into hepatocytes. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates a preference for cap-independent translation. Higher luminescence at 48 hours relative to 24 hours indicates higher oRNA stability due to inhibition of cap-dependent translation.

Example 80B: Coding Aptamers to Enhance Cap-Independent Translation

Cotransfection of oRNA and aptamers. Certain aptamers that bind to eIF4E, eIF4a, and other translation initiators are known to inhibit translation by forcing the proteins to adopt a non-functional conformation. Nucleic acid sequences containing aptamers against translation initiation factors (ie eIF4E, eIF4a) were co-transfected with a circular RNA (oRNA) containing a TIE and the coding region of a *Gaussia luciferase* reporter. Purified aptamer and oRNA were formulated into lipid nanoparticles together. LNP-oRNAs were transfected into hepatocytes. Supernatant was harvested and replaced 24 and 48 hours after transfection, and *Gaussia luciferase* expression from oRNA was determined using a coelenterazine-containing detection reagent and luminometer. Higher luminescence at 24 hours indicates a preference for cap-independent translation. Higher luminescence at 48 hours relative to 24 hours may indicate higher oRNA stability due to inhibition of cap-dependent translation.

Example 81

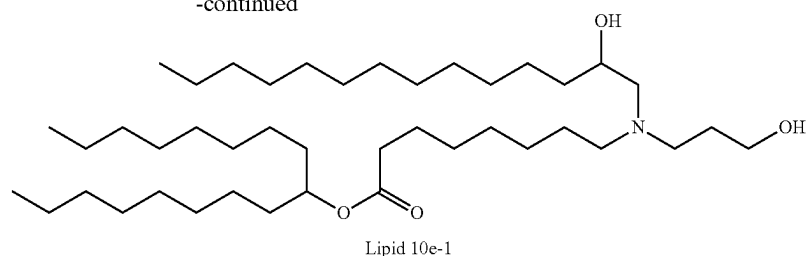
Example 81.1 Synthesis of heptadecan-9-yl 8-((3-hydroxypropyl)(2-hydroxytetradecyl)amino)octanoate (Lipid 10e-1)



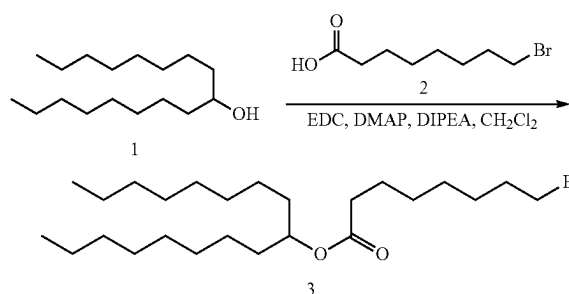
415

416

-continued



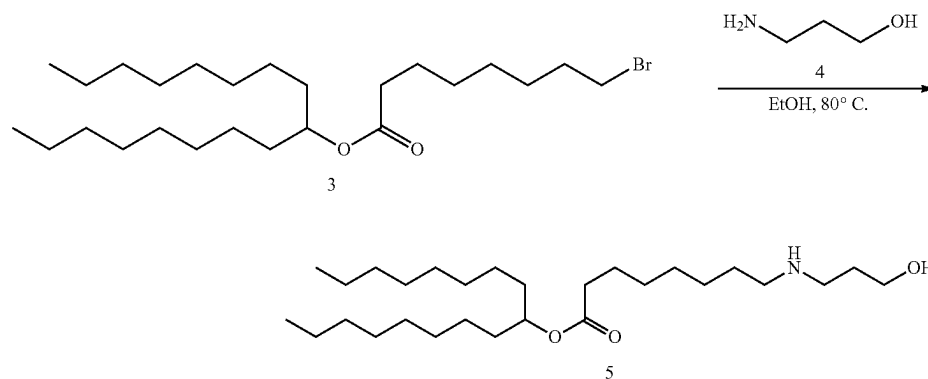
Example 81.1.1 Synthesis of heptadecan-9-yl
8-bromooctanoate (3)



To a mixture of 8-bromooctanoic acid 2 (10 g, 44.82 mmol) and heptadecan-9-ol 1 (9.6 g, 37.35 mmol) in CH_2Cl_2 (300 mL) was added DMAP (900 mg, 7.48 mmol), DIPEA (26 mL, 149.7 mmol) and EDC (10.7 g, 56.03 mmol). The reaction was stirred at room temperature overnight. After concentration of the reaction mixture, the crude residue was dissolved in ethyl acetate (300 mL), washed with 1N HCl, sat. NaHCO_3 , water and Brine. The organic layer was dried over anhydrous Na_2SO_4 . The solvent was evaporated, and the crude residue was purified by flash chromatography (SiO_2 : Hexane=100% to 30% of EtOAc in Hexane) and colorless oil product 3 was obtained (5 g, 29%).

^1H NMR (300 MHz, CDCl_3): δ ppm 4.86 (m, 1H), 3.39 (t, $J=7.0$ Hz, 2H), 2.27 (t, $J=7.6$ Hz, 2H), 1.84 (m, 2H), 1.62 (m, 2H), 1.5-1.4 (m, 8H), 1.35-1.2 (m, 26H), 0.87 (t, $J=6.7$ Hz, 6H).

Example 81.1.2 Synthesis of heptadecan-9-yl
8-((3-hydroxypropyl)amino)octanoate (5)

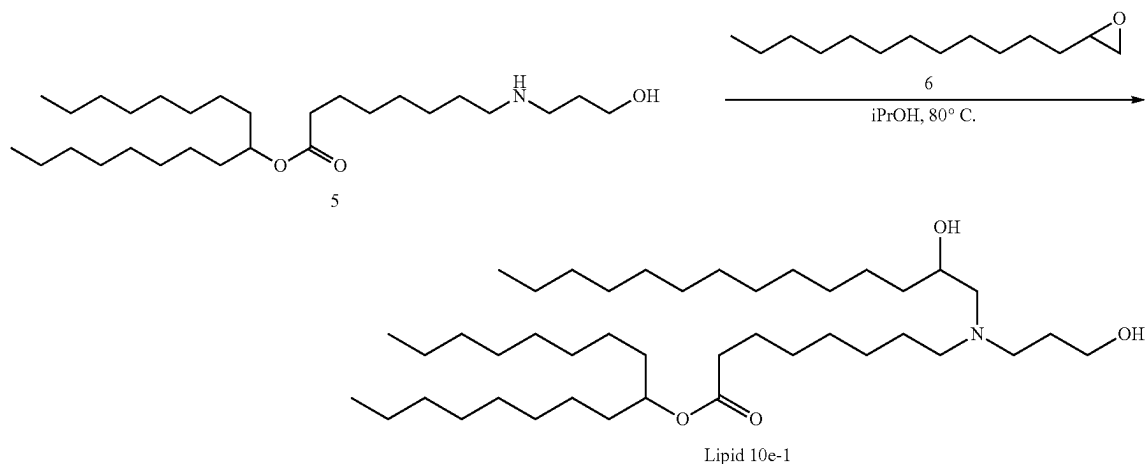


A solution of 1-octylnonyl 8-bromooctanoate 3 (7.4 g, 16.03 mmol) in EtOH (200 mL) was added 3-amino-1-propanol 4 (24.4 mL, 320 mmol) and the reaction solution was heated at 70° C. overnight. MS showed the expected product: [APCI]: [MH]⁺+456.4. After concentration of the reaction mixture, the crude residue was dissolved in methyl tert-butyl ether (500 mL), washed with sat. NaHCO_3 , water and Brine. The organic layer was dried over anhydrous Na_2SO_4 . The solvent was evaporated, and the crude residue was purified by flash chromatography (SiO_2 : CH_2Cl_2 =100% to 10% of MeOH in CH_2Cl_2 with 1% NH_4OH) and colorless oil product 5 was obtained (6.6 g, 88%).

^1H NMR (300 MHz, CDCl_3): δ ppm 4.84 (m, 1H), 3.80 (t, $J=5.5$ Hz, 2H), 2.87 (t, $J=5.76$ Hz, 2H), 2.59 (t, $J=7.2$ Hz, 2H), 2.27 (t, $J=7.6$ Hz, 2H), 1.68 (m, 2H), 1.62 (m, 2H), 1.5-1.4 (m, 5H), 1.35-1.2 (m, 32H), 0.87 (t, $J=6.7$ Hz, 6H). MS (APCI⁺): 456.4 (M+1).

417

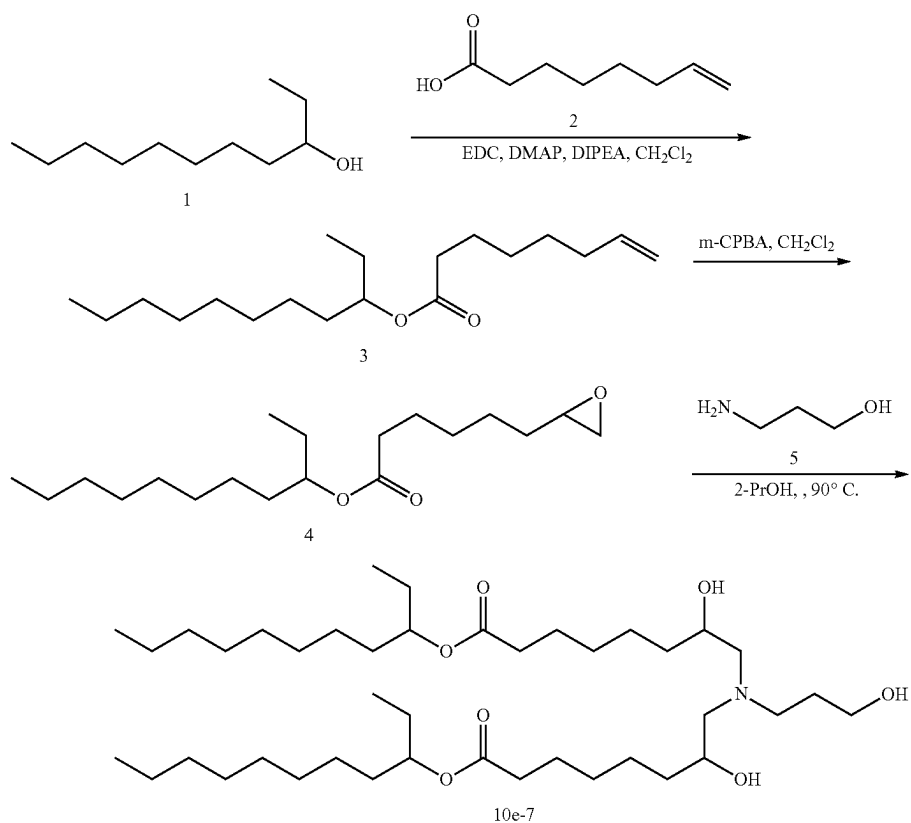
Example 81.1.3 Synthesis of heptadecan-9-yl 8-((3-hydroxypropyl)(2-hydroxytetradecyl)amino)octanoate (7)



A mixture of compound 5 (6.6 g, 14.5 mmol) and 1,2-epoxytetradecane (3.68 g, 17.4 mmol) in isopropanol (150 mL) was heated to reflux for overnight. MS showed the expected product: [APCI]: [MH]⁺668.6. The reaction mixture was concentrated, and crude product was purified flash chromatography (SiO₂: CH₂Cl₂=100% to 10% of MeOH in CH₂Cl₂ with 1% NH₄OH) to obtain Lipid 10e-1 as colorless oil (6.34 g, 65%).

¹H NMR (300 MHz, CDCl₃): δ ppm 4.85 (m, 1H), 3.76 (t, J=5.49 Hz, 2H), 3.68 (m, 1H), 2.75 (m, 1H), 2.59 (m, 2H), 2.38 (m, 3H), 2.27 (m, 2H), 1.58-1.68 (m, 2H), 1.48 (m, 6H), 1.24 (m, 56H), 0.87 (m, 9H). MS (APCI⁺): 668.6 (M+1).

Example 81.2 Synthesis of Di(undecan-3-yl) 8,8'-((3-hydroxypropyl)azanediyl)bis(7-hydroxyoctanoate) (10e-7)



419

Example 81.2.1 Synthesis of undecan-3-yl oct-7-enoate (3)

To a mixture of oct-7-enoic acid 2 (10 g, 70.3 mmol) and undecan-3-ol 1 (10 g, 58.6 mmol) in CH_2Cl_2 (300 mL) was added DMAP (1.4 g, 11.6 mmol), DIPEA (40 mL, 232 mmol) and EDC (16.9 g, 87.9 mmol). The reaction was stirred at room temperature overnight. After concentration of the reaction mixture, the crude residue was dissolved in tert-butylmethyl ether (500 mL), washed with IN HCl, sat. NaHCO_3 , water and Brine. The organic layer was dried over anhydrous Na_2SO_4 . The solvent was evaporated and the crude residue was purified by flash chromatography (SiO_2 ; Hexane=100% to 20% of EtOAc in Hexane) and colorless oil product 3 was obtained (17.2 g, 98%).

^1H NMR (300 MHz, CDCl_3): δ ppm 5.88-5.72 (m, 1H), 5.02-4.91 (m, 1H), 4.80 (m, 1H), 2.28 (t, J=7.4 Hz, 2H), 2.05-2.03 (m, 2H), 1.62-1.49 (m, 6H), 1.37-1.25 (m, 16H), 0.87 (t, J=7.4 Hz, 6H).

Example 81.2.2 Synthesis of undecan-3-yl 6-(oxiran-2-yl)hexanoate (4)

To a mixture of undecan-3-yl oct-7-enoate 3 (17.2 g, 58.1 mmol) in CH_2Cl_2 (3M) mL) was added meta-chloroperoxybenzoic acid (mCPBA, <77%) (19.5 g, 87 mmol) in one portion at 0° C. ice-water bath. The reaction was stirred at room temperature overnight. The white precipitate (meta-benzoic acid) was filtered and the filtrate was diluted with CH_2Cl_2 (200 mL), washed with 10% $\text{Na}_2\text{S}_2\text{O}_3$, sat. NaHCO_3 , water and Brine. The organic layer was dried over anhydrous Na_2SO_4 . The solvent was evaporated and the crude residue was purified by flash chromatography (SiO_2 ; Hexane=100% to 30% of EtOAc in Hexane) and colorless oil product was obtained (17.1 g, 97%).

^1H NMR (300 MHz, CDCl_3): δ ppm 4.80 (m, 1H), 2.89-2.86 (m, 1H), 3.39 (t, J=7.0 Hz, 2H), 2.74 (t, J=4.7 Hz, 1H), 2.47 (dd, J=4.9, 2.2 Hz, 1H), 2.28 (t, J=7.4 Hz, 1H), 1.74-1.46 (m, 10H), 1.35-1.2 (m, 13H) 0.87 (m, 6H).

Example 81.2.3 Synthesis of Di(undecan-3-yl) 8,8'-(3-hydroxypropyl)azanediyl)his(7-hydroxyoctanoate) (10e-7)

A solution of undecan-3-yl 6-(oxiran-2-yl)hexanoate 4 (8 g, 25.6 mmol) in isopropanol (50 mL) was added 3-amino-1-propanol (769.1 mg, 10.2 mmol) and the reaction solution was heated at 90° C. overnight. MS showed the expected product: [APCI]: [MH]⁺700.6. After concentration of the reaction mixture, the crude residue was purified by flash chromatography (SiO_2 ; CH_2Cl_2 =100% to 10% of MeOH in CH_2Cl_2) and colorless oil product was obtained (5.1 g, 71%).

^1H NMR (300 MHz, CDCl_3): δ ppm 4.81 (m, 2H), 3.80 (m, 2H), 3.73 (m, 2H), 2.78 (m, 2H), 2.52-2.43 (m, 4H), 2.28 (t, J=7.3 Hz, 2H), 1.68-1.48 (m, 15H), 1.35-1.17 (m, 37H), 0.88-0.83 (m, 12H). MS (APCI⁺): 700.6 (M+1).

Example 82

Lipid Nanoparticle Formulation Procedure

A Zetasizer Nano ZS (Malvern Instruments Ltd, Malvern, Worcestershire, UK) was used to determine the particle size, the polydispersity index (PDI), and zeta potential of the nanoparticle compositions in 1xPBS in determining particle size and 15 mM PBS in determining zeta potential. A cuvette with 1 mL of 20 $\mu\text{g}/\text{mL}$ LNPs in PBS (pH 7.4) was measured

420

for Z-average using the Malvern Panalytical Zetasizer Pro. The Z-average and polydispersity index were recorded. LNP sizes were determined by dynamic light scattering.

Ultraviolet-visible spectroscopy can be used to determine the concentration of circRNA in nanoparticle compositions. 100 μL of the diluted formulation in 1xPBS is added to 900 μL of a 4:1 (v/v) mixture of methanol and chloroform. After mixing, the absorbance spectrum of the solution is recorded, for example, between 230 nm and 330 nm on a DU 800 spectrophotometer (Beckman Coulter, Beckman Coulter, Inc., Brea, CA). The concentration of circRNA in the nanoparticle composition can be calculated based on the extinction coefficient of the circRNA used in the composition and on the difference between the absorbance at a wavelength of, for example, 260 nm and the baseline value at a wavelength of, for example, 330 nm.

For the transfer vehicle's pKa, a TNS assay was conducted. 5 μL of 60 $\mu\text{g}/\text{mL}$ 2-(p-toluidino) naphthalene-6-sulfonic acid (TNS) and 5 μL of 30 μg of RNA/mL lipid nanoparticles were added in to wells with HEPES buffer ranging from pH 2-12. The mixture was then shaken at room temperature for 5 minutes, and read for fluorescence (excitation 322 nm, emission 431 nm) using a plate reader. The inflection point of the fluorescence signal was calculated to determine the particle's pKa.

For transfer vehicle compositions including RNA, a QUANT-IT[™] RIBOGREEN[®], RNA assay (Invitrogen Corporation Carlsbad, CA) can be used to evaluate the encapsulation of RNA by the transfer vehicle composition. Nanoparticle solutions were diluted in triethylenediaminetetraacetic acid (TE) buffer at a theoretical oRNA concentration of 2 $\mu\text{g}/\text{mL}$. Standard oRNA solutions diluted in TE buffer were made ranging from 2 $\mu\text{g}/\text{mL}$ to 0.125 $\mu\text{g}/\text{mL}$. The particles and standards were plated in a black 96-well plate with both TE buffer and 4% Triton-X separately (Triton-X was used as a surfactant to lyse the nanoparticles). After an incubation (37° C. at 350 rpm for 15 minutes), Quant-iT[™] RiboGreen[™] RNA reagent was added to all wells and a second incubation was performed (37° C. at 350 rpm for 3 minutes). Fluorescence was measured using a SPECTRAMax[®] GEMINI XS microplate spectrofluorometer (Molecular Devices Corporation Sunnyvale, CA). The concentration of oRNA in each particle solution was calculated using the standard curve. The encapsulation efficiency was calculated from the ratio of oRNA detected between lysed and unlysed particles.

Example 83

Expression of mOX40L in Splenic Immune Cells.

Lipid nanoparticles comprising Lipid 1 of Table 10e and Lipid 15 of Table 10f were formulated with circular RNA encoding for mOX40L at an ionizable lipid to phosphate ratio (IL:P) of 5.7. The ionizable lipid: helper lipid: cholesterol: PEG-lipid molar ratio of these LNPs was 50:10:38.5:1.5. Dialysis of the LNPs were performed using PBS. C57BL/6 female mice (6-8 weeks, n=4) were dosed at either LNP comprising Lipid 1 of Table 10e or the Lipid 15 of Table 10f at 1 mg/kg intravenously. At 24 hours, the spleen from the mice were collected for flow cytometry analysis. mOX40L transfection of Lipid 1 of Table 10e and Lipid 15 of Table 10f were compared in T cells, myeloid cells, B cells, and NK cells.

As shown in FIG. 74, Lipid 1 of Table 10e resulted in comparable or higher levels of mOX40L transfection in splenic immune cells compared to those comprising Lipid 15 of Table 10f.

Formulation	Ionizable Lipid	Helper Lipid	PEG-Lipid	Dialysis	Z-Average (nm)	PDI	RNA Encapsulation Efficiency (%)
10e-1 (5.7A)	Table 10e, Lipid 1	DSPC	DMG-PEG(2000)	1X PBS	79	0.03	95
10f-15 (5.7A)	Table 10f, Lipid 15	DSPC	DMG-PEG(2000)	1X PBS	69	0.09	98

Example 84

Fluorescent Expression of Circular RNA Encompassed within LNP Formulations Compared to Linear RNA Encompassed within LNP Formulations.

LNPs were made to either contain circular RNAs encoding for firefly luciferase or linear RNAs (mRNA) encoding for firefly luciferase. The LNPs containing linear RNAs were also modified with 5-methoxyuridine (5-moU). These

LNPs were formulated to contain Lipid 1 or Lipid 7 of Table 10e, wherein the ionizable lipid: helper lipid: cholesterol: PEG-lipid molar ratio of these LNPs was 50:10:38.5:1.5. Dialysis of the LNP was performed using 1× PBS. Size of the LNP construct, polydispersion index (PDI), and RNA entrapment was determined.

As seen in the table below, encapsulation efficiency of circular RNA in each of the LNP formulations was greater than that of linear RNA in the same formulations.

Formulation	Ionizable Lipid	Helper Lipid	PEG-Lipid	RNA	Z-Average (nm)	PDI	RNA Encapsulation Efficiency (%)
10e-1 (5.7A)	Table 10e, Lipid 1	DSPC	DMG-PEG(2000)	FLuc oRNA	82	0.05	95
				FLuc mRNA 5MoU	89	0.05	90
10e-7 (5.7A)	Table 10e, Lipid 7	DSPC	DMG-PEG(2000)	oRNA	86	0.02	97
				FLuc mRNA 5MoU	86	0.04	94

Example 85

Formulated LNPs Undergoing Dialysis Using Either PBS or TSS.

LNPs were formulated with circular RNA at a ionizable lipid to phosphate ratio (IL:P) of 5.7 and a ionizable lipid: helper lipid: cholesterol: PEG-lipid ratio of 50:10:38.5:1.5. These LNPs then underwent dialysis using either 1× PBS or 1× TSS. Both formulated LNPs had greater than 90% encapsulation efficiency ratio of the circular RNAs.

Formulation	Ionizable Lipid	Helper Lipid	PEG-Lipid	Dialysis	Z-Average (nm)	PDI	RNA Encapsulation Efficiency (%)
10e-1 (5.7A)	Table 10e, Lipid 1	DSPC	DMG-PEG(2000)	1X PBS	87	0.06	94
				1X TSS	68	0.04	92

Example 86

Mouse Splenic Protein Expression Post-Treatment of LNP-Circular RNAs Encoding for Firefly Luciferase with Varying B-Hydroxyl Groups in the Ionizable Lipid C57BL/6 mice (female, 6-8 weeks, n=4 per group) were injected intravenously with 0.5 mg/kg circular RNA encoding for firefly luciferase encapsulated in LNPs or PBS control. The LNPs were formulated with different ionizable lipids (Table 10e, Lipid 85, 86, 89, or 90, or Table 10f Lipid 22) and formulated as described in the table below. After 6 hr, mice were injected intraperitoneally with D-luciferin (200 μ L at 15 mg/mL). After 15 minutes, mice were euthanized and their spleens were collected. Whole tissue luminescence was measured ex vivo using an IVIS Spectrum In Vivo Imaging system (PerkinElmer) and total flux was quantified using Living Image® software (PerkinElmer).

As shown in FIG. 75, increasing luciferase expression in the spleen was correlated with increasing numbers of B-hydroxyl groups present in the ionizable lipid component of the LNP.

Formulation	Ionizable Lipid	Helper Lipid	PEG-Lipid	Ionizable lipid:Helper lipid:Cholesterol:PEG-lipid (mol %)	Z-Average (nm)	PDI	RNA Encapsulation Efficiency (%)
10e-85 (5.7A)	Table 10e, Lipid 85	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	81	0.08	91
10e-89 (5.7A)	Table 10e, Lipid 89	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	82	0.04	92
10f-22 (5.7A)	Table 10f, Lipid 22	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	78	0.08	93
10e-86 (5.7A)	Table 10e, Lipid 86	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	79	0.1	92
10e-90 (5.7A)	Table 10e, Lipid 90	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	86	0.04	92

Example 87

Mouse Whole Splenic Protein Expression Post-Treatment of LNP-Circular RNAs Encoding for Firefly Luciferase and Comprising Ionizable Lipids from Table 10e

C57BL/6 mice (female, 6-8 weeks, n=4 per group) were injected intravenously with 0.5 mg/kg circular RNA encoding for firefly luciferase encapsulated in LNPs or PBS control. The LNPs were formulated with different ionizable lipids from Table 10e Lipid 1, Lipid 85, Lipid 38, Lipid 34, Lipid 45, Lipid 86, Lipid 88, Lipid 89, Lipid 90). LNPs were formulated with circular RNA at a ionizable lipid to phosphate ratio (IL:P) of 5.7 and a ionizable lipid:helper lipid: cholesterol: PEG-lipid ratio of 50:10:38.5:1.5. After 6 hr, mice were injected intraperitoneally with D-luciferin (200 μ L at 15 mg/mL). After 15 minutes, mice were euthanized and their spleens were collected. Whole tissue luminescence

of 5.7. The ionizable lipid: helper lipid: cholesterol: PEG-lipid molar ratio of these LNPs was 50:10:38.5:1.5. C57BL/6 female mice (6-8 weeks, n=4) were dosed at 1 mg/kg intravenously. At 24 hours, the spleen from the mice were collected for flow cytometry analysis, tested for weight loss and measured for serum alanine aminotransferase (ALT) after blood collection. mOX40L expression was measured in splenic T cells.

As shown in FIG. 77, Lipid 1 of Table 10e resulted in expression of mOX40L in splenic T cells. No substantial adverse effects were measured pertaining to weight loss or ALT.

Formulation	Ionizable Lipid	Helper Lipid	PEG-Lipid	Ionizable lipid:Helper lipid:Cholesterol:PEG-lipid (mol %)	Z-Average (nm)	PDI	RNA Encapsulation Efficiency (%)
10e-1 (5.7A)	Table 10e, Lipid 1	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	73	0.05	94
10e-16 (5.7A)	Table 10e, Lipid 16	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	76	0.10	95
10e-85 (5.7A)	Table 10e, Lipid 85	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	74	0.04	95
10e-34 (5.7A)	Table 10e, Lipid 34	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	72	0.03	97
10e-45 (5.7A)	Table 10e, Lipid 45	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	75	0.04	98
10e-86 (5.7A)	Table 10e, Lipid 86	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	79	0.05	94
10e-88 (5.7A)	Table 10e, Lipid 88	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	67	0.01	95
10e-89 (5.7A)	Table 10e, Lipid 89	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	73	0.01	95
10e-90 (5.7A)	Table 10e, Lipid 90	DSPC	DMG-PEG(2000)	50:10:38.5:1.5	87	0.02	94

was measured ex vivo using an IVIS Spectrum In Vivo Imaging system (PerkinElmer) and total flux was quantified using Living Image® software (PerkinElmer).

As seen in FIG. 76 the LNP-circular RNAs were able to express firefly luciferase in spleen of the mice post intravenous administration of the construct.

Example 88

Expression of mOX40L in Splenic T Cells.

Lipid nanoparticles comprising ionizable lipids from Table We (Lipid 1, 16, 85, 34, 45, 86, 88, 89, 90) or PBS (negative control) formulated with circular RNA encoding for mOX40L at an ionizable lipid to phosphate ratio (IL:P)

Example 89

Level of B Cell Depletion Post Treatment of LNP-Circular RNAs Encoding for a CD19-CAR

C57BL/6 mice (female, 6-8 weeks, n=4 per group) were injected intravenously with 1 mg/kg circular RNA encoding for anCD19-CAR encapsulated in LNPs or control circular RNA encoding for mWasabi encapsulated in LNPs on Days 0, 2, 5, and 7. The LNPs were formed with different ionizable lipids (Table 10e, Lipid 1, 16, 85, 45, 86, or 90). LNPs were formulated with circular RNA at a ionizable lipid to phosphate ratio (IL:P) of 5.7 and a ionizable lipid:helper lipid: cholesterol: PEG-lipid ratio of 50:10:38.5:1.5. On day

425

8. Cardiac punctures were performed to collect blood, and blood was stained, fixed, and lysed with BD FACS Lysis Solution per the manufacturer's protocol. To assess the frequency of B cells in the blood, single cell suspensions were stained for dead cells (LiveDead Near IR, Invitrogen) and stained with anti-mouse antibodies (CD45, 30-F11, or BUV563 [blood], BD; CD3, 17A2, APC, Biolegend; B220, RA3-6B2, PE, Biolegend; CD11b, M1/70, BV421, Biolegend) at 1:200. Flow cytometry was performed using a BD FACSSymphony flow cytometer. B Cell depletion was defined by the percentage of B220+ B cells of live, CD45+ immune cells.

Resulting blood B cell aplasia is illustrated in FIG. 78. As shown in FIG. 78, the LNP-circular RNA constructs were able to express an aCD19-CAR.

Example 90

Tumor Growth Kinetics Post Administration of LNP-oRNA Construct in a Nalm6 Model

NSG mice were engrafted with Nalm6-luciferase tumor cells and 3 days later were engrafted with human PBMCs.

426

Starting the following day, the mice were treated 4 times every other day with vehicle (PBS) or anti-CD19 LNP-oCAR compounds at a dose of 2 mg/kg. LNPs were formulated with circular RNA at a ionizable lipid to phosphate ratio (IL:P) of 5.7 and a ionizable lipid:helper lipid: cholesterol: PEG-lipid ratio of 50:10:38.5:1.5. Animals were then whole-body imaged via IVIS to monitor luciferase expression from Nalm6 cells. Nalm6 tumor burden is plotted as total flux of luciferase expression at each imaging timepoint.

As shown in FIG. 79, all three anti-CD19 oCAR LNPs each comprising of a different ionizable lipid were capable of slowing tumor growth in this Nalm6 model.

INCORPORATION BY REFERENCE

All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated as being incorporated by reference herein.

SEQUENCE LISTING

Sequence total quantity: 125

SEQ ID NO: 1 moltype = AA length = 493
 FEATURE Location/Qualifiers
 source 1..493
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 1

MALPVTALLL	PLALLLHAAR	PDIVLTQSPA	SLAVSLGERA	TINCRASESV	SVIGAHLIHW	60
YQQKPGQPPK	LLIYLASNLE	TGVPARFSGS	SGGTDFTLTI	SSLQAEDAAI	YYCLQSRIFP	120
RTFGQGTKLE	IKGSTSGSGK	PGSGEGSTKG	QVQLVQSGSE	LKKPGASVKV	SCKASGYTFT	180
DYSINWVRQA	PGQGLEWMGW	INTETREPAY	AYDFRGRFVF	SLDTSVSTAY	LQISSLKAED	240
TAVYYCARDY	SYAMDYWGQG	TLVTVSSAAA	TTTPAPRPPT	PAPTIASQPL	SLRPEACRPA	300
AGGAVHTRGL	DFACDIYIWA	PLAGTCGVLL	LSLVITLYCK	RGRKKLLYIF	KQPFMRPVQT	360
TQEDGCSGR	FPEEEGGGCE	LRVKFSRSAD	APAYQQGQNG	LYNELNLGRR	EEYDVLDRR	420
GRDPEMGKGP	RRKNPQEGLY	NELQKDKMAE	AYSEIGMKGE	RRRGKGHDGL	YQGLSTATKD	480
TYDALHMQAL	PPR					493

SEQ ID NO: 2 moltype = AA length = 207
 FEATURE Location/Qualifiers
 source 1..207
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 2

KNQVEQSPQS	LIILEGKNCT	LQCNVTVSPF	SNLRWYKQDT	GRGPVSLTIM	TFSNTKSNG	60
RYTATLDADT	KQSSSLHITAS	QLSDSASYIC	VVNHSGGSYI	PTFGRGTSLI	VHPYIQKPD	120
AVYQLRDSKS	SDKSVCLFTD	FDSQTNVSQS	KDSDVYITDK	TVLDMRSMDF	KSNSAVAWSN	180
KSDFACANAF	NNSIIPEDTF	FPSPESS				207

SEQ ID NO: 3 moltype = AA length = 246
 FEATURE Location/Qualifiers
 source 1..246
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 3

DVKVTQSSRY	LVKRTGEKVF	LECVDMDME	NMFWRQDPG	LGLRLIYFSY	DVKMKEKGDI	60
PEGYSVSREK	KERFSLILES	ASTNQTSMYL	CASSFLMTSG	DPYEQYFGPG	TRLTVTEDLK	120
NVFPPEVAVF	EPSEAEISHT	QKATLVCLAT	GFYPDHVELS	WWVNGKEVHS	GVSTDPQPLK	180
EQPALNDSRY	CLSRLRVSA	TFWQNPRNHF	RCQVQFYGLS	ENDEWTQDRA	KPVTQIVSAE	240
AWGRAD						246

SEQ ID NO: 4 moltype = AA length = 115
 FEATURE Location/Qualifiers
 source 1..115
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 4

MQEVTVIPAA	LSVPEGENLV	LNCSTFDSAI	YNLQWFRQDP	GKGLTSLLLI	QSSQREQTSG	60
RLNASLDKSS	GRSTLYIAAS	QPGDSATYLC	AVRPTSGGSY	IPTFGRGTSL	IVHPY	115

SEQ ID NO: 5 moltype = AA length = 112
 FEATURE Location/Qualifiers

-continued

```

source          1..112
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 5
MGVTQTPKFKQ VLKGTQSMTL QCAQDMNHEY MSWYRQDPGM GLRLIHYSVG AGITDQGEVP 60
NGYNVSRSTT EDFPLRLLSA APSQTSVYFC ASSYVGNTGE LFFGEGSRLT VL 112

SEQ ID NO: 6      moltype = AA length = 166
FEATURE          Location/Qualifiers
source          1..166
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 6
APPRLICDSR VLERYLLEAK EAENITTGCA EHCSLNENIT VPDTKVNFYA WKRMVGGQQA 60
VEVWQGLALL SEAVLRGQAL LVNSSQFWEP LQLHVDKAVS GLRSLTTLR ALGAQKEAIS 120
PPDAASAAPL RTITADTFRK LFRVYSNFLR GKLKLYTGEA CRTGDR 166

SEQ ID NO: 7      moltype = AA length = 452
FEATURE          Location/Qualifiers
source          1..452
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 7
MSTAVLENPG LGRKLSDFGQ ETSYIEDNCN QNGAISLIFS LKEEVGALAK VLRLFEENDV 60
NLTHIESRPS RLKKDEYEFF THLDRKSLPA LTNIKILRH DIGATVHELK RDKKKTVPW 120
FPRTIQELDR FANQILSYGA ELDADHPGPK DPVYRARRKQ FADIAYNYRH GQPIPRVEYM 180
EBEKKTWGTV FKTLKSLYKT HACYEYNHIF PLLEKYCGFH EDNIPQLEDV SQFLQTCTGF 240
RLRPVAGLLS SRDPLGGLAF RVFHCQYIR HGSKPMYTP EPDICHELGH VPLFSDRSFA 300
QFSQBIGLAS LGAPDEYIEK LATIYWFTVE FGLCKQGDSI KAYGAGLLSS FGELQYCLSE 360
KPKLLPLELE KTAIQNYTVT EFQPLYVAE SFNDAKEKVR NFAATIPRPF SVRYDPYTQR 420
IEVLNQTQQL KILADSINSE IGILCSALQK IK 452

SEQ ID NO: 8      moltype = AA length = 1462
FEATURE          Location/Qualifiers
source          1..1462
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 8
LSVKAQTAHI VLEDGTMKMG YSPGHPSVA GEVVFNTGLG GYPEAITDPA YKGQILTMAN 60
PIINGGAPD TTALDELGLS KYLESNGIKV SGLLVLDYSK DYNHWLATKS LGQWLQEEKV 120
PAIYGVDTRM LTKIIRDKGT MLGKIEFEGQ PVDVDPNPKQ NLIAEVSTKD VKVYGKGNPT 180
KVVAVDCGK NNVIRLLVLR GAHVHLVPWN HDFTKMEYDG ILIAGGPNP ALAEPLIQNV 240
RKILESDRKE PLFGISTGNL ITGLAAGAKT YKMSMANRGQ NQPVNLITNK QAFITAQNHG 300
YALDNTLPAG WKPLFVNVDN QTNNEGIMHES KPFFAVQFHP EVTPGPIDTE YLFDSFFSLI 360
KKGKATTITS VLPKPALVAS RVEVSKVLIL GSGGLSIGQA GEPDYSQSQA VKAMKEENVK 420
TVLMNPNIAS QVTNEVGLKQ ADTVYFLPIT PQFVTEVIKA EQPDGLILGM GGQTALNCGV 480
ELFKRGVLKE YGVKVLGTSV ESIMATEDRQ LFSDKLNEIN EKIAPSPFAVE SIEDALKAAD 540
TIGYPVMIRS AYALGGLGSG ICPNRETLM D LSTKAFAMTN QILVEKSVTG WKEIEYEVVR 600
DADDNCVTVC NMENVDAMGV HTGDSVVVAP AQTLNNAEFQ MLRRTSINVV RHLGIVGECN 660
IQFALHPTSM EYCIIEVNAR LSRSSALASK ATGYPLAFIA AKIALGIPLP EIKNVVSGKT 720
SACFEPSLDY MVTKIPRWDL DRPHGTSSRI GSSMSKSVGEV MAIGRTFEES FQKALRMCHP 780
STEGTPRLP SHAISEHVED LKRELSEPPS TRIYAIKAI DDNMSLDEIE KLTYIDKWFL 840
YKMRDILNME KTLKGLNSES MTEETLKRAK EIGFSDKQIS KCLGLTEAQT RELRLKKNIH 900
PWVKQIDTLA AEYPSVNTYL YVTYNGQEH D VNFDDHGMV LGCGPYHIGS SVEFDWCAVS 960
SIRTLRQLGK KTVVNVCMPE TVSTDDECD KLYFEELSLE RILDYHQEA CGGCIIISVG 1020
QIPNNLAVPL YKNGVKIMGT SPLQIDRAED RSIFS AVLDE LKVAQAPWKA VNTLNEALEF 1080
AKSVDPCLL RPSYVLSGSA MNVVFSEDEM KKFLEEATRV SQEHPVVLTK FVEGAREVEM 1140
DAVGKDRVI SHAISEHVED AGVHSGDATL MLPTQTISQG AIEKVKDATR KIAKAFASG 1200
PFNVQFLVK NDVLVIECNL RASRSPPFVS KTLGVDFIDV ATKVMIGENV DEKHLPTLDH 1260
PIIPADYVAI KAPMFSWRL RDADPILRCE MASTGEVACF GEGHTAFLK AMLSTGFKIP 1320
QKGILIGIQ SFRPRFLGVA EQLHNEGFKL FATEATSDWL NANNVPATPV AWPSQEGQNP 1380
SLSSIRKLIR DGSIDLVINL PNNNTKRVHD NYVIRRTAVD SGIPLLTNFQ VTKLFAEAVQ 1440
KSRKVDKSL FHYRQYSAGK AA 1462

SEQ ID NO: 9      moltype = AA length = 1053
FEATURE          Location/Qualifiers
source          1..1053
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 9
MKRNYILGLD IGITSVGYGI IDYETRDVID AGVRLFKKAN VENNENGRRSK RGARRLKRRL 60
RHRIQRVKKL LFDYNLLTDH SELSGINPYE ARVKGLSQKL SEEEFSAALL HLAKRGRVHN 120
VNEVEEDTGN ELSTKEQISR NSKALEEKYV AELQLERLKK DGEVRGSINR FKTSYVKEA 180
KQLLKVKQAY HQLDQSFID YIDLETRRT YVEGPGEGSP FGWKDIKEWY EMLMGHCTYF 240
PEELRSVKYA YNADLYNALN DLNNLVITRD ENEKLEYEYK FQIIENVFKQ KKKPTLKQIA 300
KEILVNEEDI KGYRVSTGK PEFTNLKVYH DIKDITARKE IENAEELLDQ IAKILTIIYS 360
SEDIQELTN LNSSELTQEEI EQISNLKGYT GTHNLSLKAI NLILDELWHT NDNQIAIFNR 420
LKLVPKKVDL SQQKEIPTL VDDFILSPVV KRSFIQSIKV INAIKKYGL PNDIIIELAR 480
EKNSKDAQKM INEMQKRNQ TNERIEEIR TTGKENAKYL IEKIKLHDMQ EGKCLYSLEA 540

```


-continued

```

IPLEDLLNPN FNYEVDHIIP RSVSFDNSFN NKVLVKQEEEN SKKGNRTPFQ YLSSSDSKIS 600
YETFKKHILN LAKGKGRIK TKKEYLLEER DINRFSVQKD FINRNLDVTR YATRGMLNLL 660
RSYFRVNNLD VKVKSINGGF TSFLRRKWKF KKERNKGYKH HAEDALI IAN ADFIFKEWKK 720
LDKAKKVMEN QMFEEKQAES MPEIETEQEY KEIFITPHQI KHIKDFKDYK YSHRVDKKPN 780
RELINDTLYS TRKDDKGNTL IVNNLNGLYD KDNDKLKKLI NKSPKLLMY HHDPQTYQKL 840
KLIMEQYGDG KNPYKYKYE TGNYLTKYSK KDNGPVIKKI KYYGNKLNH LDITDDYPNS 900
RNKVVKLSLK PYRFDVYLDN GVKFVTVKN LDVIKENY EVNSKCYEEA KKLKKISNQA 960
EFIASFYND LIKINGELR VIGVNDLLN RIEVNMIDIT YREYLENMND KRPPRIIKTI 1020
ASKTQSIKKY STDILGNLYE VKSKKHPQII KKG 1053

```

```

SEQ ID NO: 10      moltype = AA length = 1353
FEATURE           Location/Qualifiers
source            1..1353
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 10
AAGGILHLEL LVAVGPDVFQ AHQEDTERYV LTNLNIGABL LRDPISLGAQF RVHLVKMVIL 60
TEPEGAPNIT ANLTSSLLSV CGWSQTINPE DDTDPGHADL VLYITRFDLE LPDGNRQVRG 120
VTQLGGACSP TWSCLITEDT GFDLGVITIAH EIGHSFGLSH DGAPSGCGCP SGHVMSDGA 180
APRAGLAWSP CSRRQLLSLL SAGRARCVD PPRPQPGSAG HPPDAQPGLY YSANEQCRVA 240
FGPKAVACTF AREHLMCQA LSCHTDPLDQ SSCSRLLVPL LDGTECGVEK WCKGRCRSL 300
VELTPIAAVH GRWSSWGRS PCSRSCGGV VTRRRQCNNP RPAFGGRACV GADLQAEHCN 360
TQACEKTQLE FMSQQCARTD GQPLRSSPGG ASFYHWGAHV PHSQGDALCR HMCRAIGESF 420
IMKRGSFILD GTRCMPGSPR EDGTLSLCVS GSCRTFGCDG RMDSQQVWDR CQVCGDGNST 480
CSPRKGSFTA GRAREYVTFI TVTPNLTSVY IANHRPLFTH LAVRIGGRYV VAGKMSISP 540
TTYPSLLEDG RVEYRVALTE DRLPRLEEIR IWGPLQEDAD IQVYRRYGEE YGNLTRPDIT 600
FTYFQPKPRQ AWWAAVRGP CSVSCGAGLR WVNYSCLDQA RKELVETVQC QGSQQPPAWP 660
EACVLEPCPP YWAVGDFGPC SASCGGLRE RPVRCVEAQ SLLKTLPPAR CRAGAQQPAV 720
ALETCPNPQC PARWEVSEPS SCTSAGGAGL ALENETCVPG ADGLEAPVTE GPGSVDEKLP 780
APEPCVGMSC PPGWGHLDAT SAGEKAPSPW GSIRTGAQAA HVWTPAAGSC SVSCGRGLME 840
LRFLCMDSAL RVPVQEELCG LASKPGSRRE VCQAVPCPAR WQYKLAACSV SCGRGVVRI 900
LYCARAHGED DGEIILLDTQ CQGLPRPEPQ EACSLPECPP RWKVMSLGPC SASCGLTAR 960
RSVACVQLDQ QGDVEVEEAA CAALVRPEAS VPCLIACTY RWHVGTWMEC SVSCGDGIQR 1020
RRDTCGSPQA QAPVPADEFQ HLPKPVTVRG CWAGPCVGGQ TPLVPHHEA AAPGRTTATP 1080
AGASLEWSQA RGLLFSPAPQ PRRLLPGPQE NSVQSSACGR QHLEPTGTID MRGPGQADCA 1140
VAIGRPLGEV VTLRVLESSL NCSAGDMLLL WGRLTWRKMC RKLLDMTFSS KTNLTVVRQR 1200
CGRPGGGVLL RYGSQAPET FYRECQMQLF GPWGEIVSPS LSPATSNAGG CRLFINVAPH 1260
ARIAIHALAT NMAGTEGAN ASYLIRDTLH SLRTTAFHQQ QVLYWESESS QAEMEFSEGF 1320
LKAQASLRGQ YWTLQSWVPE MQDPQSWKKG EGT 1353

```

```

SEQ ID NO: 11      moltype = AA length = 431
FEATURE           Location/Qualifiers
source            1..431
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 11
MPNPRPGKPS APSLALGPSP GASPSWRAAP KASDLLGARG PGGTFQGRDL RGGAHASSSS 60
LNPMPPSQLQ LPTPLVMVA PSGARLGPLP HLQALLQDRP HFMHQLSTVD AHARTPVLQV 120
HPLESFAMIS LTPPTATGV FSLKARPLP PGINVASLEW VSREPALLCT FPNPSAPRKD 180
STLSAVPQSS YPLLAVGCK WPGCEKVFE PEDFLKHCQA DHLDEKGRA QCLLQREMVQ 240
SLEQQVLLEK EKLAMQAH AGKMAITKAS SVASSDKGSC CIVAAGSQGP VVPAWGGPRE 300
APDSLFAVR HLWGSNGHST FPEFLHNDY FKFHNMRPF TYATLIRWAI LEAPEKQRTL 360
NEIYHWFTRM FAFRRNHPAT WKNAIRHNL LHKCFVRVES EKGAVWTVDE LEPRKRSQR 420
PSRCSNPTPG P 431

```

```

SEQ ID NO: 12      moltype = AA length = 160
FEATURE           Location/Qualifiers
source            1..160
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 12
SPGQGTQSEN SCTHFPGNLP NMLRDLRDAF SRVKTFFQMK DQLDNLKKE SLEDFKGYL 60
GCQALSEMIQ FYLEEVMPQA ENQDPDIKAH VNSLGENLKT LRLRLRRCHR FLPCENKSKA 120
VEQVKNAFNK LQEKGIYKAM SEFDIFINYI EAYMTMKIRN 160

```

```

SEQ ID NO: 13      moltype = AA length = 133
FEATURE           Location/Qualifiers
source            1..133
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 13
APTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML TFKFYMPKKA TELKHLQCLE 60
EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR 120
WITFCQSIIS TLT 133

```

```

SEQ ID NO: 14      moltype = AA length = 329
FEATURE           Location/Qualifiers
source            1..329
                  mol_type = protein

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 14
MKFGSKIRRL AVAAVAGAIA LGASFAVAQA PTFFRIGTGG TAGTYYPIGG LIANAISGAG 60
EKGVPGLVAT AVSSNGSVAN INAIKSGALE SGFTQSDVAY WAYNGTGLYD GKGVKEDLRL 120
LATLYPETIH IVARLDANIK SVADLKGRV SLDEPGSGTI VDARIVLEAY GLTEDDIKAE 180
HLKPGPAGER LKDGALDAYF FVGGYPTGAI SELAISNGIS LVPISGPEAD KILEKYSFFS 240
KDVVPAGAYK DVAETPTLAV AAQWVTSKQ PDDLIYNITK VLWNEDTRKA LDAGHAKGKL 300
IKLDSATSSL GIPLHPGAER FYKEAGVLK 329

SEQ ID NO: 15      moltype = AA length = 402
FEATURE            Location/Qualifiers
source              1..402
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 15
MKKLLKSALL FAATGSALS QALPVGNP AE PSLIDGT MW EGASGDP CDP CATWCDAISI 60
RAGYYGYDYV DRVLKVDVNK TFSGMAATPT QATGNASNTN QPEANGRPNI AYGRHMQDAE 120
WFSNA AFLAL NIWDRFDIFC TLGASNGYFK ASSA AFNLVG LIGFSAASSI STDLPMLPN 180
VGITQGVVEF YTDTSFSSV GARGALWECG CATLGAEFQY AQSNPKI EML NVTSSPAQFV 240
IHKPRGYKGA SSNFPLPITA GTTEATDTKS ATIKYHEWQV GLALS YRLNM LVPYIGV NWS 300
RATFDADTIR IAQPKLKSEI LNTTTWNPSL IGSTTALPNN SGKDVLS DVL QIASIQINKM 360
KSRKACGVAV GATLIDADKW SITGEARLIN ERAAHMNAQF RF 402

SEQ ID NO: 16      moltype = AA length = 370
FEATURE            Location/Qualifiers
source              1..370
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 16
MKKLLALVLGL LLVVGSVASA KEVMPAPTPA PEKVVEYVEK PVIVYRDREV APAWRPNGSV 60
DVQYRWYGEV EKKNP KDDKD ENWATGKVNA GRLQTLTKVN FTEKQLEVR TRNHHTLNDT 120
DANNKKSNGA ADEYRLRHFY NFGKLGSSKV NATSRVEFKQ KTNDGEKSLG ASVLPDFADY 180
IYSNNFFKVD KLGLRPGYKY VWKGHGNGEE GTPTVHNEYH LAFESDFTLP FNFALNLEYD 240
LSYNRYREKF ETTDGLKKA E WYGELTAVLS NYTPLYKAGA FELGFNAEGG YDTYNNMHQYK 300
RIGGEDGTSV DRRDYELYLE PTLQVSYKPT DFKVLYAAG ADYRN RITGE SEVKRWRWQP 360
TASAGMKVTF 370

SEQ ID NO: 17      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 17
MNQHFVDVLI GAGLSGIGTA CHVTAEFPDK TIALLERER LGGTWDLFRY PGVRS DSDMF 60
TPGYKFRPWR DVKVLADGAS IRQYIADTAT EFGVDEKIH GLKVNTAEWS SRQCRWTVAG 120
VHEATGETRT YTCDYLI SCT GYNYDAGYL PDPGVHREFG GRCVHPQHWP EDLDYSGKKV 180
VVGSGATAV TLVPAMAGSN PGSAHV TML QRSPSYIFSL PAVDKISEVL GRFLPDRWVY 240
EPGRRRNIAI QRKLYQACRR WPKLMRRLLL WEVRRRLGRS VDMSNFTPNY LPWDERLCAV 300
PNGDLFKTLA SGAASVVDQ IETFTKEGIL KSGREIEAD IIVTATGLNI QMLGGMRLIV 360
DGAEYQLPEK MTYKGV LLEN APNLAWIIGY TNASWTLKSD IAGAYLCRL RHMADNGYTV 420
ATPRDAQDCA LDVGMFDQLN SGYVKRGQDI MPRQGSKHFW RVLMHYEKDA KILLEDPIDD 480
GVLHFAAAQ DHAAA 495

SEQ ID NO: 18      moltype = AA length = 95
FEATURE            Location/Qualifiers
source              1..95
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 18
MTEQQWNFAG IEAASAIQG NVTSIHSLLD EGKQSLTKLA AAWGSGSEA YQGVQKQWDA 60
TATELNALQ NLARTISEAG QAMASTE GNV TGMFA 95

SEQ ID NO: 19      moltype = AA length = 311
FEATURE            Location/Qualifiers
source              1..311
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 19
MKKSLIALTL AALPVAAMAD VTLYGTIKAG VETYRFVAHN GAQASGVETA TEIADLGSKI 60
GPKGQEDLGN GLKAIWQLEQ KAYVSGTNTG WGNRQSFGLG KGGFGKVRVG RLNSVLKDTG 120
GFNPWEGKSE YLSLSNIARP EERPISVRVD SPEFAGFSGS VQYVPNDNSG ENKSESYHAG 180
FNYKNSGFFV QYAGSYKRHN YTTEKHQIHR LVGGYDHDAL YASVAVQQQD AKLAWPDDNS 240
HNSQTEVATT VAYRFGNVTP RVSYAHGFKG SVYEANHDNT YDQVVVGAEY DFSKRTSALV 300
SAGWLQEGKG A 311

SEQ ID NO: 20      moltype = AA length = 107
FEATURE            Location/Qualifiers
source              1..107
                    mol_type = protein

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 20
FVGYPYPSQN PRDYFVPDNE LPPLVHSGFN PSFIATVSHE KGSGDTSEFE ITYGRNMDVT 60
HATRRRTTHYG NSYLEGSRH NAFVNRNYTV KYEVNWKTHE IKVKGHN 107

SEQ ID NO: 21      moltype = AA length = 301
FEATURE           Location/Qualifiers
source            1..301
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 21
EVKLSGDARM GVMYNGDDWN FSSRSRVLFT MSGTTDSGLE FGASFKAHES VGAETGEDGT 60
VFLSGAFGKI EMGDALGASE ALPGDLYEVG YTDLDDRGGN DIPYLTGDER LTAEDNPVLL 120
YTYSAGAFSV AASMSDGKVG ETSDDAQEM AVAAAYTFGN YTVGLGYEKI DSPDTALMAD 180
MEQLELAIAI KFGATNVKAY YADGELDRDF ARAVPDLTPV AAAATAVDHK AYGLSVDSTF 240
GATTVGGVYQ VLDIDTIDDV TTYGLGASYD LGGGASIVGG IADNDLPNSD MVADLGVKFK 300
F 301

SEQ ID NO: 22      moltype = AA length = 346
FEATURE           Location/Qualifiers
source            1..346
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 22
MKKTAIAIAV ALAGFATVAQ AAPKDNTWYT GAKLGWSQYH DTGFINNNGP THENQLGAGA 60
FGGYQVNPYV GFEMGYDWLG RMPYKGSVEN GAYKAQGVQL TAKLGYPTID DLDIYTRLGG 120
MVWRADTKSN VYGKNHDTGV SPVFAGGVEY AITPEIATRL EYQWNNIGD AHTIGTRPDN 180
GMLSLGVSYR FGQGEAAPVV APAPAPAPEV QTKHFTLKSD VLFNFKATL KPEGQAALDQ 240
LYSQLSNLDP KDGSVVVLGY TDRIGSDAYN QGLSERRAQS VVDYLISKGI PADKISARGM 300
GESNPVTGNT CDNVKQRAAL IDC LAPDRRV EIEVKGIKDV VTQPQA 346

SEQ ID NO: 23      moltype = AA length = 87
FEATURE           Location/Qualifiers
source            1..87
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 23
AGVATATGTK SATINYHEWQ VGASLSYRLN SLVPYIGVQW SRATFDADNI RIAQPKLPTA 60
VLNLTAWNPS LLGNATALST TDSFSDF 87

SEQ ID NO: 24      moltype = AA length = 631
FEATURE           Location/Qualifiers
source            1..631
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 24
MTTYQDDFYQ AVNGKWAETA VIPDDKPRTG GFSDLADEIE ALMLDTTDAW LAGENIPDDA 60
ILKNFVKFHR LVADYAKRDE VGVSPILPLI EEYQSLKSFS EFVANIAYKE LAGLPNEFPF 120
SVAPDFMNAQ LNVLWAEAPS ILLPDTTYVE EGNEKAEELR GIWRQSQEKL LPQFGFSTEE 180
IKDLLDKVIE LDKQLAKYVL SREEGSEYAK LYHPYVWADF KKLAPELPLD SIFEKILGQV 240
PDKVIVPEER FWTEFAATYY SEANWDLKKA NLIIVDAANAY NAYLTDDIRV ESGAYSRLS 300
GTPQAMDQKQ AAFYLAQGPF SQALGLWYAG QKFSPEAKAD VESKVARMIE VYKSRLETAD 360
WLPATREKA ITKLNVTIPH IGYPEKLPET YAKKVIDESL SLVENAQNLA KITIAHTWSK 420
WNKPVDRESE HMPAHLVNAY YDPQQNQIVF PAAILQEPFY SLDQSSSANY GGIGAVIAHE 480
ISHAFDTNGA SFDEHGLSLND WWTQEDYAAF KERTDKIVAQ FDGLESHGAK VNGKLTVSEN 540
VADLGGVACA LEAAQSEEDF SARDFFINFA TIWRMKAREE YMQMLASIDV HAPGELRTNV 600
TLTNFDAPHE TFDIKEGDAM WRAPKDRVII W 631

SEQ ID NO: 25      moltype = AA length = 341
FEATURE           Location/Qualifiers
source            1..341
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 25
MNKTLIALAV SAAAVATGAY ADGINQSGDK AGSTVYSAGK TSLEVGGRAE ARLSLKDGA 60
QNSRVRLNLF LGKAEINDSL YGVGFYEGEF TTNDQGKNAS NNSLDNRYTY AGIGGTYGEV 120
TYGKNDGALG VITDFTDIMS YHGNTAAEKI AVADRVNML AYKQFGDLG VKASYRFADR 180
NAVDAMGNVV TETNAAKYSD NGEDGYLSLA IYTFGDTGFN VGAGYADQDD QNEYMLAASY 240
RMENLYFAGL FTDGELAKDV DYTGYELAAG YKLQQAFTA TYNNAETAKE TSADNFAIDA 300
TYYPKPNFRS YISYQFNLLD SDKVGKVASE DELAIGLRYD F 341

SEQ ID NO: 26      moltype = AA length = 160
FEATURE           Location/Qualifiers
source            1..160
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 26
MKGGAGVPDL PSLDASGVRL AIVASSWHGK ICDALLDGAR KVAAGCGLDD PTVVRVLGAI 60
EIPVVAQELA RNHDAVALG VVIRGQTPHF DYVCDAVTQG LTRVSLDSST PIANGVLTNN 120

```

-continued

TEEQALDRAG LPTSAEDKGA QATVAALATA LTLRELRAHS 160

SEQ ID NO: 27 moltype = AA length = 150
 FEATURE Location/Qualifiers
 source 1..150
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 27
 MKKLTKVLLV AGSVAVLAAC GSSKKDESAG QMFGGYSVQD LQQRyntVYF GFDKYNIEGE 60
 YVQILDAHAA FLNATPATKV VVEGNTDERG TPEYNIALGQ RRADAVKHLY SAKGVQAGQV 120
 STVSYGEEKP AVLGHDEAAY SKNRRAYLAY 150

SEQ ID NO: 28 moltype = AA length = 177
 FEATURE Location/Qualifiers
 source 1..177
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 28
 MGISKASLLS LAAAGIVLAG CQSSRLGNLD NVSPPPPPAP VNAVPAQTVQ KGNLDSPTQF 60
 PNAPSTDMSA QSGTQVASLP PASAPDLTPG AVAGVWNASL GGQSCKIATP QTKYGGGYRA 120
 GPLRCFGELA NLASWAVNGK QLVLYDANGG TVASLYSSGQ GRFDGQTTGG QAVTLRSR 177

SEQ ID NO: 29 moltype = AA length = 359
 FEATURE Location/Qualifiers
 source 1..359
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 29
 MQILADLLNT IPADISTAMS RAQRHIDGLL KPVGSLGKLE VLAIQLAGMP GLNGIPHV GK 60
 KAVLVMCADH GVWEEGVAIS PKEVTAIQAE NMTRGTTGVC VLAEQAGANV HVIDVGIDTA 120
 EPIPLINMR VARGSGNIAS APAMSRQAE KLLLDVICYT QELAKNGVTL FGVGELGMAN 180
 TTPAAAIYST ITGRDPEEVV GIGANLPTDK LANKIDVVR AITLNQPNPQ DGVDVLAKVG 240
 GFDLVGIAGV MLGAASCGLP VLLDGFLSYA AALAAQMSAP AIKPYLIPSH LSAEKGARIA 300
 LSHLGLPEYL NMEMRLGEGS GAALAMPIIE AACAIYNNMG ELAASNIVLP GNTTSDLNS 359

SEQ ID NO: 30 moltype = AA length = 172
 FEATURE Location/Qualifiers
 source 1..172
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 30
 MKNARTTLIA AAIAGTLVTT SPAGIANADD AGLDPNAAAG PDAVGFDPNL PPAPDAAPVD 60
 TPPAPEDAGF DPNLPPPLAP DFLSPPAEEA PPVPVAYSVN WDAIAQCESG GNWSINTGNG 120
 YYGGLRFTAG TWRANGSGS AANASREEQI RVAENVLRSQ GIRAWPVCGR RG 172

SEQ ID NO: 31 moltype = AA length = 130
 FEATURE Location/Qualifiers
 source 1..130
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 31
 MAKLSTDELL DAFKEMTLE LSFVKKFEE TFEVTAAPV AVAAAGAAPA GAAVEAAEEQ 60
 SEFDVILEAA GDKKIGVIKV VREIVSGLGL KEAKDLVDGA PKPLLEKVAK EAADEAKAKL 120
 EAAGATVTVK 130

SEQ ID NO: 32 moltype = AA length = 199
 FEATURE Location/Qualifiers
 source 1..199
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 32
 MAENSNIDDI KAPLLAALGA ADLALATVNE LITNLRERAE ETRTDTRSRV EESRARLTKL 60
 QEDLPEQLTE LREKFTAEEL RKAAGYLEA ATSRYNELVE RGEAALERLR SQQSFEEVSA 120
 RAEGYVDQAV ELTQALGTV ASQTRAVGER AAKLVGIELP KKAAPAKKAA PAKKAAPAKK 180
 AAAKKAPAKK AAAKKVTQK 199

SEQ ID NO: 33 moltype = AA length = 594
 FEATURE Location/Qualifiers
 source 1..594
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 33
 MNKIYRIWN SALNAWAVS ELTRNHTKRA SATVATAVLA TLLFATVQAS TTDDDDLYLE 60
 PVQRTAVVLS FRSDKEGTGE KEVTEDSNWG VYFDKKGVL AGTITLKAGD NLKIKQNTNE 120
 NTNASSFTYS LKKDLTDLTS VGTEKLSFSA NSNKVNITSD TKGLNFAKKT AETNGDTTVH 180
 LNGIGSTLTD TLLNTGATTN VTNDNVTDDE KKRAASVKDV NLAGWNIGKV KPGTTASDNV 240
 DFVRTYDTVE FLSADTKTTT VNVESKDNGK RTEVKIGAKT SVIKEKDGKL VTGKDKGEND 300
 SSTDKGEGLV TAKEVIDAVN KAGWRMKTIT ANGQTGQADK FETVTSNTV TFASGKGTTA 360
 TVSKDDQONI TVMYDVNVGD ALNVNQLQNS GWNLDKAVA GSSGKVISGN VSPSKGKMDE 420

-continued

TVNINAGNNI	EITRNGKNID	IATSMTPQFS	SVSLGAGADA	PTLSVDDDEGA	LNVGSKDANK	480
PVRIITNVAPG	VKEGDVTNVA	QLKGVAQNLM	NHIDNVDGNA	RAGIAQAIAI	AGLVQAYLPG	540
KSMMAIGGGT	YRGEAGYAIG	YSSISDGGNW	IIKGTASGNS	RGHFGASASV	GYQW	594

SEQ ID NO: 34 moltype = AA length = 376
 FEATURE Location/Qualifiers
 source 1..376
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 34

MAKQDYIEIL	GVSKTAEERE	IRKAYKRLAM	KYHPDRNQGD	KEAEAKFKEI	KEAYEVLTD	60
QKRAAYDQYG	HAAFEQGGMG	GGGFGGGADF	SDIFGDVFGD	IFGGGRGRQR	AARGADLRN	120
MELTLEEAVER	GVTKEIRIPT	LEECDVCHGS	GAKPGTQPQT	CPTCHGSGQV	QMRQGFFAVQ	180
QTCPHCQGRG	TLIKDPCNKC	HGHGRVERSK	TLSVKIPAGV	DTGDRIRLAG	EGEAGEHGAP	240
AGDLVYVQVQ	KQHPIFEREG	NNLYCEVPIN	FAMAALGGEI	EVPTLDGRVK	LKVPGETQTG	300
KLFRMRGKGV	KSVRGGAGQD	LLCRVVVETP	VGLNERQKQL	LQELQESFGG	PTGEHNSPRS	360
KSFFDGVKKF	FDDLTR					376

SEQ ID NO: 35 moltype = AA length = 471
 FEATURE Location/Qualifiers
 source 1..471
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 35

MANKAVNDPI	LAMNYDKKKL	LTHQGESIEN	RFIKEGNQLP	DEFVIERKK	RLSTNTSDI	60
SVTATNDSRL	YPGALLVDE	TLENNPTLL	AVDRAPMTYS	IDLPGLASSD	SFLQVEDPSN	120
SSVRGAVNDL	LAKWHQDYGO	VNNVPARMQY	EKITAHSMEQ	LKVKFGSDFE	KTGNSLDIDF	180
NSVHSGEKQI	QIVNFKQIYY	TVSVDVKNP	GDVFQDTVTV	EDLKQRGISA	ERPLVYISSV	240
AYGRQVYLKL	ETTSKSDEVE	AAFEALIKGV	KVAPQTEWKQ	ILDNTEVKAV	ILGGDPSSGA	300
RVVTGKVDV	EDLIQEGSRF	TADHPGLPIS	YTTSLRDNV	VATFQNSTDY	VETKVTAYRN	360
GDLLLDHSGA	YVAQYYITWD	ELSYDHQGKE	VLTPKAWDRN	GQDLTAHFTT	SIPLKGVRN	420
LSVKIRECTG	LAWEWRTVY	EKTDLPLVRK	RTISIWGTTL	YPQVEDKVEN	D	471

SEQ ID NO: 36 moltype = AA length = 498
 FEATURE Location/Qualifiers
 source 1..498
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 36

MAQVINTNSL	SLITQNNINK	NQALSSSIE	RLSSGLRINS	AKDDAAGQAI	ANRFTSNIKG	60
LTQAARNAND	GISVAQTTEG	ALSEINNMLQ	RVRELTVQAT	TGTNSEDLS	SIQDEIKSRL	120
DEIDRVSGQT	QFNGVNVLAK	NGSMKIQVGA	NDNQITITIDL	KQIDAKTLGL	DGFSVKNNND	180
VTTAPVTFAP	GATTNNIKL	TGTLSTEAA	TDGTGTPAS	IEGVYTDNGN	DYYAKITGGD	240
NDGKYAVTV	ANDGVTMTAT	GATANATVTD	ANTTKATTIT	SGGTPVQIDN	TAGSATANLG	300
AVSLVKLQDS	KGNDTDYAL	KDTNGNLYAA	DVNETTGAVS	VKTITYTDSS	GAASSPTAVK	360
LGDDDGKTEV	VIDGKTYDS	ADLNGGNLQT	GLTAGGEALT	AVANGKTDTP	LKALDDAIAS	420
VDKFRSSSLG	VQNRLDASVT	NLNNTTTNLS	EAQSRIQDAD	YATEVSNMSK	AQIIQQAGNS	480
VLAKANQVPQ	QVLSLLQG					498

SEQ ID NO: 37 moltype = AA length = 189
 FEATURE Location/Qualifiers
 source 1..189
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 37

MASPFALLMV	LVLVLSCKSSC	SLGCDLPETH	SLDNRRTLML	LAQMSRISPS	SCLMDRHDFFG	60
FPQEEFDGNQ	FQKAPAISVL	HELIQQIFNL	FTTKDSSAAW	DEDLLDKFCT	ELYQQLNDLE	120
ACVMQERVG	ETPLMNADSI	LAVKKYFRRI	TLYLTEKKYS	PCAWEVVRAE	IMRSLSLSTN	180
LQERLRRKE						189

SEQ ID NO: 38 moltype = AA length = 166
 FEATURE Location/Qualifiers
 source 1..166
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 38

MKYTSYILAF	QLCIVLGLSG	CYCQDPYVKE	AENLKPYFNA	GHSVDADNGT	LFLGILKNWK	60
EESDRKIMQS	QIVSFYFKLF	KNFKDDQSIQ	KSVETIKEDM	NVKFFNSNKK	KRDDFEKLTN	120
YSVTDLNVQR	KAIHELIQVM	AELSPAATKG	KRKRQMLFR	GRRASQ		166

SEQ ID NO: 39 moltype = AA length = 153
 FEATURE Location/Qualifiers
 source 1..153
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 39

MYRMQLLSCT	ALSLALVTNS	APTSSSTKKT	QLQLEHLLLD	LQMILNGINN	YKNPKLTRML	60
TFKFMYPKKA	TELKHLQCLE	EELKPLEEVL	NLAQSKNFHL	RPRDLISIN	VIVLELKGSE	120
TTFMCEYADE	TATIVEFLNR	WITFCQSIIS	TLT			153

-continued

SEQ ID NO: 40 moltype = AA length = 39
 FEATURE Location/Qualifiers
 source 1..39
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 40
 MWPPGSASQP PPSAAATGL HPAARPVSLQ CRLSMCPAR 39

SEQ ID NO: 41 moltype = AA length = 338
 FEATURE Location/Qualifiers
 source 1..338
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 41
 MGKKQNRKTG NSKTQSASPP PKERSSSPAT EQSWMENDFD ELREEGFRRS NYSELREDIQ 60
 TKGKEVENFE KNLEECITRI SNTEKCLKEL MELKTKTREL REECRSLRSR CDQLEERVSA 120
 MEDEMNMEMR EGKPREKRIK RNEQTLQEIW DYVKRPNLRL IGVPESDVEN GTKLENTLQD 180
 IIQENFPNLA RQANVQIQEI QRTQRYSSR RATPRHIVR FTKVEMKEKM LRAAREKGRV 240
 TLKGKPIRLT ADLLAETLQA RREWGPIFNI LKGKNFQPRI SYPAKLSFIS EGEIKYFIDK 300
 QMLRDFVTRR PALKELLKEA LNMERNNRYQ LLQNHAKM 338

SEQ ID NO: 42 moltype = AA length = 162
 FEATURE Location/Qualifiers
 source 1..162
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 42
 MRISKPHLRS ISIQCYLCLL LNSHFLTEAG IHVFILGCPS AGLPKTEANW VNVISDLKKI 60
 EDLIQSMHID ATLYTESDVH PSCKVTAMKC FLLELQVISL ESGDASIHDT VENLIILANN 120
 SLSSNGNVTE SGCKECEELE EKNIKEFLQS FVHIVQMFIN TS 162

SEQ ID NO: 43 moltype = AA length = 193
 FEATURE Location/Qualifiers
 source 1..193
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 43
 MAAEPVEDNC INFVAMKFID NTLYFIAEDD ENLESDFYFK LESKLSVIRN LNDQVLFIDQ 60
 GNRPLFEDMT DSDCRDNAPR TIFIISMYKD SQPRGMAVTI SVKCEKISTL SCENKIISFK 120
 EMNPPDNIKD TKSDIIFQOR SVPGHDKMQ FESSSYEGYF LACEKERDLF KLILKKEDEL 180
 GDRSIMFTVQ NED 193

SEQ ID NO: 44 moltype = AA length = 162
 FEATURE Location/Qualifiers
 source 1..162
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 44
 MRSSPGNMER IVICLMVIFL GTLVHKSSSQ GQDRHMIRM MRQLIDIVDQLK NYVNDLVPEF 60
 LPAPEDVETN CEWSAFSCFQ KAQLKSANTG NNERIINVTI KKLKRKPPST NAGRRQKHRL 120
 TCPSCDSYEK KPPKEFLERF KSLLQKMIHQ HLSSRTHGSE DS 162

SEQ ID NO: 45 moltype = AA length = 144
 FEATURE Location/Qualifiers
 source 1..144
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 45
 MWLQSLLLLG TVACISAPAP RSPSPSTQPW EHVNAIQEAR RLLNLSRDTA AEMNETVEVI 60
 SEMFDLQEPD CLQTRLELYK QGLRGS�TKL KGPLTMMASH YKQHCPTPE TSCATQIITF 120
 ESKENLKDF LLVIPPDCWE PVQE 144

SEQ ID NO: 46 moltype = AA length = 269
 FEATURE Location/Qualifiers
 source 1..269
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 46
 MAEVPCLASE MMAYYSGNED DLFFFEADGPK QMKCSFQDLD LCPLDGGIQL RISDHHYSKG 60
 FRQAASVVVA MDKLRKMLVP CPQTFQENDL STFFPFIFEE EPIFFDTWDN EAYVHDAPVR 120
 SLNCTLRDSQ QKSLVMGPGY ELKALHLQGG DMEQQVVFMS SFVQGEESND KIPVALGLKE 180
 KNLYLSCVLK DDKPTLQLES VDPKNYPKKK MEKRFVFNKI EINKLEFES AQFPNWYIST 240
 SQAENMPVFL GGTGKGQDIT DFTMQFVSS 269

SEQ ID NO: 47 moltype = AA length = 212
 FEATURE Location/Qualifiers
 source 1..212
 mol_type = protein

-continued

```

                                organism = synthetic construct
SEQUENCE: 47
MNSFSTSAFG PVAFSLGLLL VLPAAFPAPV PPGEDSKDVA APHRQPLTSS ERIDKQIRYI 60
LDGISALRKE TCNKSNNCES SKEALAENNL NLPKMAEKDG CFQSGFNEET CLVKIITGLL 120
EFEVYLEYLQ NRPESSEEA RAVQMSTKVL IQFLQKKAKN LDAITTPDPT TNASLLTKLQ 180
AQNQWLQDMT THLILRSFKE FLQSSLRALR QM 212

SEQ ID NO: 48                moltype = AA length = 233
FEATURE                      Location/Qualifiers
source                      1..233
                             mol_type = protein
                             organism = synthetic construct

SEQUENCE: 48
MSTESMIRDV ELAEEALPKK TGGPQGSRRR LFLSLFSFLI VAGATTLFCL LHFGVIGPQR 60
EEFPRLDSL I SPLAQAVRSS SRTPSDKPVA HVVANPQAEQ QLQWLNRRAN ALLANGVELR 120
DNQLVVPSEG LYLIYSQVLF KGQGCPSHV LLTHTISRIA VSYQTKVNLL SAIKSPCQRE 180
TPEGAEAKPW YEPIYLGGVF QLEKGDRLSA EINRPDYLDF AESGQVYFGI IAL 233

SEQ ID NO: 49                moltype = AA length = 177
FEATURE                      Location/Qualifiers
source                      1..177
                             mol_type = protein
                             organism = synthetic construct

SEQUENCE: 49
MPHVSFRYIF GLPPLILVLL PVASSDCDIE GKDGGKQYESV LMVSIQQLLD SMKEIGSNCL 60
NNEFNFFKRH ICDANKEGMF LFRAARKLRQ FLKMNSTGDF DLHLLKVSEG TTILLNCTGQ 120
VKGRKPAALG EAQPTKSL EE NKLKEQKKL NDLCLKRLLL QEIKTCWNKI LMGTKEH 177

SEQ ID NO: 50                moltype = AA length = 155
FEATURE                      Location/Qualifiers
source                      1..155
                             mol_type = protein
                             organism = synthetic construct

SEQUENCE: 50
MTPGKTSLV S LLLLLLEAI VKAGITIPRN PGCPNSEDKN PPRTVMVNLN IHNRTNTNTP 60
KRSSDYNNRS TSPWNLHRE DPERYPSVIW EAKCRHLGCI NADGNVDYHM NSVPIQQEIL 120
VLRREPPHCP NSFRLKILV SVGCTCVTPI VHHVA 155

SEQ ID NO: 51                moltype = AA length = 235
FEATURE                      Location/Qualifiers
source                      1..235
                             mol_type = protein
                             organism = synthetic construct

SEQUENCE: 51
MTVLAPAWSP TTYLLLLLLL SSGLSGTQDC SFQHSPISSD FAVKIRELSD YLLQDYPVTV 60
ASNLDDEELC GGLWRLVLAQ RWMERLKTVA GSKMQGLLER VNTEIHFTK CAFQPPPSCL 120
RFVQTNISRL LQETSEQLVA LKPWITRQNF SRCLELQCQP DSSTLPPPW S PRPLEATAPT 180
APQPLLLLLL LLPVGLLLLA AAWCLHWQRT RRRTPRPGEQ VPPVPSPQDL LLVEH 235

SEQ ID NO: 52                moltype = AA length = 448
FEATURE                      Location/Qualifiers
source                      1..448
                             mol_type = protein
                             organism = synthetic construct

SEQUENCE: 52
QVQLVESGGG VVQPGSRSLRL SCAASGFTFS SYTMHWVRQA PGKGLEWVTF ISYDGNKYY 60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAIYYCARTG WLGPFDYWGQ GTLVTVSSAS 120
TKGPSVFPLA PSSKSTSGGT AALGCLVKDY FPEPVTVSWN SGALTSGVHT FPAVLQSSGL 180
YSLSSVVTVP SSSLGTQTYI CNVNHKPSNT KVDKRVPEKS CDKTHTCPPC PAPELLGGPS 240
VFLFPPPKPD TLMISRTPEV TCVVVDVSHE DPEVKFNWYV DGEVHNNAKT KPREEQYNST 300
YRVVSVLTVL HQDWLNGKEY KCKVSNKALP APIEKTISKA KGQPREPQVY TLPPSRDEL T 360
KNQVSLTCLV KGFYPSDIAV EWESNGQPEN NYKTTPPVLD SDGSFFLYSK LTVDKSRWQQ 420
GNVFSCSVMH EALHNHYTQK SLSLSPGK 448

SEQ ID NO: 53                moltype = AA length = 440
FEATURE                      Location/Qualifiers
source                      1..440
                             mol_type = protein
                             organism = synthetic construct

SEQUENCE: 53
QVQLVESGGG VVQPGSRSLRL DCKASGITFS NSGMHWVRQA PGKGLEWVAV IWYDGSKRYY 60
ADSVKGRFTI SRDNSKNTLF LQMNSLRAED TAVYYCATND DYWGQGTLT V VSSASTKGPS 120
VFLAPCSRS TSESTAALGC LVKDYFPEPV TVSWNSGALT SGVHTFPAVL QSSGLYSLSS 180
VTVVPSSSLG TKTYTCNVDH KPSNTKVDKR VESKYGPCCP PCPAPAEFLGG PSVFLFPPKP 240
KDTLMISRTP EVTCVVVDVS QEDPEVQFNW YVDGVEVHNA KTKPREEQFN STYRVVSVLT 300
VLHQDWLNGK EYKCKVSNKG LPSSIEKTIS KAKGQPREPQ VYTLPPSQEE MTKNQVSLTC 360
LVKGFYPSDI AVEWESNGQP ENNYKTTTPV LDSDGSFFLY SRLTVDKSRW QEGNVFSCSV 420
MHEALHNHYT QKSLSLSLGK 440

```

-continued

SEQ ID NO: 54 moltype = AA length = 442
 FEATURE Location/Qualifiers
 source 1..442
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 54

EVQLVQSGAE	VKKPGESLRI	SCKGSGYSFS	TYWISWVRQM	PGKGLEWMGK	IYPGDSYTNV	60
SPSFQGGVTI	SADKSISTAY	LQWSSLKASD	TAMYVCARGY	GIFDYWGQGT	LVTVSSASTK	120
GPSVFPLAPC	SRSTSESTAA	LGCLVKDYFP	EPVTVSWNSG	ALTSGVHTFP	AVLQSSGLYS	180
LSSVTVTPSS	NFGTQTYTCN	VDHKPSNTKV	DKTVERKCCV	ECPPCPAPPV	AGPSVFLFPP	240
KPKDTLMISR	TPEVTCVVVD	VSHPDPEVQF	NWYVDGVEVH	NAKTKPREEQ	FNSTFRVVS	300
LTVVHQDWLN	GKEYKCKVSN	KGLPAPIEKT	ISKTKGQPRE	PQVYTLPPSR	EEMTKNQVSL	360
TCLVKGFPYS	DIAVEWESNG	QPENNYKTP	PMLDSGDSFF	LYSKLTVDKS	RWQQGNVFS	420
SVMHEALHNN	YTQKSLSLSP	GK				442

SEQ ID NO: 55 moltype = DNA length = 625
 FEATURE Location/Qualifiers
 source 1..625
 mol_type = unassigned DNA
 organism = unidentified
 note = AP1.0 sequence

SEQUENCE: 55

attctcgggc	tacggccctg	gagccactcc	ggctcctaaa	gatttagaag	tttgagcaca	60
cccgccact	agggccccc	atccagggg	gcaacgggca	agcacttctg	tttccccggt	120
atgactctgat	aggctgtaac	cacggctgaa	acagagatta	tcgttatccg	cttcaactact	180
tcgagaagcc	tagtaaatgat	gggtgaaatt	gaatccgttg	atccgggtgc	tccccacac	240
cagaaactca	tgatgagggt	tgccatcccc	gctacggcga	cgtagcgggc	atccctgcgc	300
tggcatgagg	cctcttagga	ggacggatga	tatggatcct	gtcgtgaaga	gcctattgag	360
ctagtgtcga	ctcctccgcc	cccgtaaatg	cggctaattc	taaccccgga	gcagggtgggt	420
ccaatccagg	gcctggcctg	tcgtaaatgc	taagtctggg	acggaaccga	ctactttcgg	480
gaaggcgtgt	ttccatttgt	tcattatttg	tgtgtttatg	gtgacaactc	tgggtaaacg	540
ttctattgcg	tttattgaga	gattcccaac	aattgaacaa	acgagaacta	cctgttttat	600
taaatttaca	cagagaagaa	ttaca				625

SEQ ID NO: 56 moltype = DNA length = 549
 FEATURE Location/Qualifiers
 source 1..549
 mol_type = other DNA
 organism = unidentified
 note = CK1.0 sequence

SEQUENCE: 56

gtggccacgc	ccgggccacc	gatacttccc	ttcactcctt	cgggactggt	ggggagggaac	60
acaacagggc	tcccctgttt	tcccattcct	tcccctttt	cccaacccca	accgccgtat	120
ctgggtggcgg	caagacacac	gggtcttttc	ctctaaagca	caattgtgtg	tgtgtcccag	180
gtcctcctgc	gtacgggtgcg	ggagtgtctc	caccacaactg	ttgtaaacct	gtccaacgcg	240
tcgtcctggc	aagactatga	cgtcgcatgt	tcgctgcgg	atgcgcacgg	ggtaaccgggt	300
tccccagtg	gtgtagtgcg	atcttccagg	tctcctcgtt	tggcggtgtc	cagaaactgc	360
ttcagggtaa	tgggggtgtc	ccaatcccta	caaagggtga	ttctttcacc	accttaggaa	420
tgctccggag	gtacccacgc	aacagctggg	atctgaccgg	aggctaattg	tctacgggtg	480
gtgtttcctt	tttcttttca	cacaactcta	ctgtcgacaa	ctcactgact	atccacttgc	540
tctgtcacg						549

SEQ ID NO: 57 moltype = DNA length = 630
 FEATURE Location/Qualifiers
 source 1..630
 mol_type = unassigned DNA
 organism = unidentified
 note = PV1.0 sequence

SEQUENCE: 57

aacaaaaggc	tacaccactt	gggctacggc	ccgcgccacc	ttgtggcgca	aagacattag	60
aagaatagca	taccgcccac	tagggccctg	cagccagcag	ggtaacgggc	aagcacttct	120
gtctccccgg	tagaacggta	taggctgtac	ccacggccga	aaactgaact	atcgttaccc	180
gactccgtac	ttcgcaaaag	ttagtaggaa	actggaaagt	tcgagttatt	gacccggagt	240
gttcccccca	ctccagaaac	gcgtgatgag	ggttgccacc	ccgacctggg	cgacatgggt	300
ggcatccctg	cgctggcacg	cggcctctaa	gaggataact	cgctcctact	ggtaaccgaa	360
gagccccgtg	agctacgggt	tattcctccg	cctccctgaa	tcggcgtaat	cctaaccat	420
gagcagttgc	catagatcca	tatggtggac	tgtcgtaacg	cgtaagttgt	gggcgggaac	480
gactactttg	ggatggcggt	tttcttgtt	ttctccattt	gttggtgtat	ggtgacaagt	540
tatagatctc	gatctatagc	gtttcttgag	agatttccaa	acattttatt	aagtcgtaca	600
attcttgtgt	ttaagcagta	cagtgtaac				630

SEQ ID NO: 58 moltype = DNA length = 653
 FEATURE Location/Qualifiers
 source 1..653
 mol_type = unassigned DNA
 organism = unidentified
 note = SV1.0 sequence

SEQUENCE: 58

tctgtctcca	ccccatcttc	ccttcttttc	tgcaccgtta	cgcttactcg	catgtgcatt	60
------------	------------	------------	------------	------------	------------	----

-continued

```

gagtgggtgca cgtgcttgaa caaacagcta cactcacatg ggggcggtt tccccccct 120
gcggcctctc gcgaggccca cccctcccct tctcccata actacagtgc ttggtaggt 180
aagcatcctg atcccccgcg gaagctgctc acgtggcaac tgtggggacc cagacagggt 240
atcaaaggca cccggtcttt ccgccttcag gagtatccct gctagtgaat tctagtaggg 300
ctctgcttgg tgccaacctc ccccaaatgc gcgctgcggg agtgctcttc cccaactcac 360
cctagtatcc tctcatgtgt gtgcttggtc agcatatctg agacgatgtt ccgctgtccc 420
agaccagtcc agtaatggac gggccagtgt gcgtagtcgt ctcccggtt gtccggcgca 480
tgtttggtga accgggtggg taagggttgg gtgcccaacg cccgtacttt ggtgatacct 540
caagaccacc caggaatgcc agggagggtac ccgccttcac agcgggatct gaccctgggc 600
taattgtcta cgggtgttct tcttgcttcc acttctttct actgttcgcc acc 653

```

```

SEQ ID NO: 59      moltype = DNA length = 635
FEATURE           Location/Qualifiers
source            1..635
                  mol_type = genomic DNA
                  organism = Caprine kobuvirus
                  strain = 5delta40

```

```

SEQUENCE: 59
gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaa 60
acaacagggc tccctgtttt tccattcct tccccctttt cccaacccca accgcggtat 120
ctggtggcgg caagacacac ggggtctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtctcctgc gtacggtgcg ggagtgcctc cacccaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcgtgt tccgctgcgg atgccgaccg ggtaaccggg 300
tccccagtgt gtgtagtgcg atcttccagg tctcctcgtt tggcgttgtc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtaccccagc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta ctgctgacaa ctactgact atccacttgc 540
tctcttgcgc cttctgcctc tgggtcaagt tcttgattg tttttgactg cttttcactg 600
cttttcttct cacaactcct gctcagttca aagtc 635

```

```

SEQ ID NO: 60      moltype = DNA length = 513
FEATURE           Location/Qualifiers
source            1..513
                  mol_type = genomic DNA
                  organism = Caprine kobuvirus
                  strain = 5delta40/3delta122

```

```

SEQUENCE: 60
gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaa 60
acaacagggc tccctgtttt tccattcct tccccctttt cccaacccca accgcggtat 120
ctggtggcgg caagacacac ggggtctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtctcctgc gtacggtgcg ggagtgcctc cacccaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcgtgt tccgctgcgg atgccgaccg ggtaaccggg 300
tccccagtgt gtgtagtgcg atcttccagg tctcctcgtt tggcgttgtc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtaccccagc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactaaa gtc 513

```

```

SEQ ID NO: 61      moltype = DNA length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = genomic DNA
                  organism = Caprine kobuvirus
                  strain = 5delta40/3delta86_Distal

```

```

SEQUENCE: 61
gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaa 60
acaacagggc tccctgtttt tccattcct tccccctttt cccaacccca accgcggtat 120
ctggtggcgg caagacacac ggggtctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtctcctgc gtacggtgcg ggagtgcctc cacccaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcgtgt tccgctgcgg atgccgaccg ggtaaccggg 300
tccccagtgt gtgtagtgcg atcttccagg tctcctcgtt tggcgttgtc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtaccccagc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta ctgctgacaa ctactgact atccacttgc 540
tctaaagtc 549

```

```

SEQ ID NO: 62      moltype = DNA length = 513
FEATURE           Location/Qualifiers
source            1..513
                  mol_type = genomic DNA
                  organism = Caprine kobuvirus
                  strain = 5delta40/3delta122_Kozak

```

```

SEQUENCE: 62
gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaa 60
acaacagggc tccctgtttt tccattcct tccccctttt cccaacccca accgcggtat 120
ctggtggcgg caagacacac ggggtctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtctcctgc gtacggtgcg ggagtgcctc cacccaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcgtgt tccgctgcgg atgccgaccg ggtaaccggg 300
tccccagtgt gtgtagtgcg atcttccagg tctcctcgtt tggcgttgtc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc accttaggaa 420

```

-continued

```

tgctccggag gtacccagc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactgcc acc 513

```

```

SEQ ID NO: 63      moltype = DNA length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = genomic DNA
                  organism = Caprine kobuvirus
                  strain = 5delta40/3delta86 Proximal

```

```

SEQUENCE: 63
gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggagggaac 60
acaacagggc tcccctgttt tcccatctct tccccctttt cccaacccca accgccgtat 120
ctgggtggcgg caagacacac ggggtctttcc ctctaaagca caattgtgtg tgtgtcccag 180
gtctctctgc gtacgggtgcg ggagtgtctc cacccaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcgatg tccgctgcgg atgccgaccg ggtaaccggt 300
tccccagtggt gtgtagtgcg atcttccagg tcctcctcgt tggcgttgtc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtacccagc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactttc actgcttttc ttctcacaat ccttgctcag 540
ttcaaatgc 549

```

```

SEQ ID NO: 64      moltype = DNA length = 678
FEATURE           Location/Qualifiers
source            1..678
                  mol_type = genomic DNA
                  organism = Picornavirales sp.
                  note = isolate RtNn-PicoV/HuB2015-3 polyprotein gene-1

```

```

SEQUENCE: 64
tgaaccgtta cgcaccactc agttggtgtt tgggtggcacc aatgatggaa caaaaggcta 60
caccacttgg gctacggccc gcgccacctt gtggcgcaaa gacattagaa gaatagcata 120
ccgccacta gggccctgca gccagcaggg taacgggcaa gcacttctgt ctccccggtg 180
gaacgggata ggctgtaccc acgggcgaaa actgaactat cgttaccoga ctccgtactt 240
cgcaagcctt agtaggaaac tggaaagtgc gagttattga cccggagtgt tccccccact 300
ccagaaacgc gtgatgaggg ttgccacccc gaccatggcg acatgggtgg catccctgcg 360
ctggcacgcg gcctctaaga ggataactcg ctccactactg taaccgaaga gccccgtgag 420
ctacgggttta ttctcgcgcc tcctcgaatg cggctaatac taaccatga gcagttgcca 480
tagatccata tgggtgactg tcgtaacgcg taagtgtgtg gcggaaccga ctactttggg 540
atggcgtgtt tccttgtttt ctccatttgt tgttgtatgg tgacaagtta tagatctcga 600
tctatagcgt ttcttgagag atttccaaac atttattcaa gtctgacaat tcttgtgttt 660
aagcagtaca gtgtaagg 678

```

```

SEQ ID NO: 65      moltype = DNA length = 630
FEATURE           Location/Qualifiers
source            1..630
                  mol_type = genomic DNA
                  organism = Parabovirus sp.
                  strain = 5delta48

```

```

SEQUENCE: 65
aacaaaaggc tacaccactt gggctacggc ccgcgccacc ttgtggcgca aagacattag 60
aagaatagca taccgcccac tagggccctg cagccagcag ggtaacgggc aagcacttct 120
gtctccccgg tagaacggta taggtgtgac ccacggcgca aaactgaact atcgttaccc 180
gactccgtac ttccgaaagc ttgtaggaa actggaaagt tcgagttatt gaccocggagt 240
gttcccccca ctccagaaac gcgtgatgag ggttgccacc ccgaccatgg cgacatgggtg 300
ggcatccctg cgctggcaag cggcctctaa gaggataact cgctcctact ggtaaccgaa 360
gagccccgtg agctacgggt tattcctccg cctccctgaa tgcgggtaat cctaaccocat 420
gagcagttgc catagatcca tatggtggac tgtcgtaacg cgtaagttgt gggcggaacc 480
gactactttg ggatggcggt ttctcttgtt ttctccattt gttgtgtgat ggtgacaagt 540
tatagatctc gatctatagc gtttcttgag agatttccaa acatttattc aagtcgtaca 600
attcttgtgt ttaagcagta cagtgttaagg 630

```

```

SEQ ID NO: 66      moltype = DNA length = 611
FEATURE           Location/Qualifiers
source            1..611
                  mol_type = genomic DNA
                  organism = Parabovirus sp.
                  strain = 5delta67

```

```

SEQUENCE: 66
tgggtctacgg ccgcggccac cttgtggcgc aaagacatta gaagaatagc ataccgcca 60
ctagggccct gcagccagca gggtaacggg caagcacttc tgtctccccg gtagaacggt 120
ataggctgta cccacggcgg aaaactgaac tatcgttacc cgactccgta cttcgcaaa 180
cttagtagga aactggaaag ttcgagttat tgacccgag tggtccccc actccagaaa 240
cgcgtgatga gggttgccac ccgcaccatg gcgacatggt gggcatccct gcgtggcac 300
gcggcctcta agaggataac tcgctcctac tggtaaccga agagccccgt gagctacggt 360
ttattctctc gcctccctga atcgcgctaa tcctaaccga tgagcagttg ccatagatcc 420
atatggtgga ctgtcgtaac gcgtaagttg tgggcggaac cgactacttt gggatggcgt 480
gtttccttgt tttctcattt tgttgttgta tggtgacaag ttatagatct cgatctatag 540
cgtttcttga gagatttcca aacatttatt caagtcgtac aattcttgtg ttaagcaggt 600
acagtgaag g 611

```

-continued

SEQ ID NO: 67 moltype = DNA length = 618
 FEATURE Location/Qualifiers
 source 1..618
 mol_type = genomic DNA
 organism = Parabovirus sp.
 strain = 3delta60

SEQUENCE: 67

tgaaccgtta	cgcaaccactc	agttggtgtt	tgggtggcacc	aatgatggaa	caaaaggcta	60
caccacttgg	gctacggccc	gcgccacctt	gtggcgcaaa	gacattagaa	gaatagcata	120
ccgcccacta	gggcccgtga	gccagcaggg	taacggggcaa	gcactttctg	ctccccggt	180
gaacgggtata	ggctgtaccc	acggcgcaaa	actgaactat	cgttaccoga	ctccgtactt	240
cgcaaacgct	agtaggaaac	tggaaagtgc	gagttattga	cccgagtggt	tccccccact	300
ccagaaacgc	gtgatgaggg	ttgccacccc	gacccatggcg	acatgggtggg	catccctgcg	360
ctggcacgcg	gcctctaaga	ggataactcg	ctccactagg	taaccgaaga	gccccgtgag	420
ctacggttta	ttcctccgcc	tccttgaatg	cggtcaatcc	taaccatga	gcagttgcca	480
tagatccata	tggtggactg	tcgtaacgcg	taagttgtgg	gcggaaccga	ctactttggg	540
atggcggtgt	tccttgtttt	ctccatttgt	tggtgtatgg	tgacaagtta	tagatctcga	600
tctatagcgt	ttgtaagg					618

SEQ ID NO: 68 moltype = DNA length = 826
 FEATURE Location/Qualifiers
 source 1..826
 mol_type = genomic DNA
 organism = Apodemus agrarius picornavirus
 note = strain Longquan-Aal18 polyprotein 1

SEQUENCE: 68

tttgaagggt	gtgcgggat	catggcggtt	ctcgccatga	tatccgcaca	ttgcaaaccc	60
atattgcata	cccactgggt	atgcatttat	gggaggcccc	tttcacccct	cccccccaa	120
ttaccttttc	cccctctagt	aaccatacgc	tttactcagc	gtaactactc	cgggttacct	180
gatgaagaag	aggctacgga	gattctcggg	ctacggccct	ggagccactc	cggctcctaa	240
agatttagaa	gtttgagcac	acccgcccac	tagggccccc	catccagggg	ggcaacgggc	300
aagcactttc	gtttcccggg	tatgatctga	taggctgtaa	ccacggctga	aacagagatt	360
atcgttatcc	gcttccactc	ttcagaagac	ctagtaatga	tgggtgaaat	tgaatccggt	420
gatccggtgt	ctccccacac	ccagaaactc	atgatgaggg	ttgccatccc	ggctacggcg	480
acgtagcggg	catccctgcg	ctggcatgag	gcctcttagg	aggacgggat	atatggatct	540
tgtcgtgaag	agcctattga	gctagtgtcg	actcctcccg	ccccgtgaat	gcggctaata	600
ctaacccccg	agcaggtggg	tccaatccag	ggcctggcct	gtcgtaatgc	gtaagtctgg	660
gacggaaacc	actactttcg	ggaaggcgtg	tttccatttg	ttcattatgt	gtgtgtttat	720
ggtgacaact	ctgggtaaac	gttctattgc	gtttattgag	agattcccaa	caattgaaca	780
aacgagaact	acctgtttta	ttaaatttac	acagagaaga	attaca		826

SEQ ID NO: 69 moltype = DNA length = 721
 FEATURE Location/Qualifiers
 source 1..721
 mol_type = genomic DNA
 organism = Apodemus picornavirus
 strain = 5delta105

SEQUENCE: 69

cccctcccc	cccaattacc	ttttcccct	ctagtaacca	tacgctttac	tcagcgtaac	60
tactccgggt	tacgtgatga	agaagaggct	acggagattc	tcgggctacg	gccttgagc	120
cactccggct	cctaaagatt	tagaagtttg	agcacaccgc	cccactaggg	ccccccatcc	180
agggggggcaa	aggtcaagca	cttctgtttc	cccggatatga	tctgataggc	tgtaaccacg	240
gctgaaacag	agattatcgt	tatccgcttc	actacttcga	gaagcctagt	aatgatgggt	300
gaaattgaat	ccgttgatcc	gggtgtctcc	ccacaccaga	aaactcatgat	gagggttgcc	360
atcccggcta	cggcgacgta	gcgggcatcc	ctgcgctggc	atgaggcctc	ttaggaggac	420
ggatgatatg	gatctttctg	tgaagagcct	attgagctag	tgtcgactcc	tcgcgccccg	480
tgaatgcccc	taatcctaac	cccggagcag	gtgggtccaa	tccagggcct	ggcctgtcgt	540
aatgcgtaag	tctgggacgg	aaccgactac	tttcgggaag	gcgtgtttcc	atttgttcat	600
tatttgtgtg	tttatggtga	caactctggg	taaacggttc	attgcgttta	ttgagagatt	660
cccaacaatt	gaacaaacga	gaactacctg	ttttattaaa	tttacacaga	gaagaattac	720
a						721

SEQ ID NO: 70 moltype = DNA length = 625
 FEATURE Location/Qualifiers
 source 1..625
 mol_type = genomic DNA
 organism = Apodemus picornavirus
 strain = 5delta201

SEQUENCE: 70

attctcgggc	tacggccctg	gagccactcc	ggctcctaaa	gatttagaag	tttgagcaca	60
cccgcccact	agggccccc	atccaggggg	gcaacgggca	agcacttctg	tttccccggt	120
atgatctgat	aggctgtaac	cacggctgaa	acagagatta	tcgttatccg	cttcactact	180
tcgagaagcc	tagtaatgat	gggtgaaatt	gaatccgttg	atccgggtgc	tccccacac	240
cagaaactca	tgatgagggg	tgccatcccg	gctacggcga	cgtacggggc	atccctgcgc	300
tggcatgagg	cctctttaga	ggacggatga	tatggatctt	gtcgtgaaga	gcctattgag	360
ctagtgtcga	ctcctccgcc	cccgtaatg	cggtcaatcc	taaccccgga	gcaggtgggt	420
ccaatccagg	gcctggcctg	tcgtaatgcg	taagtctggg	acggaaccga	ctactttcgg	480
gaaggcgtgt	ttccatttgt	tcattatttg	tgtgtttatg	gtgacaactc	tggtgaaacg	540
ttctattgcg	tttattgaga	gattcccaac	aattgaacaa	acgagaacta	cctgttttat	600

-continued

taaattttaca cagagaagaa ttaca 625

SEQ ID NO: 71 moltype = DNA length = 766
 FEATURE Location/Qualifiers
 source 1..766
 mol_type = genomic DNA
 organism = Kobuvirus sp.
 note = SZAL6-KoV/2011/HUN- complete genome.1

SEQUENCE: 71

tttcacaccc	tcttttccgg	tggctccggac	ccagaccacc	gttactccat	tcagctactt	60
cggtagctgt	tcggaggaat	taaacgggca	ccctacccaa	gggttacatg	ggaccatatt	120
cctcctcccc	tgtaacttta	agttttgtgc	ccgtattctt	gactccaggc	ggatgttggtg	180
tcgcccgtcc	tgtgaacaaa	cagctagaca	ctttcctccc	ctccctctgg	gctgctccgg	240
cagtcacactc	cctcccccca	gcgtaacatg	ccccgctgga	gtgatgcacc	tggaagtcgt	300
ggacgtgggt	tagtaacttc	ggtgaaaacc	cactataatg	acaactgggt	gacccccaca	360
ctcaaaggac	tcgagctctt	ctcccttaag	gctagcccgg	ccacatgaat	ttgcagctgg	420
caactagtga	gtccaccatg	tcccgcgaac	tcggctgcgg	agtgctgttc	cccaagcgta	480
tgccctcctt	ctgtaagagt	gcgcctggca	agcacatctg	agaagtcggt	ccgctgcgtc	540
gtgccaacct	ggcgacaggt	gacccagtgt	gcgtagactt	cttccggatt	cgtccggctc	600
ttctctagga	aacatgcgtg	taaggttcat	gtgccaaagc	cctgcgcgcg	gtgttctttt	660
actgccctag	gaatgtgcgc	caggtacccc	tacttcggtg	gggatctgag	cggtagctaa	720
ttgtctacgg	gtagtttcat	tccatcttct	tcttcaggtc	gacatc		766

SEQ ID NO: 72 moltype = DNA length = 608
 FEATURE Location/Qualifiers
 source 1..608
 mol_type = genomic DNA
 organism = Kobuvirus sp.
 strain = SZAL6 5delta158

SEQUENCE: 72

ttgactccag	gcggatgttg	tgtcgcccgt	cctgtgaaca	aacagctaga	cactttcctc	60
ccctccctct	gggtgtgtcc	ggcagtcacc	tccctccccc	cagcgtaaca	tgcctcgctg	120
gagtgatgca	cctggaagtc	gtggacgtgg	gttagtaact	tcggtgaaaa	cccaataata	180
tgaaactggg	ttgaccccca	cactcaaagg	actcgagctc	ttctccctta	aggctagccc	240
ggccacatga	atttgcagct	ggcaactagt	gagtcacaca	tgtcccgcga	cctcggtctc	300
ggagtgctgt	tcccacagcg	tatgccttcc	ttctgtaaga	gtgcgcctgg	caagcacatc	360
tgagaagtgc	ttccgctgcg	tcgtgccaac	ctggcgacag	gtgaccaggt	gtgcgtagac	420
ttcttcggga	ttctgcggcg	tcttctctag	gaaacatgcg	tgtaagggtc	atgtgccaaa	480
gccctgcgcg	cgggtgttct	ctactgccct	aggaatgtgc	cgcaggtacc	cctacttcgg	540
taaggatctg	agcggtagct	aattgtctac	gggtagtttc	atttccatct	tctcttcagg	600
tcgacatc						608

SEQ ID NO: 73 moltype = DNA length = 690
 FEATURE Location/Qualifiers
 source 1..690
 mol_type = genomic DNA
 organism = Kobuvirus sp.
 strain = SZAL6 5delta76

SEQUENCE: 73

gaattaaaacg	ggcaccctac	ccaaggggta	catggggacca	tattcctcct	ccccgtaac	60
tttaagtttt	gtgcccgtat	tcttgactcc	aggcggtatg	tgtgtgcccc	gtcctgtgaa	120
caaacagcta	gacactttcc	tccctccctc	ctgggctgct	ccggcagttc	actccctccc	180
cccagcgtaa	catgccccgc	tggagtgatg	cacctggaag	tcgtggacgt	gggttagtaa	240
cttcgggtgaa	aacccactat	aatgacaact	gggtgacccc	cacactcaaa	ggactcgagt	300
ctttctccct	taaggctagc	ccggccacat	gaatttgca	ctggcaacta	gtgagtccac	360
catgtcccgc	aacctcggct	gcggagtgct	gttccccaa	cgatgcctt	ccttctgtaa	420
gagtgcgctc	ggcaagcaca	tctgagaagt	cgttcgcgtg	cgtcgtgcca	acctggcgac	480
aggtgaccca	gtgtgcgtag	acttcttcgg	gattcgtccg	gctcttctct	aggaaacatg	540
cgtgtaaggt	tcatgtgcca	aagccctgcg	cgcggtgttc	ttctactgcc	ctaggaatgt	600
gccgcaggtg	cccctaactc	ggtagggatc	tgagcggtag	ctaattgtct	acgggtagtt	660
tcatttccat	cttctcttca	ggtcgacatc				690

SEQ ID NO: 74 moltype = DNA length = 729
 FEATURE Location/Qualifiers
 source 1..729
 mol_type = genomic DNA
 organism = Kobuvirus sp.
 strain = SZAL6 3delta37

SEQUENCE: 74

tttcacaccc	tcttttccgg	tggctccggac	ccagaccacc	gttactccat	tcagctactt	60
cggtagctgt	tcggaggaat	taaacgggca	ccctacccaa	gggttacatg	ggaccatatt	120
cctcctcccc	tgtaacttta	agttttgtgc	ccgtattctt	gactccaggc	ggatgttggtg	180
tcgcccgtcc	tgtgaacaaa	cagctagaca	ctttcctccc	ctccctctgg	gctgctccgg	240
cagtcacactc	cctcccccca	gcgtaacatg	ccccgctgga	gtgatgcacc	tggaagtcgt	300
ggacgtgggt	tagtaacttc	ggtgaaaacc	cactataatg	acaactgggt	gacccccaca	360
ctcaaaggac	tcgagctctt	ctcccttaag	gctagcccgg	ccacatgaat	ttgcagctgg	420
caactagtga	gtccaccatg	tcccgcgaac	tcggctgcgg	agtgctgttc	cccaagcgta	480
tgccctcctt	ctgtaagagt	gcgcctggca	agcacatctg	agaagtcggt	ccgctgcgtc	540
gtgccaacct	ggcgacaggt	gacccagtgt	gcgtagactt	cttccggatt	cgtccggctc	600

-continued

```

ttctctagga aacatgcgtg taaggttcat gtgccaaagc cctgcgcgcg gtgtttcttct 660
actgccctag gaatgtgcgc caggtaacccc tacttcggta gggatctgag cggtagctaa 720
ttggacatc 729

```

```

SEQ ID NO: 75      moltype = DNA length = 650
FEATURE           Location/Qualifiers
source            1..650
                  mol_type = genomic DNA
                  organism = Salivirus sp.
                  strain = SZ1

```

```

SEQUENCE: 75
tctgtectca ccccatcttc ccttctttcc tgcacggtta cgcttactcg catgtgcatt 60
gagtgggtgca cgtgcttgaa caaacagcta cactcacatg ggggcggggt ttcccgccct 120
gcggcctctc gcgaggccca cccctccctc tctcccata actacagtgc ttggtagagt 180
aagcatcctg atcccccgcg gaagctgctc acgtggcaac tgtggggacc cagacagggt 240
atcaaaggca cccggtcttt ccgccttcag gagtatccct gctagtgaat tctagtaggg 300
ctctgcttgg tgccaacctc ccccaaatgc gcgctgcggg agtgctcttc cccaactcac 360
cctagtatcc tctcatgtgt gtgcttggtc agcatatctg agacgatgtt ccgctgtccc 420
agaccagtcc agtaatggac gggccagtgt gcgtagtcgt cttccggctt gtccggcgca 480
tgtttggtga accggtgggg taaggttggt gtgcccacgc cccgtacttt ggtgatacct 540
caagaccacc caggaatgcc agggagggtac cccgcttcac agcgggatct gaccctgggc 600
taattgtcta cgggtgttct tcttgcttcc acttctttct actgttcattg 650

```

```

SEQ ID NO: 76      moltype = DNA length = 508
FEATURE           Location/Qualifiers
source            1..508
                  mol_type = genomic DNA
                  organism = Crohivirus B

```

```

SEQUENCE: 76
gtataagaga cagggtgtttg ccttgtcttc ggactggcat cttgggacca accccccttt 60
tccccagcca tgggttaaat ggcaataaag gacgtaacaa cttgttaacc attagctttt 120
gtaattttgt aaccactaag ctttgtgcac ataattgtaac catcaagctt gttagtccca 180
gcaggaggtt tgcattgttg tagccgaaat ggggctcgac ccccatagt aggatacttg 240
attttgcatt ccattgtgga cctgcaaaact ctacacatag aggcctttgtc ttgcactata 300
acacctgagt acagtgtgta cctagaccct atagtacggg aggaccgttt gtttctctaa 360
taacctaca taataggcta ggtgggcatg cccaatttgc aagatccag actgggggtc 420
ggctctggga ggggttagat cctgttagct actgcctgat aggggtgtgc tcaaccatgt 480
gtagtttaaa ttgagctggt catatacc 508

```

```

SEQ ID NO: 77      moltype = DNA length = 457
FEATURE           Location/Qualifiers
source            1..457
                  mol_type = genomic DNA
                  organism = Crohivirus B
                  strain = 5delta51

```

```

SEQUENCE: 77
ccccctttt cccagccat ggggttaaat gcaataaagg acgtaacaac ttgttaacca 60
ttaagctttg taattttgta accactaagc tttgtgcaca taatgtaacc atcaagcttg 120
ttagtccag caggaggttt gcatgcttgt agccgaaatg gggctcgacc ccccatagta 180
ggatacttga ttttgcattc cattgtggac ctgcaaaactc tacacataga ggctttgtct 240
tgcattctaa cacctgagta cagtgtgtac ctagacccta tagtacggga ggaccgtttg 300
tttctcctaa aacctacat aataggctag gtgggcattg ccaatttgca agatcccaga 360
ctgggggtcg gtctgggcag gggttagatc ctggttagcta ctgcctgata ggggtgtgtc 420
caaccatgtg tagtttaaat tgagctgttc atatacc 457

```

```

SEQ ID NO: 78      moltype = DNA length = 742
FEATURE           Location/Qualifiers
source            1..742
                  mol_type = genomic DNA
                  organism = Coxsackievirus B3
                  note = clone CV-B3/28 2A C107S- complete genome.1

```

```

SEQUENCE: 78
ttaaaacagc ctgtgggttg atccccacca cagggcccat tgggcgctag cactctggta 60
tcacggtaac tttgtgcgcc tgttttatac cccctcccc aactgtaact tagaagtaac 120
acacaccgat caacagtcag cgtggcacac cagccacgtt ttgatcaagc acttctgtta 180
ccccggactg agtatcaata gactgctcac gcggttgaag gagaaagcgt tcgttatccg 240
gccaactact tgaaaaaacc tagtaacacc gtggaagttg cagagtgttt cgctcagcac 300
tacctcagtg tagatcaggt ccatgagtca cgcattccc caccggcgac cgtggcggtg 360
gtcgcgttgg cggcctgccc atggggaaac ccatgggacg ctctaataca gacatggtgc 420
gaagagtcta ttgagctagt tggtagtcct ccggccctcg aatgcggcta atcctaactg 480
cggagcacac accctcaagc cagagggcag tgtgtcgtaa cgggcaactc tgcagcgga 540
ccgactactt tgggtgtccg tgtttcattt tattcctata ctggctgctt atggtgacaa 600
ttgagagatt gttaccatatt agctatttga ttggccatcc ggtgaccaat agagctatta 660
tatatctctt tgttgggttt ataccactta gcttgaaaga ggttaaaaca ttacaattca 720
ttgttaagtt gaatacagca aa 742

```

```

SEQ ID NO: 79      moltype = DNA length = 651
FEATURE           Location/Qualifiers
source            1..651

```

-continued

```

mol_type = genomic DNA
organism = Cocksackievirus B3
strain = 3delta91

SEQUENCE: 79
ttaaaccagc ctgtgggttg atcccaccca cagggcccat tgggcgctag cactctggta 60
tcacgggtacc tttgtgccc tgttttatac cccctccccc aactgtaact tagaagtaac 120
acacaccgat caacagtcag cgtggcacac cagccacgtt ttgatcaagc acttctgtta 180
ccccggactg agtatcaata gactgctcac gcggttgaa gagaaagcgt tcgttatccg 240
gccaactact tcgaaaaacc tagtaacacc gtggaagttg cagagtgttt cgctcagcac 300
taccaccagt tagatcaggt cgatgagtca ccgcattccc cacgggcgac cgtggcggtg 360
gctgcgttgg cggcctgccc atggggaaac ccatgggacg ctctaataca gacatggtgc 420
gaagagtcta ttgagctagt tggtagtcct ccggccctcg aatgcggcta atcctaactg 480
cggagcacac accctcaagc cagagggcag tgtgtcgtaa cgggcaactc tgcagcggaa 540
ccgactactt tgggtgtccg tgtttcattt tattcctata ctggctgctt atggtgacaa 600
ttgagagatt gttaccatat agctattgga ttggccatcc ggtgaagcaa a 651

SEQ ID NO: 80      moltype = DNA length = 721
FEATURE           Location/Qualifiers
source            1..721
                  mol_type = genomic DNA
                  organism = Saffold virus

SEQUENCE: 80
cacttattta attcggcctt ttgtgacaag cccctcgggt aaagaacctc tctcttttcg 60
acgtgggttg aattgccatc atttccgacg aaagtgcata catgcctccc cgattatgtg 120
atgttttctg cctctgtggg cggagcattc tcgggttgag aaaccttgaa tctttttctt 180
tggaaccttg gttcccccg tctaaagccg ttggaatatg acagggttat tttcttgatc 240
ttattttctc ttttgccggg tctatccgta aaaagggtag gtgctgcccc ttctctctct 300
ggagaattca cacggcggtc ttccgtctc tcaacaagtg tgaatgcagc atgccggaaa 360
cggtgaaaga aacagttttc tgtggaaatt tagagtgcac atcgaaacag ctgtagcgac 420
ctcacagtag cagcggactc cctctttggc gacaagagcc tctcggccca aaagcccgct 480
ggataagatc cactgctgtg agcggtgcaa cccagcacc ctggttcgat gatcattctc 540
tatggaacca gaaaatgggt ttctcaagcc ctccggtaga gaagccaaaga atgtcctgaa 600
ggtacccccg gtgcgggatc tgatcaggag accaattggc ggtgctttac actgtcactt 660
tggttataaa attgtcacag cttctccaaa ccaagtggtc ttggttttcc aattttgttg 720
a 721

SEQ ID NO: 81      moltype = DNA length = 675
FEATURE           Location/Qualifiers
source            1..675
                  mol_type = genomic DNA
                  organism = Saffold virus
                  strain = 5delta46

SEQUENCE: 81
cctctctctt ttgcagctgg ttggaattgc catcatttcc gacgaaagtg ctatcatgcc 60
tccccgatta tgtgatgttt tctgccctgc tgggcggagc attctcgggt tgagaaacct 120
tgaattcttt tctttggaac cttggttccc ccggtctaag ccgcttgga tatgacaggg 180
ttattttctt gatcttattt ctacttttgc ggggtctatc cgtaaaaagg gtacgtgctg 240
cccttctctt ctctggagaa ttcacacggc ggtctttccg tctctcaaca agtgtgaatg 300
cagcatgccg gaaacggatg agaaaaacagt ttctgtgga aatttagagt gcacatcgaa 360
acagctgtag gcacctcaca gtagcagcgg actccctct tggcgacaag agcctctgcg 420
gccccaaagc ccgtggataa gatccactgc tgtgagcggg gcaacccag caccctggtt 480
cgatgatcat tctctatgga accagaaaat ggttttctca agcctccgg tagagaagcc 540
aagaatgtcc tgaaggatcc ccgcgtgcgg gatctgatca ggagaccaat tggcgggtct 600
ttacactgtc acttttggtt aaaaattgtc acagcttctc caaaccaagt ggtcttggtt 660
ttccaatttt gttga 675

SEQ ID NO: 82      moltype = DNA length = 628
FEATURE           Location/Qualifiers
source            1..628
                  mol_type = genomic DNA
                  organism = Saffold virus
                  strain = 5delta93

SEQUENCE: 82
gtgctatcat gccctcccga ttatgtgatg ttttctgccc tgcctggcgg agcattctcg 60
ggttgagaaa ccttgaatct ttttctttgg aaccttggtt ccccggtct aagccgcttg 120
gaatatgaca gggttatttt ctgatcttta tttctacttt tgcgggttct atccgtaaaa 180
agggtagctg ctgccccttc cttctctgga gaattcacac ggcggctttt ccgtctctca 240
acaagtgtga atgcagcatg ccggaaacgg tgaagaaaac agttttctgt ggaaatttag 300
agtgcacatc gaaacagctg tagcgacctc acagtagcag cggactcccc tcttgccgac 360
aagagcctct cgggccccaa gccccgtgga taagatccac tgcgtgtgagc ggtgcaaccc 420
cagcacccct gttcgatgat cattctctat ggaaccagaa aatgggtttt tcaagccctc 480
cggtagagaa gccaaagaat tctgaaggt accccgcgtg cgggatctga tcaggagacc 540
aattggcggg gctttacact gtcactttgg tttaaaaatt gtcacagctt ctccaaacca 600
agtgtctctg gttttccaat tttgttga 628

SEQ ID NO: 83      moltype = DNA length = 674
FEATURE           Location/Qualifiers
source            1..674
                  mol_type = genomic DNA

```

-continued

```

organism = Saffold virus
note = 3delta47

SEQUENCE: 83
cacttattta attcggcctt ttgtgacaag cccctcgggtg aaagaacctc tctcttttcg 60
acgtgggttg aattgccatc atttccgacg aaagtgcgtat catgcctccc cgattatgtg 120
atgttttctg cctcgtctgg cggagcattc tcgggttgag aaaccttgaa tctttttctt 180
tggaaccttg gttcccccggt tctaagccgc ttggaatatg acagggttat tttcttgatc 240
ttatttctac ttttgcgggt tctatccgta aaaagggtac gtgctgcccc ttccttctct 300
ggagaattca cacggcggtc tttccgtctc tcaacaagtg tgaatgcagc atgccggaaa 360
cggatgaagaa aacagttttc tgtggaatt tagagtgcac atcgaaacag ctgtagcgac 420
ctcacagtag cagcgggact cctctctggc gacaagagcc tctgcggcca aaagcccggt 480
ggataagatc cactgctgtg agcgggtgcaa ccccgagcacc ctggttcgat gatcattctc 540
tatggaacca gaaaatgggt ttctcaagcc ctccggtaga gaagccaaga atgtcctgaa 600
ggtaacccgc gtgcgggatc tgatcaggag accaattggc ggtgctttac actgtcactt 660
tggtttaatg ttga 674

SEQ ID NO: 84      moltype = DNA length = 726
FEATURE           Location/Qualifiers
source            1..726
                  mol_type = genomic DNA
                  organism = Saffold virus
                  note = Kozak

SEQUENCE: 84
cacttattta attcggcctt ttgtgacaag cccctcgggtg aaagaacctc tctcttttcg 60
acgtgggttg aattgccatc atttccgacg aaagtgcgtat catgcctccc cgattatgtg 120
atgttttctg cctcgtctgg cggagcattc tcgggttgag aaaccttgaa tctttttctt 180
tggaaccttg gttcccccggt tctaagccgc ttggaatatg acagggttat tttcttgatc 240
ttatttctac ttttgcgggt tctatccgta aaaagggtac gtgctgcccc ttccttctct 300
ggagaattca cacggcggtc tttccgtctc tcaacaagtg tgaatgcagc atgccggaaa 360
cggatgaagaa aacagttttc tgtggaatt tagagtgcac atcgaaacag ctgtagcgac 420
ctcacagtag cagcgggact cctctctggc gacaagagcc tctgcggcca aaagcccggt 480
ggataagatc cactgctgtg agcgggtgcaa ccccgagcacc ctggttcgat gatcattctc 540
tatggaacca gaaaatgggt ttctcaagcc ctccggtaga gaagccaaga atgtcctgaa 600
ggtaacccgc gtgcgggatc tgatcaggag accaattggc ggtgctttac actgtcactt 660
tggtttaaaa attgtcacag cttctccaaa ccaagtgggc ttggttttcc aattttgttg 720
accgcc 726

SEQ ID NO: 85      moltype = DNA length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = GLuc CK dCTG1

SEQUENCE: 85
gtggccacgc ccgggccacc gatacttccc ttcactcctt cgggactggt ggggaggaac 60
acaacagggc tcccctgttt tcccattcct tcccctttt cccaacccca accgcggtat 120
ctgggtggcg caagacacac ggggtctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtcctcctgc gtacgggtgc gtagtgctcc caccacaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcgatg tccgctgcgg atgcgcagcg ggtaaccgggt 300
tccccagtg gtgtagtgcg atcttccagg tctcctcggg tggcggtgtc cagaaactgc 360
ttcaggtaag tgggggtgtg ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtaccccagc aacagctggg atctgaccgg aggcataattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta cgtctgacaa ctactgact atccacttgc 540
tctaaagtc 549

SEQ ID NO: 86      moltype = DNA length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = GLuc CK dCTG1_2

SEQUENCE: 86
gtggccacgc ccgggccacc gatacttccc ttcactcctt cgggactggt ggggaggaac 60
acaacagggc tcccctgttt tcccattcct tcccctttt cccaacccca accgcggtat 120
ctgggtggcg caagacacac ggggtctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtcctcctgc gtacgggtgc gtagtgctcc caccacaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcgatg tccgctgcgg atgcgcagcg ggtaaccgggt 300
tccccagtg gtgtagtgcg atcttccagg tctcctcggg tggcggtgtc cagaaactgc 360
ttcaggtaag tgggggtgtg ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtaccccagc aacagctggg atctgaccgg aggcataattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta cgtcgtacaa ctactgact atccacttgc 540
tctaaagtc 549

SEQ ID NO: 87      moltype = DNA length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = GLuc CK dCTG1_2_3

```

-continued

SEQUENCE: 87

```

gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaa 60
acaacagggc tcccctgttt tcccatctct tccccctttt cccaacccca accgccgtat 120
ctgggtggcgg caagacacac ggggtctttcc ctctaaagca caattgtgtg tgtgtcccag 180
gtcctcctgc gtacgggtgc ggagtgtctc cacccaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcatgt tccgctgcgg atgccgaccg ggtaaccgg 300
tccccagtg gtgtagtgcg atcttccagg tctcctcgtg tggcgtgtgc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc acctaggaa 420
tgctccggag gtacccacgc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta cgtcgtacaa ctacagtact atccacttgc 540
tctaaagtc

```

SEQ ID NO: 88

FEATURE

source

moltype = DNA length = 549

Location/Qualifiers

1..549

mol_type = unassigned DNA

organism = unidentified

note = GLuc CK dAll

SEQUENCE: 88

```

gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaa 60
acaacagggc tcccctgttt tcccatctct tccccctttt cccaacccca accgccgtat 120
ctgggtggcgg caagacacac ggggtctttcc ctctaaagca caattgtgtg tgtgtcccag 180
gtcctcctgc gtacgggtgc ggagtgtctc cacccaactg ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcatgt tccgctgcgg atgccgaccg ggtaaccgg 300
tccccagtg gtgtagtgcg atcttccagg tctcctcgtg tggcgtgtgc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc acctaggaa 420
tgctccggag gtacccacgc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta cgtcgtacaa ctacagtact actcactgtc 540
tctaaagtc

```

SEQ ID NO: 89

FEATURE

source

moltype = DNA length = 549

Location/Qualifiers

1..549

mol_type = unassigned DNA

organism = unidentified

note = CK SZ1-L1S

SEQUENCE: 89

```

gggggtgggg ggggcctcgg cccctcacc ctcttttccg gtggccacgc cggggccacc 60
gatacttccc ttcactcctt cgggactgtt ggggaggaa acaacagggc tcccctgttt 120
tcccatctct tccccctttt cccaacccca accgccgtat ctgggtggcgg caagacacac 180
gggtctttcc ctctaaagca caattgtgtg tgtgtcccag gtctcctgc gtacgggtgc 240
ggagtgtctc cacccaactg ttgtaagcct gtccaacgcg tcgtcctggc aagactatga 300
cgtcgcatgt tccgctgcgg atgccgaccg ggtaaccgg tccccagtg gtgtagtgcg 360
atcttccagg tctcctcgtg tggcgtgtgc cagaaactgc ttcaggtaag tgggggtgtgc 420
ccaatcccta caaagggtga accaccagg aatgccaggg aggtaccccg cttcacacgc 480
ggatctgacc ctgggctaag tgtctacggg ggttcttctt gcttccactt ctttctactg 540
ttcgccacc

```

SEQ ID NO: 90

FEATURE

source

moltype = DNA length = 553

Location/Qualifiers

1..553

mol_type = unassigned DNA

organism = unidentified

note = CK Aichi Scan (AV-S)

SEQUENCE: 90

```

gggggtgggg ggggcctcgg cccctcacc ctcttttccg gtggccacgc cggggccacc 60
gatacttccc ttcactcctt cgggactgtt ggggaggaa acaacagggc tcccctgttt 120
tcccatctct tccccctttt cccaacccca accgccgtat ctgggtggcgg caagacacac 180
gggtctttcc ctctaaagca caattgtgtg tgtgtcccag gtctcctgc gtacgggtgc 240
ggagtgtctc cacccaactg ttgtaagcct gtccaacgcg tcgtcctggc aagactatga 300
cgtcgcatgt tccgctgcgg atgccgaccg ggtaaccgg tccccagtg gtgtagtgcg 360
atcttccagg tctcctcgtg tggcgtgtgc cagaaactgc ttcaggtaag tgggggtgtgc 420
ccaatcccta caaagggtga ttctttcacc acctaggaa tgctccggag gtacccacgc 480
aacagctggg atctgaccgg aggctaattg tctacgggtg gtgtttcatt tccaatcctt 540
ttatgtcgga gtc

```

SEQ ID NO: 91

FEATURE

source

moltype = DNA length = 667

Location/Qualifiers

1..667

mol_type = unassigned DNA

organism = unidentified

note = CK Aichi Loop (AV-L1)

SEQUENCE: 91

```

gggggtgggg ggggcctcgg cccctcacc ctcttttccg gtggccacgc cggggccacc 60
gatacttccc ttcactcctt cgggactgtt ggggaggaa acaacagggc tcccctgttt 120
tcccatctct tccccctttt cccaacccca accgccgtat ctgggtggcgg caagacacac 180
gggtctttcc ctctaaagca caattgtgtg tgtgtcccag gtctcctgc gtacgggtgc 240
ggagtgtctc cacccaactg ttgtaagcct gtccaacgcg tcgtcctggc aagactatga 300
cgtcgcatgt tccgctgcgg atgccgaccg ggtaaccgg tccccagtg gtgtagtgcg 360

```


-continued

```

atcttccagg tectctcgtg tggcggtgtc cagaaactgc ttcaggtaag tgggggtgtgc 420
ccaatcccta caaagggtga actgccctag gaatgccagg caggtacccc acctccgggt 480
gggactctgag cctgggctaa ttgtctacgg gtagttttcc tttttctttt cacacaactc 540
tactgctgac aactcaactga ctatccactt gctctctgtg gcctttctgc tctgggtcaa 600
gttccttgat tgtttttgac tgcttttcac tgcttttctt ctcacaatcc ttgctcagtt 660
caaagtc 667

```

```

SEQ ID NO: 92      moltype = DNA length = 693
FEATURE           Location/Qualifiers
source            1..693
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = CK SZ1-L2

```

```

SEQUENCE: 92
gggggtgggg ggggcctcgg cccctcacc ctcttttccg gtggccacgc cggggccacc 60
gatacttccc ttcactcctt cgggactgtt ggggaggaac acaacagggc tccccgtgtt 120
tccattcctt tccccctttt cccaacccca accgcggtat ctggtggcgg caagacacac 180
gggtctttcc ctctaaagca caattgtgtg tgtgtcccag gtctctctgc gtacggtgcg 240
ggagtgtccc caccacaactg ttgtaagcct gtccaacgca tgatcccccg cggaagctgc 300
tcacgtggca actgtgggga ccagacagg ttatcaagg caccggtctt ttccgcttc 360
aggagtatcc ctgctagtga attctagtag ggctctgctt gcgttgtcca gaaactgctt 420
caggtaagtg ggggtgtgcc aatccctaca aagggtgatt ctttcaccac cttaggaatg 480
ctccggaggt accccagcaa cagctgggat ctgaccggag gctaattgtc tacgggtggt 540
gtttcctttt tcttttcaca caactctact gctgacaact cactgactat ccacttgctc 600
tcttgctgct ttctgctctg gttcaagttc cttgattgtt tttgactgct tttcactgct 660
tttcttctca caatccttgc tcaagttcaaa gtc 693

```

```

SEQ ID NO: 93      moltype = DNA length = 671
FEATURE           Location/Qualifiers
source            1..671
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = CK Aichi TriLoop (AV-L2)

```

```

SEQUENCE: 93
gggggtgggg ggggcctcgg cccctcacc ctcttttccg gtggccacgc cggggccacc 60
gatacttccc ttcactcctt cgggactgtt ggggaggaac acaacagggc tccccgtgtt 120
tccattcctt tccccctttt cccaacccca accgcggtat ctggtggcgg caagacacac 180
gggtctttcc ctctaaagca caattgtgtg tgtgtcccag gtctctctgc gtacggtgcg 240
ggagtgtccc caccacaactg ttgtaagcct gtccaacgca tgtgcctggc aagcatatct 300
gagaagggtg tccgctgtgg ctgccaacct ggtgacaggt gccccagtggt gcgtaacctt 360
cttcgctctc cggacggtgc gttgtccaga aactgcttca ggtaagtggt gtgtgtccaa 420
tccctacaaa ggttgattct ttaccacct taggaatgct ccggaggtag ccagcaaca 480
gctgggagct gaccggaggg taattgtcta cgggtggtgt ttcctttttc ttttcacaca 540
actctactgc tgacaactca ctgactatcc acctgctctc ttgtgccttt ctgctctggt 600
tcaagttcct tgattgtttt tgactgcttt tcaactgctt tcttctcaca atccttgctc 660
agtccaaagt c 671

```

```

SEQ ID NO: 94      moltype = DNA length = 523
FEATURE           Location/Qualifiers
source            1..523
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = CK Scan Deletion (deltaS)

```

```

SEQUENCE: 94
gggggtgggg ggggcctcgg cccctcacc ctcttttccg gtggccacgc cggggccacc 60
gatacttccc ttcactcctt cgggactgtt ggggaggaac acaacagggc tccccgtgtt 120
tccattcctt tccccctttt cccaacccca accgcggtat ctggtggcgg caagacacac 180
gggtctttcc ctctaaagca caattgtgtg tgtgtcccag gtctctctgc gtacggtgcg 240
ggagtgtccc caccacaactg ttgtaagcct gtccaacgca tctgctctgc aagactatga 300
cgtcgcatgt tccgctcggg atgccgaccg ggtaaccggt tccccagtgt gtgtagtgcg 360
atcttccagg tctctcgtgt tggcgttgtc cagaaactgc ttcaggtaag tgggggtgtc 420
ccaatcccta caaagggtga tttcttcacc accttaggaa tgctccggag gtacccacag 480
aacagctggg atctgaccgg aggctaattg tctacgggtg gtc 523

```

```

SEQ ID NO: 95      moltype = DNA length = 592
FEATURE           Location/Qualifiers
source            1..592
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = CK Loop Deletion (deltaL1)

```

```

SEQUENCE: 95
gggggtgggg ggggcctcgg cccctcacc ctcttttccg gtggccacgc cggggccacc 60
gatacttccc ttcactcctt cgggactgtt ggggaggaac acaacagggc tccccgtgtt 120
tccattcctt tccccctttt cccaacccca accgcggtat ctggtggcgg caagacacac 180
gggtctttcc ctctaaagca caattgtgtg tgtgtcccag gtctctctgc gtacggtgcg 240
ggagtgtccc caccacaactg ttgtaagcct gtccaacgca tctgctctgc aagactatga 300
cgtcgcatgt tccgctcggg atgccgaccg ggtaaccggt tccccagtgt gtgtagtgcg 360
atcttccagg tctctcgtgt tggcgttgtc cagaaactgc ttcaggtaag tgggggtgtc 420
ccaatcccta caaagggtga tttctttttt cttttcacac aactctactg ctgacaactc 480

```

-continued

```
actgactatc cacttgctct cttgtgcctt tctgctctgg ttcaagtcc ttgattgttt 540
ttgactgctt ttcactgctt ttcttctcac aatccttgct cagttcaaaag tc 592
```

```
SEQ ID NO: 96      moltype = DNA length = 572
FEATURE           Location/Qualifiers
source            1..572
                  mol_type = other DNA
                  organism = unidentified
                  note = CK Triloop Deletion (deltaL2)
```

```
SEQUENCE: 96
gggggtgggg ggggctcgg cccctcacc ctcttttccg gtggccacgc cggggccacc 60
gatacttccc ttcactcctt cgggactgtt ggggaggaac acaacagggc tcccctgttt 120
tcccattcct tcccctttt cccaacccca accgcctgat ctggtggcgg caagacacac 180
gggtcttttc ctctaaagca caattgtgtg tgtgtcccag gtctcctgc gtacgggtgcg 240
ggagtgcctc caccactgt ttgtaagcct gtccaacgcg cgtgtgccag aaactgcttc 300
aggtaaagtgg ggtgtgccc atccctacaa aggttgattc tttcaccacc ttaggaatgc 360
tccggaggtta cccagcaaac agctgggatc tgaccggagg ctaattgtct acgggtggtg 420
ttctcttttt cttttcacac aactctactg ctgacaactc actgactatc cacttgctct 480
cttgtgcctt tctgctctgg ttcaagtcc ttgattgttt ttgactgctt ttcactgctt 540
ttcttctcac aatccttgct cagttcaaaag tc 572
```

```
SEQ ID NO: 97      moltype = DNA length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = GLuc CK dCTG1
```

```
SEQUENCE: 97
gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaac 60
acaacagggc tcccctgttt tcccattcct tcccctttt cccaacccca accgcctgat 120
ctggtggcgg caagacacac gggctctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtctcctgc gtacgggtgcg ggagtgcctc caccactgt ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcctgt tccgtgcgg atgcccagcg ggtaacgggt 300
tcccagtggt gtgtagtgcg atcttccagg tctcctcgtt tggcgttgtc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtaccccagc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta cgtctgacaa ctcactgact atccacttgc 540
tctaaagtc
```

```
SEQ ID NO: 98      moltype = DNA length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = other DNA
                  organism = unidentified
                  note = GLuc CK dCTG1_2
```

```
SEQUENCE: 98
gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaac 60
acaacagggc tcccctgttt tcccattcct tcccctttt cccaacccca accgcctgat 120
ctggtggcgg caagacacac gggctctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtctcctgc gtacgggtgcg ggagtgcctc caccactgt ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcctgt tccgtgcgg atgcccagcg ggtaacgggt 300
tcccagtggt gtgtagtgcg atcttccagg tctcctcgtt tggcgttgtc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtaccccagc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta cgtctgacaa ctcactgact atccacttgc 540
tctaaagtc
```

```
SEQ ID NO: 99      moltype = DNA length = 549
FEATURE           Location/Qualifiers
source            1..549
                  mol_type = unassigned DNA
                  organism = unidentified
                  note = GLuc CK dCTG1_2_3
```

```
SEQUENCE: 99
gtggccacgc cggggccacc gatacttccc ttcactcctt cgggactgtt ggggaggaac 60
acaacagggc tcccctgttt tcccattcct tcccctttt cccaacccca accgcctgat 120
ctggtggcgg caagacacac gggctctttc ctctaaagca caattgtgtg tgtgtcccag 180
gtctcctgc gtacgggtgcg ggagtgcctc caccactgt ttgtaagcct gtccaacgcg 240
tcgtcctggc aagactatga cgtcgcctgt tccgtgcgg atgcccagcg ggtaacgggt 300
tcccagtggt gtgtagtgcg atcttccagg tctcctcgtt tggcgttgtc cagaaactgc 360
ttcaggtaag tgggggtgtgc ccaatcccta caaagggtga ttctttcacc accttaggaa 420
tgctccggag gtaccccagc aacagctggg atctgaccgg aggctaattg tctacgggtg 480
gtgtttcctt tttcttttca cacaactcta cgtctgacaa ctcactgact atccacttgc 540
tctaaagtc
```

```
SEQ ID NO: 100     moltype = DNA length = 167
FEATURE           Location/Qualifiers
source            1..167
                  mol_type = other DNA
```

-continued

```

                                organism = synthetic construct
SEQUENCE: 100
ccctgcagcc gtcaccgtaa gtttgaagtt accgcataac agcctctgct tcccagcgcg 60
tccaattcct gttcttattg ttcccccctcc aggcgttacg cgtgacgacg aactgtgtcg 120
cagctaccac attattccgg agccttcatt ctgcgggctc tgatcgt 167

SEQ ID NO: 101      moltype = DNA length = 43
FEATURE            Location/Qualifiers
source             1..43
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 101
ggagaccgcg gccacgccga gtaggatcga gggtagagtc tcc 43

SEQ ID NO: 102      moltype = DNA length = 167
FEATURE            Location/Qualifiers
source             1..167
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 102
gacaccagga tcaactcttg cctgacccgc cctgtgtaga atagactcat gcttccctaa 60
gacctggatt tcttccagg cactttcacc cgcctgccct gctccttcag tggactgcac 120
ccaggaggag ggtctctgac tgccttttac tttctattct ggattgc 167

SEQ ID NO: 103      moltype = DNA length = 30
FEATURE            Location/Qualifiers
source             1..30
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 103
aaacccccct aagccgcccgc cgccgccacc 30

SEQ ID NO: 104      moltype = DNA length = 18
FEATURE            Location/Qualifiers
source             1..18
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 104
ccccccaac ccgtcagc 18

SEQ ID NO: 105      moltype = length =
SEQUENCE: 105
000

SEQ ID NO: 106      moltype = DNA length = 141
FEATURE            Location/Qualifiers
source             1..141
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 106
tctgcgcact cgtaatcagt actaaccccc ctttgtcgga cactatgcga taatcgatcc 60
gcctttttca ccgccttcgg aattttattt acctcaactg atcctggagt ctctcttggt 120
tttcacggag gcctccgcc a 141

SEQ ID NO: 107      moltype = DNA length = 43
FEATURE            Location/Qualifiers
source             1..43
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 107
ggagaccgcg gccacgccga gtaggatcga gggtagagtc tcc 43

SEQ ID NO: 108      moltype = DNA length = 141
FEATURE            Location/Qualifiers
source             1..141
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 108
ccccttgaaa cccccgcccc aggttcagtc tctcttcac cctctgtcct gcatgggtgat 60
acaaagaccc tttgtggacc ctaagccatg tagttgtcgc tccctccttc cagttgtgaa 120
tattggtttc tgtaatcac a 141

SEQ ID NO: 109      moltype = DNA length = 30
FEATURE            Location/Qualifiers
source             1..30
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 109
aaacccccct aagccgcccgc cgccgccacc 30

```

-continued

SEQ ID NO: 110	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 110		
ccccccaac ccgtcacg		18
SEQ ID NO: 111	moltype = length =	
SEQUENCE: 111		
000		
SEQ ID NO: 112	moltype = DNA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 112		
tgatagctaa ctag		14
SEQ ID NO: 113	moltype = DNA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 113		
tagtagctaa ctag		14
SEQ ID NO: 114	moltype = DNA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 114		
tgatgactga gtga		14
SEQ ID NO: 115	moltype = DNA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 115		
tagtagctag gtag		14
SEQ ID NO: 116	moltype = length =	
SEQUENCE: 116		
000		
SEQ ID NO: 117	moltype = DNA length = 14	
FEATURE	Location/Qualifiers	
source	1..14	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 117		
taatagctaa ctag		14
SEQ ID NO: 118	moltype = DNA length = 15	
FEATURE	Location/Qualifiers	
source	1..15	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 118		
taactagcta actag		15
SEQ ID NO: 119	moltype = length =	
SEQUENCE: 119		
000		
SEQ ID NO: 120	moltype = DNA length = 43	
FEATURE	Location/Qualifiers	
source	1..43	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 120		
aataagagag aaaagaagag taagaagaaa tataagagcc acc		43
SEQ ID NO: 121	moltype = DNA length = 49	
FEATURE	Location/Qualifiers	

-continued

```

source                1..49
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 121
cgaactagta ttcttctggt cccacagac tcagagagaa cccgccacc          49

SEQ ID NO: 122        moltype =   length =
SEQUENCE: 122
000

SEQ ID NO: 123        moltype = DNA length = 30
FEATURE              Location/Qualifiers
source               1..30
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 123
tataattcta ccctattgag gcattgacta          30

SEQ ID NO: 124        moltype = AA length = 5
FEATURE              Location/Qualifiers
source               1..5
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 124
GGGGS          5

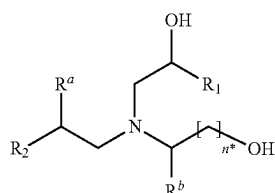
SEQ ID NO: 125        moltype = AA length = 15
FEATURE              Location/Qualifiers
source               1..15
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 125
GGGSGGGGS GGGGS          15

```

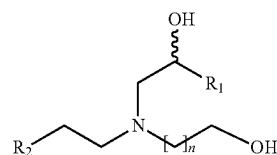
What is claimed is:

1. An ionizable lipid represented by Formula (13*):



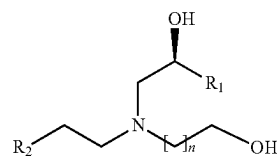
Formula (13*)

35



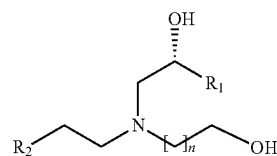
Formula (13a-1)

40



Formula (13a-2)

45



Formula (13a-3)

50

55

or a pharmaceutically acceptable salt thereof, wherein:

n^* is an integer from 1 to 7;

R^a is hydrogen;

R^b is hydrogen or C_1 - C_6 alkyl;

R_1 and R_2 are each independently unsubstituted, linear, or branched C_6 - C_{30} alkyl or $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)R^9$, wherein at least one of R_1 and R_2 is $-(CH_2)_qOC(O)(CH_2)_rCH(R^8)R^9$,

wherein:

q is an integer from 0 to 12,

r is 0,

R^8 is H or R^{10} , and

R^9 and R^{10} are independently unsubstituted linear C_1 - C_{12} alkyl or unsubstituted linear C_2 - C_{12} alk-enyl.

2. The ionizable lipid of claim 1, wherein the ionizable lipid is represented by Formula (13a-1), Formula (13a-2), or Formula (13a-3):

wherein n is 1, 2, 3, or 4.

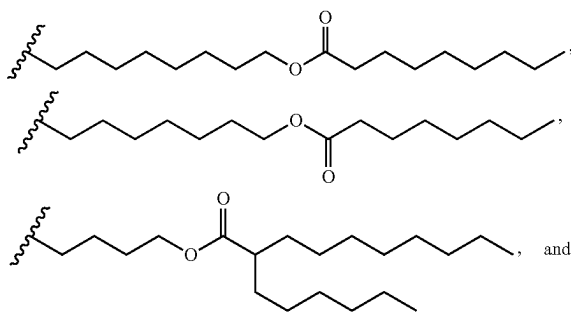
3. The ionizable lipid of claim 1, wherein q is an integer from 1 to 6.

4. The ionizable lipid of claim 3, wherein q is 3, 4, 5, or 6.

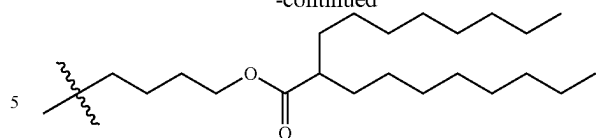
5. The ionizable lipid of claim 1, wherein R^8 is R^{10} .

6. The ionizable lipid of claim 5, wherein R^9 and R^{10} are each independently unsubstituted linear C_4 - C_8 alkyl.

7. The ionizable lipid of claim 1, wherein R_1 and R_2 are each independently selected from:

471**472**

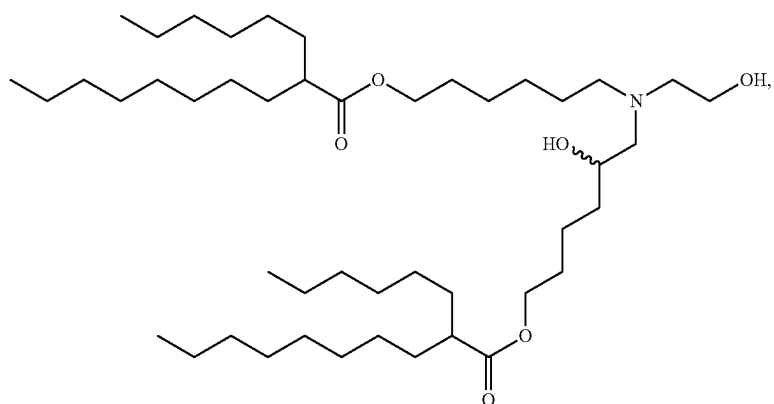
-continued



8. The ionizable lipid of claim 7, wherein R_1 and R_2 are the same.

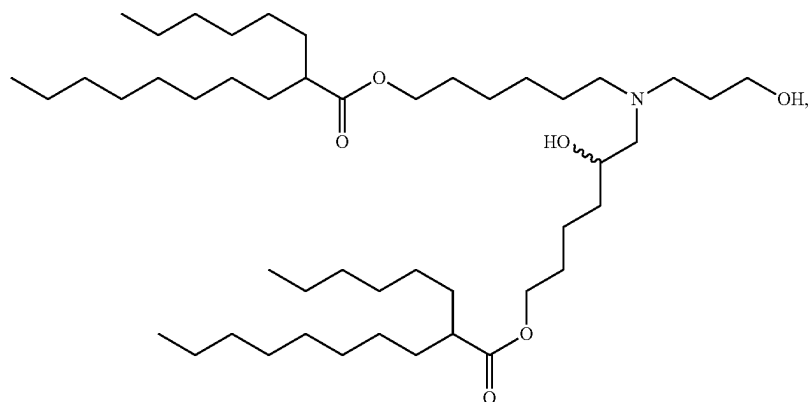
9. The ionizable lipid of claim 7, wherein R_1 and R_2 are different.

10. The ionizable lipid of claim 1, wherein the ionizable lipid is:



or is a pharmaceutically acceptable salt thereof.

11. The ionizable lipid of claim 1, wherein the ionizable lipid is:

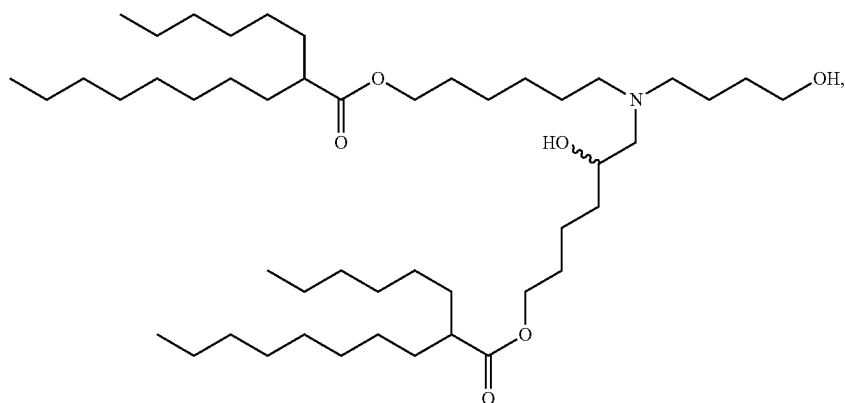


or is a pharmaceutically acceptable salt thereof.

473

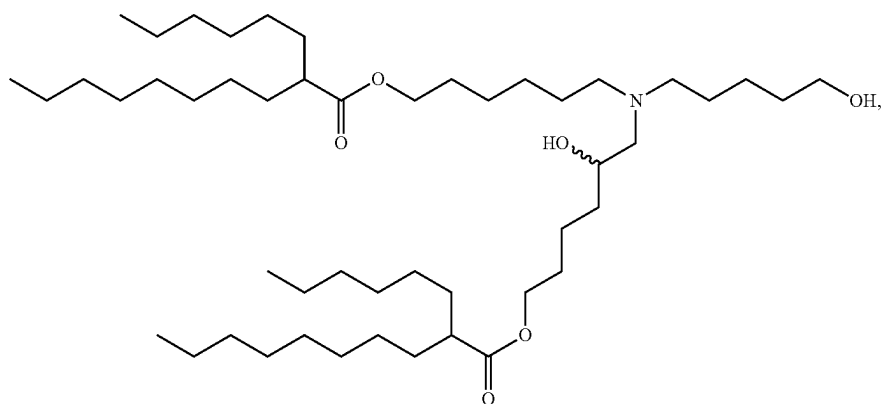
474

12. The ionizable lipid of claim 1, wherein the ionizable lipid is:



or is a pharmaceutically acceptable salt thereof.

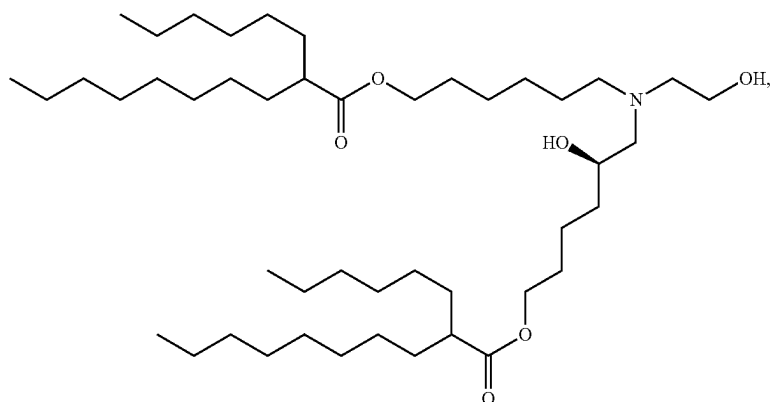
13. The ionizable lipid of claim 1, wherein the ionizable lipid is:



or is a pharmaceutically acceptable salt thereof.

45

14. The ionizable lipid of claim 1, wherein the ionizable lipid is:

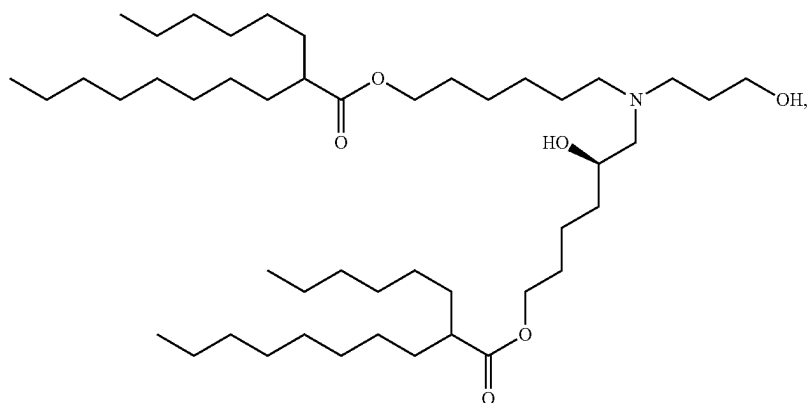


or is a pharmaceutically acceptable salt thereof.

475

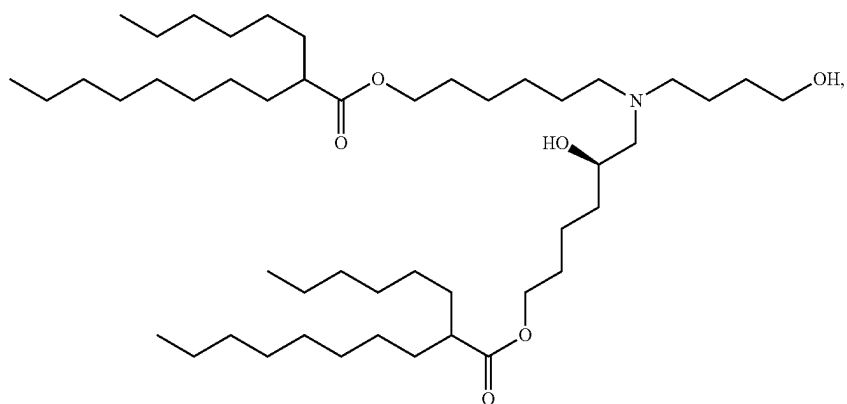
476

15. The ionizable lipid of claim 1, wherein the ionizable lipid is:



or is a pharmaceutically acceptable salt thereof.

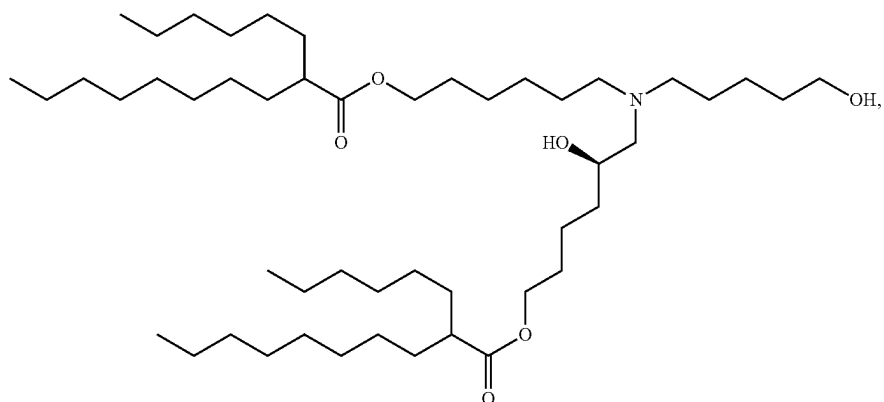
16. The ionizable lipid of claim 1, wherein the ionizable lipid is:



or is a pharmaceutically acceptable salt thereof.

45

17. The ionizable lipid of claim 1, wherein the ionizable lipid is:

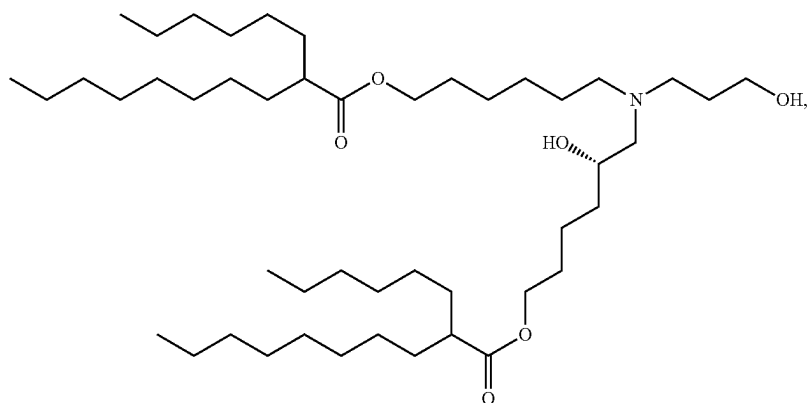


or is a pharmaceutically acceptable salt thereof.

477

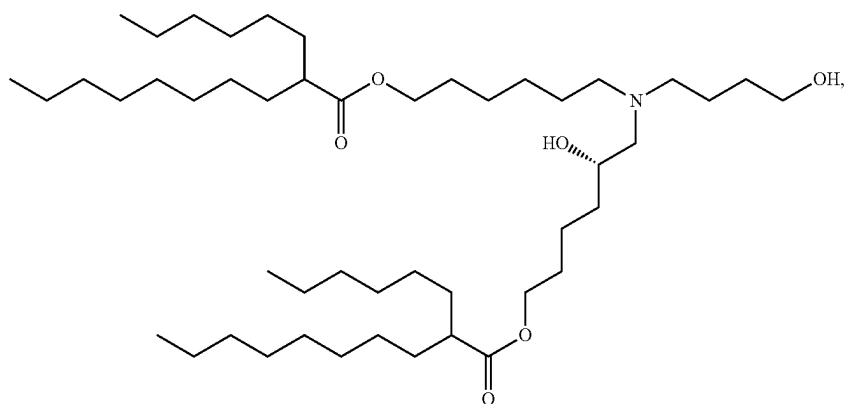
478

18. The ionizable lipid of claim 1, wherein the ionizable lipid is:



or is a pharmaceutically acceptable salt thereof.

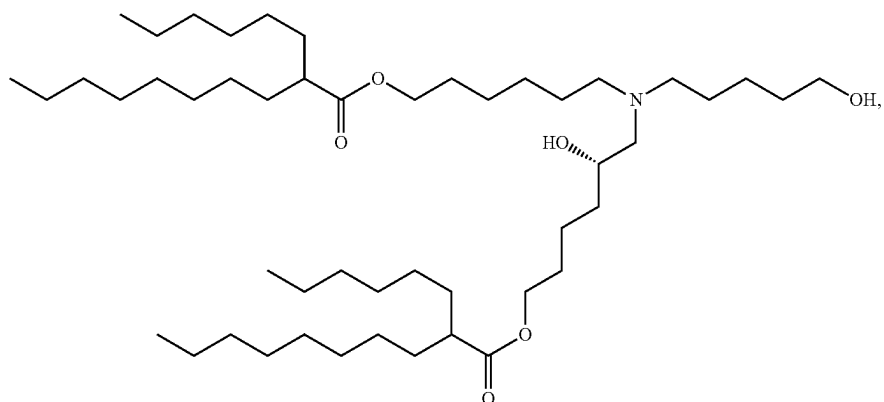
19. The ionizable lipid of claim 1, wherein the ionizable lipid is:



or is a pharmaceutically acceptable salt thereof.

45

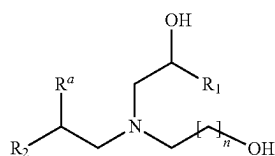
20. The ionizable lipid of claim 1, wherein the ionizable lipid is:



or is a pharmaceutically acceptable salt thereof.

479

21. An ionizable lipid represented by Formula (13):



Formula (13)

or a pharmaceutically acceptable salt thereof, wherein:

n is 1, 2, 3, or 4;

R_a is hydrogen;

R₁ and R₂ are each independently —(CH₂)_qOC(O) 15
CH₂)_rCH(R⁸)(R⁹), wherein:

q is 3, 4, 5, or 6,

r is 0,

R⁸ is R¹⁰, and

R⁹ and R¹⁰ are independently unsubstituted linear
C₄-C₈ alkyl.

480

22. The ionizable lipid of claim 21, wherein R₁ and R₂ are the same.

23. The ionizable lipid of claim 21, wherein R⁹ and R¹⁰ are different.

24. A pharmaceutical composition comprising the ionizable lipid of claim 21.

25. The pharmaceutical composition of claim 24, wherein the pharmaceutical composition further comprises a RNA polynucleotide.

26. The pharmaceutical composition of claim 25, wherein the RNA polynucleotide is a linear RNA polynucleotide.

27. The pharmaceutical composition of claim 25, wherein the RNA polynucleotide is a circular RNA polynucleotide.

28. The pharmaceutical composition of claim 24, wherein the pharmaceutical composition further comprises:
a helper lipid, wherein the helper lipid is DOPE or DSPC, cholesterol, and
a PEG-lipid, wherein the PEG-lipid is DSPE-PEG(2000) or DMG-PEG(2000).

* * * * *